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Sample Paper 01

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

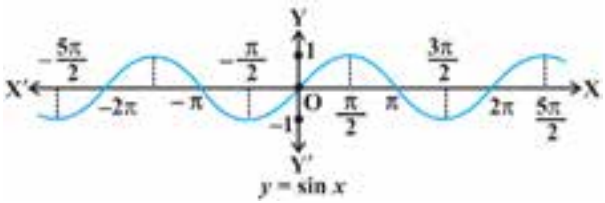
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Multiple Choice Questions each question carries 1 mark.

1. At which point does the graph of $y = \sin x$ intersect the origin in the given image?



(a) $x = -\pi$ (b) $x = 0$

(c) $x = \pi$ (d) $x = 2\pi$

Sol :

From the graph, we can see that the sine curve starts from the origin O(0,0). This means that at $x = 0$, the value of $y = \sin 0 = 0$. The curve passes exactly through this central point where the X-axis and Y-axis intersect. None of the other options $(-\pi, \pi, 2\pi)$ coincide with the origin in the graph. Hence, the correct intersection point is at (0,0).

Thus (b) is correct option.

2. If A and B are two independent events such that $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{3}$, then P (neither A nor B) is equal to

(a) $\frac{2}{3}$ (b) $\frac{1}{6}$

(c) $\frac{5}{6}$ (d) $\frac{1}{3}$

Sol :

Since, A and B are independent events.

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

Now, $$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{2} + \frac{1}{3} - \frac{1}{6} = \frac{2}{3}$$

$$P(\overline{A} \cap \overline{B}) = 1 - P(A \cup B)$$

$$= 1 - \frac{2}{3} = \frac{1}{3}$$

Thus (d) is correct option.

3. The distance of the plane $6x - 3y + 2z - 14 = 0$ from the origin is

(a) 2 (b) 1

(c) 14 (d) 8

Sol :

The distance from origin (0,0,0) to the plane $6x - 3y + 2z - 14 = 0$ is

$$d = \frac{|6(0) - 3(0) + 2(0) - 14|}{\sqrt{36 + 9 + 4}} = 2$$

Thus (a) is correct option.

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4. If $P(A \cup B) = 0.83$, $P(A) = 0.3$ and $P(B) = 0.6$, then the events will be

(a) dependent

(b) independent

(c) cannot say anything

(d) None of the above

Sol :

Given, $P(A \cup B) = 0.83,\ P(A) = 0.3$

and $P(B) = 0.6$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.83 = 0.3 + 0.6 - P(A \cap B)$$

$$P(A \cap B) = 0.9 - 0.83 = 0.07$$

Now, $P(A) \cdot P(B) = 0.3 \times 0.6 = 0.18$

$$P(A) \cdot P(B) \neq P(A \cap B)$$

Hence, events are dependent.

Thus (a) is correct option.

5. If $\vec{x}$ and $\vec{y}$ are unit vectors and $\vec{x} \cdot \vec{y} = 0$, then

(a) $|\vec{x} + \vec{y}| = 1$ (b) $|\vec{x} + \vec{y}| = \sqrt{3}$

(c) $|\vec{x} + \vec{y}| = 2$ (d) $|\vec{x} + \vec{y}| = \sqrt{2}$

Sol :

Given, $|\vec{x}| = |\vec{y}| = 1$ and $\vec{x} \cdot \vec{y} = 0$

$$|\vec{x} \cdot \vec{y}|^2 = |\vec{x}|^2 + |\vec{y}|^2 + 2(\vec{x} \cdot \vec{y})$$

$$|\vec{x} + \vec{y}|^2 = 1 + 1 + 0$$

$$|\vec{x} + \vec{y}| = \sqrt{2}$$

Thus (d) is correct option.

6. $\int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x}\, dx$ is equal to

(a) 0 (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) π

Sol :

$$I = \int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x}\, dx \quad \text{...(i)}$$

$$= \int_0^{\pi/2} \frac{\sin\left(\frac{\pi}{2} - x\right) - \cos\left(\frac{\pi}{2} - x\right)}{1 + \sin\left(\frac{\pi}{2} - x\right)\cos\left(\frac{\pi}{2} - x\right)}\, dx$$

$$= \int_0^{\pi/2} \frac{\cos x - \sin x}{1 + \cos x \sin x}\, dx \quad \text{...(ii)}$$

Adding Eqs. (i) and (ii), we get

$$2I = \int_0^{\pi/2} \frac{0}{1 + \sin x \cos x}\, dx = 0 \Rightarrow I = 0$$

Thus (a) is correct option.

7. The area bounded by the curve $y = \frac{1}{2}x^2$, the X-axis and the ordinate $x = 2$ is

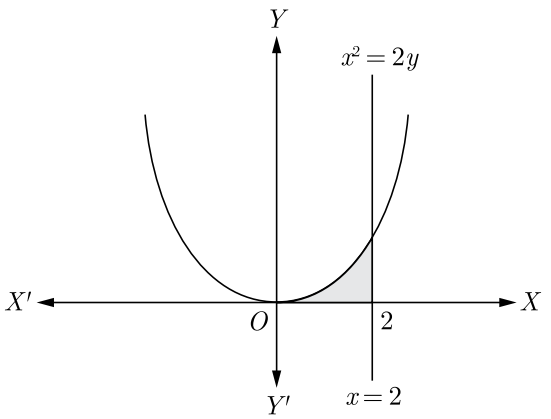
(a) $\frac{1}{3}$ sq unit (b) $\frac{2}{3}$ sq unit

(c) 1 sq unit (d) $\frac{4}{3}$ sq unit

Sol :

$$\text{Required area} = \int_0^2 y\, dx$$

$$= \int_0^2 \frac{x^2}{2}\, dx = \left[\frac{x^3}{6}\right]_0^2 = \frac{4}{3} \text{ sq units}$$



Thus (d) is correct option.

8. The solution of $\frac{dy}{dx} = \frac{ax + g}{by + f}$ represents a circle, when

(a) $a = b$

(b) $a = -b$

(c) $a = -2b$

(d) $a = 2b$

Sol :

We have $\frac{dy}{dx} = \frac{ax + g}{by + f}$

$$(by + f)\, dy = (ax + g)\, dx$$

Integrating both sides, we have

$$\frac{by^2}{2} + fy = \frac{ax^2}{2} + gx + C$$

$$ax^2 - by^2 + 2gx - 2fy + C = 0$$

which represents a circle, if $a = -b$.

Thus (b) is correct option.

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19. **Assertion :** If A is a matrix of order 2×2, then $|\text{adj A}| = |\text{A}|$

Reason : $|\text{A}| = |\text{A}^{\text{T}}|$

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

Sol :

Here, $|\text{A}| = |\text{A}^{\text{T}}|$ is true.

Since, $|\text{adj A}| = |\text{A}|^{n-1}$

here, $n = 2$

So, $|\text{adj A}| = |\text{A}|^{2-1} = |\text{A}|$

Hence, both Assertion and reason are true but reason is not a correct explanation for assertion.

Thus (b) is correct option.

20. Let us define $\tan^{-1}\text{A} + \tan^{-1}\text{B} = \tan^{-1}\left(\frac{\text{A}+\text{B}}{1-\text{AB}}\right)$

Assertion : The value of $\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{1}{7}\right)$ is $\frac{\pi}{4}$.

Reason: If $x > 0, y > 0$ than $\tan^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y-x}{y+x}\right) = \frac{\pi}{4}$

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

Sol :

As we know that, if $x > 0, y > 0$ then $\tan^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y-x}{y+x}\right) = \frac{\pi}{4}$.

So, given reason is true.

Now $\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{1}{7}\right) = \tan^{-1}\left[\frac{\frac{3}{4}+\frac{1}{7}}{1-(\frac{3}{4})(\frac{1}{7})}\right]$

$= \tan^{-1}\left[\frac{\frac{21+4}{28}}{\frac{28-3}{28}}\right]$

$= \tan^{-1}\left[\frac{\frac{25}{28}}{\frac{25}{28}}\right]$

$= \tan^{-1}(1)$

$= \frac{\pi}{4}$

Hence, both Assertion and reason are true and reason is correct explanation for assertion.

Thus (a) is correct option.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. If $\vec{a} = (2, 3, -5)$ and $\vec{b} = (2, 2, 2)$, then prove that $\vec{a}$ and $\vec{b}$ are mutually perpendicular.

Sol :

We have $\vec{a} = 2\hat{i} + 3\hat{j} - 5\hat{k}$

and $\vec{b} = 2\hat{i} + 2\hat{j} + 2\hat{k}$

$\vec{a} \cdot \vec{b} = (2\hat{i} + 3\hat{j} - 5\hat{k}) \cdot (2\hat{i} + 2\hat{j} + 2\hat{k})$

$= 2 \times 2 + 3 \times 2 + (-5) \times 2$

$= 4 + 6 - 10 = 0$

$\vec{a} \perp \vec{b}$

22. Two dice are thrown. Find the probability that the numbers appearing have a sum 8 if it is known that the second die always exhibits 4.

Sol :

Consider the following events :

A = sum of the numbers appearing on two dice is 8

$= \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$

B = Second die always exhibits 4

$= \{(1, 4), (2, 4), (3, 4), (4, 4), (5, 4), (6, 4)\}$

$A \cap B = \{(4, 4)\}$

Required probability $= P(A/B)$

$= \frac{P(A \cap B)}{P(B)}$

$= \frac{n(A \cap B)}{n(B)} = \frac{1}{6}$

23. The direction ratios of a straight line are 1,3,5. Find its direction cosines.

Sol :

If l, m, n are direction cosine of line and a, b, c are direction ratios then

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15. In Fig., the shaded region is bounded by which of the following conditions?

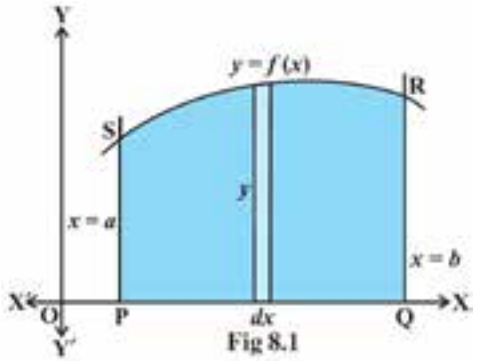


Fig 8.1

(a) x=a, x=b, the curve y=f (x), and the X-axis

(b) y=a, y=b, the curve x=f (y), and the Y-axis

(c) x=0, x=b, the line y=mx+c, and the Y-axis

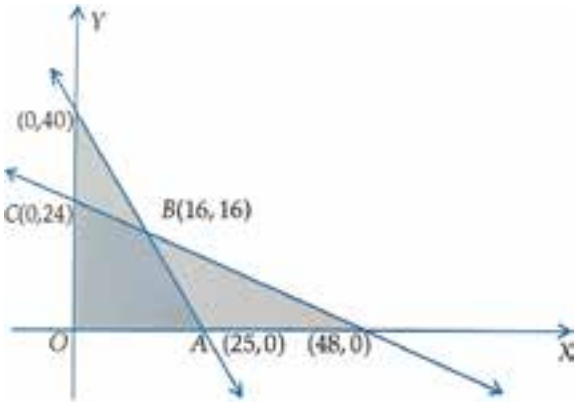
(d) y=0, y=b, the curve x=f(y), and the X-axis

Sol :

(a) x=a, x=b, the curve y=f (x), and the X-axis

The figure clearly shows the shaded area under the curve y=f(x), bounded on the left by the vertical line x=a, on the right by x=b, above by the curve y=f(x), and below by the X-axis. The boundaries are visually labeled as S, R, Q, and P, enclosing the shaded region.

16. The maximum value of $z = 4x + 3y$, if the feasible region for an LPP is as shown in the figure, is to be determined.



(a) 100 (b) 72

(c) 112 (d) None of these

Sol :

The feasible region shown in the graph is bounded by the corner points O(0,0), A(25,0), B(16,16), and C(0,24).

According to the corner point method, the value of the objective function $z = 4x + 3y$ must be calculated at each vertex of the feasible region.

At O(0,0):

$$z = 4(0) + 3(0) = 0$$

At A(25,0):

$$z = 4(25) + 3(0) = 100$$

At B(16,16):

$$z = 4(16) + 3(16) = 64 + 48 = 112$$

At C(0,24):

$$z = 4(0) + 3(24) = 72$$

Comparing all these values, the maximum value of z is 112, which occurs at the point B(16,16).

17. $+\int_2^2 \sin x\, dx + (2+2) = 4$ Which of the following function is decreasing on $(0, \pi/2)$?

(a) $\sin 2x$ (b) $\cos 3x$

(c) $\tan x$ (d) $\cos 2x$

Sol :

We have $f(x) = \cos 2x$

$$f'(x) = -2\sin 2x < 0 \text{ in } \left(0, \frac{\pi}{2}\right)$$

So, $\cos 2x$ is decreasing in $\left(0, \frac{\pi}{2}\right)$.

Thus (d) is correct option.

18. The condition that $f(x) = ax^3 + bx^2 + cx + d$ has no extreme value is

(a) $b^2 > 3ac$ (b) $b^2 = 4ac$

(c) $b^2 = 3ac$ (d) $b^2 < 3ac$

Sol :

We have $f(x) = ax^3 + bx^2 + cx + d$

Differentiating w.r.t. x, we get

$$f'(x) = 3ax^2 + 2bx + c$$

For extreme, $f'(x) = 0$, $3ax^2 + 2bx + c = 0$

Since, it has no extreme value.

$$b^2 - 4ac < 0$$

$$(2b)^2 - 4 \times 3a \times c < 0$$

$$4b^2 - 12ac < 0$$

$$b^2 - 3ac < 0$$

$$b^2 < 3ac$$

Thus (d) is correct option.

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9. If A and B are two symmetric matrices of same order. Then, the matrix $AB-BA$ is equal to

(a) a symmetric matrix

(b) a skew-symmetric matrix

(c) a null matrix

(d) the identity matrix

Sol :

We have $A = A',\ B = B'$

Now, $(AB-BA)' = (AB)'-(BA)'$

$= B'A'-A'B'$

$= BA-AB$

$= -(AB-BA)$

Hence, $AB-BA$ is a skew-symmetric matrix.

Thus (b) is correct option.

10. The family of curves $y = e^{a\sin x}$, where a is an arbitrary constant is represented by the differential equation

(a) $\log y = \tan x\dfrac{dy}{dx}$ (b) $y\log y = \tan x\dfrac{dy}{dx}$

(c) $y\log y = \sin\dfrac{dy}{dx}$ (d) $\log y = \cos x\dfrac{dy}{dx}$

Sol :

Given curve is $y = e^{a\sin x}$

On taking log both sides, we get

$$\sin x = \frac{\log y}{a}$$

$$\frac{dy}{dx} = e^{a\sin x}\cdot a\cos x \Rightarrow \frac{dy}{dx} = y\cos x\cdot\frac{\log y}{\sin x}$$

$$y\log y = \tan x\frac{dy}{dx}$$

Thus (b) is correct option.

11. The value of $\int_{-2}^{2}(x\cos x+\sin x+1)\,dx$ is

(a) 2 (b) 0

(c) -2 (d) 4

Sol :

$$\int_{-2}^{2}(x\cos x+\sin x+1)\,dx$$

$$= \int_{-2}^{2}x\cos x\,dx+\int_{-2}^{2}\sin x\,dx+\int_{-2}^{2}1\,dx$$

$$= [x\sin x-\int\sin x\,dx]_{-2}^{2}+\int_{-2}^{2}\sin x\,dx+[x]_{-2}^{2}$$

$$= 2\sin 2-2\sin 2-\int_{-2}^{2}\sin x\,dx+\int_{-2}^{2}\sin x\,dx+(2+2) = 4$$

Thus (d) is correct option.

12. $\int\dfrac{\sin 2x}{\sin^2 x+2\cos^2 x}dx$ is equal to

(a) $-\log(1+\sin^2 x)+C$

(b) $\log(1+\cos^2 x)+C$

(c) $-\log(1+\cos^2 x)+C$

(d) $\log(1+\tan^2 x)+C$

Sol :

We have $I = \int\dfrac{\sin 2x}{\sin^2 x+2\cos^2 x}dx$

$$= \int\frac{\sin 2x}{1-\cos^2 x+2\cos^2 x}dx$$

$$= \int\frac{\sin 2x}{1+\cos^2 x}dx$$

Substituting $1+\cos^2 x = t \Rightarrow -2\cos x\sin x\,dx = dt$

$$-2\sin 2x\,dx = dt$$

$$I = -\int\frac{dt}{t} = -\log|t|+C$$

$$= -\log|1+\cos^2 x|+C$$

Thus (c) is correct option.

13. If $|\vec{a}\times\vec{b}|^2+|\vec{a}\cdot\vec{b}|^2 = 144$ and $|\vec{a}| = 4$, then $|\vec{b}|$ is equal to

(a) 16 (b) 8

(c) 3 (d) 12

Sol :

$\because$ $|\vec{a}\times\vec{b}|^2+|\vec{a}\cdot\vec{b}|^2 = 144$

$$|\vec{a}|^2\cdot|\vec{b}|^2(\sin^2\theta+\cos^2\theta) = 144$$

$$16|\vec{b}|^2 = 144$$

$$|\vec{b}| = 3$$

Thus (c) is correct option.

14. If A and B are two equivalence relations defined on set C, then

(a) $A\cap B$ is an equivalence relation

(b) $A\cap B$ is not an equivalence relation

(c) $A\cap B$ is an equivalence relation

(d) $A\cap B$ is not an equivalence relation

Sol :

If A and B are equivelence relations, then $A\cap B$ is also an equivalence relation.

Thus (c) is correct option.

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$$l = \frac{a}{\sqrt{a^2+b^2+c^2}}$$

$$m = \frac{b}{\sqrt{a^2+b^2+c^2}}$$

$$n = \frac{c}{\sqrt{a^2+b^2+c^2}}$$

Given direction ratios are

$$a = 1,\ b = 3,\ c = 5$$

Then, $a^2+b^2+c^2 = 1+9+25 = 35$

Therefore, $l = \frac{1}{\sqrt{35}},\ m = \frac{3}{\sqrt{35}}$ and $n = \frac{5}{\sqrt{35}}$

24. Solve – $\frac{dy}{dx} = e^{x-y}$

OR

Write down the order and the degree of the equation:

$$8x^2\frac{d^2y}{dx^2} - 7\left(\frac{dy}{dx}\right)^2 + 9 = 0$$

Sol :

We have, $\frac{dy}{dx} = e^{x-y}$

$$\frac{dy}{dx} = e^x \cdot e^{-y}$$

$$e^y\,dy = e^x\,dx$$

$$\int e^y\,dy = \int e^x\,dx$$

$$e^y = e^x + c$$

OR

Order of a differential equation is the order of the highest differential coefficient occurring in it.

Therefore order of equation

$$8x^2\frac{d^2y}{dx^2} - 7\left(\frac{dy}{dx}\right)^2 + 9 = 0 \text{ is } 2$$

degree is power of highest order occurring in it. Therefore degree is 1.

25. Find $\frac{dy}{dx}$ if $y = \cos^{-1}\frac{1-x^2}{1+x^2}$

OR

Find $\frac{dy}{dx}$ if $y = \sqrt{\sin x}$

Sol :

We have,

$$y = \cos^{-1}\frac{1-x^2}{1+x^2}$$

$$y = 2\tan^{-1}x$$

$$\left[\because 2\tan^{-1}a = \cos^{-1}\frac{1-a^2}{1+a^2}\right]$$

$$\frac{dy}{dx} = 2\cdot\frac{d}{dx}(\tan^{-1}x)$$

$$= 2\cdot\frac{1}{1+x^2}$$

$$= \frac{2}{1+x^2}$$

OR

We have,

$$y = \sqrt{\sin x}$$

$$\frac{dy}{dx} = \frac{d}{dx}\sqrt{\sin x}$$

$$= \frac{1}{2\sqrt{\sin x}}\frac{d}{dx}(\sin x)$$

$$= \frac{1}{2\sqrt{\sin x}}\cdot\cos x$$

$$= \frac{\cos x}{2\sqrt{\sin x}}$$

SECTION - C

This section comprises of short answer-type questions (SA) of 3 marks each.

26. How fast is the volume of a ball changing with respect to its radius when radius is 3 m?

Sol :

Let V be the volume of the ball of radius r.

Then $V = \frac{4}{3}\pi r^3$

The rate of change of V with respect to

$$r = \frac{dV}{dr}$$

$$= \frac{4}{3}\pi\cdot 3r^3 = 4\pi r^2$$

$$\left(\frac{dV}{dr}\right)_{r=3\text{ m}} = 4\pi\times(3)^2$$

$$= 36\ \text{m}^3/\text{min}$$

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SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Solve the following L.P. Problem graphically.

Maximise $Z = -x + 2y$

Subject to the constraints:

$$x \geq 3$$

$$x + y \geq 5$$

$$x + 2y \geq 6$$

$$y \geq 0$$

Sol :

Maximise $Z = -x + 2y$ ...(1)

Subject to the constraints:

$$x \geq 3 \quad ...(2)$$

$$x + y \geq 5 \quad ...(3)$$

$$x + 2y \geq 6 \quad ...(4)$$

$$y \geq 0 \quad ...(5)$$

Step I

Constraint Eq. (5), $y \geq 0 \Rightarrow$ Positive side of y-axis $\Rightarrow$ Feasible region is in first and second quadrants.

Region for constraint Eq. (2) $x \geq 3$.

We know that graph of the line $x = 3$ is a vertical line parallel to y-axis at a distance 3 from origin along OX.

Hence, Region for $x \geq 3$ is the half-plane on right side of the line $x = 3$.

Table of values for line $x + y = 5$ of constraint Eq. (3).

x	0	5
y	5	0

Let us draw the line joining the points (0, 5) and (5, 0).

Let us test for origin (0,0) in constraint Eq. (2).

Putting $x = 0$ and $y = 0$ in Eq. $x + y \geq 5$, we get

$0 \geq 5$ which is not true

Hence, Region for constraint, Eq. (3) is the half plane on the non-origin side of the line $x + y = 5$.

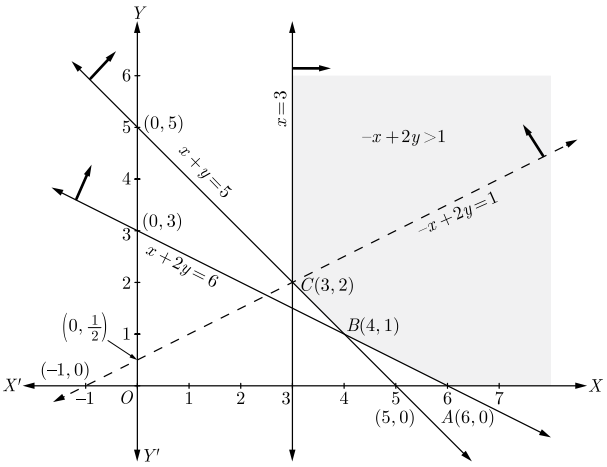
Table of values for the line $x + 2y = 6$ of constraint Eq. (4).

x	0	6
y	3	0

Let us draw the line joining the points (0, 3) and (6, 0).

Let us test for origin (0,0) in constraint Eq. (4) $x + 2y \geq 6$ which is not true.

Hence, Region for constraint Eq. (4) is again the half plane on the non-origin side of the line $x + 2y = 6$



The shaded region in the figure is the feasible region determined by the system of constraints from Eq. (2) to Eq. (5). The feasible region is unbounded.

Step II

The coordinates of the corner point A are (6, 0).

Corner point B is the intersection of the boundary lines

$$x + y = 5$$

and $$x + 2y = 6$$

Let us solve them for x and y.

Subtracting the two equations

$$2y - y = 6 - 5$$

or $$y = 1$$

Putting $y = 1$ in Eq. $x + y = 5$, we get

$$x + 1 = 5$$

$$x = 4$$

Therefore, vertex B is (4,1).

Corner point C is the intersection of the boundary lines

$$x + y = 5 \text{ and } x = 3$$

Solving for x and y;

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$$\frac{dy}{dx} = \frac{-1}{\sqrt{1-x^2}} \times \left(\frac{-1}{2}\right)$$

$$= \frac{1}{2\sqrt{1-x^2}} \quad \text{Hence Proved.}$$

30. Prove that : $\sin^{-1}\frac{3}{5} - \cos^{-1}\frac{12}{13} = \sin^{-1}\frac{16}{65}$

OR

Find the value of $\sin\left(\sin^{-1}\frac{1}{2} + \cos^{-1}\frac{1}{2}\right)$.

Sol :

$$\text{L.H.S.} = \sin^{-1}\frac{3}{5} - \cos^{-1}\frac{12}{13}$$

Let $\cos^{-1}\frac{12}{13} = \theta \Rightarrow \cos\theta = \frac{12}{13}$

$$\sin\theta = \sqrt{1-\cos^2\theta}$$

$$[\sin^2\theta + \cos^2\theta = 1]$$

$$= \sqrt{1-\left(\frac{12}{13}\right)^2}$$

$$= \sqrt{1-\frac{144}{169}}$$

$$= \sqrt{\frac{25}{169}} = \frac{5}{13}$$

Using property

$$\sin^{-1}A - \sin^{-1}B$$

$$= \sin^{-1}\left[A\sqrt{1-B^2} - B\sqrt{1-A^2}\right]$$

$$\sin^{-1}\frac{3}{5} - \sin^{-1}\frac{5}{13}$$

$$= \sin^{-1}\left[\frac{3}{5}\sqrt{1-\frac{25}{169}} - \frac{5}{13}\sqrt{1-\frac{9}{25}}\right]$$

$$= \sin^{-1}\left[\frac{3}{5}\sqrt{\frac{169-25}{169}} - \frac{5}{13}\sqrt{\frac{25-9}{25}}\right]$$

$$= \sin^{-1}\left[\frac{3}{5}\sqrt{\frac{144}{169}} - \frac{5}{13}\sqrt{\frac{16}{25}}\right]$$

$$= \sin^{-1}\left[\frac{36}{65} - \frac{20}{65}\right]$$

$$= \sin^{-1}\frac{16}{65} = \text{R.H.S.}$$

OR

We have $\sin^{-1}\frac{1}{2} = \frac{\pi}{6}$

$$\cos^{-1}\frac{1}{2} = \frac{\pi}{3}$$

Therefore,

$$\sin\left(\frac{\pi}{6} + \frac{\pi}{3}\right) = \sin\left(\frac{\pi}{2}\right) = 1$$

31. If $A = \begin{bmatrix} 9 & 10 & 11 \\ 12 & 13 & 14 \end{bmatrix}$ and $B = \begin{bmatrix} 11 & 10 & 9 \\ 8 & 7 & 6 \end{bmatrix}$,

then find $A + B =$

OR

If $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then prove that

$f(x+y) = f(x).f(y)$

Sol :

Given: $A = \begin{bmatrix} 9 & 10 & 11 \\ 12 & 13 & 14 \end{bmatrix}$

$$B = \begin{bmatrix} 11 & 10 & 9 \\ 8 & 7 & 6 \end{bmatrix}$$

Then, $A + B = \begin{bmatrix} 9 & 10 & 11 \\ 12 & 13 & 14 \end{bmatrix} + \begin{bmatrix} 11 & 10 & 9 \\ 8 & 7 & 6 \end{bmatrix}$

$$= \begin{bmatrix} 9+11 & 10+10 & 11+9 \\ 12+8 & 13+7 & 14+6 \end{bmatrix}$$

$$= \begin{bmatrix} 20 & 20 & 20 \\ 20 & 20 & 20 \end{bmatrix} = 20\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

OR

We have, $f(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$f(y) = \begin{bmatrix} \cos y & -\sin y & 0 \\ \sin y & \cos y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Hence, $f(x) \cdot f(y)$

$$= \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} \cos y & -\sin y & 0 \\ \sin y & \cos y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos x\cos y - \sin x\sin y & -\cos x\sin y - \sin x\cos y & 0 \\ \sin x\cos y + \cos x\sin y & -\sin x\sin y + \cos x\cos y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos(x+y) & -\sin(x+y) & 0 \\ \sin(x+y) & \cos(x+y) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= f(x+y)$$

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27. Is the function $f:R\to R$ onto function (where $f(x)=2x$) Give reasons.

Sol :

Let, $f(x) = y$

$2x = y$

$x = \frac{1}{2}y$

Hence, $g(y) = \frac{1}{2}y$

Hence, Domain of, $g(y) = R$

Hence, Range of, $f(x) = R$

Since codomain,

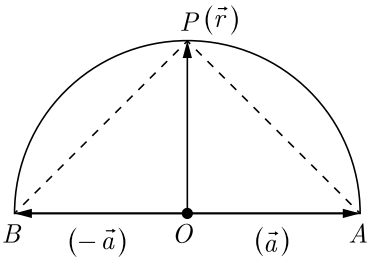
$(f(x)) =$ Range $(f(x))$.

Hence, $f(x)$ is onto function.

28. Prove, by Vector method, that the angle inscribed in a semi-circle is a right angle.

Sol :

Let O be the centre of semicircle PAB and position vector as follows



$\vec{OA} = \vec{a}$

$\vec{OB} = -\vec{a}$ (since AB is a diameters of circle

$\vec{OP} = \vec{r}$

$|\vec{OA}| = |\vec{OB}| = |\vec{OP}| = |\vec{a}| = |\vec{r}|$

$=$ Raduis of a given circle ...(1)

$\vec{AP}$ = Position vector of $\vec{OP}$-position vector of $\vec{OA}$

$= \vec{r} - \vec{a}$

$\vec{BP}$ = Position vector of $\vec{OP}$-Position vector of $\vec{OB}$

$= \vec{r} + \vec{a}$

For $\angle BPA$ to be right angled Δ

$\vec{AP}\cdot\vec{BP} = 0$

$\vec{AP}\cdot\vec{BP} = (\vec{r}-\vec{a})\cdot(\vec{r}+\vec{a})$

$= \vec{r}\cdot\vec{r} - \vec{a}\cdot\vec{r} + \vec{r}\cdot\vec{a} - \vec{a}\cdot\vec{a}$

$= |\vec{r}|^2 - |\vec{a}|^2$ $\quad [\vec{a}\cdot\vec{r} = \vec{r}\cdot\vec{a},\ \vec{r}\cdot\vec{r} = |\vec{r}|^2]$

$\vec{AP}\cdot\vec{BP} = 0$ [Using eq. (1)]

Therefore angle inscribed in a semicircle is right angled triangle.

29. If $y = \tan^{-1}\left[\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right]$, prove that

$\frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}}$.

Sol :

We have $y = \tan^{-1}\left[\frac{\sqrt{1+x}-\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right]$

Putting $x = \cos\theta$ we have

$$y = \tan^{-1}\left(\frac{\sqrt{1+\cos\theta}-\sqrt{1-\cos\theta}}{\sqrt{1+\cos\theta}+\sqrt{1-\cos\theta}}\right)$$

$$= \tan^{-1}\left[\frac{\sqrt{2\cos^2\frac{\theta}{2}}-\sqrt{2\sin^2\frac{\theta}{2}}}{\sqrt{2\cos^2\frac{\theta}{2}}+\sqrt{\sin^2\frac{\theta}{2}}}\right]$$

$\left[1+\cos A = 2\cos^2\frac{A}{2},\ 1-\cos A = 2\sin^2\frac{A}{2}\right]$

$$= \tan^{-1}\left[\frac{\sqrt{2}\left(\cos\frac{\theta}{2}-\sin\frac{\theta}{2}\right)}{\sqrt{2}\left(\cos\frac{\theta}{2}+\sin\frac{\theta}{2}\right)}\right]$$

$$= \tan^{-1}\left[\frac{1-\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}}{1+\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}}\right]$$

[Dividing numerator and denominator by $\cos\frac{\theta}{2}$]

$$= \tan^{-1}\left[\frac{1-\tan\frac{\theta}{2}}{1+\tan\frac{\theta}{2}}\right]$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{4}-\frac{\theta}{2}\right)\right) \quad \left[\tan\left(\frac{\pi}{4}-A\right) = \frac{1-\tan A}{1+\tan A}\right]$$

$$= \frac{\pi}{4}-\frac{\theta}{2}$$

Thus, $y = \frac{\pi}{4} - \frac{\cos^{-1}x}{2}$

Differentiating both sides with respect to x, we get

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Putting $x = 3$ in Eq. $x + y = 5$, we get

$$3 + y = 5$$

$$y = 2$$

Therefore corner point C is (3, 2).

Step III

Now, we evaluate Z at each corner point.

Corner Point	$Z = -x + 2y$
A (6,0)	-6
B (4,1)	-2
C (3,2)	$1 = M$ (Maximum)

From this table, we find that 1 is the maximum value of Z at (3,2).

Step IV

Since the feasible regions is unbounded, 1 may or may not be the maximum value of Z. To decide this, we graph the inequality $Z > M$ i.e., $-x + 2y > 1$.

Table of values for the line $-x + 2y = 1$ corresponding to constraint $Z > M$ i.e., $-x + 2y > 1$.

x	0	-1
y	$\frac{1}{2}$	0

Let us draw the dotted line joining the points $\left(0, \frac{1}{2}\right)$ and $(-1, 0)$.

The line is to be shown dotted because boundary of the line is to be excluded as equality sign is missing in the constraint $Z > M$.

Let us test for origin (0, 0) in this constraint $Z > M$ i.e., $-x + 2y > 1$, we have $0 > 1$ which is not true. Therefore region for this constraint is away from the origin.

We observe that the half-plane determined by $Z > M$ has points in common with the feasible region. Therefore, $Z = -x + 2y$ has no maximum value subject to the given constraints.

33. Find the values of p, so that the lines

$$l_1 : \frac{1-x}{3} = \frac{7y-14}{p} = \frac{z-3}{2} \text{ and}$$

$$l_2 : \frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5}$$

are perpendicular to each other. Also, find the equation of a line passing through a point $(3, 2, -4)$ and parallel to line l_1.

Sol :

Writing the given line in standard form as

$$\frac{x-1}{-3} = \frac{y-2}{\frac{p}{7}} = \frac{z-3}{2} = r_1 \text{ (let)} \quad ...(1)$$

and $$\frac{x-1}{\frac{-3p}{7}} = \frac{y-5}{1} = \frac{z-6}{-5} = r_2 \text{ (let)} \quad ...(2)$$

Two lines with DR's a_1, b , c_1 and a_2, b_2, c_2 are perpendicular if

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

Thus line (1) and (2) will intersect at right angle, if

$$-3\left(\frac{-3p}{7}\right) + \frac{p}{7}(1) + 2(-5) = 0$$

$$\frac{9p}{7} + \frac{p}{7} = 10$$

$$\frac{10p}{7} = 10 \Rightarrow p = 7$$

which is the required value of p.

Also, we know that, the equation of a line which passes through the point (x_1, y_1, z_1) with direction ratios a, b, c is given by

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

Since, required line is parallel to line l_1.

So, $a = -3$, $b = \frac{7}{7} = 1$ and $c = 2$

Now, equation of line passing through the point $(3, 2, -4)$ and having direction ratios $(-3, 1, 2)$ is

$$\frac{x-3}{-3} = \frac{y-2}{1} = \frac{z+4}{2}$$

$$\frac{3-x}{3} = \frac{y-2}{1} = \frac{z+4}{2}$$

34. Solve : $\sqrt{a+x}\frac{dy}{dx} + x = 0$

OR

Solve : $\frac{dy}{dx} - \frac{xy}{1-x^2} = \frac{1}{1-x^2}$

Sol :

Given: $$\sqrt{a+x}\frac{dy}{dx} + x = 0$$

or, $$\frac{dy}{dx} = \frac{-x}{\sqrt{a+x}}$$

or, $$dy = \frac{-x}{\sqrt{a+x}}dx \quad ...(1)$$

Let, $$a + x = t^2$$

or, $$x = t^2 - a \quad ...(2)$$

Differentiating both sides, we get

$$dx = 2tdt$$

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commercial trailers have built-in ramps for loading and off-loading cars, as well as power hydraulics to raise and lower ramps for stand-alone accessibility.



A transport company uses three types of trucks T_1, T_2 and T_3 to transport three types of vehicles V_1, V_2 and V_3. The capacity of each truck in terms of three types of vehicles is given below:

	V_1	V_2	V_3
T_1	1	3	2
T_2	2	2	3
T_3	3	2	2

Using matrix method find:

(i) The number of trucks of each type required to transport 85, 105 and 110 vehicles of V_1, V_2 and V_3 types respectively.

(ii) Find the number of vehicles of each type which can be transported if company has 10, 20 and 30 trucks of each type respectively.

Sol :

Let x, y and z be the number of trucks of types T_1, T_2 and T_3 respectively, then the given information can be put in the form of simultaneous linear equations as:

$$x+2y+3z = 85$$

$$3x+2y+2z = 105$$

$$2x+3y+2z = 110$$

The above system of linear equations can be written in the matrix form as:

$$\begin{bmatrix}1 & 2 & 3\\ 3 & 2 & 2\\ 2 & 3 & 2\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}85\\ 105\\ 110\end{bmatrix}$$

or $$AX = B$$

Since, $|A| = 1(4-6)-2(6-4)+3(9-4) = 9 \neq 0$, then A^{-1} exist, so that the unique solution of $AX = B$ is $X = A^{-1}B$.

Here, $$A^{-1} = \frac{\text{adj.}A}{|A|}$$

$$= \frac{1}{9}\begin{bmatrix}-2 & 5 & -2\\ -2 & -4 & 7\\ 5 & 1 & -4\end{bmatrix}$$

and $$\begin{bmatrix}x\\ y\\ z\end{bmatrix} = X = A^{-1}B$$

$$= \frac{1}{9}\begin{bmatrix}-2 & 5 & -2\\ -2 & -4 & 7\\ 5 & 1 & -4\end{bmatrix}\begin{bmatrix}85\\ 105\\ 110\end{bmatrix}$$

$$= \frac{1}{9}\begin{bmatrix}-170+525-220\\ -170-420+770\\ 425+105-440\end{bmatrix}$$

$$= \frac{1}{9}\begin{bmatrix}135\\ 180\\ 90\end{bmatrix} = \begin{bmatrix}15\\ 20\\ 10\end{bmatrix}$$

Hence the number of trucks of types T_1, T_2 and T_3 15, 20 and 10 respectively.

Now number of vehicles of types V_1, V_2 and V_3 are

$$V = \begin{matrix}\begin{matrix}T_1 & T_2 & T_3\end{matrix}\\ \begin{bmatrix}10 & 20 & 30\end{bmatrix}\end{matrix}\begin{matrix}\begin{matrix}V_1 & V_2 & V_3\end{matrix}\\ \begin{matrix}T_1\\ T_2\\ T_3\end{matrix}\begin{bmatrix}1 & 3 & 2\\ 2 & 2 & 3\\ 3 & 2 & 2\end{bmatrix}\end{matrix}$$

$$= \begin{matrix}\begin{matrix}V_1 & V_2 & V_3\end{matrix}\\ \begin{bmatrix}140 & 130 & 140\end{bmatrix}\end{matrix}$$

Hence the number of vehicles of types V_1, V_2 and V_3 are 140, 130 and 140, respectively.

37. Due to the growing need to reduce emissions across the world, most countries are replacing existing fuel-based transportation with electric mobility. India, still in the very early stages of adopting EV's can be a huge market for E-bike rental companies. As most people are now aware of traffic congestion and pollution, India is looking towards a cleaner mode of transportation to tackle the situation.



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Now, $I_2 = \int \frac{1}{2x^2+3x-2}\,dx$

$= \frac{1}{2}\int \frac{1}{x^2+\frac{3}{2}x-1}\,dx$

$= \frac{1}{2}\int \frac{1}{\left(x+\frac{3}{4}\right)^2-1-\frac{9}{16}}\,dx$

$= \frac{1}{2}\int \frac{1}{\left(x+\frac{3}{4}\right)^2-\left(\frac{5}{4}\right)^2}\,dx$

$= \frac{1}{2}\left[\frac{1}{2\left(\frac{5}{4}\right)}\log\left|\frac{\left(x+\frac{3}{4}\right)-\frac{5}{4}}{\left(x+\frac{3}{4}\right)+\frac{5}{4}}\right|\right]+C_2$

$= \frac{1}{5}\log\left|\frac{2x-1}{2x+4}\right|+C_2$

Putting values of I_1 and I_2 in (4), we get

$$I = -\frac{5}{4}\log|2x^2+3x-2|+\frac{27}{20}\log\left|\frac{2x-1}{2x+4}\right|+C$$

OR

Let $I_1 = \int_0^{\pi/2}\log\sin x\,dx$

an $I_2 = \int_0^{\pi/2}\log\cos x\,dx$

Now

$$\int_0^{\pi/2}\log\sin x\,dx = \int_0^{\pi/2}\log\sin\left(\frac{\pi}{2}-x\right)dx$$

$$= \int_0^{\pi/2}\log\cos x\,dx$$

$$\left[\int_a^b \log x\,dx = \int_a^b \log(a+b-x)\,dx\right]$$

Thus, $I_1 = I_2$

Using property $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ we have

$$I_1 = \int_0^{\pi/2}\log\sin x\,dx$$

$$= \int_0^{\pi/2}\log\sin\left(\frac{\pi}{2}-x\right)dx$$

$$= \int_0^{\pi/2}\log\cos x\,dx$$

$$2I_1 = \int_0^{\pi/2}\log\sin x\,dx+\int_0^{\pi/2}\log\cos x\,dx$$

$$= \int_0^{\pi/2}\log\sin x\,\cos x\,dx$$

$$[\log x+\log y = \log xy]$$

Multiplying and dividing by 2, we get

$$2I_1 = \int_0^{\pi/2}\log\left(\frac{2\sin x\cos x}{2}\right)dx$$

$$= \int_0^{\pi/2}\log\left(\frac{\sin 2x}{2}\right)dx$$

$$[2\sin\theta\cos\theta = \sin 2\theta]$$

$$= \int_0^{\pi/2}\log\sin 2x\,dx-\int_0^{\pi/2}\log 2\,dx$$

$$2I_1 = \int_0^{\pi/2}\log\sin 2x\,dx-\log 2\left(\frac{\pi}{2}-0\right) \quad ...(1)$$

Let, $I_3 = \int_0^{\pi/2}\log\sin 2x\,dx$

Substituting $t = 2x$ we have

$dt = 2dx$

$$[x=0 \Rightarrow t=0,\ x=\frac{\pi}{2} \Rightarrow t=\pi]$$

$$I_3 = \frac{1}{2}\int_0^{\pi}\log\sin t\,dt$$

Using property

$$\int_0^{2a} f(x)\,dx = 2\int_0^a f(a-x)\,dx$$

$$\text{if } f(2a-x) = f(x)$$

$$I_3 = \frac{1}{2}\int_0^{\pi}\log\sin t\,dt$$

$$= \int_0^{\pi/2}\sin t\,dt = I_1$$

Therefore eq.(1) becomes,

$$2I_1 = I_1-\frac{\pi}{2}\log 2$$

$$I_1 = -\frac{\pi}{2}\log 2$$

Since, $I_1 = I_2$, therefore,

$$I_1 = I_2 = -\frac{\pi}{2}\log 2$$

SECTION - E

Case study based questions are compulsory.

36. A car carrier trailer, also known as a car-carrying trailer, car hauler, or auto transport trailer, is a type of trailer or semi-trailer designed to efficiently transport passenger vehicles via truck. Commercial-size car carrying trailers are commonly used to ship new cars from the manufacturer to auto dealerships. Modern car carrier trailers can be open or enclosed. Most

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Substituting values in equation (1), we get

$$dy = \frac{-(t^2-a)2t\,dt}{t}$$

$$dy = -2(t^2-a)\,dt$$

Integrating both sides, we get

$$y = \frac{-2t^3}{3} + 2at + k$$

$$\left[\int x^n\,dx = \frac{x^{n+1}}{n+1}, \int k\,dx = kx + c\right]$$

Substituting value of t from equation (2), we get

$$y = \frac{-2}{3}(x+a)^{3/2} + 2a(x+a)^{1/2} + k$$

OR

Given: $$\frac{dy}{dx} - \frac{xy}{1-x^2} = \frac{1}{1-x^2}$$

The above equation is first order linear differential equation of the form $\frac{dy}{dx} + Py = Q$ where P and Q are functions of x whose general solution is given as

$$ye^{\int P dx} = \int Qe^{\int P dx} + C \qquad ...(1)$$

Here, $$P = \frac{-x}{1-x^2}$$

and $$Q = \frac{1}{1-x^2}$$

$$\text{I.F} = e^{\int P dx} = e^{\int \frac{-x dx}{1-x^2}}$$

Multiplying and dividing by 2

$$\text{I.F} = e^{\int \frac{-2x}{2(1-x^2)} dx}$$

using, $$\int \frac{f'(x)}{f(x)}dx = \log f(x) + C$$

We get, $$\text{I.F} = e^{\frac{1}{2}\int \frac{-2x}{1-x^2}dx}$$

$$= e^{\frac{1}{2}\ln(1-x^2)}$$

(Where $f(x) = 1-x^2$)

$$= e^{\ln(1-x^2)^{\frac{1}{2}}}$$

Thus, $$\text{I.F} = (1-x^2)^{\frac{1}{2}} \qquad (a^{\log ax} = x)$$

$$= \sqrt{1-x^2}$$

Solution of above differential equation is given by

$$y\sqrt{1-x^2} = \int \frac{\sqrt{1-x^2}}{1-x^2}dx$$

$$= \int (1-x^2)^{\frac{1}{2}-1}dx$$

$$= \int (1-x^2)^{\frac{-1}{2}}dx$$

$$= \int \frac{dx}{\sqrt{1-x^2}}$$

$$= \sin^{-1}x + C$$

$$\left[\int \frac{dx}{1-x^2} = \sin^{-1}x\right]$$

Thus, $$y\sqrt{1-x^2} = \sin^{-1}x + C$$

35. Evaluate $\int \frac{3-5x}{2x^2+3x-2}dx$

OR

Prove that : $\int_0^{\pi/2} \log\sin x\,dx = -\frac{\pi}{2}\log 2$

Sol :

Let $$I = \int \frac{3-5x}{2x^2+3x-2}dx \qquad ...(1)$$

Put $$3-5x = A\frac{d}{dx}(2x^2+3x-2) + B$$

$$3-5x = A(4x+3) + B \qquad ...(2)$$

$$3-5x = 4Ax + 3A + B$$

Comparing coefficients of x and constant terms on both sides, we get

$$-5 = 4A, \qquad 3 = 3A + B$$

Solving above equations, we get

$$A = -\frac{5}{4}, \qquad B = \frac{27}{4}$$

Putting values of A and B in (2), we get

$$3-5x = -\frac{5}{4}(4x+3) + \frac{27}{4} \qquad ...(3)$$

From (1) and (3), we get

$$I = -\frac{5}{4}\int \frac{4x+3}{2x^2+3x-2}dx + \frac{27}{4}\int \frac{1}{2x^2+3x-2}dx$$

or $$I = -\frac{5}{4}I_1 + \frac{27}{4}I_2 \qquad ...(4)$$

Now, $$I_1 = \int \frac{4x+3}{2x^2+3x-2}dx$$

Putting $2x^2+3x-2 = y$ we have

$$(4x+3)\,dx = dy$$

Thus $$I_1 = \int \frac{1}{y}dy = \log|y| + C_1$$

$$= \log|2x^2+3x-2| + C_1$$

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An owner of an electric bike rental company have determined if they charge customers ₹ x per day to rent a bike, where $50 \le x \le 200$ then number of bikes (n), they rent per day can be shown by linear function $n(x) = 2000 - 10x$. If they charge ₹ 50 per day or less, they will rent all their bikes. If they charges ₹ 200 or more per day they will not rent any bike.

Based on the above information, answer the following questions.

(i) Find the expression for total revenue R as a function of x.

(ii) Find the value of x at maximum revenue.

(iii) What is the revenue collected by the company at $x = 260$? Find the number of bikes rented per day, if $x = 105$.

OR

(iv) Find the maximum revenue, collected by company.

Sol :

(i) Let x be the charges per bike per day and n be the number of bikes rented per day.

$$R(x) = n \times x$$
$$= (2000 - 10x)\,x$$
$$= -10x^2 + 2000x$$

(ii) $$R(x) = 2000x - 10x^2$$
$$R'(x) = 2000 - 20x$$

For $R(x)$ to be maximum or minimum, $R'(x) = 0$

$$2000 - 20x = 0$$
$$x = 100$$

Also, $$R''(x) = -20 < 0$$

Thus, $R(x)$ is maximum at $x = 100$

(iii) If company charge ₹ 200 or more, they will not rent any bike. Therefore, revenue collected by him will be zero.

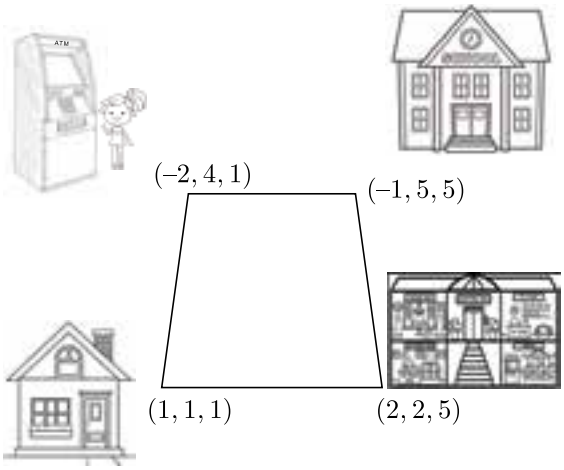
If $x = 105$, number of bikes rented per day is given by

$$n = 2000 - 10 \times 105$$
$$= 950$$

(iv) Maximum revenue will be when $R(x)$ is maximum. At $x = 100$, $R(x)$ is maximum.

$$R(100) = -10(100)^2 + 2000(100)$$
$$= ₹1,00,000$$

38. Lavanya starts walking from his house to shopping mall. Instead of going to the mall directly, she first goes to ATM, from there to her daughter's school and then reaches the mall. In the diagram, using co-ordinate geometry the location of each place is given.



Based on the above information, answer the following questions.

(i) What is the distance between House and ATM?

(ii) What is the distance between ATM and school?

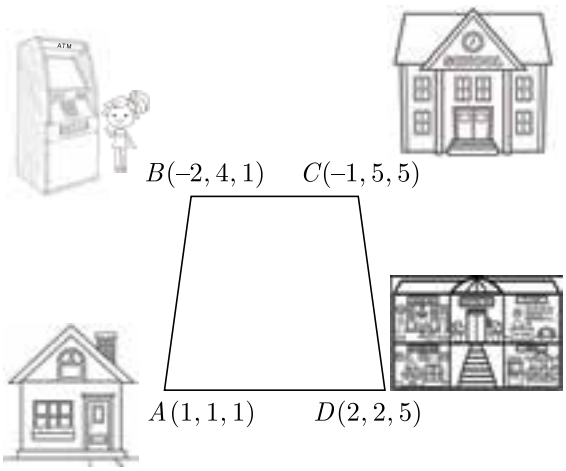
(iii) What is the total distance travelled by Lavanya?

OR

(iv) What is the extra distance travelled by Lavanya in reaching the shopping mall?

Sol :

We can redraw the given diagram as below.



(i) $$\vec{AB} = (-2\hat{i} + 4\hat{j} + \hat{k}) - (\hat{i} + \hat{j} + \hat{k})$$
$$= -3i + 3j$$

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$$\vec{AB} = \sqrt{(-3)^2 + 3^2}$$
$$= \sqrt{9+9} = \sqrt{18} = 3\sqrt{2}$$

Distance between House (A) and ATM (B) is $3\sqrt{2}$ units.

(ii) $\vec{BC} = (-\hat{i} + 5\hat{j} + 5\hat{k}) - (2\hat{i} + 4\hat{j} + \hat{k})$
$$= \hat{i} + \hat{j} + 4\hat{k}$$
$$|\vec{BC}| = \sqrt{1^2 + 1^2 + 4^2}$$
$$= \sqrt{1+1+16}$$
$$= \sqrt{18} = 3\sqrt{2}$$

Distance between ATM (B) and School (C) is $3\sqrt{2}$ units.

(iii) $\vec{CD} = (2\hat{i} + 2\hat{j} + 5\hat{k}) - (-\hat{i} + 5\hat{j} + 5\hat{k})$
$$= (3\hat{i} - 3\hat{j})$$
$$|\vec{CD}| = \sqrt{3^2 + (-3)^2}$$
$$= \sqrt{9+9} = 3\sqrt{2}$$

Distance between School (C) and Shopping mall (D) is $3\sqrt{2}$ units.

Total distance travelled by Lavanya
$$= |\vec{AB}| + |\vec{BC}| + |\vec{CD}|$$
$$= (3\sqrt{2} + 3\sqrt{2} + 3\sqrt{2}) \text{ units.}$$
$$= 9\sqrt{2}$$

(iv) Distance between house and shopping mall is $|\vec{AD}|$

Now, $|\vec{AD}| = \hat{i} + \hat{j} + 4\hat{k}$
$$|\vec{AD}| = \sqrt{1^2 + 1^2 + 4^2}$$
$$= \sqrt{1+1+16}$$
$$= \sqrt{18} = 3\sqrt{2}$$

Thus, extra distance travelled by Lavanya is reaching shopping mall $= (9\sqrt{2} - 3\sqrt{2})$ units $= 6\sqrt{2}$ units.

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Sample Paper 02

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

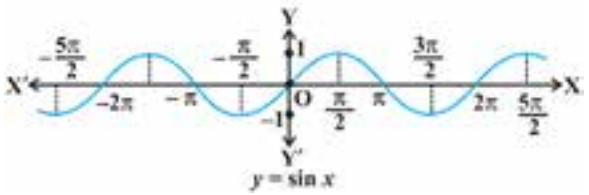
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the maximum value of $y = \sin x$ observed in the graph?



(a) 0 (b) 1

(c) –1 (d) 2

Sol :

In the graph, the sine curve reaches its highest point at several positions, such as $x = \frac{\pi}{2}$ and $x = \frac{5}{2}$. At these points, the curve touches the line parallel to the X-axis marked at $y = 1$. This shows that the maximum vertical displacement of the sine wave is exactly 1. The curve never crosses above this level in the image. Hence, the maximum value represented in the graph is 1.

Thus (b) is correct option.

2. The value of $\int_0^1 \frac{dx}{e^x + e}$ is

(a) $\frac{1}{e}\log\left(\frac{1+e}{2}\right)$ (b) $\log\left(\frac{1+e}{2}\right)$

(c) $\frac{1}{e}\log(1+e)$ (d) $\log\left(\frac{2}{1+e}\right)$

Sol :

We have $I = \int_0^1 \frac{dx}{e^x + e} = \int_0^1 \frac{dx}{e^x\left(1 + \frac{e}{e^x}\right)}$

Substituting $1 + \frac{e}{e^x} = t \Rightarrow 0 - \frac{e}{e^x}dx = dt$ we obtain

$$\frac{1}{e^x}dx = -\frac{1}{e}dt$$

$$I = -\frac{1}{e}\int_{1+e}^{2}\frac{1}{t}dt = \frac{-1}{e}[\log t]_{1+e}^{2}$$

$$= \frac{-1}{e}[\log 2 - \log(1+e)]$$

$$= \frac{-1}{e}\log\left(\frac{2}{1+e}\right)$$

$$= \frac{1}{e}\log\left(\frac{1+e}{2}\right)$$

Thus (a) is correct option.

3. If $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5, 6\}$ are two sets and function $f : A \to B$ is defined by $f(x) = x + 2$, $\forall\, x \in A$, then the function f is

(a) bijective

(b) onto

(c) one-one

(d) many-one

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Sol :

$$f(1) = 3$$
$$f(2) = 4$$
$$f(3) = 5$$
$$f(4) = 6$$

$1 \in B$, $2 \in B$ do not have any pre-image in A.
f is one-one and into.
Thus (c) is correct option.

4. If $A = \begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}$, then $A \cdot (adj\ A)$ is equal to

(a) A (b) $|A|$
(c) $|A| \cdot I$ (d) None of these

Sol :

We have $A = \begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}$

Then, $|A| = 2I - 20 = 1$

$$A\,(adj\ A) = \begin{bmatrix} 3 & 4 \\ 5 & 7 \end{bmatrix}\begin{bmatrix} 7 & -4 \\ -5 & 3 \end{bmatrix}$$
$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$= |A| \cdot I$$

Thus (c) is correct option.

5. The area of enclosed by $y = 3x - 5$, $y = 0$, $x = 3$ and $x = 5$ is

(a) 12 sq units (b) 13 sq units
(c) $13\frac{1}{2}$ sq units (d) 14 sq units

Sol :

The region is bounded by the curves $y = 3x - 5$, $y = 0$, $x = 3$ and $x = 5$.

Required area, $A = \int_3^5 (3x - 5)\,dx$

$$= \left[\frac{3x^2}{2} - 5x\right]_3^5$$
$$= \left(\frac{75}{2} - 25\right) - \left(\frac{27}{2} - 15\right)$$
$$= \frac{75}{2} - 25 - \frac{27}{2} + 15$$
$$= \frac{48}{2} - 10 = 14 \text{ sq units}$$

Thus (d) is correct option.

6. Solution of the equation $x\frac{dy}{dx} = y + x\tan\frac{y}{x}$ is

(a) $\sin\frac{x}{y} = cx$ (b) $\sin\frac{y}{x} = cx$
(c) $\sin\frac{x}{y} = cy$ (d) $\sin\frac{y}{x} = cy$

Sol :

Substituting $y = vx$ we have

$$\frac{dy}{dx} = x\frac{dv}{dx} + v$$
$$v + x\frac{dv}{dx} = v + \tan v$$
$$\cot v\,dv = \frac{dx}{x}$$
$$\log\sin v = \log cx$$
$$\sin\frac{y}{x} = cx$$

Thus (b) is correct option.

7. If $x = e^{y + e^{y + e^{y + \cdots}}}$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{1}{x}$ (b) $\frac{1 - x}{x}$
(c) $\frac{x}{1 + x}$ (d) None of these

Sol :

We have $x = e^{y + e^{y + e^{y + \cdots}}}$

$$x = e^{y + x}$$

Taking log both sides, we get

$$\log x = (x + y)\log e$$
$$\frac{1}{x} = \frac{dy}{dx} + 1 \Rightarrow \frac{dy}{dx} = \frac{1 - x}{x}$$

Thus (b) is correct option.

8. If A and B are two independent events such that $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{3}$, then P (neither A nor B) is equal to

(a) $\frac{2}{3}$ (b) $\frac{1}{6}$
(c) $\frac{5}{6}$ (d) $\frac{1}{3}$

Sol :

Since, A and B are independent events.

$$P(A \cap B) = P(A) \cdot P(B)$$
$$= \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

Now, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

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18. The degree of the differential equation satisfying $\sqrt{1-x^2}+\sqrt{1-y^2}=a(x-y)$

(a) 1 (b) 2

(c) 3 (d) 4

Sol :

Putting $x = \sin A$

$y = \sin B$

we get $\cos A+\cos B = a(\sin A-\sin B)$

$$2a\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$$

$$= 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

$$\sin^{-1}x-\sin^{-1}y = 2\cot^{-1}a = \text{constant}$$

Differentiating w.r.t. x, we get

$$\frac{1}{\sqrt{1-x^2}}-\frac{1}{\sqrt{1-y}}\frac{dy}{dx}=0$$

$$\frac{dy}{dx}=\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$$

Hence, it is of degree 1.

Thus (a) is correct option.

19. Assertion: $\int_0^{2\pi}\sin^3 x\,dx=0$

Reason: $\sin^3 x$ is an odd function

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

$$I = \int_0^{2\pi}\sin^3 x\,dx$$

$$= \int_0^{2\pi}(1-\cos^2 x)\sin x\,dx$$

Substituting $\cos x = t$ we have

$$\sin x\,dx = -dt$$

Then $$I = \int_1^1 (1-t^2)(-dt) = 0$$

Thus assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

Thus (b) is correct option.

20. Assertion: The pair of lines given by $\vec{r}=\hat{i}-\hat{j}+\lambda(2\hat{i}+\hat{k})$ and $\vec{r}=2\hat{i}-\hat{k}+\mu(\hat{i}+\hat{j}-\hat{k})$ intersect.

Reason: Two lines intersect each other, if they are not parallel and shortest distance $=0$.

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

Here, $\vec{a_1} = \hat{i}-\hat{j},\ \vec{b_1}=2\hat{i}+\hat{k}$

$\vec{a_2} = 2\hat{i}-\hat{k}, \vec{b_2}=\hat{i}+\hat{j}-\hat{k}$

$\vec{b_1} \neq \lambda\vec{b_2}$, for any scalar λ

Given lines are not parallel.

$$\vec{a_2}-\vec{a_1}\ (2\hat{i}-\hat{k})-(\hat{i}-\hat{j}) = \hat{i}+\hat{j}-\hat{k}$$

$$\vec{b_1}\times\vec{b_2} = \begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 0 & 1\\ 1 & 1 & -1\end{vmatrix} = -\hat{i}+3\hat{j}+2\hat{k}$$

$$\vec{b_1}\times\vec{b_2} = \sqrt{(-1)^2+(3)^2+(2)^2}$$

$$= \sqrt{1+9+4} = \sqrt{14}$$

$$\text{SD} = \left|\frac{(\vec{a_2}-\vec{a_1})\cdot(\vec{b_1}\times\vec{b_2})}{|(\vec{b_1}\times\vec{b_2})|}\right|$$

$$= \left|\frac{(\hat{i}+\hat{j}-\hat{k})\cdot(-\hat{i}+3\hat{j}+2\hat{k})}{\sqrt{14}}\right|$$

$$= \left|\frac{-1+3-2}{\sqrt{14}}\right| = 0$$

Thus assertion is true, Reason is true; Reason is a correct explanation for Assertion.

Thus (a) is correct option.

SECTION - B

This section comprises of very short answer-type questions (VSA) of 2 marks each.

21. Determine the value of k for which the following function is continuous at $x=3$:

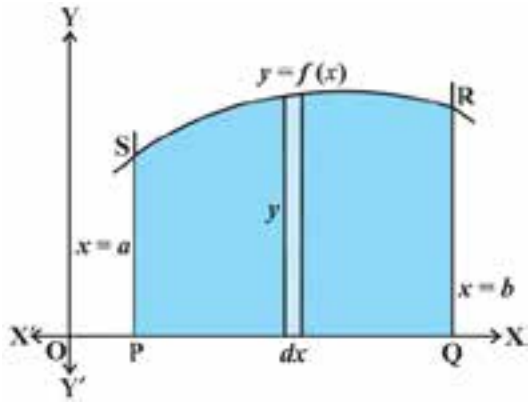
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$$f\left(x+\frac{1}{x}\right)=\left(x+\frac{1}{x}\right)^2-2$$

$$f(x)=x^2-2$$

Thus (a) is correct option.

15. In Fig, what does the element labeled "dx" represent in the shaded region?



(a) The height of the rectangle under the curve

(b) The small change in the x-coordinate (width of the strip)

(c) The slope of the curve y=f(x) at a point

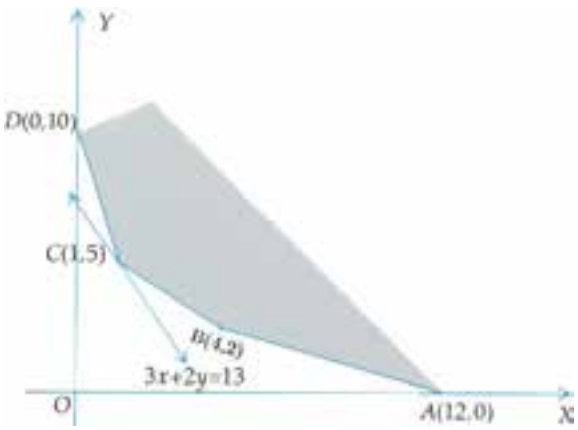
(d) The area under the curve between x=a and x=b

Sol :

The figure shows a thin vertical strip within the shaded area. This strip is drawn parallel to the Y-axis, with height y (dependent on f(x)) and an infinitesimal width labeled dx. Thus, dx represents a very small change in the x-coordinate (the width of the strip), not the height, slope, or full area.

Thus (b) is correct option.

16. The minimum value of $z=3x+2y$, if the feasible region for an LPP is as shown in Fig, is to be determined.



(a) 36 (b) 16

(c) 20 (d) 13

Ans :

The feasible region is bounded by the vertices A(12,0), B(4,2), C(1,5), and D(0,10).

According to the corner point method, we evaluate the value of the objective function z=3x+2y at each of these points.

At A(12,0):

$$z=3(12)+2(0)=36$$

At B(4,2):

$$z=3(4)+2(2)=12+4=16$$

At C(1,5):

$$z=3(1)+2(5)=3+10=13$$

At D(0,10):

$$z=3(0)+2(10)=20$$

Comparing all these values, the minimum value of z is 13, which occurs at the point C(1,5).

Thus (d) is correct option.

17. The general solution of the differential equation $\frac{dy}{dx}+\sin\frac{x+y}{2}=\sin\frac{x-y}{2}$ is

(a) $\log\tan\left(\frac{y}{2}\right)=c-2\sin x$

(b) $\log\tan\left(\frac{y}{4}\right)=c-2\sin\frac{x}{2}$

(c) $\log\tan\left(\frac{y}{2}+\frac{\pi}{4}\right)=c-2\sin x$

(d) $\log\tan\left(\frac{y}{4}+\frac{\pi}{4}\right)=c-2\sin\left(\frac{x}{2}\right)$

Sol :

$$\frac{dy}{dx}=\sin\left(\frac{x-y}{2}\right)-\sin\left(\frac{x+y}{2}\right)$$

$$=-2\cos\frac{x}{2}\sin\frac{y}{2}$$

$$\int\frac{1}{2}\operatorname{cosec}\frac{y}{2}dy=-\int\cos\frac{x}{2}dx$$

$$\log\tan\frac{y}{4}=-\frac{\sin\frac{x}{2}}{\frac{1}{2}}xc$$

$$\log\tan\left(\frac{y}{4}\right)=c-2\sin\frac{x}{2}$$

Thus (b) is correct option.

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$$= \frac{1}{2} + \frac{1}{3} - \frac{1}{6} = \frac{2}{3}$$

$$P(\overline{A} \cap \overline{B}) = 1 - P(A \cup B)$$

$$= 1 - \frac{2}{3} = \frac{1}{3}$$

Thus (d) is correct option.

9. Objective function of a linear programming problem is

(a) a constraint

(b) a function to be optimized

(c) a relation between the variables

(d) none of the above

Sol :

Objective function of linear programming problem is a function to be optimized.

Thus (b) is correct option.

10. The point of intersection of the lines $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$ is

(a) $\vec{a}$ (b) $\vec{b}$

(c) $\vec{a} + \vec{b}$ (d) $\vec{a} - \vec{b}$

Sol :

Now, $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$

$$(\vec{r} - \vec{b}) \times \vec{a} = 0$$

Hence, $\vec{r} - \vec{b} = t\vec{a}$

and $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$

$$(\vec{b} + t\vec{a}) \times \vec{b} = \vec{a} \times \vec{b}$$

$$\vec{b} \times \vec{b} + t\vec{a} = \vec{a} \times \vec{b}$$

$$t\vec{a} \times \vec{b} = \vec{a} \times \vec{b}$$

$$t = 1$$

Hence, $\vec{r} = \vec{a} + \vec{b}$

Thus (c) is correct option.

11. The foot of the perpendicular from (0, 2, 3) to the line $\frac{x+3}{5} = \frac{y=1}{2} = \frac{z+4}{3}$ is

(a) $(-2, 3, 4)$ (b) $(2, -1, 3)$

(c) $(2, 3, -1)$ (d) $(3, 2, -1)$

Sol :

Let N be the foot of the perpendicular from the point (0, 2, 3) on the given line

$$\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$$

$$= \lambda \text{ (say)} \qquad \ldots(1)$$

Any point on the line is $P(5\lambda - 3, 2\lambda + 1, 3\lambda - 4)$. If this point is N, then direction ratios of NP are

$< 5\lambda - 3 - 0, 2\lambda + 1 - 2, 3\lambda - 4 - 3 >$

i.e. $< 5\lambda - 3, 2\lambda - 1, 3\lambda - 7 >$

Since, PN is perpendicular to line (1).

Hence, $5(5\lambda - 3) + 2(2\lambda - 1) + 3(3\lambda - 7) = 0$

$$38\lambda - 38 = 0$$

$$\lambda = 1$$

Hence, required point is $(2, 3, -1)$.

Thus (c) is correct option.

12. Which of the following function is decreasing on $(0, \pi/2)$?

(a) $\sin 2x$ (b) $\cos 3x$

(c) $\tan x$ (d) $\cos 2x$

Sol :

We have $f(x) = \cos 2x$

$$f'(x) = -2\sin 2x < 0 \text{ in } \left(0, \frac{\pi}{2}\right)$$

So, $\cos 2x$ is decreasing in $\left(0, \frac{\pi}{2}\right)$.

Thus (d) is correct option.

13. $2\sin^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$ holds good for all

(a) $|x| \le 1$ (b) $1 \ge x \ge 0$

(c) $|x| \le \frac{1}{\sqrt{2}}$ (d) none of these

Sol :

Since, $2\sin^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$

Range of right hand side is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$$-\frac{\pi}{2} \le 2\sin^{-1}x \le \frac{\pi}{2}$$

$$-\frac{\pi}{4} \le \sin^{-1}x \le \frac{\pi}{4}$$

$$x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$$

Thus (c) is correct option.

14. If $f\left(x + \frac{1}{x}\right) = x^2 + \frac{1}{x^2}, x \ne 0$, then $f(x)$ is equal to

(a) $x^2 - 2$ (b) $x + 2$

(c) $x^2 + 2$ (d) $2x^2 - 5$

Sol :

Given equation can be rewritten as

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$$f(x) = \begin{cases} \frac{(x+3)^2-36}{x-3}, & x \neq 3 \\ k, & x = 3 \end{cases}$$

Sol :

We have $f(x) = \begin{cases} \frac{(x+3)^2-36}{x-3}, & x \neq 3 \\ k, & x = 3 \end{cases}$

Let $f(x)$ is continuous at $x = 3$, then we have

$$\lim_{x \to 3} f(x) = f(3)$$

$$\lim_{x \to 3} \frac{(x+3)^2-36}{x-3} = k$$

$$\lim_{x \to 3} \frac{(x+3)^2-6^2}{x-3} = k$$

$$\lim_{x \to 3} \frac{(x+3-6)(x+3+6)}{x-3} = k$$

$$\lim_{x \to 3} \frac{(x-3)(x+9)}{(x-3)} = k$$

$$\lim_{x \to 3} (x+9) = k$$

$$3+9 = k \Rightarrow k = 12$$

OR

Determine the value of constant k so that the function $f(x) = \begin{cases} \frac{kx}{|x|}, & \text{if } x < 0 \\ 3, & \text{if } x \geq 0 \end{cases}$ is continuous at $x = 0$.

Sol :

Let $f(x) = \begin{cases} \frac{kx}{|x|}, & \text{if } x < 0 \\ 3, & \text{if } x \geq 0 \end{cases}$ is continuous at $x = 0$

then, $\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} f(0-h) = f(0)$

$$\lim_{h \to 0} f(0+h) = \lim_{h \to 0} \frac{k(-h)}{|-h|} = 3$$

$$\lim_{h \to 0} \left(\frac{-kh}{h}\right) = 3 \Rightarrow \lim_{h \to 0} (-k) = 3$$

$$k = -3$$

22. If A and B are square matrices of the same order 3, such that $|A| = 2$ and $AB = 2I$. Write the values of $|B|$.

Sol :

We know that,

(i) $|kA| = k^n |A|$, if A is square matrix of nth order

(ii) $|AB| = |A| \times |B|$

Here, we have $AB = 2I$ and $n = 3$

$$|AB| = |2I|$$

$$= 2^3 |I| = 8 \cdot 1 = 8$$

$$|A||B| = 8$$

$$2 \cdot |B| = 8 \Rightarrow |B| = 4$$

23. If $P(not\ A) = 0.7$, $P(B) = 0.7$ and $P(B/A) = 0.5$, then find $P(A/B)$.

Sol :

We have $P(A') = 0.7,\ P(B) = 0.7$

and $P\left(\frac{B}{A}\right) = 0.5$

Now $P(A) = 1 - P(A')$

$$= 1 - 0.7 = 03$$

Now, $P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$

$$0.5 = \frac{P(A \cap B)}{0.3}$$

$$P(A \cap B) = 0.15$$

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{0.15}{0.7}$$

Thus $P\left(\frac{A}{B}\right) = \frac{3}{14}$

24. If $\vec{a}$ and $\vec{b}$ are perpendicular vectors, $|\vec{a} + \vec{b}| = 13$ and $|\vec{a}| = 5$, then find the value of $|\vec{b}|$.

Sol :

We have $|\vec{a} + \vec{b}| = 13$

$$|\vec{a}| = 5$$

Now, $(\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) = \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b}$

$$|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + 0 + 0 + |\vec{b}|^2$$

$$(13)^2 = (5)^2 + |\vec{b}|^2$$

$$169 = 25 + |\vec{b}|^2$$

$$169 - 25 = |\vec{b}|^2$$

$$144 = |\vec{b}|^2 \Rightarrow |\vec{b}| = 12$$

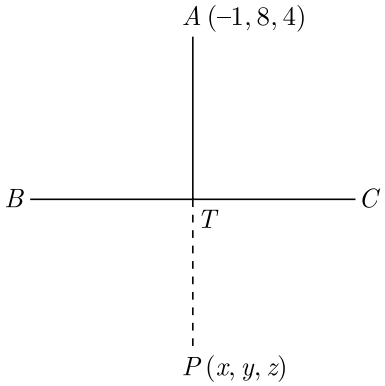
as length is always positive.

25. Find the general solution of the following differential equation $(e^x + e^{-x})dy - (e^x - e^{-x})dx = 0$

Sol :

Given differential equation is

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Now, DR's of line AT is

$$(2\lambda+1,-2\lambda-1-8,-4\lambda+3-4)$$

or $\quad (2\lambda+1,2\lambda-9,-4\lambda-1).$

Since, AT is perpendicular to BC, using $a_1a_2+b_1b_2+c_1c_2=0$ we have

$2\times(2\lambda+1)+(-2)\times(-2\lambda-9)+(-4)(-4\lambda-1)=0$

$$4\lambda+2+4\lambda+18+16\lambda+4=0$$

$$24\lambda+24=0 \Rightarrow \lambda=-1$$

Coordinate of foot of perpendicular is

$T(2\times(-1)),\ -2\times(-1)-1,\ -4\times(-1)+3$ or $T(-2,1,7)$

Let $P(x,y,z)$ be the image of a point A with respect to the line BC. So, point T is the mid-point of AP.

Coordinates of $T=$ Coordinates of mid-point of AP

$$(-2,1,7)=\left(\frac{x-1}{2},\frac{y+8}{2},\frac{z+4}{2}\right)$$

Equating the corresponding coordinates, we get

$$-2=\frac{x-1}{2},\ 1=\frac{y+8}{2} \text{ and } 7=\frac{z+4}{2}$$

$$x=-3,\ y=-6 \text{ and } z=10$$

Hence, coordinates of the foot of perpendicular is $T(-2,1,7)$ and image of the point A is $P(-3,-6,10)$

29. Evaluate $\int\frac{x^3+3x+4}{\sqrt{x}}$

Sol :

Let $\quad I=\int\frac{x^3+3x+4}{\sqrt{x}}$

$$=\int\left(\frac{x^3}{x^{1/2}}+\frac{3x}{x^{1/2}}+\frac{4}{x^{1/2}}\right)dx$$

$$=\int(x^{5/2}+3\cdot x^{1/2}+4\cdot x^{-1/2})dx$$

$$=\int x^{5/2}dx+3\int x^{1/2}dx+4\int x^{-1/2}dx$$

$$=\frac{2}{7}x^{7/2}+3\times\frac{2}{3}x^{3/2}+4\times2\cdot x^{1/2}+C$$

$$=\frac{2}{7}x^{7/2}+2x^{3/2}+8x^{1/2}+C$$

Hence, $\quad I=\frac{2}{7}x^{7/2}+2x^{3/2}+8\sqrt{x}+C$

30. Show that the relation S in the set R of real numbers defined as $S=\{(a,b):a,b\in R \text{ and } a\le b^3\}$ is neither reflexive not symmetric not transitive.

Sol :

$$S=\{(a,b):a,b\in R \text{ and } a\le b^3\}$$

Reflexive

As $\frac{1}{2}\le\left(\frac{1}{2}\right)^3$, where $\frac{1}{2}\in R$, is true.

$$\left(\frac{1}{2},\frac{1}{2}\right)\notin S$$

Thus, S is not reflexive.

Symmetric

As $-2\le(3)^3$, where $-2,3\in R$, is true but $3\le(-2)^3$ is not true.

i.e. $(-2,3)\in S$ but $(3,-2)\notin S$.

Therefore, S is not symmetric.

Transitive

As $3\le\left(\frac{3}{2}\right)^3$ and $\frac{3}{2}\le\left(\frac{4}{3}\right)^3$, where $3,\frac{3}{2},\frac{4}{3}\in R$, are true but $3\le\left(\frac{4}{3}\right)^3$ is not true.

i.e. $\left(3,\frac{3}{2}\right)\in S$ and $\left(\frac{3}{2},\frac{4}{3}\right)\in S$ but $\left(3,\frac{4}{3}\right)\in S$.

Therefore, S is not transitive.

Hence, S is neither reflexive not symmetric not transitive.

31. If the function $f(x)$ given by $f(x)=\begin{cases}3ax+b, & \text{if } x>1\\ 11, & \text{if } x=1\\ 5ax-2b, & \text{if } x<1\end{cases}$ is continuous at $x=1$, then find the values of a and b.

Sol :

Given, $f(x)=\begin{cases}3ax+b, & \text{if } x>1\\ 11, & \text{if } x=1\\ 5ax-2b, & \text{if } x<1\end{cases}$ is continuous at $x=1$.

$$\text{LHL}=RHL=f(1) \qquad ...(i)$$

Now, $\quad \text{LHL}=\lim_{x\to1^-}f(x)=\lim_{x\to1^-}(5ax-2b)$

$$=\lim_{h\to0}[5a(1-h)-2b]$$

[put $x=1-h$; when $x\to1^-$, then $h\to0$]

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Since A^{-1} exists, system has a unique solution given by

$$X = A^{-1}B.$$

$$\begin{bmatrix}x\\y\\z\end{bmatrix} = \frac{1}{4}\begin{bmatrix}-2 & 0 & 2\\ 5 & -2 & -1\\ 1 & 2 & -2\end{bmatrix}\begin{bmatrix}6\\7\\12\end{bmatrix}$$

$$= \frac{1}{4}\begin{bmatrix}-12+0+24\\ 30-14-12\\ 6+14-12\end{bmatrix}$$

$$\begin{bmatrix}x\\y\\z\end{bmatrix} = \frac{1}{4}\begin{bmatrix}12\\4\\8\end{bmatrix} = \begin{bmatrix}3\\1\\2\end{bmatrix}$$

Hence, $x=3,\ y=1,\ z=2$

OR

If $A=\begin{bmatrix}2 & -3 & 5\\ 3 & 2 & -4\\ 1 & 1 & -2\end{bmatrix}$, find A^{-1}.

Use it to solve the system of equations

$2x-3y+5z=11.3x+2y-4z=-5, x+y-2z=-3.$

Sol :

We have, $A=\begin{bmatrix}2 & -3 & 5\\ 3 & 2 & -4\\ 1 & 1 & -2\end{bmatrix}$

Here $|A| = \begin{vmatrix}2 & -3 & 5\\ 3 & 2 & -4\\ 1 & 1 & -2\end{vmatrix}$

$$= 2(-4+4)+3(-6+4)+5(3-2)$$

$$= -6+5$$

$$|A| = -1$$

Since $|A| \neq 0$, matrix A is non-singular and A^{-1} exists.

Cofactors of elements of $|A|$ are

$A_{11}=0$	$A_{21}=-1$	$A_{31}=2$
$A_{12}=2$	$A_{22}=-9$	$A_{32}=23$
$A_{13}=1$	$A_{23}=-5$	$A_{33}=13$

$$adj(A) = \begin{bmatrix}0 & 2 & 1\\ -1 & -9 & -5\\ 2 & 23 & 13\end{bmatrix}^T$$

$$= \begin{bmatrix}0 & -1 & 2\\ 2 & -9 & 23\\ 1 & -5 & 13\end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}adj(A)$$

$$= (-1)\begin{bmatrix}0 & -1 & 2\\ 2 & -9 & 23\\ 1 & -5 & 13\end{bmatrix}$$

$$= \begin{bmatrix}0 & 1 & -2\\ -2 & 9 & -23\\ -1 & 5 & -13\end{bmatrix}$$

Now, consider the given system of linear equations

$$2x-3y+5z = 11$$

$$3x+2y-4z = -5$$

and $$x+y-2z = -3$$

which can be written in matrix form as

$$AX = B \qquad ...(ii)$$

where, $A=\begin{bmatrix}2 & -3 & 5\\ 3 & 2 & -4\\ 1 & 1 & -2\end{bmatrix}$, $X=\begin{bmatrix}x\\y\\z\end{bmatrix}$ and $B=\begin{bmatrix}11\\-5\\-3\end{bmatrix}$

Since A^{-1} exists, system has a unique solution given by

$$X = A^{-1}B.$$

$$\begin{bmatrix}x\\y\\z\end{bmatrix} = \begin{bmatrix}0 & 1 & -2\\ -2 & 9 & -23\\ -1 & 5 & -13\end{bmatrix}\begin{bmatrix}11\\-5\\-3\end{bmatrix} \quad \text{[from Eq. (i)]}$$

$$= \begin{bmatrix}-5+6\\ -22-45+69\\ -11-25+39\end{bmatrix} = \begin{bmatrix}1\\2\\3\end{bmatrix}$$

Equating the corresponding elements, we get

$x=1,\ y=2$ and $z=3$

28. Find the coordinates of the foot of perpendicular drawn from the point $A(-1,8,4)$ to the line joining the points $B(0,-1,3)$ and $C(2,-3,-1)$. Hence, find the image of the point A in the line BC.

Sol :

Equation of a line joining the points $B(0,-1,3)$ and $C(2,-3,-1)$,

$$\vec{r} = (0\hat{i}-\hat{j}+3\hat{k})+\lambda[(2-0)\hat{i}+(3-1)\hat{j}+(-1-3)\hat{k}]$$

$$= (-\hat{j}+3\hat{k})+\lambda(2\hat{i}-2\hat{j}-4\hat{k})$$

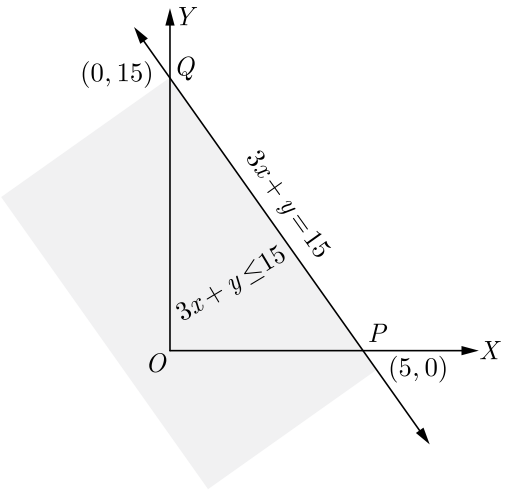
$$= (2\lambda)\hat{i}+(-2\lambda-1)\hat{j}+(-4\lambda+3)\hat{k}$$

So, any point on line BC is to the form

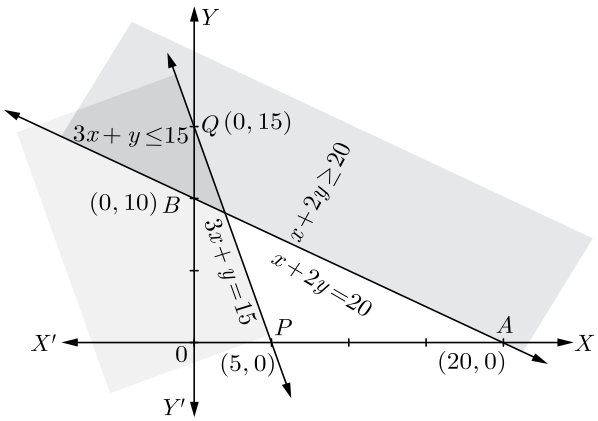
$$(2\lambda, -2\lambda-1, -4\lambda+3)$$

Let foot of the perpendicular drawn from point A to the line BC be $T(2\lambda,-2\lambda-1,-4\lambda+3)$ as shown below

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Thus, the double shaded region, common to the above two shaded regions, represented by the given inequalities, is the required solution region of the given system of inequalities. Hence, all the points in this common (double) shaded region are the solutions of the given system of linear inequalities.



SECTION - E

Case study based questions are compulsory.

36. The Vande Bharat Express, also known as Train 18, is a semi-high-speed, intercity, electric multiple unit train operated by the Indian Railways on 4 routes as of October 2022. Routes include New Delhi to Shri Mata Vaishno



In a survey at Vande Bharat Train, IRCTC asked the passenger to rate and review the food served in train. Suppose IRCTC asked 500 passenger selected at random to rate food according to price (low, medium, or high) and food (1, 2, 3, or 4 stars). The results of this survey are presented in the two-way, or contingency, table below. The numbers in this table represent frequencies. For example, in the third row and fourth column, 30 people rated the prices high and the food 4 stars. The last column contains the sum for each row, and similarly, the bottom row contains the sum for each column. These sums are often called marginal totals.

Price	**Food rating**			
Low	20	30	90	10
Medium	50	80	90	30
High	20	10	40	30

Assume that these results are representative of the entire passenger of train, so the relative frequency of occurrence is the true probability of the event. A passenger from train is randomly selected.

(i) Find the probability that the passenger rates the prices medium.

(ii) Find the probability that the passenger rates the food 2 stars.

(iii) Suppose the passenger selected rates the prices high. What is the probability that he rates the restaurants 1 star?

(iv) Suppose the passenger selected does not rate the food 4 stars. What is the probability that she rates the prices high?

Sol :

We prepare the following frequency table.

Price	**Food rating**				
	(A)	(B)	(C)	(D)	
Low (L)	20	30	90	10	150

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$$= \lim_{h \to 0}(5a - 5ah - 2b) = 5a - 2b$$

and RHL $= \lim_{x \to 1^+}(3ax + b)$

$$= \lim_{h \to 0}[3a(1+h) + b]$$

[put $x = 1 + h$; when $x \to 1^+$, then $h \to 0$]

$$= \lim_{h \to 0}(3a + 3ah + b) = 3a + b$$

Also, given that $f(1) = 11$

On substituting these values in Eq. (i), we get

$$5a - 2b = 3a + b = 11$$

$$3a + b = 11 \quad ...(ii)$$

and $$5a - 2b = 11 \quad ...(iii)$$

On substituting $3 \times$ Eq. (iii) from $5 \times$Eq. (ii), we get

$$13a + 5b - 15a + 6b = 55 - 33$$

$$11b = 22 \Rightarrow b = 2$$

On putting the value of b in Eq. (ii), we get

$$3a + 2 = 11 \Rightarrow 3a = 9 \Rightarrow a = 3$$

Hence, $a = 3$ and $b = 2$.

OR

Find the value of k, so that the functions f defined by $f(x) = \begin{cases} kx + 1, & \text{if } x \leq \pi \\ \cos x & \text{if } x > \pi \end{cases}$ is continuous at $x = \pi$.

Sol :

Let $f(x) = \begin{cases} \lambda(x^2 - 2x), & \text{if } x \leq 0 \\ 4x + 1, & \text{if } x > 0 \end{cases}$ is continuous at $x = 0$.

Then, $(\text{LHL})_{x=0} = (\text{RHL})_{x=0} = f(0)$ ...(i)

Now, $f(0) = \lambda[0 - 0] = 0$,

$$\text{LHL} = \lim_{x \to 0^-} f(x)$$

$$= \lim_{h \to 0} f(0 - h)$$

$$= \lambda \lim_{h \to 0}[(0-h)^2 - 2(0-h)]$$

$$= \lambda \times 0 = 0$$

and $$\text{RHL} = \lim_{x \to 0^+} f(x) = \lim_{h \to 0} f(0 + h)$$

$$= \lim_{h \to 0} 4(0 + h) + 1 = 1$$

LHL $\neq$ RHL, which is a contradiction to Eq. (i).

There is no value of λ for which $f(x)$ is continuous at $x = 0$.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Find the intervals in which the function given by $f(x) = \frac{3}{10}x^4 - \frac{4}{5}x^3 - 3x^2 + \frac{36x}{5} + 11$ is

(i) strictly increasing

(ii) strictly decreasing.

Sol :

We have $f(x) = \frac{3}{10}x^4 - \frac{4}{5}x^3 - 3x^2 + \frac{36x}{5} + 11$

Differentiating both sides w.r.t. x, we get

$$f'(x) = \frac{6}{5}x^3 - \frac{12}{5}x^2 - 6x + \frac{36}{5}$$

For strictly increasing or strictly decreasing, put $f'(x) = 0$, we get

$$\frac{6}{5}x^3 - \frac{12}{5}x^2 - 6x + \frac{36}{5} = 0$$

$$x^3 - 2x^2 - 5x + 6 = 0$$

$$(x-1)(x+2)(x-3) = 0$$

$$x = -2,\ 1, \text{ or } 3$$

Now, we find intervals in which $f(x)$ is strictly increasing or strictly decreasing.

Interval	$f'(x)$ $(x+2)(x-1)(x-3)$	**Sign of** $f'(x)$
$x < -2$	$(-)(-)(-)$	$-$ ve
$-2 < x < 1$	$(+)(-)(-)$	$+$ ve
$1 < x < 3$	$(+)(+)(-)$	$-$ ve
$x > 3$	$(+)(+)(+)$	$+$ ve

A function $f(x)$ is said to be strictly increasing, if $f'(x) > 0$ and it is said to be strictly decreasing, if $f'(x) < 0$. So, the given function $f(x)$ is

(i) strictly increasing on the intervals $(-2, 1)$ and $(3, \infty)$.

(ii) strictly decreasing on the intervals $(-\infty, -2)$ and $(1, 3)$.

OR

A ladder 5 m long is leaning against a wall. Bottom of ladder is pulled along the ground away from wall at the rate of 2 m/s. How fast is the height on the wall decreasing, when the foot of ladder is 4 m away from the wall?

Sol :

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$$(e^x + e^{-x})\,dy - (e^x - e^{-x})\,dx = 0$$

$$\frac{dy}{dx} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Separating the variable, we get

$$dy = \frac{e^x - e^{-x}}{e^x + e^{-x}}\,dx$$

Integrating the variables, we get

$$\int dy = \int \frac{e^x - e^{-x}}{e^x + e^{-x}}\,dx$$

Put $e^x + e^{-x} = t$

$$(e^x - e^{-x})\,dx = dt$$

$$\int dy = \int \frac{dt}{t}$$

$$y = \log|t| + C$$

$$y = \log|e^x + e^{-x}| + C$$

$$y = \log|e^x - e^{-x}| + C$$

is the required solution.

SECTION - C

This section comprises of short answer-type questions (SA) of 3 marks each.

26. Find the particular solution of the differential equation $(+ e^{2x})\,dy + (1 + y^2)\,e^x\,dx = 0$, given that $y = 1$, when $x = 0$.

Sol :

We have $(1 + e^{2x})\,dy + (1 + y^2)\,e^x\,dx = 0$

Separating the variables, we get

$$\frac{dy}{1 + y^2} = \frac{-e^x}{1 + e^{2x}}\,dx$$

Integrating both sides, we have

$$\int \frac{dy}{1 + y^2} = -\int \frac{e^x}{1 + e^{2x}}\,dx$$

Substituting $e^x = t \Rightarrow e^x\,dx = dt$ in RHS, we get

$$\tan^{-1} y = -\int \frac{1}{1 + t^2}\,dt$$

$$\tan^{-1} y = -\tan^{-1} t + C$$

$$\tan^{-1} y = -\tan^{-1}(e^x) + C \qquad ...(1)$$

Also, given that $y = 1$, when $x = 0$.

Substituting above values in Eq. (1), we get

$$\tan^{-1} 1 = -\tan^{-1}(e^0) + C$$

$$\tan^{-1} 1 = -\tan^{-1} 1 + C$$

$$2\tan^{-1} 1 = C$$

$$2\tan^{-1}\left(\tan\frac{\pi}{4}\right) = C$$

$$C = 2 \times \frac{\pi}{4} = \frac{\pi}{2}$$

Substituting $C = \frac{\pi}{2}$ in Eq. (1), we get

$$\tan^{-1} y = -\tan^{-1} e^x + \frac{\pi}{2}$$

$$y = \tan\left[\frac{\pi}{2} - \tan^{-1}(e^x)\right] = \cot\left[\tan^{-1}(e^x)\right]$$

$$= \cot\left[\cot^{-1}\left(\frac{1}{e^x}\right)\right]$$

$$y = \frac{1}{e^x}$$

which is the required solution.

27. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \\ 3 & 1 & 1 \end{bmatrix}$, find A^{-1}.

Hence, solve the system of equations $x + y + z = 6$, $x + 2z = 7$, $3x + y + z = 12$.

Sol :

We have, $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \\ 3 & 1 & 1 \end{bmatrix}$

$$|A| = 1(0 - 2) - 1(1 - 6) + 1(1 - 0)$$

$$= -2 + 5 + 1 = 4 \neq 0$$

Since $|A| \neq 0$, matrix A is non-singular and A^{-1} exists.

$$adj(A) = \begin{bmatrix} -2 & 0 & 2 \\ 5 & -2 & -1 \\ 1 & 2 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \cdot adj\ A$$

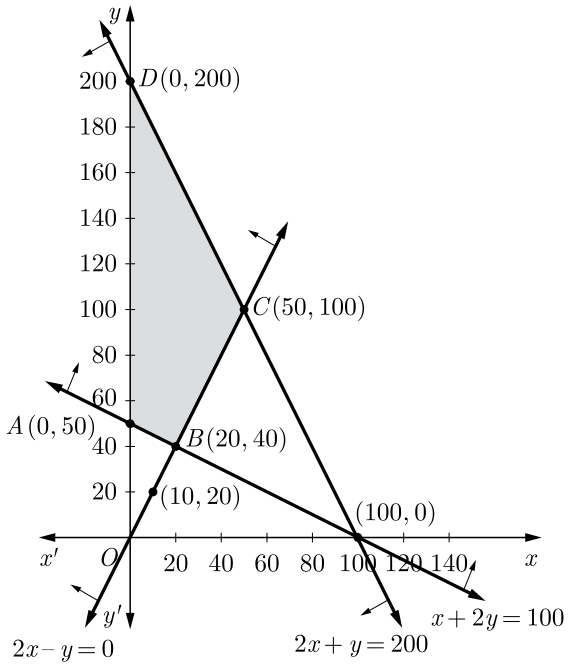
$$= \frac{1}{4}\begin{bmatrix} -2 & 0 & 2 \\ 5 & -2 & -1 \\ 1 & 2 & -1 \end{bmatrix}$$

Given, systems of the equations is $x + y + z = 6$, $x + 2z = 7$, $3x + y + z = 12$, which can be written in matrix form as

$$AX = B$$

where $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \\ 3 & 1 & 1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 6 \\ 7 \\ 12 \end{bmatrix}$

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Since, the feasible region is a bounded region, we can check the profit function at all the vertices to find the maxima. The values of Z at corner points are given below

Corner points	$Z = x + 2y$
$A(0,50)$	$Z = 0 + 2 \times 50 = 100$
$B(20,40)$	$Z = 20 + 2 \times 40 = 100$
$C(50,100)$	$Z = 50 + 2 \times 100 = 250$
$D(0,200)$	$Z = 0 + 2 \times 200 = 400$

The maximum value of Z is 400 at $D(0,200)$ and the minimum value of Z is 100 at the points on the line segment joining $A(0,50)$ and $B(20,40)$.

OR

A manufacturer has employed 5 skilled men and 10 semi-skilled men and makes two models A and B of an article. The making of one item of model A requires 2 hours work by a skilled man and 2 hours by a semi-skilled man. One item of model B requires 1 hour by a skilled man and 3 hours by semi-skilled man. No man is expected to work more than 8 hours per day. The manufacturer profit on an item of model A is Rs. 15 and on an items of model B is Rs. 10. How many of items of models should be made per day in order to maximize daily profit? Formulate the above LPP and solve it graphically and find the maximum profit.

Sol :

Let the company produce x items of A model and y items of B model. Clearly $x \geq 0$ and $y \geq 0$.

The manufacturer profit on an item of model A is Rs. 15 and on an items of model B is Rs. 10.

Profit function $Z = 15x + 10y$

Now, total time spent by 5 skilled men should be less than 40 and the total time spent by 10 semi-skilled men should be less than 80.

We tabulate the given data as given below.

	A	B	**Time Spent in 8 h**
Skilled (5)	2 h	1 h	$2x + y \leq (8 \times 5)$
Semi Skilled (10)	2 h	3 h	$2x + 3y \leq (8 \times 10)$

Now problem reduced to LPP,

Maximize $Z = 15x + 10y$

subject to $2x + y \leq 40$ ...(i)

$2x + 3y \leq 80$ ...(ii)

Clearly $x \geq 0$ and $y \geq 0$

(i) Region corresponding to $2x + y \leq 40$:

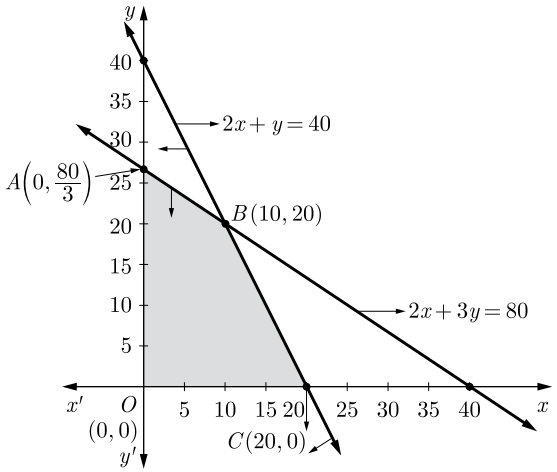
Line $2x + y = 40$			
x	0	20	Point $(0,0)$ is true for $2x + y \leq 40$. So, the region is towards the origin.
y	40	0	

(ii) Region corresponding to $2x + 3y \leq 80$:

Line $2x + 3y = 80$			
x	0	40	Point $(0,0)$ is true for $2x + 3y \leq 80$, So, the region is towards the origin.
y	$\frac{80}{3}$	0	

Since $x \geq 0$, $y \geq 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is $OABCO$. Solving equation $2x + y = 40$ and $2x + 3y = 80$ we get $B(10,20)$. The corner points of the feasible region are $A(0, \frac{80}{3})$, $B(10,20)$, $C(20,0)$, $O(0,0)$.



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Since, the feasible region is a bounded region, we can check the profit function at all the vertices to find the maxima. The values of Z at corner points are given below

Corner points	$Z = x + 2y$
$A(0, \frac{80}{3})$	$Z = 15(0) + 10(\frac{80}{3}) = \frac{800}{3}$
$B(10,20)$	$Z = 15(10) + 10(20) = 350$
$C(20,0)$	$Z = 15(20) + 10(0) = 300$
$O(0,0)$	$Z = 15(0) + 10(0) = 0$

Thus, the maximum profit of Rs 350 is at point $B(10,20)$. Hence, the manufacturer should makes 10 items of A model and 20 items of B model for maximum profit of Rs. 350.

34. (i) Evaluate $\sin^{-1}\left(\sin\frac{3\pi}{5}\right)$.

(ii) Write the value of $\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$.

(iii) Write the principal value of $\tan^{-1}\left[\sin\left(-\frac{\pi}{2}\right)\right]$.

Sol :

(i) $\sin^{-1}\left(\sin\frac{3\pi}{5}\right) = \frac{3\pi}{5} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

which is the principal value branch of $\sin^{-1}x$.

Now, $\sin^{-1}\left(\sin\frac{3\pi}{5}\right) = \sin^{-1}\left[\sin\left(\pi - \frac{2\pi}{5}\right)\right]$

$= \sin^{-1}\left(\sin\frac{2\pi}{5}\right)$

$= \frac{2\pi}{5} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Hence, $\sin^{-1}\left(\sin\frac{3\pi}{5}\right) = \frac{2\pi}{5}$

(ii) $\tan^{-1}\left(\tan\frac{3\pi}{4}\right) = \tan^{-1}\left[\tan\left(\pi - \frac{\pi}{4}\right)\right]$

$= \tan^{-1}\left(-\tan\frac{\pi}{4}\right)$

$= -\tan^{-1}\left(\tan\frac{\pi}{4}\right)$

$= -\frac{\pi}{4}$

$[\text{since } \tan^{-1}(-\theta) = -\tan^{-1}\theta]$

(iii) $\tan^{-1}\left[\sin\left(-\frac{\pi}{2}\right)\right] = \tan^{-1}\left[-\sin\frac{\pi}{2}\right]$

$[\text{since } \sin(-\theta) = -\sin\theta]$

$= \tan^{-1}(-1)$

$= -\tan^{-1}1$

$= -\frac{\pi}{4}$

$[\text{since } \tan(-\theta) = -\tan\theta$

OR

Using the principal value, evaluate the following:

(i) $\tan^{-1}1 + \sin^{-1}\left(-\frac{1}{2}\right)$

(ii) $\cos^{-1}\left(\cos\frac{2\pi}{3}\right) + \sin^{-1}\left(\sin\frac{2\pi}{3}\right)$

Sol :

(i) $\tan^{-1}1 + \sin^{-1}\left(-\frac{1}{2}\right)$

Range of the principal value branch of $\tan^{-1}$ is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and range of the principal value branch of $\sin^{-1}$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Let $\tan^{-1}1 = \theta$, then $\tan\theta = 1 = \tan\frac{\pi}{4}$

$\theta = \frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Again, let $\sin^{-1}\left(-\frac{1}{2}\right) = \phi$,

Then $\sin\phi = -\frac{1}{2} = -\sin\frac{\pi}{6}$

Now, $\sin\phi = -\frac{1}{2} = -\sin\frac{\pi}{6}$

$= \sin\left(-\frac{\pi}{6}\right)$

Hence, $\phi = -\frac{\pi}{6}$

$\sin^{-1}\left(-\frac{1}{2}\right) = -\frac{\pi}{6}$

So, $\tan^{-1}1 + \sin^{-1}\left(-\frac{1}{2}\right) = \theta + \phi$

$= \frac{\pi}{4} + \left(-\frac{\pi}{6}\right)$

$= \frac{\pi}{4} - \frac{\pi}{6}$

$= \frac{3\pi - 2\pi}{12}$

$= \frac{\pi}{12}$

(ii) $\cos^{-1}\left(\cos\frac{2\pi}{3}\right) + \sin^{-1}\left(\sin\frac{2\pi}{3}\right)$

$= \cos^{-1}\left\{\cos\left(\pi - \frac{\pi}{3}\right)\right\} + \sin^{-1}\left\{\sin\left(\pi - \frac{\pi}{3}\right)\right\}$

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$$= \cos^{-1}\left(-\cos\frac{\pi}{3}\right) + \sin^{-1}\left(\sin\frac{\pi}{3}\right)$$

$$= \cos^{-1}\left(-\frac{1}{2}\right) + \frac{\pi}{3}$$

$$= \frac{2\pi}{3} + \frac{\pi}{3} = \pi$$

$$\left[\text{Since Put } \cos^{-1}\left(-\frac{1}{2}\right) = \theta, \theta \in [0,\pi]\right]$$

$$\cos\theta = -\frac{1}{2} = -\cos\frac{\pi}{3}$$

$$= \cos\left(\pi - \frac{\pi}{3}\right)$$

$$\cos\theta = \cos\frac{2\pi}{3}$$

$$\theta = \frac{2\pi}{3}$$

$$\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$$

35. Find the shortest distance between the following lines. $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$, $\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$.

Sol :

Given equations of lines are

We have $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$ ...(1)

and $\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$ ...(2)

Comparing above equations with one point form of equation of line, i.e.

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c},$$

we get $a_1 = 1,\ b_1 = -2,\ c_1 = 1,$

$x_1 = 3,\ y_1 = 5,\ z_1 = 7$

and $a_2 = 7,\ b_2 = -6,\ c_2 = 1,\ x_2 = -1,$

$y_2 = -1,\ z_2 = -1$

Shortest distance between two lines is given by

$$d = \frac{\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1c_2-b_2c_1)^2 + (c_1a_2-c_2a_1)^2 + (a_1b_2-a_2b_1)^2}}$$

$$d = \frac{\begin{vmatrix} -1-3 & -1-5 & -1-7 \\ 1 & -2 & 1 \\ 7 & -6 & 1 \end{vmatrix}}{\sqrt{(-2-6)^2 + (7-1)^2 + (-6+14)^2}}$$

$$= \frac{\begin{vmatrix} -4 & -6 & -8 \\ 1 & -2 & 1 \\ 7 & -6 & 1 \end{vmatrix}}{\sqrt{(4)^2 + (6)^2 + (8)^2}}$$

$$= \left|\frac{-4(-2+6) + 6(1-7) - 8(-6+14)}{\sqrt{(4)^2 + (6)^2 + (8)^2}}\right|$$

$$= \left|\frac{-4(4) + 6(-6) - 8(8)}{\sqrt{16+36+64}}\right|$$

$$= \left|\frac{-16-36-64}{\sqrt{116}}\right|$$

$$= \left|\frac{-116}{\sqrt{116}}\right| = \frac{116}{\sqrt{116}} = \sqrt{116}$$

Hence, the required shortest distance is $\sqrt{116}$ units.

OR

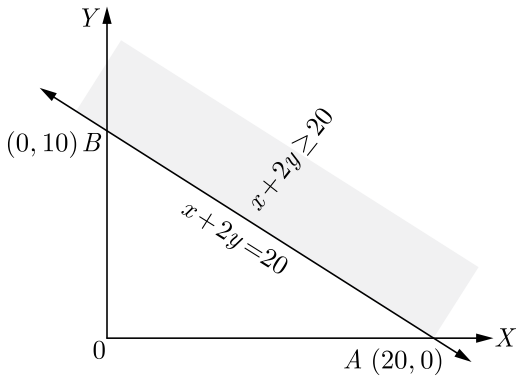
Solve graphically the following system of in equations.

$x + 2y \geq 20,\ 3x + y \leq 15$

Sol :

First we convert the inequalities $x + 2y \geq 20$ and $3x + y \leq 15$ into equations $x + 2y = 20$ and $3x + y = 15$ respectively.

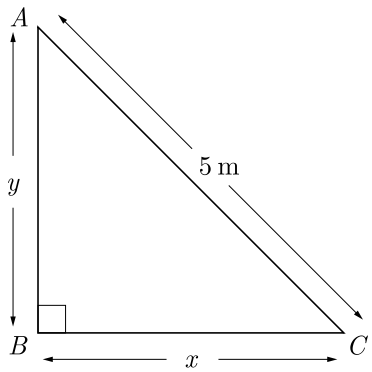
Region Represented by $x + 2y \geq 20$: The line $x + 2y = 20$ meets the axes at $A(20,0)$ and $B(0,10)$. Draw a thick line joining A and B. We find that $(0,0)$ does not satisfy the inequality $x + 2y \geq 20$. So, the portion, which does not contain the origin, represents the solution set of the inequality $x + 2y \geq 20$.



Region Represented by $3x + y \leq 15$: The line $3x + y = 15$ meets the axes at $P(5,0)$ and $Q(0,15)$. Draw a thick line joining P and Q. We find that $(0,0)$ satisfies the inequality $3x + y \leq 15$. So, the portion containing the origin, represents the solution set of the inequality $3x + y \leq 15$.

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Let AC be the ladder, $BC = x$ and height of the wall, $AB = y$. As per question the figure is shown below.



As the ladder is pulled along the ground away from the wall at the rate of 2 m/s, we have

$$\frac{dx}{dt} = 2 \text{ m/s}$$

To find $\frac{dy}{dt}$, when $x = 4$.

In right angled ΔABC, by Pythagoras theorem, we get

$$(AB)^2 + (BC)^2 = (AC)^2$$

$$x^2 + y^2 = 25 \quad ...(1)$$

$$(4)^2 + y^2 = 25$$

$$16 + y^2 = 25$$

$$y^2 = 9$$

$$y = \sqrt{9} = 3$$

Differentiating both sides of Eq. (1) w.r.t. t, we get

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

$$x\frac{dx}{dt} + y\frac{dy}{dt} = 0 \quad ...(2)$$

Substituting the values of x, y and $\frac{dx}{dt}$ in Eq. (2)

$$4 \times 2 + 3 \times \frac{dy}{dt} = 0$$

$$8 + 3 \times \frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = \frac{-8}{3} \text{ m/s}$$

Hence, height of the walls is decreasing at the rate of $\frac{8}{3}$ m/s.

NOTE : In a rate of change of a quantity, + ve sign shows that it is increasing and − ve sign shows that it is decreasing.

33. Maximise and minimise $Z = x + 2y$ subject to the constraints

$$x + 2y \geq 100$$

$$2x - y \leq 0$$

$$2x + y \leq 200$$

$$x, y \geq 0$$

Solve the above LPP graphically.

Sol :

We have to minimise and maximise

$$Z = x + 2y$$

Subject to constraints,

$$x + 2y \geq 100 \quad ...(i)$$

$$2x - y \leq 0 \quad ...(ii)$$

$$2x + y \leq 200 \quad ...(iii)$$

and $x \geq 0, y \geq 0$

(i) Region corresponding to $x + 2y \geq 100$:

Line $x + 2y = 100$			
x	0	100	Point $(0,0)$ is false for $x + 2y \geq 100$. So, the region is away from the origin.
y	50	0	

(ii) Region corresponding to $2x - y \leq 0$:

Line $2x - y = 0$			
x	0	10	Point $(0,5)$ is true for $2x - y \leq 0$, thus region includes $(0,5)$ and towards y axis.
y	0	20	

(iii) Region corresponding to $2x + y \leq 200$:

Line $2x + y = 200$			
x	0	100	Point $(0,0)$ is true for $2x + y \leq 200$. So, the region is towards the origin.
y	200	0	

Since $x \geq 0$, $y \geq 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is $ABCDA$. Solving equations $2x - y = 0$ and $x + 2y = 100$, we get $B(20,40)$. Again, solving equations $2x - y = 0$ and $2x + y = 200$, we get $C(50,100)$. The corner points of the feasible region are $A(0,50)$, $B(20,40)$, $C(50,100)$ and $D(0,200)$.

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Price	Food rating				
Medium (M)	50	80	90	30	250
High (H)	20	10	40	30	100
	90	120	220	70	500

(i) the probability that the passenger rates the prices medium,

Now $$P(M) = \frac{50+80+90+20}{500}$$
$$= \frac{250}{500} = 0.5$$

(ii) the probability that the passenger rates the food 2 stars,

$$P(B) = \frac{30+80+10}{500}$$
$$= \frac{120}{500} = 0.24$$

(iii) If the passenger selected rates the prices high, the probability that he rates the restaurants 1 star,
There are 100 passengers who rated the price high out of which 20 passengers rated the food 1 star.

Thus $$P(A \mid H) = \frac{20}{100} = 0.20$$

(iv) If passenger selected does not rate the food 4 stars, the probability that he or she rates the prices high,
There are $90+120+220=430$ passenger who doesn't rate the food 4 start. Out of which $20+10+40=70$ passenger rated the prices high.

$$P(H \mid D') = \frac{70}{430} = 0.163$$

37. Publishing is the activity of making information, literature, music, software and other content available to the public for sale or for free. Traditionally, the term refers to the creation and distribution of printed works, such as books, newspapers, and magazines.



NODIA Press is a such publishing house having two branch at Jaipur. In each branch there are three offices. In each office, there are 2 peons, 5 clerks and 3 typists. In one office of a branch, 5 salesmen are also working. In each office of other branch 2 head-clerks are also working. Using matrix notations find:

(i) the total number of posts of each kind in all the offices taken together in each branch.

(ii) the total number of posts of each kind in all the offices taken together from both branches.

Sol :

Consider the three row matrices,

$$A_1 = [2\ \ 5\ \ 3\ \ 5\ \ 0]$$
$$A_2 = [2\ \ 5\ \ 3\ \ 0\ \ 0]$$
$$A_3 = [2\ \ 5\ \ 3\ \ 0\ \ 0]$$

The matrices A_1, A_2, A_3 represent the three offices of one branch (say A), where the elements appearing in the row represents the number of peons, clerks, typists, salesman and head-clerks taken in that order working in the three offices.

Now, consider the three row matrices,

$$B_1 = [2\ \ 5\ \ 3\ \ 0\ \ 2]$$
$$B_2 = [2\ \ 5\ \ 3\ \ 0\ \ 2]$$
$$B_3 = [2\ \ 5\ \ 3\ \ 0\ \ 2]$$

The matrices B_1, B_2, B_3 represents the three offices of other branch (say B), where the elements appearing in the row represents the number of peons, clerks, typists, salesman and head-clerks taken in that order working in the three offices.

The total number of posts of each kind in all the offices of branch A are the elements of the matrix:

$$A = A_1 + A_2 + A_3$$
$$= [2\ \ 5\ \ 3\ \ 5\ \ 0] + [2\ \ 5\ \ 3\ \ 0\ \ 0] + [2\ \ 5\ \ 3\ \ 0\ \ 0]$$
$$= [2+2+2\ 5+5+5\ 3+3+3\ 5+0+0+0+0]$$
$$= [6\ \ 15\ \ 9\ \ 5\ \ 0]$$

(i) The total number of posts of each kind in all the offices of branch B are the element of the matrix:

$$B = B_1 + B_2 + B_3$$
$$= [2\ \ 5\ \ 3\ \ 0\ \ 2] + [2\ \ 5\ \ 3\ \ 0\ \ 2] + [2\ \ 5\ \ 3\ \ 0\ \ 2]$$
$$= [6\ \ 15\ \ 9\ \ 0\ \ 6]$$

(ii) The total number of posts of each kind in all the offices taken together from both branches A and B are the elements of the matrix:

$$A + B = (A_1 + A_2 + A_3) + (B_1 + B_2 + B_3)$$
$$= [12\ \ 30\ \ 18\ \ 5\ \ 6]$$

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38. Different types of drugs affect your body in different ways, and the effects associated with drugs can vary from person to person. How a drug effects an individual is dependent on a variety of factors including body size, general health, the amount and strength of the drug, and whether any other drugs are in the system at the same time. It is important to remember that illegal drugs are not controlled substances, and therefore the quality and strength may differ from one batch to another.



The concentration $C(t)$ in milligrams per cubic centimeter (mg/cm^3) of a drug in a patient's bloodstream is $0.5\,\text{mg/cm}^3$ immediately after an injection and t minutes later is decreasing at the rate

$$C'(t) = \frac{-0.01e^{0.01t}}{(e^{0.01t}+1)^2}\,\text{mg/cm}^3 \text{ per minute}$$

A new injection is given when the concentration drops below $0.05\,\text{mg/cm}^3$.

(i) Find an expression for $C(t)$.

(ii) What is the concentration after 1 hour? After 3 hours?

Sol :

(i) $$C(t) = \int C'(t)\,dt$$

$$= \int \frac{-0.01e^{0.01t}}{(e^{0.01t}+1)^2}\,dt$$

Let $u = e^{0.01t}+1$

Then $\dfrac{du}{dt} = 0.01e^{0.01t}$

or $100\,du = e^{0.01t}\,dt$

$$= -0.01\int \frac{1}{(e^{0.01t}+1)^2}e^{0.01t}\,dt$$

$$= -\int \frac{1}{u^2}\,du$$

$$= \frac{1}{e^{0.01t}+1} + C$$

When the shot is initially administered, $t = 0$ and

$$0.5 = \frac{1}{e^0+1} + C \text{ or } C = 0$$

So, $$C(t) = \frac{1}{e^{0.01t}+1}$$

(ii) After one hour, when $t = 60$ minutes, the concentration is

$$C(60) = \frac{1}{e^{0.01(60)}+1}$$

$$\approx 0.3543\,\text{mg/cm}^3$$

After three hours, when $t = 180$ minutes, the concentration is

$$C(180) = \frac{1}{e^{0.01(180)}+1}$$

$$\approx 0.1419\,\text{mg/cm}^3$$

In this book, we have provided solutions along with the papers. To download the same papers in unsolved format for printing, please visit
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Sample Paper 03

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

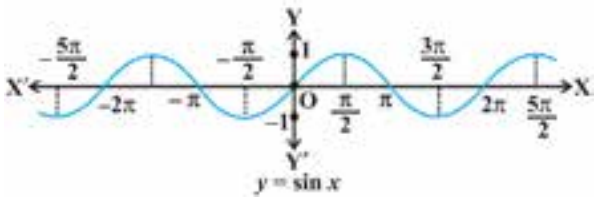
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. Which of the following points lies on the curve shown in the image?



(a) $\left(\frac{\pi}{2},0\right)$ (b) $\left(\frac{\pi}{2},1\right)$

(c) $(\pi,1)$ (d) $(2\pi,-1)$

Sol :

(b) $\left(\frac{\pi}{2},1\right)$

On the positive X side of the graph, the curve rises upward and touches the top at the point where $x=\pi/2$. At this position, the Y-coordinate reaches exactly 1. Therefore, the point $(\pi/2,1)$ lies clearly on the curve. Option (a) is incorrect because at $x=\pi/2$, the value is not 0. Option (c) is incorrect because at $x=\pi$, the value of sine returns to 0, not 1. Option (d) is incorrect because at $x=\pi/2$, the value is again 0, not –1. Hence, the correct point on the curve is $(\pi/2,1)$.

2. If $\vec{a}=\hat{i}+\hat{j}+2\hat{k}$ and $\vec{b}=3\hat{i}+2\hat{j}-\hat{k}$, then the value of $(\vec{a}+3\vec{b}).(2\vec{a}-\vec{b})$ is-

(a) 15 (b) 18

(c) -18 (d) -15

Sol :

(d) -15

We have,

$$(\vec{a}+3\vec{b}).(2\vec{a}-\vec{b}) = 2\vec{a}.\vec{a}+6\vec{a}.\vec{b}-\vec{a}.\vec{b}-3\vec{b}.\vec{b}$$
$$= 2|\vec{a}|^2+5\vec{a}.\vec{b}-3|\vec{b}|^2$$

Here $\vec{a} = \hat{i}+\hat{j}+2\hat{k}$

$\vec{b} = 3\hat{i}+2\hat{j}-\hat{k}$

$= 2(\sqrt{1^2+1^2+2^2})^2+5(1\times3+1\times2$

$+2\times(-1))-3(\sqrt{3^2+2^2+(-1)^2})^2$

$= 2\times6+5\times3-3\times14$

$= 12+15-42 = 27-42 = -15$

3. The maximum value of $\left(\frac{1}{x}\right)^{2x^2}$ is

(a) 1 (b) e

(c) $e^{1/e}$ (d) None of these

Sol :

(c) $e^{1/e}$

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Let, $f(x) = \left(\frac{1}{x}\right)^{2x^2}$

Taking logarithm on both sides, we get

$$\log f(x) = 2x^2 \log \frac{1}{x}$$

$$= -2x^2 \log x \qquad \left[\log \frac{1}{a} = -\log a\right]$$

Differentiating both sides, we get

$$\frac{f'(x)}{f(x)} = -2x^2 \times \frac{1}{x} + \log x(-4x)$$

For maximum value of $f(x)$, $f'(x)$ should be zero. Thus,

$$4x \log x = -2x$$

$$\log x = \frac{-1}{2}$$

$$x = e^{\frac{-1}{2}}$$

Thus, maximum value of $f(x)$ occur at $x = e^{\frac{-1}{2}}$.
Hence, Maximum value

$$= \left(\frac{1}{e^{-1/2}}\right)^{2(e^{-1/2})^2}$$

$$= (e^{1/2})^{2e^{-1}} = e^{e^{-1}} = e^{1/e}$$

4. Let l_1, m_1, n_1 and l_2, m_2, n_2 be the direction cosines of two straight lines. Both the lines are perpendicular to each other, if

(a) $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$

(b) $l_1 l_2 + m_1 m_2 + n_1 n_2 = 1$

(c) $\frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}$

(d) $\frac{l_1}{l_2} + \frac{m_1}{m_2} + \frac{n_1}{n_2} = 0$

Sol :

(a) $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$

The angle between two lines having direction cosine l_1, m_1, n_1 and l_2, m_2, n_2 is given by,

$$\cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

If both lines are perpendicular then $\theta = 90°$.

Hence, $\cos 90° = l_1 l_2 + m_1 m_2 + n_1 n_2$

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

5. The direction cosines of the $y-$axis are

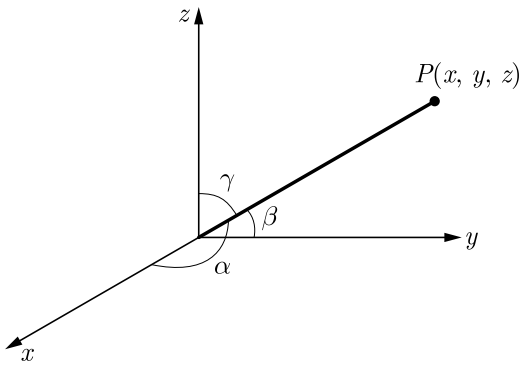
(a) (0, 0, 0) (b) (1, 0, 0)

(c) (0, 1, 0) (d) (0, 0, 1)

Sol :

(c) (0, 1, 0)

If α, β and γ are the angles which a vector $\vec{OP}$ makes with the positive direction of coordinate axes OX, OY, OZ respectively, then $\cos\alpha$, $\cos\beta$ and $\cos\gamma$ are known as the direction of cosines of $\vec{OP}$ and generally denoted by l, m and n respectively.



Clearly $\vec{PO}$ makes angles $(\pi - \alpha)$, $(\pi - \beta)$ and $(\pi - \gamma)$ with OX, OY, OZ respectively. Therefore direction cosines of $\vec{PO}$ are $\cos(\pi - \alpha)$, $\cos(\pi - \beta)$, $\cos(\pi - \gamma)$ As the X axis makes angle, 0, $\pi/2$, $\pi/2$ with OX, OY and OZ respectively. Therefore direction of cosines of X-axis are $\cos 0$; $\cos \pi/2$, $\cos \pi/2$ i.e., 1, 0, 0, similarly direction cosines of y and z axes are 0, 1, 0 and 0, 0, 1 respectively.

6. If $y = \log\cos x^2$, then $\frac{dy}{dx}$ at $x = \sqrt{\pi}$ has the value

(a) 1 (b) $\frac{\pi}{4}$

(c) 0 (d) $\sqrt{\pi}$

Sol :

(c) 0

$$y = \log\cos x^2$$

Differentiating the given function

$$y'(x) = \frac{dy}{dx} = \frac{d(\log\cos x^2)}{dx}$$

$$= \frac{1}{\cos x^2} \frac{d}{dx}(\cos x^2) \qquad \left[\frac{d}{dx}\log x = \frac{1}{x}\right]$$

$$= \frac{1}{\cos x^2} \times (-\sin x^2) \times \frac{d}{dx}(x^2)$$

$$\left[\frac{d}{dx}\cos\theta = -\sin\theta\right]$$

$$= \frac{-\sin x^2}{\cos x^2} \times 2x \qquad \left[\frac{d}{dx}x^2 = 2x\right]$$

$$= -\tan x^2 \times 2x \qquad \ldots(1)$$

Putting value of x in equation (1),

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18. Integrating factor of the differential equation $\frac{dy}{dx} + y\sec x = \tan x$ is-

(a) $\sec x + \tan x$ (b) $\sec x - \tan x$

(c) $\sec x$ (d) $\tan x \sec x$

Sol :

(a) $\sec x + \tan x$

We have,

$$\frac{dy}{dx} + y\sec x = \tan x$$

$$\text{I.f.} = e^{\int \sec x\, dx}$$

$$= e^{\log(\sec x + \tan x)}$$

$$= \sec x + \tan x$$

19. **Assertion :** If $P(E_1) = 0.3$ and $P(\overline{E_2}) = 0.6$ then $P(E_1 \cup E_2) = 0.7$

Reason : $P(\overline{E_2}) = 1 - P(E_2)$

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

Sol :

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

Given: $P(E_1) = 0.3$

$P(\overline{E}_2) = 0.6$

Assume E_1 and E_2 are two mutually exclusive events.

$$P(E_1 \cap E_2) = 0$$

$$P(E_2) = 1 - P(\overline{E}_2)$$

$$= 1 - 0.6$$

$$= 0.4 \qquad [\text{since } P(E_2) + P(\overline{E_2}) = 1]$$

We know that

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$$= 0.3 + 0.4 - 0 = 0.7$$

20. **Assertion :** If $y = \sin x^3$, then $\frac{dy}{dx} = \cos x^3 \cdot 3x^2$.

Reason: $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$.

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

Sol :

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

Given:

$$y = \sin x^3$$

Differentiating both sides with respect to x, we get

$$\frac{dy}{dx} = \frac{d}{dx}\sin x^3$$

$$= \cos x^3 \times \frac{d}{dx} x^3$$

$$= \cos x^3 \times 3x^2 = 3x^2 \cos x^3$$

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx} \text{ (Product Rule)}$$

Both Assertion and reason are true but reason is not correct explanation since we had used chain rule not the Product rule.

SECTION - B

This section comprises of very short answer-type questions (VSA) of 2 marks each.

21. Write the vector equation of a line passing through point $(1, -1, 2)$ and parallel to the line whose equation is $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z+1}{-2}$.

Sol :

Vector equation of a line passing through a point with position vector $\vec{a}$ and parallel to a given vector $\vec{b}$ is

$$\vec{r} = \vec{a} + \lambda\vec{b}, \qquad \text{where } \lambda \in R.$$

Vector for point $(1, -1, 2)$ is

$$\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$$

and vector for line $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z+1}{-2}$ is

$$\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}. \text{ DR's are } 1, 2 \text{ and } -2$$

Required vector equation of line is

$$\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \lambda(\hat{i} + 2\hat{j} - 2\hat{k}), \text{ where } \lambda \in R.$$

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$$\begin{bmatrix} 7 & 8+y \\ 10 & 2x+1 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 10 & 5 \end{bmatrix}$$

$$8+y = 0 \text{ and } 2x+1 = 5$$

$$y = -8 \text{ and } x = 2$$

14. The value of x when $\begin{bmatrix} x & 15 \\ 4 & 4 \end{bmatrix} = 0$ is-

(a) 15 (b) –15

(c) 4 (d) $4x$

Sol :

(a) 15

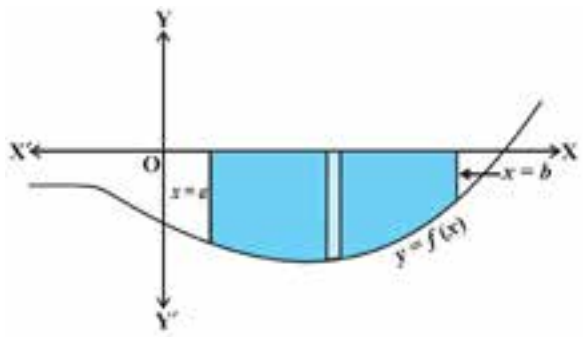
We have, $\begin{bmatrix} x & 15 \\ 4 & 4 \end{bmatrix} = 0$

$$4x - 4 \times 15 = 0$$

$$4x = 4 \times 15$$

$$x = 15$$

15. In Fig., the shaded region lies between which of the following?



(a) The curve y=f (x), the X-axis, and the vertical lines x=a and x=b

(b) The curve x=f (y), the Y-axis, and the horizontal lines y=a and y=b

(c) The curve y=f (x), the Y-axis, and the horizontal lines y=c and y=d

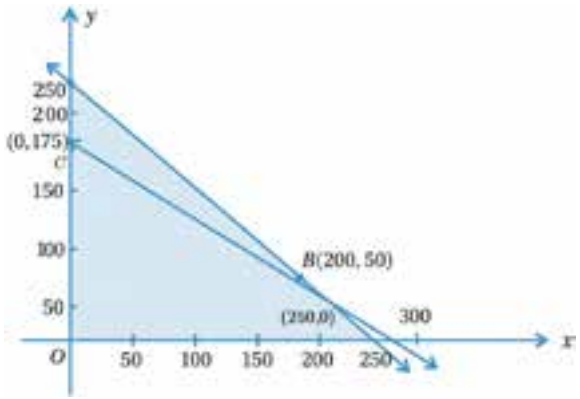
(d) The curve x=f (y), the X-axis, and the vertical lines x=a and x=b

Sol :

(a) The curve y=f(x), the X-axis, and the vertical lines x=a and x=b

The figure shows the shaded area enclosed between the X-axis (above), the curve y=f(x) (below), and the two vertical boundary lines x=a and x=b. These markings are directly visible in the diagram, making option (a) correct.

16. The corner points of the bounded feasible region of an LPP are O(0,0), A(250,0), B(200,50), and C(0,175). If the maximum value of the objective function $Z = 2ax +$ by occurs at the points A(250,0) and B(200,50), then the relation between a and b is to be determined.



(a) 2a=b (b) 2a=3b

(c) a=b (d) a=2b

Sol :

For the objective function Z=2ax+by, the maximum value occurs at two distinct points, A(250,0) and B(200,50).

This implies that the values of Z at A and B must be equal.

At A(250,0):

$$Z = 2a(250)+b(0)=500a$$

At B(200,50):

$$Z = 2a(200)+b(50)=400a+50b$$

Since both are equal at the maximum,

$$500a = 400a+50b \quad 100a = 50b \quad 2a=b$$

17. [Standard result] $\int \sqrt{1+\cos 2x}\, dx =$

(a) $\sqrt{2}\cos x + c$ (b) $\sqrt{2}\sin x + c$

(c) $-\cos x - \sin x + c$ (d) $\sqrt{2}\sin\frac{x}{2} + c$

Sol :

(b) $\sqrt{2}\sin x + c$

We have $I = \int \sqrt{1+\cos 2x}\, dx$

$$1+\cos 2x = 2\cos^2 x$$

$$I = \int \sqrt{2\cos^2 x}\, dx$$

$$= \int \sqrt{2}\cos x dx \qquad \left[\int \cos x = \sin x\right]$$

Thus, $I = \sqrt{2}\sin x + c$

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$$y'(\sqrt{\pi}) = -\tan(\sqrt{\pi})^2 \times 2(\sqrt{\pi})$$
$$= -\tan\pi \times 2\sqrt{\pi}$$

Since, $\tan\pi = 0$ then $y'(\sqrt{\pi}) = 0$

or $\left.\frac{dy}{dx}\right|_{x=\sqrt{\pi}} = 0$

7. If $-1 < x < 1$, then $2\tan^{-1}x = ?$

(a) $\sin^{-1}\frac{2x}{1+x^2}$ (b) $\sin^{-1}\frac{2x}{1-x^2}$

(c) $\sin^{-1}\frac{1-x^2}{1+x^2}$ (d) $\sin^{-1}\frac{1+x^2}{1-x^2}$

Sol :

(a) $\sin^{-1}\frac{2x}{1+x^2}$

We know that, when $-1 < x < 1$, then

$$2\tan^{-1}x = \sin^{-1}\frac{2x}{1+x^2}$$

8. $f : A \to B$ will be an onto function if-

(a) $f(A) \subset B$ (b) $f(A) = B$

(c) $f(A) \supset B$ (d) $f(A) \neq B$

Sol :

(b) $f(A) = B$

A function is said to be onto iff Range of function is equal to co-domain of the function

$f : A \to B$ is onto iff $f(A) = B$

9. Feasible region is the set of points which satisfy:

(a) The objective function

(b) Some of the given constraints

(c) All the given constraints

(d) None of these

Sol :

(c) All the given constraints

10. At time t distance of a particle moving in a straight line is given by $s = 4t^2 + 2t$. The acceleration when $t = \frac{1}{2}$, is

(a) 4 (b) 6

(c) 8 (d) 10

Sol :

(c) 8

$$v = \frac{ds}{dt} = 8t + 2$$

At $t = \frac{1}{2}$ $a = \frac{dv}{dt} = 8$

11. Solution of $\lim_{n\to\infty}\left[\frac{e^{1/n} + e^{2/n} + e^{3/n} + ... + e^{r/n}}{n}\right]$

(a) $1 - e$ (b) $e - 1$

(c) e (d) 1

Sol :

(b) $e - 1$

Let, $I = \lim_{n\to\infty}\frac{e^{1/n} + e^{2/n} + e^{3/n} + ... + e^{r/n}}{n}$

$$= \lim_{n\to\infty}\frac{\sum_{r=1}^{n} e^{r/n}}{n} = \int_0^1 e^x\,dx$$
$$= [e^x]_0^1 = e^1 - e^0 = e - 1$$

12. $\int_0^1 \frac{dx}{1+x^2} = ?$

(a) $\frac{-\pi}{4}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) $\frac{-\pi}{2}$

Sol :

(b) $\frac{\pi}{4}$

Let, $I = \int_0^1 \frac{dx}{1+x^2} = [\tan^{-1}x]_0^1$

$$= \tan^{-1}1 - \tan^{-1}0$$
$$= \frac{\pi}{4} - 0 = \frac{\pi}{4}$$

13. If $2\begin{bmatrix}3 & 4\\ 5 & x\end{bmatrix} + \begin{bmatrix}1 & y\\ 0 & 1\end{bmatrix} = \begin{bmatrix}7 & 0\\ 10 & 5\end{bmatrix}$, then-

(a) $(x = -2, y = 8)$

(b) $(x = 2, y = -8)$

(c) $(x = 3, y = -6)$

(d) $(x = -3, y = 6)$

Sol :

(b) $(x = 2, y = -8)$

Explanation :

We have $2\begin{bmatrix}3 & 4\\ 5 & x\end{bmatrix} + \begin{bmatrix}1 & y\\ 0 & 1\end{bmatrix} = \begin{bmatrix}7 & 0\\ 10 & 5\end{bmatrix}$

$$\begin{bmatrix}6+1 & 8+y\\ 10+0 & 2x+1\end{bmatrix} = \begin{bmatrix}7 & 0\\ 10 & 5\end{bmatrix}$$

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22. Two groups are computing for the positions of the Board of Directors of a corporation. The probabilities that the first and second groups will win are 0.6 and 0.4 respectively. Further, if the first group wins, the probability of introducing a new product introduced way by the second group.

Sol :

Let E_1 and E_2 denote the events that first and second group will win. Then,

$$P(E_1) = 0.6 \text{ and } P(E_2) = 0.4$$

Let E be the event of introducing the new product.

Then, $$P\left(\frac{E}{E_1}\right) = 0.7 \text{ and } P\left(\frac{E}{E_2}\right) = 0.3$$

Now, we have to find the probability that new product is introduced by second event.

$$P\left(\frac{E_2}{E}\right) = \frac{P(E_2)P\left(\frac{E}{E_2}\right)}{P(E_1)P\left(\frac{E}{E_1}\right) + P(E_2)P\left(\frac{E}{E_2}\right)}$$

$$= \frac{0.4 \times 0.3}{0.6 \times 0.7 + 0.4 \times 0.3}$$

$$= \frac{0.12}{0.42 + 0.12} = \frac{0.12}{0.54} = 0.22$$

23. Find the differential equation representing the family of curves $y = ae^{bx+5}$, where a and b are arbitrary constants.

Sol :

We have, $$y = ae^{bx+5}$$

Differentiating above equation wrt x, we have

$$\frac{dy}{dx} = ae^{bx+5} \cdot b$$

$$= y \cdot b$$

Again, differentiating above wrt x, we get

$$\frac{d^2y}{dx^2} = b\frac{dy}{dx}$$

$$= \left(\frac{1}{y} \cdot \frac{dy}{dx}\right)\left(\frac{dy}{dx}\right) \text{[using Eq. (1)]}$$

$$\frac{d^2y}{dx^2} = \frac{1}{y}\left(\frac{dy}{dx}\right)^2$$

or $$y\left(\frac{d^2y}{dx^2}\right) - \left(\frac{dy}{dx}\right)^2 = 0,$$

Which is the required differentiating equation.

24. Write the value of $\int_0^1 \frac{e^x}{1+e^{2x}}dx$.

Sol :

We have $$I = \int_0^1 \frac{e^x}{1+e^{2x}}dx$$

$$= \int_0^1 \frac{e^x}{1+(e^x)}dx$$

Substituting $e^x = t \Rightarrow e^x dx = dt$

Also, when $x = 0$, then $t = 1$ and when $x = 1$, then $t = e$

Now, $$I = \int_1^e \frac{dt}{1+t^2}$$

$$= [\tan^{-1}t]_1^e$$

$$= \tan^{-1}e - \tan^{-1}1$$

$$= \tan^{-1}\left(\frac{e-1}{1+e}\right)$$

25. Write a unit vector in the direction of the sum of the vectors $\vec{a} = 2\hat{i} + 2\hat{j} - 5\hat{k}$ and $\vec{b} = 2\hat{i} + \hat{j} - 7\hat{k}$.

OR

If $\vec{a} = \hat{i} - \hat{j} + 7\hat{k}$ and $\vec{b} = 5\hat{i} - \hat{j} + \lambda\hat{k}$, then find the value of λ, so that $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are perpendicular vectors.

Sol :

We have $$\vec{a} = 2\hat{i} + 2\hat{j} - 5\hat{k}$$

and $$\vec{b} = 2\hat{i} + \hat{j} - 7\hat{k}$$

Sum of two vectors,

$$\vec{a} + \vec{b} = (2\hat{i} + 2\hat{j} - 5\hat{k}) + (2\hat{i} + \hat{j} - 7\hat{k})$$

$$= 4\hat{i} + 3\hat{j} - 12\hat{k}$$

Required unit vector

$$\frac{\vec{a} + \vec{b}}{|\vec{a} + \vec{b}|} = \frac{4\hat{i} + 3\hat{j} - 12\hat{k}}{\sqrt{4^2 + 3^2 + 12^2}}$$

$$= \frac{4\hat{i} + 3\hat{j} - 12\hat{k}}{\sqrt{169}}$$

$$= \frac{4\hat{i} + 3\hat{j} - 12\hat{k}}{13}$$

$$= \frac{1}{13}(4\hat{i} + 3\hat{j} - 12\hat{k})$$

OR

We have $$\vec{a} = \hat{i} - \hat{j} + 7\hat{k}$$

and $$\vec{b} = 5\hat{i} - \hat{j} + \lambda\hat{k}$$

Then, $$\vec{a} + \vec{b} = (\hat{i} - \hat{j} + 7\hat{k}) + (5\hat{i} - \hat{j} + \lambda\hat{k})$$

$$= 6\hat{i} - 2\hat{j} + (7 + \lambda)\hat{k}$$

and $$\vec{a} - \vec{b} = (\hat{i} - \hat{j} + 7\hat{k}) - (5\hat{i} - \hat{j} + \lambda\hat{k})$$

$$= -4\hat{i} + (7 - \lambda)\hat{k}$$

Since, $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$ are perpendicular vectors, we have

$$(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 0$$

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$$\cos^{-1}x = \frac{\pi}{6} - \sin^{-1}\frac{x}{2}$$

$$x = \cos\left(\frac{\pi}{6} - \sin^{-1}\frac{x}{2}\right)$$

Using $\cos(x-y) = \cos x\cos y + \sin x\sin y$ we have

$$x = \cos\frac{\pi}{6}\cos\left(\sin^{-1}\frac{x}{2}\right) + \sin\frac{\pi}{6}\sin\left(\sin^{-1}\frac{x}{2}\right)$$

$$x = \frac{\sqrt{3}}{2}\cos\left(\cos^{-1}\sqrt{1-\frac{x^2}{4}}\right) + \frac{x}{4}$$

$$x = \frac{\sqrt{3}}{2}\left(\sqrt{1-\frac{x^2}{4}}\right) + \frac{x}{4}$$

$$x - \frac{x}{4} = \frac{\sqrt{3}}{2}\left(\sqrt{1-\frac{x^2}{4}}\right)$$

$$\frac{3x}{4} = \frac{\sqrt{3}}{2}\left(\sqrt{1-\frac{x^2}{4}}\right)$$

$$\frac{9x^2}{16} = \frac{3}{4}\left(1-\frac{x^2}{4}\right)$$

$$\frac{3}{4}x^2 = 1 - \frac{x^2}{4}$$

$$\frac{3}{4}x^2 + \frac{x^2}{4} = 1$$

$$\frac{4x^2}{4} = 1$$

$$x^2 = 1 \Rightarrow x = \pm 1$$

But $x = -1$, does not satisfy the given equation. Hence, $x = 1$ satisfy the given equation.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. If $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$, $B = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$ and $(A+B)^2 = A^2 + B^2$, then find the values of a and b.

OR

Find the adjoint of the matrix $A = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$ and

hence show that $A(adj\,A) = |A|I_3$.

Sol :

We have $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$

Now, $A + B = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} + \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$

$$= \begin{bmatrix} 1+a & -1+1 \\ 2+b & -1-1 \end{bmatrix} = \begin{bmatrix} 1+a & 0 \\ 2+b & -2 \end{bmatrix}$$

$$(A+B)^2 = \begin{bmatrix} 1+a & 0 \\ 2+b & -2 \end{bmatrix} \cdot \begin{bmatrix} 1+a & 0 \\ 2+b & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 1+a^2+2a & 0 \\ 2+2a+b+ab-4-2b & 4 \end{bmatrix}$$

$$= \begin{bmatrix} a^2+2a+1 & 0 \\ 2a-b+ab-2 & 4 \end{bmatrix}$$

$$\text{Also, } A^2 + B^2 = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} + \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} \cdot \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} + \begin{bmatrix} a^2+b & a-1 \\ ab-b & b+1 \end{bmatrix}$$

$$= \begin{bmatrix} a^2+b-1 & a-1 \\ ab-b & b \end{bmatrix}$$

Now, $(A+B)^2 = A^2 + B^2$

$$\begin{bmatrix} a^2+2a+1 & 0 \\ 2a-b+ab-2 & 4 \end{bmatrix} = \begin{bmatrix} a^2+b-1 & a-1 \\ ab-b & b \end{bmatrix}$$

Equating the corresponding elements, we get

$$a^2 + 2a + 1 = a^2 + b - 1$$

$$2a - b = -2 \quad \text{...(i)}$$

$$a - 1 = 0 \Rightarrow a = 1 \quad \text{...(ii)}$$

$$2a - b + ab - 2 = ab - b$$

$$2a - 2 = 0 \Rightarrow a = 1 \quad \text{...(iii)}$$

and $b = 4$ ...(iv)

Since, $a = 1$ and $b = 4$ also satisfy Eq. (i), therefore $a = 1$ and $b = 4$.

OR

We have $A = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$

Let A_{ij} be the cofactor of an element a_{ij} of $|A|$. Then, cofactors of elements of $|A|$ are

$$A_{11} = (-1)^{1+1}\begin{vmatrix} 1 & -2 \\ -2 & 1 \end{vmatrix} = (1-4) = -3$$

$$A_{12} = (-1)^{1+2}\begin{vmatrix} 2 & -2 \\ 2 & 1 \end{vmatrix} = -(2+4) = -6$$

$$A_{13} = (-1)^{1+3}\begin{vmatrix} 2 & 1 \\ 2 & -2 \end{vmatrix} = (-2-4) = -6$$

$$A_{21} = (-1)^{2+1}\begin{vmatrix} -2 & -2 \\ -2 & 1 \end{vmatrix} = -(-2-4) = 6$$

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Given, $A = \{1, 2, 3\}$

$B = \{4, 5, 6, 7\}$

and $f : A \to B$ is defined as

$f = \{(1, 4), (2, 5), (3, 6)\}$

i.e. $f(1) = 4,\ f(2) = 5\ f(3) = 6$

It can be seen that the images of distinct elements of A under f are distinct. So, f is one-one.

29. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 4\hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{c} = \hat{i} - 2\hat{j} + \hat{k}$. Find a vector of magnitude 6 units, which is parallel to the vector $2\vec{a} - \vec{b} + 3\vec{c}$.

Sol :

First, find the vector $2\vec{a} - \vec{b} + 3\vec{c}$, then find a unit vector in the direction of $2\vec{a} - \vec{b} + 3\vec{c}$. After this, the unit vector is multiplying by 6.

We have $\vec{a} = \hat{i} + \hat{j} + \hat{k}$,

$\vec{b} = 4\hat{i} - 2\hat{j} + 3\hat{k}$

and $\vec{c} = \hat{i} - 2\hat{j} + \hat{k}$

Now $\vec{d} = 2\vec{a} - \vec{b} + 3\vec{c}$

$= 2(\hat{i} + \hat{j} + \hat{k}) - (4\hat{i} - 2\hat{j} + 3\hat{k}) + + 3(\hat{i} - 2\hat{j} + \hat{k})$

$= 2\hat{i} + 2\hat{j} + 2\hat{k} - 4\hat{i} + 2\hat{j} - 3\hat{k} + 3\hat{i} - 6\hat{j} + 3\hat{k}$

Now, unit vector $\hat{d}$ in the direction of $\vec{d}$ is $\frac{\vec{d}}{|\vec{d}|}$

$$\hat{d} = \frac{2\vec{a} - \vec{b} + 3\vec{c}}{|2\vec{a} - \vec{b} + 3\vec{c}|}$$

$$= \frac{\hat{i} - 2\hat{j} + 2\hat{k}}{\sqrt{(1)^2 + (-2)^2 + (2)^2}}$$

$$= \frac{\hat{i} - 2\hat{j} + 2\hat{k}}{\sqrt{9}}$$

$$= \frac{\hat{i} - 2\hat{j} + 2\hat{k}}{3}$$

$$= \frac{1}{3}\hat{i} - \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k}$$

Hence, vector of magnitude 6 units parallel to the vector $2\vec{a} - \vec{b} + 3\vec{c}$ is given by

$$6\hat{d} = 6\left(\frac{1}{3}\hat{i} - \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k}\right)$$

$$= 2\hat{i} - 4\hat{j} + 4\hat{k}$$

30. Evaluate $\int_0^{\pi/2} x^2 \sin x\, dx$.

OR

Evaluate $\int_{\pi/4}^{\pi/2} \cos 2x \cdot \log(\sin x)\, dx$.

Sol :

We have $I = \int_0^{\pi/2} x^2 \sin x\, dx$

Using integration by parts technique

$\int \underset{\text{I}}{u} \cdot \underset{\text{II}}{v} dx = \left[u \int v dx - \int \left\{\frac{d}{dx} u \cdot \int v dx\right\} dx\right]$

and choosing its function with the help of ILATE procedure we have

$$I \int \underset{I}{x^2} \underset{II}{\sin x} dx$$

$$= -x^2 \cos x + 2 \int \underset{I}{x} \underset{II}{\cos x}\, dx$$

$$= -x^2 \cos x + 2[x(\sin x) - \int 1 \cdot (\sin x)\, dx]$$

$$= -x^2 \cos x + 2(x \sin x + \cos x)$$

$$= [-x^2 \cos x + 2(x \sin x + \cos x)]_0^{\pi/2}$$

$$= \left[-\left(\frac{\pi}{2}\right)^2 \cos\left(\frac{\pi}{2}\right) + 2\left(\frac{\pi}{2} \sin\frac{\pi}{2} + \cos\frac{\pi}{2}\right) - 2(0 + \cos 0)\right]$$

$$= -\frac{\pi^2}{4} \times 0 + 2\left(\frac{\pi}{2} + 0\right) - 2(0 + 1)$$

$$= \pi - 2$$

OR

We have $I = \int_{\pi/4}^{\pi/2} \underset{II}{\cos 2x} \log(\underset{I}{\sin x})\, dx$

Using integration by parts technique

$$\int \underset{\text{I}}{u} \cdot \underset{\text{II}}{v} dx = \left[u \int v dx - \int \left\{\frac{d}{dx} u \cdot \int v dx\right\} dx\right]$$

and choosing its function with the help of ILATE procedure we have

$$I = \left[\log(\sin x) \frac{\sin 2x}{2} - \int \frac{1}{\sin x} \cos x \cdot \frac{\sin 2x}{2} dx\right]_{\pi/4}^{\pi/2}$$

$$= \left[\frac{\log(\sin x) \cdot \sin 2x}{2}\right]_{\pi/4}^{\pi/2} - \int_{\pi/4}^{\pi/2} \frac{\cos x \cdot (2 \sin x \cos x)}{2 \sin x} dx$$

$$= \frac{1}{2}\left[\log\left(\sin\frac{\pi}{2}\right) \cdot \sin\pi - \log\left(\sin\frac{\pi}{4}\right) \cdot \sin\frac{\pi}{2}\right] - \int_{\pi/4}^{\pi/2} \cos^2 x\, dx$$

$$= \frac{1}{2}\left[0 - \log\frac{1}{\sqrt{2}}\right] - \int_{\pi/4}^{\pi/2} \frac{1 + \cos 2x}{2} dx$$

$$= \frac{1}{4}\log 2 - \frac{1}{2}\left[x + \frac{\sin 2x}{2}\right]_{\pi/4}^{\pi/2}$$

$$= \frac{1}{4}\log 2 - \frac{1}{2}\left[\left(\frac{\pi}{2} + \frac{\sin\pi}{2}\right) - \left(\frac{\pi}{4} + \frac{\sin \pi/2}{2}\right)\right]$$

$$= \frac{1}{4}\log 2 - \frac{1}{2}\left(\frac{\pi}{2} - \frac{\pi}{4} - \frac{1}{2}\right)$$

$$= \frac{1}{4}\log 2 - \frac{1}{2}\left(\frac{\pi}{4} - \frac{1}{2}\right)$$

$$= \frac{1}{4}\log 2 - \frac{\pi}{8} + \frac{1}{4}$$

31. Solve for x, $\cos^{-1} x + \sin^{-1}\left(\frac{x}{2}\right) = \frac{\pi}{6}$.

Sol :

We have $\cos^{-1} x + \sin^{-1}\left(\frac{x}{2}\right) = \frac{\pi}{6}$

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$$[6\hat{i}-2\hat{j}+(7+\lambda)\hat{k}]\cdot[-4\hat{i}+(7-\lambda)\hat{k}]=0$$
$$-24+(7+\lambda)(7-\lambda)=0$$
$$49-\lambda^2=24$$
$$\lambda^2=25$$
$$\lambda=\pm 5$$

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. Evaluate $\int\frac{x^2+x+1}{(x^2+1)(x+2)}dx$.

Sol :

We have $I=\int\frac{x^2+x-1}{(x^2+1)(x+2)}dx$

Now $\frac{x^2+x+1}{(x^2+1)(x+2)}=\frac{A}{x+2}+\frac{Bx+C}{x^2+1}$

$$x^2+x+1=A(x^2+1)+(Bx+C)(x+2) \quad ...(1)$$
$$x^2+x+1=x^2(A+B)+x(2B+C)+(A+2C) \quad ...(2)$$

Substituting $x=-2$ in eq. (1) we have

$$4-2+1=A(4+1)+0 \Rightarrow A=\tfrac{3}{5}$$

Comparing the coefficients of x^2, x and constant terms both sides in eq. (2), we get

$$A+B=1 \quad ...(3)$$
$$2B+C=1 \quad ...(4)$$
and $$A+2C=1 \quad ...(5)$$

Substituting $A=\frac{3}{5}$ in (3) and (5) we get $C=\frac{1}{5}$ and $B=\frac{2}{5}$.

Thus $\frac{x^2+x+1}{(x^2+1)(x+2)}=\frac{3}{5}\cdot\frac{1}{(x+2)}+\frac{1}{5}\frac{(2x+1)}{(x^2+1)}$

Now $I=\frac{3}{5}\int\frac{dx}{x+2}+\frac{1}{5}\int\frac{(2x+1)}{x^2+2}dx$

$$=\frac{3}{5}\int\frac{dx}{x+2}+\frac{1}{5}\int\frac{2x}{x^2+1}+\frac{1}{5}\int\frac{dx}{x^2+1}$$
$$=\tfrac{3}{5}\log|x+2|+\tfrac{1}{5}\log|x^2+1|+\tfrac{1}{5}\tan^{-1}x+C$$

27. Given $A=\begin{bmatrix}2&-3\\-4&7\end{bmatrix}$, compute A^{-1} and show that $2A^{-1}=9I-A$.

OR

If $A=\begin{bmatrix}2&3\\5&-2\end{bmatrix}$ be such that $A^{-1}=kA$, then find the value of k.

Sol :

We have, $A=\begin{bmatrix}2&-3\\-4&7\end{bmatrix}$

Here, $|A|=\begin{vmatrix}2&-3\\-4&7\end{vmatrix}$
$=14-12=2$

Since $|A|\neq 0$, therefore A^{-1} exists.

Now, $adj(A)=\begin{bmatrix}7&3\\4&2\end{bmatrix}$

$$A^{-1}=\frac{1}{|A|}adj(A)$$
$$=\frac{1}{2}\begin{bmatrix}7&3\\4&2\end{bmatrix} \quad ...(i)$$

Now $2A^{-1}=9I-A$

RHS $=9I-A$

$$=9\begin{bmatrix}1&0\\0&1\end{bmatrix}-\begin{bmatrix}2&-3\\-4&7\end{bmatrix}$$
$$=\begin{bmatrix}9&0\\0&9\end{bmatrix}-\begin{bmatrix}2&-3\\-4&7\end{bmatrix}=\begin{bmatrix}7&3\\4&2\end{bmatrix}$$
$=2A^{-1}$ [using Eq. (i)]

OR

We have, $A=\begin{bmatrix}2&3\\5&-2\end{bmatrix}$

$$|A|=\begin{vmatrix}2&3\\5&-2\end{vmatrix}$$
$=-4-15=-19$

Since $|A|\neq 0$, matrix A is non-singular and A^{-1} exists.

Now $adj\,A=\begin{bmatrix}-2&-3\\-5&2\end{bmatrix}$

$$A^{-1}=\frac{1}{|A|}adj\,A$$
$$=\frac{-1}{19}\begin{bmatrix}-2&-3\\-5&2\end{bmatrix}$$

28. If $A=\{1,2,3\}, B=\{4,5,6,7\}$ and $f=\{(1,4),(2,5),(3,6)\}$ is a function from A to B. State whether f is one-one or not.

Sol :

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$$A_{22} = (-1)^{2+2}\begin{vmatrix} -1 & -2 \\ 2 & 1 \end{vmatrix} = (-1+4) = 3$$

$$A_{23} = (-1)^{2+3}\begin{vmatrix} -1 & -2 \\ 2 & -2 \end{vmatrix} = -(2+4) = -6$$

$$A_{31} = (-1)^{3+1}\begin{vmatrix} -2 & -2 \\ 1 & -2 \end{vmatrix} = (4+2) = 6$$

$$A_{32} = (-1)^{3+2}\begin{vmatrix} -1 & -2 \\ 2 & -2 \end{vmatrix} = -(2+4) = -6$$

$$A_{33} = (-1)^{3+3}\begin{vmatrix} -1 & -2 \\ 2 & 1 \end{vmatrix} = (-1+4) = 3$$

Adjoint of the matrix A is given by

$$adj\ A = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 6 & 6 \\ -6 & 3 & -6 \\ -6 & -6 & 3 \end{bmatrix}$$

Now, $|A| = \begin{vmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{vmatrix}$

$$= -1(1-4) + 2(2+4) - 2(-4-2)$$

$$= -(-3) + 2(6) - 2(-6)$$

$$= 3 + 12 + 12 = 27$$

$$A \cdot (adj\ A) = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}\begin{bmatrix} -3 & 6 & 6 \\ -6 & 3 & -6 \\ -6 & -6 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 3+12+12 & -6-6+12 & -6+12-6 \\ -6-6+12 & 12+3+12 & 12-6-6 \\ -6+12-6 & 12-6-6 & 12+12+3 \end{bmatrix}$$

$$= \begin{bmatrix} 27 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{bmatrix}$$

$$= 27\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= 27I_3 = |A|I_3$$ Hence proved.

33. If $x = \cos t + \log\tan\left(\frac{t}{2}\right)$, $y = \sin t$, then find the values of $\frac{d^2y}{dt^2}$ and $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$.

OR

Find the values of a and b, if the function f defined by $f(x) = \begin{cases} x^2 + 3x + a, & x \le 1 \\ bx + 2, & x > 1 \end{cases}$ is differentiable at $x = 1$.

Sol :

Given that,

$$y = \sin t$$

$$\frac{dy}{dt} = \cos t \qquad [\text{differentiate w.r.t.} t] \ ...(i)$$

$$\frac{d^2y}{dt^2} = -\sin t \qquad [\text{differentiate w.r.t. } t]$$

$$\left[\frac{d^2y}{dt^2}\right]_{t=\frac{\pi}{4}} = -\sin\frac{\pi}{4} = -\frac{1}{\sqrt{2}}$$

Again, $x = \cos t + \log\tan\frac{t}{2}$

$$\frac{dx}{dt} = -\sin t + \frac{1}{\tan\frac{t}{2}} \cdot \sec^2\frac{t}{2} \cdot \frac{1}{2}$$

[differentiate w.r.t. t]

$$= -\sin t + \frac{\cos\frac{t}{2}}{2 \cdot \sin\frac{t}{2}} \cdot \frac{1}{\cos^2\frac{t}{2}}$$

$$= -\sin t + \frac{1}{\sin 2 \times \frac{t}{2}}$$

$\because\ 2\sin a\cos a = \sin 2a]$

$$= -\sin t + \operatorname{cosec} t \qquad ...(ii)$$

Now, $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\cos t}{\operatorname{cosec} t - \sin t}$

[using Eqs. (i) and (ii)]

$$= \frac{\cos t}{1 - \sin^2 t} \cdot \sin t = \frac{\sin t \cdot \cos t}{\cos^2 t}$$

$$\frac{dy}{dx} = \tan t$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{\frac{d}{dx}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}} = \frac{\sec^2 t}{\operatorname{cosec} t - \sin t}$$

$$= \frac{\sec^2 t \cdot \sin t}{\cos^2 t} = \sec^3 t \cdot \tan t$$

$$\left[\frac{d^2y}{dx^2}\right]_{t=\frac{\pi}{4}} = \sec^3\frac{\pi}{4} \cdot \tan\frac{\pi}{4} = 2\sqrt{2} \times 1 = 2\sqrt{2}$$

OR

Given,

$$f(x) = \begin{cases} x^2 + 3x + a, & x \le 1 \\ bx + 2, & x > 1 \end{cases} \text{ is differentiable at } x = 1.$$

$$Lf'(1) = Rf'(1) \qquad ...(i)$$

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SECTION - E

Case study based questions are compulsory.

36. An insurance company believes that people can be divided into two classes: those who are accident prone and those who are not. The company's statistics show that an accident-prone person will have an accident at sometime within a fixed one-year period with probability 0.6, whereas this probability is 0.2 for a person who is not accident prone. The company knows that 20% of the population is accident prone.



On the basis of above information, answer the following questions.

(i) What is the probability that a new policyholder will have an accident within a year of purchasing a policy?

(ii) Suppose that a new policy holder has an accident within a year of purchasing a policy. What is the probability that he or she is accident prone?

Sol :

Consider the following events

E_1 = The policy holder is accident prone

E_2 = The policy holder is not accident prone

E = The new policy holder has an accident within a year of purchasing a policy.

$$P(E_1) = \frac{20}{100}$$

and $$P(E_2) = \frac{80}{100}$$

$$P\left(\frac{E}{E_1}\right) = 0.6 = \frac{6}{10}$$

and $$P\left(\frac{E}{E_2}\right) = 0.2 = \frac{2}{10}$$

(i) $$P(E) = P(E_1) \times P(E/E_1) + P(E_2) \times P(E/E_2)$$

$$= \frac{20}{100} \times \frac{6}{10} + \frac{80}{100} \times \frac{2}{10}$$

$$= \frac{280}{1000} = \frac{7}{25}$$

(ii) By Baye's theorem,

$$P(E_1/E) = \frac{P(E_1) \times P(E/E_1)}{P(E)}$$

$$= \frac{\frac{20}{100} \times \frac{6}{10}}{\frac{7}{25}} = \frac{\frac{3}{25}}{\frac{7}{25}} = \frac{3}{7}$$

37. Chemical reaction, a process in which one or more substances, the reactants, are converted to one or more different substances, the products. Substances are either chemical elements or compounds. A chemical reaction rearranges the constituent atoms of the reactants to create different substances as products.



In a certain chemical reaction, a substance is converted into another substance at a rate proportional to the square of the amount of the first substance present at any time t. Initially $(t = 0)$ 50 g of the first substance was present; 1 hr later, only 10 g of it remained.

(i) Find an expression that gives the amount of the first substance present at any time t.

(ii) What is the amount present after 2 hr?

Sol :

Since rate is proportional to the square of the amount of the first substance present at any time t, we are led to the differential equation

$$\frac{dQ}{dt} = kQ^2$$

The differential equation is separable. Separating the variables and integrating, we obtain

$$\int \frac{dQ}{Q^2} = \int k\, dt$$

and $$-\frac{1}{Q} = kt + C$$

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which is a homogeneous differential equation as $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$.

Substituting $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ in eq (1) we have

$$\begin{aligned} v + x\frac{dv}{dx} &= -(v + v^2) \\ x\frac{dv}{dx} &= -v - v^2 - v \\ &= -2v - v^2 \\ \frac{dv}{v^2 + 2v} &= -\frac{dx}{x} \end{aligned}$$

Integrating both sides, we have

$$\begin{aligned} \int \frac{dv}{v^2 + 2v} &= -\int \frac{dx}{x} \\ \int \frac{dv}{v^2 + 2v + 1 - 1} &= -\int \frac{dx}{x} \\ \int \frac{dv}{(v+1)^2 - (1)^2} &= -\int \frac{dx}{x} \\ \frac{1}{2}\log\left|\frac{v+-1}{v+1+1}\right| &= -\log|x| + C \\ \frac{1}{2}\log\left|\frac{v}{v+2}\right| &= -2\log|x| + C \\ \frac{1}{2}\log\left|\frac{\frac{y}{x}}{\frac{y}{x}+2}\right| &= -\log|x| + C \\ \frac{1}{2}\log\left|\frac{y}{y+2x}\right| &= -\log|x| + C \qquad ...(2) \end{aligned}$$

Also, given that at $x = 1$, $y = 1$.

Substituting $x = y = 1$ in Eq. (2), we get

$$\begin{aligned} \frac{1}{2}\log\left|\frac{1}{1+2}\right| &= -\log 1 + C \\ \frac{1}{2}\log\left|\frac{1}{3}\right| &= -\log 1 + C \\ C &= \frac{1}{2}\log\frac{1}{3} \end{aligned}$$

Substituting the value of C in Eq. (2), we get

$$\begin{aligned} \frac{1}{2}\log\left|\frac{y}{y+2x}\right| &= \log|x| + \frac{1}{2}\log\frac{1}{3} \\ \log\left|\frac{y}{y+2x}\right| &= -2\log|x| + \log\frac{1}{3} \\ \log\left|\frac{y}{y+2x}\right| &= \log|x|^{-2} + \log\frac{1}{3} \\ \log\left|\frac{y}{y+2x}\right| &= \log\frac{1}{x^2} + \log\frac{1}{3} \\ \log\left|\frac{y}{y+2x}\right| &= \log\frac{1}{3x^2} \\ \frac{y}{y+2x} &= \frac{1}{3x^2} \Rightarrow y \cdot 3x^2 = y + 2x \\ y(3x^2 - 1) &= 2x \\ y &= \frac{2x}{3x^2 - 1} \end{aligned}$$

which is the required particular solution.

OR

Given differential equation is

We have $$x\frac{dy}{dx} - y + x\,\mathrm{cosec}\left(\frac{y}{x}\right) = 0$$

$$\begin{aligned} \frac{dy}{dx} - \frac{y}{x} + \mathrm{cosec}\left(\frac{y}{x}\right) &= 0 \\ \frac{dy}{dx} &= \frac{y}{x} - \mathrm{cosec}\left(\frac{y}{x}\right) \qquad ...(1) \end{aligned}$$

which is a homogeneous differential equation as

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

Substituting $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ in eq. (1) we have

$$\begin{aligned} v + x\frac{dv}{dx} &= \frac{vx}{x} - \mathrm{cosec}\left(\frac{vx}{x}\right) \\ v + x\frac{dv}{dx} &= v - \mathrm{cosec}\, v \\ x\frac{dv}{dx} &= -\mathrm{cosec}\, v \\ \frac{dv}{\mathrm{cosec}\, v} &= \frac{-dx}{x} \end{aligned}$$

Integrating both sides, we have

$$\begin{aligned} \int \frac{dv}{\mathrm{cosec}\, v} &= \int -\frac{dx}{x} \\ \int \sin v\, dv &= \int -\frac{dx}{x} \\ -\cos v &= -\log|x| + C \\ -\cos\frac{y}{x} &= -\log|x| + C \\ \cos\frac{y}{x} &= (\log|x| - C) \qquad ...(2) \end{aligned}$$

Also, given that $x = 1$ and $y = 0$.

Substituting above values in Eq. (2), we get

$$\begin{aligned} \cos 0 &= \log|1| - C \\ 1 &= 0 - C \Rightarrow C = -1 \\ \cos\frac{y}{x} &= \log|x| + 1 \qquad \text{[from Eq. (2)]} \end{aligned}$$

which is required particular solution of given differential equation.

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Here, $Lf'(1) = \lim_{h\to 0}\frac{f(1-h)^2 - f(1)}{-h}$

$= \lim_{h\to 0}\frac{(1-h)^2 + 3(1-h) + a - (4+a)}{-h}$

$= \lim_{h\to 0}\frac{1+h^2-2h+3-3h+a-4-a}{-h}$

$= \lim_{h\to 0}\frac{h^2-5h}{-h}$

and $Rf'(1) = \lim_{h\to 0}\frac{f(1+h)-f(1)}{h}$

$= \lim_{h\to 0}\frac{b(1+h)+2-(4+a)}{h}$

$= \lim_{h\to 0}\frac{b+bh+2-4-a}{h}$

$= \lim_{h\to 0}\frac{bh+b-a-2}{h}$

Clearly, for $Rf'(1)$ to be exist $b-a-2$ should be equal to 0, i.e.

$$b-a-2 = 0 \qquad \text{...(ii)}$$

Now, $Rf'(1) = \lim_{h\to 0}\frac{bh}{h}$

$= \lim_{h\to 0} b = b$

From Eq. (i), we have

$$Lf'(1) = Rf'(1)$$
$$5 = b$$
$$b = 5$$

Now, on substituting $b = 5$ in Eq. (ii), we get

$$5 - a - 2 = 0$$
$$a = 3$$

Hence, $a = 3$ and $b = 5$.

34. Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. Find a vector $\vec{p}$, which is perpendicular to both $\vec{a}$ and $\vec{b}$ and $\vec{p}\cdot\vec{c} = 18$.

Sol :

We have $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$,

$\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$

and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$

Let $\vec{p} = x\hat{i} + y\hat{j} + z\hat{k}$ where $\vec{p}$ is perpendicular to both $\vec{a}$ and $\vec{b}$. Thus

$$\vec{p}\cdot\vec{a} = 0$$
$$(x\hat{i} + y\hat{j} + z\hat{k})\cdot(\hat{i} + 4\hat{j} + 2\hat{k}) = 0$$
$$x + 4y + 2z = 0 \qquad \text{...(1)}$$

and $\vec{p}\cdot\vec{b} = 0$

$$(x\hat{i} + y\hat{j} + z\hat{k})\cdot(3\hat{i} - 2\hat{j} + 7\hat{k}) = 0$$
$$3x - 2y + 7z = 0 \qquad \text{...(2)}$$

Since $\vec{p}\cdot\vec{c} = 18$ we have

$$(x\hat{i} + y\hat{j} + z\hat{k})\cdot(2\hat{i} - \hat{j} + 4\hat{k}) = 18$$
$$2x - y + 4z = 18 \qquad \text{...(3)}$$

Multiplying Eq. (1) by 3 and subtracting it from Eq. (2), we get

$$-14y + z = 0 \qquad \text{...(4)}$$

Now, multiplying Eq. (1) by 2 and subtracting it from Eq. (3), we get

$$-9y = 18 \Rightarrow y = -2$$

Substituting $y = -2$ in Eq. (4), we get

$$-14(-2) + z = 0$$
$$28 + z = 0 \Rightarrow z = -28$$

Substituting $y = -2$ and $z = -28$ in Eq. (1), we get

$$x + 4(-2) + 2(-28) = 0$$
$$x - 8 - 56 = 0 \Rightarrow x = 64$$

Hence, the required vector is

$$\vec{p} = x\hat{i} + y\hat{j} + z\hat{k}$$

i.e. $\vec{p} = 64\hat{i} - 2\hat{j} - 28\hat{k}$

35. Find the particular solution of the differential equation satisfying the given condition.

$$x^2\,dy + (xy + y^2)\,dx = 0, \text{ when } y(1) = 1$$

OR

Find the particular solution of the differential equation

$$x\frac{dy}{dx} - y + x\,\text{cosec}\left(\frac{y}{x}\right) = 0$$

Sol :

We have $x^2\,dy + (xy + y^2)\,dx = 0$

$$x^2\,dy = -(xy + y^2)\,dx$$
$$\frac{dy}{dx} = \frac{-(xy + y^2)}{x^2}$$
$$\frac{dy}{dx} = -\left(\frac{y}{x} + \frac{y^2}{x^2}\right) \qquad \text{...(1)}$$

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Therefore, $Q = -\frac{1}{kt + C}$

Now, $Q = 50$ when $t = 0$, therefore, $50 = -\frac{1}{C}$ and $C = -\frac{1}{50}$.

Therefore, $Q = -\frac{1}{kt - \frac{1}{50}}$

Since $Q = 10$ when $t = 1$,

$$10 = -\frac{1}{k - \frac{1}{50}}$$

$$10\left(k - \frac{1}{50}\right) = -1$$

$$10k - \frac{1}{5} = -1$$

$$10k = -1 + \frac{1}{5} = -\frac{4}{5}$$

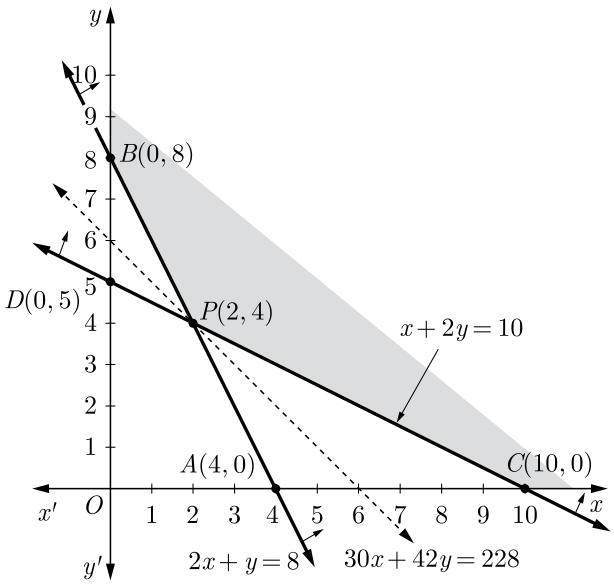
$$k = -\frac{4}{50} = -\frac{2}{25}$$

Therefore, $Q(t) = \frac{1}{\frac{2}{25}t + \frac{1}{50}}$

$$= \frac{1}{\frac{4t+1}{50}}$$

$$= \frac{50}{4t+1}$$

and $Q(2) = \frac{50}{8+1} \approx 5.56$ grams



The values of Z at corner points are given below.

Corner Points	$Z = 30x + 42y$
$C(10,\ 0)$	$Z = 30 \times 10 + 0 = 300$
$P(2,\ 4)$	$Z = 30 \times 2 + 42 \times 4 = 228$
$B(0, 8)$	$Z = 0 + 42 \times 8 = 336$

From table, the minimum value of Z is 228. As the feasible region is unbounded, therefore 228 may or may not be the minimum value of Z. For this, we draw a dotted graph of the inequality $30x + 42y < 228$ and check graph of the inequality plane has point in common with the feasible region or not.

It can seen that the feasible region has no common point with $30x + 42y < 228$. Hence, the minimum cost Rs. 38, when $x = 2$ and $y = 4$.

38. Vitamins are nutritional substances which you need in small amounts in your diet. Vitamins A and E are fat-soluble vitamins, meaning they're stored in your body's fat cells, but they need to have their levels topped up regularly. Vitamin C is a water-soluble vitamin found in citrus and other fruits and vegetables, and also sold as a dietary supplement. It is used to prevent and treat scurvy. Vitamin C is an essential nutrient involved in the repair of tissue, the formation of collagen, and the enzymatic production of certain neurotransmitters.





A dietician wishes to mix two types of foods in such a way that the vitamin contents of mixture contains atleast 8 units of vitamin A and 10 units of vitamin C. Food I contains 2 units per kg of vitamin A and 1 unit per kg of vitamin C, while food II contains 1 unit per kg of vitamin A an 2 units per kg of vitamin C. It costs Rs. 30 per kg to purchase food I and Rs. 42 per kg to purchase food II.

(i) Formulate above as an LPP and solve it graphically.

(ii) Find the minimum cost of such a mixture.

Sol :

Let the diet contains x kg of food I and y kg of food II. Clearly $x \geq 0$ and $y \geq 0$.

If costs Rs. 30 per kg to purchase food I and Rs. 42 per kg to purchase food II. Let Z be cost function.

Thus $Z = 30x + 42y$

Food I contains 2 units per kg of vitamin A and 1 unit per kg of vitamin C, while food II contains 1 unit per kg of vitamin A an 2 units per kg of vitamin C. Vitamin contents of mixture must contain atleast 8 units of vitamin A and 10 units of vitamin C.

The given data can be put in the tabular form as

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follows

Food → Vitamin ↓	**I**	**II**	**Availability**
Vit. A	2	1	$2x+y \geq 8$
Vit. C	1	2	$x+2y \geq 10$

Thus, the required problem is LPP

Minimize $Z = 30x+42y$

Subject to $2x+y \geq 8$

$x+2y \geq 10$

and $x \geq 0,\ y \geq 0$

(i) Region corresponding to $2x+y \geq 8$:

Line $2x+y=8$			
x	0	4	Point $(0,0)$ is false for $2x+y \geq 8$ So, the region is away from the origin.
y	8	0	

(ii) Region corresponding to $x+2y \geq 10$:

Line $x+2y=10$			
x	0	5	Point $(0,0)$ is false for $x+2y \geq 10$ So, the region is away from the origin.
y	10	0	

Since $x \geq 0,\ y \geq 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. The shaded region CPB represents the feasible region of the given LPP which is unbounded. The point of intersection of the lines $2x+y \geq 8$ and $x+2y=10$ is $P(2,4)$. The corner points are $C(10,\ 0)$, $P(2,\ 4)$ and $B(0,8)$.

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Sample Paper 04

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

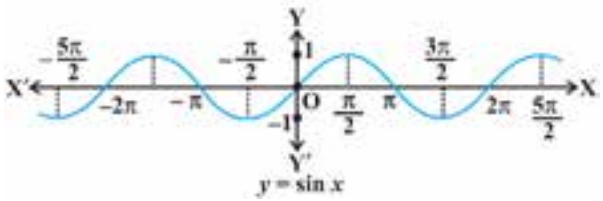
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the period of the sine curve as displayed in the image?



(a) π (b) 2π

(c) 3π (d) 4π

Ans :

(b) 2π

The graph clearly shows the repeating wave pattern of the sine function. One full cycle begins at x=0, rises to y=1 at $\pi/2$, goes down to y=-1 at $3\pi/2$, and comes back to 0 at x=2π. After this, the same pattern repeats again. This proves that the curve completes one full cycle after an interval of 2π. The options π, 3π, 4π do not match the repeating interval visible in the graph. Hence, the period is 2π.

2. The solution of the differential equation $\frac{dy}{dx} = e^{x+y}$ is

(a) $e^x + e^{-y} + k = 0$ (b) $e^{2x} = ke^y$

(c) $e^x = ke^{2y}$ (d) $e^x = ke^y$

Sol :

(a) $e^x + e^{-y} + k = 0$

Given: $\frac{dy}{dx} = e^{x+y}$

$\frac{dy}{dx} = e^x \cdot e^y$ $[e^{m+n} = e^m \cdot e^n]$

$dy\, e^{-y} = e^x\, dx$

Integrating both sides,

$-e^{-y} = e^x + k$

$e^x + e^{-y} + k = 0$

3. In the set of all straight lines in a plane, the relation R "to be Perpendicular" is-

(a) Reflexive and transitive

(b) Symmetric and transitive

(c) Symmetric

(d) None of these

Sol :

(c) Symmetric

If line L_1 is perpendicular to the line L_2, then L_2 will also be perpendicular to L_1

If $(L_1, L_2) \in R$

$(L_2, L_1) \in R$

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Hence, R is symmetric.

4. The modulus of the vector $19\hat{i}+5\hat{j}-6\hat{k}$ is.

(a) $\sqrt{322}$ (b) $\sqrt{420}$

(c) $\sqrt{421}$ (d) $\sqrt{422}$

Sol :

(d) $\sqrt{422}$

Let, $\vec{a} = 19\hat{i}+5\hat{j}-6\hat{k}$

$|\vec{a}| = \sqrt{(19^2)+(5)^2+(-6)^2}$

$= \sqrt{361+25+36} = \sqrt{422}$

5. Required distance $= \sqrt{4^2+5^2} = \sqrt{16+25} = \sqrt{41}$

If $A = \begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix}$, then which one of the following is equal to A' ?

(a) $\begin{bmatrix}2 & 1 & 3\\ 5 & 4 & 6\end{bmatrix}$ (b) $\begin{bmatrix}4 & 5 & 6\\ 1 & 2 & 3\end{bmatrix}$

(c) $\begin{bmatrix}1 & 4\\ 2 & 5\\ 3 & 6\end{bmatrix}$ (d) $\begin{bmatrix}3 & 6\\ 2 & 5\\ 1 & 4\end{bmatrix}$

Sol :

(c) $\begin{bmatrix}1 & 4\\ 2 & 5\\ 3 & 6\end{bmatrix}$

We have, $A = \begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix}$

$A' = \begin{bmatrix}1 & 4\\ 2 & 5\\ 3 & 6\end{bmatrix}$

6. $\frac{d(2^x)}{d(3^x)} =$

(a) $\left(\frac{2}{3}\right)^x$ (b) $\frac{2^{x-1}}{3^{x-1}}$

(c) $\left(\frac{2}{3}\right)^x \log_3 2$ (d) $\left(\frac{2}{3}\right)^x \log_2 3$

Sol :

(c) $\left(\frac{2}{3}\right)^x \log_3 2$

We have $f(x) = \frac{d(2^x)}{d(3^x)}$ ...(1)

$\frac{d}{dx}(2^x) = 2^x \log 2$

$\frac{d}{dx}(3^x) = 3^x \log 3$ $\left[\frac{d}{dx}(a^x) = a^x \log a\right]$

Putting values in eq.(1),

$f(x) = \frac{2^x \log 2}{3^x \log 3}$

$= \left(\frac{2}{3}\right)^x \log_3 2$ $\left[\frac{\log a}{\log b} = \log ba\right]$

7. $\int \frac{dx}{x+\sqrt{x}}$

(a) $\log x + \log(1+\sqrt{x}) + C$

(b) $2\log(1+\sqrt{x}) + C$

(c) $\log(1+\sqrt{x}) + C$

(d) $\log\sqrt{x} + C$

Sol :

(b) $2\log(1+\sqrt{x}) + C$

We have $I = \int \frac{dx}{x+\sqrt{x}}$

Let, $u^2 = x$

Hence, $2udu = dx$

Hence, $I = \int \frac{2udu}{u^2+u}$

$= \int \frac{2du}{1+u}$

$= 2\log(1+u) + C$ $\left[\int \frac{dx}{x} = \log x\right]$

Thus, $I = 2\log(1+\sqrt{x}) + C$ (from eq. (1)

8. If a particle moves in a straight line so that $s = \frac{1}{2}vt$, then acceleration is

(a) a constant (b) proportional to t

(c) proportional to v (d) proportional to s

Sol :

(a) a constant.

We have $s = \frac{1}{2}vt$

$2s = vt$

$2\frac{ds}{dt} = v + t\frac{dv}{dt}$

$2\frac{d^2s}{dt^2} = \frac{dv}{dt} + \frac{td^2v}{dt^2} + \frac{dv}{dt}$

But $\frac{ds}{dt} = a$ (acceleration), therefore

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We have $A = (2,3,4),\ B = (8,-3,8)$

Mid point of line segment joining (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$P = \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$$

Here, $x_1 = 2\ \ y_1 = 3\ \ z_1 = 4$

$x_2 = 8\ \ y_2 = -3\ \ z_2 = 8$

$$P = \left(\frac{2+8}{2}, \frac{3-3}{2}, \frac{4+8}{2}\right)$$

$$P = (5,0,6)$$

19. **Assertion :** If A and B be two events corresponding to sample space S such that $P(A) = 0.2$ and $P(B) = 0.8$, then $A \cup B$ is a sure event.

Reason : If A and B are mutually exclusive events, then $P(A) + P(B) = 1$.

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

Sol :

(d) Assertion is false, but reason is true.

We know that,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.2 + 0.8 - P(A \cap B)$$

Since $P(A \cap B)$ is not given. Therefore we can't say $P(A \cup B) = 1$ i.e., $A \cup B$ is a sure event.

For mutually exclusive events

$$P(A \cap B) = 0$$

$$P(A) + P(B) = 0.2 + 0.8 = 1$$

20. **Assertion :** $f(x) = x^3$ is continuous at $x = 2$.

Reason: $f(x)$ is continuous at $x = a$ if

$$\lim_{x \to a^+} \int f(x) = \lim_{x \to a^-} f(x) = f(a)$$

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

Sol :

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

Given: $f(x) = x^3$

Since $f(x)$ is a polynomial function, therefore it is continuous in its domain from $(-\infty, \infty)$.

Hence continuous at,

$$x = 2$$

Moreover for $f(x)$ to be continuous at any point say,

$$x = a \text{ or } 2$$

At $x = a$,

$$\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x) = f(a)$$

or, At $x = 2$,

$$\lim_{x \to 2^+} f(2) = \lim_{x \to 2^-} f(2) = f(2) = (2)^3 = 8$$

Both statement are true and statement-II is the correct explanation of statement-I.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Differentiate $\tan^{-1}\left(\frac{1+\cos x}{\sin x}\right)$ with respect to x.

Sol :

We have $y = \tan^{-1}\left(\frac{1+\cos x}{\sin x}\right)$

$$= \tan^{-1}\left(\frac{2\cos^2\frac{x}{2}}{2\sin\frac{x}{2}\cos\frac{x}{2}}\right)$$

$$= \tan^{-1}\left(\frac{\cos\frac{x}{2}}{\sin\frac{x}{2}}\right)$$

$$= \tan^{-1}\left(\cot\frac{x}{2}\right)$$

$$= \frac{\pi}{2} - \cot^{-1}\left(\cot\frac{x}{2}\right)$$

$$= \frac{\pi}{2} - \frac{x}{2}$$

Differentiating w.r.t. x, we get

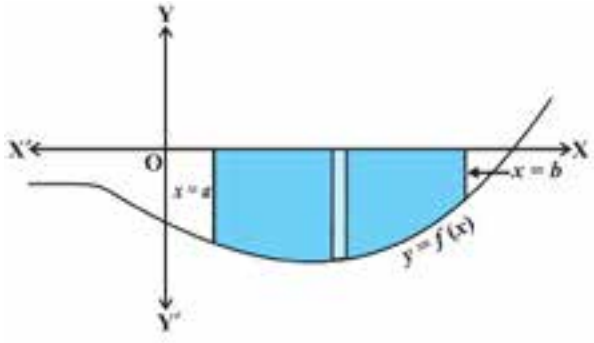
$$\frac{dy}{dx} = 0 - \frac{1}{2} = -\frac{1}{2}$$

OR

If $y = \sin^{-1}(6x\sqrt{1-9x^2})$, $-\frac{1}{3\sqrt{2}} < x < \frac{1}{3\sqrt{2}}$, then find $\frac{dy}{dx}$.

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14. In Fig., which statement correctly describes the position of the curve y=f (x)?



(a) It lies entirely above the X-axis.

(b) It lies entirely below the X-axis.

(c) It intersects the Y-axis only at one point.

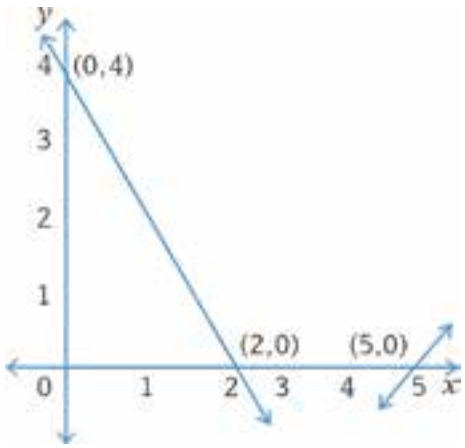
(d) It passes through both X and Y axes at multiple points.

Ans :

(b) It lies entirely below the X-axis.

In the diagram, the curve y=f(x) is clearly drawn below the X-axis between the vertical lines x=a and x=b. The shaded region represents the area trapped between the X-axis and this curve, showing that the curve remains below the X-axis in the given interval. Hence, option (b) is correct.

15. For the LPP shown in the figure, the objective is Minimise $Z = 3x + 2y$. From the graph, the minimum value of Z is obtained at the corner point (2,0). The optimal solution of this LPP:



(a) Does not exist as the feasible region is unbounded.

(b) Does not exist as the inequality 3x+2y<6 does not have any point in common with the feasible region.

(c) Exists as the inequality 3x+2y>6 has infinitely many points in common with the feasible region.

(d) Exists as the inequality 3x+2y < 6 does not have any point in common with the feasible region.

Ans :

(d) Exists as the inequality 3x+2y<6 does not have any point in common with the feasible region.

At the visible vertices, $Z(2,0) = 6, Z(0,4) = 8$, and $Z(5,0) = 15$; thus $(2,0)$ gives the least value. The region is unbounded, but a minimum can still exist. To check, test whether there is any feasible point with a smaller value than 6, i.e., satisfying $3x + 2y < 6$. From the graph, every feasible point lies on or above the line $3x + 2y = 6$; moving to $3x + 2y < 6$ leaves the feasible region immediately. Hence no feasible point satisfies$3x + 2y < 6$, and the minimum $Z\text{min} = 6$ at (2,0) is valid.

16. $\sin^{-1}\frac{1}{x} = ?$

(a) $\sec^{-1}x$ (b) $\text{cosec}^{-1}x$

(c) $\tan^{-1}x$ (d) $\sin x$

Sol :

(b) $\text{cosec}^{-1}x$

We know that,

$$\sin^{-1}\frac{1}{x} = \text{cosec}^{-1}x$$

17. $\int \frac{dx}{1+x^2} = ?$

(a) $\tan x + c$ (b) $\tan^{-1}x + c$

(c) $\cot x + c$ (d) $-\cot^{-1}x + c$

Sol :

(b) $\tan^{-1}x + c$

We have,

$$\int \frac{dx}{1+x^2} = \tan^{-1}x + c$$ (Standard formula)

18. The coordinates of the midpoint of the line segment joining the points (2, 3, 4) and $(8, -3, 8)$ are

(a) (10, 0, 12) (b) (5, 6, 0)

(c) (6, 5, 0) (d) (5, 0, 6)

Sol :

(d) (5, 0, 6)

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$$2a = a + t\frac{da}{dt} + a$$

$$\frac{da}{dt} = 0 \text{ or } t = 0$$

But $t = 0$ (impossible). Therefore acceleration is constant

9. The maximum value of $f(x) = \frac{\log x}{x}$ is.

(a) 1 (b) $\frac{2}{e}$

(c) e (d) $\frac{1}{e}$

Sol :

(d) $\frac{1}{e}$

We have, $f(x) = \frac{\log x}{x}$

$$f'(x) = \frac{\frac{1}{x}\cdot x - \log x \cdot 1}{x^2}$$

$$= \frac{1 - \log x}{x^2}$$

and $$f''(x) = \frac{-\frac{1}{x}\cdot x^2 - (1 - \log x).2x}{x^4}$$

$$= \frac{-x - 2x + 2x\log x}{x^4}$$

$$= \frac{2x\log x - 3x}{x^4}$$

$$= \frac{2\log x - 3}{x^3}$$

For maxima and minima,

$$f'(x) = 0$$

$$\frac{1 - \log x}{x^2} = 0$$

$$\log x = 1$$

$$x = e$$

$$f''(e) = \frac{2\log e - 3}{e^3}$$

$$= \frac{-1}{e^3}$$

$$= (\text{negative})$$

Maximum value of, $f(x) = f(e)$

$$= \frac{\log e}{e} = \frac{1}{e}$$

10. Evaluate $\begin{bmatrix} 1 & -1 \\ y & x \end{bmatrix}$

(a) $x + y$ (b) $x - y$

(c) $-y - x$ (d) $1 - x$

Sol :

(a) $x + y$

Let, $\Delta = \begin{bmatrix} 1 & -1 \\ y & x \end{bmatrix}$

$= x - (-y) = x + y$

11. $\frac{d}{dx}[\sin^{-1}x + \cos^{-1}x] = ?$

(a) 0 (b) $\frac{1}{\sqrt{1 - x^2}}$

(c) $-\frac{1}{\sqrt{1 - x^2}}$ (d) $\frac{1}{2}\sqrt{1 - x^2}$

Sol :

(a) 0

Let, $y = \sin^{-1}x + \cos^{-1}x$

$$y = \frac{\pi}{2}$$

$$\frac{dy}{dx} = \frac{d}{dx}\left(\frac{\pi}{2}\right) = 0$$

12. Which one of the following statements is correct

(a) Every L.P.P admits an optimal solution

(b) A L.P.P admits unique optimal solution

(c) The optimal value occurs at a corner point of the feasible region

(d) If a L.P.P admits two optimal solutions, then it has infinite optimal solutions

Sol :

(d) If a L.P.P admits two optimal solutions, then it has infinite optimal solutions

13. The distance of the point $(3, 4, 5)$ from x-axis is-

(a) 3

(b) 5

(c) $\sqrt{41}$

(d) None of these

Sol :

(c) $\sqrt{41}$

We know that, distance of a point (x, y, z) from x -axis is equal to $\sqrt{y^2 + z^2}$

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Sol :

We have $y = \sin^{-1}(6x\sqrt{1-9x^2})$

or $y = \sin^{-1}(2\cdot 3x\sqrt{1-(3x)^2})$

Substituting $3x = \sin\theta$, we have

$$y = \sin^{-1}(2\sin\theta\sqrt{1-\sin^2\theta})$$
$$= \sin^{-1}(2\sin\theta\cdot\cos\theta)$$
$$= \sin^{-1}(\sin 2\theta) = 2\theta$$

Thus $y = 2\sin^{-1}(3x)$

$$\frac{dy}{dx} = \frac{2}{\sqrt{1-9x^2}}(3)$$
$$\frac{dy}{dx} = \frac{6}{\sqrt{1-9x^2}}$$

22. Suppose a girls throws a die. If she gets 1 or 2, she tosses a coin three times and notes the number of tails. If she gets 3, 4, 5 or 6, she tosses a coin once gets notes whether a 'head' or 'tail' is obtained. If she obtained exactly one 'tail', what is the probability that she threw 3, 4 ,5 or 6 with the die?

Sol :

Let E_1 be the event that the girl gets 1 or 2, E_2 be the event that the girl gets 3, 4, 5 or 6 and A be the event that the girl gets exactly a tail.

Then, $P(E_1) = \frac{2}{6} = \frac{1}{3}$

and $P(E_2) = \frac{4}{6} = \frac{2}{3}$

Probability of getting exactly one tail when a coin is tossed three times)

$$P\left(\frac{A}{E_1}\right) = \frac{3}{8}$$

Probability of getting exactly a tail when a coin is tossed once

$$P\left(\frac{A}{E_2}\right) = \frac{1}{2}$$

Now, required probability

$$P\left(\frac{E_2}{A}\right) = \frac{P(E_2)\cdot P\left(\frac{A}{E_2}\right)}{P(E_1)\cdot P\left(\frac{A}{E_1}\right) + P(E_2)\cdot P\left(\frac{A}{E_2}\right)}$$
$$= \frac{\frac{2}{3}\cdot\frac{1}{2}}{\frac{1}{3}\cdot\frac{3}{8}+\frac{2}{3}\cdot\frac{1}{2}} = \frac{\frac{1}{3}}{\frac{1}{8}+\frac{1}{3}} = \frac{8}{11}$$

23. Find the magnitude of each of the two vectors $\vec{a}$ and $\vec{b}$, having the same magnitude such that the angle between them is 60° and their scalar product is $\frac{9}{2}$.

Sol :

Given, two vectors $\vec{a}$ and $\vec{b}$ such that $|\vec{a}| = |\vec{b}|$, $\vec{a}\cdot\vec{b} = \frac{9}{2}$ and angle between them is 60°.

If θ is angle between $\vec{a}$ and $\vec{b}$ then we have

$$\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta,$$

Substituting values we obtain

$$\frac{9}{2} = |\vec{a}|\cdot|\vec{a}|\cos 60^\circ$$
$$\frac{1}{2}\cdot|\vec{a}|^2 = \frac{9}{2} \qquad \cos 60^\circ = \frac{1}{2}$$
$$|\vec{a}|^2 = 9$$
$$|\vec{a}| = 3$$

magnitude cannot be negative]

Thus, $|\vec{a}| = |\vec{b}| = 3$

24. Find the general solution of differential equation $y = e^{2x}(a+bx)$

Sol :

Given: $y = e^{2x}(a+bx)$ ...(1)

Differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = e^{2x}\cdot b + (a+bx)\cdot 2e^{2x}$$
$$\frac{dy}{dx} = b\cdot e^{2x} + 2y \qquad \text{[From eq. (1)]}$$
$$b\cdot e^{2x} = \frac{dy}{dx} - 2y \qquad ...(2)$$

Differentiating eq. (2) w.r.t. x, we get

$$2b\cdot e^{2x} = \frac{d^2y}{dx^2} - 2\frac{dy}{dx}$$
$$2\left[\frac{dy}{dx} - 2y\right] = \frac{d^2y}{dx^2} - 2\frac{dy}{dx} \qquad \text{[From eq. (2)]}$$
$$2\frac{dy}{dx} - 4y = \frac{d^2y}{dx^2} - 2\frac{dy}{dx}$$
$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$$

Thus, $\frac{d^2y}{dx^2} - 4y\frac{dy}{dx} + 4y = 0$ is the required Differential equation.

25. Given $A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$, compute A^{-1} and show that $2A^{-1} = 9I - A$.

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28. Let R be a relation defined on the set of natural numbers N as follow:

$$R=\{(x,y): x\in N, y\in N \text{ and } 2x+y=24\}$$

Find the domain and range of the relation R. Also, find if R is an equivalence relation or not.

Sol :

$$R=(1,22),(2,20),(3,18),(4,16),(5,14),(6,12),$$
$$(7,10),(8,8),(9,6),(10,4),(11,2)$$

$$\text{Domain}=\{1,2,3,4,5,6,7,8,9,10,11\}$$

$$\text{Range}=\{2,4,6,8,10,12,14,16,18,20,22\}$$

Since $(1,22)\in R$ but $(22,1)\notin R$

Thus R is not symmetric.

Therefore R is not an equivalence relation.

29. Find a vector of magnitude 5 units and parallel to the resultant of $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}-2\hat{j}+\hat{k}$.

Sol :

We have $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$ and

$\vec{b}=\hat{i}-2\hat{j}+\hat{k}$.

Now, resultant of above vectors,

$$\vec{c}=\vec{a}+\vec{b}$$
$$=(2\hat{i}+3\hat{j}-\hat{k})+(\hat{i}-2\hat{j}+\hat{k})$$
$$=3\hat{i}+\hat{j}$$

Now, unit vector $\hat{c}$ in the direction of $\vec{c}$ is $\frac{\vec{c}}{|\vec{c}|}$

$$\hat{c}=\frac{\vec{c}}{|\vec{c}|}=\frac{3\hat{i}+\hat{j}}{\sqrt{3^2+1^2}}$$
$$=\frac{3\hat{i}+\hat{j}}{\sqrt{10}}=\frac{3}{\sqrt{10}}\hat{i}+\frac{1}{\sqrt{10}}\hat{j}$$

Hence, vector of magnitude 5 units and parallel to resultant of $\vec{a}$ and $\vec{b}$ is

$$5\left(\frac{3}{\sqrt{10}}\hat{i}+\frac{1}{\sqrt{10}}\hat{j}\right)=\frac{15}{\sqrt{10}}\hat{i}+\frac{5}{\sqrt{10}}\hat{j}$$

30. Find $\int\frac{2x}{(x^2+1)(x^2+2)^2}dx$.

Sol :

We have $I=\int\frac{2x}{(x^2+1)(x^2+2)}dx$

Substituting $x^2=t\Rightarrow 2x\,dx=dt$ we have

$$I=\int\frac{dt}{(t+1)(t+2)^2}$$

Now $\frac{1}{(t+1)(t+2)^2}=\frac{A}{t+1}+\frac{B}{t+2}+\frac{C}{(t+2)^2}$

$$\frac{1}{(t+1)(t+2)^2}=\frac{A(t+2)^2+B(t+1)(t+2)+C(t+1)}{(t+1)(t+2)^2}$$

$$1=A(t+2)^2+B(t+1)(t+2)+C(t+1) \quad ...(1)$$

Substituting $t=-1$ in eq. (1), we get

$$1=A(-1+2)^2+0+0\Rightarrow 1=A$$

Substituting $t=-2$ in eq. (i), we get

$$1=0+0+C(-2+1)\Rightarrow 1=-C\Rightarrow C=-1$$

Comparing the coefficient of x^2 of both side in eq (1) we have

$$0=A+B$$

Thus $A=-B=-1$

Now $\frac{1}{(t+1)(t+2)^2}=\frac{1}{t+1}-\frac{1}{t+2}-\frac{1}{(t+2)^2}$

$$I=\int\frac{1}{t+1}dt-\int\frac{1}{t+2}dt-\int\frac{1}{(t+2)^2}dt$$
$$=\log|t+1|-\log|t+2|-\frac{(t+2)^{-1}}{-1}+C$$
$$=\log|t+1|-\log|t+2|+\frac{1}{(t+2)}+C$$
$$=\log|x^2+1|-\log|x^2+2|+\frac{1}{(x^2+2)}+C$$

OR

Integrate w.r.t. x, $\frac{x^2-3x+1}{\sqrt{1-x^2}}$.

Sol :

We have $I=\int\frac{x^2-3x+1}{\sqrt{1-x^2}}$

$$=(-1)\int\frac{-x^2+3x-1}{\sqrt{1-x^2}}dx$$
$$=(-1)\int\frac{1-x^2+3x-2}{\sqrt{1-x^2}}dx$$
$$=(-1)\int\left[\frac{1-x^2}{\sqrt{1-x^2}}+\frac{3x-2}{\sqrt{1-x^2}}\right]dx$$
$$=(-1)\int\left[\sqrt{1-x^2}+\int\frac{3x-2}{\sqrt{1-x^2}}\right]dx$$
$$=(-1)\left[\int\sqrt{1-x^2}\,dx+\int\frac{3x-2}{\sqrt{1-x^2}}dx\right]$$

Let $I=(-1)[I_1+I_2]$

or $-I=I_1+I_2$ ...(1)

Now, $I_1=\int\sqrt{1-x^2}\,dx$

Using $\int\sqrt{a^2-x^2}\,dx=\frac{1}{2}\left[x\sqrt{a^2-x^2}+a^2\sin^{-1}\left(\frac{x}{a}\right)\right]+C$

$$I_1=\frac{1}{2}[x\sqrt{1-x^2}+\sin^{-1}x]+C_1 \quad ...(2)$$

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Sol :

We have $A = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$

Let A_{ij} be the cofactor of an element a_{ij} of $|A|$.
Then, cofactors of elements of $|A|$ are

$$A_{11} = (-1)^{1+1}\begin{vmatrix} 1 & -2 \\ -2 & 1 \end{vmatrix} = (1-4) = -3$$

$$A_{12} = (-1)^{1+2}\begin{vmatrix} 2 & -2 \\ 2 & 1 \end{vmatrix} = -(2+4) = -6$$

$$A_{13} = (-1)^{1+3}\begin{vmatrix} 2 & 1 \\ 2 & -2 \end{vmatrix} = (-2-4) = -6$$

$$A_{21} = (-1)^{2+1}\begin{vmatrix} -2 & -2 \\ -2 & 1 \end{vmatrix} = -(-2-4) = 6$$

$$A_{22} = (-1)^{2+2}\begin{vmatrix} -1 & -2 \\ 2 & 1 \end{vmatrix} = (-1+4) = 3$$

$$A_{23} = (-1)^{2+3}\begin{vmatrix} -1 & -2 \\ 2 & -2 \end{vmatrix} = -(2+4) = -6$$

$$A_{31} = (-1)^{3+1}\begin{vmatrix} -2 & -2 \\ 1 & -2 \end{vmatrix} = (4+2) = 6$$

$$A_{32} = (-1)^{3+2}\begin{vmatrix} -1 & -2 \\ 2 & -2 \end{vmatrix} = -(2+4) = -6$$

$$A_{33} = (-1)^{3+3}\begin{vmatrix} -1 & -2 \\ 2 & 1 \end{vmatrix} = (-1+4) = 3$$

Adjoint of the matrix A is given by

$$adj\ A = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix} = \begin{bmatrix} -3 & 6 & 6 \\ -6 & 3 & -6 \\ -6 & -6 & 3 \end{bmatrix}$$

Now, $$|A| = \begin{vmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{vmatrix}$$

$$= -1(1-4) + 2(2+4) - 2(-4-2)$$

$$= -(-3) + 2(6) - 2(-6)$$

$$= 3 + 12 + 12 = 27$$

$$A \cdot (adj\ A) = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}\begin{bmatrix} -3 & 6 & 6 \\ -6 & 3 & -6 \\ -6 & -6 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 3+12+12 & -6-6+12 & -6+12-6 \\ -6-6+12 & 12+3+12 & 12-6-6 \\ -6+12-6 & 12-6-6 & 12+12+3 \end{bmatrix}$$

$$= \begin{bmatrix} 27 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 27 \end{bmatrix}$$

$$= 27\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= 27I_3 = |A|I_3$$ Hence proved.

27. Find the values of a and b such that the function defined as follows is continuous.

$$f(x) = \begin{cases} x+2, & x \le 2 \\ ax+b, & 2 < x < 5 \\ 3x-2, & x \ge 5 \end{cases}$$

Sol :

Let $f(x) = \begin{cases} k(x^2+2), & if\ x \le 0 \\ 3x+1, & if\ x > 0 \end{cases}$ is continuous at $x = 0$

Then, $$LHL = RHL = f(0) \quad ...(i)$$

Here, $$LHL = \lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} k(x^2+2)$$

$$= \lim_{h \to 0} k[(0-h)^2+2]$$

[put $x = 0-h$; when $x \to 0^-$, then $h \to 0$]

$$= \lim_{h \to 0} k(h^2+2)$$

$$LHL = 2k$$

and $$RHL = \lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} (3x+1)$$

$$= \lim_{h \to 0} [3(0+h)+1]$$

[put $x = 0+h$; when $x \to 0^+$, then $h \to 0$]

$$RHL = 1$$

From Eq. (i), we have

$$LHL = RHL$$

$$2k = 1$$

$$k = \frac{1}{2}$$

Now, let us check the continuity of the given function $f(x)$ at $x = 1$.

Consider, $$\lim_{x \to 1} f(x) = \lim_{x \to 1} 3x+1$$

[$\because f(x) = 3x+1$ *for* $x > 0$]

$$= 4 = f(1)$$

[$\because f(1) = 3+1 = 4$]

$\because f(x)$ is continuous at $x = 1$

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Sol :

We have, $A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$

Here, $|A| = \begin{vmatrix} 2 & -3 \\ -4 & 7 \end{vmatrix}$

$= 14 - 12 = 2$

Since $|A| \neq 0$, therefore A^{-1} exists.

Now, $adj\,(A) = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$

$A^{-1} = \frac{1}{|A|} adj\,(A)$

$= \frac{1}{2}\begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$ ...(i)

Now $2A^{-1} = 9I - A$

RHS $= 9I - A$

$= 9\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$

$= \begin{bmatrix} 9 & 0 \\ 0 & 9 \end{bmatrix} - \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$

$= 2A^{-1}$ [using Eq. (i)]

$=$ LHS Hence proved.

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. If $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 4 \\ -2 & 2 & 1 \end{bmatrix}$, then find $(A')^{-1}$.

Sol :

We have, $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 4 \\ -2 & 2 & 1 \end{bmatrix}$

Now, $|A| = \begin{vmatrix} 1 & 2 & 3 \\ 0 & -2 & 4 \\ -2 & 2 & 1 \end{vmatrix}$

Expanding along R_1 we have

$|A| = 1(-1-8) + 2(0+8) + 3(0-2)$

$= -9 + 16 - 6 = 1$

Since $|A| \neq 0$, matrix A is non-singular and A^{-1} exists.

Cofactors of an element of $|A|$ are given by

$A_{11} = (-1)^{1+1}\begin{vmatrix} -1 & 4 \\ 2 & 1 \end{vmatrix} = (-1-8) = -9$

$A_{12} = (-1)^{1+2}\begin{vmatrix} 0 & 4 \\ -2 & 1 \end{vmatrix} = -(0+8) = -8$

$A_{13} = (-1)^{1+3}\begin{vmatrix} 0 & -1 \\ -2 & 2 \end{vmatrix} = (0-2) = -2$

$A_{21} = (-1)^{2+1}\begin{vmatrix} -2 & -1 \\ 2 & 2 \end{vmatrix} = -(-2-6) = 8$

$A_{22} = (-1)^{2+2}\begin{vmatrix} 1 & 3 \\ -2 & 1 \end{vmatrix} = (1+6) = 7$

$A_{23} = (-1)^{2+3}\begin{vmatrix} 1 & 3 \\ -2 & 1 \end{vmatrix} = -(2-4) = 2$

$A_{31} = (-1)^{3+1}\begin{vmatrix} -2 & 3 \\ -1 & 4 \end{vmatrix} = (-8+3) = -5$

$A_{32} = (-1)^{3+2}\begin{vmatrix} 1 & 3 \\ 0 & 4 \end{vmatrix} = -(4-0) = -4$

$A_{33} = (-1)^{3+3}\begin{vmatrix} 1 & -2 \\ 0 & -1 \end{vmatrix} = (-1-0) = -1$

Thus, $adj\,A = \begin{bmatrix} A_{11} & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{32} \\ A_{13} & A_{23} & A_{33} \end{bmatrix}$

$= \begin{bmatrix} -9 & 8 & -5 \\ -8 & 7 & -4 \\ -2 & 2 & -1 \end{bmatrix}$

Hence, $A^{-1} = \frac{1}{|A|} adj\,A$

$= \frac{1}{1}\begin{bmatrix} -9 & 8 & -5 \\ -8 & 7 & -4 \\ -2 & 2 & -1 \end{bmatrix}$

Now, $(A')^{-1} = (A^{-1})'$

$= \begin{bmatrix} -9 & 8 & -5 \\ -8 & 7 & -4 \\ -2 & 2 & -1 \end{bmatrix}$

$= \begin{bmatrix} -9 & -8 & -2 \\ 8 & 7 & 2 \\ -5 & -4 & -1 \end{bmatrix}$

OR

Find the adjoint of the matrix $A = \begin{bmatrix} -1 & -2 & -2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$ and hence show that $A\,(adj\,A) = |A| I_3$.

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Now, $I_2 = \int \frac{3x-2}{\sqrt{1-x^2}}dx$

$= \int \frac{3x}{\sqrt{1-x^2}}dx - 2\int \frac{dx}{\sqrt{1-x^2}}$

$= -\frac{3}{2}\int \frac{-2x}{\sqrt{1-x^2}}dx - 2\int \frac{dx}{\sqrt{1-x^2}}$

$= -\frac{3}{2} \times 2\sqrt{1-x^2} - 2\sin^{-1}x + C_2$

$= -3\sqrt{1-x^2} - 2\sin^{-1}x + C_2$ ...(3)

From eqs. (1), (2) and (3), we have

$-I = \frac{1}{2}[x\sqrt{1-x^2} + \sin^{-1}(x)] + C_1$

$-3\sqrt{1-x^2} - 2\sin^{-1}(x) + C_2$

$-I = \frac{1}{2}x\sqrt{1-x^2} - \frac{3}{2}\sin^{-1}x - 3\sqrt{1-x^2} + C_1 + C_2$

or $I = \frac{3}{2}\sin^{-1}x - \frac{x}{2}\sqrt{1-x^2} + 3\sqrt{1-x^2} + C$

where, $C = -C_1 - C_2$

31. Solve the following differential equation

$$x\frac{dy}{dx} = y - x\tan\left(\frac{y}{x}\right)$$

Sol :

We have, $x\frac{dy}{dx} = y - x\tan\left(\frac{y}{x}\right)$

$\frac{dy}{dx} = \frac{y}{x} - \tan\left(\frac{y}{x}\right)$ ...(1)

which is a homogeneous differential equation as $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$.

Substituting $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ in eq (1) we have

$v + x\frac{dv}{dx} = v - \tan v$

$x\frac{dv}{dx} = -\tan v$

$\frac{dv}{\tan v} = -\frac{dx}{x}$

$\cot v\, dv = -\frac{dx}{x}$

Integrating both sides, we have

$\int \cot v\, dv = -\int \frac{dx}{x}$

$\log|\sin v| = -\log|x| + C$

$\int \cot v dv = \log|\sin v|$

$\log|\sin v| + \log|x| = C$

$\log|x\sin v| = C$ $\log m + \log n = \log mn$

$\log\left|x\sin\frac{y}{x}\right| = C$ $v = \frac{y}{x}$

$x\sin\frac{y}{x} = e^C$

$x\sin\frac{y}{x} = A$ $e^C = A$

$\sin\frac{y}{x} = \frac{A}{x} \Rightarrow y = x\sin^{-1}\left(\frac{A}{x}\right)$.

which is the required solution.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Find the value of the following

$\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$

Sol :

Let $\tan^{-1}(1) = x_1$

$\tan x_1 = 1$

$\tan x_1 = \tan\frac{\pi}{4}$

$x_1 = \frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Let $\cos^{-1}\left(-\frac{1}{2}\right) = x_2$

$\cos x_2 = -\frac{1}{2}$

$\cos x_2 = \cos\frac{2\pi}{3}$

$x_2 = \frac{2\pi}{3} \in [0, \pi]$

Let $\sin^{-1}\left(-\frac{1}{2}\right) = x_3$

$\sin x_3 = -\frac{1}{2}$

$\sin x_3 = -\sin\frac{\pi}{6}$

$\sin x_3 = \sin\left(-\frac{\pi}{6}\right)$

$x_3 = -\frac{\pi}{6} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$x_1 + x_2 + x_3 = \frac{\pi}{4} + \frac{2\pi}{3} - \frac{\pi}{6}$

$\tan^{-1}(1) + \tan^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$

$= \frac{3\pi + 8\pi - 2\pi}{12}$

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A manufactured component has its quality graded on its performance, appearance, and cost. Each of these three characteristics is graded as either pass or fail. There is a probability of 0.40 that a component passes on both appearance and cost. There is a probability of 0.35 that a component passes on both performance and appearance. There is a probability of 0.31 that a component passes on all three characteristics. There is a probability of 0.64 that a component passes on performance. There is a probability of 0.19 that a component fails on all three characteristics. There is a probability of 0.06 that a component passes on appearance but fails on both performance and cost.

(i) What is the probability that a component passes on cost but fails on both performance and appearance?

(ii) If a component passes on both appearance and cost, what is the probability that it passes on all three characteristics?

(iii) If a component passes on both performance and appearance, what is the probability that it passes on all three characteristics?

Sol :

Let E be the event that the 'component passes on performance',

Let A be the event that the 'component passes on appearance', and let C be the event that the 'component passes on cost'.

$$P(A \cap C) = 0.4$$

$$P(E \cap A) = 0.35$$

(i) $$P(E \cap A \cap C) = 0.31$$

$$P(E) = 0.64$$

$$P(E' \cap A' \cap C') = 0.19$$

$$P(E' \cap A' \cap C') = 0.06$$

Therefore,

$$\begin{aligned} P(E' \cap A' \cap C') &= P(E' \cap A') - P(E' \cap A' \cap C') \\ &= P(E') - P(E' \cap A) - 0.19 \\ &= 1 - P(E) - P(E' \cap A \cap C) \\ &\quad - P(E' \cap A \cap C') - 0.19 \\ &= 1 - 0.64 - P(A \cap C) \\ &\quad + P(E \cap A \cap C) - 0.06 - 0.19 \\ &= 1 - 0.64 - 0.40 + 0.31 - 0.06 - 0.19 \\ &= 0.02 \end{aligned}$$

(ii) $$P(E \cap A \cap C \mid A \cap C) = \frac{P(E \cap A \cap C)}{P(A \cap C)} = \frac{0.31}{0.40} = 0.775$$

(iii) $$P(E \cap A \cap C \mid E \cap A) = \frac{P(E \cap A \cap C)}{P(E \cap A)} = \frac{0.31}{0.35} = 0.886$$

38. Commodity prices are primarily determined by the forces of supply and demand in the market. For example, if the supply of oil increases, the price of one barrel decreases. Conversely, if demand for oil increases (which often happens during the summer), the price rises. Gasoline and natural gas fall into the energy commodities category.



The price p (dollars) of each unit of a particular commodity is estimated to be changing at the rate

$$\frac{dp}{dx} = \frac{-135x}{\sqrt{9+x^2}}$$

where x (hundred) units is the consumer demand (the number of units purchased at that price). Suppose 400 units $(x = 4)$ are demanded when the price is \$30 per unit.

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So, $0, \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$ and $\frac{7\pi}{4}$ are the points of maxima and minima.

At $x=0$, $f(0) = \sin 2(0) = \sin 0 = 0$

At $x=\frac{\pi}{4}$, $f\left(\frac{\pi}{4}\right) = \sin 2\left(\frac{\pi}{4}\right) = \sin\frac{\pi}{2} = 1$

At $x=\frac{3\pi}{4}$, $f\left(\frac{3\pi}{4}\right) = \sin 2\left(\frac{3\pi}{4}\right)$

$= \sin\frac{3\pi}{2} = -1$

At $x=\frac{5\pi}{4}$, $f\left(\frac{5\pi}{4}\right) = \sin 2\left(\frac{5\pi}{4}\right) = \sin\frac{5\pi}{2}$

$= \sin\left(2\pi+\frac{\pi}{2}\right)\sin\frac{\pi}{2} = 1$

At $x=\frac{7\pi}{4}$, $f\left(\frac{7\pi}{4}\right) = \sin 2\left(\frac{7\pi}{4}\right)$

$= \sin\left(\frac{7\pi}{4}\right)$

$= \sin\left(4\pi-\frac{\pi}{2}\right)$

$= -\sin\frac{\pi}{2} = -1$

Hence, the maximum value is 1 at $x=\frac{\pi}{4}$ and $x=\frac{5\pi}{4}$ and the minimum value is -1 at $x=\frac{3\pi}{4}$ and $x=\frac{7\pi}{4}$.

34. Maximize $Z=-x+2y$, Subject to the constraints:

$$x \geq 3,\ x+y \geq 5,\ x+2y \geq 6,\ y \geq 0$$

Sol :

Here objective function is $Z=-x+2y$

Subject to the constraints:

$$x \geq 3,\ x+y \geq 5,\ x+2y \geq 6,\ y \geq 0$$

Let us consider the following equations:

(i) $x+y=5$

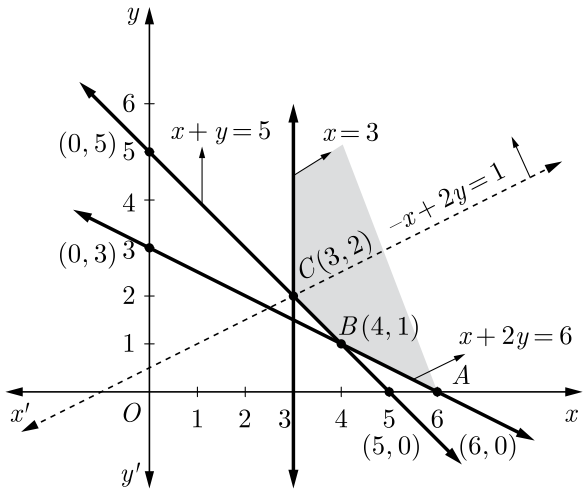
x	0	5
y	5	0

(ii) $x+2y=6$

x	0	6
y	3	0

(iii) $x=3,\ y=0$

On plotting the graph of the given inequalities, we get an open unbounded feasible region.



Corner points $Z = -x+2y$

$A(6,0)$ $Z = -6+2\times 0 = -6$

$B(4,1)$ $Z = -4+2\times 1 = -2$

$C(3,2)$ $Z = -3+2\times 2 = 1$

Since the feasible region is unbounded, therefore, $Z=1$ may or may not be the maximum value. For this, we graph the inequality $-x+2y>1$ and check weather the resulting half plane has points in common with the feasible region or not.

The resulting feasible region has points in common with the feasible region. Hence, $Z=1$ is not the maximum value. Or Z has no maximum value.

OR

Minimise $Z=x+2y$ subject to $2x+y\geq 3$, $x+2y\geq 6$, $x,\ y\geq 0$. Show that the minimum of Z occurs at more than two points.

Sol :

Minimise $Z = x+2y$

Subject to $2x+y \geq 3$ ...(i)

$x+2y \geq 6$ ...(ii)

and $x\geq 0,\ y\geq 0$

(i) Region corresponding to $2x+y \geqslant 3$:

Line $2x+y=3$			
x	0	$\frac{3}{2}$	Point (0,0) is false for $2x+y \geqslant 3$. So, the region is away from the origin.
y	3	0	

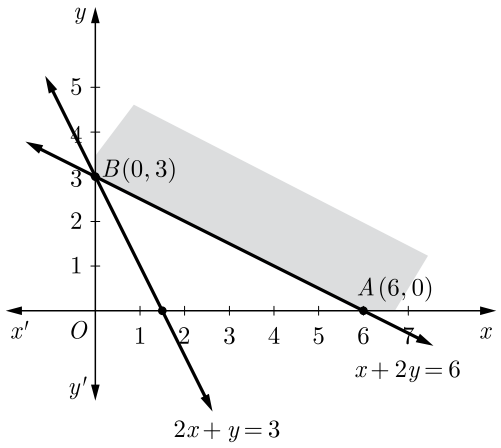
(ii) Region corresponding to $x+2y \geqslant 6$:

Line $x+2y=6$			
x	0	6	Point (0,0) is false for $x+2y \geqslant 6$. So, the region is away from the origin.
y	3	0	

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Since $x \geq 0$, $y \geq 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is ABC which is open and unbounded region. The corner points of the feasible region are $A(6,0)$ and $B(0,3)$.



Since, the feasible region is a bounded region, we can check the objective function at all the corner to find the maxima. The values of objective function Z at these points are as follows.

Corner points	$Z = x + 2y$
$A(6,0)$	$Z = 6 + 0 = 6$
$B(0,3)$	$Z = 0 + 2 \times 3 = 6$

The minimum value of Z is 6 at all points on the line segment joining the points $A(6,0)$ and $B(0,3)$.

35. Find the value of λ, so that the lines $\frac{1-x}{3} = \frac{7y-14}{\lambda} = \frac{z-3}{2}$ and $\frac{7-7x}{3\lambda} = \frac{y-5}{1} = \frac{6-z}{5}$ are at right angles. Also, find whether the lines are intersecting or not.

Sol :

Writing the given line in standard form as

$$\frac{x-1}{-3} = \frac{y-2}{\frac{\lambda}{7}} = \frac{z-3}{2} = r_1 \text{ (let)} \qquad ...(1)$$

and $$\frac{x-1}{\frac{-3\lambda}{7}} = \frac{y-5}{1} = \frac{z-6}{-5} = r_2 \text{ (let)} \qquad ...(2)$$

Two lines with DR's a_1, b, c_1 and a_2, b_2, c_2 are perpendicular if

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

Thus line (1) and (2) will intersect at right angle, if

$$-3\left(\frac{-3\lambda}{7}\right) + \frac{\lambda}{7}(1) + 2(-5) = 0$$

$$\frac{9\lambda}{7} + \frac{\lambda}{7} = 10$$

$$\frac{10\lambda}{7} = 10 \Rightarrow \lambda = 7$$

which is the required value of λ.

Now, let us check whether the lines are intersecting or not.

Coordinates of any point on line (1) are

$$(-3r_1 + 1, r_1 + 2, 2r_1 + 3)$$

and coordinates of any point on line (2) are

$$(-3r_1 + 1, r_1 + 5, -5r_2 + 6)$$

Clearly, the line will intersect if

$$(-3r_1 + 1, r_1 + 2, 2r_1 + 3) = (-3r_2 + 1, r_2 + 5, -5r_2 + 6)$$

$$-3r_1 + 1 = -3r_2 + 1;\ r_1 + 2 = r_2 = 5$$

and $$2r_1 + 3 = -5r_2 + 6$$

$$r_1 = r_2;\ r_1 - r_2 = 3;\ 2r_1 + 5r_2 = 3$$

which is not possible simultaneously for any r_1, $r_2 \in R$

Hence, the lines are not intersecting.

OR

Find the shortest distance between the lines

$$\vec{r} = (4\hat{i} - \hat{j}) + \lambda(\hat{i} + 2\hat{j} - 3\hat{k})$$

and $$\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(2\hat{i} + 4\hat{j} - 5\hat{k}).$$

Sol :

We have $$\vec{r} = (4\hat{i} - \hat{j}) + \lambda(\hat{i} + 2\hat{j} - 3\hat{k})$$

and $$\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(2\hat{i} + 4\hat{j} - 5\hat{k})$$

Comparing Eqs. (1) and (2), with $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$ respectively, we get

$$\vec{a}_1 = 4\hat{i} - \hat{j},\ \vec{b}_1 = \hat{i} + 2\hat{j} - 3\hat{k}$$

and $$\vec{a}_2 = \hat{i} - \hat{j} + 2\hat{k},\ \vec{b}_2 = 2\hat{i} + 4\hat{j} - 5\hat{k}$$

Here $\vec{a}_2 - \vec{a}_1 = -3\hat{i} + 2\hat{k}$

and $$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 2 & 4 & -5 \end{vmatrix}$$

$$= \hat{i}(-10 + 12) - \hat{j}(-5 + 6) + \hat{k}(4 - 4)$$

$$= 2\hat{i} - \hat{j}$$

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{2^2 + (-1)^2}$$

$$= \sqrt{4 + 1} = \sqrt{5}$$

Now, the shortest distance between the given lines,

$$d = \frac{5(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|}$$

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$$= \frac{|(2\hat{i} - \hat{j}) \cdot (-3\hat{i} + 2\hat{k})|}{\sqrt{5}}$$

$$= \frac{|-6|}{\sqrt{5}} = \frac{6}{\sqrt{5}} \text{ units}$$

SECTION - E

Case study based questions are compulsory.

36. Sun Pharmaceutical Industries Limited is an Indian multinational pharmaceutical company headquartered in Mumbai, Maharashtra, that manufactures and sells pharmaceutical formulations and active pharmaceutical ingredients in more than 100 countries across the globe.

Sun Pharmaceutical produces three final chemical products P_1, P_2 and P_3 requiring mixup of three raw material chemicals M_1, M_2 and M_3. The per unit requirement of each product for each material (in litres) is as follows:

$$A = \begin{matrix} \\ P_1 \\ P_2 \\ P_3 \end{matrix} \begin{matrix} M_1 & M_2 & M_3 \end{matrix} \begin{bmatrix} 2 & 3 & 1 \\ 4 & 2 & 5 \\ 2 & 4 & 2 \end{bmatrix}$$



(i) Find the total requirement of each material if the firm produces 100 litres of each product,

(ii) Find the per unit cost of production of each product if the per unit of materials M_1, M_2 and M_3 are ₹ 5, ₹ 10 and ₹ 5 respectively, and

(iii) Find the total cost of production if the firm produces 200 litres of each product.

Sol :

When the firm produces 100 litres of each product, then the production matrix B is given by:

$$B = \begin{matrix} P_1 & P_2 & P_3 \end{matrix} \begin{bmatrix} 100 & 100 & 100 \end{bmatrix}$$

The total requirement of each material is:

$$B = \begin{matrix} P_1 & P_2 & P_3 \end{matrix} \begin{bmatrix} 100 & 100 & 100 \end{bmatrix}_{1\times 3} \begin{matrix} \\ P_1 \\ P_2 \\ P_3 \end{matrix} \begin{matrix} M_1 & M_2 & M_3 \end{matrix} \begin{bmatrix} 2 & 3 & 1 \\ 4 & 2 & 5 \\ 2 & 4 & 2 \end{bmatrix}_{3\times 3}$$

$$= \begin{matrix} M_1 & M_2 & M_3 \end{matrix} \begin{bmatrix} 200+400+200 & 300+200+400 & 100+500+200 \end{bmatrix}$$

$$= \begin{matrix} M_1 & M_2 & M_3 \end{matrix} \begin{bmatrix} 800 & 900 & 800 \end{bmatrix}$$

When the cost per unit of materials M_1, M_2 and M_3 are ₹ 5, ₹ 10 and ₹ 5 respectively, then the cost materials matrix C is given by:

$$C = \begin{matrix} M_1 \\ M_2 \\ M_3 \end{matrix} \begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix}$$

The cost of production of each product is given by the matrix

$$AC = \begin{matrix} \\ P_1 \\ P_2 \\ P_3 \end{matrix} \begin{matrix} M_1 & M_2 & M_3 \end{matrix} \begin{bmatrix} 2 & 3 & 1 \\ 4 & 2 & 5 \\ 2 & 4 & 2 \end{bmatrix} \begin{matrix} \text{Cost} \\ \text{(per unit)} \\ M_1 \\ M_2 \\ M_3 \end{matrix} \begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix}$$

$$= \begin{matrix} \\ P_1 \\ P_2 \\ P_3 \end{matrix} \begin{matrix} \text{Cost of} \\ \text{(Production)} \end{matrix} \begin{bmatrix} 10+30+5 \\ 20+20+25 \\ 10+40+10 \end{bmatrix}$$

$$= \begin{matrix} P_1 \\ P_2 \\ P_3 \end{matrix} \begin{bmatrix} 45 \\ 65 \\ 60 \end{bmatrix}$$

When the firm produces 200 litres of each product, then the matrix D is given by:

$$D = \begin{matrix} P_1 & P_2 & P_3 \end{matrix} \begin{bmatrix} 200 & 200 & 200 \end{bmatrix}$$

The total cost of production is given by:

$$D(AC) = \begin{matrix} P_1 & P_2 & P_3 \end{matrix} \begin{bmatrix} 200 & 200 & 200 \end{bmatrix} \begin{matrix} P_1 \\ P_2 \\ P_3 \end{matrix} \begin{bmatrix} 45 \\ 65 \\ 60 \end{bmatrix}$$

$$= \begin{bmatrix} 200 \times 45 + 200 \times 65 + 200 \times 60 \end{bmatrix}$$

$$= \begin{bmatrix} 34000 \end{bmatrix}$$

37. Quality assurance (QA) testing is the process of ensuring that manufactured product is of the highest possible quality for customers. QA is simply the techniques used to prevent issues with product and to ensure great user experience for customers.

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$$= \frac{9\pi}{12} = \frac{3\pi}{4}$$

Hence, the required value is $\frac{3\pi}{4}$.

Alternative :Short cut Method:

$$\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right) = \frac{\pi}{4} + \frac{\pi}{2} = \frac{3\pi}{4}$$

$$\left[\because \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}\right]$$

Hence, the required value is $\frac{3\pi}{4}$.

OR

Find the value of the following $\cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right)$

Sol :

Let $\cos^{-1}\left(\frac{1}{2}\right) = x_1$

$$\cos x_1 = \frac{1}{2}$$

$$\cos x_1 = \cos\frac{\pi}{3}$$

$$x_1 = \frac{\pi}{3} \in [0, \pi]$$

Let $2\sin^{-1}\left(\frac{1}{2}\right) = x_2$

$$\sin^{-1}\left(\frac{1}{2}\right) = \frac{x_2}{2}$$

$$\sin\left(\frac{x_2}{2}\right) = \frac{1}{2}$$

$$\sin\left(\frac{x_2}{2}\right) = \sin\frac{\pi}{6}$$

$$\frac{x_2}{2} = \frac{\pi}{6} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$x_2 = \frac{\pi}{3}$$

So, $x_1 + x_2 = \frac{\pi}{3} + \frac{\pi}{3}$

$$\cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right) = \frac{2\pi}{3}$$

Hence, the require value is $\frac{2\pi}{3}$.

33. Find both the maximum value and minimum value of $3x^4 - 8x^3 + 12x^2 - 48x + 25$ on the interval $[0,3]$.

Sol :

Let $f(x) = 3x^4 - 8x^3 + 12x^2 - 48x + 25$

$$f'(x) = 12x^3 - 24x^2 + 24x - 48$$

For maximas and minimas, $f'(x) = 0$

$$12x^3 - 24x^2 + 24x - 48 = 0$$

$$x^3 - 2x^2 + 2x - 4 = 0$$

$$x^2(x-2) + 2(x-2) = 0$$

$$(x-2)(x^2+2) = 0$$

$$x - 2 = 0$$

$$x = 2$$

$$x^2 + 2 = 0$$

$$x^2 = -2 \text{ (Rejected)}$$

[$\because$ Square of a number cannot be a negative number]

So, $x = 0$, $x = 2$ and $x = 3$ are the points of maxima and minima.

At $x = 0$,

$$f(0) = 3(0)^4 - 8(2)^3 + 12(0)^2 - 48(0) + 25$$

$$= 25$$

At $x = 2$,

$$f(2) = 3(2)^4 - 8(2)^3 + 12(2)^2 - 48(2) + 25$$

$$= 48 - 64 + 48 - 96 + 25$$

$$= -39$$

At $x = 3$,

$$f(3) = 3(3)^4 - 8(3)^3 + 12(3)^2 - 48(3) + 25$$

$$= 243 - 216 + 108 - 144 + 25$$

$$= 376 - 360 = 16$$

Hence, the maximum value is 25 at $x = 0$ and the minimum value is -39 at $x = 2$.

OR

At what points in the interval $[0, 2\pi]$, does the function $\sin 2x$ attain its maximum value?

Sol :

Let $f(x) = \sin 2x$

$$f'(x) = 2\cos 2x$$

For maxima and minima, $f'(x) = 0$

$$2\cos 2x = 0$$

$$\cos 2x = 0$$

$$2x = (2\pi + 1)\frac{\pi}{2},\ n \in I$$

$$2x = \frac{\pi}{3}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

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(i) Find the demand function $p(x)$.
(ii) At what price will 300 units be demanded? At what price will no units be demanded?
(iii) How many units are demanded at a price of \$20 per unit?

Sol :

(i) $$p(x) = \int \frac{dp}{dx}\,dx$$

Let $u = 9 + x^2$ then we have $du = 2x\,dx$ or $x\,dx = \frac{1}{2}\,du$

Thus
$$p(x) = \int \frac{-135x}{\sqrt{9+x^2}}\,dx$$
$$= \int \frac{-135}{u^{1/2}}\left(\frac{1}{2}\right)du$$
$$= \frac{-135}{2}\int u^{-1/2}\,du$$
$$= \frac{-135}{2}\left(\frac{u^{1/2}}{\frac{1}{2}}\right) + C$$
$$= -135\sqrt{9+x^2} + C$$

Since $p = 30$ when $x = 4$, we have
$$30 = -135\sqrt{9+4^2} + C$$
$$C = 30 + 135\sqrt{25} = 705$$

So, $$p(x) = -135\sqrt{9+x^2} + 705$$

(ii) When 300 units are demanded, $x = 3$ and the corresponding price is
$$p(3) = -135\sqrt{9+3^2} + 705$$
$$= \$132.24 \text{ per unit}$$

No units are demanded when $x = 0$ and the corresponding price is
$$p(0) = -135\sqrt{9+0} + 705$$
$$= \$300 \text{ per unit}$$

(iii) Number of units demanded at a unit price of \$20 per unit,
$$20 = -135\sqrt{9+x^2} + 705$$
$$-135\sqrt{9+x^2} + 705 = 20$$
$$135\sqrt{9+x^2} = 685$$
$$\sqrt{9+x^2} = \frac{685}{135}$$
$$9 + x^2 \approx 25.75$$
$$x^2 \approx 16.75$$
$$x \approx 4.09$$

That is, roughly 409 units will be demanded when the price is \$20 per unit.

In this book, we have provided solutions along with the papers. To download the same papers in unsolved format for printing, please visit
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Sample Paper 05

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

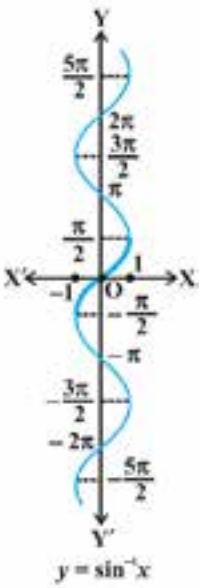
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the range of the graph of $y = \sin^{-1}x$ as shown in the image?



(a) $[-1,1]$ (b) $[-\pi/2,\pi/2]$

(c) $[0,\pi]$ (d) $(-\infty,\infty)$

Ans :

From the graph, the curve of $y = \sin^{-1}x$ extends only between $-\pi/2$ at the bottom and $\pi/2$ at the top on the Y-axis. This is the restricted range of the inverse sine function. The X-values are between -1 and 1, but the Y-values are confined to this interval. Hence, the correct range is $[-\pi/2,\pi/2]$.

Thus (b) is correct option.

2. If $dy/dx = e^{-2y}$ and $y = 0$, when $x = 5$, then the value of x, when $y = 3$ is

(a) e^5 (b) $e^6 + 1$

(c) $\frac{e^6 + 9}{2}$ (d) $\log_e 6$

Sol :

$$\frac{dy}{dx} = e^{-2y}$$

$$\int e^{2y}\,dy = \int dx + c$$

$$\frac{e^{2y}}{2} = x - \frac{9}{2}$$

When $x = 5$, $y = 0$

$$\frac{1}{2} = 5 + c$$

$$c = -\frac{9}{2}$$

$$\frac{e^{2y}}{2} = x - \frac{9}{2}$$

When $y = 3$, $\frac{e^6}{2} = x - \frac{9}{2}$

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$$x = \frac{9+e^6}{2}$$

Thus (c) is correct option.

3. A constant force $\vec{F} = 2\vec{i} - 3\vec{j} + 2\vec{k}$ is acting on a particle such that the particle is displaced from the point $A(1,2,3)$ to the point $B(3,4,5)$. The work done by the force is

(a) 2 (b) 3

(c) $\sqrt{17}$ (d) $2\sqrt{51}$

Sol :

Displacement vector

$$\vec{d} = (3-1)\hat{i} + (4-2)\hat{j} + (5-3)\hat{k}$$
$$= 2\hat{i} + 2\hat{j} + 2\hat{k}$$

$$\text{Work done} = \vec{F} \cdot \vec{d}$$
$$= \left(2\hat{i} - 3\hat{j} + 2\hat{k}\right) \cdot \left(2\hat{i} + 2\hat{j} + 2\hat{k}\right)$$
$$= 4 - 6 + 4 = 2$$

Thus (a) is correct option.

4. The points (5, 2, 4), $(6,-1,2)$ and $(8,-7,k)$ are collinear, if k is equal to

(a) -2 (b) 2

(c) 3 (d) -1

Sol :

Since, points are collinear.

Hence, $$\begin{vmatrix} 5 & 2 & 4 \\ 6 & -1 & 2 \\ 8 & -7 & k \end{vmatrix} = 0$$

$$8(4+4) + 7(10-24) + k(-5-12) = 0$$
$$64 - 98 - 17k = 0$$
$$k = -2$$

Thus (a) is correct option.

5. If x-coordinate of a point P of line joining the points and $R(5,2,-2)$ is 4, then the z-coordinate of P is

(a) -2 (b) -1

(c) 1 (d) 2

Sol :

Suppose P divides QR in the ratio $\lambda:1$. Then coordinates of P are $\left(\frac{5\lambda+2}{\lambda+1}, \frac{2\lambda+2}{\lambda+1}, \frac{-2\lambda+1}{\lambda+1}\right)$.

Since, the x-coordinate of P is 4.

Hence, $$\frac{5\lambda+2}{\lambda+1} = 4$$
$$\lambda = 2$$

So, z-coordinate of P is

$$\frac{-2\lambda+1}{\lambda+1} = -1$$

Thus (b) is correct option.

6. Minimum value of the function $f(x) = x^2 + x + 1$ is

(a) 1 (b) 3

(c) $\frac{3}{4}$ (d) 4

Sol :

$$f(x) = x^2 + x + 1$$
$$= \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4},\ \forall\, x \in R$$

Thus (c) is correct option.

7. If $P(A) = \frac{1}{12}$, $P(B) = \frac{5}{12}$ and $P\left(\frac{B}{A}\right) = \frac{1}{15}$, then $P(A \cup B)$ is equal to

(a) $\frac{89}{180}$ (b) $\frac{90}{180}$

(c) $\frac{91}{180}$ (d) $\frac{92}{180}$

Sol :

We have $$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$$
$$\frac{1}{15} = \frac{P(A \cap B)}{\frac{1}{12}}$$
$$P(A \cap B) = \frac{1}{180}$$

Also, $$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$P(A \cup B) = \frac{1}{12} + \frac{5}{12} - \frac{1}{180} = \frac{89}{180}$$

Thus (a) is correct option.

8. Range of the function $f(x) = \frac{x}{1+x^2}$ is

(a) $(-\infty, \infty)$ (b) $[-1,1]$

(c) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (d) $[-\sqrt{2}, \sqrt{2}]$

Sol :

Let, $$y = \frac{x}{1+x^2}$$
$$x^2 y - x + y = 0$$

For x to be real, $$1 - 4y^2 \geq 0$$
$$(1-2y)(1+2y) \geq 0$$

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19. **Assertion :** $I=\int_0^{\frac{\pi}{2}}\sqrt{\tan x}\,dx=\frac{\pi}{\sqrt{2}}$

Reason : $\tan x=t^2$ makes the integrand in I as a rational function.

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

$$I=\int_0^{\frac{\pi}{2}}\sqrt{\tan x}\,dx,$$

Substituting $\tan x=t^2$ we have

$$dx=\frac{2tdt}{1+t^4}$$

If $x=0,\quad t=0$

If $x=\frac{\pi}{2}\quad t=\infty$

$$\begin{aligned}I&=\int_0^{\infty}\frac{2t^2\,dt}{1+t^4}\\&=\int_0^{\infty}\frac{t^2+1+t^2-1}{1+t^4}dt\\&=\int_0^{\infty}\frac{1+\frac{1}{t^2}}{t^2+\frac{1}{t^2}}dt+\int_0^{\infty}\frac{1-\frac{1}{t^2}}{t^2+\frac{1}{t^2}}dt\\&=\int_0^{\infty}\frac{d(t-\frac{1}{t})}{(t-\frac{1}{t})^2+2}+\int_0^{\infty}\frac{d(t+\frac{1}{t})}{(t+\frac{1}{t})^2-2}\\&=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{t-\frac{1}{t}}{\sqrt{2}}\right)\Bigg|_0^{\infty}+\frac{1}{2\sqrt{2}}1n\left(\frac{t+\frac{1}{t}-\sqrt{2}}{t+\frac{1}{t}+\sqrt{2}}\right)\Bigg|_0^{\infty}\\&=\frac{\pi}{\sqrt{2}}+0=\frac{\pi}{\sqrt{2}}\end{aligned}$$

Thus (a) is correct option.

20. **Assertion:** $f(\theta)=\begin{vmatrix}\cos(\theta+\alpha) & \cos(\theta+\beta) & \cos(\theta+\gamma)\\ \sin(\theta+\alpha) & \sin(\theta+\beta) & \sin(\theta+\gamma)\\ \sin(\beta-\gamma) & \sin(\gamma-\alpha) & \sin(\alpha-\beta)\end{vmatrix}$ is independent of θ

Reason: If $f(\theta)=c$ then $f(\theta)$ is independent of θ

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

Let,

$$f(\theta)=\begin{vmatrix}\cos(\theta+\alpha) & \cos(\theta+\beta) & \cos(\theta+\gamma)\\ \sin(\theta+\alpha) & \sin(\theta+\beta) & \sin(\theta+\gamma)\\ \sin(\beta-\gamma) & \sin(\gamma-\alpha) & \sin(\alpha-\beta)\end{vmatrix}$$

$$\begin{aligned}f'(\theta)=&\begin{vmatrix}-\sin(\theta+\alpha) & -\sin(\theta+\beta) & -\sin(\theta+\gamma)\\ \sin(\theta+\alpha) & \sin(\theta+\beta) & \sin(\theta+\gamma)\\ \sin(\beta-\gamma) & \sin(\gamma-\alpha) & \sin(\alpha-\beta)\end{vmatrix}\\&+\begin{vmatrix}\cos(\theta+\alpha) & \cos(\theta+\beta) & \cos(\theta+\gamma)\\ \cos(\theta+\alpha) & \cos(\theta+\beta) & \cos(\theta+\gamma)\\ \sin(\beta-\gamma) & \sin(\gamma-\alpha) & \sin(\alpha-\beta)\end{vmatrix}\\&+\begin{vmatrix}\cos(\theta+\alpha) & \cos(\theta+\beta) & \cos(\theta+\gamma)\\ \sin(\theta+a) & \sin(\theta+\beta) & \sin(\theta+\gamma)\\ 0 & 0 & 0\end{vmatrix}\\&=0+0+0=0\end{aligned}$$

$f'(\theta)=0$

$f(\theta)=c$

Thus (a) is correct option.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Write the direction ratios of the vector $3\vec{a}+2\vec{b}$, where $\vec{a}=\hat{i}+\hat{j}-2\hat{k}$ and $\vec{b}=2\hat{i}-4\hat{j}+5\hat{k}$.

OR

Find the unit vector in the direction of the sum of vectors $2\hat{i}+3\hat{j}-\hat{k}$ and $4\hat{i}-3\hat{j}+2\hat{k}$.

Sol :

The given vector $3\vec{a}+2\vec{b}$ can be given as

$$\begin{aligned}3\vec{a}+2\vec{b}&=3(\hat{i}+\hat{j}-2\hat{k})+2(2\hat{i}-4\hat{j}+5\hat{k})\\&=(3\hat{i}+3\hat{j}-6\hat{k})+(4\hat{i}-8\hat{j}+10\hat{k})\\&=7\hat{i}-5\hat{j}+4\hat{k}\end{aligned}$$

Hence, direction ratios of vectors $3\vec{a}+2\vec{b}$ are $7,-4$ and 4.

OR

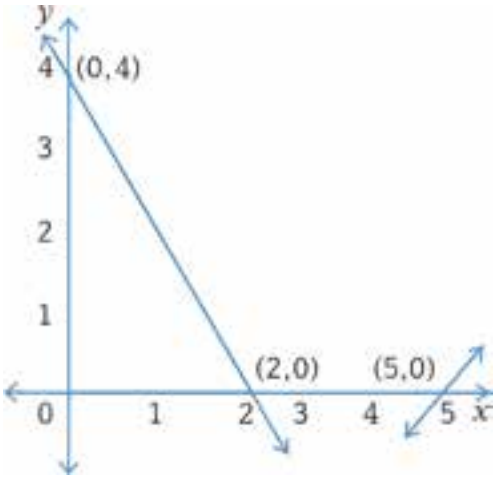
We have $\vec{a}=2\hat{i}+3\hat{j}-\hat{k}$

and $\vec{b}=4\hat{i}-3\hat{j}+2\hat{k}$

Sum of two vectors,

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14. For the same LPP with objective function $Z = 3x + 2y$, what is the minimum value of Z and at which point does it occur?



(a) Z=8 at (0,4) (b) Z=15 at (5,0)

(c) Z=6 at (2,0) (d) No minimum exists

Sol :

(c) Z=6 at (2,0)

By evaluating Z at all corner points of the feasible region:

At (2,0), Z=6; at (0,4), Z=8; at (5,0), Z=15.

The least value is 6, obtained at (2,0).

15. The existence of the unique solution of the system of equations $x + y + z = \beta$; $5x - y + \alpha z = 10$ and $2x + 3y = 6$ depends on

(a) α only (b) β only

(c) Both α and β (d) Neither β nor α

Sol :

Given system of equations is

$x + y + z = \beta$, $5x - y + az = 10$ and $2x + 3y - z = 6$

For unique solution, $\begin{vmatrix} 1 & 1 & 1 \\ 5 & -1 & \alpha \\ 2 & 3 & -1 \end{vmatrix} \neq 0$

$1(1 - 3\alpha) - 1(-5 - 2\alpha) + 1(15 + 2) \neq 0$

$1 - 3\alpha + 5 + 2\alpha + 17 = 0 \Rightarrow -\alpha + 23 \neq 0$, $\alpha \neq 23$

Hence, for the existence of the unique solution, system of equations depends on α only.

Thus (a) is correct option.

16. At $x = \frac{3}{2}$, the function $f(x) = \frac{|2x-3|}{2x-3}$ is

(a) continuous (b) discontinuous

(c) differentiable (d) non-zero

Sol :

We have $f(x) = \begin{cases} \frac{2x-3}{2x-3}, & if\ x > \frac{3}{2} \\ \frac{-(2x-3)}{2x-3}, & if\ x < \frac{3}{2} \end{cases}$

$= \begin{cases} 1, & if\ x > \frac{3}{2} \\ -1, & if\ x < \frac{3}{2} \end{cases}$

Now, $RHL = \lim_{x \to 3/2^+} f(x) = \lim_{x \to 3/2^+} 1 = 1$

and $LHL = \lim_{x \to 3/2^-} f(x) = \lim_{x \to 3/2^-} (-1) = -1$

$RHL \neq LHL$

So, $f(x)$ is discontinuous at $x = \frac{3}{2}$.

Thus (b) is correct option.

17. If $f(x) = \begin{cases} \frac{\log x}{x-1}, & if\ x \neq 1 \\ k, & if\ x = 1 \end{cases}$ is continuous at $x = 1$, then the value of k is

(a) 0 (b) -1

(c) 1 (d) e

Sol :

At $x = 1$,

$$\lim_{x \to 1^+} f(x) = \lim_{h \to 0} \frac{\log(1+h)}{1+h-1}$$

$$= \lim_{h \to 0} \frac{\log(1+h)}{h} = 1$$

As $f(x)$ is continuous at $x = 1$.

$$\lim_{x \to 1^+} f(x) = f(1) \Rightarrow 1 = k$$

Thus (c) is correct option.

18. Given two vectors

$\vec{a} = 2\hat{i} - 3\hat{j} + 6\hat{k}$, $\vec{b} = -2\hat{i} + 2\hat{j} - \hat{k}$

and $\lambda = \frac{\text{the projection of } \vec{a} \text{ on } \vec{b}}{\text{the projectin of } \vec{b} \text{ on } \vec{a}}$

then the value of λ is

(a) $\frac{3}{7}$ (b) $\frac{7}{3}$

(c) 3 (d) 7

Sol :

$$\lambda = \frac{|\vec{a}|\cos\theta}{|\vec{b}|\cos\theta} = \frac{|\vec{a}|}{|\vec{b}|}$$

$$= \frac{\sqrt{4+9+36}}{\sqrt{4+4+1}} = \frac{7}{3}$$

Thus (b) is correct option.

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$$\left(\frac{1}{2}-y\right)\left(\frac{1}{2}+y\right) \geq 0$$

$$-\frac{1}{2} \leq y \leq \frac{1}{2}$$

$$y = f(x) \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

Thus (c) is correct option.

9. If $A=\{1,2,3\}$ and $B=\{2,3,4\}$, then which of the following relations is a function from A to B?

(a) $\{(1,2),(2,3),(3,4),(2,2)\}$

(b) $\{(1,2),(2,3),(1,3)\}$

(c) $\{(1,3),(2,3),(3,3)\}$

(d) $\{(1,1),(2,3),(3,4)\}$

Sol :

In the given options, only option (c) satisfies the condition of a function.

Thus (c) is correct option.

10. The order of the differential equation $\left[1+\left(\frac{dy}{dx}\right)\right]^{3/2}=\frac{d^2y}{dx^2}$ is

(a) 1 (b) 2

(c) 3 (d) 4

Sol :

Given differential equation can be rewritten as

$$\left[1+\frac{dy}{dx}\right]^3 = \left(\frac{d^2y}{dx^2}\right)^2$$

Here, degree is 2.

Thus (b) is correct option.

11. The function $f(x)=2+4x^2+6x^4+8x^6$ has

(a) only one maxima

(b) only one minima

(c) no maxima and minima

(d) many maxima and minima

Sol :

$$f'(x) = 8x+24x^2+48x^5, \forall\, x \in R$$

and $$f''(x) = 8+72x^2+240x^4 \geq 8, \forall\, x \in R$$

For maxima and minima, put

$$f'(x) = 0$$

$$x(8+24x^2+48x^4) = 0$$

$$x = 0$$

Since, $f''(x) > 0$, therefore, f has a local minima at $x=0$.

Moreover, f has no other critical point and $f'(x)$ exists at all $x \in R$.

Hence, there can be only one minima.

Thus (b) is correct option.

12. $2\sin^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$ holds good for all

(a) $|x| \leq 1$ (b) $1 \geq x \geq 0$

(c) $|x| \leq \frac{1}{\sqrt{2}}$ (d) none of these

Sol :

Since, $2\sin^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$

Range of right hand side is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

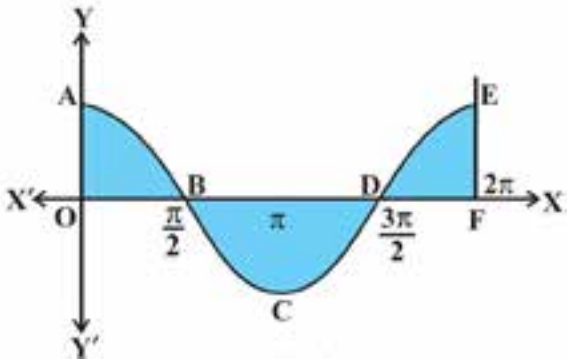
$$-\frac{\pi}{2} \leq 2\sin^{-1}x \leq \frac{\pi}{2}$$

$$-\frac{\pi}{4} \leq \sin^{-1}x \leq \frac{\pi}{4}$$

$$x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$$

Thus (c) is correct option.

13. In Fig., the shaded regions are bounded by which of the following?



(a) The curve passing through points A,B,C,D,E and the X-axis between x=0 and x=2π

(b) Only the straight line y=0 and the X-axis

(c) The vertical lines x=-π and x=π

(d) A semicircle between 0 and 2π

Ans :

(a) The curve passing through points A,B,C,D,E and the X-axis between x=0 and x=2π

The shaded areas are enclosed between the wavy curve (which dips below the X-axis at point C) and the X-axis. The boundaries are clearly marked: vertical line at x=0 (point A), vertical line at x=2π (point E), and the horizontal axis between them. Thus, option (a) is correct.

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$$\vec{a}+\vec{b} = (2\hat{i}+2\hat{j}-\hat{k})+(4\hat{i}-3\hat{j}+2\hat{k})$$
$$= 6\hat{i}+\hat{k}$$

Required unit vector

$$\frac{\vec{a}+\vec{b}}{|\vec{a}+\vec{b}|} = \frac{6\hat{i}+\hat{k}}{\sqrt{6^2+1^2}} = \frac{6\hat{i}+\hat{k}}{\sqrt{36+1}}$$
$$= \frac{6\hat{i}+\hat{k}}{\sqrt{37}}$$
$$= \frac{6}{\sqrt{37}}\hat{i}+\frac{\hat{k}}{\sqrt{37}}$$

22. Let R is the equivalence relation in the set $A=\{0,1,2,3,4,5\}$ given by $R=\{(a,b):2 \text{ divides } (a-b)\}$. Write the equivalence class [0].

Sol :

We have $R = \{(a,b):2 \text{ divides } (a-b)\}$

and $A = \{0,1,2,3,4,5\}$

Clearly, $[0] = \{b \in A:(0,b) \in R\}$
$= \{b \in A:2 \text{ divides } (0-b)\}$
$= \{b \in A:2 \text{ divides } (-b)\}$
$= \{0,2,4\}$

23. Find the vector equation of the line which passes through the point (3,4,5) and is parallel to the vector $2\hat{i}+2\hat{j}-3\hat{k}$.

Sol :

Equation of a line passing through a point with position vector $\vec{a}$ and parallel to a vector $\vec{b}$ is given by

$$\vec{r} = \vec{a}+\lambda\vec{b}$$

Since, line passes through (3,4,5), is

$$\vec{a} = 3\hat{i}+4\hat{j}+5\hat{k}$$

Since, lines is parallel to $2\hat{i}+2\hat{j}-3\hat{k}$, we get

$$\vec{b} = 2\hat{i}+2\hat{j}-3\hat{k}$$

Equation of line is $\vec{r}=\vec{a}+\lambda\vec{b}$, i.e.

$$\vec{r} = (3\hat{i}+4\hat{j}+5\hat{k})+\lambda(2\hat{i}+2\hat{j}-3\hat{k})$$

Which is the required vector equation.

24. Prove that if E and F are independent events, then the events E and F' are also independent.

Sol :

Since event E and F are independent, we have

$$P(E \cap F) = P(E)\cdot P(F)$$

Now, $P(E \cap F)+P(E \cap F') = P(E)$

$$P(E \cap F') = P(E)-P(E \cap F)$$
$$= P(E)-P(E)P(F)$$
$$= P(E)[1-P(F)]$$

or $P(E \cap F') = P(E)P(F')$

Thus E and F' are also independent events.

Hence proved.

25. Find $\int \frac{\sin^6 x}{\cos^8 x}\,dx$.

OR

Evaluate $\int \frac{dx}{\sin^2 x \cos^2 x}$.

Sol :

We have $I = \int \frac{\sin^6 x}{\cos^8 x}\,dx = \int \tan^6 x \sec^2 x\,dx$

Substituting $\tan x = t \Rightarrow \sec^2 x\,dx = dt$ we have

$$I = \int t^6\,dt$$
$$= \frac{t^7}{7}+C$$
$$= \frac{\tan^7 x}{7}+C$$

OR

We have $I = \int \frac{dx}{\sin^2 x \cos^2 x}$

$$= \int \frac{(\sin^2 x+\cos^2 x)}{\sin^2 x \cdot \cos^2 x}\,dx \qquad \sin^2\theta+\cos^2\theta = 1$$
$$= \int (\sec^2 x+\operatorname{cosec}^2 x)\,dx$$
$$= \int \sec^2 x\,dx+\int \operatorname{cosec}^2 x\,dx$$
$$= \tan x-\cot x+C$$

ALTERNATIVE METHOD :

Dividing the numerator and denominator by $\cos^4 x$, we get

$$I = \int \frac{\sec^2 x \cdot \sec^2 x}{\tan^2 x}\,dx$$
$$= \int \frac{(1+\tan^2 x)\cdot \sec^2 x}{\tan^2 x}\,dx$$

Substituting $\tan x = t \Rightarrow \sec^2 x\,dx = dt$ we have

$$I = \int \frac{1+t^2}{t^2}\,dt = \int 1\,dt+\int \frac{1}{t^2}\,dt$$
$$= t-\frac{1}{t}+C$$

Substituting $t=\tan x$ we obtain

$$I = \tan x-\cot x+C$$

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Sol :

We have $\left[x\sin^2\left(\frac{y}{x}\right)-y\right]dx+x\,dy=0$

$$\frac{dy}{dx}=\frac{y}{x}-\sin^2\left(\frac{y}{x}\right) \quad ...(1)$$

Which is a homogeneous differential equation as

$$\frac{dy}{dx}=F\left(\frac{y}{x}\right)$$

Substituting $y=vx \Rightarrow \frac{dy}{dx}=v+x\frac{dv}{dx}$ in eq (1) we have

$$v+x\frac{dv}{dx}=v-\sin^2 v$$

$$x\frac{dv}{dx}=-\sin^2 v$$

$$\operatorname{cosec}^2 v\,dv=-\frac{dx}{x}$$

Integrating both sides, we have

$$\int \operatorname{cosec}^2 v\,dv+\int\frac{dx}{x}=0$$

$$-\cot v+\log|x|=C$$

$$-\cot\left(\frac{y}{x}\right)+\log|x|=C$$

$$y=x\cdot\cot^{-1}(\log x-C) \quad ...(2)$$

which is the required solution.

Also, given that $y=\frac{\pi}{4}$, when $x=1$.

Substituting $x=1$ and $y=\frac{\pi}{4}$ in Eq. (2), we get

$$-\cot\left(\frac{\pi}{4}\right)+\log 1=C \Rightarrow C=-1$$

Substituting this value of C in Eq. (2), we get

$$-\cot\left(\frac{y}{x}\right)+\log|x|=-1$$

$$1+\log|x|-\cot\left(\frac{y}{x}\right)=0$$

Which is the required particular solution of given differential equation.

33. Maximise $Z=5x+3y$ subject to the constraints: $3x+5y\le 15$; $5x+2y\le 10$, $x\ge 0$, $y\ge 0$.

OR

Minimize $Z=3x+5y$ such that $x+3y\ge 3$, $x+y\ge 2$, x, $y\ge 0$.

Sol :

Maximize $Z=5x+3y$

Subject to $3x+5y\le 15$ (i)

$5x+2y\le 10$ (ii)

and $x\ge 0$, $y\ge 0$

(i) Region corresponding to $3x+5y\le 15$:

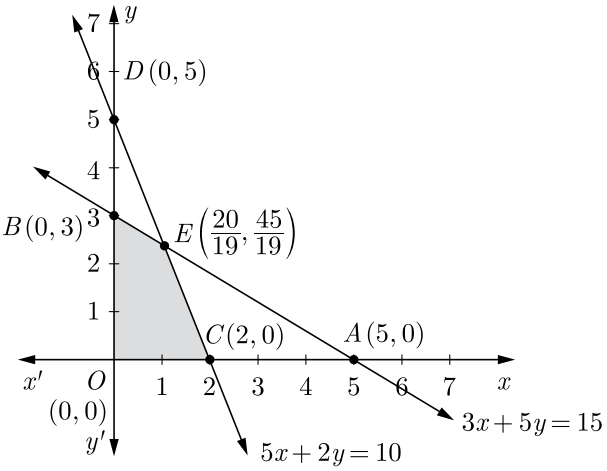
Line $3x+5y=15$			
x	0	5	Point $(0,0)$ is true for $3x+5y\le 15$. So, the region is towards the origin.
y	3	0	

(ii) Region corresponding to $5x+2y\le 10$:

Line $5x+2y=10$			
x	0	2	Point $(0,0)$ is true for $5x+2y\le 10$. So, the region is towards the origin.
y	5	0	

Since $x\ge 0$, $y\ge 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is $OBECO$ which is bounded. Solving equations $3x+5y=15$ and $5x+2y=10$, we get $E\left(\frac{20}{19},\frac{45}{19}\right)$. The corner points of the feasible region are $O(0,0)$, $B(0,3)$, $E\left(\frac{20}{19},\frac{45}{19}\right)$ and $B(2,0)$.



Since, the feasible region is a bounded region, we can check the objective function at all the corner to find the maxima. The values of objective function Z at these points are as follows.

Corner	$Z=5x+3y$
$O(0,0)$	$Z=0+0=0$
$B(0,3)$	$Z=5\times 0+3\times 3=9$
$E\left(\frac{20}{19},\frac{45}{19}\right)$	$Z=5\times\frac{20}{19}+3\times\frac{45}{19}=12.37$
$C(2,0)$	$=5\times 2+3\times 0=10$

Hence the maximum value of Z is 12.37 is at $E\left(\frac{20}{19},\frac{45}{19}\right)$ and the optimal solution is $x=\frac{20}{19}$ and $y=\frac{45}{19}$.

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For any $x \in R$, $3(x-1)^2+3>0$ because, a perfect square number cannot be negative. Thus given function $f(x)$ is an increasing function on R.

28. Write a unit vector perpendicular to both the vectors $\vec{a}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$.

Sol :

First, determine perpendicular vectors of $\vec{a}$ and $\vec{b}$, i.e. $\vec{a}\times\vec{b}$. Further, determine perpendicular unit vector by using formula $\dfrac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$.

We have $\vec{a}=\hat{i}+\hat{j}+\hat{k}$

and $\vec{b}=\hat{i}+\hat{j}$

Now $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&1&1\\1&1&0\end{vmatrix}$

$=\hat{i}(0-1)-\hat{j}(0-1)+\hat{k}(1-1)$

$=-\hat{i}+\hat{j}$

Unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ is given by

$$\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}=\frac{-\hat{i}+\hat{j}}{\sqrt{(-1)^2+(1)^2}}$$

$$=\frac{-\hat{i}+\hat{j}}{\sqrt{2}}=\frac{-\hat{i}}{\sqrt{2}}+\frac{\hat{j}}{\sqrt{2}}$$

29. Write the value of $\cos^{-1}\left(\frac{1}{2}\right)-2\sin^{-1}\left(-\frac{1}{2}\right)$.

Sol :

$$\cos^{-1}\left(\frac{1}{2}\right)-2\sin^{-1}\left(\frac{-1}{2}\right)=\cos^{-1}\left(\frac{1}{2}\right)+2\sin^{-1}\left(\frac{1}{2}\right)$$

$$=\frac{\pi}{3}+2\sin^{-1}\left(\sin\frac{\pi}{6}\right)$$

$$=\frac{\pi}{3}+2\times\frac{\pi}{6}$$

$$=\frac{\pi}{3}+\frac{\pi}{3}=\frac{2\pi}{3}$$

30. Find the maximum value of $\begin{vmatrix}1&1&1\\1&1+\sin\theta&1\\1&1&1+\cos\theta\end{vmatrix}$

Sol :

We have $\Delta=\begin{vmatrix}1&1&1\\1&1+\sin\theta&1\\1&1&1+\cos\theta\end{vmatrix}$

Expanding along first row i.e. R_1, we get

$$\Delta=1\begin{vmatrix}1+\sin\theta&1\\1&1+\cos\theta\end{vmatrix}-1\begin{vmatrix}1&1\\1&1+\cos\theta\end{vmatrix}+1\begin{vmatrix}1&1+\sin\theta\\1&1\end{vmatrix}$$

$$=[(1+\sin\theta)(1+\cos\theta)-1]-[1+\cos\theta-1]+[1-1-\sin\theta]$$

$$=1+\cos\theta+\sin\theta+\sin\theta\cos\theta-1-\cos\theta-\sin\theta$$

$$=\sin\theta\cos\theta$$

$$=\left|\frac{1}{2}\right|(2\sin\theta\cos\theta)=\frac{1}{2}\sin 2\theta$$

Since maximum value of $\sin 2\theta$ is 1.

$$\Delta_{\max}=\frac{1}{2}\times 1=\frac{1}{2}.$$

31. Evaluate $\int\frac{2\cos x}{3\sin^2 x}dx$.

Sol :

We have $I=\int\frac{2\cos x}{3\sin^2 x}dx$

Substituting $\sin x=t\Rightarrow\cos x\,dx=dt$ we have

$$I=\int\frac{2}{3t^2}dt$$

$$=\frac{2}{3}\int\frac{dt}{t^2}$$

$$=\frac{2t^{-1}}{3(-1)}+C$$

Substituting $t=\sin x$ we have

$$I=\frac{-2}{3}(\sin x)^{-1}+C$$

$$=\frac{-2}{3\sin x}+C$$

$$=-\frac{2}{3}\operatorname{cosec}x+C$$

ALTERNATIVE METHOD :

We have $I=\int\frac{2\cos x}{3\sin^2 x}dx$

$$=\frac{2}{3}\int\frac{1}{\sin x}\cdot\frac{\cos x}{\sin x}dx$$

$$=\frac{2}{3}\int\operatorname{cosec}x\cot x\,dx$$

$$=\frac{2}{3}(-\operatorname{cosec}x)+C$$

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Show that the differential equation $\left[x\sin^2\left(\frac{y}{x}\right)-y\right]dx+x\,dy=0$ is homogeneous.

Find the particular solution of this differential equation, given that $y=\frac{\pi}{4}$, when $x=1$.

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$$= \frac{A(2+t)+B(1+t)}{(1+t)(2+t)}$$

$$1 = 2A + tA + B + Bt$$

$$1 = 1(2A+B) + t(A+B)$$

Comparing the coefficients of t and constant term on both sides, we get

$$2A + B = 1 \text{ and } A + B = 0$$

$$A = 1 \text{ and } B = -1$$

Thus $\frac{1}{(1+t)(2+t)} = \frac{1}{1+t} - \frac{1}{2+t}$

and $I = \int \left(\frac{1}{1+t} - \frac{1}{2+t}\right) dt$

$$= \int \frac{1}{(1+t)} dt - \int \frac{1}{(2+t)} dt$$

$$= \log|1+t| - \log|2+t| + C$$

$$= \log\left|\frac{1+t}{2+t}\right| + C$$

Now substituting $t = \sin x$ we have

$$I = \log\left|\frac{1+\sin x}{2+\sin x}\right| + C$$

OR

We have $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

Consider RHS $\int_0^a f(a-x)\,dx$

Substituting $t = a - x$, then $dt = -dx$

Also, when $x = 0$, then $t = a$ and when $x = a$, then $t = 0$

Thus RHS $\int_0^a f(a-x)\,dx = -\int_a^0 f(t)\,dt$

$$= \int_0^a f(t)\,dt$$

$$= \text{LHS Hence proved.}$$

Now, $I = \int_0^{\pi/2} \frac{x}{(\sin x + \cos x)} dx$ ...(1)

$$= \int_0^{\pi/2} \frac{\left(\frac{\pi}{2} - x\right)}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)} dx$$

$$I = \int_0^{\pi/2} \frac{\left(\frac{\pi}{2} - x\right)}{(\cos x + \sin x)} dx$$

$$= \int_0^{\pi/2} \frac{\left(\frac{\pi}{2} - x\right)}{(\sin x + \cos x)} dx \quad \text{...(2)}$$

Adding Eqs. (1) and (2), we get

$$2I = \frac{\pi}{2}\int_0^{\pi/2} \frac{dx}{\sin x + \cos x}$$

$$I = \frac{\pi}{4}\int_0^{\pi/2} \frac{dx}{\sin x + \cos x}$$

$$= \frac{\pi}{4}\int_0^{\pi/2} \frac{dx}{\left[\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}} + \frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right]}$$

$$= \frac{\pi}{4}\int_0^{\pi/2} \frac{\sec^2\frac{x}{2}}{1 - \tan^2\frac{x}{2} + 2\tan\frac{x}{2}} dx$$

Substituting $t = \tan\frac{x}{2} \Rightarrow \frac{1}{2}\sec^2\frac{x}{2}\,dx = dt$

and $x = 0 \Rightarrow t = 0$ and $x = \frac{\pi}{2} \Rightarrow t = 1$

$$I = \frac{\pi}{4}\int_0^1 \frac{2\,dt}{(1 - t^2 + 2t)}$$

$$= \frac{\pi}{4}\int_0^1 \frac{dt}{[(\sqrt{2})^2 - (t-1)^2]} dt$$

$$= \frac{\pi}{2} \cdot \frac{1}{2\sqrt{2}} \log\left|\frac{\sqrt{2} + (t-1)}{\sqrt{2} - (t-1)}\right|_0^1$$

$$= \frac{\pi}{4\sqrt{2}} \log\left|\frac{\sqrt{2}+1}{\sqrt{2}-1}\right|$$

SECTION - E

Case study based questions are compulsory.

36. The U.S. Constitution directs the government to conduct a census of the population every 10 years. Population totals are used to allocate congressional seats, electoral votes, and funding for many government programs. The U.S. Census Bureau also compiles information related to income and poverty, living arrangements for children, and marital status. The following joint probability table lists the probabilities corresponding to marital status and sex of persons 18 years and over.14

Sex	Marital Status			
	(R)	(N)	(W)	(D)
(M)	0.282	0.147	0.013	0.043
(F)	0.284	0.121	0.050	0.060

$R \Rightarrow$ Married
$N \Rightarrow$ Never married
$W \Rightarrow$ Widowed
$D \Rightarrow$ Discovered or separated
$M \Rightarrow$ Male
$F \Rightarrow$ Male

Suppose a U.S. resident 18 years or older is selected at random.

(i) Find the probability that the person is female and widowed.

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OR

Minimize $Z = 3x + 5y$

Subject to $x + 3y \geq 3$ (i)

$x + y \geq 2$ (ii)

and $x \geq 0,\ y \geq 0$

(i) Region corresponding to $x + 3y \geqslant 3$:

Line $x + 3y = 3$			
x	0	3	Point $(0,0)$ is false for $x + 3y \geqslant 3$. So, the region is away from the origin.
y	1	0	

(ii) Region corresponding to $x + y \geqslant 2$:

Line $x + y = 2$			
x	0	2	Point $(0,0)$ is false for $x + y \geqslant 2$. So, the region is away from the origin.
y	2	0	

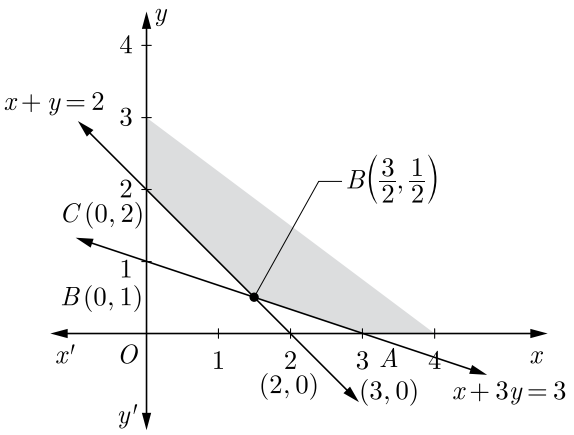
Since $x \geq 0,\ y \geq 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is ABC which is open and unbounded region. Solving equations $x + 3y = 3$ and $x + y = 2$, we get $B(\frac{3}{2}, \frac{1}{2})$. The corner points of the feasible region are $A(3,0)$, $B(\frac{3}{2}, \frac{1}{2})$ and $C(0,2)$.

The values of objective function Z at these points are as follows.

Corner points	$Z = 3x + 5y$
$A(3,0)$	$Z = 3 \times 3 + 0 = 9$
$B(\frac{3}{2}, \frac{1}{2})$	$Z = 3 \times \frac{3}{2} + 5 \times \frac{1}{2} = 7$
$C(0,2)$	$Z = 3 \times 0 + 5 \times 2 = 10$

Hence the minimum value of Z is 7 at $B(\frac{3}{2}, \frac{1}{2})$.



34. Show that the line lines

$\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + \lambda(3\hat{i} - \hat{j})$ and

$\vec{r} = (4\hat{i} - \hat{k}) + \mu(2\hat{i} + 3\hat{k})$

intersect. Also, find their point of intersection.

Sol :

Given lines can be rewritten as

$\vec{r} = (3\lambda + 1)\hat{i} + (1 - \lambda)\hat{j} - \hat{k}$ ...(1)

and $\vec{r} = (4 + 2\mu)\hat{i} + 0\hat{i} + (3\mu - 1)\hat{k}$ ...(2)

Clearly, any point on line (1) is of the form $P(3\lambda + 1, -\lambda, -1)$ and any point on line (2) is of the form $Q(4 + 2\mu, 0, 3\mu - 1)$

If line (1) and (2) intersect, then these points must coincide for some λ and μ.

Consider, $3\lambda + 1 = 4 + 2\mu$

$2\lambda - 2\mu = 3$ ...(3)

$1 - \lambda = 0$ ...(4)

and $3\mu - 1 = -1$ ...(5)

From Eq. (4), we get $\lambda = 1$ and put the value of λ in Eq. (3), we get

$3(1) - 2\mu = 3$

$-2\mu = 3 - 3 \Rightarrow \mu = 0$

Substituting the value of μ in Eq. (5), we get

$3(0) - 1 = -1 \Rightarrow 0 - 1 = -1$

$-1 = -1$, which is true

Hence, both lines intersect each other.

The point of intersection of both lines can be obtained by putting $\lambda = 1$ in coordinates of P. So, the point of intersection is $(3 + 1, 1 - 1, -1)$, i.e. $(4, 0, -1)$.

35. Find the value of $\int \frac{\cos x}{(1 + \sin x)(2 + \sin x)}\, dx$.

OR

Prove that $\int_0^a f(x)\, dx = \int_0^a f(a - x)\, dx$. and hence evaluate $\int_0^{\pi/2} \frac{x}{\sin x + \cos x}\, dx$.

Sol :

We have $I = \int \frac{\cos x}{(1 + \sin x)(2 + \sin x)}\, dx$

Substituting $\sin x = t \Rightarrow \cos x\, dx = dt$ we have

$I = \int \frac{dt}{(1 + t)(2 + t)}$ ...(i)

Now $\frac{1}{(1 + t)(2 + t)} = \frac{A}{1 + t} + \frac{B}{2 + t}$

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SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. Find the value of k, so that the following functions is continuous at $x=2$.

$$f(x)=\begin{cases}\dfrac{x^3-x^2-16x+20}{(x-2)^2}, & x\neq 2\\ k, & x=2\end{cases}$$

OR

Find $\dfrac{dy}{dx}$ at $x=1,\ y=\dfrac{\pi}{4}$ if $\sin^2 y+\cos xy=K$.

Sol :

We have $$f(x)=\begin{cases}\dfrac{x^3-x^2-16x+20}{(x-2)^2}, & x\neq 2\\ k, & x=2\end{cases}$$

If $f(x)$ is continuous at $x=0$ then we have

$$\text{LHL}=\text{RHL}=f(0) \qquad ...(i)$$

Now $$\lim_{x\to 2} f(x)=\lim_{x\to 2}\frac{x^3+x^2-16x+20}{(x-2)^2}$$

$$=\lim_{x\to 2}\frac{(x-2)(x^2+3x-10)}{(x-2)^2}$$

$$=\lim_{x\to 2}\frac{(x-2)(x+5)(x-2)}{(x-2)^2}$$

$$=\lim_{x\to 2}(x+5)=2+5=7$$

Since, $f(x)$ is continuous at $x=2$.

$$\lim_{x\to 2} f(x)=f(2)\Rightarrow 7=k\Rightarrow k=7$$

OR

We have, $$\sin^2 y+\cos xy=K$$

Differentiating w.r.t. x, we get

$$\frac{d}{dx}(\sin^2 y+\cos xy)=\frac{d}{dx}(K)$$

$$\frac{d}{dx}(\sin^2 y)+\frac{d}{dx}(\cos xy)=0$$

$$2\sin y\cos y\frac{dy}{dx}+(-\sin xy)\frac{d}{dx}(xy)=0$$

$$\sin 2y\frac{dy}{dx}-\sin xy\left(x\frac{dy}{dx}+y\cdot 1\right)=0$$

$$\sin 2y\frac{dy}{dx}-x\sin xy\frac{dy}{dx}=y\sin xy$$

$$\frac{dy}{dx}=\frac{y\sin(xy)}{\sin 2y-x\sin(xy)}$$

At $x=1,\ y=\dfrac{\pi}{4}$

$$\left(\frac{dy}{dx}\right)=\frac{\frac{\pi}{4}\sin\left(1\cdot\frac{\pi}{4}\right)}{\sin\left(2\cdot\frac{\pi}{4}\right)-1\sin\left(1\cdot\frac{\pi}{4}\right)}$$

$$=\frac{\frac{\pi}{4}\sin\left(\frac{\pi}{4}\right)}{\sin\left(\frac{\pi}{2}\right)-\sin\left(\frac{\pi}{2}\right)}$$

$$=\frac{\frac{\pi}{4}\left(\frac{1}{\sqrt{2}}\right)}{1-\frac{1}{\sqrt{2}}}$$

$$=\frac{\pi}{4(\sqrt{2}-1)}$$

27. The volume of a sphere is increasing at the rate of 8 cm^3/s. Find the rate of which its surface area is increasing when the radius of the sphere is 12 cm.

OR

Show that the function $f(x)=x^3-3x^2+6x-100$ is increasing on R.

Sol :

Let r be the radius, V be the volume and S be the surface area of sphere.

We have $$\frac{dV}{dt}=8\ \text{cm}^3/\text{s}$$

Now $$V=\frac{4}{3}\pi r^3$$

$$\frac{dV}{dt}=\frac{4}{3}\pi\cdot 3r^2\cdot\frac{dr}{dt}$$

$$8=4\pi\times r^2\frac{dr}{dt}$$

$$\frac{dr}{dt}=\frac{2}{\pi r^2}\ \text{cm/s} \qquad ...(1)$$

Now, $$S=4\pi r^2$$

$$\frac{dS}{dt}=\frac{d}{dt}(4\pi r^2)$$

$$=4\pi\times 2r\cdot\frac{dr}{dt}$$

$$=8\pi r\times\frac{2}{\pi r^2} \qquad \text{Using eq. (1)}$$

$$=\frac{16}{r}$$

$$\left(\frac{dS}{dt}\right)_{r=12}=\frac{16}{12}=\frac{4}{3}cm^2/s$$

OR

We have $$f(x)=x^3-3x^2+6x-100$$

Differentiating both sides w.r.t. x, we get

$$f'(x)=3x^2-6x+6$$

$$=3x^2-6x+3+3$$

$$=3(x^2-2x+1)+3$$

$$=3(x-1)^2+3$$

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(ii) Suppose the person is male. What is the probability that he was never married?

(iii) Suppose the person is married. What is the probability that the person is female?

Sol :

First we prepare following table

Sex	Marital Status				Total
	(R)	(N)	(W)	(D)	
Male (M)	0.282	0.147	0.013	0.043	0.485
Female (F)	0.284	0.121	0.050	0.060	0.515
Total	0.566	0.268	0.063	0.103	1.000

(i) The keyword is and, which means intersection. The probability of female (F) and widowed (W) is found by reading the appropriate cell.

$$P(F \cap W) = 0.050$$

(ii) The keyword is suppose. That suggests conditional probability. The extra information is male.

$$P(N \mid M) = \frac{P(N \cap M)}{P(M)}$$

$$= \frac{0.147}{0.485} = 0.303$$

(iii) This is another conditional probability. This time, the event R is given.

$$P(F \mid R) = \frac{P(F \cap R)}{P(R)}$$

$$= \frac{0.284}{0.566} = 0.502$$

37. A market analysis is a quantitative and qualitative assessment of a market. It looks into the size of the market both in volume and in value, the various customer segments and buying patterns, the competition, and the economic environment in terms of barriers to entry and regulation.



Based on the past marketing trends and his own experience, marketing expert suggested to the concerned the segments of market for their products as follows:

The first segment consisted of lower income class, the second segment that of middle income and the third segment that of high income. The data based on the income of the consumers was readily available. During a particular month in particular year, the agent reported that for three products of the company the following were the sales: There were 200 customers who bought all the three products, 240 customers who bought I and III, 60 customers only products II and II and 80 customers only products only III regardless of the market segmentation groups.

Based on the market segmentation analysis, for product I, the percentage for the income groups are given as (40%, 20% and 40%), for product II (30%, 20% and 50%), for product III (10%, 50% and 40%).

(i) Taking the suitable variable form the system of equation that represent given problem.

(ii) Using matrix method, find out the number of persons in the lower income, middle income and higher income class in the region referred.

Sol :

The number of persons purchasing product I, II and III can be put in the tabular form as:

I $\quad (200 + 240 + 80 + 120) = 640$

II $\quad (200 + 60 + 80 + 220) = 560$

III $\quad (200 + 60 + 240 + 80) = 580$

Let the number of persons in the lower income group, middle income group and higher income group be x, y and z, respectively. Then the given information can be put in the form of simultaneous linear equations as:

$$0.4x + 0.2y + 0.4z = 640$$

$$2x + y + 2z = 3200 \quad \text{(i)}$$

$$0.3x + 0.2y + 0.5z = 560$$

$$3x + 2y + 5z = 5600 \quad \text{(ii)}$$

$$0.1x + 0.5y + 0.4z = 580$$

$$x + 5y + 4z = 5800 \quad \text{(iii)}$$

The above system of equations can be written in matrix form as:

$$\begin{bmatrix} 2 & 1 & 2 \\ 3 & 2 & 5 \\ 1 & 5 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3200 \\ 5600 \\ 5800 \end{bmatrix}$$

or $\quad AX = B$

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Here $|A| = 2(8-25) - 1(12-5) + 2(15-2)$

$= -34 - 7 + 26 = -15$

Since $|A| \neq 0$ thus A^{-1} exists, so that the unique solution of $AX = B$ is $X = A^{-1}B$.

Here, $A^{-1} = \frac{\text{adj.}A}{|A|}$

$$= \frac{1}{-15}\begin{bmatrix} -17 & 6 & 1 \\ -7 & 6 & -4 \\ 13 & -9 & 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = X = A^{-1}B$$

$$= \frac{1}{-15}\begin{bmatrix} -17 & 6 & 1 \\ -7 & 6 & -4 \\ 13 & -9 & 1 \end{bmatrix}\begin{bmatrix} 3200 \\ 5600 \\ 5800 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{-1}{15}\begin{bmatrix} -54400 & +33600 & +5800 \\ -22400 & +33600 & -23200 \\ 41600 & -50400 & 5800 \end{bmatrix}$$

$$= \frac{-1}{15}\begin{bmatrix} -15000 \\ -12000 \\ -3000 \end{bmatrix} = \begin{bmatrix} 1000 \\ 800 \\ 200 \end{bmatrix}$$

$x = 1000,\ y = 800$ and $z = 200$

Hence, the number of persons in the lower income, middle income and higher income class in the region referred are 1000, 800 and 200 respectively.

38. In apparels industries retailers have an interesting conundrum facing them. On one hand, consumers are more drawn to hot promotional deals than ever before. The result of this is that they sell more units (of product) for less money, and this adversely impacts comp store sales.



Arvind Fashions knows that the it can sell 1000 shirts when the price is ₹ 400 per shirt and it can sell 1500 shirts when the price is ₹ 200 a shirt. Determine

(i) The price function

(ii) The revenue function

(iii) The marginal revenue function.

Sol :

(i) The price function

Since the demanded price is assumed to be linear, let it be given by

$$p = a + bx$$

Where p is the price per shirt and x is the quantity demanded at this price.

Since, $x = 1000$ when $p = 400$

and $x = 1500$ when $p = 200$

We have, $400 = a + 1000b$ and $200 = a + 1500b$

Solving these equations simultaneously for a and b, we obtain

$$a = 800 \text{ and } b = -\frac{2}{5}$$

Hence the demanded price is,

$$p = 800 - \frac{2x}{5}$$

(ii) The revenue function

The revenue function is

$$R(x) = px$$

$$= \left(800 - \frac{2x}{5}\right)x$$

$$= 800x - \frac{2x^2}{5}$$

(iii) The marginal revenue function.

The marginal revenue function is,

$$MR = \frac{dR(x)}{dx}$$

$$= 800 - \frac{4x}{5}$$

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Sample Paper 06

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

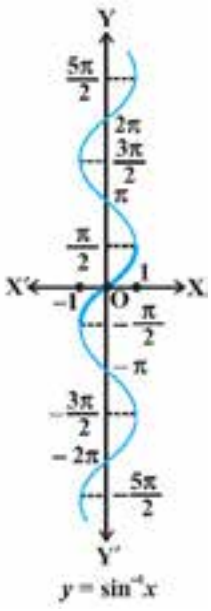
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the domain of the function $y = \sin^{-1} x$ as represented in the graph?



(a) $(-\infty,\infty)$ (b) $[0,\pi]$

(c) $[-1,1]$ (d) $[-\pi,\pi]$

Sol :

On the X-axis, the curve is only drawn from $x = -1$ to $x = 1$. This means the function is defined only in this closed interval. Any X-values outside [-1,1] are not part of the graph, which confirms the domain of $y = \sin^{-1} x$.

Thus (c) is correct option.

2. Two dice are thrown n times in succession. The probability of obtaining a doublet six atleast once is

(a) $\left(\frac{1}{36}\right)^n$ (b) $1-\left(\frac{35}{36}\right)^n$

(c) $\left(\frac{1}{12}\right)^n$ (d) None of these

Sol :

p = Probability of getting double six in two dice

$= \frac{1}{6^2} = \frac{1}{36}$ and $q = \frac{35}{36}$

Required probability

= 1 − (Probability of not getting double six)n

$= 1-\left(\frac{35}{36}\right)^n$

Thus (b) is correct option.

3. The degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 = x\sin\left(\frac{d^2y}{dx^2}\right)$

(a) 1 (b) 3

(c) 2 (d) none of these

Sol :

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Since the given equation is not a polynomial equation in differential coefficients. Hence, its degree is not defined.

Thus (d) is correct option.

4. The solution of the differential equation $x^2\frac{dy}{dx} - xy = 1 + \cos\frac{y}{x}$ is

(a) $\tan\frac{y}{2x} = c - \frac{1}{2x^2}$ (b) $\tan\frac{y}{x} = c + \frac{1}{x}$

(c) $\cos\frac{y}{x} = 1 + \frac{c}{x}$ (d) $x^2 = (c + x^2)\tan\frac{y}{x}$

Sol :

The given differential equation can be rewritten as

$$\frac{dy}{dx} - \frac{y}{x} = \frac{1}{x^2} + \frac{1}{x^2}\cos\frac{y}{x}$$

Put $y = vx$

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

$$v + x\frac{dv}{dx} - v = \frac{1}{x^2} + \frac{1}{x^2}\cos v$$

$$x^3\frac{dv}{dx} = 1 + \cos v$$

$$\frac{dv}{2\cos^2\frac{v}{2}} = \frac{dx}{x^3}$$

$$\int \frac{1}{2}\sec^2\frac{v}{2}dv = \int x^{-3}dx + c$$

$$\tan\frac{v}{2} = \frac{x^{-2}}{2} + c$$

$$\tan\frac{y}{2x} = -\frac{1}{2x^2} + c$$

Thus (a) is correct option.

5. Let G be the centroid of a triangle ABC and O be any other point, then $\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}$ is equal to

(a) $\vec{0}$ (b) $\overrightarrow{OG}$

(c) $3\overrightarrow{OG}$ (d) none of these

Sol :

Since, $\overrightarrow{OG} = \frac{\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}}{3}$

$$\overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC} = 3\overrightarrow{OG}$$

Thus (c) is correct option.

6. The projections of a line segment on x, y, z axes are 12, 4, 3. The length and the direction cosines of the line segment are

(a) $13, < \frac{12}{13}, \frac{4}{13}, \frac{3}{13} >$ (b) $19, < \frac{12}{19}, \frac{4}{19}, \frac{3}{19} >$

(c) $11, < \frac{12}{11}, \frac{14}{11}, \frac{3}{11} >$ (d) none of these

Sol :

The length of line segment $= \sqrt{12^2 + 4^2 + 3^2} = 13$

Hence, DC's of a line are $< \frac{12}{13}, \frac{4}{13}, \frac{3}{13} >$.

Thus (c) is correct option.

7. If A and B are mutually exclusive events with $P(B) \neq 1$, then $P(A/\overline{B})$ is equal to (here, $\overline{B}$ is the complement of the event B)

(a) $\frac{1}{P(B)}$ (b) $\frac{1}{1 - P(B)}$

(c) $\frac{P(A)}{P(B)}$ (d) $\frac{P(A)}{1 - P(B)}$

Sol :

Given, A and B are mutually exclusive events.

$$P(A \cap B) = 0$$

Now, $$P(A/\overline{B}) = \frac{P(A \cap \overline{B})}{P(\overline{B})}$$

$$= \frac{P(A) - P(A \cap B)}{P(\overline{B})}$$

$$= \frac{P(A)}{1 - P(B)}$$

Thus (d) is correct option.

8. Equation to the curve through (2,1) whose slope at the point (x, y) is $\frac{x^2 + y^2}{2xy}$, is

(a) $2(x^2 - y^2) = 3x$ (b) $2(y^2 - x^2) = 6y$

(c) $x(x^2 - y^2) = 6$ (d) none of these

Sol :

We have $$\frac{dy}{dx} = \frac{x^2 + y^2}{2xy}$$

Put $y = vx$,

$$v + x\frac{dv}{dx} = \frac{1 + v^2}{2v}$$

$$v + x\frac{dv}{dx} = \frac{1 + v^2}{2v}$$

$$x\frac{dv}{dx} = \frac{1 + v^2}{2v} - v$$

$$\frac{2v}{1 - v^2}dv = \frac{dv}{dx}$$

$$\log(1 - v^2) + \log x = \log c$$

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19. Assertion: If $A=\begin{bmatrix}\sqrt{3} & 1\\ 2 & 2\end{bmatrix}$ and $b=\begin{bmatrix}x & y & z\\ a & b & c\end{bmatrix}$, then orders of $(A+B)$ is 2×3

Reason: If $[aij]$ and $[bij]$ are two matrics of the same order, then order of $A+B$ is the same as the order of A or B

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

We know that $A+B$ is defined only if A and B have same order.

Here, A is of order 2×2 and B is order of 2×3. As A and B are not of the same order, so $(A+B)$ is not defined, therefore it is not possible to find the sum of A and B.

Hence, Assertion is false and reason is true.

Thus (d) is correct option.

20. Assertion: The relation R in a set $A=\{1,2,3,4\}$ defined by $R=\{(x,y):3x-y=0)\}$ have the domain $=\{1,2,3,4\}$ and range $=\{3,6,9,12\}$

Reason: Domain & range of the relation (R) is respectively the set of all first & second entries of the distinct ordered pair of the relation.

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

$$R=\{(x,y):y=3x, x\in A\}$$

$$R=\{(1,3),(2,6),(3,9),(4,12)\}$$

$$\text{Domain of the relation}=\{1,2,3,4\}$$

and $\quad\text{Range of the relation}=\{3,6,9,12\}$

Thus (a) is correct option.

SECTION - B

Section B consists of 5 questions of 2 marks each.

21. If a line makes angles 90° and 60°, respectively with the positive directions of X and Y-axes, find the angle which it makes with the positive direction of Z-axis.

Sol :

Let a line makes angles α, β and γ with the X-axis, Y-axis and Z-axis, respectively.

Now $\quad \cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$

$$\cos^2 90^\circ+\cos^2 60^\circ+\cos^2\gamma=1$$

$$0+\left(\frac{1}{2}\right)^2+\cos^2\gamma=1$$

$$\cos^2\gamma=1-\frac{1}{4}=\frac{3}{4}$$

$$\cos\gamma=\pm\frac{\sqrt{3}}{2}$$

Thus $\quad \gamma=30^\circ, 150^\circ$

22. Evaluate $\int\frac{2}{1+\cos 2x}dx$.

OR

Write the value of $\int\frac{x+\cos 6x}{3x^2+\sin 6x}dx$.

Sol :

We have $\quad I=\int\frac{2}{1+\cos 2x}dx$

$$=\int\frac{2}{2\cos^2 x}dx \qquad \cos 2\theta=2\cos^2\theta-1$$

$$=\int\sec^2 x dx=\tan x+C$$

OR

We have $\quad I=\int\frac{x+\cos 6x}{3x^2+\sin 6x}dx$

Substituting $3x^2+\sin 6x=t$ we have

$$6x+6\cos 6x=\frac{dt}{dx}$$

$$(x+\cos 6x)\,dx=\frac{dt}{6}$$

Thus $\quad I=\int\frac{dt}{6t}=\frac{1}{6}\log|t|+C \qquad \int\frac{1}{x}dx=\log|x|+C$

Now Substituting $t=3x^2+\sin 6x$ we get

$$I=\frac{1}{6}[\log|(3x^2+\sin 6x)|]+C$$

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Sol :

We have $I = \int \frac{dx}{x(x^7+1)}$

Substituting $x^7 = t \Rightarrow dx = \frac{1}{7x^6}dt$ we obtain

$$I = \frac{1}{7}\int \frac{dt}{t(t+1)} = \frac{1}{7}\int\left(\frac{1}{t} - \frac{1}{t+1}\right)dt$$

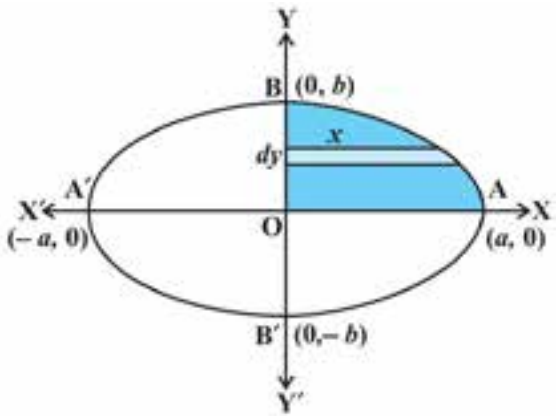
$$= \frac{1}{7}[\log t - \log(t+1)] + C$$

$$= \frac{1}{7}\log\left(\frac{t}{t+1}\right) + C$$

$$= \frac{1}{7}\log\left(\frac{x^7}{x^7+1}\right) + C$$

Thus (b) is correct option.

16. In Fig., the shaded region lies inside which type of curve?



(a) Circle with radius a

(b) Ellipse centered at O with semi-major axis a and semi-minor axis b

(c) Hyperbola with transverse axis on the X-axis

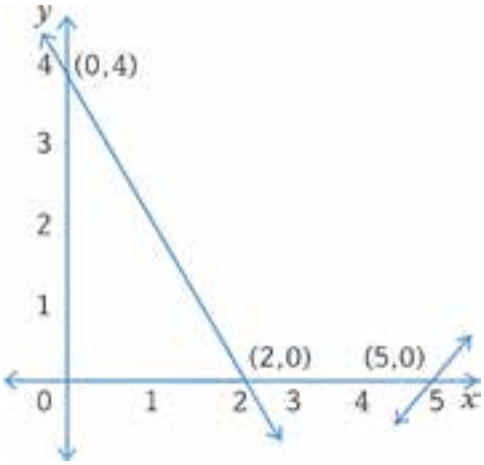
(d) Parabola opening along the Y-axis

Sol :

The figure shows an ellipse centered at O, with vertices A(a,0) and A'(-a,0) on the X-axis, and co-vertices B(0,b) and B'(0,-b) on the Y-axis. These labeled points confirm it is an ellipse with semi-major axis a and semi-minor axis b.

Thus (b) is correct option.

17. Why does the minimum value of $Z = 3x + 2y$ exist for this problem even though the feasible region is unbounded?



(a) Because the feasible region is bounded by all constraints.

(b) Because the inequality 3x+2y<6 has no point in common with the feasible region.

(c) Because the inequality 3x+2y<6 has infinitely many points in common with the feasible region.

(d) Because every LPP always has both a minimum and maximum solution.

Sol :

The feasible region extends infinitely, but still the minimum exists because the line $3x + 2y = 6$ touches the feasible region at (2,0) and all other feasible points give values ≥ 6. Since no feasible point satisfies $3x + 2y < 6$, the minimum value is valid and finite.

Thus (b) is correct option.

18. The length of the largest interval in which the function $3\sin x - 4\sin^3 x$ is increasing, is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$

(c) $\frac{3\pi}{2}$ (d) π

Sol :

Let $f(x) = 3\sin x - 4\sin^3 x$

$= \sin 3x$

Since, $\sin x$ is increasing in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

$$-\frac{\pi}{2} \leq 3x \leq \frac{\pi}{2}$$

$$-\frac{\pi}{6} \leq x \leq \frac{\pi}{6}$$

Hence, the length of interval $= \left|\frac{\pi}{6} - \left(-\frac{\pi}{6}\right)\right| = \frac{\pi}{3}$

Thus (a) is correct option.

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$$\log x\left(1-\frac{y^2}{x^2}\right) = \log c$$

$$x^2 - y^2 = cx$$

It passes through (2, 1), we get,

$$4 - 1 = 2c$$

$$c = \frac{3}{2}$$

$$2(x^2 - y^2) = 3x$$

Thus (a) is correct option.

9. If $x = e^{y+e^{y+e^{y+\cdots}}}$, then $\frac{dy}{dx}$ is equal to

(a) $\frac{1}{x}$ (b) $\frac{1-x}{x}$

(c) $\frac{x}{1+x}$ (d) None of these

Sol :

We have $x = e^{y+e^{y+e^{y+\cdots}}}$

$$x = e^{y+x}$$

Taking log both sides, we get

$$\log x = (x+y)\log e$$

$$\frac{1}{x} = \frac{dy}{dx} + 1 \Rightarrow \frac{dy}{dx} = \frac{1-x}{x}$$

Thus (b) is correct option.

10. The point of discontinuous of $\tan x$ are

(a) $n\pi,\ n \in I$ (b) $2n\pi,\ n \in I$

(c) $(2n+1)\frac{\pi}{2},\ n \in I$ (d) None of these

Sol :

The points of discontinuity of $f(x)$ are those points, where $\tan x$ is infinite.

i.e. $\tan x = \tan\infty \Rightarrow x = (2n+1)\frac{\pi}{2},\ n \in I$

Thus (c) is correct option.

11. The area bounded by the parabola $y^2 = 8x$ and its latus rectum is

(a) 16/3 sq units (b) 32/3 sq units

(c) 8/3 sq units (d) 64/3 sq units

Sol :

Required area, $A = 2\int_0^2 \sqrt{8x}\,dx$

$$= 4\sqrt{2}\left[\frac{x^{3/2}}{3/2}\right]_0^2$$

$$= 4\sqrt{2}\left[\frac{2\sqrt{2}}{3/2}\right] = \frac{32}{3} \text{ sq units}$$

Thus (b) is correct option.

12. If $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|$, then $\vec{a}, \vec{b}$ are

(a) Perpendicular (b) Like parallel

(c) Unlike parallel (d) Coincident

Sol :

Given that, $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|$

$$\cos\theta = 1$$

$$\theta = 0^\circ$$

Vectors are like parallel.

Thus (b) is correct option.

13. Let $f(x) = x^3 + \frac{3}{2}x^2 + 3x + 3$, then $f(x)$ is

(a) Am even function

(b) An odd function

(c) An increasing function

(d) A decreasing function

Sol :

$$f'(x) = 3x^2 + 3x + 3 = 3(x^2 + x + 1)$$

$$= 3\left\{\left(x+\frac{1}{2}\right)^2 + \frac{3}{4}\right\} \geq \frac{9}{4}$$

Hence, $f(x)$ is an increasing, function.

Thus (c) is correct option.

14. If $P = \begin{bmatrix}1 & 2 & 1\\ 1 & 3 & 1\end{bmatrix}$ and $Q = PP^T$, then the value of Q is

(a) 2 (b) -2

(c) 1 (d) 0

Sol :

We have $P = \begin{bmatrix}1 & 2 & 1\\ 1 & 3 & 1\end{bmatrix}$ and $Q = PP^T$

Clearly, $Q = \begin{bmatrix}1 & 2 & 1\\ 1 & 3 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\ 2 & 3\\ 1 & 1\end{bmatrix} = \begin{bmatrix}6 & 8\\ 8 & 11\end{bmatrix}$

$$|Q| = \begin{vmatrix}6 & 8\\ 8 & 11\end{vmatrix} = 66 - 64 = 2$$

Thus (a) is correct option.

15. $\int \frac{dx}{x(x^7+1)}$ is equal to

(a) $\log\left(\frac{x^7}{x^7+1}\right) + C$ (b) $\frac{1}{7}\log\left(\frac{x^7}{x^7+1}\right) + C$

(c) $\log\left(\frac{x^7+1}{x^7}\right) + C$ (d) $\frac{1}{7}\log\left(\frac{x^7+1}{x^7}\right) + C$

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23. Prove that if E and F are independent events, then the events E and F' are also independent.

Sol :

Since event E and F are independent, we have

$$P(E \cap F) = P(E) \cdot P(F)$$

Now, $P(E \cap F') + P(E \cap F) = P(E)$

$$P(E \cap F') = P(E) - P(E \cap F)$$
$$= P(E) - P(E)P(F)$$
$$= P(E)[1 - P(F)]$$

or $$P(E \cap F') = P(E)P(F')$$

Thus E and F' are also independent events.

Hence proved.

24. Find λ and μ, if $(\hat{i} + 3\hat{j} + 9\hat{k}) \times (3\hat{i} - \lambda\hat{j} + \mu\hat{k}) = \vec{0}$.

OR

If $\vec{a}$, $\vec{b}$, $\vec{c}$ are unit vectors such that $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then write the value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$.

Sol :

We have $(\hat{i} + 3\hat{j} + 9\hat{k}) \times (3\hat{i} - \lambda\hat{j} + \mu\hat{k}) = \vec{0}$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & 9 \\ 3 & -\lambda & \mu \end{vmatrix} = \vec{0}$$

$$\hat{i}(3\mu + 9\lambda) - \hat{j}(\mu - 27) + \hat{k}(-\lambda - 9) = 0\hat{i} + 0\hat{j} + 0\hat{k}$$

Comparing the coefficients of $\hat{i}$, $\hat{j}$ and $\hat{k}$, we have

$$3\mu + 9\lambda = 0,\ -\mu + 27 = 0 \text{ and } -\lambda - 9 = 0$$

We obtain $\mu = 27$ and $\lambda = -9$

These values of μ and λ also satisfy the equation $3\mu + 9\lambda = 0$.

Hence, $\mu = 27$ and $\lambda = -9$.

OR

We have $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$

and $\vec{a} + \vec{b} + \vec{c} = \vec{0}$

Consider $\vec{a} + \vec{b} + \vec{c} = \vec{0}$

$$(\vec{a} + \vec{b} + \vec{c})^2 = (\vec{0})^2$$

$$(\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b} + \vec{c}) = \vec{0} \cdot \vec{0}$$

$$\vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} + \vec{c} \cdot \vec{b} + \vec{c} \cdot \vec{c} = 0$$

$$|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$1 + 1 + 1 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = \frac{-3}{2}$$

25. If $R = \{(x, y): x + 2y = 8\}$ is a relation on N, then write the range of R.

Sol :

Given, the relation R is defined on the set of natural numbers, i.e. N as

$$R = \{(x, y): x + 2y = 8\}$$

To find the range of R, $x + 2y = 8$ can be rewritten as

$$y = \frac{8 - x}{2}$$

Substituting $x = 2$ we get

$$y = \frac{8 - 2}{2} = 3$$

Substituting $x = 4$ we get

$$y = \frac{8 - 4}{2} = 2$$

Substituting $x = 6$ we get

$$y = \frac{8 - 6}{2} = 1$$

As, $x, y \in N$, therefore $R = \{(2, 3), (4, 2), (6, 1)\}$.

Hence, the range of relation R is $\{3, 2, 1\}$.

Note- For $x = 1, 3, 5, 7, 9, \ldots$, we do not get y as natural number.

SECTION - C

Section C consists of 6 questions of 3 marks each.

26. If $y = e^{\tan^{-1}x}$, prove that

$$(1 + x^2)\frac{d^2y}{dx^2} + (2x - 1)\frac{dy}{dx} = 0.$$

Sol :

We have, $y = e^{\tan^{-1}x}$

Differentiating w.r.t. x, we get

$$\frac{dy}{dx} = e^{\tan^{-1}x} \frac{d}{dx}(\tan^{-1}x)$$

$$\frac{dy}{dx} = e^{\tan^{-1}x} \times \frac{1}{(1 + x^2)}$$

$$(1 + x^2)\frac{dy}{dx} = e^{\tan^{-1}x} \quad \ldots(i)$$

Again differentiating both sides w.r.t. x, we get

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$$4x + 2\pi r = 28$$

$$2x + \pi r = 14$$

$$x = \frac{14 - \pi r}{2} \quad \text{...(i)}$$

Let A denotes the combined area of circle and square.

Then, $A = \pi r^2 + x^2$

$$A = \pi r^2 + \left(\frac{14 - \pi r}{2}\right)^2 \quad x = \frac{14 - \pi r}{2}$$

Differentiating both sides w.r.t. r, we get

$$\frac{dA}{dr} = 2\pi r + 2\left(\frac{14 - \pi r}{2}\right)\left(-\frac{\pi}{2}\right)$$

$$= 2\pi r - \left(\frac{14\pi - \pi^2 r}{2}\right)$$

For maxima or minima, $\frac{dA}{dr} = 0$

$$2\pi r - \left(\frac{14\pi - \pi^2 r}{2}\right) = 0$$

$$2\pi r = \frac{14\pi - \pi^2 r}{2}$$

$$r = \frac{14}{\pi + 4}$$

Also, $\frac{d^2A}{dr^2} = \frac{d}{dr}\left(\frac{dA}{dr}\right)$

$$= \frac{d}{dr}\left[2\pi r - \left(\frac{14\pi - \pi^2 r}{2}\right)\right]$$

$$\frac{d^2A}{dr^2} = 2\pi + \frac{\pi^2}{2} > 0$$

Since $\frac{d^2A}{dr^2} > 0$, A is minimum.

Now, substituting $r = \frac{14}{\pi + 4}$ in Eq. (i), we get

$$x = \frac{14 - \pi\frac{14}{\pi+4}}{2}$$

$$= \frac{14\pi + 56 - 14\pi}{2(\pi + 4)} = \frac{28}{\pi + 4}$$

$$x = \frac{28}{\pi + 4} \text{ and } r = \frac{14}{\pi + 4}$$

Now, length of circular part

$$2\pi r = 2\pi \times \frac{14}{\pi + 4} = \frac{28\pi}{\pi + 4}$$

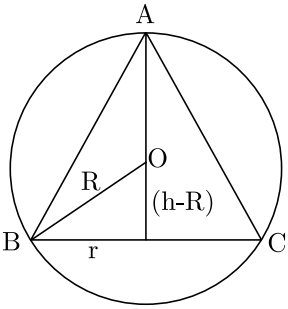
and length of square part

$$4x = 4 \times \frac{28}{\pi + 4} = \frac{112}{\pi + 4}$$

Which are the required length of two pieces.

OR

Let the centre of the sphere be O and radius be R.
Let the height and radius of the variable cone inside the sphere be h and r respectively.
As per question figure is shown below.



Now in figure $OA = OB = R$

$$AD = h, \; BD = r$$

$$OD = AD - OA = h - R$$

In ΔOBD, by Pythagoras theorem, we have

$$OB^2 = OD^2 + BD^2$$

$$R^2 = (h - R)^2 + r^2$$

$$R^2 = h^2 - 2hR + R^2 + r^2$$

$$r^2 = 2hR - h^2 \quad \text{...(i)}$$

Volume of V of the cone,

$$V = \frac{1}{3}\pi r^2 h$$

$$= \frac{1}{3}\pi(2hR - h^2)h \quad \text{[from Eq. (i)]}$$

$$= \frac{1}{3}\pi(2Rh^2 - h^3) \quad \text{...(ii)}$$

Differentiating both sides of Eq. (ii) w.r.t. h, we get

$$\frac{dV}{dh} = \frac{1}{3}\pi(4Rh - 3h^2) \quad \text{...(iii)}$$

For maxima or minima, putting $\frac{dV}{dh} = 0$ we have

$$\frac{1}{3}\pi(4Rh - 3h^2) = 0$$

$$4Rh = 3h^2$$

$$4R = 3h$$

$$h = \frac{4R}{3} \quad [\because h \neq 0]$$

Again, on differentiating Eq. (iii) w.r.t. h, we get

$$\frac{d^2V}{dh^2} = \frac{1}{3}\pi(4R - 6h)$$

At $h = \frac{4R}{3}$, $\left(\frac{d^2V}{dh^2}\right)_{h=\frac{4R}{3}} = \frac{1}{3}\pi\left(4R - 6 \times \frac{4R}{3}\right)$

$$= \frac{\pi}{3}(4R - 8R)$$

$$= -\frac{4\pi R}{3} < 0$$

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31. Find the intervals in which the function $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$ is

(i) strictly increasing

(ii) strictly decreasing.

OR

Find the intervals in which the function given by $f(x) = \frac{3}{10}x^4 - \frac{4}{5}x^3 - 3x^2 + \frac{36x}{5} + 11$ is

(i) strictly increasing

(ii) strictly decreasing.

Sol :

We have $f(x) = 3x^4 - 4x^3 - 12x^2 + 5$

Differentiating both sides w.r.t. x, we get

$$f'(x) = 12x^3 - 12x^2 - 24x$$

For strictly increasing or strictly decreasing, put $f'(x) = 0$, we get

$$12x^3 - 12x^2 - 24x = 0$$

$$12x(x^2 - x - 2) = 0$$

$$12x[x^2 - 2x + x - 2] = 0$$

$$12x(x+1)(x-2) = 0$$

$$x = 0, -1 \text{ or } 2$$

Now, we find intervals in which $f(x)$ is strictly increasing or strictly decreasing.

Interval	$f'(x) = 12(x+1)x(x-2)$	**Sign of** $f'(x)$
$x < -1$	$(-)(-)(-)$	$-$ ve
$-1 < x < 0$	$(+)(-)(-)$	$+$ ve
$0 < x < 2$	$(+)(+)(-)$	$-$ ve
$x > 2$	$(+)(+)(+)$	$+$ ve

A function $f(x)$ is said to be strictly increasing, if $f'(x) > 0$ and it is said to be strictly decreasing, if $f'(x) < 0$. So, the given function $f(x)$ is

(i) strictly increasing on the intervals $(-1, 0)$ and $(2, \infty)$.

(ii) strictly decreasing on the intervals $(-\infty, -1)$ and $(0, 2)$.

OR

We have $f(x) = \frac{3}{10}x^4 - \frac{4}{5}x^3 - 3x^2 + \frac{36x}{5} + 11$

Differentiating both sides w.r.t. x, we get

$$f'(x) = \frac{6}{5}x^3 - \frac{12}{5}x^2 - 6x + \frac{36}{5}$$

For strictly increasing or strictly decreasing, put $f'(x) = 0$, we get

$$\frac{6}{5}x^3 - \frac{12}{5}x^2 - 6x + \frac{36}{5} = 0$$

$$x^3 - 2x^2 - 5x + 6 = 0$$

$$(x-1)(x+2)(x-3) = 0$$

$$x = -2,\ 1, \text{ or } 3$$

Now, we find intervals in which $f(x)$ is strictly increasing or strictly decreasing.

Interval	$f'(x)$ $(x+2)(x-1)(x-3)$	**Sign of** $f'(x)$
$x < -2$	$(-)(-)(-)$	$-$ ve
$-2 < x < 1$	$(+)(-)(-)$	$+$ ve
$1 < x < 3$	$(+)(+)(-)$	$-$ ve
$x > 3$	$(+)(+)(+)$	$+$ ve

A function $f(x)$ is said to be strictly increasing, if $f'(x) > 0$ and it is said to be strictly decreasing, if $f'(x) < 0$. So, the given function $f(x)$ is

(i) strictly increasing on the intervals $(-2, 1)$ and $(3, \infty)$.

(ii) strictly decreasing on the intervals $(-\infty, -2)$ and $(1, 3)$.

SECTION - D

Section D consists of 4 questions of 5 marks each.

32. A wire of length 28 m is to be cut into two pieces. One of the two pieces is to be made into a square and the other into a circle. What should be the lengths of two pieces, so that the combined area of circle and square is minimum?

OR

Show that the volume of the largest cone that can be inscribed in a sphere of radius R is $\frac{8}{27}$ of the volume of the sphere.

Sol :

Let x m be the side of the square and r be the radius of circular part. Then,

Length of square part = perimeter of square

$$= 4x$$

and length of circular part = perimeter of circle

$$= 2\pi r$$

Since length of wire is 28 m,

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$$(1+x^2)\frac{d^2y}{dx^2}+2x\frac{dy}{dx} = e^{\tan^{-1}x}\times\frac{1}{(1+x^2)}$$

$$(1+x^2)\frac{d^2y}{dx^2}+(2x)\frac{dy}{dx} = \frac{dy}{dx} \quad \text{[from Eq. (i)]}$$

$$(1+x^2)\frac{d^2y}{dx^2}+(2x-1)\frac{dy}{dx} = 0 \quad \text{Hence proved.}$$

27. If $A=\begin{bmatrix}1 & 2\\ 3 & -1\end{bmatrix}$ and $B=\begin{bmatrix}1 & 3\\ -1 & 1\end{bmatrix}$, write the value of $|AB|$.

Sol :

$$|A| = \begin{vmatrix}1 & 2\\ 3 & -1\end{vmatrix} = -1-6 = -7$$

and $$|B| = \begin{vmatrix}1 & 3\\ -1 & 1\end{vmatrix} = 1+3 = 4$$

$$|AB| = |A|\cdot|B| = (-7)(4) = -28$$

28. If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a}+\vec{b}$ is also a unit vector, then find the angle between $\vec{a}$ and $\vec{b}$.

Sol :

Since $\vec{a}$, $\vec{b}$ and $\vec{a}+\vec{b}$ are unit vector, we have

$$|\vec{a}| = 1,\ |\vec{b}| = 1 \text{ and } |\vec{a}+\vec{b}| = 1$$

Now, $$|\vec{a}+\vec{b}|^2 = (\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b})$$

$$1 = \vec{a}\cdot\vec{a}+\vec{b}\cdot\vec{a}+\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{b}$$

$$1 = |\vec{a}|^2+2\vec{a}\cdot\vec{b}+|\vec{b}|^2$$

$$1 = 1+2\vec{a}\cdot\vec{b}+1$$

$$2\vec{a}\cdot\vec{b} = -1$$

$$|\vec{a}||\vec{b}|\cos\theta = -\frac{1}{2} \qquad \vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$$

$$\cos\theta = -\frac{1}{2} \qquad |\vec{a}| = |\vec{b}| = 1$$

$$\cos\theta = \cos\frac{2\pi}{3} \Rightarrow \theta = \frac{2\pi}{3}$$

Hence, the angle between $\vec{a}$ and $\vec{b}$ is $\frac{2\pi}{3}$.

29. Write the value of $\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$.

Sol :

Here angle $\frac{3\pi}{4}$ doesn't lie in the principal value branch $[-\frac{\pi}{2},\frac{\pi}{2}]$. So we reduce the angle such that, it lies in principal value branch.

We write as $\frac{3\pi}{4} = \pi-\frac{\pi}{4}$

Now, $\tan^{-1}\left(\tan\frac{3\pi}{4}\right) = \tan^{-1}\left[\tan\left(\pi-\frac{\pi}{4}\right)\right]$

$$= \tan^{-1}\left(-\tan\frac{\pi}{4}\right)$$

$$= \tan^{-1}\left[\tan\left(-\frac{\pi}{4}\right)\right]$$

$$= -\frac{\pi}{4}$$

Thus required value is $-\frac{\pi}{4}$.

30. Evaluate $\int\frac{\cos 2x-\cos 2\alpha}{\cos x-\cos\alpha}dx$.

OR

Evaluate $\int\frac{dx}{x(x^5+3)}$.

Sol :

We have $$I = \int\frac{\cos 2x-\cos 2\alpha}{\cos x-\cos\alpha}dx$$

Using the fact $\cos 2\theta = 2\cos^2\theta-1$ we have

$$I = \int\frac{(2\cos^2 x-1)-(2\cos^2\alpha-1)}{\cos x-\cos\alpha}dx$$

$$= \int\frac{2(\cos^2 x-\cos^2\alpha)}{(\cos x-\cos\alpha)}dx$$

$$= \int\frac{2(\cos x-\cos\alpha)(\cos x+\cos\alpha)}{(\cos x-\cos\alpha)}dx$$

$$= \int 2(\cos x+\cos\alpha)\,dx$$

$$= 2\left[\int\cos x\,dx+\cos\alpha\int dx\right]$$

or $$I = 2(\sin x+\cos\alpha)+C$$

OR

We have $$I = \int\frac{dx}{x(x^5+3)}$$

$$= \int\frac{x^4}{x^5(x^5+3)}dx$$

Substituting $t = x^5 \Rightarrow dt = 5x^4\,dx$ we have

$$I = \int\frac{dt}{5t(t+3)}$$

$$= \frac{1}{5}\int\frac{1}{3}\left[\frac{1}{t}-\frac{1}{t+3}\right]dt$$

$$= \frac{1}{15}[\log|t|-\log|t+3|]+C$$

$$= \frac{1}{15}\log\left|\frac{t}{t+3}\right|+C$$

$$= \frac{1}{15}\log\left|\frac{x^5}{x^5+3}\right|+C \qquad t = x^5$$

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Thus V is maximum h $h=\frac{4R}{3}$.

Substituting the value of h in Eq. (ii), we get

$$\begin{aligned}V &= \frac{1}{3}\pi\left[2R\left(\frac{4R}{3}\right)^2-\left(\frac{4R}{3}\right)^3\right]\\ &= \frac{\pi}{3}\left[\frac{32}{9}R^3-\frac{64}{27}R^3\right] = \frac{\pi}{3}R^3\left[\frac{32}{9}-\frac{64}{27}\right]\\ &= \frac{\pi}{3}R^3\left[\frac{96-64}{27}\right] = \frac{\pi}{3}R^3\left(\frac{32}{27}\right)\\ &= \frac{8}{27}\times\left(\frac{4}{3}\pi R^3\right)\\ &= \frac{8}{27}\times \text{ (Volume of sphere)}\end{aligned}$$

Hence, maximum volume of the cone is $\frac{8}{27}$ of the volume of the sphere.

33. Find $\int\frac{4}{(x-2)(x^2+4)}dx$.

OR

Find $\int\frac{2x}{(x^2+1)(x^2+2)^2}dx$.

Sol :

We have $I = \int\frac{4}{(x-2)(x^2+4)}dx$

Now $\frac{4}{(x-2)(x^2+4)} = \frac{A}{(x-2)}+\frac{Bx+C}{x^2+4}$

$$4 = A(x^2+4)+(Bx+C)(x-2)$$

$$4 = x^2(A+B)+x(-2B+C)+4A-2C$$

Equating the coefficients of x^2, x and constant in both sides, we get

$$A+B = 0 \quad ...(1)$$

$$-2B+C = 0 \quad ...(2)$$

and $$4A-2C = 4 \quad ...(3)$$

Solving eqs. (1), (2) and (3), we get

$$A = \frac{1}{2},\ B=-\frac{1}{2} \text{ and } C=1$$

Thus $\frac{4}{(x-2)(x^2+4)} = \frac{\frac{1}{2}}{(x-2)}+\frac{-\frac{1}{2}x+1}{x^2+4}$ Now

$$\begin{aligned}I &= \int\frac{4}{(x-2)(x^2+4)}dx\\ &= \int\frac{\frac{1}{2}}{(x-2)}dx+\int\frac{-\frac{1}{2}x+1}{x^2+4}dx\\ &= \frac{1}{2}\int\frac{dx}{(x-2)}-\int\frac{x-2}{2(x^2+4)}dx\\ &= \frac{1}{2}\log|x-2|-\frac{1}{2}\log|x^2+4|+\frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right)+C\end{aligned}$$

OR

We have $I = \int\frac{2x}{(x^2+1)(x^2+2)}dx$

Substituting $x^2=t\Rightarrow 2x\,dx=dt$ we have

$$I = \int\frac{dt}{(t+1)(t+2)^2}$$

Now $\frac{1}{(t+1)(t+2)^2} = \frac{A}{t+1}+\frac{B}{t+2}+\frac{C}{(t+2)^2}$

$$\frac{1}{(t+1)(t+2)^2} = \frac{A(t+2)^2+B(t+1)(t+2)+C(t+1)}{(t+1)(t+2)^2}$$

$$1 = A(t+2)^2+B(t+1)(t+2)+C(t+1) \quad (1)$$

Substituting $t=-1$ in eq. (1), we get

$$1 = A(-1+2)^2+0+0\Rightarrow 1=A$$

Substituting $t=-2$ in eq. (i), we get

$$1 = 0+0+C(-2+1)\Rightarrow 1=-C\Rightarrow C=-1$$

Comparing the coefficient of x^2 of both side in eq (1) we have

$$0 = A+B$$

Thus $$A = -B=-1$$

Now $\frac{1}{(t+1)(t+2)^2} = \frac{1}{t+1}-\frac{1}{t+2}-\frac{1}{(t+2)^2}$

$$\begin{aligned}I &= \int\frac{1}{t+1}dt-\int\frac{1}{t+2}dt-\int\frac{1}{(t+2)^2}dt\\ &= \log|t+1|-\log|t+2|-\frac{(t+2)^{-1}}{-1}+C\\ &= \log|t+1|-\log|t+2|+\frac{1}{(t+2)}+C\\ &= \log|x^2+1|-\log|x^2+2|+\frac{1}{(x^2+2)}+C\end{aligned}$$

34. An urn contains 4 white and 6 red balls. Four balls are drawn at random (without replacement) from the urn. Find the probability distribution of the number of white balls?

OR

If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three mutually perpendicular vectors of the same magnitude, then prove that $\vec{a}+\vec{b}+\vec{c}$ is equally inclined with the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$.

Sol :

Let random variable X denotes the number of white balls in a draw of 4 balls. Then, X can take values of 0, 1, 2, 3 and 4.

Probability of getting no white ball,

$$P(X=0) = \frac{{}^6C_4}{{}^{10}C_4} = \frac{{}^6C_2}{{}^{10}C_4}$$

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economy class passengers and Wilbur carry 80 first-class and 120 economy-class passengers. Also, no more than 52 flights of Orville and 30 flights of Wilbur are allowed.

Now, according to question

$$20x+80y \geq 1600$$

or $$x+4y \geq 80$$

$$80x+120y \geq 4800$$

or $$2x+3y \geq 120$$

$$x \leq 52$$

$$y \leq 30$$

Now, the given LPP reduced to

$$\text{Minimize } z = 12000x+18000y$$

Subject to the constraints

$$x+4y \geq 80$$

$$2x+3y \geq 120$$

$$x \leq 52$$

$$y \leq 30$$

$$x \geq 0$$

$$y \geq 0$$

(i) Region corresponding to $x+4y \geq 80$:

Line $x+4y \geq 80$			
x	0	80	Point (0, 0) is not true for $x+4y \geq 80$. So, the region is opposite to the origin.
y	20	0	

(ii) Region corresponding to $2x+3y \geq 120$:

Line $2x+3y \geq 120$			
x	0	60	Point (0, 0) is not true for $2x+3y \geq 120$. So, the region is opposite to the origin.
y	40	0	

Region $x \leq 52$, $y \leq 30$ is the region corresponding to the rectangle made by both line.

Since $x \geq 0$, $y \geq 0$, the feasible region lies in the fist quadrant.

Now, we draw all line on the graph and find the common area. Clearly feasible region is $ABCD$. Solving $x+4y=80$ and $2x+3y=120$ we get $A(48,8)$. Solving $x=52$ and $x+4y=80$ we get $B(52,7)$. Solving $y=30$ and $2x+3y=120$ we get $D(15,30)$. The coordinates of the corner points of the region are $A(48,8)$, $B(52,7)$, $C(52,30)$, $D(15,30)$.

Since the feasible region is a bounded region, we can check the cost function at all the vertices to find the minima. The values of z at corner points are given below.

Corner points	$z = 12000x+18000y$
$A(48,8)$	$z = 12000 \times 48 + 18000 \times 8 = 720000$
$B(52,7)$	$z = 12000 \times 52 + 18000 \times 7 = 750000$
$C(52,30)$	$z = 12000 \times 52 + 18000 \times 30 = 1164000$
$D(15,30)$	$z = 12000 \times 15 + 18000 \times 30 = 720000$

The minimum value of z is 720000 at $A(48,8)$ and $D(15,30)$. Hence minimum value of z lie on the points which are on the line joining (48, 8) and (15, 30).

37. Chemical reaction, a process in which one or more substances, the reactants, are converted to one or more different substances, the products. Substances are either chemical elements or compounds. A chemical reaction rearranges the constituent atoms of the reactants to create different substances as products.



In a certain chemical reaction, a substance is converted into another substance at a rate proportional to the square of the amount of the first substance present at any time t. Initially $(t=0)$ 50 g of the first substance was present; 1 hr later, only 10 g of it remained.

(i) Find an expression that gives the amount of the first substance present at any time t.

(ii) What is the amount present after 2 hr?

Sol :

Since rate is proportional to the square of the amount of the first substance present at any time t, we are led to the differential equation

$$\frac{dQ}{dt} = kQ^2$$

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and $a_2 = -1,\ b_2 = 2,\ c_2 = 4$

Angle between two lines is given by

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \cdot \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

$$= \frac{(2)(-1) + (7)(2) + (-3)(4)}{\left[\sqrt{(2)^2 + (7)^2 + (-3)^2} \cdot \sqrt{(-1)^2 + (2)^2 + (4)^2}\right]}$$

$$\cos\theta = \frac{-2 + 14 - 12}{\sqrt{62} \times \sqrt{21}} = \frac{0}{\sqrt{62} \times \sqrt{21}} = 0$$

$$\cos\theta = \cos\frac{\pi}{2}$$

$$\theta = \frac{\pi}{2}$$

Hence, the angle between them is $\frac{\pi}{2}$. Therefore, the given pair of lines are perpendicular to each other.

OR

We have $\vec{r} = (1-t)\,\hat{i} + (t-2)\,\hat{j} + (3-2t)\,\hat{k}$ ...(1)

and $\vec{r} = (s+1)\,\hat{i} + (2s-1)\,\hat{i} - (2s+1)\,\hat{k}$ ...(2)

Firstly, we convert both equations in the vector form as

$$\vec{r} = \vec{a} + \lambda\vec{b} \quad ...(3)$$

So, Eq. (1) can be written as

$$\vec{r} = (\hat{i} - 2\hat{j} + 3\hat{k}) + t(-\hat{i} + \hat{j} - 2\hat{k}) \quad ...(4)$$

and Eq. (2) can be written as

$$\vec{r} = (\hat{i} - 2\hat{j} - \hat{k}) + s(\hat{i} + 2\hat{j} - 2\hat{k}) \quad ...(5)$$

From Eqs. (3), (4) and (5), we get

$$\vec{a}_1 = \hat{i} - 2\hat{j} + 3\hat{k},\ \vec{b}_1 = -\hat{i} + \hat{j} - 2\hat{k}$$

$$\vec{a}_2 = \hat{i} - \hat{j} - \hat{k},\ \vec{b}_2 = \hat{i} + 2\hat{j} - 2\hat{k}$$

$$\text{Now, } \vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & -2 \\ 1 & 2 & -2 \end{vmatrix}$$

$$= \hat{i}(-2+4) - \hat{j}(2+2) + \hat{k}(-2-1)$$

$$\vec{b}_1 \times \vec{b}_2 = 2\hat{i} - 4\hat{j} - 3\hat{k}$$

$$\left|\vec{b}_1 \times \vec{b}_2\right| = \sqrt{(2)^2 + (-4)^2 + (-3)^2}$$

$$= \sqrt{4 + 16 + 9} = \sqrt{29}$$

$$\text{Also, } \vec{a}_2 - \vec{a}_1 = (\hat{i} - \hat{j} - \hat{k}) - (\hat{i} - 2\hat{j} + 3\hat{k})$$

$$= \hat{j} - 4\hat{k}$$

Shortest distance between the lines,

$$d = \left|\frac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{\left|\vec{b}_1 \times \vec{b}_2\right|}\right|$$

$$= \left|\frac{(2\hat{i} - 4\hat{j} - 3\hat{k}) \cdot (\hat{j} - 4\hat{k})}{\sqrt{29}}\right|$$

$$= \left|\frac{0 - 4 + 12}{\sqrt{29}}\right| = \frac{8}{\sqrt{29}}$$

Thus $d = \frac{8\sqrt{29}}{29}$ units

SECTION - E

Case study based questions are compulsory.

36. Global Air Lines has contracted with a tour group to transport a minimum of 1,600 first-class passengers and 4,800 economy-class passengers from New York to London during a 6-month time period. Global Air has two types of airplanes, the Orville 606 and the Wilbur W-1112. The Orville 606 carries 20 first-class passengers and 80 economy-class passengers and costs \$12,000 to operate. The Wilbur W-1112 carries 80 first-class passengers and 120 economy-class passengers and costs \$18,000 to operate.



During the time period involved, Global Air can schedule no more than 52 flights on Orville 606s and no more than 30 flights on Wilbur W-1112s.

(i) How should Global Air Lines schedule its flights to minimize its costs?

(ii) What operating costs would this schedule entail?

Sol :

Let x be number of Orville flights and y be number of Wilbur flights. Clearly $x \geq 0,\ y \geq 0$.

Each flight of Orville costs \$12000 and each flight of Wilbur costs \$18000.

Therefore,

Cost function $z = 12000x + 18000y$

Each flight of Orville carry 20 first-class and 80

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$$= \frac{\frac{6\times5}{2\times1}}{\frac{10\times9\times8\times7}{4\times3\times2\times1}} = \frac{15}{210} = \frac{1}{14}$$

Probability of getting one white ball,

$$P(X=1) = \frac{{}^4C_1 \times {}^6C_3}{{}^{10}C_4} = \frac{4\times\frac{6\times5\times4}{3\times2\times1}}{\frac{10\times9\times8\times7}{4\times3\times2\times1}}$$

$$= \frac{4\times20}{210} = \frac{80}{210} = \frac{8}{21}$$

Probability of getting two white ball,

$$P(X=2) = \frac{{}^4C_2 \times {}^6C_2}{{}^{10}C_4} = \frac{\frac{4\times3}{2\times1}\times\frac{6\times5}{2\times1}}{\frac{10\times9\times8\times7}{4\times3\times2\times1}}$$

$$= \frac{6\times15}{210} = \frac{90}{210} = \frac{3}{7}$$

Probability of getting three white ball,

$$P(X=3) = \frac{{}^4C_3 \times {}^6C_1}{{}^{10}C_4} = \frac{4\times6}{210} = \frac{24}{210} = \frac{4}{35}$$

Probability of getting four white ball,

$$P(X=4) = \frac{{}^4C_4}{{}^{10}C_4} = \frac{1}{210}$$

The probability distribution of X is

X	0	1	2	3	4
$P(X=x)$	$\frac{1}{14}$	$\frac{8}{21}$	$\frac{3}{7}$	$\frac{4}{35}$	$\frac{1}{210}$

OR

If three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are mutually perpendicular to each other, then $\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{c} = \vec{c}\cdot\vec{a} = 0$ and if all three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are equally inclined with the vector $(\vec{a}+\vec{b}+\vec{c})$, that means each vector $\vec{a}$, $\vec{b}$ and $\vec{c}$ makes equal angle with $(\vec{a}+\vec{b}+\vec{c})$ by using formula $\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

We have $|\vec{a}| = |\vec{b}| = |\vec{c}| = \lambda$ (say) ...(1)

and $\vec{a}\cdot\vec{b} = 0,\ \vec{b}\cdot\vec{c} = 0$ and $\vec{c}\cdot\vec{a} = 0$ ...(2)

Now,

$$|\vec{a}+\vec{b}+\vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$$

$$= \lambda^2 + \lambda^2 + \lambda^2 + 2(0+0+0)$$

$$= 3\lambda^2$$

$$|\vec{a}+\vec{b}+\vec{c}| = \sqrt{3}\lambda \quad \text{as length cannot be negative.}$$

Suppose $(\vec{a}+\vec{b}+\vec{c})$ is inclined at angles θ_1, θ_2 and θ_3 respectively with vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$, then

$$(\vec{a}+\vec{b}+\vec{c})\cdot\vec{a} = |\vec{a}+\vec{b}+\vec{c}||\vec{a}|\cos\theta_1$$

$\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$

$$|\vec{a}|^2 + \vec{a}\cdot\vec{b} + \vec{a}\cdot\vec{c} = \sqrt{3}\lambda\times\lambda\cos\theta_1$$

$$\lambda^2 + 0 + 0 = \sqrt{3}\lambda^2\cos\theta_1 \quad \text{from eq (1) and 2)}$$

$$(\vec{a}\cdot\vec{b}\cdot\vec{c})\cdot\vec{b} = |\vec{a}+\vec{b}+\vec{c}||\vec{b}|\cos\theta_2$$

$$\vec{a}\cdot\vec{b} + |\vec{b}|^2 + \vec{c}\cdot\vec{b} = \sqrt{3}\lambda\cdot\lambda\cos\theta_2$$

$$0 + \lambda^2 + 0 = \sqrt{3}\lambda^2\cos\theta_2 \quad \text{from eq (1) and (2)}$$

$$\cos\theta_2 = \frac{1}{\sqrt{3}}$$

Similarly,

$$(\vec{a}+\vec{b}+\vec{c})\cdot\vec{c} = |\vec{a}+\vec{b}+\vec{c}||\vec{c}|\cos\theta_3$$

$$\cos\theta_1 = \frac{1}{\sqrt{3}}$$

Thus, $\cos\theta_1 = \cos\theta_2 = \cos\theta_3 = \frac{1}{\sqrt{3}}$

Hence, it is proved that $(\vec{a}+\vec{b}+\vec{c})$ is equally inclined with the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$.

35. Find the angle between following pair of lines.

$\frac{-x+2}{-2} = \frac{y-1}{7} = \frac{z+3}{-3}$ and $\frac{x+2}{-1} = \frac{2y-8}{4} = \frac{z-5}{4}$

and check whether the lines are parallel or perpendicular.

OR

Find the shortest distance between lines whose vector equations are

$$\vec{r} = (1-t)\hat{i} + (t-2)\hat{j} + (3-2t)\hat{k}$$

$$\vec{r} = (s+1)\hat{i} + (2s-1)\hat{j} - (2s+1)\hat{k}.$$

Sol :

We have $\frac{-x+2}{-2} = \frac{y-1}{7} = \frac{z+3}{-3}$

and $\frac{x+2}{-1} = \frac{2y-8}{4} = \frac{z-5}{4}$

Above equations can be written as

$$\frac{x-2}{2} = \frac{y-1}{7} = \frac{z+3}{-3} \quad ...(1)$$

and $\frac{x+2}{-1} = \frac{y-4}{2} = \frac{z-5}{4}$ ...(2)

Comparing Eqs. (1) and (2) with one point form of equations of line

$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$, we get

$$a_1 = 2,\ b_1 = 7,\ c_1 = -3$$

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The differential equation is separable. Separating the variables and integrating, we obtain

$$\int \frac{dQ}{Q^2} = \int k\, dt$$

and $$-\frac{1}{Q} = kt + C$$

Therefore, $$Q = -\frac{1}{kt + C}$$

Now, $Q = 50$ when $t = 0$, therefore, $50 = -\frac{1}{C}$ and $C = -\frac{1}{50}$.

Therefore, $$Q = -\frac{1}{kt - \frac{1}{50}}$$

Since $Q = 10$ when $t = 1$,

$$10 = -\frac{1}{k - \frac{1}{50}}$$

$$10\left(k - \frac{1}{50}\right) = -1$$

$$10k - \frac{1}{5} = -1$$

$$10k = -1 + \frac{1}{5}$$

$$= -\frac{4}{5}$$

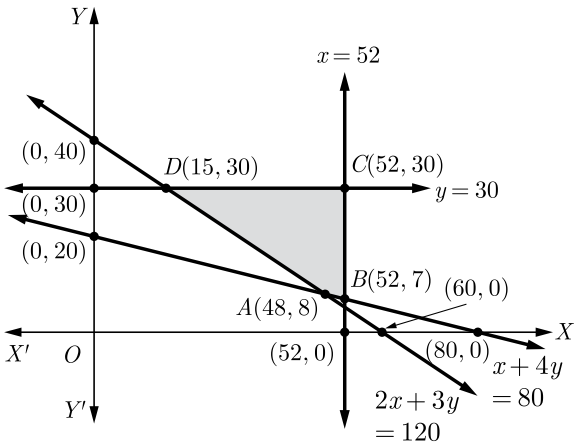
$$k = -\frac{4}{50} = -\frac{2}{25}$$

Therefore, $$Q(t) = \frac{1}{\frac{2}{25}t + \frac{1}{50}}$$

$$= \frac{1}{\frac{4t+1}{50}}$$

$$= \frac{50}{4t+1}$$

and $$Q(2) = \frac{50}{8+1} \approx 5.56 \text{ grams}$$



38. Rice is a nutritional staple food which provides instant energy as its most important component is carbohydrate (starch). On the other hand, rice is poor in nitrogenous substances with average composition of these substances being only 8 per cent and fat content or lipids only negligible, i.e., 1 per cent and due to this reason it is considered as a complete food for eating. Rice flour is rich in starch and is used for making various food materials.



Two farmers Ramkishan and Gurcharan Singh cultivate only three varieties of rice namely Basmati, Permal and Naura. The sale (in ₹) of these varieties of rice by both the farmers in the month of September and October are given by the following matrices A and B.

September Sales (in ₹)

$$A = \begin{array}{c} \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \end{matrix} \\ \begin{bmatrix} 10000 & 20000 & 30000 \\ 50000 & 30000 & 10000 \end{bmatrix} \end{array} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

October Sales (in ₹)

$$B = \begin{array}{c} \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \end{matrix} \\ \begin{bmatrix} 5000 & 10000 & 6000 \\ 20000 & 10000 & 10000 \end{bmatrix} \end{array} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

(i) Find the combined sales in September and October for each farmer in each variety.

(ii) Find the decrease in sales from September to October.

(iii) If both farmers receive 2% profit on gross sales, compute the profit for each farmer and for each variety sold in October.

Sol :

The sale (in ₹) of these varieties of rice by both the farmers in the month of September and October are given by the following matrices A and B.

September Sales (in ₹)

$$A = \begin{array}{c} \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \end{matrix} \\ \begin{bmatrix} 10000 & 20000 & 30000 \\ 50000 & 30000 & 10000 \end{bmatrix} \end{array} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

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October Sales (in ₹)

$$B = \begin{matrix} \\ \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix} \begin{matrix} \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \end{matrix} \\ \begin{bmatrix} 5000 & 10000 & 6000 \\ 20000 & 10000 & 10000 \end{bmatrix} \end{matrix}$$

(i) Combined sales in September and October for each farmer in each variety is given by

$$A + B = \begin{matrix} \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \end{matrix} \\ \begin{bmatrix} 10000 + 5000 & 20000 + 10000 & 30000 + 6000 \\ 50000 + 20000 & 30000 + 10000 & 10000 + 10000 \end{bmatrix} \end{matrix}$$

$$= \begin{matrix} \\ \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix} \begin{matrix} \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \end{matrix} \\ \begin{bmatrix} 15000 & 30000 & 36000 \\ 70000 & 40000 & 20000 \end{bmatrix} \end{matrix}$$

(ii) Change in sales from September to October is given by

$$A - B = \begin{matrix} \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \end{matrix} \\ \begin{bmatrix} 10000 - 5000 & 20000 - 10000 & 30000 - 6000 \\ 50000 - 20000 & 30000 - 10000 & 10000 - 10000 \end{bmatrix} \end{matrix}$$

$$= \begin{matrix} \\ \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix} \begin{matrix} \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \end{matrix} \\ \begin{bmatrix} 5000 & 10000 & 24000 \\ 30000 & 20000 & 0 \end{bmatrix} \end{matrix}$$

(iii) $$2\% \text{ of } B = \frac{2}{100} \times B$$

$$= 0.02 \times B$$

$$= 0.02 \begin{matrix} \\ \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix} \begin{matrix} \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \end{matrix} \\ \begin{bmatrix} 5000 & 10000 & 6000 \\ 20000 & 10000 & 10000 \end{bmatrix} \end{matrix}$$

$$= \begin{matrix} \\ \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix} \begin{matrix} \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \end{matrix} \\ \begin{bmatrix} 100 & 200 & 120 \\ 400 & 200 & 200 \end{bmatrix} \end{matrix}$$

Thus, in October Ramkishan receives ₹ 100, ₹ 200 and ₹ 120 as profit in the sale of each variety of rice, respectively, and Gurcharan Singh receives profit of ₹ 400, ₹ 200 and ₹ 200 in the sale of each variety of rice, respectively.

In this book, we have provided solutions along with the papers. To download the same papers in unsolved format for printing, please visit
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Sample Paper 07

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

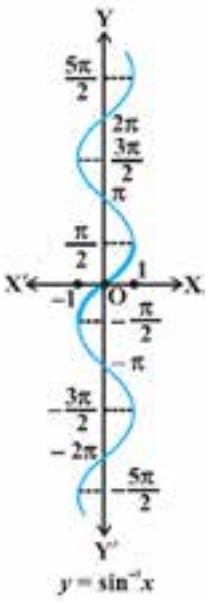
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. At which point does the graph of $y = \sin^{-1} x$ intersect the origin?



(a) (-1,0) (b) (0,0)

(c) (1,0) (d) (0,1)

Sol :

From the image, it is clear that the graph passes through the origin, where both X and Y are zero. This shows that $\sin^{-1}(0) = 0$. Thus, the correct intersection point is (0,0).

Thus (b) is correct option.

2. The order of the differential equation of all circles of radius a is

(a) 2 (b) 3

(c) 4 (d) 1

Sol :

The general equation of all circles of radius a is

$$(x-h)^2 + (y-k)^2 = a^2$$

Since, it has two arbitrary constants.

Hence, order of the differential equation obtained is 2.

Thus (a) is correct option.

3. Two dice are thrown n times in succession. The probability of obtaining a doublet six atleast once is

(a) $\left(\frac{1}{36}\right)^n$ (b) $1-\left(\frac{35}{36}\right)^n$

(c) $\left(\frac{1}{12}\right)^n$ (d) None of these

Sol :

p = Probability of getting double six in two dice

$= \frac{1}{6^2} = \frac{1}{36}$ and $q = \frac{35}{36}$

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Required probability

$= 1 -$ (Probability of not getting double six)n

$= 1 - \left(\frac{35}{36}\right)^n$

Thus (b) is correct option.

4. Solution of $(x + 2y^3)\,dy = y\,dx$ is

(a) $x = y^3 + cy$ (b) $x + y^3 = cy$

(c) $y^2 - x = cy$ (d) none of these

Sol :

Given equation can be rewritten as

$$\frac{dx}{dy} = \frac{x}{y} + 2y^2$$

$$\frac{dx}{dy} - \frac{x}{y} = 2y^2$$

$$\text{IF} = e^{\int p dx} = e^{\int -\frac{1}{y} dy} = \frac{1}{y}$$

Solution is, $\frac{x}{y} = \int 2y\,dy + c = y^2 + c$

$$x = y^3 + cy$$

Thus (a) is correct option.

5. $\int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x} dx$ is equal to

(a) 0 (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) π

Sol :

$$I = \int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x} dx \quad \text{...(i)}$$

$$= \int_0^{\pi/2} \frac{\sin\left(\frac{\pi}{2} - x\right) - \cos\left(\frac{\pi}{2} - x\right)}{1 + \sin\left(\frac{\pi}{2} - x\right)\cos\left(\frac{\pi}{2} - x\right)} dx$$

$$= \int_0^{\pi/2} \frac{\cos x - \sin x}{1 + \cos x \sin x} dx \quad \text{...(ii)}$$

Adding Eqs. (i) and (ii), we get

$$2I = \int_0^{\pi/2} \frac{0}{1 + \sin x \cos x} dx = 0 \Rightarrow I = 0$$

Thus (a) is correct option.

6. If the position vectors of the vertices A, B, C of a triangle ABC are $7\hat{j} + 10\hat{k}$, $-\hat{i} + 6\hat{j} + 6\hat{k}$ and $-4\hat{i} + 9\hat{j} + 6\hat{k}$ respectively, then triangle is

(a) Equilateral

(b) Isosceles

(c) Scalene

(d) Right angled and isosceles also

Sol :

Now, $\overrightarrow{AB} = -\hat{i} - \hat{j} - 4\hat{k}$,

$\overrightarrow{BC} = -3\hat{i} + 3\hat{j}$

and $\overrightarrow{CA} = 4\hat{i} - 2\hat{j} - 4\hat{k}$

$$|\overrightarrow{AB}| = \sqrt{(-1)^2 + (-1)^2 + (-4)^2} = 3\sqrt{2}$$

$$|\overrightarrow{BC}| = \sqrt{(-3)^2 + 3^2} = 3\sqrt{2}$$

and $|\overrightarrow{CA}| = \sqrt{4^2 + (-2)^2 + (-4)^2} = 6$

Now, $|\overrightarrow{AB}|^2 + |\overrightarrow{BC}|^2 = |\overrightarrow{CA}|^2$

$18 + 18 = 36$

$36 = 36$

Hence, $\triangle ABC$ is right angled and isosceles also.

Thus (d) is correct option.

7. The derivative of $\log|x|$ is

(a) $\frac{1}{x}$, $x > 0$ (b) $\frac{1}{|x|}$, $x \neq 0$

(c) $\frac{1}{x}$, $x \neq 0$ (d) None of these

Sol :

We have $y = \log|x|$

$$= \begin{cases} \log x, & x > 0 \\ \log(-x), & x < 0 \end{cases}$$

$$\frac{dy}{dx} = \begin{cases} \frac{1}{x}, & x > 0 \\ \frac{1}{-x}(-1) = \frac{1}{x}, & x < 0 \end{cases}$$

$$\frac{dy}{dx} = \frac{1}{x},\ x \neq 0$$

Thus (c) is correct option.

8. $\cos^{-1}\left[\cos\left(-\frac{\pi}{3}\right)\right]$ is equal to

(a) $-\frac{\pi}{3}$ (b) $\frac{\pi}{3}$

(c) $\frac{2\pi}{3}$ (d) $\frac{5\pi}{3}$

Sol :

$$\cos^{-1}\left[\cos\left(-\frac{\pi}{3}\right)\right] = \cos^{-1}\left(\cos\frac{\pi}{3}\right) = \frac{\pi}{3}$$

Thus (b) is correct option.

9. If $A = \begin{bmatrix} 1 & x \\ x^2 & 4y \end{bmatrix}$, $B = \begin{bmatrix} -3 & 1 \\ 1 & 0 \end{bmatrix}$ and $adj\,(A) + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, then the values of x and y are

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If $x>0$, then $f'(x)=2x>0$, it is an increasing function and if $x<0$, then $f(x)=2x<0$, it is a decreasing function.

Hence, it is neither decreasing nor increasing.

Thus (c) is correct option.

19. Assertion: If $A=\frac{1}{3}\begin{bmatrix}1 & -2 & 2\\ -2 & 1 & 2\\ -2 & -2 & -1\end{bmatrix}$, then $(A^T)\,A=I$

Reason: For any square matrix A, $(A^T)^T=A$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

$$AA^T=\frac{1}{3}\begin{bmatrix}1 & -2 & 2\\ -2 & 1 & 2\\ -2 & -2 & -1\end{bmatrix}\cdot\frac{1}{3}\begin{bmatrix}1 & -2 & -2\\ -2 & 1 & -2\\ 2 & 2 & -1\end{bmatrix}$$

$$=\frac{1}{9}\begin{bmatrix}9 & 0 & 0\\ 0 & 9 & 0\\ 0 & 0 & 9\end{bmatrix}=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}=I$$

Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

Thus (b) is correct option.

20. Assertion: Let $A=\{-1,1,2,3\}$ and $B=\{1,4,9\}$ where $f:A\to B$ given by $f(x)=x^2$, then f is a many-one function.

Reason: If $x_1\neq x^2\Rightarrow f(x_1)\neq f(x_2)$, for every $x_1,x_2\in$ domain then f is one-one or else many

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

Here, $f(-1)=1,\ f(1)=1,$

$f(2)=4,\ f(3)=9$

Two elements 1 and -1 have the same image $1\in B$.

So, f is a many-one function .

Assertion and reason are true and reason is the correct explanation of assertion.

Thus (a) is correct option.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. What is the range of the function

$$f(x)=\frac{|x-1|}{x-1},\ x\neq 1?$$

Sol :

Firstly, redefine the function by using the definition of modulus function, i.e. by using $|x|=\begin{cases}x, & x\geq 0\\ -x, & x<0\end{cases}$

Further, simplify it to get the range.

Given, function is $f(x)=\frac{|x-1|}{x-1}, x\neq 1.$

The above function can be written as

$$f(x)=\begin{cases}\frac{x-1}{x-1}, & \text{if } x>1\\ -\frac{(x-1)}{x-1}, & \text{if } x<1\end{cases}$$

$$f(x)=\begin{cases}1, & \text{if } x>1\\ -1, & \text{if } x<1\end{cases}$$

Hence, the range of $f(x)$ is $\{-1,1\}$.

22. If $|\vec{a}\times\vec{b}|^2+|\vec{a}\cdot\vec{b}|^2=400$ and $|\vec{a}|=5$, then write the value of $|\vec{b}|$.

OR

If $\vec{a}=7\hat{i}+\hat{j}-4\hat{k}$ and $\vec{b}=2\hat{i}+6\hat{j}+3\hat{k}$, then find the projection of $\vec{a}$ on $\vec{b}$.

Sol :

Given,

We have $|\vec{a}\times\vec{b}|^2+|\vec{a}\cdot\vec{b}|^2=400$

and $|\vec{a}|=5$

Now $|\vec{a}\times\vec{b}|^2+|\vec{a}\cdot\vec{b}|^2=400$

$$|\vec{a}|^2|\vec{b}|^2\sin^2\theta+|\vec{a}|^2|\vec{b}|^2\cos^2\theta=400$$

$$|\vec{a}|^2|\vec{b}|^2(\sin^2\theta+\cos^2\theta)=400$$

$$25\cdot|\vec{b}|^2=400\qquad[\because|\vec{a}|=5]$$

$$|\vec{b}|^2=16\Rightarrow|\vec{b}|=4$$

as length cannot be negative.

Here we have used the fact that

$$|\vec{a}\times\vec{b}|=|\vec{a}||\vec{b}|\sin\theta$$

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and $P(B) = 0.6$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.83 = 0.3 + 0.6 - P(A \cap B)$$

$$P(A \cap B) = 0.9 - 0.83 = 0.07$$

Now, $P(A) \cdot P(B) = 0.3 \times 0.6 = 0.18$

$$P(A) \cdot P(B) \neq P(A \cap B)$$

Hence, events are dependent.

Thus (a) is correct option.

15. The function $f(x) = x^3$ has a

(a) local minima at $x = 0$

(b) local maxima at $x = 0$

(c) point of inflexion at $x = 0$

(d) none of the above

Sol :

(a) local minima at $x = 0$

local minima at $x = 0$

$$f'(x) = 3x^2$$

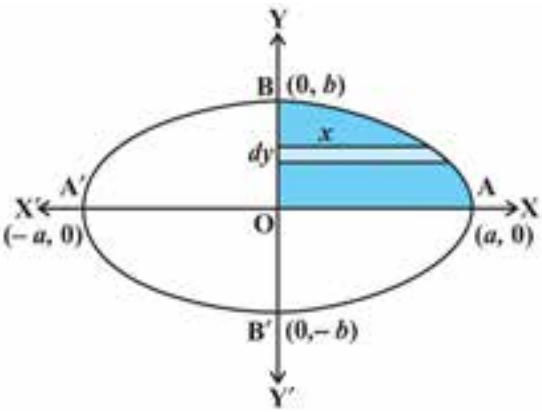
$$f'(0) = 0$$

Similarly, $f''(0) = 0$ and $f'''(0) = 6$

Hence, f has a point of inflexion at $x = 0$.

Thus (a) is correct option.

16. In Fig., the element marked "dy" in the shaded strip represents:



(a) The semi-minor axis of the ellipse

(b) A small change in the y-coordinate (thickness of the strip)

(c) A small change in the x-coordinate (width of the strip)

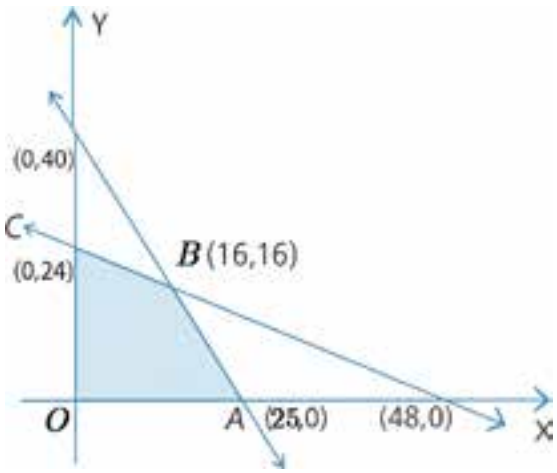
(d) The total height of the ellipse along the Y-axis

Sol :

(b) A small change in the y-coordinate (thickness of the strip)

The shaded portion shows a thin horizontal strip parallel to the X-axis. Its vertical thickness is marked as "dy", which represents an infinitesimal change in the y-coordinate. This strip has length x and thickness dy, confirming that option (b) is correct.

17. For the given feasible region in the figure, bounded by the lines shown, consider the objective function $z = 2x + y$. The feasible region includes the corner points $O(0,0), A(25,0), B(16,16)$, and $C(0,24)$. At the vertex $B(16,16)$, what is the value of z?



(a) 32 (b) 48

(c) 64 (d) 72

Sol :

(b) 48

To evaluate the value of the objective function at a specific point, we substitute the coordinates of that point into the given equation. At B(16,16), we calculate: $z = 2x + y = 2(16) + 16 = 32 + 16 = 48$

Therefore, the value of the function at this vertex is 48. Since this is exactly equal to option (b), the correct answer is 48.

18. The function $f(x) = x^2$, for all real x, is

(a) decreasing

(b) increasing

(c) neither decreasing nor increasing

(d) none of the above

Sol :

Given that, $f(x) = x^2$

$$f'(x) = 2x \; \forall \, x \in R$$

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(a) 1, 1 (b) ± 1, 1

(c) 1, 0 (d) None of these

Sol :

We have $adj\,(A) = \begin{bmatrix} 4y & -x \\ -x^2 & 1 \end{bmatrix}$

$$adj\,(A) + B = \begin{bmatrix} 4y & -x \\ -x^2 & 1 \end{bmatrix} + \begin{bmatrix} -3 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4y-3 & -x+1 \\ -x^2+1 & 1+0 \end{bmatrix}$$

$$4y - 3 = 1 \Rightarrow y = 1$$

and $-x + 1 = 0 \Rightarrow x = 1$

Thus (a) is correct option.

10. If $f(x) = \begin{cases} ax+3, & x \le 2 \\ a^2x - 1, & x > 2 \end{cases}$, then the values of a for which f is continuous for all x are

(a) 1 and -2 (b) 1 and 2

(c) -1 and 2 (d) -1 and -2

Sol :

We have $f(x) = \begin{cases} ax+3, & x \le 2 \\ a^2x - 1, & x > 2 \end{cases}$

Continuity at $x = 2$,

$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2} (ax+3) = 2a+3$$

$$\lim_{x \to 2^+} f(x) = \lim_{x \to 2} (a^2x - 1) = 2a^2 - 1$$

Since, $f(x)$ is continuous for all values of x.

$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x)$$

$$2a + 3 = 2a^2 - 1$$

$$2a^2 - 2a - 4 = 0$$

$$a^2 - a - 2 = 0$$

$$a^2 - 2a + a - 2 = 0$$

$$a(a-2) + 1(a-2) = 0$$

$$(a+1)(a-2) = 0$$

$$a = -1, 2$$

Thus (c) is correct option.

11. Integrating factor of differential equation $\cos x \dfrac{dy}{dx} + y \sin x = 1$

(a) $\cos x$ (b) $\tan x$

(c) $\sec x$ (d) $\sin x$

Sol :

The given differential equation can be rewritten as

$$\frac{dy}{dx} + \tan x\, y = \sec x$$

Here, $P = \tan x$

$$\text{IF} = e^{\int p\,dx}$$

$$= e^{\int \tan x\,dx}$$

$$= \sec x$$

Thus (c) is correct option.

12. If α, β, γ are the angles which a half ray makes with the positive directions of the axes, then $\sin^2\alpha + \sin^2\beta + \sin^2\gamma$ is equal to

(a) 2 (b) 1

(c) 0 (d) -1

Sol :

Since, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Hence, $1 - \sin^2\alpha + 1 - \sin^2\beta + 1 - \sin^2\gamma = 1$

$$\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$$

Thus (a) is correct option.

13. $\int \dfrac{1 + x + \sqrt{x + x^2}}{\sqrt{x} + \sqrt{1+x}}\,dx$ is equal to

(a) $\frac{1}{2}\sqrt{1+x} + C$ (b) $\frac{2}{3}(1+x)^{3/2} + C$

(c) $\sqrt{1+x} + C$ (d) $2(1+x)^{3/2} + C$

Sol :

$$\int \frac{1 + x + \sqrt{x + x^2}}{\sqrt{x} + \sqrt{1+x}}\,dx = \int \frac{\sqrt{(1+x)^2} + \sqrt{x}\sqrt{x+1}}{\sqrt{x} + \sqrt{1+x}}\,dx$$

$$= \int \frac{\sqrt{1+x}\,(\sqrt{1+x} + \sqrt{x})}{(\sqrt{1+x} + \sqrt{x})}\,dx$$

$$= \int \sqrt{1+x}\,dx$$

$$= \frac{2}{3}(1+x)^{3/2} + C$$

Thus (b) is correct option.

14. If $P(A \cup B) = 0.83$, $P(A) = 0.3$ and $P(B) = 0.6$, then the events will be

(a) dependent

(b) independent

(c) cannot say anything

(d) None of the above

Sol :

We have $P(A \cup B) = 0.83$, $P(A) = 0.3$

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$$|\vec{a}\cdot\vec{b}| = |\vec{a}||\vec{b}|\cos\theta$$

OR

We have $\vec{a} = 7\hat{i}+\hat{j}-4\hat{k}$

and $\vec{b} = 2\hat{i}+6\hat{j}+3\hat{k}$

Now, the projection of $\vec{a}$ and $\vec{b}$ is given by

$$\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = \frac{(7\hat{i}+\hat{j}-4\hat{k})\cdot(2\hat{i}+6\hat{j}+3\hat{k})}{\sqrt{2^2+6^2+3^2}}$$

$$= \frac{14+6-12}{\sqrt{49}} = \frac{8}{7}$$

23. Find the general solution of following equation $e^x\tan y\,dx+(1-e^x)\sec^2 y\,dy=0$

Sol :

Given differential equation is

$$e^x\tan y\,dx+(1-e^x)\sec^2 y\,dy = 0$$

$$e^x\tan y\,dx = (1-e^x)\sec^2 y\,dy$$

$$(1-e^x)\sec^2 y\,dy = -e^x\tan y\,dx$$

Separating the variables, we get

$$\frac{\sec^2 y}{\tan y}dy = -\frac{e^x}{1-e^x}dx$$

Integrating both sides, we get

$$\int\frac{\sec^2 y}{\tan y}dy = \int\frac{e^x}{e^x-1}dx + C$$

$$\log|\tan y| = \log|e^x-1|+\log C$$

$$\log|\tan y| = \log|e^x-1|+\log C$$

$$\log\left|\frac{\tan y}{e^x-1}\right| = \log C$$

$$\frac{\tan y}{e^x-1} = C$$

$$\tan y = C(e^x-1)$$

$\tan y = C(e^x-1)$ is the required solution.

24. If a line makes angles 90°, 60° and θ with X, Y and Z-axis respectively, where θ is acute angle, then find θ.

Sol :

If a line makes angles α, β and γ with the x-axis, y-axis and z-axis, respectively, the we have

$$\cos^2\alpha+\cos^2\beta+\cos^2\gamma = 1$$

$$\cos^2 90^\circ+\cos^2 60^\circ+\cos^2\theta = 1$$

$$0+\left(\frac{1}{2}\right)^2+\cos^2\theta = 1$$

$$\cos^2\theta = 1-\frac{1}{4} = \frac{3}{4}$$

$$\cos\theta = \pm\frac{\sqrt{3}}{2}$$

$$= \cos 30^\circ$$

$$\theta = 30^\circ$$

Since $\cos\theta$ cannot be negative as θ is an acute angle.

25. Write the value of $\int\frac{\sec^2 x}{\operatorname{cosec}^2 x}dx$.

OR

Write the value of $\int\frac{dx}{x^2+16}$.

Sol :

We have $I = \int\frac{\sec^2 x}{\operatorname{cosec}^2 x}dx = \int\frac{\frac{1}{\cos^2 x}}{\frac{1}{\sin^2 x}}dx$

$$= \int\frac{\sin^2 x}{\cos^2 x}dx$$

$$= \int\tan^2 x\,dx \qquad \tan^2 x = \sec^2 x-1$$

$$= \int(\sec^2 x-1)\,dx$$

$$= \int\sec^2 x\,dx-\int 1\,dx = \tan x-x+C$$

$$= \int\sec^2 x\,dx-\int 1\,dx = \tan x-x+C$$

OR

We have $I = \int\frac{dx}{x^2+16}$

$$= \int\frac{dx}{x^2+(4)^2} \qquad \int\frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}+C$$

$$= \frac{1}{4}\tan^{-1}\frac{x}{4}+C$$

SECTION - C

This section comprises of short answer-type questions (SA) of 3 marks each.

26. In the interval $\frac{\pi}{2} < x < \pi$, find the value of x for which the matrix $\begin{bmatrix}2\sin x & 3\\ 1 & 2\sin x\end{bmatrix}$ is singular.

Sol :

Let

We have $A = \begin{bmatrix}2\sin x & 3\\ 1 & 2\sin x\end{bmatrix}$

Since A is a singular matrix, we have

$$|A| = 0$$

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$$I = \int \underset{I}{t} \underset{II}{\sin t}\, dt$$

$$= t\int \sin t\, dt - \int\left[\frac{d}{dt}(t)\cdot \int \sin t\, dt\right]dt$$

$$= -t\cos t - \int (1\times -\cos t)\, dt$$

$$= -t\cos t + \int \cos t dt$$

$$= -t\cos t + \sin t + C$$

$$= -t\sqrt{1-\sin^2 t} + \sin t + C$$

Substituting $t = \sin^{-1}x \Rightarrow x = \sin t$ we have

$$I = -\sin^{-1}x\sqrt{1-x^2} + x + C$$

33. Maximise and minimise $Z = x + 2y$ subject to the constraints

$$x + 2y \geq 100$$

$$2x - y \leq 0$$

$$2x + y \leq 200$$

$$x,\ y \geq 0$$

Solve the above LPP graphically.

OR

Maximise $Z = 8x + 9y$ subject to the constraints given below

$$2x + 3y \leq 6$$

$$3x - 2y \leq 6$$

$$y \leq 1$$

$$x,\ y \geq 0.$$

Sol :

We have to minimise and maximise

$$Z = x + 2y$$

Subject to constraints,

$$x + 2y \geq 100 \quad \text{...(i)}$$

$$2x - y \leq 0 \quad \text{...(ii)}$$

$$2x + y \leq 200 \quad \text{...(iii)}$$

and $x \geq 0,\ y \geq 0$

(i) Region corresponding to $x + 2y \geq 100$:

Line $x + 2y = 100$			
x	0	100	Point $(0,0)$ is false for $x + 2y \geq 100$. So, the region is away from the origin.
y	50	0	

(ii) Region corresponding to $2x - y \leq 0$:

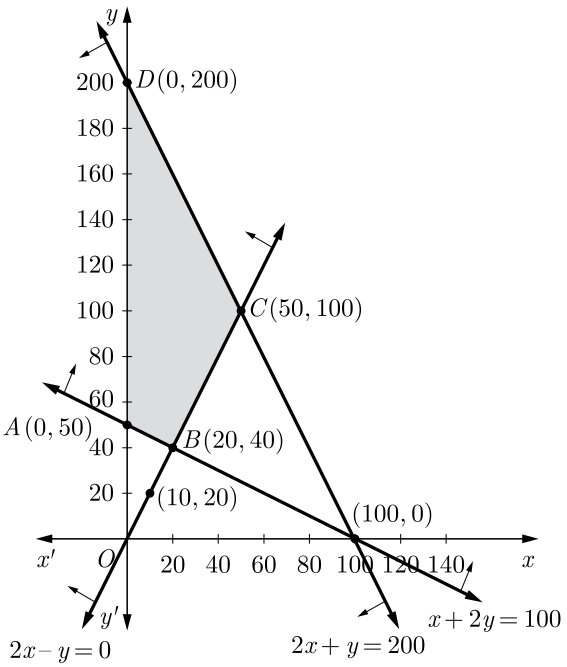
Line $2x - y = 0$			
x	0	10	Point $(0,5)$ is true for $2x - y \leq 0$, thus region includes $(0,5)$ and towards y axis.
y	0	20	

(iii) Region corresponding to $2x + y \leq 200$:

Line $2x + y = 200$			
x	0	100	Point $(0,0)$ is true for $2x + y \leq 200$. So, the region is towards the origin.
y	200	0	

Since $x \geq 0,\ y \geq 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is $ABCDA$. Solving equations $2x - y = 0$ and $x + 2y = 100$, we get $B(20,40)$. Again, solving equations $2x - y = 0$ and $2x + y = 200$, we get $C(50,100)$. The corner points of the feasible region are $A(0,50)$, $B(20,40)$, $C(50,100)$ and $D(0,200)$.



Since, the feasible region is a bounded region, we can check the profit function at all the vertices to find the maxima. The values of Z at corner points are given below

Corner points	$Z = x + 2y$
$A(0,50)$	$Z = 0 + 2\times 50 = 100$
$B(20,40)$	$Z = 20 + 2\times 40 = 100$
$C(50,100)$	$Z = 50 + 2\times 100 = 250$
$D(0,200)$	$Z = 0 + 2\times 200 = 400$

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$$= \pi\left[\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right] = \frac{\pi^2}{2}$$

$$I = \frac{\pi^2}{4}$$

OR

We have $I = \int_0^{\pi} \frac{x \tan x}{\sec x + \tan x}\, dx$ ...(1)

Using the fact $\int_0^a f(x)\, dx = \int_0^a f(a-x)\, dx$ we have

$$I = \int_0^{\pi} \frac{(\pi - x)\tan(\pi - x)}{\sec(\pi - x) + \tan(\pi - x)}$$

$$= \int_0^{\pi} \frac{(\pi - x)\tan x}{\sec x + \tan x}\, dx \quad ...(2)$$

Adding eqs. (1) and (2), we get

$$2I = \int_0^{\pi} \frac{\pi \tan x}{\sec x + \tan x}\, dx$$

$$I = \frac{\pi}{2}\int_0^{\pi} \frac{\tan x(\sec x - \tan x)}{(\sec x + \tan x)(\sec x - \tan x)}\, dx$$

$$= \frac{\pi}{2}\int_0^{\pi} \frac{(\tan x \sec x - \tan^2 x)\, dx}{(\sec^2 x - \tan^2 x)}$$

$$= \frac{\pi}{2}\int_0^{\pi} \frac{\tan x \sec x - \sec^2 x + 1}{1}\, dx \quad \tan^2 x = \sec^2 x - 1$$

$$= \frac{\pi}{2}[\sec x - \tan x + x]_0^{\pi}$$

$$= \frac{\pi}{2}[(\sec \pi - \tan \pi + \pi) - (\sec 0 - \tan 0 + 0)]$$

$$= \frac{\pi}{2}[\sec \pi - \sec 0 - \tan \pi + \tan 0 + \pi - 0]$$

$$= \frac{\pi}{2}[-1 - 1 - 0 + 0 + \pi - 0] = \frac{\pi}{2}[\pi - 2]$$

31. The sides of an equilateral triangle are increasing at the rate of 2 cm/s. Find the rate at which the area increases, when the side is 10 cm?

OR

Find the value(s) of x for which $y = [x(x-2)]^2$ is an increasing function.

Sol :

Let a be the side of an equilateral triangle and A be the area of an equilateral triangle.

We have $\frac{da}{dt} = 2$ cm/s

Area of an equilateral triangle,

$$A = \frac{\sqrt{3}}{4}a^2$$

Differentiating both sides w.r.t. t, we get

$$\frac{dA}{dt} = \frac{\sqrt{3}}{4} \times 2a \times \frac{da}{dt}$$

$$= \frac{\sqrt{3}}{4} \times 2 \times 10 \times 2 \quad [\text{given, } a = 10]$$

$$= 10\sqrt{3} \text{ cm}^2/\text{s}$$

Thus, the rate of area increasing is $10\sqrt{3}$ cm^2/s.

OR

We have $y = [x(x-2)]^2 = [x^2 - 2x]^2$.

Differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = 2(x^2 - 2x)\frac{d}{dx}(x^2 - 2x)$$

$$= 2(x^2 - 2x)(2x - 2)$$

$$= 4x(x-2)(x-1)$$

Substituting $\frac{dy}{dx} = 0$, we get

$$4x(x-2)(x-1) = 0$$

$$x = 0, 1 \text{ and } 2$$

Now, we find interval in which $f(x)$ is strictly increasing or strictly decreasing.

Interval	$\frac{dy}{dx} = 4x(x-1)(x-2)$	**Sign of** $f'(x)$
$(-\infty, 0)$	$(-)(-)(-)$	$-$ ve
$(0,1)$	$(+)(-)(-)$	$+$ ve
$(1,2)$	$(+)(+)(-)$	$-$ ve
$(2,\infty)$	$(+)(+)(+)$	$+$ ve

Hence, y is strictly increasing in $(0,1)$ and $(2,\infty)$. Also, y is a polynomial function, so it continuous at $x = 0$, 1 and 2. Hence, y is increasing in $[0,1] \cup [2,\infty)$.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Evaluate $\int \frac{x \sin^{-1} x}{\sqrt{1-x^2}}\, dx$.

Sol :

We have $I = \int \frac{x \sin^{-1} x}{\sqrt{1-x^2}}\, dx$

Substituting $\sin^{-1} x = t$ we get

$$\frac{1}{\sqrt{1-x^2}}\, dx = dt$$

Thus $I = \int t \sin t\, dt$

Using integration by parts, taking t as the first function and $\sin t$ as the second function, we get

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$$\begin{vmatrix} 2\sin x & 3 \\ 1 & 2\sin x \end{vmatrix} = 0$$

$$4\sin^2 x - 3 = 0$$

$$\sin^2 x = \frac{3}{4}$$

Taking positive square root because $\frac{\pi}{2} < x < \pi$,

$$\sin x = \frac{\sqrt{3}}{2}$$

$$x = \frac{2\pi}{3}$$

27. Evaluate $P(A \cup B)$, if $2P(A) = P(B) = \frac{5}{13}$ and $P\left(\frac{A}{B}\right) = \frac{2}{5}$.

Sol :

We have

$$2P(A) = P(B) = \frac{5}{13}$$

$$P(A) = \frac{5}{26},\ P(B) = \frac{5}{13} \text{ and } P\left(\frac{A}{B}\right) = \frac{2}{5}$$

Now $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$

$$\frac{2}{5} = \frac{P(A \cap B)}{\frac{5}{13}}$$

$$P(A \cap B) = \frac{2}{5} \times \frac{5}{13} = \frac{2}{13}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{5}{26} + \frac{5}{13} - \frac{2}{13}$$

$$= \frac{5 + 10 - 4}{26} = \frac{11}{26}$$

28. If $e^y(x+1) = 1$, then show that $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$.

Sol :

We have $e^y(x+1) = 1$

Taking log both sides, we get

$$\log[e^y(x+1)] = \log 1$$

$$\log e^y + \log(x+1) = 0$$

$$y + \log(x+1) = 0 \qquad [\because \log e^y = y]$$

Differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} + \frac{1}{x+1} = 0 \qquad \ldots(1)$$

Again, differentiating both sides w.r.t. x, we get

$$\frac{d^2y}{dx^2} - \frac{1}{(x+1)^2} = 0$$

$$\frac{d^2y}{dx^2} - \left(-\frac{dy}{dx}\right)^2 = 0 \qquad \text{from Eq. (1)}$$

$$\frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^2 = 0$$

$$\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2 \text{ Hence proved.}$$

29. Write the value of $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$.

Sol :

Here angle $\frac{7\pi}{6}$ doesn't lie in the principal value branch $[0, \pi]$. So we reduce the angle such that, it lies in principal value branch.

We write as $\frac{7\pi}{6} = 2\pi - \frac{5\pi}{6}$

Now, $\cos^{-1}\left(\cos\frac{7\pi}{6}\right) = \cos^{-1}\left[\cos\left(2\pi - \frac{5\pi}{6}\right)\right]$

$$= \cos^{-1}\left(\cos\frac{5\pi}{6}\right)$$

$$= \frac{5\pi}{6}$$

Thus required value is $\frac{5\pi}{6}$.

30. Evaluate $\int_0^\pi \frac{x\sin x}{1+\cos^2 x}\,dx$.

OR

Evaluate $\int_0^\pi \frac{x\tan x}{\sec x + \tan x}\,dx$.

Sol :

We have $I = \int_0^\pi \frac{x\sin x}{1+\cos^2 x}\,dx$ ...(i)

$$= \int_0^\pi \frac{(\pi - x)\sin(\pi - x)}{1+\cos^2(\pi - x)}\,dx$$

$$= \int_0^\pi \frac{(\pi - x)\sin x}{1+\cos^2 x}\,dx \qquad \ldots(ii)$$

Adding Eqs. (i) and (ii), we get

$$2I = \int_0^\pi \frac{\pi\sin x}{(1+\cos^2 x)}\,dx$$

Substituting $\cos x = t \Rightarrow -\sin x\,dx = dt$

$$\sin x\,dx = -dt$$

when $x = 0$, then $t = 1$ and $x = \pi$, then $t = -1$

Thus $2I = -\pi\int_1^{-1} \frac{dt}{1+t^2}$,

$$= \pi\int_{-1}^{1} \frac{dt}{(1+t^2)}$$

$$= \pi[\tan^{-1} t]_{-1}^{1}$$

$$= \pi[\tan^{-1}1 - \tan^{-1}(-1)]$$

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The maximum value of Z is 400 at $D(0,200)$ and the minimum value of Z is 100 at the points on the line segment joining $A(0,50)$ and $B(20,40)$.

OR

We have the following LPP,

Maximise $Z = 8x+9y$

Subject to the constraints

$$2x+3y \le 6 \quad \text{(i)}$$

$$3x-2y \le 6 \quad \text{(ii)}$$

$$y \le 1 \quad \text{(iii)}$$

and $x,\ y \ge 0$

(i) Region corresponding to $2x+3y \le 6$:

Line $2x+3y=6$			
x	0	3	Point $(0,0)$ is true for $2x+3y \le 6$
y	2	0	So, the region is towards the origin.

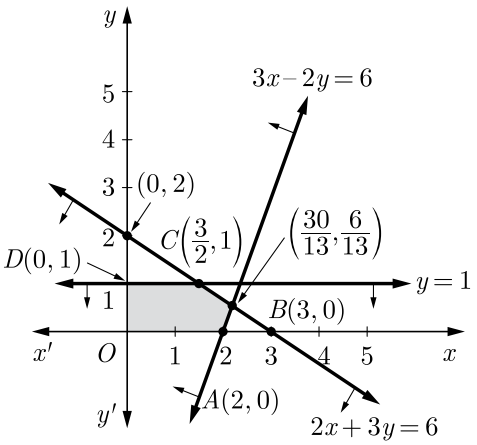
(ii) Region corresponding to $3x-2y \le 6$:

Line $3x-2y=6$			
x	2	0	Point $(0,0)$ is true for $3x-2y \le 6$
y	0	-3	So, the region is towards the origin.

(iii) For $y \le 1$ the line $y=1$ is perpendicular the x axis. On putting $(0,0)$ in the Inequality $y \le 1$ is true for point $(0,0)$, thus the region is towards the origin.

Since $x \ge 0,\ y \ge 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. The shaded region $OABCDO$ represents the feasible region of the given LPP. The point of intersection of the lines $2x+3y=6$ and $3x-2y=6$ is $B(\frac{30}{13},\frac{6}{13})$. The point of intersection of the lines $3x-2y=6$ and $y=1$ is $C(\frac{3}{2},1)$. The corner points are $O(0,\ 0)$, $A(2,\ 0)$, $B(\frac{30}{13},\frac{6}{13})$, $C(\frac{3}{2},1)$ and $D(0,1)$.



Since, the feasible region is a bounded region, we can check the profit function at all the vertices to find the maxima. The values of Z at corner points are given below.

Corner points	$Z=8x+9y$
$O(0,\ 0)$	$Z=0+0=0$
$A(2,\ 0)$	$Z=8\times 2+0=16$
$B(\frac{30}{13},\frac{6}{13})$	$Z=8\times\frac{30}{13}+\frac{9\times 6}{13}=\frac{294}{13}=22.62$
$C(\frac{3}{2},1)$	$Z=8\times\frac{3}{2}+1=13$
$D(0,1)$	$Z=0+9\times 1=9$

In the table, we find that maximum value of Z is 22.62, when $x=\frac{30}{13}$ and $y=\frac{6}{13}$.

34. Solve the following differential equation.

$$\left[y-x\cos\left(\frac{y}{x}\right)\right]dy+\left[y\cos\left(\frac{y}{x}\right)2x\sin\left(\frac{y}{x}\right)\right]dx=0$$

OR

Find the particular solution of the differential equation $(3xy+y^2)\,dx+(x^2+xy)\,dy=0$, for $x=1$ and $y=1$.

Sol :

We have

$$\left[y-x\cos\left(\frac{y}{x}\right)\right]dy+\left[y\cos\left(\frac{y}{x}\right)2x\sin\left(\frac{y}{x}\right)\right]dx=0$$

$$\frac{dy}{dx}=\frac{2x\sin\left(\frac{y}{x}\right)-y\cos\left(\frac{y}{x}\right)}{y-x\cos\left(\frac{y}{x}\right)}$$

$$\frac{dy}{dx}=\frac{2\sin\left(\frac{y}{x}\right)-\frac{y}{x}\cos\left(\frac{y}{x}\right)}{\frac{y}{x}-\cos\left(\frac{y}{x}\right)} \quad ...(1)$$

which is a homogeneous differential equation as

$$\frac{dy}{dx}=F\left(\frac{y}{x}\right)$$

Substituting $y=vx \Rightarrow \frac{dy}{dx}=v+x\frac{dv}{dx}$ in eq (1) we have

$$v+x\frac{dv}{dx}=\frac{2\sin v-v\cos v}{v-\cos v}$$

$$x\frac{dv}{dx}=\frac{2\sin v-v\cos v}{v-\cos v}-v$$

$$x\frac{dv}{dx}=\frac{2\sin v-v\cos v-v^2+v\cos v}{v-\cos v}$$

$$x\frac{dv}{dx}=\frac{2\sin v-v^2}{v-\cos v}$$

$$\left(\frac{v-\cos v}{v^2-2\sin v}\right)dv=-\frac{dx}{x}$$

Integrating both sides, we have

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$$P(B \cap C) = \frac{1}{6}$$

and $P(A \cup B) = \frac{4}{6} = \frac{2}{3}$

Now, $P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{1/6}{1/3} = \frac{1}{2}$

and $P(B/A) = \frac{P(B \cap A)}{P(A)} = \frac{1/6}{1/2} = \frac{1}{3}$

(ii) $P(A/C) = \frac{P(A \cap C)}{P(C)} = \frac{1/3}{2/3} = \frac{1}{2}$

$$P(A \cap B/C) = \frac{P\{(A \cap B) \cap C\}}{P(C)}$$

$$= \frac{P(A \cap B \cap C)}{P(C)} = \frac{1/6}{2/3} = \frac{1}{4}$$

and $P(B/C) = \frac{P(B \cap C)}{P(C)} = \frac{1/3}{2/3} = \frac{1}{2}$

Now, $P(A \cup B/C) = P(A/C) + (B/C) - P(A \cap B/C)$

$$= \frac{1}{2} + \frac{1}{2} - \frac{1}{4} = \frac{3}{4}$$

37. Sachin Mehara is a final year student of civil engineering at IIT Delhi. As a final year real time project, he has got the job of designing a auditorium for cultural activities purpose. The shape of the floor of the auditorium is rectangular and it has a fixed perimeter, say. P.



Based on the above information, answer the following questions.

(i) If l and b represents the length and breadth of the rectangular region, then find the relationship between l ,b, p.

(ii) Find the area (A) of the floor, as a function of is l

(iii) College authority is interested in maximising the area of the floor A. For this purpose, find the value of l.

OR

Find the maximum area of the floor.

Sol :

(i) Perimeter of rectangular floor

$$= 2 \text{ (Length + breadth)}$$

$$P = 2(l + b)$$

(ii) Area A = length × breadth

$$A = l \times b$$

$$P = 2(l + b)$$

$$\frac{P}{2} = l + b$$

$$\frac{P}{2} - l = b$$

$$\frac{P - 2l}{2} = b$$

$$A = \frac{Pl - 2l^2}{2}$$

(iii) We have,

$$A = \frac{Pl - 2l^2}{2}$$

Differentiating w.r.t. l, we get

$$\frac{dA}{dl} = \frac{1}{2}(P - 4l)$$

For maximum area of floor, $\frac{dA}{dl} = 0$

$$\frac{1}{2}(P - 4l) = 0$$

$$P - 4l = 0$$

$$l = \frac{P}{4}$$

Clearly at $l = \frac{P}{4}$, $\frac{d^2A}{dl^2} = -2 < 0$

Area is maximum at $l = \frac{P}{4}$

OR

(iii) We have,

$$A = l \times b$$

For maximum area,

$$l = \frac{P}{4}$$

Now, $b = \frac{P - 2l}{2}$

$$= \frac{P}{2} - l = \frac{P}{2} - \frac{P}{4} = \frac{P}{4}$$

$$(A)_{\max} = l \times b = \frac{P}{4} \times \frac{P}{4} = \frac{P^2}{16} \text{ sq. units.}$$

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$$d = \frac{5(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|}$$

$$= \frac{|(2\hat{i} - \hat{j}) \cdot (-3\hat{i} + 2\hat{k})|}{\sqrt{5}}$$

$$= \frac{|-6|}{\sqrt{5}} = \frac{6}{\sqrt{5}} \text{ units}$$

OR

We have $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$

and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$

Comparing the given equations of lines with standard form we have

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$$

and $\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$.

we get $x_1 = 1,\ y_1 = 2,\ z_1 = 3;$

$a_1 = 2,\ b_1 = 3,\ c_1 = 4$

and $x_2 = 2,\ y_2 = 4,\ z_2 = 5;$

$a_2 = 3,\ b_2 = 4,\ c_2 = 5$

Now $\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = \begin{vmatrix} 2-1 & 4-2 & 5-3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$

$$= \begin{vmatrix} 1 & 2 & 2 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$$

$$= 1(15-16) - 2(10-12) + 2(8-9)$$

$$= -1 + 4 - 2 = 1$$

Now, $\sqrt{(b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 + (a_1b_2 - a_2b_1)^2}$

$= \sqrt{(3\times5 - 4\times4)^2 + (4\times3 - 5\times2)^2 + (2\times4 - 3\times3)^2}$

$= \sqrt{(15-16)^2 + (12-10)^2 + (8-9)^2}$

$= \sqrt{(-1)^2 + (2)^2 + (-1)^2}$

$= \sqrt{1+4+1} = \sqrt{6}$

$$\text{SD} = \frac{\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 + (a_1b_2 - a_2b_1)^2}}$$

$= \frac{1}{\sqrt{6}}$ units,

which is the required shortest distance.

SECTION - E

Case study based questions are compulsory.

36. At its simplest, a fair die states that each of the faces has a similar probability of landing facing up. A standard fair six-sided die, for example, can be regarded as "fair" if each of the faces consists of a probability of 1/6.



A fair die is rolled. Consider the events $A = \{1,3,5\}$, $B = \{2,3\}$, and $C = \{2,3,4,5,\}$

On the basis of above information, answer the following questions.

(i) Find the probability $P(A/B)$ and $P(B/A)$.

(ii) Find the probability $P(A/C)$, $P(A \cap B/C)$ and $P(A \cup B/C)$

Sol :

(i) We have,

$$S = \{1,2,3,4,5,6\}$$

$$A = \{1,3,5\}$$

$$B = \{2,3\}$$

and $C = \{2,3,4,5\}$

$$n(S) = 6$$

$$n(A) = 3$$

$$n(B) = 2$$

and $n(C) = 4$

$$P(A) = \frac{3}{6} = \frac{1}{2}, P(B)$$

$$= \frac{2}{6} = \frac{1}{3} P(C) = \frac{4}{6} = \frac{2}{3}$$

$$P(A \cap B) = \frac{1}{6}$$

and $P(A \cap C) = \frac{2}{6} = \frac{1}{3}$

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$$\int\left(\frac{v-\cos v}{v^2-2\sin v}\right)dv = -\int\frac{dx}{x}$$

$$\frac{1}{2}\log|v^2-2\sin v| = -\log|x|+\log C_1$$

$$\log\sqrt{v^2-2\sin v} = \log\left|\frac{C_1}{x}\right|$$

$$\sqrt{v^2-2\sin v} = \frac{C_1}{x}$$

$$\sqrt{\frac{y^2}{x^2}-2\sin\left(\frac{y}{x}\right)} = \frac{C_1}{x}$$

$$\sqrt{y^2-2\sin\left(\frac{y}{x}\right)} = C_1$$

$$y^2-2x^2\sin\left(\frac{y}{x}\right) = C_1^2$$

$$y^2-2x^2\sin\left(\frac{y}{x}\right) = C, \quad \text{where } C=C_1^2$$

OR

We have $(3xy+y^2)\,dx+(x^2+xy)\,dy = 0$

$$\frac{dy}{dx} = -\frac{3xy+y^2}{x^2+xy} \qquad ...(1)$$

$$\frac{dy}{dx} = -\frac{3\frac{y}{x}+\frac{y^2}{x^2}}{1+\frac{y}{x}} \qquad ...(1)$$

Which is a homogeneous differential equation as

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

Substituting $y=vx \Rightarrow \frac{dy}{dx} = v+x\frac{dv}{dx}$ in eq (1) we have

$$v+x\frac{dv}{dx} = -\frac{3v+v^2}{1+v}$$

$$x\frac{dv}{dx} = -\left(\frac{3v+v^2+v+v^2}{1+v}\right)$$

$$x\frac{dv}{dx} = -\left(\frac{2v^2+4v}{1+v}\right)$$

$$\frac{(1+v)\,dv}{2(v^2+2v)} = -\frac{dx}{x}$$

Integrating both sides, we get

$$\int\frac{1+v}{2(v^2+2v)}dv = -\int\frac{dx}{x}$$

Again, substituting $v^2+2v=z \Rightarrow (2v+2dv)=dz$

$$(1+v)\,dv = \frac{dz}{2}$$

Then, Eq. (2) becomes,

$$\int\frac{1}{2}\times\frac{dz}{2z} = -\int\frac{dx}{x}$$

$$\frac{1}{4}\log|z| = -\log|x|+\log|C|$$

$$\frac{1}{4}[\log|z|+4\log|x|] = \log|C|$$

$$\log|zx^4| = 4\log|C|$$

$$zx^4 = C^4 \Rightarrow zx^4 = C_1,$$

Where $C_1 = C^4$

$$x^4(v^2+2v) = C_1 \qquad z=v^2+2v$$

$$x^4\left(\frac{y^2}{x^2}+\frac{2y}{x}\right) = C_1 \qquad [\text{put } v=\frac{y}{x}] \ ...(3)$$

Also, given that $y=1$ for $x=1$.

Substituting $x=1$ and $y=1$ in Eq. (3), we get

$$1\left(\frac{1}{1}+\frac{2}{1}\right) = C_1 \Rightarrow C_1=3$$

Thus substituting $C_1=3$ in Eq. (3), we get

$$x^4\left(\frac{y^2}{x^2}+\frac{2y}{x}\right) = 3$$

$$y^2x^2+2yx^3 = 3$$

Which is the required particular solution.

35. Find the shortest distance between the lines

$$\vec{r} = (4\hat{i}-\hat{j})+\lambda(\hat{i}+2\hat{j}-3\hat{k})$$

and $$\vec{r} = (\hat{i}-\hat{j}+2\hat{k})+\mu(2\hat{i}+4\hat{j}-5\hat{k}).$$

OR

Find the shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}$.

Sol :

We have $\vec{r} = (4\hat{i}-\hat{j})+\lambda(\hat{i}+2\hat{j}-3\hat{k})$

and $\vec{r} = (\hat{i}-\hat{j}+2\hat{k})+\mu(2\hat{i}+4\hat{j}-5\hat{k})$

Comparing Eqs. (1) and (2), with $\vec{r}=\vec{a}_1+\lambda\vec{b}_1$ and $\vec{r}=\vec{a}_2+\mu\vec{b}_2$ respectively, we get

$$\vec{a}_1 = 4\hat{i}-\hat{j},\ \vec{b}_1=\hat{i}+2\hat{j}-3\hat{k}$$

and $$\vec{a}_2 = \hat{i}-\hat{j}+2\hat{k},\ \vec{b}_2=2\hat{i}+4\hat{j}-5\hat{k}$$

Here $\vec{a}_2-\vec{a}_1 = -3\hat{i}+2\hat{k}$

and $$\vec{b}_1\times\vec{b}_2 = \begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 2 & -3\\ 2 & 4 & -5\end{vmatrix}$$

$$= \hat{i}(-10+12)-\hat{j}(-5+6)+\hat{k}(4-4)$$

$$= 2\hat{i}-\hat{j}$$

$$|\vec{b}_1\times\vec{b}_2| = \sqrt{2^2+(-1)^2}$$

$$= \sqrt{4+1} = \sqrt{5}$$

Now, the shortest distance between the given lines,

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38. Pfizer Inc. is an American multinational pharmaceutical and biotechnology corporation headquartered on 42nd Street in Manhattan, New York City. The company was established in 1849 in New York by two German immigrants, Charles Pfizer and his cousin Charles F. Erhart. Pfizer develops and produces medicines and vaccines for immunology, oncology, cardiology, endocrinology, and neurology.



The purchase officer of the Pfizer informs the production manger that during the month, following supply of three chemicals, Asprin (A), Caffieine (C) and Decongestant (D) used in the production of three types of pain-killing tablet will be 16, 10 and 16 kg respectively. According to the specification, each strip of 10 tables of Paingo requires 2 gm of A, 3 gm of C and 1 gm of D. The requirements for other tables are:

X-prene	4 gm of A	1 gm of C	3 gm of D
Relaxo	1 gm of A	2 gm of C	3 gm of D

(i) Taking the suitable variable form the system of equation that represent given problem.

(ii) Use matrix inversion method to find the number of strips of each type so that raw materials are consumed entirely.

Sol :

Let the production manager produces x number of strips (of 10 tablets) of Paingo, y number of strips (of 10 tables) X-prene and z number of strips (of 10 tablets) Relaxo.

We enter the given information in following table.

	Paingo	**X-prene**	**Relaxo**	**Availability**
A	$2x$	$4y$	z	16000
C	$3x$	y	$2z$	10000
D	x	$3y$	$3z$	16000

From above table we have

$$2x + 4y + z = 16000 \quad ...(1)$$

$$3x + y + 2z = 10000 \quad ...(2)$$

and $$x + 3y + 3z = 16000 \quad ...(3)$$

The above system of equations can be written in matrix form as

$$\begin{bmatrix} 2 & 4 & 1 \\ 3 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16000 \\ 10000 \\ 16000 \end{bmatrix}$$

or $$AX = B$$

where $$A = \begin{bmatrix} 2 & 4 & 1 \\ 3 & 1 & 2 \\ 1 & 3 & 3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 16000 \\ 10000 \\ 16000 \end{bmatrix}$$

Since, $$|A| = 2(3-6) - 4(9-2) + 1(9-1)$$

$$= -6 - 28 + 8 = -26 \neq 0,$$

Thus A^{-1} exists, so that the unique solution of $AX = B$ is $X = A^{-1}B$.

Here, adj $A = \begin{bmatrix} -3 & -9 & 7 \\ -7 & 5 & -1 \\ 8 & -2 & -10 \end{bmatrix}$

and $$A^{-1} = \frac{\text{adj.}A}{|A|} = \frac{1}{-26}\begin{bmatrix} -3 & -9 & 7 \\ -7 & 5 & -1 \\ 8 & -2 & -10 \end{bmatrix}$$

Now $$X = A^{-1}B$$

$$= \frac{1}{-26}\begin{bmatrix} -3 & -9 & 7 \\ -7 & 5 & -1 \\ 8 & -2 & -10 \end{bmatrix}\begin{bmatrix} 16000 \\ 10000 \\ 16000 \end{bmatrix}$$

$$= \frac{1}{-26}\begin{bmatrix} -48000 & -900000 & +112000 \\ -1120000 & +50000 & -16000 \\ 128000 & -20000 & -160000 \end{bmatrix}$$

$$= \frac{-1}{26}\begin{bmatrix} -26000 \\ -78000 \\ -52000 \end{bmatrix} = \begin{bmatrix} 1000 \\ 3000 \\ 2000 \end{bmatrix}$$

Thus $x = 1000$, $y = 3000$ and $z = 2000$. Hence, the number of strips produced are: Paingo-1000, X -prene-3000 and Relaxo-2000.

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Sample Paper 08

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

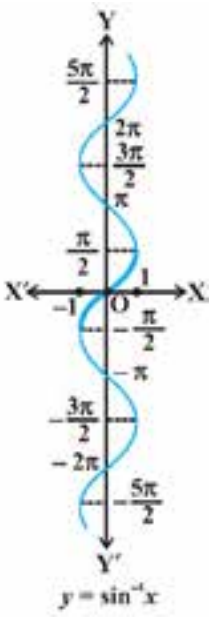
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What are the extreme points on the X-axis visible in the graph of $y = \sin^{-1}x$?



(a) $-\pi$ and π (b) $-\pi/2$ and $\pi/2$

(c) –1 and 1 (d) –2 and 2

Sol :

The graph extends horizontally from $x = -1$ to $x = 1$. At these points, the Y-values are $-\pi/2$ and $\pi/2$, respectively. This confirms that the extreme points on the X-axis are –1 and 1, as shown in the diagram.

Thus (c) is correct option.

2. The area of enclosed by $y = 3x - 5$, $y = 0$, $x = 3$ and $x = 5$ is

(a) 12 sq units

(b) 13 sq units

(c) $13\frac{1}{2}$ sq units

(d) 14 sq units

Sol :

The region is bounded by the curves $y = 3x - 5$, $y = 0$, $x = 3$ and $x = 5$.

Required area, $A = \int_3^5 (3x - 5)\,dx$

$$= \left[\frac{3x^2}{2} - 5x\right]_3^5$$

$$= \left(\frac{75}{2} - 25\right) - \left(\frac{27}{2} - 15\right)$$

$$= \frac{75}{2} - 25 - \frac{27}{2} + 15$$

$$= \frac{48}{2} - 10 = 14 \text{ sq units}$$

Thus (d) is correct option.

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3. The general solution of the differential equation $x(1+y^2)\,dx + y(1+x^2)\,dy = 0$ is

(a) $(1+x^2)(1+y^2) = 0$

(b) $(1+x^2)(1+y^2) = c$

(c) $(1+y^2) = c(1+x^2)$

(d) $(1+x^2) = c(1+y^2)$

Sol :

Given equation can be rewritten as

$$\int \frac{2x}{1+x^2}dx + \int \frac{2y}{1+y^2}dy = \log c$$

$$\log(1+x^2) + \log(1+y^2) = \log c$$

$$(1+x^2)(1+y^2) = c$$

Thus (b) is correct option.

4. The foot of the perpendicular from (0, 2, 3) to the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$ is

(a) $(-2, 3, 4)$ (b) $(2, -1, 3)$

(c) $(2, 3, -1)$ (d) $(3, 2, -1)$

Sol :

Let N be the foot of the perpendicular from the point (0, 2, 3) on the given line

$$\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$$

$$= \lambda \text{ (say)} \qquad ...(1)$$

Any point on the line is $P(5\lambda-3, 2\lambda+1, 3\lambda-4)$. If this point is N, then direction ratios of NP are $< 5\lambda-3-0, 2\lambda+1-2, 3\lambda-4-3 >$

i.e. $< 5\lambda-3, 2\lambda-1, 3\lambda-7 >$

Since, PN is perpendicular to line (1).

Hence, $5(5\lambda-3) + 2(2\lambda-1) + 3(3\lambda-7) = 0$

$$38\lambda - 38 = 0$$

$$\lambda = 1$$

Hence, required point is $(2, 3, -1)$.

Thus (c) is correct option.

5. If $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.3$, then $P\left(\frac{A'}{B}\right)$ is equal to

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$

(c) $\frac{2}{3}$ (d) $\frac{3}{4}$

Sol :

$$P(A'/B') = \frac{P(A' \cap B')}{P(B')} = \frac{P(A \cup B)'}{P(B')}$$

$$= \frac{1 - P(A \cup B)}{1 - P(B)}$$

$$= \frac{1 - [P(A) + P(B) - P(A \cap B)]}{1 - P(B)}$$

$$= \frac{1 - [0.5 + 0.4 - 0.3]}{1 - 0.4} = \frac{0.4}{0.6} = \frac{2}{3}$$

Thus (c) is correct option.

6. The value of $\sin\left[2\cos^{-1}\left(-\frac{3}{5}\right)\right]$ is

(a) $\frac{24}{25}$ (b) $-\frac{24}{25}$

(c) $\frac{7}{25}$ (d) none of these

Sol :

$$\sin\left[2\cos^{-1}\left(-\frac{3}{5}\right)\right] = 2\sin\left[\cos^{-1}\left(-\frac{3}{5}\right)\right]\cos\left[\cos^{-1}\left(-\frac{3}{5}\right)\right]$$

$$= 2\sqrt{1-\left(-\frac{3}{5}\right)^2}\left(-\frac{3}{5}\right) = -\frac{24}{25}$$

Thus (b) is correct option.

7. $\int \frac{\sin 2x}{\sin^2 x + 2\cos^2 x}dx$ is equal to

(a) $-\log(1+\sin^2 x) + C$

(b) $\log(1+\cos^2 x) + C$

(c) $-\log(1+\cos^2 x) + C$

(d) $\log(1+\tan^2 x) + C$

Sol :

We have $I = \int \frac{\sin 2x}{\sin^2 x + 2\cos^2 x}dx$

$$= \int \frac{\sin 2x}{1 - \cos^2 x + 2\cos^2 x}dx$$

$$= \int \frac{\sin 2x}{1 + \cos^2 x}dx$$

Substituting $1 + \cos^2 x = t \Rightarrow -2\cos x \sin x\,dx = dt$

$$-2\sin 2x\,dx = dt$$

$$I = -\int \frac{dt}{t} = -\log|t| + C$$

$$= -\log|1 + \cos^2 x| + C$$

Thus (c) is correct option.

8. $\int_0^{\pi} \sqrt{\frac{1+\cos 2x}{2}}\,dx$ is equal to

(a) 0 (b) 2

(c) 4 (d) -2

Sol :

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(a) y=c, y=d, the curve x=g(y), and the Y-axis

(b) x=c, x=d, the curve y=g(x), and the X-axis

(c) y=0, y=d, the curve x=g(y), and the X-axis

(d) x=0, x=d, the curve y=g(x), and the Y-axis

Sol :

(a) y=c, y=d, the curve x=g(y), and the Y-axis

The shaded region is clearly enclosed between the horizontal lines y=c (below) and y=d (above), the vertical Y-axis on the left, and the curve x=g(y) on the right. This makes option (a) correct, as all boundaries are marked directly in the figure.

18. If $R=\{(3,3),(6,6),(9,9),(12,12),(6,12),(3,9),(3,12),(3,6)\}$ is a relation on the set $A=\{3,6,9,12\}$. Then, the relation is

(a) An equivalence relation

(b) Reflexive and symmetric

(c) Reflexive and symmetric

(d) Only reflexive

Sol :

R is reflexive as (3, 3), (6, 6), (9, 9), (12, 12) $\in R$. R is not symmetric as (6, 12) $\in R$ but (12, 6) $\notin R$. R is transitive as the only pair which needs verification is (3, 6) and (6, 12) $\in R$. (3,12) $\in R$.

Thus (d) is correct option.

19. For any square matrix A with real number entries consider the following statements.

Assertion: $A+A'$ is a symmetric matrix.

Reason: $A-A'$ is a skew-symmetric matrix.

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

Let $B = A+A'$ then

$$B' = (A+A')' = A'+(A')' = A'+A = A+A' = B$$

Therefore,

$B = A+A'$ is a symmetric matrix.

Now, let $C = A-A'$

$$C' = (A-A')' = A'-(A')' = A'-A = -(A-A') = -C$$

Therefore, $C=A-A'$ is a skew-symmetric matrix.

Here assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

Thus (b) is correct option.

20. Let A and B be two events associated with an experiment such that $P(A\cap B)=P(A)P(B)$

Assertion: $P(A\mid B)=P(A)$ and $P(B\mid A)=P(B)$

Reason: $P(A\cup B)=P(A)+P(B)$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

Assertion is true; Reason is false.

Since, $P(A\cap B)=P(A)P(B)$,

Therefore, A and B are independent events.

$$P(A\mid B)=\frac{P(A\cap B)}{P(B)}=\frac{P(A)P(B)}{P(B)}=P(A)$$

Similarly, $P(B\mid A)=P(B)$

Thus, assertion is true. However, reason is not true for independent events. For example, when a dice is rolled once, then the events

A : 'an even number' shows up

and B : 'a multiple of 3' show up are independent as

$$P(A)P(B)=\frac{3}{6}\times\frac{2}{6}=\frac{1}{6}$$

$$=P(A\cap B)(A=\{2,4,6,\},B=\{3,6\})$$

But, $P(A\cup B)=(\{2,3,4,6\})$

$$=\frac{4}{6}$$

$$\neq P(A)+P(B)$$

$$\left(P(A)+P(B)=\frac{3}{6}+\frac{2}{6}=\frac{5}{6}\neq\frac{4}{6}\right)$$

Thus (c) is correct option.

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$$= t\left(\hat{i} + 3\hat{j} - \hat{k}\right)$$

$$|\vec{r}| = t\sqrt{(1+9+1)} = \sqrt{11}\,t$$

$$\frac{\vec{r}}{|\vec{r}|} = \frac{\hat{i} + 3\hat{j} - \hat{k}}{\sqrt{11}}$$

$$= \frac{1}{\sqrt{11}}\left(\hat{i} + 3\hat{j} - \hat{k}\right)$$

Thus (a) is correct option.

14. A mapping $f : n \to N$, where N is the set of natural numbers is define as

$$f(n) = \begin{cases} n^2, & \text{for } n \text{ odd} \\ 2n+1, & \text{for } n \text{ even} \end{cases}$$

for $n \in N$. Then, f is

(a) Surjective but not injective

(b) Injective but not surjective

(c) Bijective

(d) neither injective nor surjective

Sol :

Since, $f(n) = \begin{cases} n^2, & \text{if } n \text{ is odd.} \\ 2n+1, & \text{if } n \text{ is even} \end{cases}$

$$f(1) = 1^2 = 1$$

$$f(2) = 2(2) + 1 = 5$$

$$f(3) = 3^2 = 9$$

$$f(4) = 2(4) + 1 = 9$$

$$f(3) = f(4)$$

So, f is not injective.

Also, f is not surjective as some element of N (co-domain) is not the image of any element of N.

Thus (d) is correct option.

15. The function $f(x) = x^2 - 2x$ is increasing in the interval

(a) $x \neq -1$ (b) $x \geq -1$

(c) $x \neq 1$ (d) $x \geq 1$

Sol :

Given, $f(x) = x^2 - 2x$

$$f'(x) = 2x - 2$$

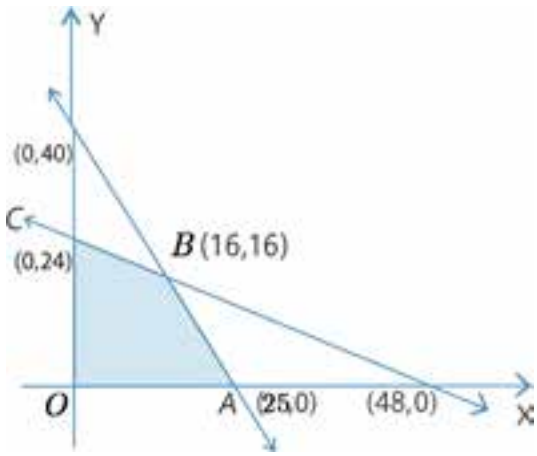
$$= 2(x-1)$$

So, $f(x)$ is increasing, if

$$2(x-1) \geq 0, \text{ie}, x \geq 1$$

Thus (d) is correct option.

16. Using the same feasible region as shown in the figure, determine the minimum value of the objective function $z = 5x + 2y$. The feasible region has vertices at O(0,0),A(25,0),B(16,16), and C(0,24). At which point is this minimum value attained, and what is it equal to?



(a) 50 (b) 80

(c) 100 (d) 0

Sol :

(d) 0

The method to solve such problems is called the corner point method. According to this method, the value of the objective function must be calculated at all corner points of the feasible region.

At O(0,0) : z = 5(0)+2(0) = 0

At A(25,0) : z = 5(25)+2(0) = 125

At B(16,16) : z = 5(16)+2(16) = 80+32 = 112

At C(0,24) : z = 5(0)+2(24) = 48

Among these values, the smallest is clearly 0, which occurs at the origin (0,0). Hence, the minimum value of the objective function is 0.

17. In the given figure, the shaded region is bounded by which of the following?

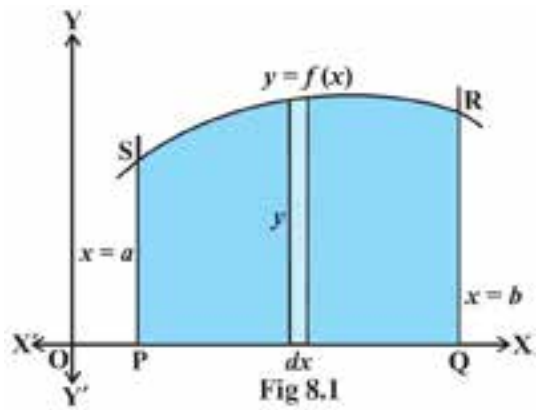


Fig 8.1

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$$\int_0^{\pi}\sqrt{\frac{1+\cos 2x}{2}}\,dx = \int_0^{\pi}\sqrt{\frac{2\cos^2 x}{2}}\,dx$$
$$= \int_0^{\pi}\sqrt{\cos^2 x}\,dx$$
$$= \int_0^{\pi}|\cos x|\,dx \quad [\because \sqrt{x^2} = |x|]$$
$$= \int_0^{\pi/2}\cos x\,dx - \int_{\pi/2}^{\pi}\cos x\,dx$$
$$= [\sin x]_0^{\pi/2} - [\sin x]_{\pi/2}^{\pi}$$
$$= [1-0]-[0-1] = 1+1 = 2$$

Thus (b) is correct option.

9. If $AB = A$ and $BA = B$, then B^2 is equal to

(a) B (b) A

(c) $-B$ (d) B^3

Sol :

Given, $AB = A$ and $BA = B$

Now, $BA = B$

$$BAB = BB$$
$$B(A) = B^2$$
$$B = B^2$$

Thus (a) is correct option.

10. The solution of $e^{dy/dx} = x+1, y(0) = 3$, is

(a) $y = x\log x - x + 2$

(b) $y = (x+1)\log(x+1) - x + 3$

(c) $y = (x+1)\log(x+1) + x + 3$

(d) $y = x\log x + x + 3$

Sol :

$$e^{dy/dx} = x+1$$
$$\frac{dy}{dx} = \log(x+1)$$
$$y = (x+1)\log(x+1) - x + c$$

At $x = 0,\ y = 3,$

then $c = 3$

$$y = (x+1)\log(x+1) - x + 3$$

Thus (b) is correct option.

11. $(x^2+xy)\,dy = (x^2+y^2)\,dx$ is

(a) $\log x = \log(x-y) + \frac{y}{x} + c$

(b) $\log x = 2\log(x-y) + \frac{y}{x} + c$

(c) $\log x = \log(x-y) + \frac{x}{y} + c$

(d) none of the above

Sol :

We have $\frac{dy}{dx} = \frac{x^2+y^2}{x^2+xy}$

Putting $y = vx$ we have

$$v + x\frac{dv}{dx} = \frac{1+v^2}{1+v}$$
$$x\frac{dv}{dx} = \frac{1+v^2}{1+v} - v = \frac{1-v}{1+v}$$
$$\left(-1 + \frac{2}{1-v}\right)dv = \frac{dx}{x}$$

On integrating both sides, we get

$$-\frac{y}{x} - 2\log\left(1-\frac{y}{x}\right) + c = \log x$$
$$\frac{y}{x} + 2\log(x-y) + c = \log x$$

Thus (b) is correct option.

12. If $\vec{a}$ and $\vec{b}$ are position vectors of A and B respectively, then the position vector of a point C in AB produced such that $\overrightarrow{AC} = 3\overrightarrow{AB}$, is

(a) $3\vec{a} - \vec{b}$ (b) $3\vec{b} - \vec{a}$

(c) $3\vec{a} - 2\vec{b}$ (d) $3\vec{b} - 2\vec{a}$

Sol :

Given that,

$$\overrightarrow{AC} = 3\overrightarrow{AB}$$
$$\vec{c} - \vec{a} = 3(\vec{b} - \vec{a})$$
$$\vec{c} = 3\vec{b} - 2\vec{a}$$

Thus (d) is correct option.

13. If $\vec{a} = \hat{i} + \hat{j}, \vec{b} = 2\hat{j} - \hat{k}$ and $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$, $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$, then $\frac{\vec{r}}{|\vec{r}|}$ is equal to

(a) $\frac{1}{\sqrt{11}}(\hat{i} + 3\hat{j} - \hat{k})$

(b) $\frac{1}{\sqrt{11}}(\hat{i} - 3\hat{j} + \hat{k})$

(c) $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$

(d) none of these

Sol :

Since, $\vec{r} \times \vec{a} + \vec{r} \times \vec{b} = \vec{b} \times \vec{a} + \vec{a} \times \vec{b} = \vec{0}$

$$\vec{r} \times (\vec{a} + \vec{b}) = \vec{0}$$

$\vec{r}$ is parallel to $\vec{a} + \vec{b}$

$$\vec{r} = t(\vec{a} + \vec{b}), \quad \text{where } t \text{ is some scalar}$$

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SECTION - B

Section B consists of 5 questions of 2 marks each.

21. Find $|\vec{x}|$, if for a unit vector $\hat{a}$, $(\vec{x}-\vec{a})\cdot(\vec{x}+\vec{a})=15$.

Sol :

If $\hat{a}$ is a unit vector, then $|\hat{a}|=1$

Now, we have $(\vec{x}-\hat{a})\cdot(\vec{x}+\hat{a}) = 15$

$$\vec{x}\cdot\vec{x}-\hat{a}\cdot\vec{x}+\vec{x}\cdot\hat{a}-\hat{a}\cdot\hat{a} = 15$$

$$\vec{x}\cdot\vec{x}-\hat{a}\cdot\vec{x}+\hat{a}\cdot\vec{x}-\hat{a}\cdot\hat{a} = 15$$

$$|\vec{x}|^2-|\hat{a}|^2 = 15$$

$$|\vec{x}|^2-1 = 15 \qquad |\hat{a}|=1$$

$$|\vec{x}|^2 = 16$$

$$|\vec{x}| = 4$$

as length is always positive.

22. If $A=\{1,2,3\}, B=\{4,5,6,7\}$ and $f=\{(1,4),(2,5),(3,6)\}$ is a function from A to B. State whether f is one&one or not.

Sol :

Given, $A = \{1,2,3\}$

$B = \{4,5,6,7\}$

and $f: A \to B$ is defined as

$f = \{(1,4),(2,5),(3,6)\}$

i.e. $f(1)=4,\ f(2)=5,\ f(3)=6$

It can be seen that the images of distinct elements of A under f are distinct. So, f is one-one.

23. It is given that the events A and B are such that $P(A)=\frac{1}{4}$, $P(\frac{A}{B})=\frac{1}{2}$ and $P(\frac{B}{A})=\frac{2}{3}$. Then, find $P(B)$?

OR

Find the probability distribution of X, the number of heads in a simultaneous toss of two coins.

Sol :

We have $P\left(\frac{A}{B}\right) = \frac{P(A\cap B)}{P(B)}$

and $P\left(\frac{B}{A}\right) = \frac{P(B\cap A)}{P(A)}$

$$P(B) = \frac{P\left(\frac{B}{A}\right)\cdot P(A)}{P\left(\frac{A}{B}\right)} = \frac{\frac{2}{5}\cdot\frac{1}{4}}{\frac{1}{2}} = \frac{1}{3}$$

OR

When two coins are tossed, there may be 1 head, 2 heads or on head at all. Thus, the possible values of X are 0, 1, 2.

Now, P (Getting no head)

$$P(X=0) = P(TT) = \frac{1}{4}$$

P (Getting one head)

$$P(X=1) = P\ (HT \text{ or } TH) = \frac{2}{4} = \frac{1}{2}$$

P (Getting two head)

$$P(X=2) = P(HH) = \frac{1}{4}$$

Thus, the required probability distribution of X is

X	0	1	2
$P(X)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

24. Find the vector equation of the line passing through the point $A(1,2,-1)$ and parallel to the line $5x-25=14-7y=35z$.

Sol :

We have $5x-25 = 14-7y = 35z$.

$$\frac{x-5}{\frac{1}{5}} = \frac{2-y}{\frac{1}{7}} = \frac{z}{\frac{1}{35}}$$

$$\frac{x-5}{\frac{1}{5}} = \frac{y-2}{-\frac{1}{7}} = \frac{z}{\frac{1}{35}}$$

Direction ratio of the given line are $\frac{1}{5}, -\frac{1}{7}, \frac{1}{35}$.

Direction ratio of a line parallel to the given line are $\frac{1}{5}, -\frac{1}{7}, \frac{1}{35}$.

The required equation of a line passing through the point $A(1,2,-1)$ and parallel to the given line is

$$\frac{x-1}{\frac{1}{5}} = \frac{y-2}{-\frac{1}{7}} = \frac{z+1}{\frac{1}{35}}$$

25. Evaluate $\int (ax-b)^3\,dx$.

OR

Evaluate $\int \frac{(1+\log x)^2}{x}\,dx$.

Sol :

We have $I = \int (ax+b)^3\,dx$

Substituting $t = ax+b \Rightarrow \frac{dt}{dx} = a \Rightarrow \frac{dt}{a} = dx$ we have

$$I = \int \frac{t^3}{a}\,dt = \frac{1}{a}\cdot\frac{t^4}{4}+C = \frac{(ax+b)^4}{4a}+C$$

$$= \int \frac{t^3}{a}\,dt = \frac{1}{a}\cdot\frac{t^4}{4}+C = \frac{(ax+b)^4}{4a}+C$$

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31. If $(x-y)\,e^{\frac{x}{x-y}} = a$, prove that $y\dfrac{dy}{dx} + x = 2y$.

Sol :

We have $(x-y)\cdot e^{\frac{x}{x-y}} = a$

Taking log both sides, we get

$$\log\left[(x-y)\cdot e^{\frac{x}{x-y}}\right] = \log a$$

$$\log(x-y) + \log e^{\frac{x}{x-y}} = \log a$$

$$\log(x-y) + \frac{x}{x-y}\log_e e = \log a$$

$$\log(x-y) + \frac{x}{x-y} = \log a$$

Differentiating both sides w.r.t. x, we get

$$\frac{d}{dx}[\log(x-y)] + \frac{d}{dx}\left(\frac{x}{x-y}\right) = \frac{d}{dx}(\log a)$$

$$\frac{1}{x-y}\frac{d}{dx}(x-y) + \frac{(x-y)\dfrac{d}{dx}(x) - x\dfrac{d}{dx}(x-y)}{(x-y)^2} = 0$$

$$\frac{1}{x-y}\cdot(1-y') + \frac{(x-y)-x(1-y')}{(x-y)^2} = 0$$

$$(x-y)(1-y') + x - y - x(1-y') = 0$$

$$yy' + x - 2y = 0$$

$$y\frac{dy}{dx} + x = 2y$$

SECTION - D

Section D consists of 4 questions of 5 marks each.

32. Find the particular solution of the differential equation $\log\left(\dfrac{dy}{dx}\right) = 3x + 4y$, given that $y = 0$, when $x = 0$.

OR

Find the particular solution of the differential equation $x(1+y^2)\,dx - y(1+x^2)\,dy = 0$, given that $y = 1$, when $x = 0$.

Sol :

We have $\log\left(\dfrac{dy}{dx}\right) = 3x + 4y$

$$\frac{dy}{dx} = e^{3x+4y}$$

$$\frac{dy}{dx} = e^{3x}\,e^{4y}$$

Separating the variables, we get

$$\frac{1}{e^{4y}}dy = e^{3x}\,dx$$

Integrating both sides, we have

$$\int e^{-4y}\,dy = \int e^{3x}\,dx \qquad \ldots(1)$$

$$\int e^{-4y}\,dy = \int e^{3x}\,dx$$

$$= \frac{e^{3x}}{3} + C \qquad \ldots(1)$$

Also, given that $y = 0$, when $x = 0$.

Substituting $y = 0$ and $x = 0$ in Eq. (1), we get

$$\frac{e^{-4(0)}}{-4} = \frac{e^{3(0)}}{3} + C$$

$$-\frac{1}{4} = \frac{1}{3} + C$$

$$C = -\frac{1}{4} - \frac{1}{3}$$

$$= -\frac{7}{12}$$

Substituting the value of C in Eq. (1), we get

$$\frac{e^{-4y}}{-4} = \frac{e^{3x}}{3} - \frac{7}{12}$$

$$4e^{3x} + 3e^{-4y} - 7 = 0$$

Which is the required particular solution of given differential equation.

OR

We have $x(1+y^2) - y(1+x^2)\,dy = 0$

$$x(1+y^2)\,dx = y(1+x^2)\,dy$$

Separating the variables, we get

$$\frac{y}{(1+y^2)}dy = \frac{x}{(1+x^2)}dx$$

Integrating both sides, we have

$$\int \frac{y}{1+y^2}dy = \int \frac{x}{(1+x^2)}dx$$

$$\frac{1}{2}\log|1+y^2| = \frac{1}{2}\log|1+x^2| + C \qquad \ldots(1)$$

Also, given that $y = 1$, when $x = 0$.

Substituting the values of x and y in Eq. (1), we get

$$\frac{1}{2}\log|1+(1)^2| = \frac{1}{2}|1+(0)^2| + C$$

$$\frac{1}{2}\log 2 = C$$

Substituting $C = \dfrac{1}{2}\log 2$ in Eq. (1), we get

$$\frac{1}{2}\log|1+y^2| = \frac{1}{2}\log|1+x^2| + \frac{1}{2}\log 2$$

$$\log|1+y^2| = \log|1+x^2| + \log 2$$

$$\log|1+y^2| - \log|1+x^2| = \log 2$$

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(i) Strictly increasing in $(-3,0)$ and $(5,\infty)$.

(ii) Strictly decreasing in $(-\infty,-3)$ and $(0,5)$.

28. Let $\vec{a}=4\hat{i}+5\hat{j}-\hat{k}$, $\vec{b}=\hat{i}-4\hat{j}+5\hat{k}$ and $\vec{c}=3\hat{i}+\hat{j}-\hat{k}$. Find a vector $\vec{d}$ which is perpendicular to both $\vec{c}$ and $\vec{b}$ and $\vec{d}\cdot\vec{a}=21$.

Sol :

We have, $\vec{a}=4\hat{i}+5\hat{j}-\hat{k}$,

$\vec{b}=\hat{i}-4\hat{j}+5\hat{k}$

and $\vec{c}=3\hat{i}+\hat{j}-\hat{k}$

Since, $\vec{d}$ is perpendicular to both $\vec{c}$ and $\vec{b}$.

$$\vec{d}=\lambda(\vec{c}\times\vec{d})$$

$$=\lambda\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\3&1&-1\\1&-4&5\end{vmatrix}$$

$$=\lambda[\hat{i}(5-4)-\hat{j}(15+1)+\hat{k}(-12-1)]$$

$$\vec{d}=\lambda(\hat{i}-16\hat{j}-13\hat{k}) \quad ...(1)$$

Since $\vec{d}\cdot\vec{a}=21$ we have

$$\lambda(\hat{i}-16\hat{j}-13\hat{k})\cdot(4\hat{i}+5\hat{j}-\hat{k})=21$$

$$\lambda(4-80+13)=21$$

$$\lambda(-63)=21$$

$$\lambda=\frac{-1}{3}$$

Now, from Eq. (1), we get

$$\vec{d}=-\frac{1}{3}(\hat{i}-16\hat{j}-13\hat{k})$$

29. Evaluate $\int\frac{x\cos^{-1}x}{\sqrt{1-x^2}}dx$.

OR

Evaluate $\int\frac{\sin^6x+\cos^6x}{\sin^2x\cos^2x}dx$.

Sol :

We have $I=\int\frac{x\cos^{-1}x}{\sqrt{1-x^2}}dx$

Substituting $\cos^{-1}x=t$ we get

$$-\frac{1}{\sqrt{1-x^2}}dx=dt$$

Thus $I=-\int t\cos t\,dt$

Using integration by parts, taking t as the first function and $\cos t$ as the second function, we get

$$I=-\int\underset{I}{t}\,\underset{II}{\cos t}\,dt$$

$$=-t\int\cos t\,dt+\int\left[\frac{d}{dt}(t)\cdot\int\cos dt\right]dt$$

$$=-t\sin t+\int(1\times\sin t)\,dt$$

$$=-t\sin t+\int\sin t\,dt$$

$$=-t\sin t-\cos t+C$$

$$=-t\sqrt{1-\cos^2t}-\cos t+C$$

Substituting $t=\cos^{-1}x\Rightarrow x=\cos t$ we have

$$I=-\cos^{-1}x\sqrt{1-x^2}-x+C$$

OR

We have $I=\int\frac{\sin^6x+\cos^6x}{\sin^2x\cos^2x}dx$

$$=\int\frac{(\sin^2x)^3+(\cos^2x)^3}{\sin^2x\cos^2x}dx$$

Using the fact $a^3+b^3=(a+b)^3-3ab(a+b)$ we have

$$I=\int\frac{(\sin^2x+\cos^2x)^3-3\sin^2x\cos^2x(\sin^2x+\cos^2x)}{\sin^2x\cos^2x}dx$$

$$=\int\frac{(1)^3-3\sin^2x\cos^2x}{\sin^2x\cos^2x}dx \quad \sin^2x+\cos^2x=1$$

$$=\int\frac{1}{\sin^2x\cos^2x}dx-3\int\frac{\sin^2x\cos^2x}{\sin^2x\cos^2x}dx$$

$$=\int\frac{\sin^2x+\cos^2x}{\sin^2x\cos^2x}dx-3\int1\,dx$$

$$=\int\left[\frac{\sin^2x}{\sin^2x\cos^2x}+\frac{\cos^2x}{\sin^2x\cos^2x}\right]dx-3x$$

$$=\int(\sec^2x+\operatorname{cosec}^2x)\,dx-3x$$

$$=\int\sec^2x\,dx+\int\operatorname{cosec}^2x-3x$$

$$=\tan x-\cot x-3x+C$$

30. Write the value of $\tan^{-1}\left[2\sin\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right]$.

Sol :

$$\tan^{-1}\left[2\sin\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right]$$

$$=\tan^{-1}\left[2\sin\left\{2\cos^{-1}\left(\cos\frac{\pi}{6}\right)\right\}\right]$$

$$=\tan^{-1}\left[2\sin\left\{2\times\frac{\pi}{6}\right\}\right]$$

$$=\tan^{-1}\left(2\sin\frac{\pi}{3}\right)$$

$$=\tan^{-1}\left(2\cdot\frac{\sqrt{3}}{2}\right)$$

$$=\tan^{-1}(\sqrt{3})$$

$$=\tan^{-1}\left(\tan\frac{\pi}{3}\right)=\frac{\pi}{3}$$

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$$= \int \frac{t^3}{a}\,dt = \frac{1}{a}\cdot\frac{t^4}{4} + C = \frac{(ax+b)^4}{4a} + C$$

Substituting $t = (ax+b)$ we have

$$I = \frac{(ax+b)^4}{4a} + C$$

OR

We have $I = \int \frac{(1+\log x)^2}{x}\,dx$

Substituting $1+\log x = t \Rightarrow \frac{1}{x}\,dx = dt$ we have

$$I = \int \frac{(1+\log x)^2}{x}\,dx = \int t^2\,dt = \frac{t^3}{3} + C$$

$$= \int \frac{(1+\log x)^2}{x}\,dx = \int t^2\,dt = \frac{t^3}{3} + C$$

$$= \int \frac{(1+\log x)^2}{x}\,dx = \int t^2\,dt = \frac{t^3}{3} + C$$

Substituting $t = 1+\log x$ we have

$$I = \frac{(1+\log x)^3}{3} + C$$

SECTION - C

Section C consists of 6 questions of 3 marks each.

26. Find the maximum value of $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+\sin\theta & 1 \\ 1 & 1 & 1+\cos\theta \end{vmatrix}$

Sol :

We have $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+\sin\theta & 1 \\ 1 & 1 & 1+\cos\theta \end{vmatrix}$

Expanding along first row i.e. R_1, we get

$$\Delta = 1\begin{vmatrix} 1+\sin\theta & 1 \\ 1 & 1+\cos\theta \end{vmatrix} - 1\begin{vmatrix} 1 & 1 \\ 1 & 1+\cos\theta \end{vmatrix} + 1\begin{vmatrix} 1 & 1+\sin\theta \\ 1 & 1 \end{vmatrix}$$

$$= [(1+\sin\theta)(1+\cos\theta) - 1] - [1+\cos\theta - 1] + [1 - 1 - \sin\theta]$$

$$= 1 + \cos\theta + \sin\theta + \sin\theta\cos\theta - 1 - \cos\theta - \sin\theta$$

$$= \sin\theta\cos\theta$$

$$= \left|\frac{1}{2}\right|(2\sin\theta\cos\theta) = \frac{1}{2}\sin 2\theta$$

Since maximum value of $\sin 2\theta$ is 1.

$$\Delta_{\max} = \frac{1}{2}\times 1 = \frac{1}{2}.$$

27. Show that the function $f(x) = x^3 - 3x^2 + 3x$, $x \in R$ is increasing on R.

OR

Find the intervals in which the function

$$f(x) = \frac{3}{2}x^4 - 4x^3 - 45x^2 + 51 \text{ is}$$

(i) strictly increasing

(ii) strictly decreasing.

Sol :

We know that, a continuous function $y = f(x)$ is said to be increasing on R, if $\frac{dy}{dx} \geq 0$, $\forall\, x \in R$.

We have $y = x^3 - 3x^2 + 3x$

Differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = 3x^2 - 6x + 3$$

$$= 3(x^2 - 2x + 1)$$

$$= 3(x-1)^2$$

Here $3(x-1)^2 \geq 0$ for all real values of x, i.e. $\forall\, x \in R$. Thus $\frac{dy}{dx} \geq 0$, $\forall\, x \in R$. Hence, the given function is increasing on R.

OR

We have $f(x) = \frac{3}{2}x^4 - 4x^3 - 45x^2 + 51$

Differentiating both sides w.r.t. x, we get

$$f'(x) = 6x^3 - 12x^2 - 90x$$

For strictly increasing or strictly decreasing, put $f'(x) = 0$, we get

$$6x^3 - 12x^2 - 90x = 0$$

$$6x(x^2 - 2x - 15) = 0$$

$$6x(x+3)(x-5) = 0$$

$$x = 0,\ -3,\ 5$$

Now, we find intervals in which $f(x)$ is strictly increasing or strictly decreasing.

Interval	$f'(x) = 6(x+3)x(x-5)$	**Sing of** $f'(x)$
$x < -3$	$(-)(-)(-)$	$-$ ve
$-3 < x < 0$	$(+)(-)(-)$	$+$ ve
$0 < x < 5$	$(+)(+)(-)$	$-$ ve
$x > 5$	$(+)(+)(+)$	$+$ ve

A function $f(x)$ is said to be strictly increasing, if $f'(x) > 0$ and it is said to be strictly decreasing, if $f'(x) < 0$. So, the given function $f(x)$ is

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$$\log\left|\frac{1+y^2}{1+x^2}\right| = \log 2$$

$$\frac{1+y^2}{1+x^2} = 2$$

$$1+y^2 = 2+2x^2$$

$$y^2-2x^2-1 = 0$$

Which is the required particular solution of given differential equation.

33. Maximize $Z=3x+2y$ subject to $x+2y\le 0$, $3x+y\le 15$, x, $y\ge 0$.

OR

Minimise $Z=x+2y$ subject to $2x+y\ge 3$, $x+2y\ge 6$, x, $y\ge 0$. Show that the minimum of Z occurs at more than two points.

Sol :

Maximize $Z = 3x+2y$

Subject to $x+2y \le 10$ ...(i)

$3x+y \le 15$ ...(ii)

and $x\ge 0$, $y\ge 0$

(i) Region corresponding to $x+2y\le 10$:

Line $x+2y=10$			
x	0	10	Point $(0,0)$ is true for $x+2y\le 10$. So, the region is towards the origin.
y	5	0	

(ii) Region corresponding to $3x+y\le 15$:

Line $3x+y=15$			
x	0	5	Point $(0,0)$ is true for $3x+y\le 15$. So, the region is towards the origin.
y	15	0	

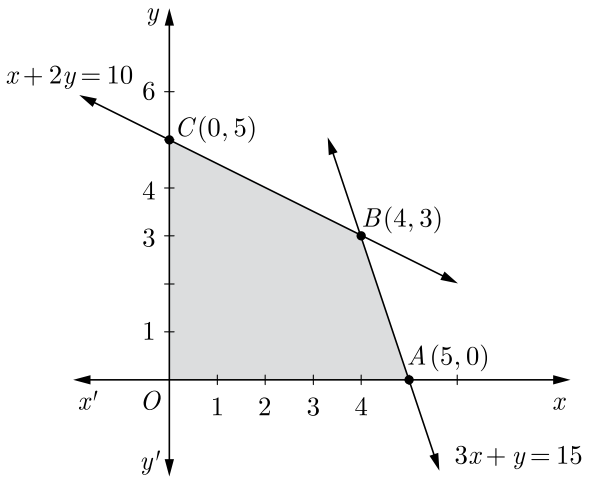
Since $x\ge 0$, $y\ge 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is $OABCO$ which is bounded. Solving equations $x+2y=10$ and $3x+y=15$, we get $B(4,3)$ The corner points of the feasible region are $O(0,0)$, $A(5,0)$, $B(4,3)$ and $C(0,5)$.

Since, the feasible region is a bounded region, we can check the objective function at all the corner to find the maxima. The values of objective function Z at these points are as follows.

Corner points	$Z=3x+2y$
$O(0,0)$	$Z=0+0=0$
$A(5,0)$	$Z=3\times 5+0=15$
$B(4,3)$	$Z=3\times 4+2\times 3=18$
$C(0,5)$	$Z=0+2\times 5=10$

Thus, Z is maximum at $B(4,3)$ and the maximum value is 18.



OR

Minimise $Z = x+2y$

Subject to $2x+y \ge 3$ ...(i)

$x+2y \ge 6$ ...(ii)

and $x\ge 0$, $y\ge 0$

(i) Region corresponding to $2x+y\geqslant 3$:

Line $2x+y=3$			
x	0	$\frac{3}{2}$	Point $(0,0)$ is false for $2x+y\geqslant 3$. So, the region is away from the origin.
y	3	0	

(ii) Region corresponding to $x+2y\geqslant 6$:

Line $x+2y=6$			
x	0	6	Point $(0,0)$ is false for $x+2y\geqslant 6$. So, the region is away from the origin.
y	3	0	

Since $x\ge 0$, $y\ge 0$ the feasible region lies in the first quadrant.

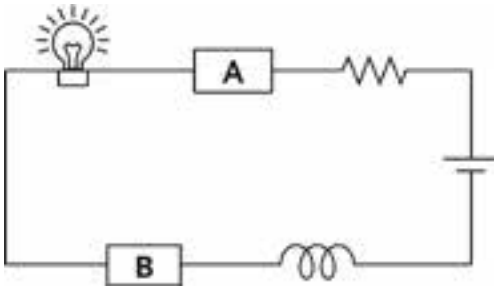
Now we draw all line on the graph and find the common area. Clearly, feasible region is ABC which is open and unbounded region. The corner points of the feasible region are $A(6,0)$ and $B(0,3)$.

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37. An electro-mechanical assembler basically makes machines or/and other assemblies that contain electronic components like wires or microchips. Typically assemblers use blueprints, work instructions, and computer software to manufacture whatever they are working on.



An electronic assembly consists of two sub-systems say A and B as shown below.



From previous testing procedures, the following probabilities are assumed to be known $P(A \text{ fails}) = 0.2$, $P(B \text{ fails alone}) = 0.15$, $P(A \text{ and } B \textit{ fail}) = 0.15$.

On the basis of above information, answer the following questions.

(i) Find the probability P(B fails) and the probability P(A fails alone).

(ii) Find the probability P(whole system fail) and the probability P(A fails/B has failed).

Sol :

Consider the following events

Let E be the event that sub-system A fails and F be the event that sub-system B fails.

(i) We have,

$$P(\overline{E} \cap F) = 0.15$$

$$P(F) - P(E \cap F) = 0.15$$

$$P(F) = 0.15 + P(E \cap F)$$

$$= 0.15 + 0.15 = 0.30$$

$$P(B \text{ fails}) = P(F) = 0.30$$

Now, $P(E \cap \overline{F}) = P(E) - P(E \cap F)$

$$= 0.2 - 0.15 = 0.05$$

$$P\ (A \text{ fails alone}) = P(E \cap \overline{F}) = 0.05$$

(ii) P (whole system fail) $= P(E \cup F)$

$$= P(E) + P(F) - P(E \cap F)$$

$$= 0.2 + 0.3 - 0.15$$

$$= 0.5 - 0.15 = 0.35$$

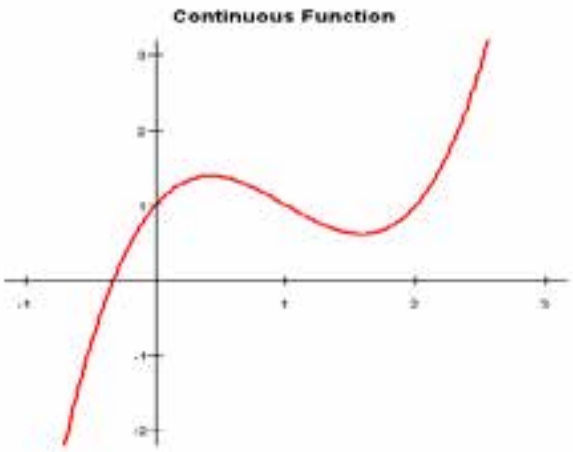
Now, $P\ (A \text{ fails}/B \text{ has failed}) = P(E/F)$

$$= \frac{P(E \cap F)}{P(F)}$$

$$= \frac{0.15}{0.30}$$

$$= \frac{1}{2} = 0.5$$

38. In mathematics, a continuous function is a function such that a continuous variation of the argument induces a continuous variation of the value of the function. This means that there are no abrupt changes in value, known as discontinuities.



Let $f(x)$ be a continuous function defined on $[a, b]$, then

$$\int_a^b f(x)\,dx = \int_0^b f(a+b-x)\,dx$$

On the basis of above information, answer the following questions.

(i) Evaluate $\int_1^2 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{3-x}}\,dx$

(ii) Evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \log \tan x\,dx$

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point intersection.

Sol :

We have $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{-7} = \lambda$ (let) ...(1)

and $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5} = \mu$ (let) ...(2)

Then, any point on line (1) is of the form

$$P(3\lambda - 1, 5\lambda - 3, 7\lambda - 5) \quad ...(3)$$

and any point on line (2) is of the form

$$Q(u + 2, 3\mu + 4, 5\mu + 6) \quad ...(4)$$

If lines (1) and (2) intersect, then these points must coincide for some λ and μ.

Consider, $3\lambda - 1 = \mu + 2$

$$5\lambda - 3 = 3\mu + 4$$

and $7\lambda - 5 = 5\mu + 6$

$$3\lambda - \mu = 3 \quad ...(5)$$

$$5\lambda - 3\mu = 7 \quad ...(6)$$

and $7\lambda - 5\mu = 11 \quad ...(7)$

Multiplying Eq. (5) by 3 and then subtracting Eq, (6), we get

$$9\lambda - 3\mu - 5\lambda + 3\mu = 9 - 7$$

$$4\lambda = 2 \Rightarrow \lambda = \frac{1}{2}$$

Substituting the value of λ in Eq. (5), we get

$$3 \times \frac{1}{2} - \mu = 3$$

$$\frac{3}{2} - \mu = 3 \Rightarrow \mu = -\frac{3}{2}$$

Substituting the values of λ and μ in Eq. (7), we get

$$7 \times \frac{1}{2} - 5\left(-\frac{3}{2}\right) = 11$$

$$\frac{7}{2} + \frac{15}{2} = 11 \Rightarrow \frac{22}{2} = 11$$

$11 = 11$, which is true.

Hence, lines (1) and (2) intersect and substituting $\lambda = \frac{1}{2}$ in Eq. (3) we get their point of intersection as

$$P\left(3 \times \frac{1}{2} - 1, 5 \times \frac{1}{2} - 3, 7 \times \frac{1}{2} - 5\right)$$

i.e. $$P\left(\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}\right)$$

SECTION - E

Case study based questions are compulsory.

36. Bob is taking a learning test in which the time he takes to memorize items from a given list is recorded. Let $M(t)$ be the number of items he can memorize in t minutes. His learning rate is found to be

$$M'(t) = 0.4t - 0.005t^2$$



(i) How many items can Bob memorize during the first 10 minutes?

(ii) How many additional items can he memorize during the next 10 minutes (from time $t = 10$ to $t = 20$)?

Sol :

We have $M'(t) = 0.4t - 0.005t^2$

(i) $M(t) = \int M'(t)\,dt$

$$= \int (0.4t - 0.005t^2)\,dt$$

$$= 0.4\int t\,dt - 0.005\int t^2\,dt$$

$$= 0.4\frac{t^2}{2} - 0.005\frac{t^3}{3} + C$$

$$= 0.2t^2 - \frac{0.005}{3}t^3 + C$$

Since $M(t) = 0$ when $t = 0$, $C = 0$.

So, $M(t) = 0.2t^2 - \frac{0.005}{3}t^3$

In ten minutes, Bob can memorize

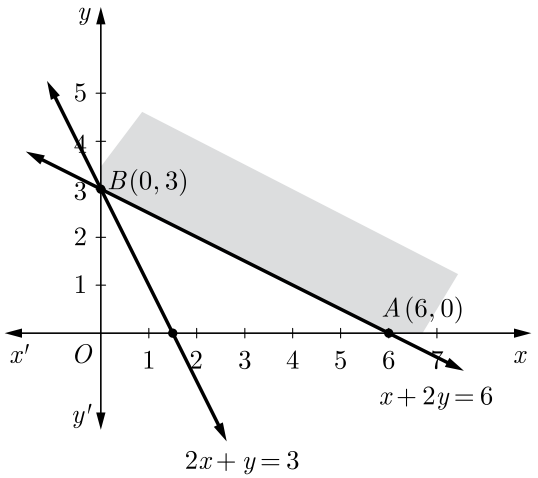
$$M(10) = 0.2(10)^2 - \frac{0.005}{3}(10)^3$$

$$= 18\frac{1}{3} \text{ items}$$

(ii) $M(20) - M(10) = \left[0.2(20)^2 - \frac{0.005}{3}(20)^3\right] - 18\frac{1}{3}$

$$= 66\frac{2}{3} - 18\frac{1}{3} = 48\frac{1}{3} \text{ items}$$

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Since, the feasible region is a bounded region, we can check the objective function at all the corner to find the maxima. The values of objective function Z at these points are as follows.

Corner points	$Z = x + 2y$
$A(6,0)$	$Z = 6 + 0 = 6$
$B(0,3)$	$Z = 0 + 2 \times 3 = 6$

The minimum value of Z is 6 at all points on the line segment joining the points $A(6,0)$ and $B(0,3)$.

34. If $A = \begin{bmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -20 \end{bmatrix}$, find A^{-1}.

Using A^{-1} solve the system of equations $\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 2, \frac{4}{x} - \frac{6}{y} + \frac{5}{z} = 5$ and $\frac{6}{x} + \frac{9}{y} - \frac{20}{z} = -4$

Sol :

Let $\frac{1}{x} = p$, $\frac{1}{y} = q$ and $\frac{1}{z} = r$

Thus, the given equations becomes

$$2p + 3q + 10r = 2$$

$$4p - 6q + 5r = 5$$

$$6p + 9q - 20r = -4$$

Which can be written in matrix form as

$$AX = B$$

Where $A = \begin{bmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 5 & 9 & -20 \end{bmatrix}$,

$$X = \begin{bmatrix} p \\ q \\ r \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 \\ 5 \\ -4 \end{bmatrix}$$

Here, $|A| = \begin{vmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -20 \end{vmatrix}$

$$= 2(120 - 45) - 3(-80 - 30) + 10(36 + 36)$$

$$= 150 + 330 + 720 = 1200 \neq 0$$

Since $|A| \neq 0$, matrix A is non-singular and A^{-1} exists.

Therefore, the above system has a unique solution given by

$$X = A^{-1}B$$

Cofactors of A are

$A_{11} = 75,$	$A_{21} = 150,$	$A_{31} = 75$
$A_{12} = 110,$	$A_{22} = -100,$	$A_{32} = 30$
$A_{13} = 72,$	$A_{23} = 0,$	$A_{33} = -24$

$$adj(A) = \begin{bmatrix} 75 & 110 & 72 \\ 150 & -100 & 0 \\ 75 & 30 & -24 \end{bmatrix}$$

$$= \begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|}(adj\, A)$$

$$= \frac{1}{1200}\begin{bmatrix} 75 & 150 & 75 \\ 110 & -100 & 30 \\ 72 & 0 & -24 \end{bmatrix}$$

Since A^{-1} exists, system has a unique solution given by

$$X = A^{-1}B.$$

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \frac{1}{1200}\begin{bmatrix} 150 + 750 - 300 \\ 220 - 500 - 120 \\ 144 + 0 + 96 \end{bmatrix}$$

$$= \frac{1}{1200}\begin{bmatrix} 900 - 300 \\ 220 - 620 \\ 144 + 96 \end{bmatrix}$$

$$= \frac{1}{1200}\begin{bmatrix} 600 \\ -400 \\ 240 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{3} \\ \frac{1}{5} \end{bmatrix}$$

Now $p = \frac{1}{2}, p = -\frac{1}{3}, r = \frac{1}{5}$

Thus $x = 2, y = -3$ and $z = 5$

35. Show that the lines $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5}$ intersect. Also, find their

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iii) Evaluate $\int_a^b \frac{x^{\frac{1}{n}}}{x^{\frac{1}{n}} + (a+b-x)^{\frac{1}{n}}} dx$]

OR

Evaluate $\int_a^b \frac{f(x)}{f(x)+f(a+b-x)} dx$

Sol :

(i) Let, $I = \int_1^2 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{3-x}} dx$ ...(i)

$$I = \int_1^2 \frac{\sqrt{3-x}}{\sqrt{3-x}+\sqrt{x}} dx \quad ...(ii)$$

Adding Eqs. (i) and (ii), we get

$$2I = \int_1^2 1\,dx = [x]_1^2 = (2-1) = 1$$

$$I = \frac{1}{2}$$

(ii) Let, $I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \log(\tan x)\, dx$ ...(i)

$$I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \log \tan\left(\frac{\pi}{2} - x\right) dx$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \log \tan\left(\frac{\pi}{2} - x\right) dx$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \log \cot x\, dx \quad ...(ii)$$

Adding Eqs. (i) and (ii), we get

$$2I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \log[(\tan x)(\cot x)]\, dx$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 0\,dx = 0 \qquad [\log 1 = 0]$$

$$I = 0$$

(iii) Let, $I = \int_a^b \frac{x^{1/n}}{x^{1/n} + (a+b-x)^{1/n}} dx$ ...(i)

$$I = \int_a^b \frac{(a+b-x)^{1/n}}{(a+b-x)^{1/n} + x^{1/n}} dx \quad ...(ii)$$

Adding Eqs. (i) and (ii), we get

$$2I = \int_a^b 1\,dx = [x]_a^b$$

$$2I = b - a$$

$$I = \frac{b-a}{2}$$

OR

Let, $I = \int_a^b \frac{f(x)}{f(x)+f(a+b-x)} dx$ ...(i)

$$I = \int_a^b \frac{f(a+b-x)}{f(a+b-x)+f(a+b-(a+b-x)} dx$$

$$I = \int_a^b \frac{f(a+b-x)}{f(a+b-x)+f(x)} dx \quad ...(ii)$$

On adding Eqs. (i) and (ii), we get

$$2I = \int_a^b 1\,dx = [x]_a^b = b - a$$

$$I = \frac{1}{2}(b-a)$$

In this book, we have provided solutions along with the papers. To download the same papers in unsolved format for printing, please visit
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Sample Paper 09

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

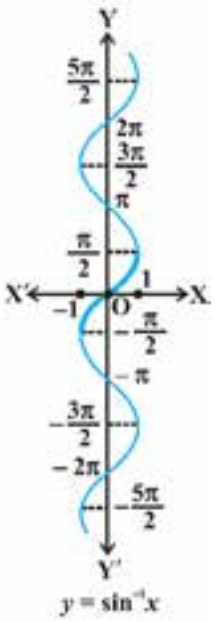
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the value of y when $x = 1$ in the graph of $y = \sin^{-1} x$?



(a) 0 (b) $\pi/2$

(c) 1 (d) π

Sol :

From the image, when $x = 1$, the curve reaches the topmost point on the Y-axis at $\pi/2$. This matches the property of the inverse sine function, where $\sin^{-1}(1) = \pi/2$.

Thus (b) is correct option.

2. If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b}$ is a vector such that $\vec{a} \cdot \vec{b} = |\vec{b}|^2$ and $|\vec{a} - \vec{b}| = \sqrt{7}$, then $|\vec{b}|$ is equal to

(a) $\sqrt{7}$ (b) $\sqrt{3}$

(c) 7 (d) 3

Sol :

Given, $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$

$\vec{a} \cdot \vec{b} = |\vec{b}|^2$

and $|\vec{a} - \vec{b}| = \sqrt{7} \Rightarrow |\vec{a} - \vec{b}|^2 = 7$

$|\vec{a}|^2 + |\vec{b}|^2 - 2\vec{a} \cdot \vec{b} = 7$

$(\sqrt{1+4+9})^2 + |\vec{b}|^2 - 2|\vec{b}|^2 = 7$

$14 - |\vec{b}|^2 = 7$

$|\vec{b}|^2 = 7$

$|\vec{b}| = \sqrt{7}$

Thus (a) is correct option.

3. If $f(x) = \log_e(\log_e x)$, then $f'(e)$ is equal to

(a) e^{-1} (b) e

(c) 1 (d) 0

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Sol :

Given,

$$f(x) = \log_e(\log_e x)$$

$$f'(x) = \frac{1}{x\log_e(x)} \Rightarrow f'(e) = \frac{1}{e}$$

Thus (a) is correct option.

4. The degree of the differential equation $x = 1 + \left(\frac{dy}{dx}\right) + \frac{1}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{1}{3!}\left(\frac{dy}{dx}\right)^3 + \ldots$, is

(a) 3 (b) 2

(c) 1 (d) not defined

Sol :

Given differential equation is

$$x = 1 + \frac{dy}{dx} + \frac{1}{2!}\left(\frac{dy}{dx}\right)^2 + \frac{1}{3!}\left(\frac{dy}{dx}\right)^3 + \ldots$$

$$x = e^{\frac{dy}{dx}} \Rightarrow \frac{dy}{dx} = \log_e x$$

Hence, degree of differential equation is 1.
Thus (c) is correct option.

5. The least, value of the function $f(x) = ax + b/x$, $a > 0$, $b > 0$, $x > 0$ is

(a) $\sqrt{ab}$ (b) $2\sqrt{\frac{a}{b}}$

(c) $2\sqrt{\frac{b}{a}}$ (d) $2\sqrt{ab}$

Sol :

We have $f(x) = ax + \frac{b}{x}$, a, b, $x > 0$

$$f'(x) = a - \frac{b}{x^2}$$

For maxima or minima, putting $f'(x) = 0$ we have

$$x^2 = \frac{b}{a},\ x = \pm\sqrt{\frac{b}{a}}$$

At $x = \sqrt{\frac{b}{a}}$, $\frac{d^2y}{dx^2} > 0$

So, $f(x)$ attains its minima at $x = \sqrt{b/a}$ and the minimum value is $2\sqrt{ab}$.
Thus (d) is correct option.

6. If the lines $\frac{1-x}{3} = \frac{y-2}{2\alpha} = \frac{z-3}{2}$ and $\frac{x-1}{3\alpha} = y - 1 = \frac{6-z}{5}$ are perpendicular, then the value of α is

(a) $\frac{-10}{7}$ (b) $\frac{10}{7}$

(c) $\frac{-10}{11}$ (d) $\frac{10}{11}$

Sol :

Given lines can be rewritten as

$$\frac{x-1}{-3} = \frac{y-2}{3a} = \frac{z-3}{2} \text{ and}$$

$$\frac{x-1}{3a} = \frac{y-1}{1} = \frac{z-6}{-5}$$

Since, lines are perpendicular.

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

$$(-3)(3\alpha) + 2\alpha(1) + 2(-5) = 0$$

$$-9\alpha + 2\alpha - 10 = 0$$

$$\alpha = -\frac{10}{7}$$

Thus (a) is correct option.

7. A bag A contains 4 green and 3 red balls and bog B contains 4 red and 3 green balls. One bag is taken at random and a ball is drawn and noted to be green. The probability that it comes from bag B is

(a) $\frac{2}{7}$

(b) $\frac{2}{3}$

(c) $\frac{3}{7}$

(d) $\frac{1}{3}$

Sol :

$$P\left(\frac{G}{A}\right) = \frac{{}^4C_1}{{}^7C_1} = \frac{4}{7}$$

and $P\left(\frac{G}{B}\right) = \frac{{}^3C_1}{{}^7C_1} = \frac{3}{7}$

Now, $P\left(\frac{B}{G}\right) = \frac{P(B)\cdot P\left(\frac{G}{B}\right)}{P(A)P\left(\frac{G}{A}\right) + P(B)P\left(\frac{G}{B}\right)}$

$$= \frac{\frac{1}{2}\cdot\frac{3}{7}}{\frac{1}{2}\cdot\frac{4}{7} + \frac{1}{2}\cdot\frac{3}{7}} = \frac{3}{7}$$

Thus (c) is correct option.

8. Find the area of a curve $xy = 4$, bounded by the lines $x = 1$ and $x = 3$ and X-axis.

(a) $\log 12$

(b) $\log 64$

(c) $\log 81$

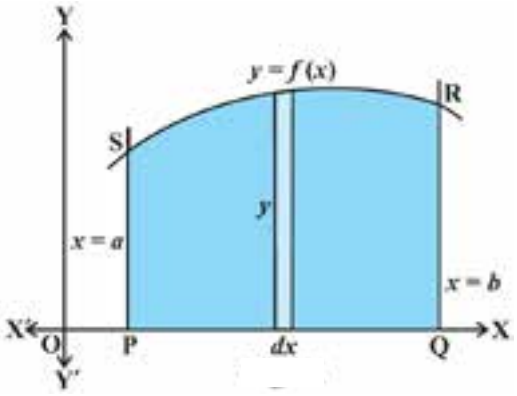
(d) $\log 27$

Sol :

Given curve is $xy = 4$.

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16. In the figure, the element labeled "dy" represents:



(a) The length of the horizontal strip

(b) The small change in the y-coordinate (thickness of the strip)

(c) The slope of the curve x=g(y)

(d) The total height of the shaded region

Sol :

(b) The small change in the y-coordinate (thickness of the strip)

The figure shows a thin horizontal strip drawn parallel to the X-axis within the shaded region. This strip has length x and a very small thickness labeled dy. Therefore, dy indicates a small change in the y-coordinate, not the slope or the entire height.

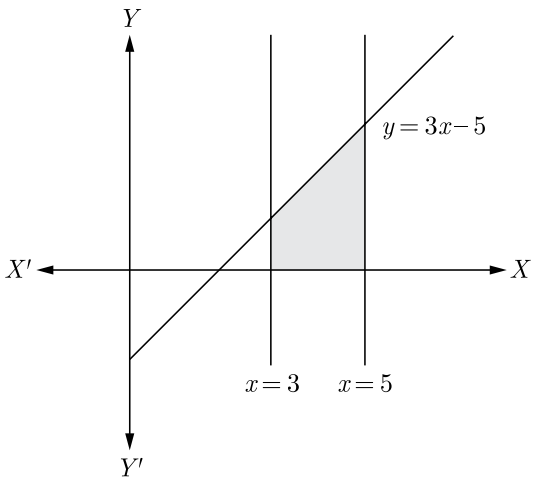
17. The area of enclosed by $y = 3x - 5$, $y = 0$, $x = 3$ and $x = 5$ is

(a) 12 sq units (b) 13 sq units

(c) $13\frac{1}{2}$ sq units (d) 14 sq units

Sol :

The region is bounded by the curves $y = 3x - 5$, $y = 0$, $x = 3$ and $x = 5$.



$$\text{Required area} = \int_3^5 (3x-5)\,dx$$

$$= \left[\frac{3x^2}{2} - 5x\right]_3^5$$

$$= \left(\frac{75}{2} - 25\right) - \left(\frac{27}{2} - 15\right)$$

$$= \frac{75}{2} - 25 - \frac{27}{2} + 15$$

$$= \frac{48}{2} - 10 = 14 \text{ sq units}$$

Thus (d) is correct option.

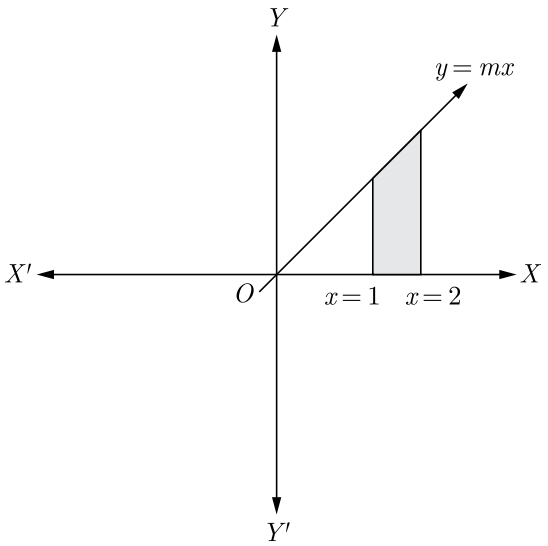
18. The area of the region bounded by the lines $y = mx$, $x = 1$, $x = 2$ and X-axis is 6 sq units, then m is equal to

(a) 3 (b) 1

(c) 2 (d) 4

Sol :

Given, equation of line is $y = mx$ and bounded by $x = 1$, $x = 2$ and X-axis.



$$\text{Required area} = \int_1^2 mx\,dx$$

$$6 = m\left[\frac{x^2}{2}\right]_1^2 \Rightarrow 6 = m\left(\frac{4}{2} - \frac{1}{2}\right)$$

$$6 = mx\frac{3}{2}$$

$$m = 4$$

Thus (d) is correct option.

19. **Assertion :** $\int \frac{dx}{e^x + e^{-x} + 2} = \frac{1}{e^x + 1} + C$

Reason : $\int \frac{d\{f(x)\}}{\{f(x)\}^2} = -\frac{1}{f(x)} + C$

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12. The value of $\int\left(\sqrt{x}-\frac{1}{\sqrt{x}}\right)^2 dx$ is

(a) $\frac{x^2}{2}+\log|x|-2x+C$

(b) $\frac{x^2}{2}+\log|x|+2x+C$

(c) $\frac{x^2}{2}-\log|x|-2x+C$

(d) None of these

Sol :

$$\int\left(\sqrt{x}-\frac{1}{\sqrt{x}}\right)^2 dx = \int\left(x+\frac{1}{x}-2\times\sqrt{x}\times\frac{1}{\sqrt{x}}\right)dx$$

$$= \int\left(x+\frac{1}{x}-2\right)dx$$

$$= \int x\,dx+\int\frac{1}{x}dx-2\int 1\,dx$$

$$= \frac{x^2}{2}+\log|x|-2x+C$$

Thus (a) is correct option.

13. The solution of $\frac{dy}{dx}=\frac{ax+g}{by+f}$ represents a circle, when

(a) $a=b$ (b) $a=-b$

(c) $a=-2b$ (d) $a=2b$

Sol :

We have,

$$\frac{dy}{dx} = \frac{ax+g}{by+f}$$

$$(by+f)\,dy = (ax+g)\,dx$$

Integrating both sides, we have

$$\frac{by^2}{2}+fy = \frac{ax^2}{2}+gx+C$$

$$ax^2-by^2+2gx-2fy+C = 0$$

Which represents a circle, if $a=-b$.

Thus (b) is correct option.

14. If $y=\tan^{-1}\sqrt{\frac{1-\sin x}{1+\sin x}}$, then the value of $\frac{dy}{dx}$ at $x=\frac{\pi}{6}$ is

(a) $-\frac{1}{2}$ (b) $\frac{1}{2}$

(c) 1 (d) -1

Sol :

Given,

$$y = \tan^{-1}\sqrt{\frac{1-\sin x}{1+\sin x}}$$

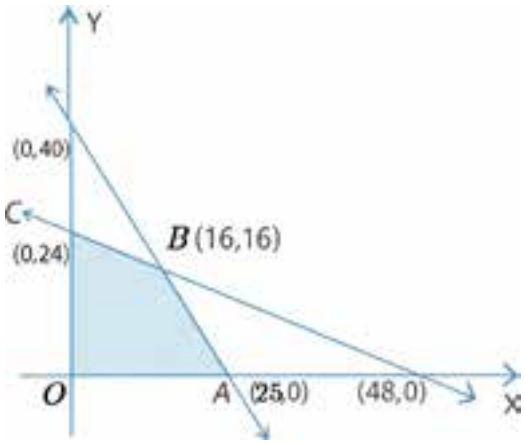
$$= \tan^{-1}\sqrt{\frac{1-\cos\left(\frac{\pi}{2}-x\right)}{1+\cos\left(\frac{\pi}{2}-x\right)}}$$

$$= \tan^{-1}\left|\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)\right| = \frac{\pi}{4}-\frac{x}{2}$$

$$\frac{dy}{dx} = -\frac{1}{2}$$

Thus (a) is correct option.

15. In the same figure, if the objective function is given as $z=4x+y$, then at which vertex of the feasible region does this function attain its maximum value?



(a) O(0,0)

(b) A(25,0)

(c) B(16,16)

(d) C(0,24)

Sol :

(b) A(25,0)

We substitute the coordinates of each vertex into the given objective function $z=4x+y$:

At O(0,0) : z=4(0)+0=0

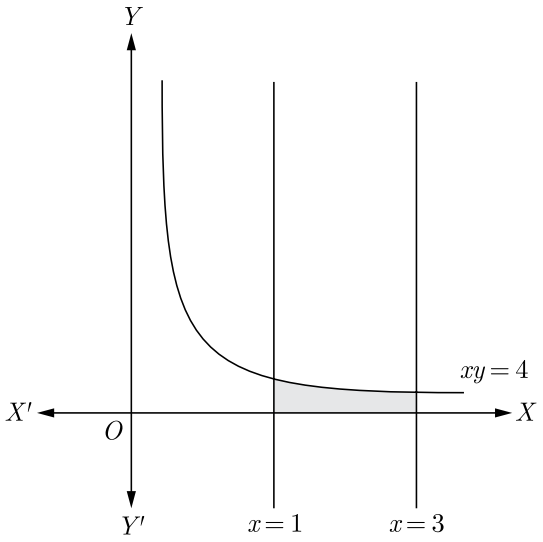
At A(25,0) : z=4(25)+0=100

At B(16,16) : z=4(16)+16=64+16=80

At C(0,24) : z=4(0)+24=24

Clearly, the largest value among these is 100, which occurs at A(25,0). Hence the maximum value is obtained at vertex A.

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$$\text{Required area} = \int_1^3 \frac{4}{x}\,dx = 4\cdot[\log x]_1^3$$
$$= 4(\log 3 - \log 1)$$
$$= 4\log 3 = \log 81$$

Thus (c) is correct option.

9. If $P(A)=\frac{4}{5}$, and $Q(A\cap B)=\frac{7}{10}$, then $P\left(\frac{B}{A}\right)$ is equal to

(a) $\frac{1}{10}$ (b) $\frac{1}{8}$

(c) $\frac{7}{8}$ (d) $\frac{17}{20}$

Sol :

We have, $P(A)=\frac{4}{5}$, $P(A\cap B)=\frac{7}{10}$

$$P(B/A)=\frac{P(A\cap B)}{P(A)}=\frac{\frac{7}{10}}{\frac{4}{5}}=\frac{7}{8}$$

Thus (c) is correct option.

10. The point on the curve $x^2=2y$ which is nearest to the point $(0,5)$ is

(a) $\left(2\sqrt{2},4\right)$

(b) $\left(2\sqrt{2},0\right)$

(c) $(0,0)$

(d) $(2,2)$

Sol :

Let $A(x_1,y_1)$ be any point on the curve $x^2=2y$.

$$x_1^2 = 2y_1 \qquad ...(1)$$

Let $B(0,5)$ be the given point.

$$AB = S = \sqrt{(x_1-0)^2+(y_2-5)^2}$$

$$S^2 = x_1^2+(y_1-5)^2$$
$$S^2 = z \text{ (Let)} = x_1^2+(y_1-5)^2$$
$$z = 2y_1+(y_1-5)^2$$
$$\frac{dz}{dy_1} = 2+2(y_1-5)$$

For maxima and minima, $\frac{dz}{dy_1}=0$

$$2+2(y_1-5) = 0$$
$$1+y_1-5 = 0$$
$$y_1-4 = 0$$
$$y_1 = 4$$

From eq. (1), we get

$$x_1^2 = 2\times 4$$
$$x_1^2 = 8$$
$$x_1 = \pm 2\sqrt{2}$$
$$\frac{d^2z}{dy_1^2} = 2 > 0 \text{ (minima)}$$

Thus, required points are $\left(2\sqrt{2},4\right)$ and $\left(-2\sqrt{2},4\right)$.
Thus (a) is correct option.

11. The symmetric part of the matrix $A=\begin{bmatrix}1 & 2 & 4\\ 6 & 8 & 2\\ 2 & -2 & 7\end{bmatrix}$ is equal to

(a) $\begin{bmatrix}0 & -2 & -1\\ -2 & 0 & -2\\ -1 & -2 & 0\end{bmatrix}$ (b) $\begin{bmatrix}1 & 4 & 3\\ 2 & 8 & 0\\ 3 & 0 & 7\end{bmatrix}$

(c) $\begin{bmatrix}0 & -2 & 1\\ 2 & 0 & 2\\ -1 & 2 & 0\end{bmatrix}$ (d) $\begin{bmatrix}1 & 4 & 3\\ 4 & 8 & 0\\ 3 & 0 & 7\end{bmatrix}$

Sol :

We have $A=\begin{bmatrix}1 & 2 & 4\\ 6 & 8 & 2\\ 2 & -2 & 7\end{bmatrix}$

Symmetric part of A,

$$\frac{1}{2}[A+A'] = \frac{1}{2}\left\{\begin{bmatrix}1 & 2 & 4\\ 6 & 8 & 2\\ 2 & -2 & 7\end{bmatrix}+\begin{bmatrix}1 & 6 & 2\\ 2 & 8 & -2\\ 4 & 2 & 7\end{bmatrix}\right\}$$
$$= \frac{1}{2}\left\{\begin{bmatrix}2 & 8 & 6\\ 8 & 16 & 0\\ 6 & 0 & 14\end{bmatrix}+\begin{bmatrix}1 & 4 & 3\\ 4 & 8 & 0\\ 3 & 0 & 7\end{bmatrix}\right\}$$

Thus (b) is correct option.

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(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

Sol :

As we know that, $\int \frac{d\{f(x)\}}{\{f(x)\}^2} = -\frac{1}{f(x)} + C$

So, given Reason is true.

Now $\int \frac{dx}{e^x + e^{-x} + 2} = \int \frac{e^x\,dx}{e^{2x} + 2e^x + 1}$

$= \int \frac{e^x\,dx}{(e^x+1)^2}$

$= \int \frac{d(e^x+1)}{(e^x+1)^2}$

$= \frac{-1}{e^x+1} + C$

Hence, Assertion is false; reason is true.

Thus (d) is correct option.

20. Assertion : $\int \frac{xe^x}{(x+1)^2}dx = \frac{e^x}{x+1} + C$

Reason : $\int e^x\{f(x) + f'(x)\}\,dx = e^x f(x) + C$

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

Sol :

As we know that, $\int e^x\{f(x) + f'(x)\}\,dx = e^x f(x) + C$

So, given reason is true.

Now, $\int \frac{xe^x}{(x+1)^2}dx = \int \frac{(x+1-1)}{(x+1)^2}e^x\,dx$

$= fe^x\left\{\frac{1}{(x+1)} - \frac{1}{(x+1)^2}\right\}dx$

Let $f(x) = \frac{1}{(x+1)}$

then, $f'(x) = \frac{-1}{(x+1)^2}$

So, $\int \frac{xe^x}{(x+1)^2}dx = \frac{e^x}{(x+1)} + C$

Hence, Both Assertion and reason are true and reason is correct explanation for assertion

Thus (a) is correct option.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Find distance between (4, 3, 7) and $(1, -1, -5)$?

Sol :

Distance between two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by distance formula

$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

Here, $x_1 = 4,\ y_1 = 3,\ z_1 = 7$

$x_2 = 1,\ y_2 = -1,\ z_2 = -5$

Using distance formula,

$d = \sqrt{(4-1)^2 + (3-(-1))^2 + (7-(-5))^2}$

$= \sqrt{3^2 + 4^2 + 12^2}$

$= \sqrt{9 + 16 + 144}$

$= \sqrt{144 + 25} = \sqrt{169} = 13$

22. If $P(A) = \frac{2}{5}, P(B) = \frac{3}{5}, P(A \cup B) = \frac{3}{4}$, then find $P\left(\frac{A}{B}\right)$ and $P\left(\frac{B}{A}\right)$

Sol :

We have,

$P(A) = \frac{2}{5},\ P(B) = \frac{3}{5}, P(A \cup B) = \frac{3}{4}$

We know that,

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$\frac{3}{4} = \frac{2}{5} + \frac{3}{5} - P(A \cap B)$

$P(A \cap B) = 1 - \frac{3}{4} = \frac{1}{4}$

$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{3/5} = \frac{5}{12}$

and $P(B/A) = \frac{P(A \cap B)}{P(A)} = \frac{1/4}{2/5} = \frac{5}{8}$

23. If $\vec{a} = 2\hat{i} - 3\hat{j} - 5\hat{k}$ and $\vec{b} = -7\hat{i} + 6\hat{j} + 8\hat{k}$ find $\vec{a} \times \vec{b}$

Sol :

We have $\vec{a} = 2\hat{i} - 3\hat{j} - 5\hat{k}$

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Find the value of x and y if-

$$2\begin{bmatrix}1 & 3\\ 0 & x\end{bmatrix}+\begin{bmatrix}y & 0\\ 1 & 2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$$

Sol :

We have, $A=\begin{bmatrix}5 & 4\\ 2 & 3\end{bmatrix}$

and $B=\begin{bmatrix}3 & 5 & 1\\ 6 & 8 & 4\end{bmatrix}$

$$AB=\begin{bmatrix}5 & 4\\ 2 & 3\end{bmatrix}\begin{bmatrix}3 & 5 & 1\\ 6 & 8 & 4\end{bmatrix}$$

$$=\begin{bmatrix}5\times3+4\times6 & 5\times5+4\times8 & 5\times1+4\times4\\ 2\times3+3\times6 & 2\times5+3\times8 & 2\times1+3\times4\end{bmatrix}$$

$$=\begin{bmatrix}15+24 & 25+32 & 5+16\\ 6+18 & 10+24 & 2+12\end{bmatrix}=\begin{bmatrix}39 & 57 & 21\\ 24 & 34 & 14\end{bmatrix}$$

OR

We have,

$$2\begin{bmatrix}1 & 3\\ 0 & x\end{bmatrix}+\begin{bmatrix}y & 0\\ 1 & 2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$$

$$\begin{bmatrix}2 & 6\\ 0 & 2x\end{bmatrix}+\begin{bmatrix}y & 0\\ 1 & 2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$$

$$\begin{bmatrix}2+y & 6\\ 1 & 2x+2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$$

Thus $2+y=5$ and $2x+2=8$

$y=3$ and $x=3$

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Solve graphically the following L.P. problem:

Maximise $Z=x+y$

Subject to the linear constraints

$$y-2x-1\le 0$$

$$x\le 2$$

$$x+y\le 3$$

and $x,\ y\ge 0$

Sol :

Step I.

$$x\ge 0$$

$$y\ge 0$$

Feasible region is in first quadrant.

Let us graph the first constraint,

$$y-2x-1\le 0 \qquad ...(1)$$

Replacing $\le$ by $=$ the corresponding equation of the line is

$$y-2x-1=0$$

i.e., $y=2x+1$

Table of values for $y=2x+1$

x	0	$-\frac{1}{2}$
y	1	0

Hence, Graph of $y-2x-1=0$ is the line passing through the points $(-\frac{1}{2},0)$ and $(0,1)$.

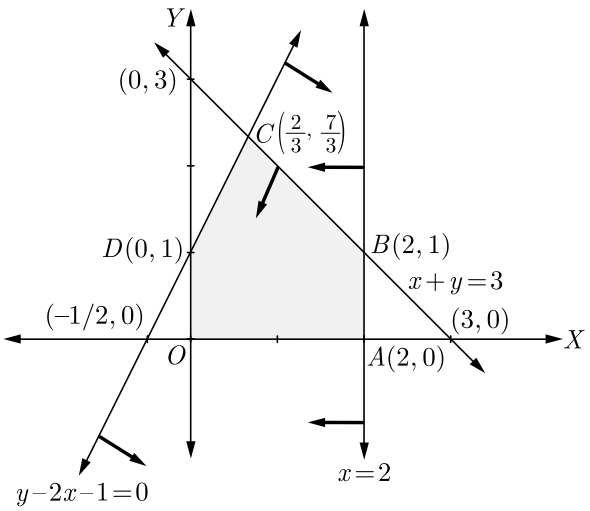
Test for Origin in (1)

Putting $x=0$ and $y=0$ in Eq. (1), we get

$-1\le 0$ which is true

Therefore, origin lies in the region determined by in equation (1).

Hence, Graph of in equation (1) is the region containing the origin (as shown by arrows in the figure) in the first quadrant.



The second constraint is

$$x\le 2 \qquad ...(2)$$

$x=2$ is a line parallel to y-axis at a distance 2 to the right of origin.

Text for origin in (2)

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27. Find the rate of change of the area of a circle with respect to its radius r when (i) $r = 3$ cm and (ii) $r = 5$ cm.

Sol :

Let A be the area of the circle of radius r, then $A = \pi r^2$

Then rate of change of A wrt r is

$$\frac{dA}{dr} = \pi \cdot 2r$$

$$\frac{dA}{dr} = 2\pi r$$

(i) $\left(\frac{dA}{dr}\right)_{r=3\text{ cm}} = 2\pi(3) = 6\pi \text{ cm}^2$

(ii) $\left(\frac{dA}{dr}\right)_{r=5\text{ cm}} = 2\pi(5) = 10\pi \text{ cm}^2$

28. If $y = \sin^{-1}[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}]$, find $\frac{dy}{dx}$.

Sol :

Given:

$$y = \sin^{-1}[x\sqrt{1-x} - \sqrt{x}\sqrt{1-x^2}]$$

Let, $\sin^{-1}x = \theta$ and $\sin^{-1}\sqrt{x} = \phi$ ...(1)

and $\sin(\theta - \phi) = \sin\theta\cos\phi - \sin\phi\cos\theta$

Using equation (1),

$$x = \sin\theta,\ \sqrt{x} = \sin\phi$$

Squaring both sides, we get

$$x^2 = \sin^2\theta,\ x = \sin^2\phi$$

$$1 - x^2 = 1 - \sin^2\theta,\ \sin\phi = \sqrt{x}$$

$$\cos^2\theta = 1 - x^2,\ \cos^2\phi = 1 - x$$

$$\cos\theta = \sqrt{1-x^2},$$

$$\cos\phi = \sqrt{1-x}$$

Therefore, $y = \sin^{-1}[\sin\theta\cos\phi - \sin\phi\cos\theta]$

$$= \sin^{-1}[\sin(\theta - \phi)]$$

$$= \theta - \phi$$

$$y = \sin^{-1}x - \sin^{-1}\sqrt{x}$$

(using equation)

Differentiating both sides we get

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x}} \times \frac{1}{2\sqrt{x}}$$

$$= \frac{1}{\sqrt{1-x^2}} - \frac{1}{2\sqrt{x-x^2}}$$

29. Examine whether the function $f : R \to R$ is one-one (injective) if $f(x) = x^3$, $x \in R$.

Sol :

Let $f: R \to R$ is said to be one-one if and only if there exist a unique image of $x \in R$.

Here, $f(x_1) = f(x_2)$

$$f(x) = x^3$$

$$x_1^3 = x_2^3 \Rightarrow x_1 = x_2$$

Therefore unique image exist.

Thus $f(x)$ is one-one function.

30. Prove that $\tan^{-1}\sqrt{x} - \frac{1}{z}\cos^{-1}\frac{1-x}{1+x}$

OR

Prove that $\sec^2(\tan^{-1}2) + \text{cosec}^2(\cot^{-1}3) = 15$

Sol :

$$\text{RHS} = \frac{1}{2}\cos^{-1}\left(\frac{1-x}{1+x}\right)$$

$$= \frac{1}{2}\cos^{-1}\left(\frac{1-(\sqrt{x})^2}{1+(\sqrt{x})^2}\right)$$

$$= \frac{1}{2} \cdot 2\tan^{-1}\sqrt{x}$$

$$\left[2\tan^{-1}a = \cos^{-1}\frac{1-a^2}{1+a^2}\right]$$

$$= \tan^{-1}\sqrt{x} \text{ (LHS)}$$

OR

We have,

$$\text{LHS} = \sec^2(\tan^{-1}2) + \text{cosec}^2(\cot^{-1}3)$$

As we know that,

$$\sec^2\theta = 1 + \tan^2\theta$$

$$\text{cosec}^2\theta = 1 + \cot^2\theta$$

Hence, $= 1 + \tan^2(\tan^{-1}2) + 1 + \cot^2(\cot^{-1}3)$

$$= 2 + (\tan\tan^{-1}2)^2 + (\cot\cot^{-1}3)^2$$

$$= 2 + (2)^2 + (3)^2$$

$$= 2 + 4 + 9$$

$$= 15 \text{ (RHS)}$$

31. If $A = \begin{bmatrix} 5 & 4 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 5 & 1 \\ 6 & 8 & 4 \end{bmatrix}$, find AB.

OR

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and $\vec{b} = -7\hat{i} + 6\hat{j} + 8\hat{k}$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & -5 \\ -7 & 6 & 8 \end{vmatrix}$$

$= \hat{i}[-3 \times 8 - 6 \times (-5)] - \hat{j}[2 \times 8 - (-7) \times (-5)] + \hat{k}[2 \times 6 - (-7)(-3)]$

$= \hat{i}(-24 + 30) - \hat{j}(16 - 35) + \hat{k}(12 - 21)$

$= 6\hat{i} + 29\hat{j} - 9\hat{k}$

24. Find $\frac{dy}{dx}$ if $y = \tan(x + y)$

OR

Find $\frac{dy}{dx}$ if $y = \sin^3 x \cos^5 x$

Sol :

We have,

$$y = \tan(x + y)$$

$$\frac{dy}{dx} = \sec^2(x + y)\left(1 + \frac{dy}{dx}\right)$$

$$\frac{dy}{dx} = \sec^2(x + y) + \sec^2(x + y)\frac{dy}{dx}$$

$$(1 - \sec^2(x + y))\frac{dy}{dx} = \sec^2(x + y)$$

$$\frac{dy}{dx} = \frac{\sec^2(x + y)}{1 - \sec^2(x + y)}$$

OR

We have,

$$y = \sin^3 x \cos^5 x$$

$$\frac{dy}{dx} = \cos^5 x \frac{d}{dx}\sin^3 x + \sin^3 x \frac{d}{dx}\cos^5 x$$

$$= \cos^5 x(3\sin^2 x \cos x) + \sin^3 x(5\cos^4 x(-\sin x))$$

$$= 3\sin^2 x \cos^6 x - 5\cos^4 x \sin^4 x$$

$$= \sin^2 x \cos^4 x(3\cos^2 x - 5\sin^2 x)$$

25. Solve – $\frac{dy}{dx} + y\cot x = 2\cos x$

OR

Solve – $xdy + ydx = 0$

Sol :

We have, $\frac{dy}{dx} + y\cot x = 2\cos x$

$$I.F. = e^{\int \cot x\, dx}$$

$$= e^{\int \log \sin x} = \sin x$$

So, solution of the given differential equation is

$$y \cdot IF = \int Q \cdot IF\, dx + c$$

$$y \sin x = \int 2\cos x . \sin x\, dx + c$$

$$y \sin x = \int \sin 2x dx + c$$

$$y \sin x = \frac{-1}{2}\cos 2x + c$$

OR

We have, $xdy + ydx = 0$

$$\frac{1}{y}dy + \frac{1}{x}dx = 0$$

$$\log y + \log x = c$$

$$\log xy = c$$

$$xy = e^c$$

$$xy = k$$

Where, $k = e^c$

SECTION - C

This section comprises of short answer-type questions (SA) of 3 marks each.

26. Prove that – $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - |\vec{a}\ \vec{b}|^2$

Sol :

We know that,

$$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta$$

$$|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2\sin^2\theta \quad ...(1)$$

Also, $|\vec{a} \cdot \vec{b}| = |\vec{a}||\vec{b}|\cos\theta$

$$|\vec{a} \cdot \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2\cos^2\theta \quad ...(2)$$

on adding equation (1) and (2), we get

$$|\vec{a} \times \vec{b}|^2 + |\vec{a} \cdot \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2(\sin^2\theta + \cos^2\theta)$$

$$|\vec{a} \times \vec{b}|^2 + |\vec{a} \cdot \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2$$

$$|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2 - |\vec{a} \cdot \vec{b}|^2$$

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Putting $x = 0$ and $y = 0$ in Eq. (2), we get

$$0 \leq 2 \text{ which is true}$$

Hence, Origin lies in the region determined by in equation (2).

Hence, The graph of in equation $x \leq 2$ is the region (containing the origin) to the left of the line $x = 2$ (as shown by arrows in the figure) in the first quadrant.

The third constraint is

$$x + y \leq 3 \qquad ...(3)$$

Let us first draw the graph of the line $x + y = 3$

Table of values for $x + y = 3$

x	0	3
y	3	0

Hence, the graph of $x + y = 3$ is the line passing through the points (3, 0) and (0, 3).

Test for origin in (3)

Putting, $x = 0$ and $y = 0$ in Eq. (3), we get

$$0 \leq 3 \text{ which is true}$$

Hence, Origin lies in the region determined by in equation (3).

Hence, Graph of in equation (3) is the region containing the origin (as shown by arrows in the figure), in the first quadrant.

The intersection (common points) of the three regions corresponding to in equations (1), (2) and (3) is the feasible region as shown in the figure and it is the region **bounded** by the pentagon $OABCD$.

Corner Points are O, A, B, C and D. (See Note at the end of solution of Example 1).

Step II

To find the co-ordinates of vertices O, A, B, C, D.

Vertex O is (0, 0). Vertex A is (2, 0).

For vertex B, let us solve $x + y = 3$ and $x = 2$ to find their point of intersection B.

Putting $x = 2$ in $x + y = 3$, we get

$$y = 1$$

Hence, Vertex B (2, 1).

To find vertex C.

Let us solve $y - 2x = 1$

and $x + y = 3$ for x and y.

Subtracting $-3x = -2$

or, $x = \frac{2}{3}$

$$y = 3 - x = 3 - \frac{2}{3} = \frac{7}{3}$$

Hence, Vertex C is $\left(\frac{2}{3}, \frac{7}{3}\right)$

Vertex D is (0,1).

Step III

Let us find the value of the objective function $Z = x + y$ at each of the vertices O, A, B, C, D.

At O (0,0);	$Z = 0 + 0 = 0$
At A (2,0);	$Z = 2 + 0 = 2$
At B (2,1);	$Z = 2 + 1 = 3$ (Maximum)
At $C\left(\frac{2}{3}, \frac{7}{3}\right)$	$Z = \frac{2}{3} + \frac{7}{3} = 3$ (Maximum)
At D (0,1)	$Z = 0 + 1 = 1$

We observe that maximum value of Z is $M = 3$ at two corner points B and C.

Hence, Z is maximum at an infinite number of points (on every point of the line segment BC, $x + y = 3$ as is clear from the graph) and these infinite points are all points of line segment BC.

Note :

Whenever Z (objective function) is maximum at two corner points say B and C, then it (Z) has maximum value at all the infinite points of the line segment BC.

Similar result is true for minimum also.

33. Solve $(x^2 - y^2)\frac{dy}{dx} = 2xy$

OR

Solve $x\cos x\frac{dy}{dx} + y(x\sin x + \cos x) = 1$

Sol :

$$(x^2 - y^2)\frac{dy}{dx} = 2xy \qquad ...(1)$$

Let, $y = vx$

Differentiating w.r.t. x

$$\frac{dy}{dx} = v + \frac{xdv}{dx}$$

From eq. (1),

$$\frac{dy}{dx} = \frac{2xy}{(x^2 - y^2)}$$

$$v + x\frac{dv}{dx} = \frac{2xvx}{x^2 - v^2x^2} = \frac{2vx^2}{(1 - v^2)x^2}$$

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In a certain city there are 50 colleges and 400 schools. Each school and college has 18 peons, 5 clerks and 1 cashier. Each college in addition has 1 section officer and one librarian. The monthly salary of each of them is as follows:

Peon-₹ 3000, Clerk-₹ 5000, Cashier-₹ 6000, Section Officer-₹ 7000 and Librarian-₹ 9000

Using matrix notation, find

(a) total number of posts of each kind in schools and colleges taken together.

(b) the total monthly salary bill of all the schools and colleges taken together.

Sol :

The number of colleges and schools can be represented by the matrix A as:

$$A = \begin{matrix} \text{College} & \text{School} \end{matrix}$$
$$\begin{bmatrix} 50 & 400 \end{bmatrix}$$

The number of posts of each kind in each college and school can be represented by the matrix B as:

$$B = \begin{bmatrix} 18 & 5 & 1 & 1 & 1 \\ 18 & 5 & 1 & 0 & 0 \end{bmatrix}$$

The monthly salary of each of them can be represented by the matrix C as

$$C = \begin{matrix} \\ \text{Peon} \\ \text{Clerk} \\ \text{Cashier} \\ \text{Section Officer} \\ \text{Libratian} \end{matrix} \begin{matrix} \text{Salary} \\ \begin{bmatrix} 3000 \\ 5000 \\ 6000 \\ 7000 \\ 9000 \end{bmatrix} \end{matrix}$$

The total number of posts of each kind in colleges and schools taken together can be obtained by the matrix product AB as:

$$AB = \begin{bmatrix} 50 & 400 \end{bmatrix} \begin{bmatrix} 18 & 5 & 1 & 1 & 1 \\ 18 & 5 & 1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 8100 & 2250 & 450 & 50 & 50 \end{bmatrix}$$

Thus there are 8100 peons, 2250 clerks, 450 cashiers, 50 section officers and 50 librarians in all colleges and schools taken together.

The total monthly salary bill of all the colleges and schools taken together is given by the product $(AB)\,C$ as

$$(AB)\,C = \begin{bmatrix} 8100 & 2250 & 450 & 50 & 50 \end{bmatrix} \begin{bmatrix} 3000 \\ 5000 \\ 6000 \\ 7000 \\ 9000 \end{bmatrix}$$

$$= [39050000]$$

Hence total monthly salary bill of all the colleges and schools taken together .

(a) There are 8100 peons, 2250 clerks, 450 cashiers, 50 section officers and 50 librarians in all colleges and schools taken together.

(b) Total monthly salary bill of all the colleges and schools taken together = ₹ 39050000.

38. Minimum Support Price (MSP) is a form of market intervention by the Government of India to insure agricultural producers against any sharp fall in farm prices. The minimum support prices are announced by the Government of India at the beginning of the sowing season for certain crops on the basis of the recommendations of the Commission for Agricultural Costs and Prices (CACP). MSP is price fixed by Government of India to protect the producer - farmers - against excessive fall in price during bumper production years. The minimum support prices are a guarantee price for their produce from the Government. The major objectives are to support the farmers from distress sales and to procure food grains for public distribution. In case the market price for the commodity falls below the announced minimum price due to bumper production and glut in the market, government agencies purchase the entire quantity offered by the farmers at the announced minimum price.



The Government declare that farmers can get `300 per quintal for onions on 1st July and after that, the price will be dropped by `3 per quintal per extra day. Ramawatar has 80 quintal of onions in the field on 1st July and he estimated that crop is increasing at the rate of 1 quintal per day.

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$$= \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

$$P(X=2) = P(H)\cdot P(H)$$

$$= \frac{1}{2}\times\frac{1}{2} = \frac{1}{4}$$

Probability distribution is

X	0	1	2
$P(X)$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{4}$

(ii) Here $X = 0, 1, 2, 3$

$$P(T) = \frac{1}{2},\ P(H) = \frac{1}{2}$$

$$P(X=0) = P(H)\cdot P(H)\cdot P(H)$$

$$= \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} = \frac{1}{8}$$

$$P(X=1) = P(T)\cdot P(H)\cdot P(H) + P(H)\cdot P(T)\cdot P(H) + P(H)\cdot P(H)\cdot P(T)$$

$$= \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}$$

$$= \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$$

$$P(X=2) = P(T)\cdot P(T)\cdot P(H) + P(T)\cdot P(H)\cdot P(T) + P(H)\cdot P(T)\cdot P(T)$$

$$= \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}$$

$$= \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$$

$$P(X=3) = P(T)\cdot P(T)\cdot P(T)$$

$$= \frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} = \frac{1}{8}$$

Probability distribution is

X	0	1	2	3
$P(X)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

(iii) Here $X = 0, 1, 2, 3, 4$

$$P(H) = \frac{1}{2},\ P(T) = \frac{1}{2}$$

$$P(X=0) = P(T)\cdot P(T)\cdot P(T)\cdot P(T)$$

$$= {}^4C_0\cdot\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} = \frac{1}{16}$$

$$P(X=1) = {}^4C_0 P(H)\cdot P(T)\cdot P(T)\cdot P(T)$$

$$= 4\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} = \frac{4}{16}$$

$$P(X=2) = {}^4C_2 P(H)\cdot P(H)\cdot P(T)\cdot P(T)$$

$$= \frac{4\cdot 3}{2\cdot 1}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} = \frac{6}{16}$$

$$P(X=3) = {}^4C_3 P(H)\cdot P(H)\cdot P(H)\cdot P(T)$$

$$= 4\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} = \frac{4}{16}$$

$$P(X=4) = {}^4C_4 P(H)\cdot P(H)\cdot P(H)\cdot P(H)$$

$$= 1\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2}\times\frac{1}{2} = \frac{1}{16}$$

Probability distribution

X	0	1	2	3	4
$P(X)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

35. Evaluate $\int \frac{5x-2}{1+2x+3x^2}\,dx$

OR

Evaluate: $\int_0^{\frac{\pi}{4}} (\tan x - x)\tan^2 x\,dx$.

Sol :

Let $$I = \int \frac{5x-2}{1+2x+3x^2}\,dx \qquad ...(1)$$

Put $$5x-2 = A\frac{d}{dx}(1+2x+3x^2) + B$$

$$5x-2 = A(2+6x) + B \qquad ...(2)$$

$$5x-2 = 2A + 6Ax + B$$

Comparing coefficients of x and constant terms on both sides, we get

$$5 = 6A, \qquad -2 = 2A + B$$

Solving above equations, we get

$$A = \frac{5}{6}, \qquad B = -\frac{11}{3}$$

Putting values of A and B in (2), we get

$$5x-2 = \frac{5}{6}(2+6x) - \frac{11}{3} \qquad ...(3)$$

From (1) and (3), we get

$$I = \frac{5}{6}\int \frac{2+6x}{1+2x+3x^2}\,dx - \frac{11}{3}\int \frac{1}{1+2x+3x^2}\,dx$$

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or $I = \frac{5}{6}I_1 - \frac{11}{3}I_2$ ...(4)

Now, $I_1 = \int \frac{2+6x}{1+2x+3x^2}\,dx$

Putting $1+2x+3x^2 = y$ we have

$(2+6x)\,dx = dy$

Thus $I_1 = \int \frac{1}{y}\,dy$

$= \log|y| + C_1$

$= \log|1+2x+3x^2| + C_1$

Now, $I_2 = \int \frac{1}{1+2x+3x^2}\,dx$

$= \frac{1}{3}\int \frac{1}{x^2+\frac{2}{3}x+\frac{1}{3}}\,dx$

$= \frac{1}{3}\int \frac{1}{\left(x+\frac{1}{3}\right)^2+\frac{1}{3}-\frac{1}{9}}\,dx$

$= \frac{1}{3}\int \frac{1}{\left(x+\frac{1}{3}\right)^2+\left(\frac{\sqrt{2}}{3}\right)^2}\,dx$

$= \frac{1}{3}\left[\frac{1}{\left(\frac{\sqrt{2}}{3}\right)}\tan^{-1}\left\{\frac{\left(x+\frac{1}{3}\right)}{\left(\frac{\sqrt{2}}{3}\right)}\right\}\right] + C_2$

$= \frac{1}{\sqrt{2}}\tan^{-1}\left[\frac{3x+1}{\sqrt{2}}\right] + C_2$

Putting values of I_1 and I_2 in (4), we get

$I = \frac{5}{6}\log|1+2x+3x^2|$

$= -\frac{11}{3\sqrt{2}}\tan^{-1}\left[\frac{3x+1}{\sqrt{2}}\right] + C$

OR

Let, $I = \int_0^{\frac{\pi}{4}} (\tan x - x)\tan^2 x\,dx$

$= \int_0^{\frac{\pi}{4}} \tan^3 x\,dx - \int_0^{\frac{\pi}{4}} x\tan^2 x\,dx$

$= I_1 - I_2$ ...(1)

$I_1 = \int_0^{\frac{\pi}{4}} \tan^3 x\,dx$

$= \int_0^{\frac{\pi}{4}} \tan x\cdot\tan^2 x\,dx$

$= \int_0^{\frac{\pi}{4}} \tan x\cdot(\sec^2 x - 1)\,dx$

$[1+\tan^2 x = \sec^2 x]$

$= \int_0^{\frac{\pi}{4}} \tan x\cdot\sec^2 x\,dx - \int_0^{\frac{\pi}{4}} \tan x\,dx$

$= I_3 - I_4$ ...(2)

$I_3 = \int_0^{\frac{\pi}{4}} \tan x\cdot\sec^2 x\,dx$

$= \int_0^{\frac{\pi}{4}} \sec x(\tan x\cdot\sec x)\,dx$

Let $\sec x = t$ then we have

$\sec x\tan x\,dx = dt$

We have $x = 0 : \Rightarrow t = 1$

$x = \frac{\pi}{4} : \Rightarrow t = \sqrt{2}$

$I_3 = \int_1^{\sqrt{2}} t\,dt = \frac{t^2}{2}\Big|_1^{\sqrt{2}}$

$= \frac{2}{2} - \frac{1}{2} = 1 - \frac{1}{2}$

or $I_3 = \frac{1}{2}$ ...(3)

$I_4 = \int_0^{\frac{\pi}{4}} \tan x\,dx = \ln\sec x\Big|_0^{\frac{\pi}{4}}$

$= \ln\sec\frac{\pi}{4} - \ln\sec 0$

$= \ln\sqrt{2} - \ln 1 = \frac{1}{2}\ln 2$ ...(4)

Hence, $I_1 = \frac{1}{2}(1-\ln 2)$ ...(5)

Putting values from equation (3) and (4) in equation (2)

$I_2 = \int_0^{\frac{\pi}{4}} x\tan^2 x\,dx$

$= \int_0^{\frac{\pi}{4}} x(\sec^2 x - 1)\,dx \qquad [1+\tan^2\theta = \sec^2\theta]$

$= \int_0^{\frac{\pi}{4}} x\sec^2 x\,dx - \int_0^{\frac{\pi}{4}} x\,dx$

Now doing integrating by parts we get

$I_2 = x\int_0^{\pi/4} \sec^2 x\,dx - \int_0^{\pi/4} \tan x\,dx - \frac{x^2}{2}\Big|_0^{\frac{\pi}{4}}$

$= x\tan x - \ln\sec x\Big|_0^{\frac{\pi}{4}} - \left(\frac{\pi^2}{32} - 0\right)$

$= \frac{\pi}{4} - \ln\sqrt{2} - \frac{\pi^2}{32}$

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Thus, $I_2 = \frac{\pi}{4} - \frac{1}{2}\ln 2 - \frac{\pi^2}{32}$ ...(6)

Putting values from equation (5) and (6) in equation (1), we get

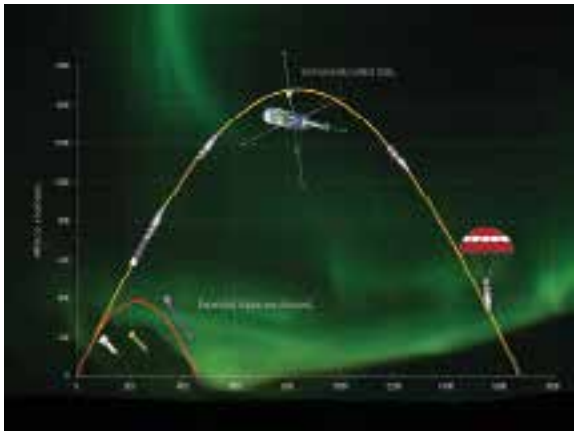
$$I = \frac{1}{2} - \frac{1}{2}\ln 2 - \frac{\pi}{4} + \frac{1}{2}\ln 2 + \frac{\pi^2}{32}$$

$$= \frac{1}{2} - \frac{\pi}{4} + \frac{\pi^2}{32}$$

SECTION - E

Case study based questions are compulsory.

36. Rocket motion is based on Newton's third law, which states that "for every action there is an equal and opposite reaction". Hot gases are exhausted through a nozzle of the rocket and produce the action force. The reaction force acting in the opposite direction is called the thrust force.



The equation of motion of a rocket are : $x = 2t$, $y = -4t$, $z = 4t$, where the time t is given in seconds, and the distance measured is in kilometres. Based on the above information, answer the following questions.

(i) What is the path of the rocket? Which of the following points lie on the path of the rocket?

(ii) At what distance will the rocket be from the starting point (0, 0, 0) in seconds?

(iii) At certain instant of time, if the rocket is above sea level, where equation of surface of sea is given by $3x - y + 4z = 2$ and position of rocket at that instant of time is (1, −2, 2), then find the image of position of rocket in the sea.

Sol :

(i) Eliminating t form the given equations, we get equations of path as,

$$\frac{x}{2} = \frac{y}{-4} = \frac{z}{4}$$

or $$\frac{x}{1} = \frac{y}{-2} = \frac{z}{2}$$

Thus, the path of the rocket represent a straight line.

Since, only $(1, -2, 2)$ satisfy the equation of path of rocket therefore $(1, -2, 2)$ lie on the path of rocket.

(ii) For $t = 10\,\text{sec}$, we have $x = 20$, $y = -40$, $z = 40$

Now, required distance

$$= \sqrt{x^2 + y^2 + z^2}$$

$$= \sqrt{20^2 + (-40)^2 + (40)^2}$$

$$= \sqrt{400 + 1600 + 1600}$$

$$= \sqrt{3600} = 60 \text{ km}$$

(iii) Let Q be the image of point $P(1, -2, 2)$ in the plane $3x - y + 4z = 2$. Then equation of PQ is

$$\frac{x-1}{3} = \frac{y+2}{-1} = \frac{z-2}{4}$$

Let the coordinate of Q be $(3r+1, -r-2, 4r+2)$.

Let R be the mid-point of PQ. Then, coordinate of R are

$$\left(\frac{3r+2}{2}, \frac{-r-4}{2}, \frac{4r+4}{2}\right) \text{ or } \left(\frac{3}{2}r+1, \frac{-r}{2}-2, 2r+2\right)$$

Since, R lies on $3x - y + 4z = 2$

$$3\left(\frac{3}{2}r+1\right) - \left(\frac{-r}{2}-2\right) + 4(2r+2) = 2$$

$$\frac{9r}{2} + 3 + \frac{r}{2} + 2 + 8r + 8 = 2$$

$$13r + 13 = 2$$

$$r = \frac{-11}{13}$$

Hence, the coordinate of Q are

$$\left(\frac{-133}{13}+1, \frac{11}{13}-2, \frac{-44}{13}+2\right) \text{ i.e., } \left(\frac{-20}{13}, \frac{-15}{13}, \frac{-18}{13}\right)$$

37. The D.A.V. College Managing Committee, familiarly known as DAVCMC, is a non-governmental educational organisation in India and overseas with over 900 schools. 75 colleges and a university. It is based on the ideals of Maharishi Dayanand Saraswati. Full Form of DAV is Dayanand Anglo Vedic.

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$$= \frac{2v}{1-v^2}$$

$$\frac{xdv}{dx} = \frac{2v}{1-v^2} - v$$

$$= \frac{2v - v + v^3}{1-v^2}$$

$$= \frac{v+v^3}{1-v^2} = \frac{v(1+v^2)}{1-v^2}$$

$$\frac{xdv}{dx} = \frac{v(1+v^2)}{1-v^2}$$

$$\frac{1-v^2}{v(1+v^2)}dv = \frac{dx}{x}$$

$$\frac{dx}{x} = \frac{1+v^2-2v^2}{v(1+v^2)}dv$$

$$= \frac{1+v^2}{v(1+v^2)}dv - \frac{2v^2\,dv}{v(1+v^2)}$$

$$\frac{dx}{x} = \frac{dv}{v} - \frac{2vdv}{1+v^2}$$

Integrating Both sides

$$\int \frac{dx}{x} = \int \frac{dv}{v} - \int \frac{2vdv}{1+v^2}$$

$$\ln x = \ln v - \ln(1+v^2) + \ln k \qquad \left[\int \frac{f'(x)}{f(x)} = \ln f(x)\right]$$

$$\ln x = \ln \frac{vk}{1+v^2} \qquad \left[\begin{matrix}\ln(xy) = \ln x + \ln y \\ \ln\left(\frac{x}{y}\right) = \ln x - \ln y\end{matrix}\right]$$

Comparing L.H.S. and R.H.S.

$$x = \frac{vk}{1+v^2}$$

Putting $v = \frac{y}{x}$, $\quad x = \frac{\frac{y}{x}k}{1+\left(\frac{y}{x}\right)^2} = \frac{\frac{yk}{x}}{\frac{x^2+y^2}{x^2}}$

$$= \frac{kxy}{x^2+y^2}$$

$$x^2 + y^2 = ky \qquad \text{(Required solution)}$$

OR

$$x\cos x\frac{dy}{dx} + y(x\sin x + \cos x) = 1 \qquad ...(1)$$

Simplifying the given eq.

$$\frac{dy}{dx} + y\left(\frac{x\sin x + \cos x}{x\cos x}\right) = \frac{1}{x\cos x} \quad \left[\frac{1}{\cos x} = \sec x\right]$$

$$\frac{dy}{dx} + y\left(\tan x + \frac{1}{x}\right) = \frac{\sec x}{x}$$

For equation of the form

$$\frac{dy}{dx} + Py = Q$$

Where P and Q are function of x

General solution is given by

$$ye^{\int P dx} = \int Qe^{\int P dx} + c$$

Here, $\quad P = \left(\tan x + \frac{1}{x}\right)$

and $\quad Q = \frac{\sec x}{x}$

$$\int P dx = \int \left(\tan x + \frac{1}{x}\right)dx$$

$$= \int \tan x dx + \int \frac{dx}{x}$$

$$= \ln\sec x + \ln x$$

$$= \ln\sec x \cdot x$$

Putting values in general solution

$$ye^{\ln(\sec x \cdot x)} = \int \frac{\sec x}{x} e^{\ln(\sec x \cdot x)} dx$$

$$y\sec x \cdot x = \int \frac{\sec x}{x} \times \sec x \times x dx \qquad [a^{\log_a x} = x]$$

$$= \int \sec^2 x dx \qquad \left[\int \sec^2 x dx = \tan x\right]$$

$$= \tan x + c$$

Thus, $\quad xy\sec x = \tan x + c$

34. Find the probability distribution of

(i) Number of heads in two tosses of a coin.

(ii) Number of tails in the simultaneous tosses of three coins.

(iii) Number of heads in four tosses of a coin.

Sol :

(i) Here $\quad X = 0, 1, 2$

Probability of getting head $= P(H) = \frac{1}{2}$

Probability of getting tail $= P(T) = 1 - \frac{1}{2} = \frac{1}{2}$

$$P(X=0) = P(T)\cdot P(T)$$

$$= \frac{1}{2}\times\frac{1}{2} = \frac{1}{4}$$

$$P(X=1) = P(H)\cdot P(T) + P(T)\cdot P(H)$$

$$= \frac{1}{2}\times\frac{1}{2} + \frac{1}{2}\times\frac{1}{2}$$

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Based on the above information, answer the following questions.

(i) If x is the number of days after 1st July then find the price and quantity of onion in terms of x.

(ii) Find the expression for the revenue as a function of x.

(iii) Find the number of days after 1st July, when Ramawatar attain maximum revenue.

OR

(iv) On which day should Ramawatar harvest the onions to maximum his revenue? What is this maximum revenue?

Sol :

(i) Let x be the number of extra days after 1st July.

$$\text{Price} = ₹(300 - 3 \times x) = ₹(300 - 3x)$$

$$\text{Quantity} = 80 \text{ quintals} + x \text{ (1 quintal per day)}$$

$$= (80 + x) \text{ quintals}$$

(ii) $$R(x) = \text{Quantity} \times \text{Price}$$

$$= (80 + x)(300 - 3x)$$

$$= 24000 - 240x + 300x - 3x^2$$

$$= 24000 + 60x - 3x^2$$

(iii) $$R(x) = 24000 + 60x - 3x^2$$

$$R'(x) = 60 - 6x$$

$$R''(x) = -6$$

For $R(x)$ to be maximum, $R'(x) = 0$ and $R''(x) < 0$

$$60 - 6x = 0$$

$$x = 10$$

(iv) Ramawatar will attain maximum revenue after 10 days.

So, he should harvest the onions after 10 days of 1st July i.e., on 11th July

Maximum revenue is collected by Ramawatar when $x = 10$

Maximum revenue

$$R(10) = 24000 + 60(10) - 3(10)^2$$

$$= 24000 + 600 - 300 = 24300$$

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Sample Paper 10

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

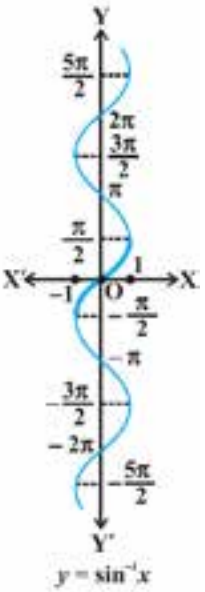
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the domain of the function $y = \cos^{-1}x$ as shown in the graph?



(a) $(-\infty,\infty)$ (b) $[-1,1]$

(c) $[0,\pi]$ (d) $[-\pi,\pi]$

Sol :

The graph clearly shows that the X-values are limited between –1 and 1. Beyond these values, the curve is not defined. Thus, the domain of the inverse cosine function is exactly the interval [-1,1]. Thus (b) is correct option.

2. Solution of $(x+2y^3)\,dy = y\,dx$ is

(a) $x = y^3 + cy$ (b) $x + y^3 = cy$

(c) $y^2 - x = cy$ (d) none of these

Sol :

Given equation can be rewritten as

$$\frac{dx}{dy} = \frac{x}{y} + 2y^2$$

$$\frac{dx}{dy} - \frac{x}{y} = 2y^2$$

$$\text{IF} = e^{\int p\,dx} = e^{\int -\frac{1}{y}dy} = \frac{1}{y}$$

Solution is, $$\frac{x}{y} = \int 2y\,dy + c = y^2 + c$$

$$x = y^3 + cy$$

Thus (a) is correct option.

3. If $\vec{a}$ is perpendicular to $\vec{b}$ and $\vec{p}$ is a non-zero vector such that $p\vec{r} + (\vec{r}\cdot\vec{b})\vec{a} = \vec{c}$, then $\vec{r}$ is equal to

(a) $\frac{\vec{c}}{p} - \frac{(\vec{b}\cdot\vec{c})\vec{a}}{p^2}$ (b) $\frac{\vec{a}}{p} - \frac{(\vec{c}\cdot\vec{a})\vec{b}}{p^2}$

(c) $\frac{\vec{b}}{p} - \frac{(\vec{a}\cdot\vec{b})\vec{c}}{p^2}$ (d) $\frac{\vec{c}}{p^2} - \frac{(\vec{b}\cdot\vec{c})\vec{a}}{p}$

Sol :

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Given that, $p\vec{r}+(\vec{r}\cdot\vec{b})\vec{a}=\vec{c}$ ...(i)

$$p(\vec{r}\cdot\vec{b})+(\vec{r}\cdot\vec{b})(\vec{a}\cdot\vec{b})=\vec{c}\cdot\vec{b}$$

$$p(\vec{r}\cdot\vec{b})=\vec{c}\cdot\vec{b}$$

$$\vec{r}\cdot\vec{b}=\frac{\vec{c}\cdot\vec{b}}{p}$$

From Eq. (i), we get

$$p\vec{r}+\frac{(\vec{c}\cdot\vec{b})}{p}\vec{a}=\vec{c}$$

$$\vec{r}=\frac{\vec{c}}{p}-\frac{\vec{c}\cdot\vec{b}}{p^2}\vec{a}$$

Thus (a) is correct option.

4. The lines $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3}$

and $\frac{x}{2}=\frac{y+2}{2}=\frac{z-3}{-2}$ are

(a) Parallel (b) Intersecting

(c) At right angle (d) None of these

Sol :

Here, $a_1=1,\ b_1=2,\ c_1=3$

and $a_2=2,\ b_2=2,\ c_2=-2$

Now, $a_1a_2+b_1b_2+c_1c_2=1\cdot2+2\cdot2+3\cdot(-2)$

$$=2+4-6=0$$

Hence, lines are at right angle.

Thus (c) is correct option.

5. A coin and six faced die, both unbiased, are thrown simultaneously. The probability of getting a head on the coin and an odd number on the die is

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$

(c) $\frac{1}{4}$ (d) $\frac{2}{3}$

Sol :

Let the following event be defined

E = Event of getting a head on a coin and

F = Event of getting an odd number $\{1,3,5\}$, on a die

$$P(E)=\frac{1}{2},\ P(F)=\frac{3}{6}=\frac{1}{2}$$

Since, E and F are independent events.

$$P(E\cap F)=P(E)\cdot P(F)=\frac{1}{2}\times\frac{1}{2}=\frac{1}{4}$$

Thus (c) is correct option.

6. The number of solutions of $y'=\frac{y+1}{x-1}, y(1)=2$ is

(a) Zero (b) One

(c) Two (d) Infinite

Sol :

We have $\frac{dy}{dx}=\frac{y+1}{x-1}$

$$\frac{dy}{y+1}=\frac{dx}{x-1}$$

On integrating, we get

$$\log(y+1)=\log(x-1)+\log c$$

$$y+1=c(x-1)$$

For $x=1$, $y=2$

$$3=c(0)$$

$$3=0$$

Which is not possible.

Hence, given equation has zero solution.

Thus (a) is correct option.

7. Solution set of the inequality $x\geq0$ is

(a) half plane on the left of y-axis

(b) half plane on the right of y-axis excluding the points of y-axis

(c) half plane on the right of y-axis including the points on y-axis

(d) none of the above

Sol :

Solution set of the given inequality is $\{(x,y): x\geq0\}$ i.e., the set of all points whose abscissa are non-negative. All these points lie either on y-axis or on the right of y-axis.

Thus (c) is correct option.

8. Find the area of a curve $xy=4$, bounded by the lines $x=1$ and $x=3$ and X-axis.

(a) $\log12$ (b) $\log64$

(c) $\log81$ (d) $\log27$

Sol :

Given curve is $xy=4$.

Required area, $A=\int_1^3\frac{4}{x}dx=4\cdot[\log x]_1^3$

$$=4(\log3-\log1)$$

$$=4\log3=\log81$$

Thus (c) is correct option.

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Also, f is not subjective as some element of N (co-domain) is not the image of any element of N.

Thus (d) is correct option.

19. **Assertion:** $\int \sin 3x \cos 5x \, dx = \frac{-\cos 8x}{16} + \frac{\cos 2x}{4} + C$

Reason: $2\cos A \sin B = \sin(A+B) - \sin(A-B)$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

$$\begin{aligned} I &= \int \sin 3x \cos 5x \, dx \\ &= \frac{1}{2}\int 2\cos 5x \sin 3x \, dx \\ &= \frac{1}{2}\int (\sin 8x - \sin 2x)\, dx \\ &= \frac{1}{2}\Big[\int \sin 8x \, dx - \int \sin 2x \, dx\Big] \\ &= \frac{1}{2}\Big[\frac{-\cos 8x}{8}\Big] - \frac{1}{2}\Big[\frac{-\cos 2x}{4}\Big] + C \\ &= \frac{-\cos 8x}{16} + \frac{\cos 2x}{4} + C \end{aligned}$$

Both Assertion and reason are true and reason is the correct explanation of assertion.

Thus (a) is correct option.

20. **Assertion:** If $x = at^2$ and $y = 2at$, then $\left.\frac{d^2y}{dx^2}\right|_{t=2} = \frac{-1}{16a}$

Reason: $\frac{d^2y}{dx^2} = \left(\frac{dy}{dt}\right)^2 \times \left(\frac{dt}{dx}\right)^2$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

We have,

$$\begin{aligned} x &= at^2, \; y = 2at \\ \frac{dx}{dt} &= 2at \text{ and } \frac{dy}{dt} = 2a \\ \frac{dy}{dx} &= \frac{dy}{dt} \times \frac{dt}{dx} = \frac{1}{t} \end{aligned}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= \frac{-1}{t^2} \times \frac{dt}{dx} \\ &= \frac{-1}{t^2} \times \frac{1}{2at} = \frac{-1}{2at^3} \\ \left.\frac{d^2y}{dx^2}\right|_{t=2} &= \frac{-1}{2 \times a \times (2)^3} = \frac{-1}{16a} \end{aligned}$$

Thus (c) is correct option.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Find a vector in the direction of vector $2\hat{i} - 3\hat{j} + 6\hat{k}$ which has magnitude of 21 units.

OR

Find the angle between X-axis and the vector $\hat{i} + \hat{j} + \hat{k}$.

Sol :

We have $\vec{a} = 2\hat{i} - 3\hat{j} + 6\hat{k}$

Now $|\vec{a}| = \sqrt{2^2 + (-3)^2 + 6^2}$

$= \sqrt{4 + 9 + 36}$

$= \sqrt{49} = 7$ units

The unit vector in the direction of the given vector $\vec{a}$ is

$$\begin{aligned} \hat{a} &= \frac{\vec{a}}{|\vec{a}|} = \frac{1}{7}(2\hat{i} - 2\hat{j} + 6\hat{k}) \\ &= \frac{2}{7}\hat{i} - \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k} \end{aligned}$$

The vector of magnitude equal to 21 units and in the direction of $\vec{a}$ is given by

$$\begin{aligned} 21\hat{a} &= 21\Big(\frac{2}{7}\hat{i} - \frac{3}{7}\hat{j} + \frac{6}{7}\hat{k}\Big) \\ &= 6\hat{i} - 9\hat{j} + 18\hat{k} \end{aligned}$$

OR

We have $\vec{a} = \hat{i} + \hat{j} + \hat{k}$

Now, unit vector in the direction of $\vec{a}$ is

$$\begin{aligned} \hat{a} &= \frac{\vec{a}}{|\vec{a}|} = \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{1^2 + 1^2 + 1^2}} \\ &= \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}} \\ &= \frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \frac{1}{\sqrt{3}}\hat{k} \end{aligned}$$

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Sol :

Since, $0 \le x \le \frac{\pi}{2}$

$0 \le \frac{x}{2} \le \frac{\pi}{4}$

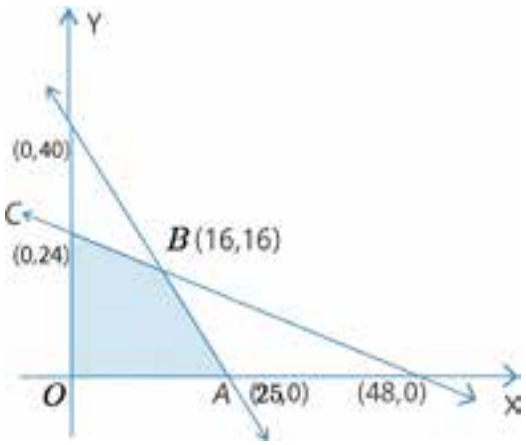
$0 \le \sin\frac{x}{2} \le \frac{1}{\sqrt{2}}$

$\left(0, \frac{1}{\sqrt{2}}\right) \subset [0, \infty)$

Hence, function is injective.

Thus (a) is correct option.

16. For the feasible region in the figure, determine the maximum value of the objective function $z = 3x + 5y$.



(a) 72

(b) 80

(c) 112

(d) 128

Sol :

By the corner point method, calculate values of the objective function at all corner points:

At O(0,0) : z=3(0)+5(0)=0

At A(25,0) : z=3(25)+5(0)=75

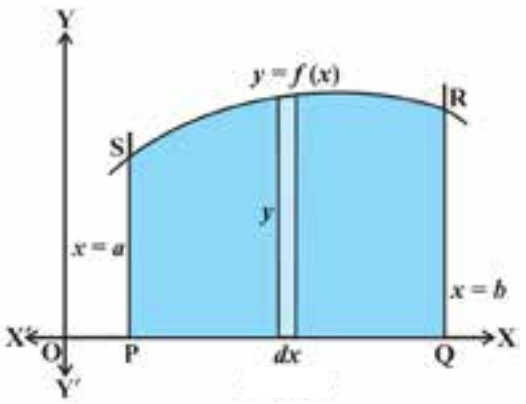
At B(16,16) : z=3(16)+5(16)=48+80=128

At C(0,24) : z=3(0)+5(24)=120

Among these, the maximum value is 128, which occurs at vertex B(16,16). Therefore, the correct answer is 128.

Thus (d) is correct option.

17. In the given figure, the shaded region represents which part of the circle?



(a) A quarter of the circle in the first quadrant bounded by OA, OB, and the arc AB

(b) A semicircle bounded by the X-axis and Y-axis

(c) The entire circle centered at O with radius a

(d) A quarter of the circle in the second quadrant bounded by OA, OB, and the arc AB

Sol :

(a) A quarter of the circle in the first quadrant bounded by OA, OB, and the arc AB

The circle is centered at O with radius a. The shaded region lies in the first quadrant, bounded by the X-axis, Y-axis, and the arc joining A(a,0) and B(0,a). Thus, it clearly represents a quarter of the circle in the first quadrant.

18. A mapping $f : n \to N$, where N is the set of natural numbers is define as

$$f(n) = \begin{cases} n^2, & \text{for } n \text{ odd} \\ 2n+1, & \text{for } n \text{ even} \end{cases}$$

for $n \in N$. Then, f is

(a) Subjective but not injective

(b) Injective but not subjective

(c) Bijective

(d) neither injective nor subjective

Sol :

Since, $f(n) = \begin{cases} n^2, & \text{if } n \text{ is odd.} \\ 2n+1, & \text{if } n \text{ is even} \end{cases}$

$f(1) = 1^2 = 1$

$f(2) = 2(2) + 1 = 5$

$f(3) = 3^2 = 9$

$f(4) = 2(4) + 1 = 9$

$f(3) = f(4)$

So, f is not injective.

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9. The derivative of $\cos^3 x$ with respect to $\sin^3 x$ is

(a) $-\cot x$ (b) $\cot x$

(c) $\tan x$ (d) $-\tan x$

Sol :

Let $u = \cos^3 x$ and $v = \sin^3 x$

$$\frac{du}{dx} = -3\cos^2 x \sin x \text{ and } \frac{dv}{dx} = 3\sin^2 x \cos x$$

Now, $$\frac{du}{dv} = \frac{-3\cos^2 x \sin x}{3\sin^2 x \cos x} = -\cot x$$

Thus (a) is correct option.

10. Let $f(x) = x - \cos x, x \varepsilon R$, then f is

(a) A decreasing function

(b) An odd function

(c) An increasing function

(d) None of the above

Sol :

$$f'(x) = 1 + \sin x \geq 0, \text{ for all } x \varepsilon R.$$

$$(-1 \leq \sin x \leq 1 \; 0 \leq 1 + \sin x \leq 2)$$

Hence, $f(x)$ is a decreasing function.

Thus (c) is correct option.

11. The length of the longest interval, in which $f(x) = 3\sin x - 4\sin^3 x$ is increasing is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$

(c) $\frac{3\pi}{2}$ (d) π

Sol :

We have $f(x) = 3\sin x - 4\sin^3 x = \sin 3x$

Since, $\sin x$ is increasing in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

$$-\frac{\pi}{2} \leq 2x \leq \frac{\pi}{2}$$

$$-\frac{\pi}{6} \leq x \leq \frac{\pi}{6}$$

Thus, the length of interval is $\frac{\pi}{6} - \left(-\frac{\pi}{6}\right) = \frac{\pi}{3}$.

Thus (a) is correct option.

12. $\int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x} dx$ is equal to

(a) 0 (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) π

Sol :

$$I = \int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x} dx \quad ...(i)$$

$$= \int_0^{\pi/2} \frac{\sin\left(\frac{\pi}{2} - x\right) - \cos\left(\frac{\pi}{2} - x\right)}{1 + \sin\left(\frac{\pi}{2} - x\right)\cos\left(\frac{\pi}{2} - x\right)} dx$$

$$= \int_0^{\pi/2} \frac{\cos x - \sin x}{1 + \cos x \sin x} dx \quad ...(ii)$$

Adding Eqs. (i) and (ii), we get

$$2I = \int_0^{\pi/2} \frac{0}{1 + \sin x \cos x} dx = 0 \Rightarrow I = 0$$

Thus (a) is correct option.

13. If $f(\theta) = \begin{vmatrix} \cos^2\theta & \cos\theta\sin\theta & -\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta & \cos\theta \\ \sin\theta & -\cos\theta & 0 \end{vmatrix}$, then for all values of θ

(a) $f(\theta) = 0$ (b) $f(\theta) = 1$

(c) $f(\theta) = -1$ (d) None of these

Sol :

We have

$$f(\theta) = \begin{vmatrix} \cos^2\theta & \cos\theta\sin\theta & -\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta & \cos\theta \\ \sin\theta & -\cos\theta & 0 \end{vmatrix}$$

$$= \cos^2\theta(0 + \cos^2\theta) - \cos\theta\sin\theta(0 - \sin\theta\cos\theta) - \sin\theta(-\cos^2\theta\sin\theta - \sin^3\theta)$$

$$= \cos^4\theta + 2\sin\theta\cos^2\theta + \sin^4\theta)$$

$$= (\cos^2\theta + \sin^2\theta)^2 = (1)^2 = 1$$

For all θ, $f(\theta) = 1$

Thus (b) is correct option.

14. If $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$ and A^2 is the identity matrix, then x is equal to

(a) -1 (b) 0

(c) 1 (d) 2

Sol :

We have $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$

Now, $$A^2 = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} x^2 + 1 & x \\ x & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Equating the corresponding elements, we get $x = 0$.

Thus (b) is correct option.

15. If $f : [0, \frac{\pi}{2}] \to [0, \infty]$ be a function defined by $y = \sin\left(\frac{\pi}{2}\right)$, then f is

(a) injective (b) subjective

(c) bijective (d) none of these

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If angle between X-axis and the vector $\hat{i}+\hat{j}+\hat{k}$ is α then

$$\cos\alpha = \frac{1}{\sqrt{3}} \Rightarrow \alpha = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

22. Evaluate $\int \cos^{-1}(\sin x)\,dx$.

OR

Write the anti-derivative of $\left(3\sqrt{x}+\frac{1}{\sqrt{x}}\right)$.

Sol :

We have $I = \int \cos^{-1}(\sin x)\,dx$

$$= \int \cos^{-1}\left[\cos\left(\frac{\pi}{2}-x\right)\right]dx$$

$$= \int \left(\frac{\pi}{2}-x\right)dx \qquad \cos^{-1}(\cos\theta)=\theta$$

$$= \frac{\pi}{2}\int dx - \int x\,dx$$

$$= \frac{\pi}{2}x - \frac{x^2}{2} + C$$

OR

Anti-derivative of $f(x)$ is $\int f(x)$. Thus anti-derivative of $\left(3\sqrt{x}+\frac{1}{\sqrt{x}}\right)$, is the integration of

$$I = \int \left(3\sqrt{x}+\frac{1}{\sqrt{x}}\right)dx$$

$$= 3\int \sqrt{x}\,dx + \int \frac{1}{\sqrt{x}}\,dx$$

$$= \int \left(3\sqrt{x}+\frac{1}{\sqrt{x}}\right)dx$$

$$= 3\int \sqrt{x}\,dx + \int \frac{1}{\sqrt{x}}\,dx$$

$$= 3\left(\frac{x^{1/2+1}}{1/2+1}\right) + \left[\frac{x^{-1/2+1}}{-1/2+1}\right] + C$$

$$= 2\left(x^{3/2}+x^{1/2}\right) + C$$

23. What are the direction consines of a line which makes equal angles with the coordinate axes?

Sol :

Let α, β and γ be the angles made by the line with coordinate axes. Since line makes equal angles with coordinate axes, we have

$$\alpha = \beta = \gamma \Rightarrow \cos\alpha = \cos\beta = \cos\gamma$$

Since $l=\cos\alpha, m=\cos\beta$ and $n=\cos\gamma$ we have

$$l = m = n \qquad ...(1)$$

Now $l^2+m^2+n^2 = 1$

Substituting $l=m=n$ in above we have

$$l^2+l^2+l^2 = 1$$

$$3l^2 = 1$$

$$l^2 = \frac{1}{3} \Rightarrow l = \pm\frac{1}{\sqrt{3}}$$

From Eq. (1), direction cosines of a line as

$$\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right) \text{ or } \left(\frac{-1}{\sqrt{3}},\frac{-1}{\sqrt{3}},\frac{-1}{\sqrt{3}}\right)$$

24. If $A=\{1,2,3\}, B=\{4,5,6,7\}$ and $f=\{(1,4),(2,5),(3,6)\}$ is a function from A to B. State whether f is one-one or not.

Sol :

Given, $A = \{1,2,3\}$

$B = \{4,5,6,7\}$

and $f: A \to B$ is defined as

$f = \{(1,4),(2,5),(3,6)\}$

i.e. $f(1)=4,\ f(2)=5\ f(3)=6$

It can be seen that the images of distinct elements of A under f are distinct. So, f is one-one.

25. Two groups are computing for the positions of the Board of Directors of a corporation. The probabilities that the first and second groups will win are 0.6 and 0.4 respectively. Further, if the first group wins, the probability of introducing a new product introduced way by the second group.

Sol :

Let E_1 and E_2 denote the events that first and second group will win. Then,

$$P(E_1) = 0.6 \text{ and } P(E_2) = 0.4$$

Let E be the event of introducing the new product.

Then, $P\left(\frac{E}{E_1}\right) = 0.7$ and $P\left(\frac{E}{E_2}\right) = 0.3$

Now, we have to find the probability that new product is introduced by second event.

$$P\left(\frac{E_2}{E}\right) = \frac{P(E_2)P\left(\frac{E}{E_2}\right)}{P(E_1)P\left(\frac{E}{E_1}\right)+P(E_2)P\left(\frac{E}{E_2}\right)}$$

$$= \frac{0.4\times 0.3}{0.6\times 0.7+0.4\times 0.3}$$

$$= \frac{0.12}{0.42+0.12}$$

$$= \frac{0.12}{0.54} = 0.22$$

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and $\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5} = \mu$ (let)...(2)

Then, any point on line (1) is of the form

$$P(3\lambda - 1, 5\lambda - 3, 7\lambda - 5) \quad ...(3)$$

and any point on line (2) is of the form

$$Q(u + 2, 3\mu + 4, 5\mu + 6) \quad ...(4)$$

If lines (1) and (2) intersect, then these points must coincide for some λ and μ.

Consider, $3\lambda - 1 = \mu + 2$

$$5\lambda - 3 = 3\mu + 4$$

and $7\lambda - 5 = 5\mu + 6$

$$3\lambda - \mu = 3 \quad ...(5)$$

$$5\lambda - 3\mu = 7 \quad ...(6)$$

and $7\lambda - 5\mu = 11 \quad ...(7)$

Multiplying Eq. (5) by 3 and then subtracting Eq, (6), we get

$$9\lambda - 3\mu - 5\lambda + 3\mu = 9 - 7$$

$$4\lambda = 2 \Rightarrow \lambda = \frac{1}{2}$$

Substituting the value of λ in Eq. (5), we get

$$3 \times \frac{1}{2} - \mu = 3$$

$$\frac{3}{2} - \mu = 3 \Rightarrow \mu = -\frac{3}{2}$$

Substituting the values of λ and μ in Eq. (7), we get

$$7 \times \frac{1}{2} - 5\left(-\frac{3}{2}\right) = 11$$

$$\frac{7}{2} + \frac{15}{2} = 11 \Rightarrow \frac{22}{2} = 11$$

$11 = 11$, which is true.

Hence, lines (1) and (2) intersect and substituting $\lambda = \frac{1}{2}$ in Eq. (3) we get their point of intersection as

$$P\left(3 \times \frac{1}{2} - 1, 5 \times \frac{1}{2} - 3, 7 \times \frac{1}{2} - 5\right)$$

i.e. $P\left(\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}\right)$.

33. Find $\int \frac{x^2 + x + 1}{(x+2)(x^2+1)} dx$.

OR

Evaluate $\int_0^{\pi} e^{2x} \cdot \sin\left(\frac{\pi}{4} + x\right) dx$.

Sol :

We have $I = \int \frac{x^2 + x + 1}{(x+2)(x^2+1)} dx$

Here we will make given expression simplify using partial fraction.

Let $\frac{x^2 + x + 1}{(x^2+1)(x+2)} = \frac{A}{(x+2)} + \frac{Bx + C}{x^2+1}$

$$x^2 + x + 1 = A(x^2 + 1) + (Bx + C)(x + 2)$$

Substituting $x = -2$ we have,

$$4 - 2 + 1 = A(5) + 0 \Rightarrow 5A = 3 \Rightarrow A = \frac{3}{5}$$

Substituting $x = 0$, we have

$$0 + 0 + 1 = A(0 + 1) + (0 + C)(0 + 2)$$

$$1 = A + 2C \Rightarrow 1 = \frac{3}{5} + 2C$$

$$2C = \frac{2}{5} \Rightarrow C = \frac{1}{5}$$

and substituting $x = 1$ we obtain,

$$1 + 1 + 1 = 2A + (B + C)(3)$$

$$3 = 2A + 3(B + C)$$

$$3 = 2\left(\frac{3}{2}\right) + 3\left(B + \frac{1}{5}\right)$$

$$3 - \frac{6}{5} = 3\left(B + \frac{1}{5}\right)$$

$$\frac{9}{5} = 3\left(B + \frac{1}{5}\right)$$

$$\frac{3}{5} - \frac{1}{5} = B \Rightarrow B = \frac{2}{5}$$

Thus, $\frac{x^2 + x + 1}{(x+2)(x^2+1)} = \frac{3}{5(x+2)} + \frac{\left(\frac{2}{5}x + \frac{1}{5}\right)}{(x^2+1)}$

Now $I = \int \frac{x^2 + x + 1}{(x+2)(x^2+1)} dx$

$$= \int \frac{3}{5(x+2)} dx + \frac{1}{5}\int \frac{2x}{x^2+1} dx + \frac{1}{5}\int \frac{dx}{x^2+1}$$

$$= \frac{3}{5}\log|x + 2| + \frac{1}{5}\log|x^2 + 1| + \frac{1}{5}\tan^{-1}(x) + C$$

OR

We have $I = \int_0^{\pi} e^{2x} \sin\left(\frac{\pi}{4} + x\right) dx$

Using integration by parts technique

$$\int \underset{I}{u} \cdot \underset{II}{v} dx = \left[u \int v dx - \int \left\{\frac{d}{dx} u \cdot \int v dx\right\} dx\right]$$

and choosing its function with the help of ILATE procedure we have

$$I_1 = \int \underset{II}{e^{2x}} \sin\left(\frac{\pi}{4} + \underset{I}{x}\right) dx \quad ...(i)$$

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$$\cos^{-1}\left(\frac{1}{2}\right)+2\sin^{-1}\left(\frac{1}{2}\right)=\cos^{-1}\left(\cos\frac{\pi}{3}\right)+2\sin^{-1}\left(\sin\frac{\pi}{6}\right)$$
$$=\frac{\pi}{3}+2\times\frac{\pi}{6}$$
$$=\frac{\pi}{3}+\frac{\pi}{3}=\frac{2\pi}{3}$$

30. If $y=\sin^{-1}(6x\sqrt{1-9x^2})$, $-\frac{1}{3\sqrt{2}}<x<\frac{1}{3\sqrt{2}}$, then find $\frac{dy}{dx}$.

OR

If $(\cos x)^y=(\cos y)^x$, then find $\frac{dy}{dx}$.

Sol :

We have $y=\sin^{-1}(6x\sqrt{1-9x^2})$

or $y=\sin^{-1}(2\cdot 3x\sqrt{1-(3x)^2})$

Substituting $3x=\sin\theta$, we have

$$y=\sin^{-1}(2\sin\theta\sqrt{1-\sin^2\theta})$$
$$=\sin^{-1}(2\sin\theta\cdot\cos\theta)$$
$$=\sin^{-1}(\sin 2\theta)\ =2\theta$$

Thus $y=2\sin^{-1}(3x)$

$$\frac{dy}{dx}=\frac{2}{\sqrt{1-9x^2}}(3)$$
$$\frac{dy}{dx}=\frac{6}{\sqrt{1-9x^2}}$$

OR

We have, $(\cos x)^y=(\cos y)^x$

Taking log both sides, we get

$$\log(\cos x)^y=\log(\cos y)^x$$
$$y\log(\cos x)=x\log(\cos y)$$

Differentiating w.r.t. x, we get

$$y\cdot\frac{d}{dx}\log(\cos x)+\log\cos x\cdot\frac{d}{dx}(y)$$
$$=x\frac{d}{dx}\log(\cos y)+\log(\cos y)\frac{d}{dx}(x)$$
$$y\cdot\frac{1}{\cos x}\frac{d}{dx}(\cos x)+\log(\cos x)\frac{dy}{dx}$$
$$=x\cdot\frac{1}{\cos y}\frac{d}{dx}(\cos y)+\log\cos y\cdot 1$$
$$y\cdot\frac{1}{\cos x}(-\sin x)+\log(\cos x)\frac{dy}{dx}$$
$$=-x\tan y\frac{dy}{dx}+\log(\cos y)$$
$$[x\tan y+\log(\cos x)]\frac{dy}{dx}=\log(\cos y)+y\tan x$$
$$\frac{dy}{dx}=\frac{\log(\cos y)+y\tan x}{x\tan y+\log(\cos x)}$$
$$\frac{dy}{dx}=\frac{-1}{(1+x)^2}$$ Hence proved.

31. If $\begin{vmatrix} x & \sin\theta & \cos\theta \\ -\sin\theta & -x & 1 \\ \cos\theta & 1 & x \end{vmatrix}=8$, write the value of x.

Sol :

We have, $\begin{vmatrix} x & \sin\theta & \cos\theta \\ -\sin\theta & -x & 1 \\ \cos\theta & 1 & x \end{vmatrix}=8$

Expanding along R_1, we get

$$x(-x^2-1)-\sin\theta(-x\sin\theta-\cos\theta)$$
$$+\cos\theta(-\sin\theta+x\cos\theta)=8$$
$$-x^3-x+x\sin^2\theta+\sin\theta\cos\theta-\sin\theta\cos\theta$$
$$+x\cos^2\theta=8$$
$$-x^3-x+x(\sin^2\theta+\cos^2\theta)=8$$
$$-x^3-x+x=8$$
$$-x^3=8$$
$$x^3+8=0$$
$$x^3+2^3=0$$
$$(x+2)(x^2-2x+4)=0$$
$$x=-2$$

Here $x^2-2x+4=0$, gives imaginary values.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Show that the lines $\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}$ and $\frac{x-2}{1}=\frac{y-4}{3}=\frac{z-6}{5}$ intersect. Also, find their point intersection.

Sol :

We have $\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{-7}=\lambda$ (let)...(1)

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SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. The volume of a sphere is increasing at the rate of 3 cubic centimeter per second. Find the rate of increase of its surface area, when the radius is 2 cm.

OR

Show that the function $f(x) = 4x^3 - 18x^2 + 27x - 7$ is always increasing on R.

Sol :

Let r be the radius, V be the volume and S be the surface area of sphere.

We have $\frac{dV}{dt} = 3 \text{ cm}^3/\text{s}$

Now $V = \frac{4}{3}\pi r^3$

$\frac{dV}{dt} = \frac{4}{3}\pi \cdot 3r^2 \cdot \frac{dr}{dt}$

$3 = 4\pi \times r^2 \frac{dr}{dt}$

$\frac{dr}{dt} = \frac{3}{4\pi r^2} \text{ cm/s}$ ...(1)

Now, $S = 4\pi r^2$

$\frac{dS}{dt} = \frac{d}{dt}(4\pi r^2)$

$= 4\pi \times 2r \cdot \frac{dr}{dt}$

$= 8\pi r \times \frac{3}{4\pi r^2}$ Using eq. (1)

$= \frac{6}{r}$

$\left(\frac{dS}{dt}\right)_{r=2} = \frac{6}{2} = 2 \text{ cm}^2/\text{s}$

OR

We have, $f(x) = 4x^3 - 18x^2 + 27x - 7$

Differentiating both sides w.r.t. x, we get

$f'(x) = 12x^2 - 36x + 27$

$= 3(4x^2 - 12x + 9)$

$= 3(2x-3)^2$

For any $x \in R$, $(2x-3)^2 \geq 0$ because, a perfect square number cannot be negative. Thus given function $f(x)$ is an increasing function on R.

27. Evaluate $\int \frac{2\cos x}{\sin^2 x} dx$.

Sol :

We have $I = \int \frac{2\cos x}{\sin^2 x} dx$

Substituting $\sin x = t \Rightarrow \cos x\, dx = dt$ we have

$I = \int \frac{2}{t^2} dt$

$= 2\int \frac{dt}{t^2}$

$= \frac{2t^{-1}}{-1} + C$

Substituting $t = \sin x$ we have

$I = -2(\sin x)^{-1} + C$

$= \frac{-2}{\sin x} + C$

$= -2\operatorname{cosec} x + C$

ALTERNATIVE METHOD :

We have $I = \int \frac{2\cos x}{\sin^2 x} dx$

$= 2\int \frac{1}{\sin x} \cdot \frac{\cos x}{\sin x} dx$

$= 2\int \operatorname{cosec} x \cot x\, dx$

$= 2(-\operatorname{cosec} x) + C$

$= -2\operatorname{cosec} x + C$

28. Find the area of a parallelogram whose adjacent sides represented by the vectors $2\hat{i} - 3\hat{k}$ and $4\hat{i} + 2\hat{k}$.

Sol :

We have $\vec{a} = 2\hat{i} - 3\hat{k}$

and $\vec{b} = 4\hat{j} + 2\hat{k}$

where $\vec{a}$ and $\vec{b}$ are sides of a parallelogram

Area of parallelogram is give by

$|\vec{a} \times \vec{b}| = |(2\hat{i} - 3\hat{k}) \times (4\hat{j} + 2\hat{k})|$

$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & -3 \\ 0 & 4 & 2 \end{vmatrix}$

$= |\hat{i}(0+12) - \hat{j}(4+0) + \hat{k}(8-0)|$

$= |12\hat{i} - 4\hat{j} + 8\hat{k}|$

$= \sqrt{12^2 + (-4)^2 + (8)^2}$

$= \sqrt{144 + 16 + 64}$

$= \sqrt{224} = 4\sqrt{14}$ sq units

29. Using the principal values, write the value of $\cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right)$.

Sol :

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$$= \sin\left(\frac{\pi}{4}+x\right)\int e^{2x}\,dx - \int\left\{\frac{d}{dx}\sin\left(\frac{\pi}{4}+x\right)\int e^{2x}\,dx\right\}dx$$

$$= \sin\left(\frac{\pi}{4}+x\right)\frac{e^{2x}}{2} - \int\cos\left(\frac{\pi}{4}+x\right)\frac{e^{2x}}{2}\,dx$$

Again using integration by parts technique

$$I_1 = \frac{e^{2x}}{2}\sin\left(\frac{\pi}{4}+x\right) - \frac{1}{2}\int \underset{II}{e^{2x}}\cos\left(\frac{\pi}{4}+x\right)\underset{I}{dx}$$

$$= \frac{e^{2x}}{2}\sin\left(\frac{\pi}{4}+x\right) - \frac{1}{2}\left[\cos\left(\frac{\pi}{4}+x\right)\frac{e^{2x}}{2} - \int -\sin\left(\frac{\pi}{4}+x\right)\frac{e^{2x}}{2}dx\right]$$

$$= \frac{e^{2x}}{2}\sin\left(\frac{\pi}{4}+x\right) - \frac{e^{2x}}{4}\cos\left(\frac{\pi}{4}+x\right) - \frac{1}{4}\int e^{2x}\sin\left(\frac{\pi}{4}+x\right)dx$$

$$I_1 = \frac{e^{2x}}{4}\left\{2\sin\left(\frac{\pi}{4}+x\right) - \cos\left(\frac{\pi}{4}+x\right)\right\} - \frac{1}{4}I_1$$

$$I_1 + \frac{1}{4}I_1 = \frac{e^{2x}}{4}\left\{2\sin\left(\frac{\pi}{4}+x\right) - \cos\left(\frac{\pi}{4}+x\right)\right\}$$

$$\frac{5}{4}I_1 = \frac{e^{2x}}{4}\left\{2\sin\left(\frac{\pi}{4}+x\right) - \cos\left(\frac{\pi}{4}+x\right)\right\}$$

$$I_1 = \frac{e^{2x}}{5}\left\{2\sin\left(\frac{\pi}{4}+x\right) - \cos\left(\frac{\pi}{4}+x\right)\right\}$$

$$I = [I_1]_0^{\pi}$$

$$= \left[\frac{e^{2x}}{5}\left\{2\sin\left(\frac{\pi}{4}+x\right) - \cos\left(\frac{\pi}{4}+x\right)\right\}\right]_0^{\pi}$$

$$= \frac{1}{5}\left[e^{2\pi}\left\{2\sin\left(\frac{\pi}{4}+\pi\right) - \cos\left(\frac{\pi}{4}+\pi\right)\right\}\right] - e^{0}\left\{2\sin\left(\frac{\pi}{4}+0\right) - \cos\left(\frac{\pi}{4}+0\right)\right\}$$

$$= \frac{1}{5}\left[e^{2\pi}\left\{-2\sin\frac{\pi}{4} + \cos\frac{\pi}{4}\right\} - e^{0}\left\{2\sin\frac{\pi}{4} - \cos\frac{\pi}{4}\right\}\right]$$

$$= \frac{1}{5}\left[e^{2\pi}\left\{-2\times\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right\} - 1\left\{2\times\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right\}\right]$$

$$= \frac{1}{5}\left[e^{2\pi}\left\{-\frac{1}{\sqrt{2}}\right\} - \frac{1}{\sqrt{2}}\right] = -\frac{1}{5\sqrt{2}}[e^{2\pi}+1]$$

34. Solve the differential equation $\frac{dy}{dx} - 3y\cot x = \sin 2x$, given $y = 2$ when $x = \frac{\pi}{2}$.

Sol :

We have, $\frac{dy}{dx} - 3y\cot x = \sin 2x$ ...(1)

This is a linear differential equation of the form $\frac{dy}{dx} = Py = Q$, where $P = -3\cot x$ and $Q = \sin 2x$.

$$\text{IF} = e^{\int P\,dx} = e^{-3\int \cot x\,dx}$$

$$= e^{-3\log|\sin x|} = e^{\log|\sin x|^{-3}}$$

$$= |\sin x|^{-3}$$

Using $y\cdot(IF) = \int Q\cdot(IF)\,dx + C$ the general solution is

$$y\cdot(\sin x)^{-3} = \int(\sin x)^{-3}(\sin 2x)\,dx + C$$

$$= \int\frac{2\sin x\cos x}{\sin^3 x}\,dx + C$$

$$y\cdot(\sin x)^{-3} = \int\frac{2\cos x}{\sin^2 x}\,dx + C \quad ...(1)$$

Substituting $\sin x = t \Rightarrow \cos x\,dx = dt$ in Eq. (1), we get

$$y\cdot(\sin x)^{-3} = 2\int\frac{dt}{t^2} + C = 2\times\frac{t^{-1}}{-1} + C$$

$$y(\sin x)^{-3} = -\frac{2}{t} + C$$

$$y(\sin x)^{-3} = \frac{-2}{\sin x} + C$$

$$y = -2\sin^2 x + C\sin^3 x \quad ...(2)$$

Substituting $x = \frac{\pi}{2}$ and $y = 2$ in Eq. (2), we get

$$2 = -2\sin^2\frac{\pi}{2} + C\sin^3\frac{\pi}{2}$$

$$2 = -2\cdot 1 + C\cdot 1$$

$$C = 4$$

$$y = -2\sin^2 x + 4\sin^3 x,$$

Which is required particular solution.

35. Maximize $Z = 3x + 2y$ subject to $x + 2y \le 0$, $3x + y \le 15$, $x, y \ge 0$.

OR

Minimise $Z = x + 2y$ subject to $2x + y \ge 3$, $x + 2y \ge 6$, $x, y \ge 0$. Show that the minimum of Z occurs at more than two points.

Sol :

Maximize $Z = 3x + 2y$

Subject to $x + 2y \le 10$ (i)

$3x + y \le 15$ (ii)

and $x \ge 0,\ y \ge 0$

(i) Region corresponding to $x + 2y \le 10$:

Line $x + 2y = 10$			
x	0	10	Point (0,0) is true for $x + 2y \le 10$. So, the region is towards the origin.
y	5	0	

(ii) Region corresponding to $3x + y \le 15$:

Line $3x + y = 15$			
x	0	5	Point (0,0) is true for $3x + y \le 15$. So, the region is towards the origin.
y	15	0	

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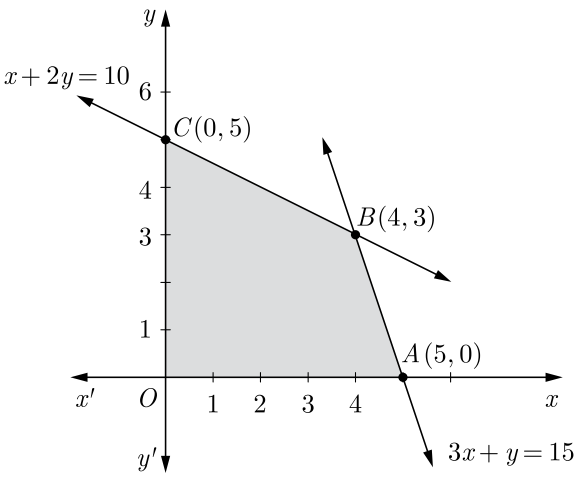
Since $x \geq 0$, $y \geq 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is $OABCO$ which is bounded. Solving equations $x+2y=10$ and $3x+y=15$, we get $B(4,3)$ The corner points of the feasible region are $O(0,0)$, $A(5,0)$, $B(4,3)$ and $C(0,5)$.

Since, the feasible region is a bounded region, we can check the objective function at all the corner to find the maxima. The values of objective function Z at these points are as follows.

Corner points	$Z=3x+2y$
$O(0,0)$	$Z=0+0=0$
$A(5,0)$	$Z=3\times5+0=15$
$B(4,3)$	$Z=3\times4+2\times3=18$
$C(0,5)$	$Z=0+2\times5=10$

Thus, Z is maximum at $B(4,3)$ and the maximum value is 18.



OR

Minimise $Z = x+2y$

Subject to $2x+y \geq 3$ (i)

$x+2y \geq 6$ (ii)

and $x \geq 0,\ y \geq 0$

(i) Region corresponding to $2x+y \geqslant 3$:

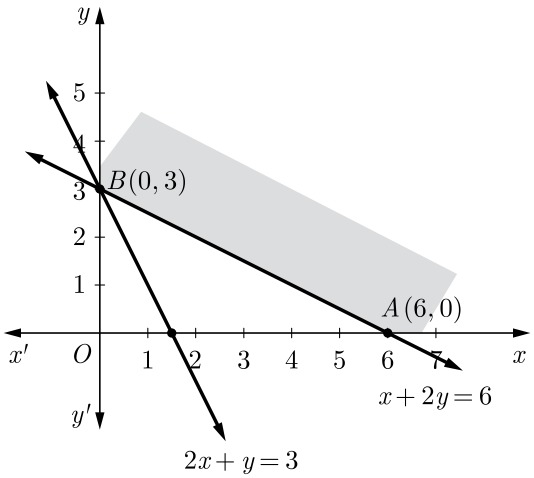
Line $2x+y=3$			
x	0	$\frac{3}{2}$	Point $(0,0)$ is false for $2x+y \geqslant 3$. So, the region is away from the origin.
y	3	0	

(ii) Region corresponding to $x+2y \geqslant 6$:

Line $x+2y=6$			
x	0	6	Point $(0,0)$ is false for $x+2y \geqslant 6$. So, the region is away from the origin.
y	3	0	

Since $x \geq 0$, $y \geq 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is ABC which is open and unbounded region. The corner points of the feasible region are $A(6,0)$ and $B(0,3)$.



Since, the feasible region is a bounded region, we can check the objective function at all the corner to find the maxima. The values of objective function Z at these points are as follows.

Corner points	$Z=x+2y$
$A(6,0)$	$Z=6+0=6$
$B(0,3)$	$Z=0+2\times3=6$

The minimum value of Z is 6 at all points on the line segment joining the points $A(6,0)$ and $B(0,3)$.

SECTION - E

Case study based questions are compulsory.

36. Ravindra Manch was established in 1963 to commemorate the 100th birth anniversary of Ravindra Nath Tagore. Ravindra Manch is one of the myriad places in Jaipur that hold a historical significance. The auditorium was among the seventeen cultural centers that were envisioned by Pandit Jawaharlal Nehru and was thrown open to the public on Independence Day in the year 1963. Since then, the place has hosted a wide number of cultural shows and events. Some of the most

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renowned artists, dancers and actors have displayed their talent at this prestigious venue.



Last year, 300 people attended the Ravindra Manch Drama Club's winter play. The ticket price was ₹ 70. The advisor estimates that 20 fewer people would attend for each ₹ 10 increase in ticket price.

(i) What ticket price would give the most income for the Drama Club?

(ii) If the Drama Club raised its tickets to this price, how much income should it expect to bring in?

Sol :

Let x represent the number of Rs 10 fare increases. Then $70+10x$ is the price per passenger and $300-20x$ is the number of passengers. The income is the number of passengers multiplied by the price per ticket. Let $R(x)$ represent income as a function of x.

$$\text{Now} \quad R(x) = (300-20x)(70+10x)$$
$$= 20(15-x)(10)(7+x)$$
$$= 200(105+8x-x^2)$$
$$= 21000+1600x-200x^2$$

Differentiating wrt x we have

$$\frac{dR(x)}{dx} = 1600-400x$$

For $R(x)$ to be maximum, $\frac{dR(x)}{dx}=0$

$$1600-400x = 0 \Rightarrow x=4.$$

Now, $\frac{d^2R}{dx^2} = -400$ which is negative for all x.

Thus, $\frac{d^2R}{dx^2}<0$, $R(x)$ is maximum when $x=4$.

(i) Ticket price,

Thus ticket price,

$$p = 70+10x = 70+10\times 4 = 110$$

(ii) Income

Maximum Revenue,

$$R(4) = 21000+1600x-200x^2$$
$$= 21000+1600\times 4-200\times 4^2$$
$$= 24200$$

37. Cross holding, also referred to as cross shareholding, describes a situation where one publicly-traded company holds a significant number of shares of another publicly-traded company. The shares owned of the second publicly-traded company are referred to as a cross-holding of the first company.



Two companies A and B are holding shares in each other. A is holding 20% shares of B and B is holding 10% shares. of A. The separately earned profits of the two companies are ₹ 98000 and ₹ 49000 respectively.

(i) Find total profit of each company using matrix notations.

(ii) Show that the total of the profits allocated to outside shareholders is equal to the total of separately earned profit.

Sol :

Let x and y be the total profits of companies A and B, respectively, then the given information can be put in the form of simultaneous linear equations as:

$$x = 98000+0.2y$$
$$x-0.2y = 98000 \quad ...(1)$$
$$y = 49000+0.1x$$
$$0.1x+y = 9000 \quad ...(2)$$

The above system of equations can be written in matrix form as:

$$\begin{bmatrix} 1 & -0.2 \\ -0.1 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 98000 \\ 49000 \end{bmatrix} \text{ or } AX=B$$

Here $|A| = 1-0.02 = 0.98$

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Since $|A| \neq 0$ thus A^{-1} exists, so that the unique solution of $AX = B$ is $X = A^{-1}B$.

Here, $A^{-1} = \frac{1}{|A|}\text{adj}.A = \frac{1}{0.98}\begin{bmatrix} 1 & 0.2 \\ 0.1 & 1 \end{bmatrix}$

$$\begin{bmatrix} x \\ y \end{bmatrix} = X = A^{-1}B = \frac{1}{0.98}\begin{bmatrix} 1 & 0.2 \\ 0.1 & 1 \end{bmatrix}\begin{bmatrix} 98000 \\ 49000 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{0.98}\begin{bmatrix} 98000 + 9800 \\ 9800 + 49000 \end{bmatrix}$$

$$= \frac{1}{098}\begin{bmatrix} 107800 \\ 58800 \end{bmatrix} = \begin{bmatrix} 110000 \\ 60000 \end{bmatrix}$$

Thus, $x = ₹\,110000$ and $y = ₹\,60000$

In company A, company B hold 10 % shares and outside holed 90 % shares. In company B, company A hold 20 % shares and outside hold 80 % shares Total of the profits allocated to outside shareholders:

$$P = [0.9 \;\; 0.8]\begin{bmatrix} 110000 \\ 60000 \end{bmatrix}$$

$$= [99000 + 48000] = [147000]$$

Total of separately earned profits:

$$[98000 + 49000] = [147000]$$

Hence, the total of the profits allocated to outside shareholders equal to the total of separately earned profits.

38. Federal health officials have reported that the proportion of children (ages 19 to 35 months) who received a full series of inoculations against vaccine-preventable diseases, including diphtheria, tetanus, measles, and mumps, increased up until 2006, but has stalled since. The CDC reports that 14 states have achieved a vaccination coverage rate of at least 80% for the 4:3:1:3:3:1 series.26 The probability that a randomly selected toddler in Alabama has received a full set of inoculations is 0.792, for a toddler in Georgia, 0.839, and for a toddler in Utah, 0.711.27 Suppose a toddler from each state is randomly selected.



(i) Find the probability that all three toddlers have received these inoculations.

(ii) Find the probability that none of the three has received these inoculations.

Sol :

We define the following three events:

A = toddler A from Alabama has received these inoculations;

G = toddler G from Georgia has received these inoculations; and

U = toddler U from Utah has received these inoculations.

(i) Assume these three events are independent

$$P(A \cap G \cap U) = P(A) \cdot P(G) \cdot P(U)$$

$$= (0.792)(0.839)(0.711)$$

$$= 0.4725$$

(ii) The probability that all three toddlers have received these inoculations is 0.4725.

None of the three has received inoculations means toddler A has not received the inoculations and toddler G has not received the inoculations and toddler U has not received the inoculations.

$$P(A' \cap G' \cap U') = P(A') \cdot P(G') \cdot P(|U')$$

$$= [1 - P(A)] \cdot [1 - P(G)] \cdot [1 - P(U)]$$

$$= (1 - 0.792)(1 - 0.839)(1 - 0.711)$$

$$= (0.208(0.161)(0.289)$$

$$= 0.0097$$

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Sample Paper 11

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

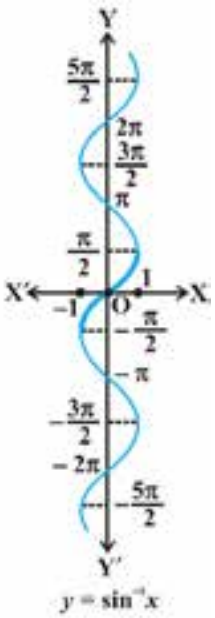
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the range of $y = \cos^{-1}x$ as represented in the graph?



(a) $[0,\pi]$ (b) $[-1,1]$

(c) $[-\pi/2,\pi/2]$ (d) $(-\infty,\infty)$

Sol :

The Y-values of the curve extend from 0 (at $x=1$) up to π (at $x=-1$). No part of the graph exists outside this range. Hence, the range of $y=\cos^{-1}x$ is $[0,\pi]$. Thus (a) is correct option.

2. The solution of the differential equation $\frac{dy}{dx}+1e^{x+y}$

(a) $(x+c)\,e^{x+y}=0$ (b) $(x+y)\,e^{x+y}=0$

(c) $(x-c)\,e^{x+y}=1$ (d) $(x-c)\,e^{x+y}=0$

Sol :

Given equation can be rewritten as

$$e^{-y}\frac{dy}{dx}+e^{-y} = e^{x}$$

Put $e^{-y} = z$

$$-e^{-y}\frac{dy}{dx} = \frac{dz}{dx}$$

$$-\frac{dz}{dx}+z = e^{x}$$

$$\frac{dz}{dx}-z = -e^{x}$$ which is linear in z.

Here, $P=-1$, thus

$$\text{IF} = e^{\int -1dx} = e^{-x}$$

Solution is, $z\cdot e^{-x} = \int -e^{x}\cdot e^{-x}dx+c$

$$= -x+c$$

$$e^{-(x+y)} = -(x-c)$$

$$(x-c)\,e^{x+y}+1 = 0$$

Thus (d) is correct option.

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3. Solution of $\frac{dy}{dx} + y\sec x = \tan x$ is

(a) $y(\sec x + \tan x) = \sec x + \tan x - x + c$

(b) $y = \sec x + \tan x - x + c$

(c) $y(\sec x + \tan x) = \sec x + \tan x + x + c$

(d) None of the above

Sol :

Here, $P = \sec x,$

$Q = \tan x$

$\text{IF} = e^{\int P dx} = e^{\log(\sec x + \tan x)}$

$= \sec x + \tan x$

Solution is,

$y(\sec x + \tan x) = \int (\sec x \tan x + \tan^2 x)\, dx$

$= \int \sec x \tan x\, dx + \int \sec^2 x\, dx - \int dx$

$= \sec x + \tan x - x + c$

Thus (a) is correct option.

4. The vector $\vec{a}$ is equal to

(a) $(\vec{a}\cdot\hat{k})\hat{i} + (\vec{a}\cdot\hat{i})\hat{j} + (\vec{a}\cdot\hat{j})\hat{k}$

(b) $(\vec{a}\cdot\hat{i})\hat{i} + (\vec{a}\cdot\hat{j})\hat{j} + (\vec{a}\cdot\hat{k})\hat{k}$

(c) $(\vec{a}\cdot\hat{j})\hat{i} + (\vec{a}\cdot\hat{k})\hat{j} + (\vec{a}\cdot\hat{i})\hat{k}$

(d) $(\vec{a}\cdot\vec{a})\left(\hat{i}+\hat{j}+\hat{k}\right)$

Sol :

Let, $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$

$\vec{a}\cdot\hat{i} = x\hat{i}\cdot\hat{i} = x$

Similarly, $y = \vec{a}\cdot\hat{j}$ and $z = \vec{a}\cdot\hat{k}$

$\vec{a} = (\vec{a}\cdot\hat{i})\hat{i} + (\vec{a}\cdot\hat{j})\hat{j} + (\vec{a}\cdot\hat{k})\hat{k}$

Thus (b) is correct option.

5. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be vectors of magnitude 3, 4 and 5 respectively. If $\vec{a}$ is perpendicular to $(\vec{b}+\vec{c}), \vec{b}$ is perpendicular to $(\vec{c}+\vec{a})$ and $\vec{c}$ is perpendicular to $(\vec{a}+\vec{b})$, then the magnitude of the vector $\vec{a}+\vec{b}+\vec{c}$ is

(a) 5 (b) $5\sqrt{2}$

(c) $5\sqrt{3}$ (d) 2

Sol :

Now, $\left[\vec{a}+\vec{b}+\vec{c}\right]^2$

$= \vec{a}^2 + \vec{b}^2 + \vec{c}^2 + 2\vec{a}\cdot\vec{b} + 2\vec{b}\cdot\vec{c} + 2\vec{c}\cdot\vec{a}$

$= a^2 + b^2 + c^2 + \vec{a}\cdot(\vec{b}+\vec{c}) + \vec{b}\cdot(\vec{c}+\vec{a}) + \vec{c}\cdot(\vec{a}+\vec{b})$

$= 9 + 16 + 25 + 0 + 0 + 0 = 50$

$\left|\vec{a}+\vec{b}+\vec{c}\right| = 5\sqrt{2}$

Thus (b) is correct option.

6. The least, value of the function $f(x) = ax + b/x$, $a > 0$, $b > 0$, $x > 0$ is

(a) $\sqrt{ab}$ (b) $2\sqrt{\frac{a}{b}}$

(c) $2\sqrt{\frac{b}{a}}$ (d) $2\sqrt{ab}$

Sol :

We have $f(x) = ax + \frac{b}{x},\ a,\ b,\ x > 0$

$f'(x) = a - \frac{b}{x^2}$

For maxima or minima, putting $f'(x) = 0$ we have

$x^2 = \frac{b}{a},\ x = \pm\sqrt{\frac{b}{a}}$

At $x = \sqrt{\frac{b}{a}},\ \frac{d^2y}{dx^2} > 0$

So, $f(x)$ attains its minima at $x = \sqrt{b/a}$ and the minimum value is $2\sqrt{ab}$.

Thus (d) is correct option.

7. A bag A contains 4 green and 3 red balls and bog B contains 4 red and 3 green balls. One bag is taken at random and a ball is drawn and noted to be green. The probability that it comes from bag B is

(a) $\frac{2}{7}$ (b) $\frac{2}{3}$

(c) $\frac{3}{7}$ (d) $\frac{1}{3}$

Sol :

$P\left(\frac{G}{A}\right) = \frac{{}^4C_1}{{}^7C_1} = \frac{4}{7}$

and $P\left(\frac{G}{B}\right) = \frac{{}^3C_1}{{}^7C_1} = \frac{3}{7}$

Now, $P\left(\frac{B}{G}\right) = \frac{P(B)\cdot P\left(\frac{G}{B}\right)}{P(A)P\left(\frac{G}{A}\right) + P(B)P\left(\frac{G}{B}\right)}$

$= \frac{\frac{1}{2}\cdot\frac{3}{7}}{\frac{1}{2}\cdot\frac{4}{7} + \frac{1}{2}\cdot\frac{3}{7}} = \frac{3}{7}$

Thus (c) is correct option.

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19. Consider the function $f(x)=\begin{cases} x^2, & x \geq 1 \\ x+1, & x<1 \end{cases}$

Assertion: f is not derivable at $x=1$ as $\lim\limits_{x\to 1^-} f(x) \neq \lim\limits_{x\to 1^+} f(x)$

Reason: If a function f is derivable at a point a then it is continuous at a

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

Reason is a standard result.

Also $\lim\limits_{x\to 1^-} f(x) = \lim\limits_{x\to 1^-}(x+1) = 2$

and $\lim\limits_{x\to 1^+} f(x) = \lim\limits_{x\to 1^+} x^2 = 1$

$\lim\limits_{x\to 1^-} f(x) \neq \lim\limits_{x\to 1^+} f(x)$

f is not continuous at $x=1$

f is not derivable at $x=1$

Thus both Assertion and reason are true and Reason is the correct explanation of assertion.

Thus (a) is correct option.

20. **Assertion:** Scalar matrix $A=[a_{ij}]=\begin{cases} k: & i=j \\ 0: & i\neq j \end{cases}$ where k is a scalar, in an identity matrix when $k=1$

Reason: Every identity matrix is not a scalar matrix.

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

A scalar matrix $A=[a_{ij}]$ $\begin{cases} k, & i=j \\ 0 & i\neq j \end{cases}$

Is an identity matrix when $k=1$. But every identity matrix is clearly a scalar matrix.

Thus (c) is correct option.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. A line passes through the point with position vector $2\hat{i}-\hat{j}+4\hat{k}$ and is the direction of the vector $\hat{i}+\hat{j}-2\hat{k}$. Find the equation of the line in cartesian form.

Sol :

Since given line passes through the point A having position vector $\vec{a}_1=2\hat{i}-\hat{j}+4\hat{k}$ and is parallel to the vector $\vec{b}=(\hat{i}+\hat{j}-2\hat{k})$, we may write equation of the given line as

$$\vec{r} = \vec{a}_1+\lambda\vec{b}$$

$$= (2\hat{i}-\hat{j}+4\hat{k})+\lambda(\hat{i}+\hat{j}-2\hat{k})$$

Substituting $\vec{r}=x\hat{i}+y\hat{j}+z\hat{k}$ in above equation we have

$$(x\hat{i}+y\hat{j}+z\hat{k}) = (2\hat{i}-\hat{j}+4\hat{k})+\lambda(\hat{i}+\hat{j}-2\hat{k})$$

$$x\hat{i}+y\hat{j}+z\hat{k} = (2+\lambda)\hat{i}+(\lambda-1)\hat{j}+(4-2\lambda)\hat{k}$$

Comparing coefficient we have

$$x = 2+\lambda,\ y=\lambda-1 \text{ and } z=4-2\lambda$$

$$\frac{x-2}{1}=\frac{y+1}{1}=\frac{z-4}{-2}=\lambda$$

Hence, $\frac{x-2}{1}=\frac{y+1}{1}=\frac{z-4}{-2}$

is the required equation of the given line in cartesian form.

22. State the reason for the relation R in the set $\{1,2,3\}$ given by $R=\{(1,2),(2,1)\}$ not to be transitive.

Sol :

We know that for a relation to be transitive,

$(x,y)\in R$ and $(y,z)\in R$ $(x,z)\in R$.

Here, $(1,2)\in R$ and $(2,1)\in R$ but $(1,1)\notin R$.

R is not transitive.

23. Write the vector in the direction of the vector $\hat{i}-2\hat{j}+2\hat{k}$ that has magnitude 9 units.

OR

Find the magnitude of each of the two vectors $\vec{a}$ and $\vec{b}$, having the same magnitude such that the angle between them is 60° and their scalar product is $\frac{9}{2}$.

Sol :

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$$6 = m\left[\frac{x^2}{2}\right]_1^2$$

$$6 = m\left(\frac{4}{2} - \frac{1}{2}\right)$$

$$6 = mx\frac{3}{2}$$

$$m = 4$$

Thus (d) is correct option.

15. The value of $\sec^2(\tan^{-1}2) + \text{cosec}^2(\cot^{-1}3)$ is

(a) 15 (b) 5

(c) 13 (d) 14

Sol :

$$\sec^2(\tan^{-1}2) + \text{cosec}^2(\cot^{-1}3)$$

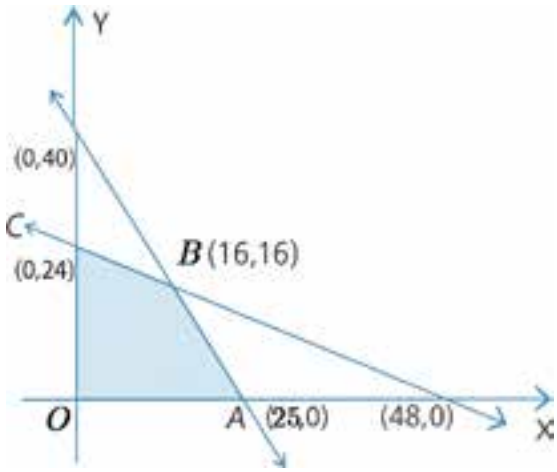
$$= \left\{1 + \tan^2(\tan^{-1}2)\right\} + \left\{1 + \cot^2(\cot^{-1}3)\right\}$$

$$= 1 + \left\{\tan(\tan^{-1}2)\right\}^2 + 1 + \left\{\cot(\cot^{-1}3)\right\}^2$$

$$= 1 + 2^2 + 1 + 3^2 = 15$$

Thus (a) is correct option.

16. Consider the objective functions $z_1 = 4x + 3y$ and $z_2 = 5x + 2y$. At which vertex of the feasible region do these two functions give the same value?



(a) A(25,0) (b) B(16,16)

(c) C(0,24) (d) None of these

Sol :

We evaluate both functions at each vertex:

At A(25,0) : $z_1 = 100, z_2 = 125$ → not equal.

At B(16,16) : $z_1 = 4(16) + 3(16) = 64 + 48 = 112$.

Also, $z_2 = 5(16) + 2(16) = 80 + 32 = 112$. → equal.

At C(0,24) : $z_1 = 72, z_2 = 48$ → not equal.

Thus, the two objective functions have the same value only at vertex B(16,16).

Thus (b) is correct option.

17. In the figure, what does the element labeled "dx" represent?

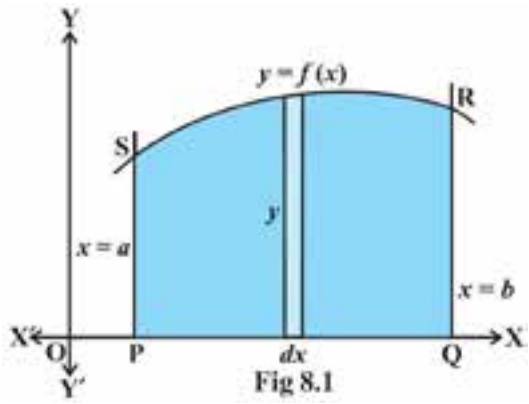


Fig 8.1

(a) The infinitesimal height of the strip parallel to the Y-axis

(b) The infinitesimal width of the strip along the X-axis

(c) The radius of the circle

(d) The area of the shaded region

Sol :

Inside the shaded quarter circle, a thin vertical strip is drawn parallel to the Y-axis. Its small horizontal width is marked as dx. This width corresponds to a very small change in the x-coordinate, not the height, radius.

Thus (b) is correct option.

18. If matrix $\begin{bmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ \lambda & -3 & 0 \end{bmatrix}$ is singular, then λ is equal to

(a) -2 (b) -1

(c) 1 (d) 2

Sol :

Given matrix is singular.

Therefore, $\begin{vmatrix} 0 & 1 & -2 \\ -1 & 0 & 3 \\ \lambda & -3 & 0 \end{vmatrix} = 0$

Expanding from R_2 we have

$$1(0-6) + \lambda(3) = 0$$

$$-6 + 3\lambda = 0$$

$$\lambda = 2$$

Thus (d) is correct option.

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8. The equation of the line through the point $(2,3,-5)$ and equally inclined to the axes are

(a) $x-2=y-3=z+5$

(b) $\frac{x-1}{2}=\frac{y-1}{3}=\frac{z-1}{-5}$

(c) $\frac{x}{2}=\frac{y}{3}=\frac{z}{-5}$

(d) none of these

Sol :

Since, DR's of a line which is equally inclined to the coordinate axes are $<1,1,1>$.

Hence, required equation of line is

$$\frac{x-2}{1}=\frac{y-3}{1}=\frac{z+5}{1}$$

Thus (a) is correct option.

9. Let $f(x)=x|x|$, then $f(x)$ has a

(a) local maxima at $x=0$

(b) local minima at $x=0$

(c) point of inflexion at $x=0$

(d) none of the above

Sol :

$$f(x) = x|x|$$

$$f'(x) = x\cdot\frac{x}{|x|}+|x| = 2|x|$$

$$\Rightarrow \quad f'(x) = 0,$$

only, if $|x|=0$, ie, if $x=0$

Since, $f'(x)$ does not change sign as we move from left to right through origin.

Hence, f has a point of inflexion at $x=0$.

Thus (c) is correct option.

10. The value of $\int_{-2}^{2}(x\cos x+\sin x+1)\,dx$ is

(a) 2 (b) 0

(c) -2 (d) 4

Sol :

$$\int_{-2}^{2}(x\cos x+\sin x+1)\,dx$$

$$=\int_{-2}^{2}x\cos x\,dx+\int_{-2}^{2}\sin x\,dx+\int_{-2}^{2}1\,dx$$

$$=[x\sin x-\int\sin x\,dx]_{-2}^{2}+\int_{-2}^{2}\sin x\,dx+[x]_{-2}^{2}$$

$$=2\sin 2-2\sin 2-\int_{-2}^{2}\sin x\,dx+\int_{-2}^{2}\sin x\,dx+(2+2)$$

$$=4$$

Thus (d) is correct option.

11. If x is measured in degrees, then $\frac{d}{dx}(\cos x)$ is equal to

(a) $-\sin x$ (b) $-\frac{180}{\pi}\sin x$

(c) $-\frac{\pi}{180}\sin x$ (d) $\sin x$

Sol :

$$\frac{d}{dx}(\cos x)=-\frac{\pi}{180}\sin x$$

Thus (c) is correct option.

12. The image of the interval $[1,3]$ under the mapping $f:R\to R$ given by $f(x)=2x^3-24x+107$ is

(a) $[75,89]$ (b) $[74,89]$

(c) $[0,75]$ (d) none of these

Sol :

Since, the given function has minimum value 75 which is attained at $x=2$ and maximum value 89 which is attained at $x=3$. Hence, the range of f is $[75,89]$.

Thus (a) is correct option.

13. $\int_{0}^{\pi/2}\left|\cos\left(\frac{x}{2}\right)\right|dx$ is equal to

(a) 1 (b) -2

(c) $\sqrt{2}$ (d) 0

Sol :

$$\int_{0}^{\pi/2}\left|\cos\left(\frac{x}{2}\right)\right|dx = \int_{0}^{\pi/2}\cos\left(\frac{x}{2}\right)dx$$

$$= 2\left[\sin\left(\frac{x}{2}\right)\right]_{0}^{\pi/2}]$$

$$= 2\left[\sin\frac{\pi}{4}-\sin 0\right]$$

$$= 2\sin\frac{\pi}{4}$$

$$= 2\times\frac{1}{\sqrt{2}}=\sqrt{2}$$

Thus (c) is correct option.

14. The area of the region bounded by the lines $y=mx$, $x=1$, $x=2$ and X-axis is 6 sq units, then m is equal to

(a) 3 (b) 1

(c) 2 (d) 4

Sol :

Given, equation of line is $y=mx$ and bounded by $x=1$, $x=2$ and X-axis.

Required area, $A = \int_{1}^{2}mx\,dx$

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We have $\vec{a} = \hat{i} - 2\hat{j} + 2\hat{k}$

Now $|\vec{a}| = \sqrt{1^2 + (-2)^2 + 2^2}$

$= \sqrt{1+4+4}$

$= \sqrt{9} = 3$ units

The unit vector in the direction of the given vector $\vec{a}$ is

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{1}{3}(\hat{i} - 2\hat{j} + 3\hat{k})$$

The vector of magnitude equal to 21 units and in the direction of $\vec{a}$ is given by

$$21\hat{a} = 9 \times \frac{1}{3}(\hat{i} - 2\hat{j} + 3\hat{k})$$

$$= 3\hat{i} - 6\hat{j} + 6\hat{k}$$

OR

Given, two vectors $\vec{a}$ and $\vec{b}$ such that $|\vec{a}| = |\vec{b}|$, $\vec{a} \cdot \vec{b} = \frac{9}{2}$ and angle between them is 60°.

If θ is angle between $\vec{a}$ and $\vec{b}$ then we have

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta,$$

Substituting values we obtain

$$\frac{9}{2} = |\vec{a}| \cdot |\vec{a}| \cos 60^\circ$$

$$\frac{1}{2} \cdot |\vec{a}|^2 = \frac{9}{2} \qquad \cos 60^\circ = \frac{1}{2}$$

$$|\vec{a}|^2 = 9$$

$$|\vec{a}| = 3 \quad \text{magnitude cannot be negative]}$$

Thus, $|\vec{a}| = |\vec{b}| = 3$

24. Evaluate $\int (1-x)\sqrt{x}\, dx$.

OR

Given, $\int e^x(\tan x + 1)\sec x\, dx = e^x f(x) + C$. Write $f(x)$ satisfying above.

Sol :

We have $I = \int (1-x)\sqrt{x}\, dx = \int (\sqrt{x} - x\sqrt{x})\, dx$

$= \int (x^{1/2} - x^{3/2})\, dx = \frac{2}{3}x^{3/2} - \frac{2}{5}x^{5/2} + C$

OR

$$\int e^x(\tan x + 1)\sec x dx = e^x f(x) + C$$

$$\int e^x(\tan x \sec x + \sec x)\, dx = e^x f(x) + C$$

$$\int e^x(\sec x + \sec x \tan x)\, dx = e^x f(x) + C$$

It may be easily seen that $f(x) = \sec x$ and $f'(x) = \sec x \tan x$. Thus we may use

$$\int e^x[f(x) + f'(x)]\, dx = e^x f(x) + C$$

and given integral reduce to

$$e^x \cdot \sec x + C = e^x f(x) + C$$

Comparing both sides, we get

$$f(x) = \sec x$$

25. If $P(not\ A) = 0.7$, $P(B) = 0.7$ and $P(B/A) = 0.5$, then find $P(A/B)$.

Sol :

We have $P(A') = 0.7$, $P(B) = 0.7$

and $P\left(\frac{B}{A}\right) = 0.5$

Now $P(A) = 1 - P(A') = 1 - 0.7 = 03$

Now, $P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$

$$0.5 = \frac{P(A \cap B)}{0.3}$$

$$P(A \cap B) = 0.15$$

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{0.15}{0.7}$$

Thus $P\left(\frac{A}{B}\right) = \frac{3}{14}$

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. Find $\int \frac{dx}{5 - 8x - x^2}$.

Sol :

We have $I = \int \frac{dx}{5 - 8x - x^2}$

$= \int \frac{dx}{5 - 2 \cdot 4 \cdot x - x^2 - (4)^2 + (4)^2}$

$= \int \frac{dx}{5 + 16 - [x^2 + (4)^2 + 2 \cdot 4 \cdot x]}$

$= \int \frac{dx}{21 - (x+4)^2}$

$= \int \frac{dx}{(\sqrt{21})^2 - (x+4)^2}$

Using the fact $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + C$ we have

$$I = \frac{1}{2\sqrt{21}}\log\left|\frac{\sqrt{21} + x + 4}{\sqrt{21} - x - 4}\right| + C$$

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Probability that item is produced by operator A if it is defective, i.e.

$$P\left(\frac{A}{D}\right) = \frac{P(A)\cdot P\left(\frac{D}{A}\right)}{P(A)\cdot P\left(\frac{D}{A}\right)+P(B)\cdot\left(\frac{D}{B}\right)+P(C)\cdot\left(\frac{D}{C}\right)}$$
$$= \frac{0.5\times 0.01}{0.5\times 0.01+0.3\times 0.05+0.2\times 0.07}$$
$$= \frac{0.005}{0.005+0.015+0.014}$$
$$= \frac{0.005}{0.034} = \frac{5}{34}$$

Therefore required probability is $\frac{5}{34}$.

OR

Let us define the events as

E_1 : Event that insured person is a scooter driver

E_2 : Event that insured person is a car driver

E_3 : Event that insured person is a truck driver

A : Event that insured person meets with an accident

Now $n(E_1) = 2000,\ n(E_2) = 4000$

and $n(E_3) = 6000$

Here, total insured person, $n(S) = 12000$

Probability that the insured person is a scooter driver

$$P(E_1) = \frac{n(E_1)}{n(S)} = \frac{2000}{12000} = \frac{1}{6}$$

Probability that the insured person is a car driver,

$$P(E_2) = \frac{n(E_2)}{n(S)} = \frac{4000}{12000} = \frac{1}{3}$$

Probability that the insured person is a truck driver,

$$P(E_3) = \frac{n(E_3)}{n(S)} = \frac{6000}{12000} = \frac{1}{2}$$

Probability that scooter driver meets with an accident

$$P\left(\frac{A}{E_1}\right) = 0.01$$

Probability that car driver meets with an accident

$$P\left(\frac{A}{E_2}\right) = 0.03$$

Probability that truck driver meets with an accident

$$P\left(\frac{A}{E_3}\right) = 0.15$$

The probability that the person met with an accident was a scooter driver.

$$P\left(\frac{E_1}{A}\right) = \frac{P(E_1)\cdot P\left(\frac{A}{E_1}\right)}{P(E_1)\cdot P\left(\frac{A}{E_1}\right)+P(E_2)\cdot P\left(\frac{A}{E_2}\right)+P(E_3)\cdot\left(\frac{A}{E_3}\right)}$$
$$= \frac{0.01\times\frac{1}{6}}{0.01\times\frac{1}{6}+0.03\times\frac{1}{3}+0.15\times\frac{1}{2}}$$
$$= \frac{\frac{1}{6}}{\frac{1}{6}+1+\frac{15}{2}} = \frac{1}{1+6+45} = \frac{1}{52}$$

The probability that the person met with an accident was a car driver, $P(E_2/A)$

$$P\left(\frac{E_2}{A}\right) = \frac{0.03\times\frac{1}{3}}{0.01\times\frac{1}{6}+0.03\times\frac{1}{3}+0.15\times\frac{1}{2}}$$
$$= \frac{1}{\frac{1}{6}+1+\frac{15}{2}} = \frac{6}{1+6+45} = \frac{6}{52}$$

Hence, the required probability is $P(E_1 \cup E_2/A)$

$$P\left(\frac{E_1\cup E_2}{A}\right) = P\left(\frac{E_1}{A}\right)+P\left(\frac{E_2}{A}\right)$$
$$= \frac{1}{52}+\frac{6}{52} = \frac{7}{52}$$

33. Find the values of p, so that the lines

$$l_1 : \frac{1-x}{3} = \frac{7y-14}{p} = \frac{z-3}{2} \text{ and}$$

$$l_2 : \frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5}$$

are perpendicular to each other. Also, find the equation of a line passing through a point $(3, 2, -4)$ and parallel to line l_1.

Sol :

Writing the given line in standard form as

$$\frac{x-1}{-3} = \frac{y-2}{\frac{p}{7}} = \frac{z-3}{2} = r_1 \text{ (let)} \quad ...(1)$$

and $$\frac{x-1}{\frac{-3p}{7}} = \frac{y-5}{1} = \frac{z-6}{-5} = r_2 \text{ (let)} \quad ...(2)$$

Two lines with DR's a_1, b, c_1 and a_2, b_2, c_2 are perpendicular if

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

Thus line (1) and (2) will intersect at right angle, if

$$-3\left(\frac{-3p}{7}\right)+\frac{p}{7}(1)+2(-5) = 0$$
$$\frac{9p}{7}+\frac{p}{7} = 10$$
$$\frac{10p}{7} = 10 \Rightarrow p = 7$$

which is the required value of p.

Also, we know that, the equation of a line which passes through the point (x_1, y_1, z_1) with direction ratios a, b, c is given by

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

Since, required line is parallel to line l_1.

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$$\frac{dx}{dy} = \frac{\cos(a+y)\frac{d}{dy}(\sin y) - \sin y \frac{d}{dy}\cos(a+y)}{\cos^2(a+y)}$$

$$= \frac{\cos(a+y)\cos y + \sin y \sin(a+y)}{\cos^2(a+y)}$$

$$= \frac{\cos(a+y-y)}{\cos^2(a+y)}$$

$$= \frac{\cos a}{\cos^2(a+y)}$$

$$\frac{dy}{dx} = \frac{\cos^2(a+y)}{\cos a}$$

Substituting $x = 0$ in Eq. (1), we get $y = 0$

Now, $\frac{dy}{dx} = \frac{\cos^2(a+0)}{\cos a} = \frac{\cos^2 a}{\cos a}$

or $\frac{dy}{dx} = \cos a$ Hence proved.

OR

We have $y = a\tan^3\theta$ and $x = a\sec^3\theta$

Differentiating w.r.t. θ, we get

$$\frac{dy}{d\theta} = 3a\tan^2\theta\frac{d}{d\theta}(\tan\theta)$$

$$= 3a\tan^2\theta\sec^2\theta$$

and $\frac{dx}{d\theta} = 3a\sec^2\theta\frac{d}{d\theta}(\sec\theta)$

$$= 3a\sec^2\theta\sec\theta\tan\theta$$

Now, $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$

$$= \frac{3a\tan^2\theta\sec^2\theta}{3a\sec^2\theta\sec\theta\tan\theta}$$

$$= \frac{\tan\theta}{\sec\theta}$$

$$\frac{dy}{dx} = \sin\theta$$

Again differentiating both sides w.r.t. x, we get

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}(\sin\theta)$$

$$= \frac{d}{d\theta}(\sin\theta)\frac{d\theta}{dx}$$

$$= \cos\theta\frac{1}{3a\sec^3\theta\tan\theta} = \frac{\cos^5\theta}{3a\sin\theta}$$

At $\theta = \frac{\pi}{3}$,

$$\frac{d^2y}{dx^2} = \frac{\cos^5\frac{\pi}{3}}{3a\sin\frac{\pi}{3}}$$

$$= \frac{\left(\frac{1}{2}\right)^5}{3a\left(\frac{\sqrt{3}}{2}\right)}$$

$$= \frac{1\times 2}{2^5\times 3a\sqrt{3}} = \frac{1}{48\sqrt{3}\,a}$$

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. A manufacturer has three machine operators A, B and C. The first operator A produces 1% of defective items, whereas the other two operators B and C produces 5% and 7% defective items respectively. A is on the job for 50% of the time, B on the job 30% of the time and C on the job for 20% of the time. All the items are put into one stockpile and then one items is chosen at random from this and is found to be defective. What is the probability that it was produced by A?

OR

An insurance company insured 2000 scooter driver, 4000 car drivers and 6000 truck drivers. The probabilities of an accident for them are 0.01, 0.03 and 0.15 respectively. One of the insured persons meets with an accident. What is the probability that he is a scooter driver or a car driver?

Sol :

Let the following event be:

A : Event that item produced by operator A

B : Event that item produced by operator B

C : Event that item produced by operator C

D : Event that item produced is defective

Probability of item is produced by operator A,

$$P(A) = 50\% = \frac{50}{100} = 0.5$$

Probability of item is produced by operator B,

$$P(B) = 30\% = \frac{30}{100} = 0.3$$

Probability of item is produced by operator C,

$$P(C) = 20\% = \frac{20}{100} = 0.2$$

Probability of defective item produced by operator A,

$$P\left(\frac{D}{A}\right) = 1\% = \frac{1}{100} = 0.01$$

Probability of defective item produced by operator B,

$$P\left(\frac{D}{B}\right) = 5\% = \frac{5}{100} = 0.05$$

Probability of defective item produced by operator C,

$$P\left(\frac{D}{C}\right) = 7\% = \frac{7}{100} = 0.07$$

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27. If $\vec{a}$ and $\vec{b}$ are perpendicular vectors, $|\vec{a}+\vec{b}|=13$ and $|\vec{a}|=5$, then find the value of $|\vec{b}|$.

Sol :

We have $|\vec{a}+\vec{b}|=13$

$|\vec{a}|=5$

Now,

$$(\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b}) = \vec{a}\cdot\vec{a}+\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{a}+\vec{b}\cdot\vec{b}$$

$$|\vec{a}+\vec{b}|^2 = |\vec{a}|^2+0+0+|\vec{b}|^2$$

$$(13)^2 = (5)^2+|\vec{b}|^2$$

$$169 = 25+|\vec{b}|^2$$

$$169-25 = |\vec{b}|^2$$

$$144 = |\vec{b}|^2 \Rightarrow |\vec{b}| = 12$$

as length is always positive.

28. The volume of a cube is increasing at the rate of 8 cm^3/s. How fast is the surface area increasing when the length of its edge is 12 cm?

OR

The side of an equilateral triangle is increasing at the rate of 2 cm/s. At what rate is its area increasing, when the side of the triangle is 20 cm?

Sol :

Let x be the length of an edge of the cube, V be the volume and S be the surface area at any time t. Then, $V=x^3$ and $S=6x^2$.

It is given that,

We have $\dfrac{dV}{dt} = 8\,\text{cm}^3/\text{sec}$

or $\dfrac{d}{dt}(x^3) = 8$

$$3x^2\frac{dx}{dt} = 8$$

$$\frac{dx}{dt} = \frac{8}{3x^2}$$

Now, $S = 6x^2$

$$\frac{dS}{dt} = 12\frac{dx}{dt} = 12x\times\frac{8}{3x^2}$$

$$\frac{dS}{dt} = \frac{32}{x}$$

$$\left(\frac{dS}{dt}\right)_{x=12} = \frac{32}{12} = \frac{8}{3}\text{cm}^2/\text{sec}$$

OR

Let a be the side of an equilateral triangle and A be the area of an equilateral triangle.

We have $\dfrac{da}{dt} = 2$ cm/s

Area of an equilateral triangle,

$$A = \frac{\sqrt{3}}{4}a^2$$

Differentiating both sides w.r.t. t, we get

$$\frac{dA}{dt} = \frac{\sqrt{3}}{4}\times 2a\times\frac{da}{dt}$$

$$= \frac{\sqrt{3}}{4}\times 2\times 20\times 2 \quad [\text{given, } a=20]$$

$$= 20\sqrt{3}\ \text{cm}^2/\text{s}$$

Thus, the rate of area increasing is $20\sqrt{3}$ cm^2/s.

29. If $A=\begin{bmatrix}5 & 6 & -3\\ -4 & 3 & 2\\ -4 & -7 & 3\end{bmatrix}$, then write the cofactor of the element a_{21} of its 2nd row.

Sol :

We have $A=\begin{bmatrix}5 & 6 & -2\\ -4 & 3 & 3\\ -4 & -7 & 3\end{bmatrix}$

Cofactor of a_{21} $A_{21} = (-1)\begin{vmatrix}6 & -3\\ -7 & 3\end{vmatrix}$

$$= -(18-21) = 3$$

30. Write the value of $\sin\left[\frac{\pi}{3}-\sin^{-1}\left(-\frac{1}{2}\right)\right]$.

Sol :

$$\sin\left[\frac{\pi}{3}-\sin^{-1}\left(-\frac{1}{2}\right)\right] = \sin\left[\frac{\pi}{3}+\sin^{-1}\left(\frac{1}{2}\right)\right]$$

$$= \sin\left[\frac{\pi}{3}+\sin^{-1}\left(\sin\frac{\pi}{6}\right)\right]$$

$$= \sin\left[\frac{\pi}{3}+\frac{\pi}{6}\right]$$

$$= \sin\frac{\pi}{2} = 1$$

31. If $y=x\cos(a+y)$, then show that $\dfrac{dy}{dx}=\dfrac{\cos^2(a+y)}{\cos a}$

Also, show that $\dfrac{dy}{dx}=\cos a$, when $x=0$.

OR

If $x=a\sec^3\theta$ and $y=a\tan^3\theta$, find $\dfrac{d^2y}{dx^2}$ at $\theta=\dfrac{\pi}{3}$.

Sol :

We have $\sin y = x\cos(a+y)$ ...(1)

$$x = \frac{\sin y}{\cos(a+y)}$$

Differentiating w.r.t. x, we get

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So, $a = -3,\ b = \frac{7}{7} = 1$ and $c = 2$

Now, equation of line passing through the point $(3,2,-4)$ and having direction ratios $(-3,1,2)$ is

$$\frac{x-3}{-3} = \frac{y-2}{1} = \frac{z+4}{2}$$

$$\frac{3-x}{3} = \frac{y-2}{1} = \frac{z+4}{2}$$

34. Show that the differential equation $\frac{dy}{dx} = \frac{y^2}{xy - x^2}$ is homogeneous and also solve it.

Sol :

We have $\frac{dy}{dx} = \frac{y^2}{xy - x^2}$ ...(1)

Let $F(x,y) = \frac{y^2}{xy - x^2}$

Now, replacing x by λx and y by λy, we get

$$F(\lambda x, \lambda y) = \frac{\lambda^2 y^2}{\lambda^2 (xy - x^2)}$$

$$= \lambda^0 \frac{y^2}{xy - x^2} = \lambda^0 F(x,y)$$

Which is a homogeneous differential equation as

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

Substituting $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ in eq (1) we have

From Eq. (1), we get

$$v + x\frac{dv}{dx} = \frac{v^2 x^2}{vx^2 - x^2} = \frac{v^2}{v-1}$$

$$x\frac{dv}{dx} = \frac{v^2}{v-1} - v = \frac{v^2 - v^2 + v}{v-1}$$

$$x\frac{dv}{dx} = \frac{v}{v-1}$$

$$\frac{v-1}{v}dv = \frac{dx}{x}$$

Integrating both sides, we get

$$\int \left(1 - \frac{1}{v}\right)dv = \int \frac{dx}{x}$$

$$v - \log|v| = \log|x| + C$$

$$\frac{y}{x} - \log\left|\frac{y}{x}\right| = \log|x| + C$$

$$\frac{y}{x} - \log|y| + \log|x| = \log|x| + C$$

$$\frac{y}{x} - \log|y| = C$$

Which is the required solution.

35. Find $\int \frac{2\cos x}{(1-\sin x)(2-\cos^2 x)}dx$.

OR

Find $\int \frac{2\cos x}{(1-\sin x)(1+\sin^2 x)}dx$.

Sol :

We have $I = \int \frac{2\cos x}{(1-\sin x)(2-\cos^2 x)}dx$

$$= \int \frac{2\cos x}{(1-\sin x)(1+\sin^2 x)}dx$$

Substituting $\sin x = t \Rightarrow \cos x\,dx = dt$ we have

$$I = \int \frac{2}{(1-t)(1+t^2)}dt$$

Now, let

$$\frac{2}{(1-t)(1+t^2)} = \frac{A}{1-t} + \frac{Bt+C}{1+t^2}$$

$$2 = A(1+t^2) + (Bt+C)(1-t) \quad ...(1)$$

Substituting $t = 1$ in eq. (1), we get

$$2 = A(1+1) + 0 \Rightarrow A = 1 \quad ...(2)$$

Substituting $t = 0$ in eq. (2), we get

$$2 = A + C \Rightarrow 2 = 1 + C \Rightarrow C = 1$$

Substituting $t = -1$ in eq. (1), we get

$$2 = 2A + (-B + C)(2)$$

$$2 = 2 - 2B + 2$$

$$2B = 2 \Rightarrow B = 1$$

Thus $\frac{2}{(1-t)(1+t^2)} = \frac{1}{1-t} + \frac{t+1}{1+t^2}$

Now

$$I = 1\int \frac{1}{1-t}dt + \int \frac{t+1}{1+t^2}dt$$

$$= \int \frac{1}{1-t}dt + \frac{1}{2}\int \frac{2t}{1+t^2}dt + \int \frac{1}{1+t^2}dt$$

$$= -\log|1-t| + \frac{1}{2}\log|1+t^2| + \tan^{-1}t + C$$

$$= -\log|1-\sin x| + \frac{1}{2}\log|1+\sin^2 x| + \tan^{-1}(\sin x) + C$$

$$= \tan^{-1}(\sin x) + \log\left|\frac{\sqrt{1+\sin^2 x}}{1-\sin x}\right| + C$$

Here we have used the fact $\log m - \log n = \log\left(\frac{m}{n}\right)$ and $n\log m = \log m^n$

OR

We have $I = \int \frac{2\cos x}{(1-\sin x)(1+\sin^2 x)}dx$

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We have $x = 75 - 3p$

Rearranging the above equation we get

$$p = 25 - \frac{1}{3}x$$

Revenue function,

$$R(x) = px$$

$$= \left(25 - \frac{1}{3}x\right)x$$

$$= 25x - \frac{1}{3}x^2$$

Cost function, $C(x) = \frac{x^2}{25} + 3x + 100$

Profit function, $P(x) = R(x) - C(x)$

$$= 25x - \frac{1}{3}x^2 - \frac{x^2}{25} - 3x - 100$$

Differentiating wrt x we have

$$\frac{dP(x)}{dx} = 25 - \frac{2}{3}x - \frac{2}{25}x - 3$$

$$= 22 - \frac{56}{75}x$$

For maximum profit, $\frac{dP(x)}{dx} = 0$, thus

$$22 - \frac{56}{75}x = 0 \Rightarrow x = 29.46 \simeq 30 \text{ sets}$$

Now, $\frac{d^2P}{dx^2} = -\frac{56}{75} < 0$

which is negative for all value of x

Thus $x = 30$ gives the maximum profit..

At $x = 30$,

$$P(30) = 25 - \frac{1}{3}x$$

$$= 25 - 10 = ₹15 \text{ per toy.}$$

38. A manufacturing company has two service departments, S_1, S_2 and four production departments P_1, P_2, P_3 and P_4.

Overhead is allocated to the production departments for inclusion in the stock valuation. The analysis of benefits received by each department during the last quarter and the overhead expense incurred by each department were:

Service Department	Percentages to be allocated to departments					
	S_1	S_2	P_1	P_2	P_3	P_4
S_1	0	20	30	25	15	10
S_2	30	0	10	35	20	5
Direct overhead expense ₹'000	20	40	25	30	20	10



You are required to find out following using matrix method.

(i) Express the total overhead of the service departments in the form of simultaneous equations.

(ii) Express these equations in a matrix form and solve for total overhead of service departments using matrix inverse method.

(iii) Determine the total overhead to be allocated from each of S_1 and S_2 to the production department.

Sol :

Let S_1 be total overhead of service department and S_2 be total overhead of service department.

Then, $S_1 = 20000 + 0.3S_2$

and $S_2 = 40000 + 0.2S_1$

The total overhead of the service departments can be expressed in the form of simultaneous equations as:

$$S_1 - 0.3S_2 = 20000$$

$$-0.2S_1 + S_2 = 40000$$

The above simultaneous equations can be expressed in the matrix form as:

$$\begin{bmatrix} 1 & -0.3 \\ -0.2 & 1 \end{bmatrix}\begin{bmatrix} S_1 \\ S_2 \end{bmatrix} = \begin{bmatrix} 20000 \\ 40000 \end{bmatrix},$$

or $AS = B$

Here $|A| = 1 - 0.06 = 0.94$

Since $|A| \neq 0$ thus A^{-1} exists, so that the unique solution of $AX = B$ is $X = A^{-1}B$.

Here, $A^{-1} = \frac{\text{adj.}A}{|A|}$

$$= \frac{1}{0.94}\begin{bmatrix} 1 & 0.3 \\ 0.2 & 1 \end{bmatrix}$$

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$$25x+20y \leq 900$$

or $$5x+4y \leq 180 \quad ...(ii)$$

and $$x \geq 0$$

$$y \geq 0$$

(i) Region corresponding to $3x+2y \leq 96$:

Line $3x+2y \leq 96$			
x	0	32	Point (0, 0) is true for $3x+2y \leq 96$. So, the region is towards the origin.
y	48	0	

(ii) Region corresponding to $5x+4y \leq 180$:

Line $5x+4y \leq 180$			
x	0	36	Point (0, 0) is true for $5x+4y \leq 180$. So, the region is towards the origin.
y	45	0	

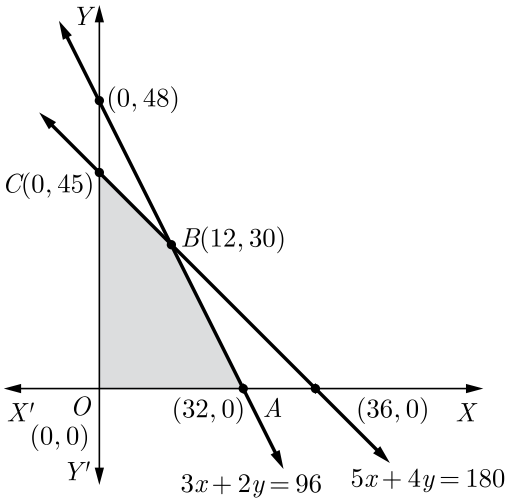
Since $x \geq 0$, $y \geq 0$ the feasible region lies in the first quadrant.

Now, we draw all line on the graph and find the common area. Clearly feasible region is $OABC$. Solving equation $3x+2y=96$ and $5x+4y=180$ we get $B(12,30)$. The coordinates of the corner points of the region are $O(0,0)$, $A(32,0)$, $B(12,30)$, $C(0,45)$.

Since the feasible region is a bounded region, we can check the profit function at all the vertices to find the maxima. The values of P at corner points are given below.

Corner points	$P=39x+33y$
$O(0,0)$	$P=39\times 0+33\times 0=0$
$A(32,0)$	$P=39\times 32+33\times 0=1248$
$B(12,30)$	$P=39\times 12+33\times 30=1458$
$C(0,45)$	$P=39\times 0+33\times 45=1485$

The maximum value of P is 1485 at $C(0,45)$. Hence the manufacturer should produce 0 units of floor lamps and 45 units of table lamp to get a maximum profit of \$1485.



37. The Indian toy industry is estimated to be worth US\$1.5 billion, making up 0.5% of the global market share. The toy manufacturers in India can mostly be found in NCR, Mumbai, Karnataka, Tamil Nadu, and several smaller towns and cities across central states such as Chhattisgarh and Madhya Pradesh. The sector is fragmented with 90% of the market being unorganised. The toys industry has been predicted to grow to US\$2-3 billion by 2024. The Indian toy industry only represents 0.5% of the global industry size indicating a large potential growth opportunity for Indian consumer product companies who will develop exciting innovations to deliver international quality standards at competitive prices.



Fisher Price is a leading toy manufacturer in India. Fisher Price produces x set per week at a total cost of $\frac{1}{25}x^2+3x+100$. The produced quantity for his market is $x=75-3p$ where p is the price set.

(i) Show that the maximum profit is obtained when about 30 toys are produced per week.

(ii) What is the price at maximum profit?

Sol :

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Substituting $\sin x = t \Rightarrow \cos x\,dx = dt$ we have

$$I = \int \frac{2}{(1-t)(1+t^2)}\,dt$$

Now, let $$\frac{2}{(1-t)(1+t^2)} = \frac{A}{1-t} + \frac{Bt+C}{1+t^2}$$

$$2 = A(1+t^2) + (Bt+C)(1-t) \quad ...(1)$$

Substituting $t = 1$ in eq. (1), we get

$$2 = A(1+1) + 0 \Rightarrow A = 1 \quad ...(2)$$

Substituting $t = 0$ in eq. (2), we get

$$2 = A + C \Rightarrow 2 = 1 + C \Rightarrow C = 1$$

Substituting $t = -1$ in eq. (1), we get

$$2 = 2A + (-B + C)(2)$$

$$2 = 2 - 2B + 2$$

$$2B = 2 \Rightarrow B = 1$$

Thus $$\frac{2}{(1-t)(1+t^2)} = \frac{1}{1-t} + \frac{t+1}{1+t^2}$$

Now $$I = 1\int \frac{1}{1-t}\,dt + \int \frac{t+1}{1+t^2}\,dt$$

$$= \int \frac{1}{1-t}\,dt + \frac{1}{2}\int \frac{2t}{1+t^2}\,dt + \int \frac{1}{1+t^2}\,dt$$

$$= -\log|1-t| + \frac{1}{2}\log|1+t^2| + \tan^{-1}t + C$$

$$= -\log|1-\sin x| + \frac{1}{2}\log|1+\sin^2 x| + \tan^{-1}(\sin x) + C$$

$$= \tan^{-1}(\sin x) + \log\left|\frac{\sqrt{1+\sin^2 x}}{1-\sin x}\right| + C$$

Here we have used the fact $\log m - \log n = \log\left(\frac{m}{n}\right)$ and $n\log m = \log m^n$

SECTION - E

Case study based questions are compulsory.

36. A craftswoman produces two products: floor lamps and table lamps. Production of one floor lamp requires 75 minutes of her labor and materials that cost \$25. Production of one table lamp requires 50 minutes of labor, and the materials cost \$20. The craftswoman wishes to work no more than 40 hours each week, and her financial resources allow her to pay no more than \$900 for materials each week.



(i) If she can sell as many lamps as she can make and if her profit is \$39 per floor lamp and \$33 per table lamp, how many floor lamps and how many table lamps should she make each week to maximize her weekly profit?

(ii) What is that maximum profit?

Sol :

Let the craftswoman produces x units of floor lamps and y units of table lamps. Clearly $x \geq 0$ and $y \geq 0$. Each unit of floor lamp is sold at profit of \$39 and table lamp at a profit of \$33. Thus

Profit function $P = 39x + 33y$

Each unit of floor lamp requires 75 minutes of labour time and material of cost \$25 and each unit of table lamp requires 50 minutes of labour time and material of cost \$20 we tabulate the given data as given below

	Floor lamp	Table lamp	Availability
Labour time	75 minutes	50 minutes	$75x + 50y \leq 2400$
Material cost	25	20	$25x + 20y \leq 900$

Now, the given LPP reduced to

Maximize $P = 39x + 33y$

Subject to the constraints,

$$75x + 50y \leq 2400$$

or $$3x + 2y \leq 96 \quad ...(i)$$

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$$\begin{bmatrix} S_1 \\ S_2 \end{bmatrix} = S = A^{-1}B$$

$$= \frac{1}{0.94}\begin{bmatrix} 1 & 0.3 \\ 0.2 & 1 \end{bmatrix}\begin{bmatrix} 20000 \\ 40000 \end{bmatrix}$$

$$= \begin{bmatrix} 34043 \\ 46809 \end{bmatrix}$$

$S_1 = ₹\,34043$ and $S_2 = ₹\,46809$

The allocation of overhead from S_1 and S_2 to the production department becomes:

$$[P_1 \ P_2 \ P_3 \ P_4] = [34043][0.3 \ 0.25 \ 0.15 \ 0.1]$$

$$= [10213 \ 8511 \ 5106 \ 3404]$$

and $$[P_1 \ P_2 \ P_3 \ P_4] = [46809][0.1 \ 0.35 \ 0.2 \ 0.05]$$

$$= [4681 \ 16383 \ 9362 \ 2340]$$

The final allocation becomes :

Department	Total	P_1	P_2	P_3	P_4
S_1	27234	10213	8511	5106	3404
S_2	32766	4681	16383	9362	2340
Total	60000	14894	24894	14468	5744

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Sample Paper 12

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

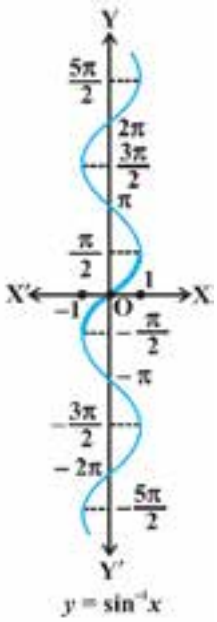
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. At which point does the graph of $y = \cos^{-1} x$ intersect the Y-axis?



(a) $(0,0)$ (b) $(0,\pi/2)$

(c) $(0,\pi)$ (d) $(0,1)$

Sol :

From the graph, the curve passes through the point $(0,\pi/2)$. This means that when $x = 0$, the value of $y = \cos^{-1}(0) = \pi/2$. This is the exact point where the graph touches the Y-axis.

Thus (b) is correct option.

2. Integrating factor (IF) of the differential equation

$$\frac{dy}{dx} - y\cos x = \sin x\cos x$$

(a) $e^{-\sin x}$ (b) $e^{\sin x}$

(c) $e^{-\cos x}$ (d) $e^{\cos x}$

Sol :

We have,

$$\frac{dy}{dx} - y\cos x = \sin x\cos x$$

Thus IF $e^{\int -\cos x\,dx} = e^{-\sin x}$

Thus (a) is correct option.

3. The solution of the differential equation $\frac{dy}{dx} = \frac{x}{y}$ is

(a) $x - y = k$ (b) $x^2 - y^2 = k$

(c) $x^3 - y^3 = k$ (d) $xy = k$

Sol :

We have $\frac{dy}{dx} = \frac{x}{y}$

Separating x and y terms in L.H.S. and R.H.S, we get,

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$$y\,dy = x\,dx$$

Integrating both sides, we get

$$\int y\,dy = \int x\,dx$$

$$\frac{y^2}{2} = \frac{x^2}{2} + c$$

$$y^2 = x^2 + 2c$$

$$y^2 - x^2 = 2c$$

or $$x^2 - y^2 = -2c$$

$$x^2 - y^2 = k \qquad [-2c = k]$$

Where, k is constant of integration.

Thus (b) is correct option.

4. $\frac{d}{dx}(\sin x) =$

(a) $\cos x$ (b) $-\sin x$

(c) $-\cos x$ (d) $\tan x$

Sol :

$$\frac{d}{dx}(\sin x) = \cos x \qquad \text{(Standard formula)}$$

Thus (a) is correct option.

5. $\hat{i} \times (\hat{i} \times \hat{j}) + \hat{j} \times (\hat{j} \times \hat{k}) + \hat{k} \times (\hat{k} \times \hat{i}) =$

(a) $\hat{i} + \hat{j} + \hat{k}$ (b) 0

(c) 1 (d) $-(\hat{i} + \hat{j} + \hat{k})$

Sol :

Let, $$\vec{x} = \hat{i} \times (\hat{i} \times \hat{j}) + \hat{j} \times (\hat{j} \times \hat{k}) + \hat{k} \times (\hat{k} \times \hat{i}) \qquad ...(1)$$

Since, $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$

Putting values in eq. (1)

$$\vec{x} = \hat{i} \times \hat{k} + \hat{j} \times \hat{i} + \hat{k} \times \hat{j}$$

$$[\hat{i} \times \hat{k} = -\hat{j}, \hat{j} \times \hat{i} = -\hat{k}, \hat{k} \times \hat{j} = \hat{i}]$$

$$= -\hat{j} - \hat{k} - \hat{i} = -(\hat{i} + \hat{j} + \hat{k})$$

Thus (d) is correct option.

6. The real number which most exceeds its cube is

(a) $\frac{1}{\sqrt{3}}$ (b) $\frac{1}{2}$

(c) $\frac{1}{\sqrt{2}}$ (d) None of these

Sol :

Let the number be x, then

$$f(x) = x - x^3$$

$$f'(x) = 1 - 3x^2$$

and $$f''(x) = -6x$$

Putting $f'(x) = 0$ we have $x = \pm\frac{1}{\sqrt{3}}$

Now $$f''\left(\frac{1}{\sqrt{3}}\right) < 0.$$

Hence $$x = \frac{1}{\sqrt{3}}$$

Thus (a) is correct option.

7. If the direction cosines of two straight lines are l_1, m_1, n_1 and l_2, m_2, n_2 then the cosine of the angle θ between them or $\cos\theta$ is

(a) $(l_1 + m_1 + n_1)(l_2 + m_2 + n_2)$

(b) $\frac{l_1}{l_2} + \frac{m_1}{m_2} + \frac{n_1}{n_2}$

(c) $l_1 l_2 + m_1 m_2 + n_1 n_2$

(d) $\frac{l_1 + m_1 + n_1}{l_2 + m_2 + n_2}$

Sol :

Direction cosine of line

$$L_1 = (l_1, m_1, n_1)$$

Therefore, vector parallel to

$$L_1 = \left(l_1\hat{i} + m_1\hat{j} + n_1\hat{k}\right) = \vec{a}$$

Similarly direction cosine of line

$$L_2 = (l_2, m_2, n_2)$$

Vector parallel to $L_2 = (l_2\hat{i} + m_2\hat{j} + n_2\hat{k}) = \vec{b}$

$$\vec{a} \cdot \vec{b} = (l_1\hat{i} + m_1\hat{j} + n_1\hat{k}).(l_2\hat{u} + m_2\hat{j} + n_2\hat{k})$$

$$= \sqrt{l_1^2 + m_1^2 + n_1^2}\sqrt{l_2^2 + m_2^2 + n_2^2}\cos\theta$$

Where θ = Angle between L_1 and L_2

Hence $$l_1 l_2 + m_1 m_2 + n_1 n_2 = \cos\theta$$

Since magnitude of vector with direction cosine as their component = 1.

Thus (c) is correct option.

8. If $\lambda \in R$ and $\Delta = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$ then $\lambda\Delta =$

(a) $\begin{vmatrix} \lambda a & \lambda b \\ \lambda c & \lambda d \end{vmatrix}$ (b) $\begin{vmatrix} \lambda a & b \\ c & d \end{vmatrix}$

(c) $\begin{vmatrix} \lambda a & b \\ \lambda c & d \end{vmatrix}$ (d) None of these

Sol :

Multiplying a determinant by λ means multiplying the elements of only one row (or one column) by λ.

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19. **Assertion :** If A and B are two independent events $P(A \cup B) = 1 - P(A')\,P(B')$

Reason : $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

and $P(A \cap B) = P(A)\,P(B)$

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

Sol :

A and B are two independent event

$P(A \cap B) = P(A) \cdot P(B)$

We know that

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= P(A) + P(B) - P(A) \cdot P(B) \\ &= P(A) + P(B) - \big(1 - P(\overline{A})\big)\big(1 - P(\overline{B})\big) \\ &= P(A) + P(B) - \big[1 - P(\overline{B}) - P(\overline{A}) \\ &\qquad + P(\overline{A}) \cdot P(\overline{B})\big] \\ &= P(A) + P(B) - 1 + P(\overline{B}) + P(\overline{A}) \\ &\qquad - P(\overline{A}) \cdot P(\overline{B}) \\ &= 1 - 1 + 1 - P(\overline{A}) \cdot P(\overline{B}) \end{aligned}$$

(Since $P(A) + P(\overline{A}) = 1$ and $P(B) + P(\overline{B}) = 1$)

$$= 1 - P(\overline{A}) \cdot P(\overline{B}) = \text{R.H.S.}$$

Thus both assertion and reason are true and reason is the correct explanation of assertion.

Thus (a) is correct option.

20. **Assertion :** $f(x)$ is defined as $f(x) = \begin{cases} x^3 - 3, x \le 2 \\ x^2 + 1, x > 2 \end{cases}$ is continuous at $x = 2$.

Reason : $f(2) = \lim\limits_{x \to 2} f(x)$.

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

Sol :

Given, $f(x) = \begin{cases} x^3 - 3, & x \le 2 \\ x^2 + 1, & x > 2 \end{cases}$

Function $f(x)$ is continuous at $x = 2$, if and only if

$$\lim_{x \to 2^+} f(x) = \lim_{x \to 2^-} f(x) = f(2)$$

Now, $$\begin{aligned} \lim_{x \to 2^+} f(x) &= \lim_{h \to 0} f(2 + h) \\ &= \lim_{h \to 0} (2 + h)^2 + 1 \\ &= \lim_{h \to 0} 4 + h^2 + 4h + 1 \\ &= 4 + 1 = 5 \end{aligned}$$

and $$\begin{aligned} \lim_{x \to 2^-} f(x) &= \lim_{h \to 0} f(2 - h) \\ &= \lim_{h \to 0} (2 - h)^3 - 3 \\ &= \lim_{h \to 0} 2^3 - h^3 - 6h^2 + 12h - 3 \\ &= 8 - 3 = 5 \end{aligned}$$

$$f(2) = 2^3 - 3 = 8 - 3 = 5$$

Since, $$f(2) = \lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x)$$

Therefore, given function is continuous at $x = 2$

Thus (a) is correct option.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Find $\frac{dy}{dx}$ if $y = \cos\sqrt{\sin x}$

OR

Find $\frac{dy}{dx}$, when $x = y\log(xy)$

Sol :

We have, $y = \cos\sqrt{\sin x}$

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}\cos\sqrt{\sin x} \\ &= -\sin\sqrt{\sin x} \cdot \frac{d}{dx}\sqrt{\sin x} \\ &= -\sin\sqrt{\sin x} \cdot \frac{1}{2\sqrt{\sin x}} \cdot \frac{d}{dx}(\sin x) \\ &= -\frac{1}{2}\frac{\sin\sqrt{\sin x}}{\sqrt{\sin x}} \cdot \cos x \\ &= \frac{-1}{2\sqrt{\sin x}}\sin\sqrt{\sin x}\cos x \end{aligned}$$

OR

$$x = y\log(xy) \qquad \ldots(1)$$

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Sol :

Relation which is reflexive, symmetric and transitive is called equivalence relation.

Reflexive: If every element of A is related to itself, i.e. $(a,a) \in R$

Symmetric: If $(a,b) \in R \Rightarrow (b,a) \in R$ for all $a,b \in A$.

Transitive: If $(a,b) \in R$ and $(b,c) \in R \Rightarrow (a,c) \in R$ for all $a,b,c \in A$.

Therefore only two relations are possible.
Thus (a) is correct option.

15. $\frac{d}{dx}[\log x] = ?$

(a) $\frac{1}{x}$ (c) $-\frac{1}{x^2}$

(c) 1 (d) $\frac{1}{x^2}$

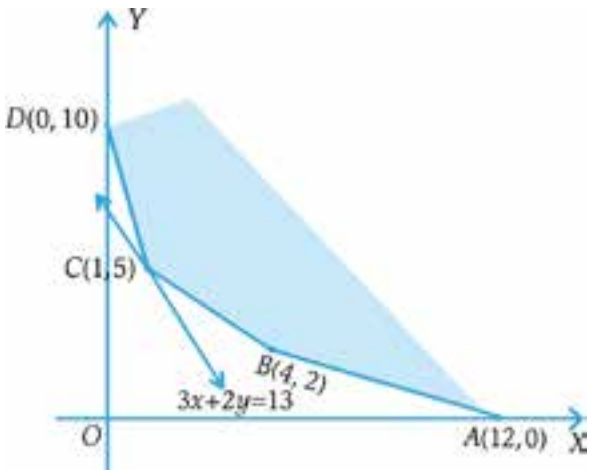
Sol :

Let, $f(x) = \log x$

$$\frac{d}{dx}f(x) = \frac{d}{dx}(\log x) = \frac{1}{x}$$

Thus (a) is correct option.

16. Using the feasible region shown in Fig., determine the maximum value of the objective function $z = 2x + 3y$. The feasible region has corner points A(12,0), B(4,2), C(1,5), and D(0,10).



(a) 24 (b) 14

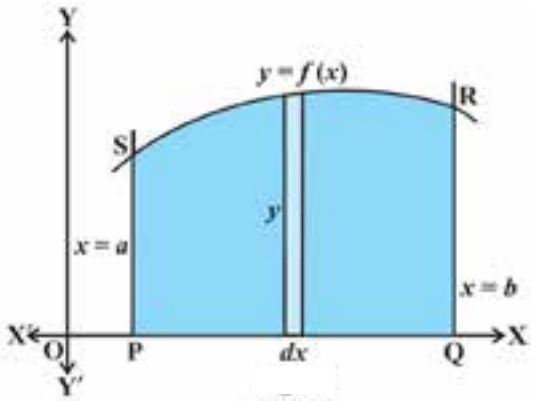
(c) 17 (d) 30

Sol :

For a linear programming problem, an optimum (maximum or minimum) value of the objective function when it exists occurs at a corner (vertex) of the feasible region. Substituting the coordinates of each vertex into $z = 2x + 3y$ gives the following values: at $A(12,0), z = 2(12) + 3(0) = 24$; at $B(4,2)$, $z = 2(4) + 3(2) = 14$; at $C(1,5), z = 2(1) + 3(5) = 17$; at $D(0,10), z = 2(0) + 3(10) = 30$. Among these, the largest value is 30, which is attained at the vertex D(0,10). Hence the maximum value of the objective function is 30.

Thus (d) is correct option.

17. In the given figure, the shaded region lies inside which curve?



(a) Circle with center O and radius a

(b) Ellipse with semi-major axis a and semi-minor axis b

(c) Parabola opening along the X-axis

(d) Hyperbola with foci on the X-axis

Sol :

The figure clearly shows an ellipse centered at O, with vertices A(a,0) and A'(-a,0) on the X-axis, and B(0,b) and B'(0,-b) on the Y-axis. These points confirm that the curve is an ellipse with semi-major axis a and semi-minor axis b.

Thus (b) is correct option.

18. $\int_2^1 \frac{dx}{x} = ?$

(a) $\log\frac{2}{3}$ (b) $\log\frac{3}{2}$

(c) $\log\frac{1}{2}$ (d) $\log\frac{x}{2}$

Sol :

Let, $I = \int_2^1 \frac{dx}{x}$

$= \left[\log|x|\right]_2^1$

$= \log 1 - \log 2$

$= \log\frac{1}{2}$

Thus (c) is correct option.

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Hence, $\lambda\Delta = \lambda\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} \lambda a & b \\ \lambda c & d \end{vmatrix}$

Thus (c) is correct option.

9. The direction ratios of a straight line are 1,3,5. Its direction cosines are

(a) $\frac{1}{\sqrt{35}}, \frac{3}{\sqrt{35}}, \frac{5}{\sqrt{35}}$ (b) $\frac{1}{9}, \frac{1}{3}, \frac{5}{9}$

(c) $\frac{3}{\sqrt{35}}, \frac{5}{\sqrt{35}}, \frac{1}{\sqrt{35}}$ (d) $\frac{5}{\sqrt{35}}, \frac{3}{\sqrt{35}}, \frac{1}{\sqrt{35}}$

Sol :

If l, m, n are direction cosine of line and a, b, c are direction ratios then

$$l = \frac{a}{\sqrt{a^2+b^2+c^2}}$$
$$m = \frac{b}{\sqrt{a^2+b^2+c^2}}$$
$$n = \frac{c}{\sqrt{a^2+b^2+c^2}}$$

Given direction ratios are

$$a = 1, \quad b = 3, \quad c = 5$$

Then, $a^2+b^2+c^2 = 1+9+25 = 35$

Therefore, $l = \frac{1}{\sqrt{35}}$, $m = \frac{3}{\sqrt{35}}$

and $n = \frac{5}{\sqrt{35}}$

Thus (a) is correct option.

10. $\int \frac{xe^x}{(x+1)^2}dx =$

(a) $\frac{e^x}{(x+1)^2}+c$ (b) $\frac{-e^x}{x+1}+c$

(c) $\frac{e^x}{x+1}+c$ (d) $\frac{-e^x}{(x+1)^2}+c$

Sol :

We have $I = \int \frac{xe^x}{(1+x)^2}dx$

Adding and Subtracting e^x in numerator

$$I = \int \frac{xe^x+e^x-e^x}{(1+x)^2}$$
$$= \int e^x\left[\frac{1}{1+x} - \frac{1}{(1+x)^2}\right]$$

Let, $f(x) = \frac{1}{1+x}$

$$f'(x) = \frac{-1}{(1+x)^2}$$

Hence, $I = \frac{e^x}{1+x} + c \quad \left[\int e^x\big(f(x)+f'(x)\big) = e^x f(x)\right]$

Thus (c) is correct option.

11. Maximum value of $f(x) = \sin x + \cos x$ is

(a) 1 (b) 2

(c) $\frac{1}{\sqrt{2}}$ (d) $\sqrt{2}$

Sol :

Max value $= \sqrt{a^2+b^2} = \sqrt{1^2+1^2} = \sqrt{2}$

Thus (d) is correct option.

12. $f(x) = 2x^3 - 15x^2 + 36x + 4$ is

(a) increasing in $(-\infty, 2]$

(b) increasing in $[2, 3]$

(c) decreasing in $(3, \infty)$

(d) None of these

Sol :

Function $y = f(x)$ is increasing where $f'(x) > 0$ and decreasing where $f'(x) < 0$

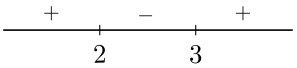
Now, $f(x) = 2x^3 - 15x^2 + 36x + 4$

Hence, $f'(x) = 6x^2 - 30x + 36$

$$= 6(x^2 - 5x + 6) = 6(x-2)(x-3)$$

Therefore, $f'(x) > 0$ for $x \in (-\infty, 2) \cup (3, \infty)$

$f'(x) < 0$ for $x \in (2, 3)$



Hence, $f(x)$ is increasing for $x \in (-\infty, 2) \cup (3, \infty)$ and decreasing for $x \in (2, 3)$.

Thus (d) is correct option.

13. The optimal value of the objective function is attached at the point:

(a) given by intersection of in-equations with axes only.

(b) given by intersection of in-equations with x-axis only.

(c) given by corner points of the feasible region.

(d) none of the above.

Sol :

Thus (c) is correct option.

14. If $A = \{1, 2, 3\}$, then how many equivalence relation can be defined on A containing (1, 2) :

(a) 2 (b) 3

(c) 8 (d) 6

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Differentiating both sides using chain rules with respect to x

$$\begin{aligned}1 &= y\frac{dy}{dx}\{\log xy\}+(\log xy)\frac{dy}{dx}\\ &= y\left[\frac{1}{xy}\right]\times\frac{d}{dx}(xy)+\frac{dy}{dx}(\log xy)\\ &= \frac{1}{x}\left[\frac{xdy}{dx}+y\right]+\log xy\frac{dy}{dx}\\ &= \frac{dy}{dx}+\frac{y}{x}+\log xy\frac{dy}{dx}\\ &= \frac{y}{x}+[\log xy+1]\frac{dy}{dx}\end{aligned}$$

Thus, $1-\frac{y}{x} = [\log xy+1]\frac{dy}{dx}$

$$\frac{x-y}{x(\log xy+1)} = \frac{dy}{dx}$$

Putting $x = y\log(xy)$ from (1)

or $\frac{x}{y} = \log(xy)$

$$\frac{x-y}{x\left(\frac{x}{y}+1\right)} = \frac{dy}{dx}$$

$$\frac{y(x-y)}{x(x+y)} = \frac{dy}{dx}$$

22. Show that the line joining the points (4, 7, 8), (2, 3, 4) is parallel to the line joining the points (2, 4, 10), (–2, –4, 2).

Sol :

Two lines are said to be parallel if direction ratio of points forming the line are proportional to each other.

Direction Ratio of line joining $P_1(4,7,8)$ and $P_2(2,3,4)$ is

$$\begin{aligned}DR_1 &= (4-2,7-3,8-4)\\ &= (2,4,4)\end{aligned}$$

Direction Ratio of line joining $P_3(2,4,10)$ and $P_4(-2,-4,2)$ is

$$\begin{aligned}DR_2 &= (2+2,4+4,10-2)\\ &= (4,8,8)\end{aligned}$$

Since direction ratio are proportional to each other with ratio 2, hence lines formed by (P_1,P_2) and (P_3,P_4) are parallel to each other.

23. Prove that – $|\vec{a}\times\vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2-|\vec{a}\,\vec{b}|^2$

Sol :

We know that,

$$|\vec{a}\times\vec{b}| = |\vec{a}||\vec{b}|\sin\theta$$

$$|\vec{a}\times\vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2\sin^2\theta \quad ...(1)$$

Also, $|\vec{a}\cdot\vec{b}| = |\vec{a}||\vec{b}|\cos\theta$

$$|\vec{a}\cdot\vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2\cos^2\theta \quad ...(2)$$

on adding equation (1) and (2), we get

$$|\vec{a}\times\vec{b}|^2+|\vec{a}\cdot\vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2(\sin^2\theta+\cos^2\theta)$$

$$|\vec{a}\times\vec{b}|^2+|\vec{a}\cdot\vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2$$

$$|\vec{a}\times\vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2-|\vec{a}\cdot\vec{b}|^2$$

24. If A and B are two independent events then prove that : $P(A\cup B) = 1-P(A')\cdot P(B')$

Sol :

Let $P(A)$ denotes probability of occurrence of Event A. $P(B)$ denotes probability of occurrence of Event B.

Since A and B are two independent Events

$$\Rightarrow \quad P(A\cap B) = P(A)\cdot P(B) \quad ...(1)$$

Using property

$$\begin{aligned}P(A\cup B) &= P(A)+P(B)-P(A\cup B)\\ &= P(A)+P(B)-P(A)\cdot P(B) \quad \text{[Using (1)]}\\ &= P(A)+P(B)[1-P(A)]\\ &= P(A)+P(B)\cdot P(A') \quad [1-P(A)=P(A')]\\ &= 1-P(A')+P(B)\cdot P(A')\\ &= 1-P(A')[1-P(B)] \quad [1-P(B)=P(B')]\\ &= 1-P(A')P(B') \quad \text{Hence proved.}\end{aligned}$$

25. Solve : $\frac{dy}{dx}+2y\tan x = \sin x$.

OR

Show that the function $1+8y^2\tan x = ay^2$ is a solution of differential equation

$$\cos^2 x\frac{dy}{dx} = 4y^3$$

Sol :

Given: $\frac{dy}{dx}+2y\tan x = \sin x$

This is first order linear differential equation of the form

$$\frac{dy}{dx}+Py = Q$$

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$$\left[\cot\left(\frac{\pi}{2}-\theta\right)=\tan\theta\right]$$

$$=\cot^{-1}\left(\cot\frac{5\pi}{14}\right)$$

$$=\frac{5\pi}{14} \qquad \left[\cot^{-1}\cot\theta=\theta\right]$$

OR

Let, $f = 4(\cot^{-1}3+\operatorname{cosec}^{-1}\sqrt{5})$

and $\cot^{-1}3 = \alpha$

or $\cot\alpha = 3$

$\Rightarrow$ $\tan\alpha = \frac{1}{3}$

and $\operatorname{cosec}^{-1}\sqrt{5} = \beta$

or $\operatorname{cosec}\beta = \sqrt{5}$

$\Rightarrow$ $\tan\beta = \frac{1}{\cot\beta}$

$$=\frac{1}{\sqrt{\operatorname{cosec}^2\beta-1}}$$

$$=\frac{1}{\sqrt{5-1}}=\frac{1}{2}$$

Therefore, $\tan\alpha = \frac{1}{3} \Rightarrow \alpha = \tan^{-1}\frac{1}{3}$

$\tan\beta = \frac{1}{2} \Rightarrow \beta = \tan^{-1}\frac{1}{2}$

Therefore, $f = 4(\alpha+\beta)$

$$=4\left(\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{2}\right)$$

$$=4\left(\tan^{-1}\frac{\frac{1}{3}+\frac{1}{2}}{1-\frac{1}{3}\times\frac{1}{2}}\right)$$

$$\left[\tan^{-1}A+\tan^{-1}B=\tan^{-1}\left(\frac{A+B}{1-AB}\right)\right]$$

$$=4\left(\tan^{-1}\frac{\frac{5}{6}}{\frac{6-1}{5}}\right)$$

$$=4\times\tan^{-1}1$$

$$=4\times\tan^{-1}\left(\tan\frac{\pi}{4}\right)$$

$$=4\times\frac{\pi}{4}=\pi \quad \text{Hence proved.}$$

31. In the set Q of all rational numbers, a binary operation $o: Q\times Q\to Q$ is defined by $o(x,y) = x\,o\,y = x+y-xy$ then show that o is commutative.

Sol :

Given : $o\,Q\times Q\to Q$

$o(x,y) = xoy = x+y-xy$ ...(1)

$o(y,x) = yox = y+x-yx$ ...(2)

Therefore, $xoy = yox$

Hence, Binary operation o is commutative.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Evaluate $\int\frac{-7x+2}{\sqrt{16x^2-9}}dx$

OR

Prove that $\int_0^{\pi/2}\log\sin x\,dx=\int_0^{\pi/2}\log\cos x\,dx=-\frac{\pi}{2}\log 2$

Sol :

Let $I = \int\frac{-7x+2}{\sqrt{16x^2-9}}dx$

$$=-7\int\frac{x}{\sqrt{16x^2-9}}dx+2\int\frac{1}{\sqrt{16x^2-9}}dx$$

or $I = -7I_1+2I_2$ ...(1)

Now, $I_1 = \int\frac{x}{\sqrt{16x^2-9}}dx$

Putting $\sqrt{16x^2-9} = y$ we have

$\frac{x}{\sqrt{16x^2-9}}dx = \frac{1}{16}dy$

Thus $I_1 = \frac{1}{16}\int 1\,dy$

$$=\frac{1}{16}(y)+C_1$$

$$=\frac{1}{16}\sqrt{16x^2-9}+C_1$$

Now, $I_2 = \int\frac{1}{\sqrt{16x^2-9}}dx$

$$=\int\frac{1}{\sqrt{16\left(x^2-\frac{9}{16}\right)}}dx$$

$$=\frac{1}{4}\int\frac{1}{\sqrt{x^2-\left(\frac{3}{4}\right)^2}}dx$$

$$=\frac{1}{4}\log\left|x+\sqrt{x^2-\left(\frac{3}{4}\right)^2}\right|+C_2$$

$$=\frac{1}{4}\log\left|x+\sqrt{x^2-\frac{9}{16}}\right|+C_2$$

Putting values of I_1 and I_2 in (1), we get

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$$+(-x-2)\begin{vmatrix} 9x & -45 \\ 2x^2+3x & -10x-15 \end{vmatrix}$$

$2x[90x+135-(90x+135)]-(-5x-10)[-18x^2-27x-(-18x^2-27x)]$

$$+(-x-2)\left[-90x^2-135x-(-90x^2-135x)\right]$$

$2x[90x+135-90x-135]+(5x+10)[-18x^2-27x+18x^2+27x]$

$$-(x+2)\left[-90x^2-135x+90x^2+135x\right]=0$$

Hence, for any value of x, multiplication of matrix is zero (0).

OR

We have $A=\begin{bmatrix}2 & 5\\ 3 & 1\end{bmatrix}$ and $B=\begin{bmatrix}1 & 5\\ 6 & 2\end{bmatrix}$

Then, $A+B=\begin{bmatrix}2 & 5\\ 3 & 1\end{bmatrix}+\begin{bmatrix}1 & 5\\ 6 & 2\end{bmatrix}$

$$=\begin{bmatrix}2+1 & 5+5\\ 3+6 & 1+2\end{bmatrix}=\begin{bmatrix}3 & 10\\ 9 & 3\end{bmatrix}$$

$A-B=\begin{bmatrix}2 & 5\\ 3 & 1\end{bmatrix}-\begin{bmatrix}1 & 5\\ 6 & 2\end{bmatrix}$

$$=\begin{bmatrix}2-1 & 5-5\\ 3-6 & 1-2\end{bmatrix}$$

$$=\begin{bmatrix}1 & 0\\ -3 & -1\end{bmatrix}$$

Therefore, $A+B=\begin{bmatrix}3 & 10\\ 9 & 3\end{bmatrix}$

$A-B=\begin{bmatrix}1 & 0\\ -3 & -1\end{bmatrix}$

28. The radius of a circle is increasing uniformly at the rate of 3 cm/sec. Find the rate at which the area of the circle is increasing when the radius is 10 cm.

Sol :

Let r be the radius and A be the area of circle at time t.

Rate of increase of radius wrt time

$$t=\frac{dr}{dt}=3\,\text{cm/sec}$$

Area of circle $A=\pi r^2$

Rate of change of area wrt time

$$t=\frac{dA}{dt}$$

$$=\frac{d}{dt}(\pi r^2)$$

$$=\pi\cdot 2r\frac{dr}{dt}$$

$$=2\pi r\frac{dr}{dt}=2\pi r(3)\quad\left[\frac{dr}{dt}=3\,\text{cm/sec}\right]$$

$$=6\pi r$$

Rate of increase of area wrt t when r is 10 cm $6\pi\times 10=60\pi\,\text{cm}^2/\text{sec}$.

29. If $f(x)=x\sin\frac{1}{x}$, when $x\neq 0$; and, $f(x)=0$, when $x=0$, then test the continuity of $f(x)$ at $x=0$.

Sol :

Given:

$$f(x)=\begin{cases}x\sin\frac{1}{x} & x\neq 0\\ 0 & x=0\end{cases}$$

Now,

$\text{LHL}=\lim\limits_{x\to 0^-}f(x)$

$=\lim\limits_{h\to 0}f(0-h)=\lim\limits_{h\to 0}f(-h)$

$=\lim\limits_{h\to 0}-h\sin(\frac{1}{-h})$

$=\lim\limits_{h\to 0}h\sin\frac{1}{h}$

$=0\times$ (finite quantity between -1 and 1)

$=0$

$\text{RHL}=\lim\limits_{x\to 0^+}f(x)$

$=\lim\limits_{h\to 0}f(0+h)=\lim\limits_{h\to 0}f(h)$

$=\lim\limits_{h\to 0}h\times\sin\left(\frac{1}{h}\right)$

$=0\times$ (finite quantity between -1 and 1)

$=0$

Since,

$$\lim_{x\to 0^-}f(x)=\lim_{x\to 0^+}f(x)$$

$$=f(0)=0$$

Hence, $f(x)$ is continuous at $x=0$.

30. Find the value of $\cot^{-1}\left(\tan\frac{\pi}{7}\right)$?

OR

Prove that $4(\cot^{-1}3+\text{cosec}^{-1}\sqrt{5})=\pi$.

Sol :

Let, $I=\cot^{-1}\left(\tan\frac{\pi}{7}\right)$

Therefore, $I=\cot^{-1}\left(\cot\left(\frac{\pi}{2}-\frac{\pi}{7}\right)\right)$

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Whose general solution is given by

$$\text{I}\cdot\text{F}\,y = \int \text{I}\cdot\text{F}\,Q\,dx + c$$

Where, $\text{I}\cdot\text{F} = e^{\int P dx}$

Here, $P = 2\tan x$ and $Q = \sin x$

$$\text{I}\cdot\text{F} = e^{\int 2\tan x\,dx}$$

$$= e^{2\int \tan x\,dx} \left[\int \tan x = \log\sec x\right]$$

$$= e^{2\log\sec x} = \sec^2 x \quad [a^{\log_a x} = x]$$

Thus general solution is

$$y\sec^2 x = \int \sec^2 x \sin x\,dx$$

$$= \int \frac{\sin x}{\cos^2 x}dx \quad \left[\frac{1}{\cos x} = \sec x\right]$$

$$= \int \tan x \sec x\,dx = \sec x + c$$

Thus solution of given differential equation is

$$y\sec^2 x = \sec x + c$$

OR

Given function is

$$1 + 8y^2\tan x = ay^2$$

$$\frac{1}{y^2} + 8\tan x = a$$

Differentiating both sides w.r.t. x, we get

$$-2y^{-3}\frac{dy}{dx} + 8\sec^2 x = 0$$

$$-\frac{2}{y^3}\frac{dy}{dx} = -8\sec^2 x$$

$$\frac{1}{y^3}\frac{dy}{dx} = \frac{4}{\cos^2 x}$$

$$\cos^2 x\frac{dy}{dx} = 4y^3$$

Which is the given differential equation.
Hence, given function is a solution of the given differential equation.

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. Find the value of p, if
$(2\hat{i} + 6\hat{j} + 27\hat{k}) \times (\hat{i} + 3\hat{j} + p\hat{k}) = 0$

Sol :

$$(2\vec{i} + 6\vec{j} + 27\vec{k}) \times (\vec{i} + 3\vec{j} + p\vec{k}) = 0$$

Expressing it in determinant form

$$\Delta = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 6 & 27 \\ 1 & 3 & p \end{bmatrix}$$

Solving Determinant

$$\Delta = \hat{i}(6p - 27\times 3) - \hat{j}(2p - 27) + \hat{k}(6 - 6)$$

$$= \hat{i}(6p - 81) - \hat{j}(2p - 27) + 0$$

Equating $\Delta = 0$ with $\hat{i}$ and $\hat{j}$ terms

$$6p - 81 = 0$$

$$p = \frac{27}{2}$$

Thus $\vec{a}\times\vec{b} = 0$ when $p = \frac{27}{2}$.

27. Find the value of x, such that

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}\begin{bmatrix} x \\ 4 \\ 1 \end{bmatrix}\begin{bmatrix} x & -5 & -1 \end{bmatrix} = 0$$

OR

If $A = \begin{bmatrix} 2 & 5 \\ 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 5 \\ 6 & 2 \end{bmatrix}$ then find $(A + B)$ and $(A - B)$.

Sol :

From L.H.S.

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}\begin{bmatrix} x \\ 4 \\ 1 \end{bmatrix}[x \ -5 \ -1] = 0$$

$$\begin{bmatrix} x+0+2 \\ 0+8+1 \\ 2x+0+3 \end{bmatrix}[x \ -5 \ -1] = 0$$

$$\begin{bmatrix} x+2 \\ 9 \\ 2x+3 \end{bmatrix}[x \ -5 \ -1] = 0$$

$$\begin{bmatrix} x(x+2) & -5(x+2) & -1(x+2) \\ 9x & -45 & -9 \\ x(2x+3) & -5(2x+3) & -1(2x+3) \end{bmatrix} = 0$$

$$\begin{bmatrix} x^2+2x & -5x-10 & -x-2 \\ 9x & -45 & -9 \\ 2x^2+3x & -10x-15 & -2x-3 \end{bmatrix} = 0$$

$$2x\begin{vmatrix} -45 & -9 \\ -10x-15 & -2x-3 \end{vmatrix} - (-5x-10)\begin{vmatrix} 9x & -9 \\ 2x^2+3x & -1(2x+3) \end{vmatrix}$$

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$$I = -\frac{7}{16}\sqrt{16x^2-9} + \frac{1}{2}\log\left|x+\sqrt{x^2-\frac{9}{16}}\right| + C$$

OR

Let $I_1 = \int_0^{\pi/2} \log\sin x\,dx$

and $I_2 = \int_0^{\pi/2} \log\cos x\,dx$

$$\int_0^{\pi/2} \log\sin x\,dx = \int_0^{\pi/2} \log\sin\left(0+\frac{\pi}{2}-x\right)dx$$

$$= \int_0^{\pi/2} \log\cos x\,dx$$

$$\left[\int_a^b \log x\,dx = \int_a^b \log(a+b-x)\,dx\right]$$

$\Rightarrow \quad I_1 = I_2$

Using property

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

$$I_1 = \int_0^{\pi/2} \log\sin x\,dx$$

$$= \int_0^{\pi/2} \log\sin\left(\frac{\pi}{2}-x\right)dx$$

$$= \int_0^{\pi/2} \log\cos x\,dx$$

$$2I_1 = \int_0^{\pi/2} \log\sin x\,dx + \int_0^{\pi/2} \log\cos x\,dx$$

$$= \int_0^{\pi/2} \log\sin x\cos x\,dx$$

$$[\log x + \log y = \log xy]$$

Multiplying and dividing by 2 in R.H.S., we get

$$2I_1 = \int_0^{\pi/2} \log\left(\frac{2\sin x\cos x}{2}\right)dx$$

$$= \int_0^{\pi/2} \log\left(\frac{\sin 2x}{2}\right)dx$$

$$[2\sin\theta\cos\theta = \sin 2\theta]$$

$$= \int_0^{\pi/2} \log\sin 2x\,dx - \int_0^{\pi/2} \log 2\,dx$$

$$2I_1 = \int_0^{\pi/2} \log\sin 2x\,dx - \log 2\left(\frac{\pi}{2}-0\right) \ldots(1)$$

Let $I_3 = \int_0^{\pi/2} \log\sin 2x\,dx$

Substituting $t = 2x$,

$dt = 2dx$

$dx = \frac{dt}{2}$ $\qquad [x=0 \Rightarrow t=0$

$I_3 = \frac{1}{2}\int_0^{\pi} \log\sin t\,dt \qquad x=\frac{\pi}{2} \Rightarrow t=\pi]$

Using property

$$\int_0^{2a} f(x)\,dx = 2\int_0^a f(a-x)\,dx \quad \text{if } f(2a-x) = f(x)$$

$$I_3 = \frac{1}{2}\int_0^{\pi} \log\sin t\,dt$$

$$= \int_0^{\pi/2} \sin t\,dt = I_1$$

Therefore equation (1) becomes,

$$2I_1 = I_1 - \frac{\pi}{2}\log 2$$

$$I_1 = -\frac{\pi}{2}\log 2$$

Since, $I_1 = I_2$, therefore,

$$I_1 = I_2 = -\frac{\pi}{2}\log 2$$

33. If $x = \cos t + \log\tan\left(\frac{t}{2}\right)$, $y = \sin t$, then find the values of $\frac{d^2y}{dt^2}$ and $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$.

OR

Find the values of a and b, if the function f defined by $f(x) = \begin{cases} x^2+3x+a, & x \le 1 \\ bx+2, & x > 1 \end{cases}$ is differentiable at $x = 1$.

Sol :

Given that,

$$y = \sin t$$

$$\frac{dy}{dt} = \cos t \qquad [\text{differentiate w.r.t.} t] \ldots(\text{i})$$

$$\frac{d^2y}{dt^2} = -\sin t \qquad [\text{differentiate w.r.t. } t]$$

$$\left[\frac{d^2y}{dt^2}\right]_{t=\frac{\pi}{4}} = -\sin\frac{\pi}{4} = -\frac{1}{\sqrt{2}}$$

Again, $x = \cos t + \log\tan\frac{t}{2}$

$$\frac{dx}{dt} = -\sin t + \frac{1}{\tan\frac{t}{2}} \cdot \sec^2\frac{t}{2} \cdot \frac{1}{2}$$

[differentiate w.r.t. t]

$$= -\sin t + \frac{\cos\frac{t}{2}}{2\cdot\sin\frac{t}{2}} \cdot \frac{1}{\cos^2\frac{t}{2}}$$

$$= -\sin t + \frac{1}{\sin 2\times\frac{t}{2}}$$

$\because\ 2\sin a\cos a = \sin 2a]$

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$$R(x) = p \times x = \left(15 - \frac{x}{3000}\right)x$$

$$= 15x - \frac{x^2}{3000}$$

(ii) $R(x) = 15x - \frac{x^2}{3000}$

$$R'(x) = 15 - \frac{x}{1500}$$

For maxima/minima, put $R'(x) = 0$

$$15 - \frac{x}{1500} = 0$$

$$x = 22500$$

Also, $R''(x) = -\frac{1}{1500} < 0$

(iii) Maximum revenue will be at

$$\text{Price of ticket} = 15 - \frac{22500}{3000}$$

$$= 15 - 7.5 = ₹7.5$$

OR

(iv) Number of spectators will be equal to number of tickets sold.

Required number of spectators = 22500

38. If two vectors are represented by the two sides of a triangle taken in order, then their sum is represented by the third side of the triangle taken in opposite order and this is known as triangle law of vector addition.

Based on the above information, answer the following questions.

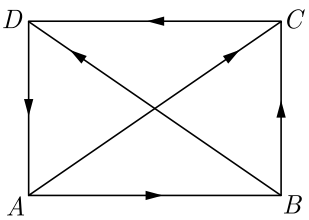
(i) If $\hat{p}$, $\hat{q}$, $\hat{r}$ are the vectors represented by the side of a triangle taken in order, then find $\vec{q} + \vec{r}$.

(ii) If $ABCD$ is a parallelogram and AC and BD are its diagonals, then find $\overrightarrow{AC} + \overrightarrow{BD}$.

(iii) If $ABCD$ is a parallelogram, where $\overrightarrow{AB} = 2\vec{a}$ and $\overrightarrow{BC} = 2\vec{b}$, then find $\overrightarrow{AC} - \overrightarrow{BD}$.

OR

(iv) If ABCD is a quadrilateral, whose diagonals are $\overrightarrow{AC}$ and $\overrightarrow{BD}$, then find $\overrightarrow{BA} + \overrightarrow{CD}$.



Sol :

(i) Let OAB be a triangle such that

$$\overrightarrow{AO} = -\vec{p}, \overrightarrow{AB}$$

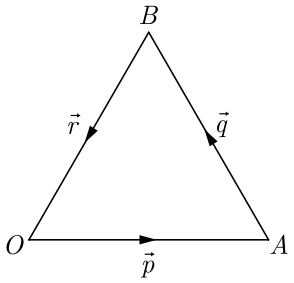
$$= \vec{q}, \overrightarrow{BO} = \vec{r}$$

Now, $\vec{q} + \vec{r} = \overrightarrow{AB} + \overrightarrow{BO}$

$$= \overrightarrow{AO} = -\vec{p}$$

(ii) From the triangle law of vector addition,

$$\overrightarrow{AC} + \overrightarrow{BD} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{BC} + \overrightarrow{CD}$$



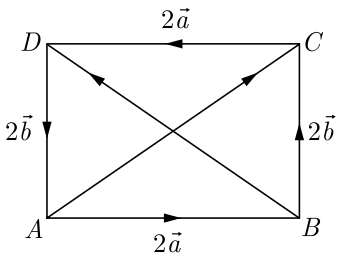
$$= \overrightarrow{AB} + 2\overrightarrow{BC} + \overrightarrow{CD}$$

$$\overrightarrow{AB} + 2\overrightarrow{BC} + -\overrightarrow{AB}$$

$$= 2\overrightarrow{BC} \qquad [\overrightarrow{AB} = -\overrightarrow{CD}]$$

(iii) In ΔABC,

$$\overrightarrow{AC} = 1\vec{a} + 2\vec{b} \qquad ...(1)$$



and in ΔABD,

$$2\vec{b} = 2\vec{a} + \overrightarrow{BD} \qquad ...(2)$$

[By triangle of law of addition]

Addition (1) and (2), we have

$$\overrightarrow{AC} + 2\vec{b} = 4\vec{a} + \overrightarrow{BD} + 2\vec{b}$$

$$\overrightarrow{AC} - \overrightarrow{BD} = 4\vec{a}$$

OR

(iv) In ΔABC,

$$\overrightarrow{BC} + \overrightarrow{CD} = \overrightarrow{BC} \qquad ...(1)$$

[By triangle law]

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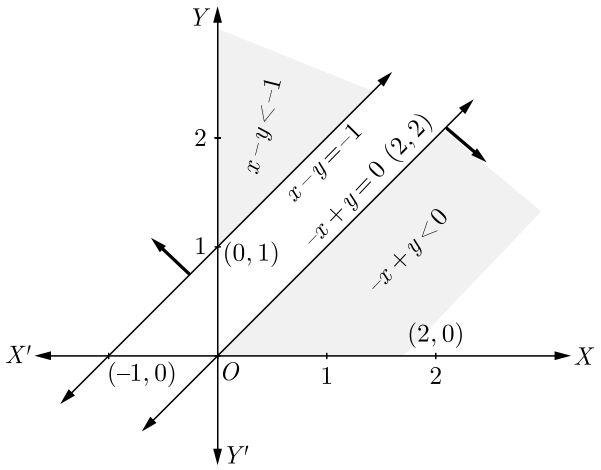
Table of values for the line $-x+y=0$ i.e., $y=x$ of constraint Eq. (3).

x	0	2
y	0	2

Let us draw the line joining the points (0, 0) and (2, 2).

Let us test for the point (2, 0) (say) [and not origin as line passes through (0, 0)] in constraint Eq. (3) $-x+y\leq 0$, we have $-2\leq 0$ which is true.

Hence, Region for constraint Eq. (3) is towards the point (2, 0) side of the line (shown shaded in the figure).



From the figure, we observe that there is no point common in the two shaded regions. Thus, the problem has no feasible region and hence no feasible solution i.e., no maximum value of Z.

Note :

When there is no feasible region in a problem, then there is no optimal solution (i.e., Neither maximum nor minimum value corner point) and hence the objective function has no optimal value. Such type of Linear Programming problems are said to have **infeasible** solution.

35. Solve : $\frac{dy}{dx}-\frac{2y}{x}=y^4$

OR

Solve $y^2\,dx+(x^2+xy)\,dy=0$

Sol :

Given: $$\frac{dy}{dx}-\frac{2y}{x}=y^4$$

Dividing the above differential equation by y^4 we get

$$\frac{1}{y^4}\frac{dy}{dx}-\frac{2}{y^3x}=1 \qquad ...(1)$$

Let $$-\frac{1}{y^3}=z$$

Differentiating both side with respect to x

$$-(-3)y^{-3-1}\frac{dy}{dx}=\frac{dz}{dx}$$

$$\frac{3}{y^4}\frac{dy}{dx}=\frac{dz}{dx}$$

$$\frac{1}{y^4}\frac{dy}{dx}=\frac{1}{3}\frac{dz}{dx}$$

Substituting the values of $-\frac{1}{y^3}$ and $\frac{1}{y^4}\frac{dy}{dx}$ in equation (1)

$$\frac{1}{3}\frac{dz}{dx}+\frac{2z}{x}=1$$

or, $$\frac{dz}{dx}+\frac{6z}{x}=3$$

This is linear differential equation of first order of the form $\frac{dz}{dx}+Pz=Q$ where P and Q are function of x.

The general solution of this equation is given by

$$ze^{\int P dx}=\int Qe^{\int P dx}dx+c \qquad ...(2)$$

Here, $$P=\frac{6}{x} \text{ and } Q=3 \qquad ...(3)$$

$$e^{\int P dx}=e^{\int \frac{6}{x}dx}$$

$$\left[\int \frac{k}{x}dx=k\int\frac{1}{x}dx=k\ln x+c\right]$$

$$=e^{6\ln x}$$

$$=e^{\ln x^6} \qquad [a\ln b=\ln b^a]$$

$$=x^6 \quad [a^{\log_a x}=x] \qquad ...(4)$$

Putting values from equation (3) and (4) in equation (2),

$$zx^6=\int 3x^6\,dx \quad \left[\int x^n dx=\frac{x^{n+1}}{n+1}+c\right]$$

$$zx^6=\frac{3x^7}{7}+c$$

Substituting $z=-\frac{1}{y^3}$

$$-\frac{1}{y^3}\times x^6=\frac{3x^7}{7}+c$$

Therefore required solution is

$$\frac{3x^7}{7}+\frac{x^6}{y^3}=c$$

where c is constant of integration.

OR

$$y^2\,dx+(x^2+xy)\,dy=0$$

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$$(x^2+xy)\,dy = -y^2\,dx$$

$$\frac{dy}{dx} = \frac{-y^2}{x^2+xy} \qquad ...(1)$$

Let $y = vx$ ...(2)

Differentiating both side

$$\frac{dy}{dx} = v + x\frac{dv}{dx}$$

Substituting value in equation (1),

$$v + x\frac{dv}{dx} = \frac{-(vx)^2}{x^2+x(vx)}$$

$$= -\frac{v^2x^2}{x^2(1+v)}$$

Therefore,

$$v + x\frac{dv}{dx} = -\frac{v^2}{1+v}$$

$$x\frac{dv}{dx} = -\frac{v^2}{1+v} - v$$

$$= \frac{-v^2-v-v^2}{1+v}$$

$$= \frac{-2v^2-v}{1+v}$$

$$x\frac{dv}{dx} = -\left(\frac{2v^2+v}{1+v}\right)$$

$$-x\frac{dv}{dx} = \frac{2v^2+v}{v+1}$$

$$\frac{(v+1)\,dv}{v(2v+1)} = -\frac{dx}{x}$$

Adding and subtracting v in numerator of L.H.S., we get

$$\frac{(v+1-v+v)\,dv}{v(2v+1)} = -\frac{dx}{x}$$

$$\frac{(2v+1)\,dv - v\,dv}{v(2v+1)} = -\frac{dx}{x}$$

Separating the numerator value with denominator we get

$$-\frac{dx}{x} = \frac{dv}{v} - \frac{dv}{2v+1}$$

Integrating both sides

$$-\log x = \log v - \frac{\log(2v+1)}{2} + \log c$$

$$\log x = -\log v + \frac{\log(2v+1)}{2} - \log c$$

$$\left[\log xy = \log x + \log y, \log\frac{x}{y} = \log x - \log y\right]$$

$$\log x + \log v - \log(2v+1)^{1/2} + \log c = 0$$

$$[a\log x = \log x^a]$$

$$\log\frac{cxv}{(2v+1)^{1/2}} = 1 \qquad [\log_e e = 1]$$

$$cxv = \sqrt{2v+1}$$

Substituting $v = \frac{y}{x}$ using (2),

$$cx\frac{y}{x} = \sqrt{2\left(\frac{y}{x}\right)+1}$$

Squaring both sides,

$$c^2y^2x = 2y + x$$

Therefore, $\frac{2y+x}{xy^2} = c^2 = k$ [Let, $c^2 = k$]

Where k is constant of integration.

SECTION - E

Case study based questions are compulsory.

36. Rice is a nutritional staple food which provides instant energy as its most important component is carbohydrate (starch). On the other hand, rice is poor in nitrogenous substances with average composition of these substances being only 8 per cent and fat content or lipids only negligible, i.e., 1per cent and due to this reason it is considered as a complete food for eating. Rice flour is rich in starch and is used for making various food materials.



Two farmers Ramkishan and Gurcharan Singh cultivate only three varieties of rice namely Basmati, Permal and Naura. The sale (in ₹) of these varieties of rice by both the farmers in the month of September and October are given by the following matrices A and B.

September Sales (in ₹)

$$A = \begin{matrix} \\ \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}\begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ \end{matrix}\begin{bmatrix} 10000 & 20000 & 30000 \\ 50000 & 30000 & 10000 \end{bmatrix}$$

October Sales (in ₹)

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$$B = \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ \end{matrix}$$
$$\begin{bmatrix} 5000 & 10000 & 6000 \\ 20000 & 10000 & 10000 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

(i) Find the combined sales in September and October for each farmer in each variety.

(ii) Find the decrease in sales from September to October.

(iii) If both farmers receive 2% profit on gross sales, compute the profit for each farmer and for each variety sold in October.

Sol :

The sale (in ₹) of these varieties of rice by both the farmers in the month of September and October are given by the following matrices A and B.

September Sales (in ₹)

$$A = \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ \end{matrix}$$
$$\begin{bmatrix} 10000 & 20000 & 30000 \\ 50000 & 30000 & 10000 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

October Sales (in ₹)

$$B = \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ \end{matrix}$$
$$\begin{bmatrix} 5000 & 10000 & 6000 \\ 20000 & 10000 & 10000 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

(i) Combined sales in September and October for each farmer in each variety is given by

$$A + B = \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ \end{matrix}$$
$$\begin{bmatrix} 10000+5000 & 20000+10000 & 30000+6000 \\ 50000+20000 & 30000+10000 & 10000+10000 \end{bmatrix}$$
$$= \begin{bmatrix} 15000 & 30000 & 36000 \\ 70000 & 40000 & 20000 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

(ii) Change in sales from September to October is given by

$$A - B = \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ \end{matrix}$$
$$\begin{bmatrix} 10000-5000 & 20000-10000 & 30000-6000 \\ 50000-20000 & 30000-10000 & 10000-10000 \end{bmatrix}$$
$$= \begin{bmatrix} 5000 & 10000 & 24000 \\ 30000 & 20000 & 0 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

(iii) $$2\% \text{ of } B = \frac{2}{100} \times B = 0.02 \times B$$

$$= 0.02 \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ \end{matrix}$$
$$\begin{bmatrix} 5000 & 10000 & 6000 \\ 20000 & 10000 & 10000 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

$$= \begin{matrix} \text{Basmati} & \text{Permal} & \text{Naura} \\ \end{matrix}$$
$$\begin{bmatrix} 100 & 200 & 120 \\ 400 & 200 & 200 \end{bmatrix} \begin{matrix} \text{Ramkishan} \\ \text{Gurcharan Singh} \end{matrix}$$

Thus, in October Ramkishan receives ₹100, ₹200 and ₹120 as profit in the sale of each variety of rice, respectively, and Gurcharan Singh receives profit of ₹400, ₹200 and ₹200 in the sale of each variety of rice, respectively.

37. Western music is a form of country music composed by and about the people who settled and worked throughout the Western United States and Western Canada. Western music celebrates the lifestyle of the cowboy on the open ranges, Rocky Mountains, and prairies of Western North America.



Western music is organised every year in the stadium that can hold 36000 spectators. With ticket price of `10, the average attendance has been 24000. Some financial expert estimated that price of a ticket should be determined by the function $p(x) = 15 - \frac{x}{3000}$, where x is the number of ticket sold.

Bases on the above information, answer of the following questions.

(i) Find the expression for total revenue R as a function of x.

(ii) Find the value of x for which revenue is maximum.

(iii) When the revenue is maximum, what will be the price of the ticket?

OR

(iv) How many spectators should be present to maximum the revenue?

Sol :

(i) Let p be the price per ticket and x be the number of tickets sold.

Then, revenue function

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$$= -\sin t + \operatorname{cosec} t \quad \ldots(ii)$$

Now, $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\cos t}{\operatorname{cosec} t - \sin t}$

[using Eqs. (i) and (ii)]

$$= \frac{\cos t}{1 - \sin^2 t} \cdot \sin t$$

$$= \frac{\sin t \cdot \cos t}{\cos^2 t}$$

$$\frac{dy}{dx} = \tan t$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

$$= \frac{\frac{d}{dx}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

$$= \frac{\sec^2 t}{\operatorname{cosec} t - \sin t}$$

$$= \frac{\sec^2 t \cdot \sin t}{\cos^2 t}$$

$$= \sec^3 t \cdot \tan t$$

$$\left[\frac{d^2y}{dx^2}\right]_{t=\frac{\pi}{4}} = \sec^3 \frac{\pi}{4} \cdot \tan \frac{\pi}{4}$$

$$= 2\sqrt{2} \times 1 = 2\sqrt{2}$$

OR

Given,

$$f(x) = \begin{cases} x^2 + 3x + a, & x \le 1 \\ bx + 2, & x > 1 \end{cases}$$ is differentiable at $x = 1$.

$$Lf'(1) = Rf'(1) \quad \ldots(i)$$

Here, $Lf'(1) = \lim_{h \to 0} \frac{f(1-h)^2 - f(1)}{-h}$

$$= \lim_{h \to 0} \frac{(1-h)^2 + 3(1-h) + a - (4+a)}{-h}$$

$$= \lim_{h \to 0} \frac{1 + h^2 - 2h + 3 - 3h + a - 4 - a}{-h}$$

$$= \lim_{h \to 0} \frac{h^2 - 5h}{-h}$$

and $Rf'(1) = \lim_{h \to 0} \frac{f(1+h) - f(1)}{h}$

$$= \lim_{h \to 0} \frac{b(1+h) + 2 - (4+a)}{h}$$

$$= \lim_{h \to 0} \frac{b + bh + 2 - 4 - a}{h}$$

$$= \lim_{h \to 0} \frac{bh + b - a - 2}{h}$$

Clearly, for $Rf'(1)$ to be exist $b - a - 2$ should be equal to 0, i.e.

$$b - a - 2 = 0 \quad \ldots(ii)$$

Now, $Rf'(1) = \lim_{h \to 0} \frac{bh}{h}$

$$= \lim_{h \to 0} b = b$$

From Eq. (i), we have

$$Lf'(1) = Rf'(1)$$

$$5 = b$$

$$b = 5$$

Now, on substituting $b = 5$ in Eq. (ii), we get

$$5 - a - 2 = 0$$

$$a = 3$$

Hence, $a = 3$ and $b = 5$.

34. Solve the following L.P. Problem graphically:

Maximise $Z = x + y$

Subject to $x - y \le -1$

$-x + y \le 0$

$x, y \ge 0$

Sol :

Maximise $Z = x + y \quad \ldots(1)$

subject to $x - y \le -1 \quad \ldots(2)$

$-x + y \le 0 \quad \ldots(3)$

$x, y \ge 0 \quad \ldots(4)$

Step I

Constraint Eq. (4) $x, y \ge 0$

$\Rightarrow$ Feasible region is in first quadrant.

Table of vales for the line $x - y = -1$ of constraint Eq. (2)

x	0	-1
y	1	0

Let us draw the straight line joining the points (0, 1) and $(-1, 0)$.

Let us test for origin (0,0) in constraint Eq. (2) $x - y \le -1$, we have $0 \le -1$ which is not true.

Therefore region for constrain Eq. (2) is the region on that side of the line which is away from the origin (as shown shaded in the figure)

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In ΔBCD,

$$\overrightarrow{BC} + \overrightarrow{CD} = \overrightarrow{BD} \quad \ldots(2)$$

From (1) and (2), we have

$$\overrightarrow{BA} + \overrightarrow{AC} = \overrightarrow{BD} - \overrightarrow{CD}$$

$$\overrightarrow{BA} + \overrightarrow{CD} = \overrightarrow{BD} - \overrightarrow{AC}$$

$$= \overrightarrow{BD} + \overrightarrow{CA}$$

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Sample Paper 13

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

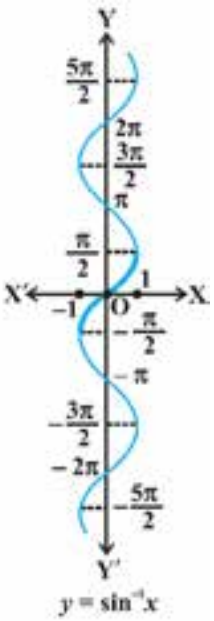
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the value of y when $x = 1$ in the graph of $y = \cos^{-1} x$?



(a) 0 (b) $\pi/2$

(c) 1 (d) π

Sol :

The graph shows that at $x = 1$, the curve touches the origin point on the X-axis. This means $y = \cos^{-1}(1) = 0$. Therefore, the lowest point of the graph occurs at this coordinate.

Thus option (a) is correct.

2. If $\int_0^a f(2a - x)\,dx = m$ and $\int_0^a f(x)\,dx = n$, then $\int_0^{2a} f(x)\,dx$ is equal to

(a) $2m + n$

(b) $m + 2n$

(c) $m - n$

(d) $m + n$

Sol :

$$\int_0^{2a} f(x)\,dx = \int_0^a \{f(2a - x) + f(x)\}\,dx$$

$$= \int_0^a f(2a - x)\,dx + \int_0^a f(x)\,dx = m + n$$

Thus option (d) is correct.

3. The area bounded by $y = |\sin x|$, X-axis and the lines $|x| = \pi$ is

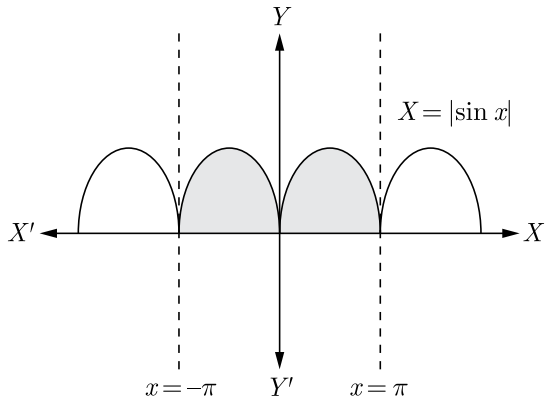
(a) 2 sq units (b) 3 sq units

(c) 4 sq units (d) None of these

Sol :

Required area $= 2\int_0^{\pi} \sin x\,dx$

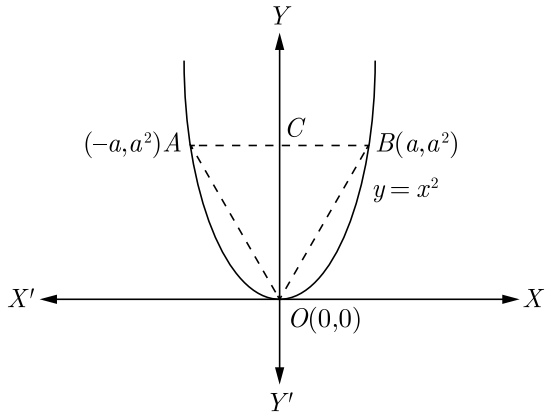
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$= 2[-\cos x]_0^{\pi} = 2[1+1] = 4$ sq unit

Thus option (c) is correct.

4. The given figure shows a ΔAOB and the parabola $y = x^2$. The ratio of the area of the ΔAOB to the area of the region AOB of the parabola $y = x^2$ is equal to

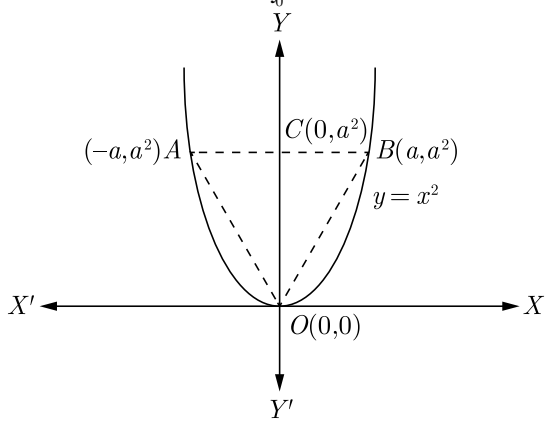


(a) $\frac{3}{5}$ (b) $\frac{3}{4}$

(c) $\frac{7}{8}$ (d) $\frac{5}{6}$

Sol :

Area of curve $AOB = 2\int_0^{a^2} x\,dy$



$= 2\int_0^{a^2} \sqrt{y}\,dy$

$= 2\left[\frac{y^{3/2}}{3/2}\right]_0^{a^2}$

$= \frac{4}{3}[a^3]$

Now, area of $\Delta AOB = \frac{1}{2} \times AB \times OC$

$= \frac{1}{2} \times 2a \times a^2 = a^3$

$\frac{\text{Area of } \Delta AOB}{\text{Area of curve } AOB} = \frac{a^3}{\frac{4}{3}a^3} = \frac{3}{4}$

Thus option (b) is correct.

5. The angle between the lines $x = 1$, $y = 2$ and $y = -1$, $z = 0$ is

(a) 30° (b) 60°

(c) 90° (d) 0°

Sol :

Given lines are

$\frac{x-1}{0} = \frac{y-2}{0} = \frac{z}{1}$ and

$\frac{x}{1} = \frac{y+1}{0} = \frac{z}{0}$

$\cos\theta = 0 \cdot 1 + 0 \cdot 0 + 1 \cdot 0$

$= 0 \Rightarrow \theta = 90^\circ$

Thus option (c) is correct.

6. Order of the equation $\left(1 + 5\frac{dy}{dx}\right)^{3/2} = 10\frac{d^3y}{dx^3}$ is

(a) 2 (b) 3

(c) 1 (d) 0

Sol :

Given,

$\left(1 + 5\frac{dy}{dx}\right)^{3/2} = 10\frac{d^3y}{dx^3}$

On squaring both sides, we get

$\left(1 + 5\frac{dy}{dx}\right)^3 = 100\left(\frac{d^3y}{dx^3}\right)^2$

$1 + 125\left(\frac{dy}{dx}\right)^3 + 15\frac{dy}{dx}\left(1 + 5\frac{dy}{dx}\right) = 100\left(\frac{d^3y}{dx^3}\right)^2$

Clearly, the order of highest derivative occurring in the differential equation is 3. Hence, the order of given differential equation is 3.

Thus option (b) is correct.

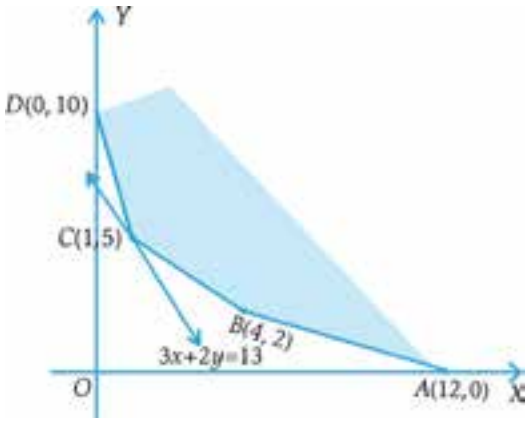
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Sol :

From the left branch of the graph, when x=-1, the curve reaches the value y=-$\pi/2$. This agrees with the definition that $\csc^{-1}(-1) = -\pi/2$.

Thus option (c) is correct.

17. With the same feasible region, find the minimum value of $z = 4x + y$.



(a) 48 (b) 18

(c) 9 (d) 10

Sol :

Evaluate z=4x+y at the vertices. At A(12,0), z=48; at B(4,2), z=18; at C(1,5), z=9; at D(0,10), z=10. The least of these is 9, achieved at C(1,5). Therefore the minimum value is 9.

Thus option (c) is correct.

18. $3a\int_0^1 \left(\frac{ax-1}{a-1}\right)^2 dx$ is equal to

(a) $a-1+(a-1)^{-2}$ (b) $a+a^{-2}$

(c) $a-a^2$ (d) $a^2+\frac{1}{a^2}$

Sol :

$$3a\int_0^1 \left(\frac{ax-1}{a-1}\right)^2 dx = \frac{3a}{(a-1)^2}\left[\frac{(ax-1)^3}{3}\times\frac{1}{a}\right]_0^1$$

$$= \frac{1}{(a-1)^2}[(a-1)^3+1]$$

$$= (a-1)+(a-1)^{-2}$$

Thus option (a) is correct.

19. **Assertion :** The matrix $A = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}$ is an orthogonal matrix.

Reason : If A and B are orthogonal, then AB is also orthogonal.

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

Sol :

As we know that, if A and B are orthogonal matrix, then AB is also orthogonal matrix. So given Reason is true.

given that. $A = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}$

now, $A^1 = \frac{1}{3}\begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ 2 & 2 & -1 \end{bmatrix}$

So, $AA' = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}\cdot\frac{1}{3}\begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ 2 & 2 & -1 \end{bmatrix}$

$$= \frac{1}{9}\begin{bmatrix} 1+4+4 & -2-2+4 & -2+4-2 \\ -2-2+4 & 4+1+4 & 4-2-2 \\ -2+4-2 & 4-2-2 & 4+4+1 \end{bmatrix}$$

$$= \frac{1}{9}\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

So, A is orthogonal matrix.

Hence, Both Assertion and reason are true but reason is not a correct explanation for assertion.

Thus option (b) is correct.

20. **Assertion :** If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, then $\text{adj}(\text{adj}\,A) = A$.

Reason : $|\text{adj}(\text{adj}\,A)| = |A|^{(n-1)^2}$, where A be n rowed non-singular matrix.

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

Sol :

As we know that, $\text{adj}(\text{adj}\,A) = |A|^{n-2}A$

So given Reason is false.

Given that, $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}_{3\times 3}$

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12. For two events A and B, if $P(A)=P\left(\frac{A}{B}\right)=\frac{1}{4}$ and $P\left(\frac{B}{A}\right)=\frac{1}{2}$, then

(a) A and B are independent events

(b) $P\left(\frac{A'}{B}\right)=\frac{3}{4}$

(c) $P\left(\frac{B'}{A}\right)=\frac{1}{2}$

(d) All of the above

Sol :

Given,

$$P\left(\frac{B}{A}\right)=\frac{1}{2}\Rightarrow P(B\cap A)=\frac{1}{8}$$

and $$P\left(\frac{A}{B}\right)=\frac{1}{4}\Rightarrow P(B)=\frac{1}{2}$$

$$P(A\cap B)=\frac{1}{8}=P(A)\cdot P(B)$$

So, events are independent.

Now, $P\left(\frac{A'}{B}\right)=\frac{P(A'\cap B)}{P(B)}=\frac{P(A\cap B)}{P(B)}=\frac{3}{4}$

and $P\left(\frac{B'}{A}\right)=\frac{P(A\cap B')}{P(A)}=\frac{P(A)-P(A\cap B)}{P(A)}=\frac{1}{2}$

Thus option (d) is correct.

13. The projection of $\vec{a}=3\hat{i}-\hat{j}+5\hat{k}$ on $\vec{b}=2\hat{i}+3\hat{j}+\hat{k}$ is

(a) $\frac{8}{\sqrt{35}}$ (b) $\frac{8}{\sqrt{39}}$

(c) $\frac{8}{\sqrt{14}}$ (d) $\sqrt{14}$

Sol :

The projection of $\vec{a}$ on $\vec{b}=\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$

$$=\frac{(3\hat{i}-\hat{j}+5\hat{k})\cdot(2\hat{i}+3\hat{j}+\hat{k})}{\sqrt{2^2+3^2+1^2}}=\frac{8}{\sqrt{4}}$$

Thus option (c) is correct.

14. If $A=\begin{bmatrix}x & 1\\ 1 & 0\end{bmatrix}$ and A^2 is the identity matrix, then x is equal to

(a) -1 (b) 0

(c) 1 (d) 2

Sol :

We have $$A=\begin{bmatrix}x & 1\\ 1 & 0\end{bmatrix}$$

Now, $$A^2=\begin{bmatrix}x & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}x & 1\\ 1 & 0\end{bmatrix}$$

$$=\begin{bmatrix}x^2+1 & x\\ x & 1\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$$

Equating the corresponding elements, we get $x=0$

Thus option (b) is correct.

15. $\int_0^{\pi/2}\frac{\sin x-\cos x}{1+\sin x\cos x}dx$ is equal to

(a) 0 (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{2}$ (d) π

Sol :

We have

$$I=\int_0^{\pi/2}\frac{\sin x-\cos x}{1+\sin x\cos x}dx \quad ...(i)$$

$$I=\int_0^{\pi/2}\frac{\sin\left(\frac{\pi}{2}-x\right)-\cos\left(\frac{\pi}{2}-x\right)}{1+\sin\left(\frac{\pi}{2}-x\right)\cos\left(\frac{\pi}{2}-x\right)}dx$$

$$\left[\because \int_0^a f(x)\,dx=\int_0^a f(a-x)\,dx\right]$$

$$=\int_0^{\pi/2}\frac{\cos x-\sin x}{1+\cos x\sin x}dx \quad ...(ii)$$

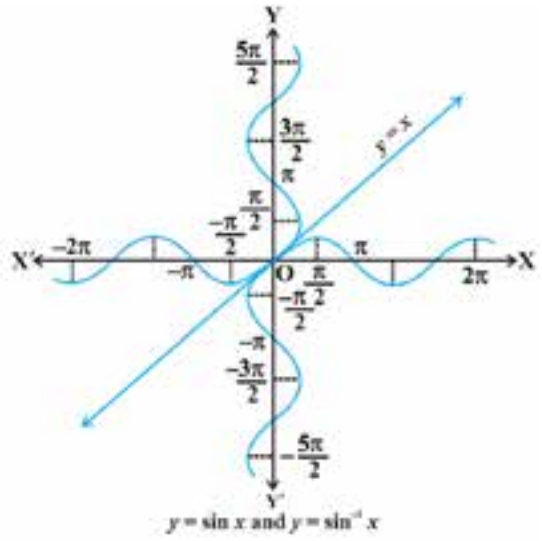
$$\left[\because \sin\left(\frac{\pi}{2}-x\right)=\cos x \text{ and } \cos\left(\frac{\pi}{2}-x\right)=\sin x\right]$$

On adding Eqs. (i) and (ii), we get

$$2I=\int_0^{\pi/2}\frac{0}{1+\sin x\cos x}dx=0\Rightarrow I=0$$

Thus option (a) is correct.

16. What is the value of y when $x=-1$ in the graph of $y=\csc^{-1}x$?



(a) 0 (b) $\pi/2$

(c) $-\pi/2$ (d) $-\pi/2$ or $\pi/2$

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7. If x is real, then minimum value of $x^2-8x+17$ is

(a) -1 (b) 0

(c) 1 (d) 2

Sol :

Let $f(x) = x^2-8x+17$

$f'(x) = 2x-8$

So, $f'(x) = 0$, gives $x=4$

Now, $f''(x) = 2 > 0,\ \forall x$

So, $x=4$ is the point of local minima.
Minimum value of $f(x)$ at $x=4$,

$$f(4) = 4\times 4-8\times 4+17 = 1$$

Thus option (c) is correct.

8. If $P(A)=0.5$, $P(B)=0.4$ and $P(A\cap B)=0.3$, then $P\left(\frac{A'}{B}\right)$ is equal to

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$

(c) $\frac{2}{3}$ (d) $\frac{3}{4}$

Sol :

$$P(A'/B') = \frac{P(A'\cap B')}{P(B')} = \frac{P(A\cup B)'}{P(B')}$$
$$= \frac{1-P(A\cup B)}{1-P(B)}$$
$$= \frac{1-\left[P(A)+P(B)-P(A\cap B)\right]}{1-P(B)}$$
$$= \frac{1-[0.5+0.4-0.3]}{1-0.4} = \frac{0.4}{0.6} = \frac{2}{3}$$

Thus option (c) is correct.

9. If $\hat{a}, \hat{b}$ and $\hat{c}$ are unit vectors such that $\vec{a}+\vec{b}+\vec{c}=0$. Then, which one of the following is correct?

(a) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a} = 0$

(b) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a} \neq 0$

(c) $\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{a}\times\vec{c} = 0$

(d) $\vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}$ are mutually perpendicular.

Sol :

Given, $\vec{a}+\vec{b}+\vec{c} = 0$

Taking cross product of both sides, we get

$$(\vec{a}+\vec{b}+\vec{c})\times\vec{a} = 0\times\vec{a}$$
$$\vec{a}\times\vec{a}+\vec{b}\times\vec{a}+\vec{c}\times\vec{a} = 0$$
$$0+\vec{b}\times\vec{a}+\vec{c}\times\vec{a} = 0$$
$$-\vec{a}\times\vec{b}+\vec{c}\times\vec{a} = 0$$
$$\vec{a}\times\vec{b} = \vec{c}\times\vec{a}$$

Similarly, $\vec{b}\times\vec{c} = \vec{c}\times\vec{a}$

$$\vec{a}\times\vec{b} = \vec{b}\times\vec{c} = \vec{c}\times\vec{a} \neq 0$$

Thus option (b) is correct.

10. The line joining the points $(1,1,2)$ and $(3,-2,1)$ meets the planes $3x+2y+z=6$ at the point

(a) $(1,1,2)$ (b) $(3,-2,1)$

(c) $(2,-3,1)$ (d) $(3,2,1)$

Sol :

The straight line joining the points $(1,1,2)$ and $(3,-2,1)$ is

$$\frac{x-1}{2} = \frac{y-1}{-3} = \frac{z-2}{-1} = r \qquad \text{[say]}$$

Then, the point is $(2r+1, 1-3r, 2-r)$ which lies on

$$3x+2y+z = 6$$
$$3(2r+1)+2(1-3r)+2-r = 6 \Rightarrow r=1$$

So, the required point is $(3,-2,1)$.
Thus option (b) is correct.

11. If $f(x) = \begin{cases} \frac{3\sin\pi x}{5x}, & x\neq 0 \\ 2k, & x=0 \end{cases}$ is continuous at $x=0$, then the value of k is

(a) $\frac{\pi}{10}$ (b) $\frac{3\pi}{10}$

(c) $\frac{3\pi}{2}$ (d) $\frac{3\pi}{5}$

Sol :

Given,

$$f(x) = \begin{cases} \frac{3\sin\pi x}{5x}, & x\neq 0 \\ 2k, & x=0 \end{cases}$$

Now, $\lim_{x\to 0} f(x) = \lim_{x\to 0}\left(\frac{3\sin x}{5x}\right)$

$$= \frac{3}{5}\lim_{x\to 0}\left(\sin\frac{\pi x}{\pi x}\right)\times\pi = \frac{3}{5}\times 1\times\pi = \frac{3}{5}\pi$$

Also, $f(0) = 2k$

Since, $f(x)$ is continuous at $x=0$.

$$f(0) = \lim_{x\to 0} f(x) \Rightarrow 2k = \frac{3}{5}\pi \Rightarrow k = \frac{3\pi}{10}$$

Thus option (b) is correct.

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Here, $n = 3$

So. $\text{adj}(\text{adj A}) = |\text{A}|^{3-2}\text{A}$

$= |\text{A}|\text{A}$

Now, $|\text{A}| = 3(-3+4)+3(2-0)+4(-2)$

$= 3+6-8 = 1$

Now, $\text{adj}(\text{adj A}) = (1)\text{A}$

$= \text{A}$

Hence, Assertion is true; reason is false.

Thus option (c) is correct.

SECTION - B

Section B consists of 5 questions of 2 marks each.

21. If a line has direction ratios $(2,-1,-2)$, then what are its direction cosines?

Sol :

If (a,b,c) are DR's of a line then direction cosines of line is given by

$$l = \pm\frac{a}{\sqrt{a^2+b^2+c^2}}$$

$$m = \pm\frac{b}{\sqrt{a^2+b^2+c^2}}$$

$$n = \pm\frac{c}{\sqrt{a^2+b^2+c^2}}$$

Here DR's of the line are $(2,-1,-2)$.

Now $\sqrt{2^2+(-1)^2+(-2)^2} = \sqrt{9} = 3$

Direction cosines of the give line are $\left(\frac{2}{3},\frac{-1}{3},\frac{-2}{3}\right)$

22. Prove that if E and F are independent events, then the events E and F' are also independent.

Sol :

Since event E and F are independent, we have

$$P(E \cap F) = P(E)\cdot P(F)$$

Now, $P(E \cap F') + P(E \cap F) = P(E)$

$$P(E \cap F') = P(E) - P(E \cap F)$$

$$= P(E) - P(E)P(F)$$

$$= P(E)[1 - P(F)]$$

or $$P(E \cap F') = P(E)P(F')$$

Thus E and F' are also independent events.

23. The total cost $C(x)$ associated with the production of x units of an item is given by $C(x) = 0.005x^3 - 0.02x^2 + 30x + 5000$.

Find the marginal cost when 3 units are produced, where by marginal cost we mean the instantaneous rate of change of total cost at any level of output.

Sol :

We have,

$$C(x) = 0.005x^3 - 0.02x^2 + 30x + 5000$$

Marginal cost,

$$MC(x) = \frac{d}{dx}C(x)$$

$$= \frac{d}{dx}(0.005x^3 - 0.02x^2 + 30x + 5000)$$

$$= 0.005 \times 3x^2 - 0.02 \times 2x + 30 + 0$$

$$= 0.015x^2 - 0.04x + 30$$

When 3 units are produced, Marginal cost

$$MC(3) = 0.015(9) - 0.04(3) + 30$$

$$= 0.135 - 0.12 + 30 = 30.015$$

24. Find the general solution of the differential equation $\sec^2 x \tan y\, dx + \sec^2 y \tan x\, dy = 0$

Sol :

The given differential equation is

$$\sec^2 x \tan y\, dx + \sec^2 y \tan x\, dy = 0$$

$$\sec^2 x \tan y\, dx = -\sec^2 y \tan x\, dy$$

Separating the variables, we get

$$\frac{\sec^2 y}{\tan y}dy = -\frac{\sec^2 x}{\tan x}dx$$

Integrating both sides, we get

$$\int \frac{\sec^2 y}{\tan y}dy = -\int \frac{\sec^2 x}{\tan x}dx$$

$$\log|\tan y| = -\log|\tan x| + \log|C|$$

$$\left[\because \int \frac{f'(x)}{f(x)}dx = \log|f(x)| + C\right]$$

$$\log|\tan y| + \log|\tan x| = \log C$$

$$\log|\tan x \tan y| = \log|C|$$

$$\tan x \tan y = C$$

which is the required general solution.

OR

Find the general solution of the differential equation $\frac{dy}{dx} = e^{x+y}$.

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$$I = \int \frac{2}{(1-t)(1+t^2)}\,dt$$

Now, let

$$\frac{2}{(1-t)(1+t^2)} = \frac{A}{1-t} + \frac{Bt+C}{1+t^2}$$

$$2 = A(1+t^2) + (Bt+C)(1-t) \qquad ...(1)$$

Substituting $t = 1$ in eq. (1), we get

$$2 = A(1+1) + 0 \Rightarrow A = 1 \qquad ...(2)$$

Substituting $t = 0$ in eq. (2), we get

$$2 = A + C \Rightarrow 2 = 1 + C \Rightarrow C = 1$$

Substituting $t = -1$ in eq. (1), we get

$$2 = 2A + (-B + C)(2)$$

$$2 = 2 - 2B + 2$$

$$2B = 2 \Rightarrow B = 1$$

Thus $\frac{2}{(1-t)(1+t^2)} = \frac{1}{1-t} + \frac{t+1}{1+t^2}$

Now $I = 1\int \frac{1}{1-t}\,dt + \int \frac{t+1}{1+t^2}\,dt$

$$= \int \frac{1}{1-t}\,dt + \frac{1}{2}\int \frac{2t}{1+t^2}\,dt + \int \frac{1}{1+t^2}\,dt$$

$$= -\log|1-t| + \frac{1}{2}\log|1+t^2| + \tan^{-1}t + C$$

$$= -\log|1-\sin x| + \frac{1}{2}\log|1+\sin^2 x| + \tan^{-1}(\sin x) + C$$

$$= \tan^{-1}(\sin x) + \log\left|\frac{\sqrt{1+\sin^2 x}}{1-\sin x}\right| + C$$

Here we have used the fact $\log m - \log n = \log\left(\frac{m}{n}\right)$ and $n\log m = \log m^n$

29. Find the value of

$$\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) + \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) + \tan^{-1}\left[\sin\left(\frac{-\pi}{2}\right)\right].$$

Sol :

We have, $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) + \cot^{-1}\left(\frac{1}{\sqrt{3}}\right) + \tan^{-1}\left[\sin\left(\frac{-\pi}{2}\right)\right]$

$$= \tan^{-1}\left(\tan\frac{5\pi}{6}\right) + \cot^{-1}\left(\cot\frac{\pi}{3}\right) + \tan^{-1}(-1)$$

$$= \tan^{-1}\left[\tan\left(\pi - \frac{\pi}{6}\right)\right] + \cot^{-1}\left[\cot\left(\frac{\pi}{3}\right)\right] + \tan^{-1}\left[\tan\left(\pi - \frac{\pi}{4}\right)\right]$$

$$= \tan^{-1}\left(-\tan\frac{\pi}{6}\right) + \cot^{-1}\left(\cot\frac{\pi}{3}\right) + \tan^{-1}\left(-\tan\frac{\pi}{4}\right)$$

$$\left[\because \tan^{-1}(\tan x) = x, x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)\right.$$
$$\left.\cot^{-1}(\cot x) = x, x \in (0,\pi)\right.$$
$$\left.\text{and } \tan^{-1}(-x) = -\tan^{-1}x\right]$$

$$= -\frac{\pi}{6} + \frac{\pi}{3} - \frac{\pi}{4} = \frac{-2\pi + 4\pi - 3\pi}{12}$$

$$= \frac{-5\pi + 4\pi}{12} = -\frac{\pi}{12}$$

30. Find the vector and Cartesian equations of the line through the point $(1,2,-4)$ and perpendicular to the lines

$$\vec{r} = (8\hat{i} - 19\hat{j} + 10\hat{k}) + \lambda(3\hat{i} - 16\hat{j} + 7\hat{k})$$

and $\vec{r} = (15\hat{i} + 29\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})$.

Sol :

We have $\vec{r} = (8\hat{i} - 19\hat{j} + 10\hat{k}) + \lambda(3\hat{i} - 16\hat{j} + 7\hat{k})$

and $\vec{r} = (15\hat{i} + 29\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})$

Comparing with $\vec{r} = \vec{a} + \lambda\vec{b}$ we get

$$\vec{b}_1 = 3\hat{i} - 16\hat{j} + 7\hat{k}$$

and $\vec{b}_2 = 3\hat{i} + 8\hat{j} - 5\hat{k}$

Now $\vec{b} = \vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -16 & 7 \\ 3 & 8 & -5 \end{vmatrix}$

$$= \hat{i}(80 - 56) - \hat{j}(-15 - 21) + \hat{k}(24 + 48)$$

$$= 24\hat{i} + 36\hat{j} + 72\hat{k} = 12(2\hat{i} + 3\hat{j} + 6\hat{k})$$

Since, the required line is perpendicular to the given lines. So, it is parallel to $\vec{b}_1 \times \vec{b}_2$. Now, the equation of a line passing through the point $(1,2,-4)$ and parallel to $24\hat{i} + 36\hat{j} + 72\hat{k}$ or $(2\hat{i} + 3\hat{j} + 6\hat{k})$ is

$$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$$

which is required vector equation of a line.

For Cartesian equation, substituting $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, we get

$$x\hat{i} + y\hat{j} + z\hat{k} = (1 + 2\lambda)\hat{i} + (2 + 3\lambda)\hat{j} + (-4 + 6\lambda)\hat{k}$$

Comparing the coefficients of $\hat{i}$, $\hat{j}$ and $\hat{k}$, we get

$$x = 1 + 2\lambda,\ y = 2 + 3\lambda$$

and $z = -4 + 6\lambda$

Thus $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$

which is the required Cartesian equation of a line.

ALTERNATIVE METHOD :

Let the equation of line passing through $(1,2,-4)$ is

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We have $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$

$$A^2 = A \times A$$

$$= \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix} \times \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 4+0+1 & 0+0-1 & 2+0+0 \\ 4+2+3 & 0+1-3 & 2+3+0 \\ 2-2+0 & 0-1+0 & 1-3-0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix}$$

Now,

$$A^2 - 5A = \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix} - 5\begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & -1 & 2 \\ 9 & -2 & 5 \\ 0 & -1 & -2 \end{bmatrix} - \begin{bmatrix} 10 & 0 & 5 \\ 10 & 5 & 15 \\ 5 & -5 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 5-15 & -1-0 & 2-5 \\ 9-10 & -2-5 & 5-15 \\ 0-5 & -1+5 & -2-0 \end{bmatrix} = \begin{bmatrix} -5 & -1 & -3 \\ -1 & -7 & -10 \\ -5 & 4 & -2 \end{bmatrix}$$

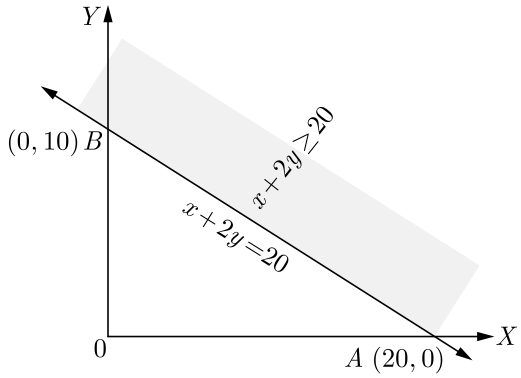
27. Solve graphically the following system of in equations.

$x + 2y \geq 20,\ 3x + y \leq 15$

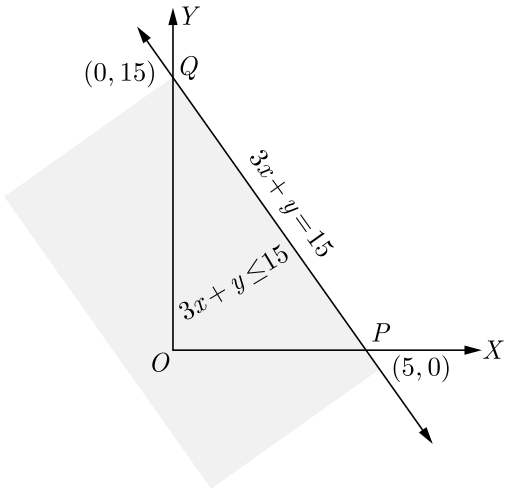
Sol :

First we convert the inequalities $x + 2y \geq 20$ and $3x + y \leq 15$ into equations $x + 2y = 20$ and $3x + y = 15$ respectively.

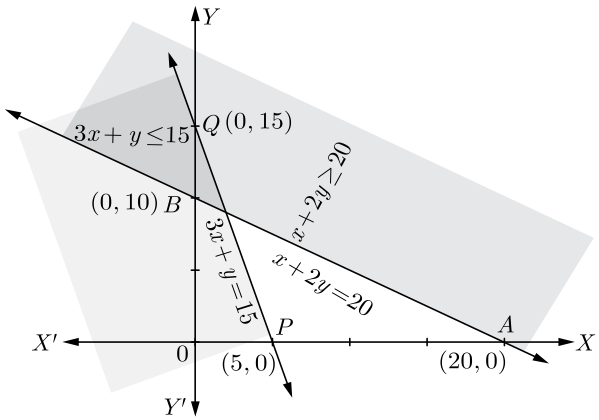
Region Represented by $x + 2y \geq 20$: The line $x + 2y = 20$ meets the axes at $A(20,0)$ and $B(0,10)$. Draw a thick line joining A and B. We find that $(0,0)$ does not satisfy the inequality $x + 2y \geq 20$. So, the portion, which does not contain the origin, represents the solution set of the inequality $x + 2y \geq 20$.



Region Represented by $3x + y \leq 15$: The line $3x + y = 15$ meets the axes at $P(5,0)$ and $Q(0,15)$. Draw a thick line joining P and Q. We find that $(0,0)$ satisfies the inequality $3x + y \leq 15$. So, the portion containing the origin, represents the solution set of the inequality $3x + y \leq 15$.



Thus, the double shaded region, common to the above two shaded regions, represented by the given inequalities, is the required solution region of the given system of inequalities. Hence, all the points in this common (double) shaded region are the solutions of the given system of linear inequalities.



28. Find $\int \frac{2\cos x}{(1-\sin x)(2-\cos^2 x)}\,dx$.

Sol :

We have $I = \int \frac{2\cos x}{(1-\sin x)(2-\cos^2 x)}\,dx$

$$= \int \frac{2\cos x}{(1-\sin x)(1+\sin^2 x)}\,dx$$

Substituting $\sin x = t \Rightarrow \cos x\,dx = dt$ we have

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Sol :

We have $\frac{dy}{dx} = e^{x+y} \Rightarrow \frac{dy}{dx} = e^x \cdot e^y$

$$dy = e^x \cdot e^y \, dx$$

$$e^{-y} \, dy = e^x \, dx$$

$$\int e^{-y} \, dy = \int e^x \, dx$$

$$-e^{-y} = e^x + C$$

which is the required solution.

25. Minimize $Z = 3x + 5y$ such that $x + 3y \geq 3$, $x + y \geq 2$, $x, y \geq 0$.

Sol :

Minimize $Z = 3x + 5y$

Subject to $x + 3y \geq 3$ (i)

$x + y \geq 2$ (ii)

and $x \geq 0, \; y \geq 0$

(i) Region corresponding to $x + 3y \geqslant 3$:

Line $x + 3y = 3$			
x	0	3	Point (0,0) is false for $x + 3y \geqslant 3$. So, the region is away from the origin.
y	1	0	

(ii) Region corresponding to $x + y \geqslant 2$:

Line $x + y = 2$			
x	0	2	Point (0,0) is false for $x + y \geqslant 2$. So, the region is away from the origin.
y	2	0	

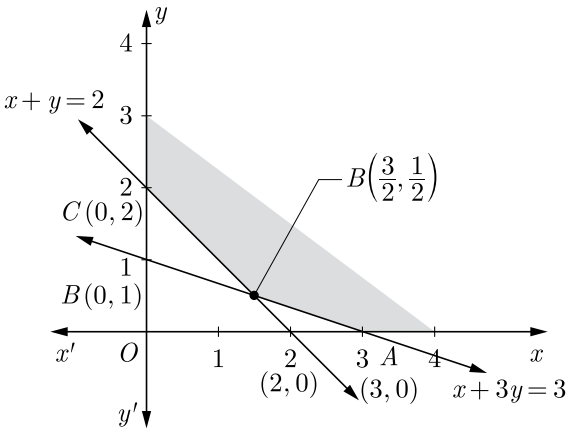
Since $x \geq 0$, $y \geq 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is ABC which is open and unbounded region. Solving equations $x + 3y = 3$ and $x + y = 2$, we get $B(\frac{3}{2}, \frac{1}{2})$. The corner points of the feasible region are $A(3, 0)$, $B(\frac{3}{2}, \frac{1}{2})$ and $C(0, 2)$.

The values of objective function Z at these points are as follows.

Corner points	$Z = 3x + 5y$
$A(3, 0)$	$Z = 3 \times 3 + 0 = 9$
$B(\frac{3}{2}, \frac{1}{2})$	$Z = 3 \times \frac{3}{2} + 5 \times \frac{1}{2} = 7$
$C(0, 2)$	$Z = 3 \times 0 + 5 \times 2 = 10$

Hence the minimum value of Z is 7 at $B(\frac{3}{2}, \frac{1}{2})$.



SECTION - C

Section C consists of 6 questions of 3 marks each.

26. Find a matrix A such that $2A - 3B + 5C = O$, where $B = \begin{bmatrix} -2 & 2 & 0 \\ 3 & 1 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} 2 & 0 & -2 \\ 7 & 1 & 6 \end{bmatrix}$.

Sol :

We have

$$2A - 3B + 5C = O$$

$$2A = 3B - 5C$$

$$2A = 3\begin{bmatrix} -2 & 2 & 0 \\ 3 & 1 & 4 \end{bmatrix} - 5\begin{bmatrix} 2 & 0 & -2 \\ 7 & 1 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} -6 & 6 & 0 \\ 9 & 3 & 12 \end{bmatrix} - \begin{bmatrix} 10 & 0 & -10 \\ 35 & 5 & 30 \end{bmatrix}$$

$$= \begin{bmatrix} -6-10 & 6-0 & 0-(-10) \\ 9-35 & 3-5 & 12-30 \end{bmatrix}$$

$$2A = \begin{bmatrix} -16 & 6 & 10 \\ -26 & -2 & -18 \end{bmatrix}$$

$$A = \frac{1}{2}\begin{bmatrix} -16 & 6 & 10 \\ -26 & -2 & -18 \end{bmatrix}$$

$$= \begin{bmatrix} -8 & 3 & 5 \\ -13 & -1 & -9 \end{bmatrix}$$

OR

If $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$, then find the values of $(A^2 - 5A)$.

Sol :

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$$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda_1(b_1\hat{i} + b_2\hat{j} + b_3\hat{k}) \quad ...(1)$$

Since, the line (1) is perpendicular to the given lines

$$\vec{r} = (8\hat{i} - 19\hat{j} + 10\hat{k}) + \lambda(3\hat{i} - 16\hat{j} + 7\hat{k})$$

and $$\vec{r} = (15\hat{i} + 29\hat{j} + 5\hat{k}) + \mu(3\hat{i} + 8\hat{j} - 5\hat{k})$$

Therefore, we have

$$(b_1\hat{i} + b_2\hat{j} + b_3\hat{k}) \cdot (3\hat{i} - 16\hat{j} + 7\hat{k}) = 0$$

$$3b_1 - 16b_2 + 7b_3 = 0 \quad ...(2)$$

and $$(b_1\hat{i} + b_2\hat{j} + b_3\hat{k}) \cdot (3\hat{i} + 8\hat{j} - 5\hat{k}) = 0$$

$$3b_1 + 8b_2 - 5b_3 = 0 \quad ...(3)$$

Since if two lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \lambda\vec{b}_2$ are perpendicular, then $\vec{b}_1 \cdot \vec{b}_2 = 0$.

Solving Eqs. (2) and (3), we obtain

$$\frac{b_1}{80 - 56} = \frac{b_2}{21 + 15} = \frac{b_3}{24 + 48}$$

$$\frac{b_1}{24} = \frac{b_2}{36} = \frac{b_3}{72}$$

$$\frac{b_1}{2} = \frac{b_2}{3} = \frac{b_3}{6}$$

$b_1 = 2k$, $b_2 = 3k$ and $b_3 = 6k$, for some constant k.

Thus, the required vector equation of line is

$$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k}),$$

where $\lambda = \lambda_1 k$ is any constant.

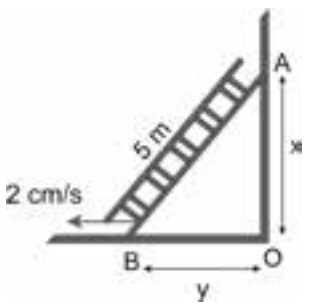
Now, for Cartesian equation do same as in above method.

31. A ladder 5 m is leaning against a wall .The bottom of the ladder is pulled along the ground anyway from the wall, at the rate 2 cm/s. How fast is its height on the wall decreasing, when the foot of the ladder is 4 meter away from the wall?

Sol :

Let the bottom of the ladder is x m away from the wall and upper end be at y m on the wall such that $BC = x$ m, $AB = y$ m and $AC = 5$ m

$$\frac{dx}{dt} = 2\,\text{cm/s} = 0.02\,\text{m/s} \text{ (given)}$$



$$BC^2 + AB^2 = AC^2$$

$$x^2 + y^2 = 25 \quad ...(1)$$

Differentiating both sides, w.r.t. x, we get

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

$$x\frac{dx}{dt} + y\frac{dy}{dt} = 0$$

$$x\frac{dx}{dt} + \sqrt{25 - x^2}\frac{dy}{dt} = 0 \quad [\because x^2 + y^2 = 25]$$

$$4 \times 0.02 + \sqrt{25 - (4)^2}\frac{dy}{dt} = 0 \quad [\because x = 4\,\text{cm}]$$

$$.08 + \sqrt{25 - 16}\frac{dy}{dt} = 0$$

$$.08 + 3\frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = -\frac{0.08}{3}\ \text{m/s}$$

$$\frac{dy}{dt} = -\frac{8}{3}\ \text{cm/s}$$

$(-)$ sign shows that the heigh on the wall is decreasing.

Hence, the rate of decrease is height on the wall $= \frac{8}{3}$ cm/s.

OR

A balloon, which always remains spherical on inflation, is being inflated by pumping is 900 cubic centimetres of gas per second. Find the rate at which the radius of the balloon increases when the radius is 15 cm.

Sol :

Let r be the radius and V be the volume of a spherical balloon at any time t.

$$\frac{dV}{dt} = 900\ \text{cm}^3/\text{s} \quad \text{(given)}$$

$$V = \frac{4}{3}\pi r^3$$

Differentiating both sides, w.r.t. t, we get

$$\frac{dV}{dt} = \frac{4}{3} \cdot 3\pi r^2 \cdot \frac{dr}{dt}$$

$$900 = 4\pi \times 15 \times 15 \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{900}{4\pi \times 15 \times 15}$$

$$= \frac{1}{\pi}\ \text{cm/s}$$

Hence, the required rate of change $= \frac{1}{\pi}$ cm/s.

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while indirect tax includes goods and service tax i.e. GST and any local tax.



A company earns before tax profits of ₹100000. It is committed to making a donation to the Red Cross 10% of its after-tax profits. The Central Government levies income taxes of 50% of profits after deducing charitable donations and any local taxes. The company must also pay local taxes of 10% of its profit less the donation to the Red Cross.

(i) Taking the suitable variable form the system of equation that represent given problem.

(ii) Compute how much the company pays in income taxes, local taxes and as a donation to the Red Cross, using Creamer's Rule.

Sol :

Let the amount donated to the Red Cross be ₹x, the amount paid in income taxes be ₹y and the amount paid in local taxes be ₹z, respectively.

Donation to the Red Cross is 10% of its after-tax profits, thus

$$x = 10\% \text{ of } (100000 - y - z)$$

$$x = 0.10(100000 - y - z)$$

$$10x + y + z = 100000 \quad (1)$$

The central government levies income taxes of 50% of profits after deducing charitable donations and any local taxes. Thus

$$y = 50\% \text{ of } (100000 - x - z)$$

$$y = 0.5(100000 - x - z)$$

$$x + 2y + z = 100000 \quad (2)$$

The company must also pay local taxes of 10% of its profit less the donation to the Red Cross. Thus

$$z = 10\% \text{ of } (100000 - x)$$

$$z = 0.10(100000 - x)$$

$$x + 0y + 10z = 100000 \quad (3)$$

Rewriting the system of equations we have

$$10x + y + z = 100000 \quad (1)$$

$$x + 2y + z = 100000 \quad (2)$$

$$x + 0y + 10z = 100000 \quad (3)$$

Here, $$D = \begin{vmatrix} 10 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 0 & 10 \end{vmatrix}$$

$$= 10(20 - 0) - 1(10 - 1) + 1(0 - 2)$$

$$= 200 - 9 - 2 = 189 \neq 0$$

Since $D \neq 0$, then the system of equations has a unique solution by Cramer's Rule and is given by

$$x = \frac{D_1}{D};\ y = \frac{D_2}{D};\ z = \frac{D_3}{D}$$

Now, $$D_1 = \begin{vmatrix} 100000 & 1 & 1 \\ 100000 & 2 & 1 \\ 100000 & 0 & 10 \end{vmatrix}$$

$$= 100000 \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 0 & 10 \end{vmatrix}$$

$$= 100000[20 - 9 - 2] = 900000$$

$$D_2 = \begin{vmatrix} 10 & 100000 & 1 \\ 1 & 100000 & 1 \\ 1 & 100000 & 10 \end{vmatrix} = 100000 \begin{vmatrix} 10 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 10 \end{vmatrix}$$

$$= 100000[90 - 9 + 0] = 8100000$$

$$D_3 = \begin{vmatrix} 10 & 1 & 100000 \\ 1 & 2 & 100000 \\ 1 & 0 & 100000 \end{vmatrix}$$

$$= 100000[20 - 0 - 12] = 1800000$$

$$x = \frac{D_1}{D} = \frac{900000}{189} = \frac{100000}{21}$$

$$y = \frac{D_2}{D} = \frac{8100000}{189} = \frac{300000}{7}$$

$$z = \frac{D_3}{D} = \frac{1800000}{189} = \frac{200000}{21}$$

Hence the company pay donation to the Red Cross ₹$\frac{100000}{21}$, in income taxes ₹$\frac{300000}{7}$ and in local tax ₹$\frac{200000}{21}$ respectively.

37. Chemical reaction, a process in which one or more substances, the reactants, are converted to one or more different substances, the products. Substances are either chemical elements or compounds. A chemical reaction rearranges the constituent atoms of the reactants to create different substances as products.

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$\Rightarrow |a-c|$ is divisible by 4 $\therefore (a,c) \in R$

$\Rightarrow R$ is transitive

Hence R is an equivalence relations in A

Set of elements related to 1 is $(1,5,9)$

and $[2] = \{2,6,10\}$

Detailed Answer :

Given $R = \{(a,b): a,b \in A, |a,b| \text{ is ivied by } 4\}$

Where $A = \{x \in Z: 0 \le x \le 12\}$

$= \{1,2,3,4,5,6,7,8,9,10,11,12\}$

Reflexivity : For any $a \in A$, e have $|a-a| = 0$ is divisible by 4.

OR

Let $A = R - \{2\}$, $B = R - \{1\}$. If $f: A \to B$ is a function defined by $f(x) = \left(\frac{x-1}{x-2}\right)$, show that f is one-one and onto. Hence find f^{-1}

Sol :

Given $f(x) = \frac{x-1}{x-2}$

To show f is one-one :

Let $f(x_1) = f(x_2)$,

then $\frac{x_1-1}{x_1-2} = \frac{x_2-1}{x_2-2}$

or $(x_1-1)(x_2-2) = (x_1-2)(x_2-1)$

or $x_1x_2 - 2x_1 - x_2 + 2 = x_1x_2 - x_1 - 2x_2 + 2$

or $-2x_1 - x_2 = -x_1 - 2x_2$

or $-2x_1 + x_1 = x_2 - 2x_2$

or $-x_1 = -x_2$

or $x_1 = x_2$

$\therefore f$ is one-one.

To show f is noto :

Let $y \in B$

$\therefore y = f(x)$

or, $y = \frac{x-1}{x-2}$

or, $y(x-2) = x-1$

or, $xy - 2y = x - 1$

or $xy - x = 2y - 1$

or, $x(y-1) = 2y - 1$

or, $x = \frac{2y-1}{y-1} \in B = R - \{1\}$

$\therefore$ For every value of y except $y = 1$, there is a pre image $x = \frac{2y-1}{y-1} = g(y), g(y) \ne 2$.

$f(x) = y$

$x = f^{-1}(y)$

$f^{-1}(y) = \frac{2y-1}{y-1}; y \ne 1, y \in B$

$\therefore f$ is onto.

To find f^{-1}

Since, f is one-one and onto, therefore f is invertible.

Thus $f^{-1}(x) = \frac{2x-1}{x-1} = g(x)$

34. The scalar product of the vector $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ with a unit vector along the sum of vectors $\vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{c} = \lambda\hat{i} + 2\hat{j} + 3\hat{k}$ is equal to 1. Find the value of λ and hence find unit vector along $\vec{b} \times \vec{c}$.

Sol :

We have $\vec{a} = \hat{i} + \hat{j} + \hat{k}$,

$\vec{b} = 2\hat{i} + 4\hat{j} - 5\hat{k}$

and $\vec{c} = \lambda\hat{i} + 2\hat{j} + 3\hat{k}$

Now $\vec{b} + \vec{c} = (2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}$

Let $\hat{r}$ denote the unit vector along $\vec{b} + \vec{c}$.

Then, $\hat{r} = \frac{\vec{b} + \vec{c}}{|\vec{b} + \vec{c}|}$

$= \frac{(2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{(2+\lambda)^2 + 36 + 4}}$

$= \frac{(2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{(2+\lambda)^2 + 40}}$ ...(1)

Now, according to given condition, we have

$(\hat{i} + \hat{j} + \hat{k}) \cdot \hat{r} = 1$

$(\hat{i} + \hat{j} + \hat{k}) \cdot \frac{(2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}}{\sqrt{(2+\lambda)^2 + 40}} = 1$

$(\hat{i} + \hat{j} + \hat{k}) \cdot \{(2+\lambda)\hat{i} + 6\hat{j} - 2\hat{k}\} = \sqrt{(2+\lambda)^2 + 40}$

$2 + \lambda + 6 - 2 = \sqrt{(2+\lambda)^2 + 40}$

$(\lambda + 6)^2 = (2+\lambda)^2 + 40$

$8\lambda = 8 \Rightarrow \lambda = 1$

Substituting $\lambda = 1$ in eq (1), we get

$\hat{r} = \frac{1}{7}(3\hat{i} + 6\hat{j} - 2\hat{k})$

OR

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If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three mutually perpendicular vectors of the same magnitude, then prove that $\vec{a}+\vec{b}+\vec{c}$ is equally inclined with the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$.

Sol :

If three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are mutually perpendicular to each other, then $\vec{a}\cdot\vec{b}=\vec{b}\cdot\vec{c}=\vec{c}\cdot\vec{a}=0$ and if all three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are equally inclined with the vector $(\vec{a}+\vec{b}+\vec{c})$, that means each vector $\vec{a}$, $\vec{b}$ and $\vec{c}$ makes equal angle with $(\vec{a}+\vec{b}+\vec{c})$ by using formula $\cos\theta=\dfrac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$

We have $|\vec{a}|=|\vec{b}|=|\vec{c}|=\lambda$ (say) ...(1)

and $\vec{a}\cdot\vec{b}=0,\ \vec{b}\cdot\vec{c}=0$ and $\vec{c}\cdot\vec{a}=0$ ...(2)

Now,

$$|\vec{a}+\vec{b}+\vec{c}|^2=|\vec{a}|^2+|\vec{b}|^2+|\vec{c}|^2+2(\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{c}+\vec{c}\cdot\vec{a})$$
$$=\lambda^2+\lambda^2+\lambda^2+2(0+0+0)$$
$$=3\lambda^2$$
$$|\vec{a}+\vec{b}+\vec{c}|=\sqrt{3}\lambda \text{ as length cannot be negative.}$$

Suppose $(\vec{a}+\vec{b}+\vec{c})$ is inclined at angles θ_1, θ_2 and θ_3 respectively with vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$, then

$$(\vec{a}+\vec{b}+\vec{c})\cdot\vec{a}=|\vec{a}+\vec{b}+\vec{c}||\vec{a}|\cos\theta_1$$

$\vec{a}\cdot\vec{b}=|\vec{a}||\vec{b}|\cos\theta$

$$|\vec{a}|^2+\vec{a}\cdot\vec{b}+\vec{a}\cdot\vec{c}=\sqrt{3}\lambda\times\lambda\cos\theta_1$$
$$\lambda^2+0+0=\sqrt{3}\lambda^2\cos\theta_1 \quad \text{from eq (1) and 2)}$$
$$(\vec{a}\cdot\vec{b}\cdot\vec{c})\cdot\vec{b}=|\vec{a}+\vec{b}+\vec{c}||\vec{b}|\cos\theta_2$$
$$\vec{a}\cdot\vec{b}+|\vec{b}|^2+\vec{c}\cdot\vec{b}=\sqrt{3}\lambda\cdot\lambda\cos\theta_2$$
$$0+\lambda^2+0=\sqrt{3}\lambda^2\cos\theta_2 \quad \text{from eq (1) and (2)}$$
$$\cos\theta_2=\frac{1}{\sqrt{3}}$$

Similarly,

$$(\vec{a}+\vec{b}+\vec{c})\cdot\vec{c}=|\vec{a}+\vec{b}+\vec{c}||\vec{c}|\cos\theta_3$$
$$\cos\theta_1=\frac{1}{\sqrt{3}}$$

Thus, $\cos\theta_1=\cos\theta_2=\cos\theta_3=\dfrac{1}{\sqrt{3}}$

Hence, it is proved that $(\vec{a}+\vec{b}+\vec{c})$ is equally inclined with the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$.

35. Using integration, find the area of the following region.
$\{(x,y)\ :\ |x-1|\le y\le\sqrt{5-x^2}\}$

Sol :

Given region is $\{(x,y)\ :\ |x-1|\le y\le\sqrt{5-x^2}\}$
Above region has two equations

$$y=|x-1| \text{ and } y=\sqrt{5-x^2}$$

Now, $y=|x-1|=\begin{cases}x-1, & \text{if } x-1\ge 0\\ -(x-1), & \text{if } x-1<0\end{cases}$

$$y=\begin{cases}x-1, & \text{if } x\ge 1\\ 1-x, & \text{if } x<1\end{cases}$$

Also, other curve is $y=\sqrt{5-x^2}$
Squaring both sides, we get

$$y^2=5-x^2$$
$$x^2+y^2=5$$

which represents equation of circle with centre $(0,0)$ and radius, $r=\sqrt{5}$. But the actual equation of curve is $y=\sqrt{5-x^2}$ which represents a semi-circle whose centre is $(0,0)$ and radius $r=\sqrt{5}$.

Now we have $y=1-x$ ...(1)

$y=x-1$ ...(2)

and $y=\sqrt{5-x^2}$ ...(3)

Substituting $y=1-x$ from Eq. (1) in Eq. (3), we get

$$(1-x)=\sqrt{5-x^2}$$
$$x^2+(1-x)^2=5$$
$$x^2+1+x^2-2x=5$$
$$2x^2-2x-4=0$$
$$x^2-x-2=0$$
$$(x+1)(x-2)=0\ \Rightarrow\ x=-1 \text{ or } 2$$

Now, when $x=-1$, then

$$y=\sqrt{5-x^2}=\sqrt{5-1}=\sqrt{4}\Rightarrow y=2$$

and when $x=2$, we have

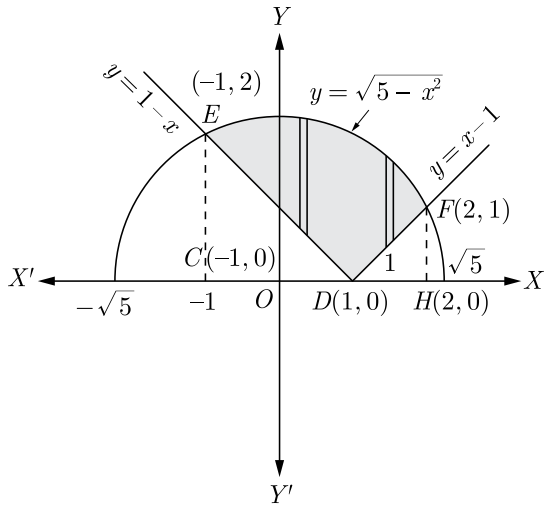
$$y=\sqrt{5-x^2}$$
$$=\sqrt{5-4}=1\Rightarrow y=1$$

So, point of intersection of Eqs. (1) and (2) are $(-1,2)$ and $(2,1)$.
Similarly, solving Eq. (2) and Eqs. (3), we get

$$x=-1 \text{ or } 2$$

From Eq. (3), at $x=-1$, $y=2$ an at $x=2$, $y=1$.
Hence, the two curves intersect at $(-1,2)$ and $(2,1)$.
Now, the given both curve can be sketched shown below.

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Now, required area

$$\text{Area} = \int_{-1}^{2} y_{(circle)}\,dx - \int_{-1}^{1} y_{(line\ DE)}\,dx - \int_{1}^{2} y_{(line\ DF)}\,dx$$

$$= \int_{-1}^{2} \sqrt{5-x^2}\,dx - \int_{-1}^{1}(1-x)\,dx - \int_{1}^{2}(x-1)\,dx$$

$$= \left[\frac{x}{2}\sqrt{5-x^2} + \frac{5}{2}\sin^{-}\frac{x}{\sqrt{5}}\right]_{-1}^{2} - \left[x - \frac{x}{2}\right]_{-1}^{1} - \left[\frac{x^2}{2} - x\right]_{1}^{2}$$

$$= \left[\left(1 + \frac{5}{2}\sin^{-1}\frac{2}{\sqrt{5}}\right) - \left\{-1 + \frac{5}{2}\sin^{-1}\left(-\frac{1}{\sqrt{5}}\right)\right\}\right]$$

$$-\left[\left(1-\frac{1}{2}\right) - \left(-1-\frac{1}{2}\right)\right] - \left[\left(\frac{4}{2}-2\right) - \left(\frac{1}{2}-1\right)\right]$$

$$= 1 + \frac{5}{2}\sin^{-1}\frac{2}{\sqrt{5}} + 1 - \frac{5}{2}\sin^{-1}\left(-\frac{1}{\sqrt{5}}\right)$$

$$-\left(\frac{1}{2}+\frac{3}{2}\right) - \left(0+\frac{1}{2}\right)$$

$$= 1 + \frac{5}{2}\sin^{-1}\frac{2}{\sqrt{5}} + 1 + \frac{5}{2}\sin^{-1}\frac{1}{\sqrt{5}} - 2 - \frac{1}{2}$$

$$= -\frac{1}{2} + \frac{5}{2}\left(\sin^{-1}\frac{2}{\sqrt{5}} + \sin^{-1}\frac{1}{\sqrt{5}}\right)$$

Hence, required area

$$= \left[\frac{5}{2}\left(\sin^{-1}\frac{2}{\sqrt{5}} + \sin^{-1}\frac{1}{\sqrt{5}}\right) - \frac{1}{2}\right] \text{ sq units}$$

OR

Find the area of the smaller region bounded by the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ and the line $\frac{x}{3}+\frac{y}{2}=1$.

Using integration, find the area of the following region. $\left\{(x,y) : \frac{x^2}{9}+\frac{y^2}{4} \le 1 \le \frac{x}{3}+\frac{y}{2}\right\}$

Sol :

Given equation of ellipse is

$$\frac{x^2}{9}+\frac{y^2}{4} = 1 \qquad ...(1)$$

and equation of line is

$$\frac{x}{3}+\frac{y}{2} = 1 \qquad ...(2)$$

Substituting the value of x from Eq. (2) in Eq. (1), we get

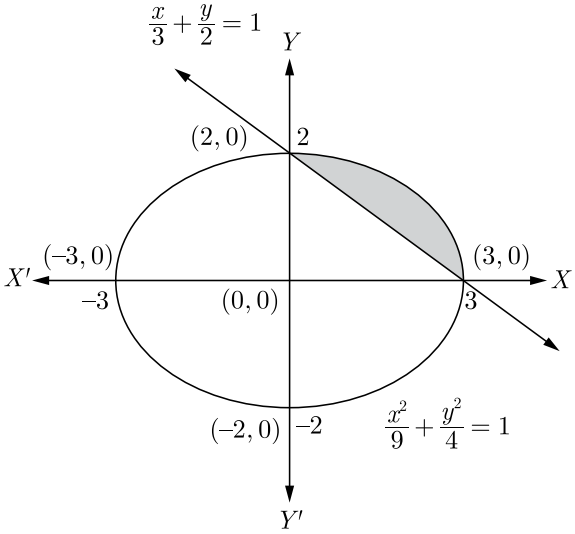
$$\left(1-\frac{y^2}{2}\right)+\frac{y^2}{4} = 1 \Rightarrow 1+\frac{y^2}{4}-y+\frac{y^2}{4}=1$$

$$y^2 - 2y = 0 \Rightarrow y = 0, 2$$

When $y = 0$, then $x = 3$ and point is $A(3,0)$.

When $y = 2$, then $x = 0$ and point is $B(0,2)$.

Now, the Eqs. (1) and (2) can be sketched shown below. Here required area is the area of shaded region.



Clearly, required area

$$\text{Area} = \int_{0}^{3} y_{(\text{ellipse})}\,dx - \int_{0}^{3} y_{(\text{line})}\,dx$$

$$= \int_{0}^{3} 2\sqrt{1-\frac{x^2}{9}}\,dx - \int_{0}^{3} 2\left(1-\frac{x}{3}\right)dx$$

$$= \frac{2}{3}\int_{0}^{3}\sqrt{3^2-x^2}\,dx - \frac{2}{3}\int_{0}^{3}(3-x)\,dx$$

$$= \frac{2}{3}\left[\frac{x}{2}\sqrt{3^2-x^2} + \frac{9}{2}\sin^{-1}\frac{x}{3}\right]_{0}^{3} - \frac{2}{3}\left[3x - \frac{x^2}{2}\right]_{0}^{3}$$

$$= \frac{2}{3}\left[0 + \frac{9}{2}\sin^{-1}(1) - 0\right] - \frac{2}{3}\left[9 - \frac{9}{2} - 0\right]$$

$$= \frac{2}{3}\times\frac{9}{2}\cdot\frac{\pi}{2} - \frac{2}{3}\times\frac{9}{2} = \frac{3}{2}(\pi-2) \text{ sq units}$$

SECTION - E

Case study based questions are compulsory.

36. When it comes to taxes, there are two types of taxes in India - Direct and Indirect tax. The direct tax includes income tax, gift tax, capital gain tax, etc

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SECTION - D

Section D consists of 4 questions of 5 marks each.

32. If $y = x^x$, then prove that $\frac{d^2y}{dx^2} - \frac{1}{y}\left(\frac{dy}{dx}\right)^2 - \frac{y}{x} = 0$.

Sol :

We have $y = x^x$

Taking log both sides, we get

$$\log y = \log x^x$$

$$\log y = x \log x$$

Differentiating w.r.t. x, we get

$$\frac{1}{y}\frac{dy}{dx} = x\frac{d}{dx}(\log x) + \log x \frac{d}{dx}(x)$$

$$\frac{1}{y}\frac{dy}{dx} = x \times \frac{1}{x} + \log x \cdot 1$$

$$\frac{1}{y}\frac{dy}{dx} = (1 + \log x)$$

$$\frac{dy}{dx} = y(1 + \log x) \quad \ldots(i)$$

Again, differentiating both sides w.r.t. x, we get

$$\frac{d^2y}{dx^2} = y\frac{d}{dx}(1 + \log x) + (1 + \log x)\frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = y \times \frac{1}{x} + (1 + \log x)\frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{y}{x} + (1 + \log x)\frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{y}{x} + \frac{1}{y}\left(\frac{dy}{dx}\right)\left(\frac{dy}{dx}\right) \quad \text{[using Eq. (i)]}$$

$$\frac{d^2y}{dx^2} - \frac{1}{y}\left(\frac{dy}{dx}\right)^2 - \frac{y}{x} = 0$$

OR

For what value of λ, the function defined by $f(x) = \begin{cases} \lambda(x^2+2), & \text{if } x \le 0 \\ 4x+6, & x > 0 \end{cases}$ is continuous at $x = 0$? Hence, check the differentiability of $f(x)$ at $x = 0$.

Sol :

Given function is

$$f(x) = \begin{cases} \lambda(x^2+2), & \text{if } x \le 0 \\ 4x+6, & \text{if } x > 0 \end{cases}$$

Let $f(x)$ is continuous at $x = 0$.

Then, LHL = RHL = $f(0)$ ...(i)

Here, $\text{RHL} = \lim_{x \to 0^+} f(x) = \lim_{x \to 0^+}(4x+6)$

$$= \lim_{h \to 0}[4(0+h)+6]$$

[put $x = 0 + h$; when $x \to 0^+$, then $h \to 0$]

$$= \lim_{h \to 0}(4h+6)$$

$$= 4 \times 0 + 6 = 6$$

and $f(0) = \lambda(0^2+2) = 2\lambda$

From Eq. (i), RHL = $f(0)$

$2\lambda = 6 \Rightarrow \lambda = 3$

Now, given function becomes

$$f(x) = \begin{cases} 3(x^2+2), & \text{if } x \le 0 \\ 4x+6, & \text{if } x > 0 \end{cases}$$

Now, let us check the differentiability at $x = 0$.

$$\text{LHD} = \lim_{h \to 0}\frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \to 0}\frac{3[(0-h)^2+2] - 3(0+2)}{-h}$$

$$= \lim_{h \to 0}\frac{3[h^2+2] - 6}{-h} = \lim_{h \to 0}(-3h) = 0$$

and $$\text{RHD} = \lim_{h \to 0}\frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \to 0}\frac{[4(0+h)+6] - 3(0+2)}{h}$$

$$= \lim_{h \to 0}\frac{4h}{h} = 4$$

$\text{LHD} \neq \text{RHD}$

$\therefore$ $f(x)$ is not differentiable at $x = 0$.

33. Let $A = \{x \in Z : 0 \le x \le 12\}$. Show that $R = \{(a, b) : a, b \in A, |a - b| \text{ is divisible by } 4\}$ is an equivalence relation. Find the set of all elements related to 1. Also write the equivalence class [2].

Sol :

Reflexive : $|a - a| = 0$, which is divisible by 4, $\forall\, a \in A$

$\therefore$ $(a, a) \in R$, $a \in A$

$\therefore$ R is reflexive

Symmetric : Let $(a, b) \in R$

$\Rightarrow |a - b|$ is divisible by 4

$\Rightarrow |b - a|$ is divisible by 4 $\quad (\because |a - b| = |b - a|)$

Transitive : Let $(a, b), (b, c) \in R$

$\Rightarrow |a - b| \& |b - c|$ are divisible by 4

$\Rightarrow a - b = \pm 4m,\ b - c = \pm 4n,\ n \in Z$

Adding e get, $a - c = 4(\pm m \pm n)$

$\Rightarrow (a - c)$ is divisible by 4.

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In a certain chemical reaction, a substance is converted into another substance at a rate proportional to the square of the amount of the first substance present at any time t. Initially $(t=0)$ 50 g of the first substance was present; 1 hr later, only 10 g of it remained.

(i) Find an expression that gives the amount of the first substance present at any time t.

(ii) What is the amount present after 2 hr?

Sol :

Since rate is proportional to the square of the amount of the first substance present at any time t, we are led to the differential equation

$$\frac{dQ}{dt} = kQ^2$$

The differential equation is separable. Separating the variables and integrating, we obtain

$$\int \frac{dQ}{Q^2} = \int k\,dt$$

and $$-\frac{1}{Q} = kt + C$$

Therefore, $$Q = -\frac{1}{kt+C}$$

Now, $Q=50$ when $t=0$, therefore, $50 = -\frac{1}{C}$ and $C = -\frac{1}{50}$.

Therefore, $$Q = -\frac{1}{kt-\frac{1}{50}}$$

Since $Q=10$ when $t=1$,

$$10 = -\frac{1}{k-\frac{1}{50}}$$

$$10\left(k-\frac{1}{50}\right) = -1$$

$$10k - \frac{1}{5} = -1$$

$$10k = -1+\frac{1}{5} = -\frac{4}{5}$$

$$k = -\frac{4}{50} = -\frac{2}{25}$$

Therefore, $$Q(t) = \frac{1}{\frac{2}{25}t+\frac{1}{50}}$$

$$= \frac{1}{\frac{4t+1}{50}} = \frac{50}{4t+1}$$

and $$Q(2) = \frac{50}{8+1} \approx 5.56 \text{ grams}$$

38. Joint Entrance Examination – Advanced, is an academic examination held annually in India. It is organised by one of the seven zonal IITs under the guidance of the Joint Admission Board on a round-robin rotation pattern for the qualifying candidates of the JEE-Main. A candidate can attempt JEE (Advanced) maximum of two times in two consecutive years. A successful candidates get the admission in any IITs of India based on merit.



Applicants have a 0.26 probability of passing IIT advanced test when they take it for the first time, and if they pass it they can get admission in IIT. However, if they fail the test the first time, they must take the test a second time, and when applicants take the test for the second time there is a 0.43 chance that they will pass and be allowed to get admission. Applicants are rejected if the test is failed on the second attempt.

(i) What is the probability that an applicant gets admission in IIT but needs two attempts at the test?

(ii) What is the probability that an applicant gets admission in IIT?

(iii) If an applicant gets admission in IIT, what is the probability that he or she passed the test on the first attempt?

Sol :

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Let F be the event that applicant pass the advanced in first time.

Let S be the event that applicant pass the advanced in second time.

Let A be the event that applicant pass the advanced.

(i) Now $P(F) = 0.26$

$P(S) = 0.43$

(i) Probability that an applicant gets admission in IIT but needs two attempts at the test.

$$P(F' \cup S) = P(F') \times P(S)$$
$$= (1 - 0.26) \times 0.43$$
$$= 0.3182$$

(ii) Probability that an applicant gets admission in IIT,

$$P(A) = 1 - P(F' \cup S')$$
$$= 1 - (1 - 0.26) \times (1 - 0.43)$$
$$= 0.5782$$

(iii) If an applicant gets admission in IIT, the probability that he or she passed the test on the first attempt,

$$P(F \mid A) = \frac{P(F)}{P(A)}$$
$$= \frac{0.26}{0.5782} = 0.45$$

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Sample Paper 14

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

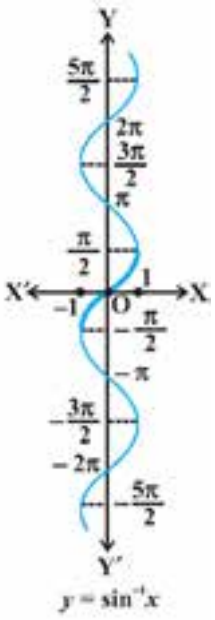
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the value of y when $x = -1$ in the graph of $y = \cos^{-1}x$?



(a) 0 (b) $\pi/2$

(c) π (d) –1

Sol :

On the left side of the X-axis, the graph extends up to $x = -1$. At this point, the Y-value reaches the topmost value shown, which is $y=\pi$. This corresponds to the property $\cos^{-1}(-1) = \pi$.

Thus (c) is correct option.

2. For what values of x, function $f(x) = x^4 - 4x^3 + 4x^2 + 40$ is monotonic decreasing?

(a) $0 < x < 1$ (b) $1 < x < 2$

(c) $2 < x < 3$ (d) $4 < x < 5$

Sol :

We have $f(x) = x^4 - 4x^3 + 4x^2 + 40$

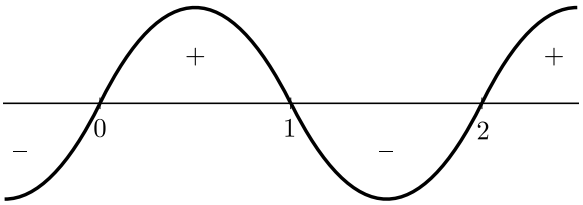
$$f'(x) = 4x^3 - 12x^2 + 8x$$

For monotonic decreasing, $f'(x) < 0$

$$x(4x^2 - 12x + 8) < 0$$

$$x(x^2 - 3x + 2) < 0$$

$$x(x-1)(x-2) < 0$$



$$x \in (-\infty, 0) \cup (1, 2)$$

Thus (b) is correct option.

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3. The solution of $\frac{dy}{dx}=\frac{ax+g}{by+f}$ represents a circle, when

(a) $a=b$ (b) $a=-b$

(c) $a=-2b$ (d) $a=2b$

Sol :

We have, $\frac{dy}{dx}=\frac{ax+g}{by+f}$

$$(by+f)\,dy=(ax+g)\,dx$$

Integrating both sides, we have

$$\frac{by^2}{2}+fy=\frac{ax^2}{2}+gx+C$$

$$ax^2-by^2+2gx-2fy+C=0$$

Which represents a circle, if $a=-b$.

Thus (b) is correct option.

4. If $\int_0^a f(2a-x)\,dx=m$ and $\int_0^a f(x)\,dx=n$, then $\int_0^{2a} f(x)\,dx$ is equal to

(a) $2m+n$ (b) $m+2n$

(c) $m-n$ (d) $m+n$

Sol :

$$\int_0^{2a} f(x)\,dx=\int_0^a\{f(2a-x)+f(x)\}\,dx$$

$$=\int_0^a f(2a-x)\,dx+\int_0^a f(x)\,dx=m+n$$

Thus (d) is correct option.

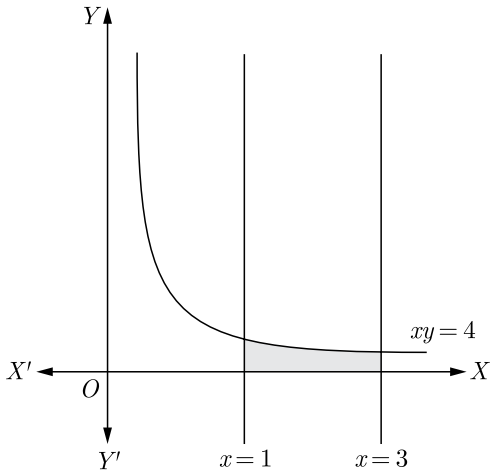
5. Find the area of a curve $xy=4$, bounded by the lines $x=1$ and $x=3$ and X-axis.

(a) $\log 12$ (b) $\log 64$

(c) $\log 81$ (d) $\log 27$

Sol :

Given curve is $xy=4$.



$$\text{Required area}=\int_1^3\frac{4}{x}\,dx=4\cdot[\log x]_1^3$$

$$=4(\log 3-\log 1)$$

$$=4\log 3=\log 81$$

Thus (c) is correct option.

6. Order of the equation $\left(1+5\frac{dy}{dx}\right)^{3/2}=10\frac{d^3y}{dx^3}$ is

(a) 2 (b) 3

(c) 1 (d) 0

Sol :

Given,

$$\left(1+5\frac{dy}{dx}\right)^{3/2}=10\frac{d^3y}{dx^3}$$

On squaring both sides, we get

$$\left(1+5\frac{dy}{dx}\right)^3=100\left(\frac{d^3y}{dx^3}\right)^2$$

$$1+125\left(\frac{dy}{dx}\right)^3+15\frac{dy}{dx}\left(1+5\frac{dy}{dx}\right)=100\left(\frac{d^3y}{dx^3}\right)^2$$

Clearly, the order of highest derivative occurring in the differential equation is 3. Hence, the order of given differential equation is 3.

Thus (b) is correct option.

7. The figure formed by four points $\hat{i}+\hat{j}+\hat{k}$, $2\hat{i}+3\hat{j}$, $3\hat{i}-5\hat{j}-2\hat{k}$, $\hat{k}-\hat{j}$ is a

(a) Parallelogram

(b) Rectangle

(c) Trapezium

(d) Square

Sol :

Let the vertices be $\overrightarrow{OA}=\hat{i}+\hat{j}+\hat{k}$,

$$\overrightarrow{OB}=2\hat{i}+3\hat{j},\ \overrightarrow{OC}=3\hat{i}+5\hat{j}-2\hat{k} \text{ and}$$

$$\overrightarrow{OD}=\hat{k}-\hat{j}$$

$$\overrightarrow{AB}=\hat{i}+2\hat{j}-\hat{k},\ \overrightarrow{BC}=\hat{i}+2\hat{j}-2\hat{k}$$

$$\overrightarrow{CD}=-3\hat{i}-6\hat{j}+3\hat{k}=-(\hat{i}+2\hat{j}-\hat{k})$$

$$\overrightarrow{DA}=\hat{i}+2\hat{j}+\hat{k}$$

It is clear that, $|\overrightarrow{AB}|\neq|\overrightarrow{BC}|\neq|\overrightarrow{CA}|\neq|\overrightarrow{DA}|$

Also, $\overrightarrow{AB}\parallel\overrightarrow{CD}$

Hence, figure formed by four points is a trapezium.

Thus (c) is correct option.

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Sol :

The feasible region is bounded by the vertices A(12,0), B(4,2), C(1,5), and D(0,10).

According to the corner point method, we evaluate the value of the objective function z=3x+2y at each of these points.

At A(12,0) : z =3(12)+2(0)=36

At B(4,2) : z =3(4)+2(2)=12+4=16

At C(1,5) : z =3(1)+2(5)=3+10=13

At D(0,10) : z =3(0)+2(10)=20

Comparing all these values, the minimum value of z is 13, which occurs at the point C(1,5).

Thus (d) is correct option.

18. If $f(x)=\begin{cases}\frac{\log x}{x-1}, & \text{if } x\neq 1\\ k, & \text{if } x=1\end{cases}$ is continuous at $x=1$, then the value of k is

(a) 0 (b) -1

(c) 1 (d) e

Sol :

At $x=1$,

$$\text{LHL}=\lim_{x\to 1^-}f(x)=\lim_{h\to 0}\frac{\log(1+h)}{1+h-1}$$

$$=\lim_{h\to 0}\frac{\log(1+h)}{h}=1$$

As $f(x)$ is continuous at $x=1$.

$$\lim_{x\to 1^-}f(x)=f(1)\Rightarrow 1=k$$

Thus (c) is correct option.

19. **Assertion :** area of the parallelogram whose adjacent sides are $\hat{i}+\hat{j}-\hat{k}$ and $2\hat{i}-j+\hat{k}$ is $3\sqrt{2}$ square units.

Reason : area of the parallelogram whose adjacent sides are represented by the vectors $\vec{a}$ and $\vec{b}$ is $|\vec{a}-\vec{b}|$

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

Sol :

As we know that area of the parallelogram whose adjacent side are represented by the vectors $\vec{a}$ is $\vec{b}$ is $|\vec{a}\times\vec{b}|$ so, given Reason is false.

Let $\vec{a}=\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=2\hat{i}-\hat{j}+\hat{k}$

then $\vec{a}\times\vec{b}=\begin{bmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 1 & -1\\ 2 & -1 & 1\end{bmatrix}$

$$=\hat{i}(1-1)-\hat{j}(1+2)+\hat{k}(-1-2)$$

$$=3\hat{j}-3\hat{k}$$

now $|\vec{a}\times\vec{b}|=|-3\hat{j}-3\hat{k}|$

$$=\sqrt{(-3)^2+(-3)^2}=\sqrt{9+9}$$

$$=\sqrt{18}=3\sqrt{2}\text{ square units}$$

Hence, Assertion is true; reason is false,

Thus (c) is correct option.

20. **Assertion:** if $2P(A)=P(B)=\frac{5}{13}$ and $P\left(\frac{A}{B}\right)=\frac{2}{5}$, then $P(A\cup B)$ is $\frac{11}{26}$

Reason: J, F, E_1 and E_2 are two events. then

$$P\left(\frac{E_1}{E_2}\right)=\frac{P(E_1\cup E_2)}{P(E_2)},\ 0<P(E_2)\leq 1$$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

Given that,

$$2P(A)=P(B)=\frac{5}{13}$$

$$P(A)=\frac{P(A\cup B)}{P(B)}$$

$$\frac{2}{5}=\frac{P(A\cup B)}{\frac{5}{13}}$$

$$P(A\cap B)=\frac{2}{5}\times\frac{5}{13}=\frac{2}{13}$$

Again,

$$P(A\cup B)=P(A)+P(B)-P(A\cap B)$$

$$=\frac{5}{26}+\frac{5}{13}-\frac{2}{13}$$

$$=\frac{5+10-4}{26}$$

$$=\frac{11}{26}$$

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13. The area bounded by the curve $y = \frac{1}{2}x^2$, the X-axis and the ordinate $x = 2$ is

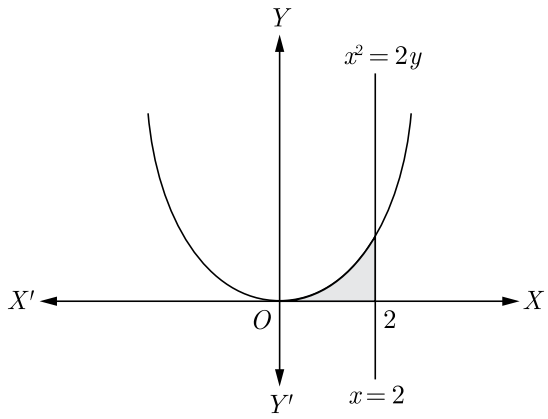
(a) $\frac{1}{3}$ sq unit (b) $\frac{2}{3}$ sq unit

(c) 1 sq unit (d) $\frac{4}{3}$ sq unit

Sol :

$$\text{Required area} = \int_0^2 y\,dx$$

$$= \int_0^2 \frac{x^2}{2}\,dx = \left[\frac{x^3}{6}\right]_0^2 = \frac{4}{3} \text{ sq units}$$



Thus (d) is correct option.

14. If $A = [a_{ij}]_{2\times 2}$, where $a_{ij} = i + j$, then A is equal to

(a) $\begin{bmatrix}1 & 1\\ 2 & 2\end{bmatrix}$ (b) $\begin{bmatrix}1 & 2\\ 1 & 2\end{bmatrix}$

(c) $\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$ (d) $\begin{bmatrix}2 & 3\\ 3 & 4\end{bmatrix}$

Sol :

We have $A = \begin{bmatrix}a_{11} & a_{12}\\ a_{21} & x+2\end{bmatrix}$

$$= \begin{bmatrix}1+1 & 1+2\\ 2+1 & 2+2\end{bmatrix} = \begin{bmatrix}2 & 3\\ 3 & 4\end{bmatrix}$$

Thus (d) is correct option.

15. Discuss the continuity of the function $f(x) = \sin 2x - 1$ at the point $x = 0$ and $x = \pi$

(a) Continuous at $x = 0,\ \pi$

(b) Discontinuous at $x = 0$ but continuous at $x = \pi$

(c) Continuous at $x = 0$ but discontinuous at $x = \pi$

(d) Discontinuous at $x = 0,\ \pi$

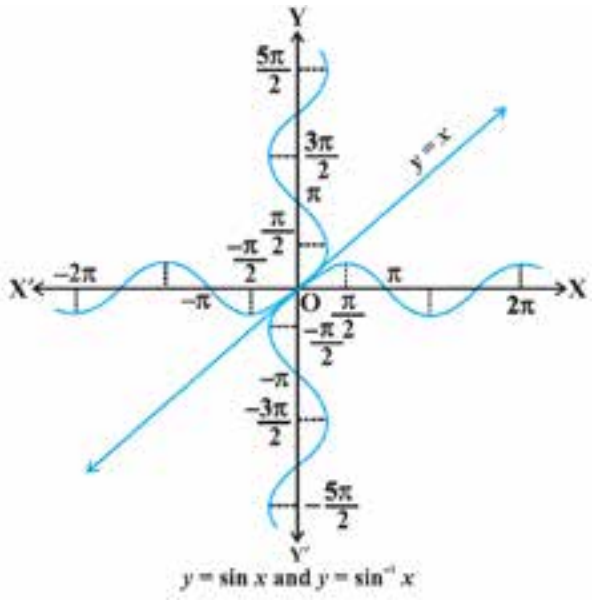
Sol :

$$f(x) = \sin 2x - 1$$

Being sine function, $f(x)$ is defined and continuous at $x = 0,\ \pi$.

Thus (a) is correct option.

16. At which points does the graph of $y = \tan x$ intersect the X-axis?



(a) $-\pi, -\pi/2, \pi/2, \pi$ (b) $-\pi, 0, \pi$

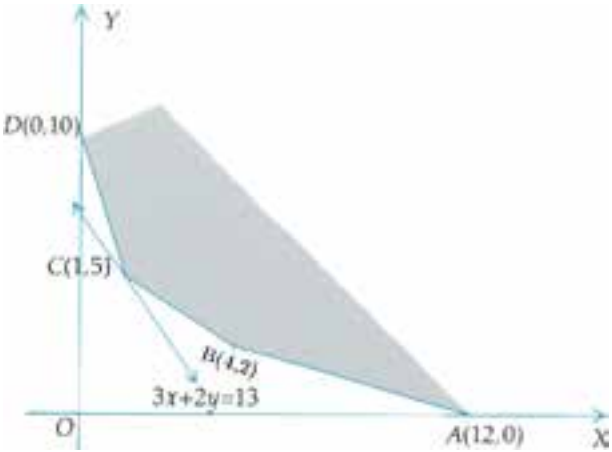
(c) $-\pi/2, 0\pi/2$ (d) $0, \pi/2\pi$

Sol :

From the graph, the curve crosses the X-axis at multiples of π (e.g., $-\pi, 0, \pi$). This is because $\tan x = 0$ when $x = n\pi$, where n is an integer.

Thus (b) is correct option.

17. The minimum value of $z = 3x + 2y$, if the feasible region for an LPP is as shown in Fig., is to be determined.



(a) 36 (b) 16

(c) 20 (d) 13

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8. If $P(A \cup B) = 0.83$, $P(A) = 0.3$ and $P(B) = 0.6$, then the events will be

(a) Dependent

(b) Independent

(c) Cannot say anything

(d) None of the above

Sol :

Given,

$$P(A \cup B) = 0.83,\ P(A) = 0.3$$

and $$P(B) = 0.6$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.83 = 0.3 + 0.6 - P(A \cap B)$$

$$P(A \cap B) = 0.9 - 0.83 = 0.07$$

Now, $$P(A) \cdot P(B) = 0.3 \times 0.6 = 0.18$$

$$P(A) \cdot P(B) \neq P(A \cap B)$$

Hence, events are dependent.

Thus (a) is correct option.

9. Find the equation of the line which passes through the point $(1,2,3)$ and is parallel to the vector $3\hat{i} + 2\hat{j} - 2\hat{k}$.

(a) $\left(\hat{i} + \hat{j} + 3\hat{k}\right) + \lambda\left(3\hat{i} + 2\hat{j} - 2\hat{k}\right)$

(b) $\left(\hat{i} + 2\hat{j} + 3\hat{k}\right) + \lambda\left(3\hat{i} + 2\hat{j} - 2\hat{k}\right)$

(c) $\left(\hat{i} + 2\hat{j} + 5\hat{k}\right) + \lambda\left(3\hat{i} + 2\hat{j} - 2\hat{k}\right)$

(d) $\left(\hat{i} + 2\hat{j} + 3\hat{k}\right) + \lambda\left(5\hat{i} + 2\hat{j} - 2\hat{k}\right)$

Sol :

We know that the equation of the line passing through the position vector $\vec{a}$ and parallel to the vector $\vec{b}$ is

$$\vec{r} = \vec{a} + \lambda\vec{b}$$

Here $$\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k} \text{ and } \vec{b} = 3\hat{i} + 2\hat{j} - 2\hat{k}$$

$$\vec{r} = \left(\hat{i} + 2\hat{j} + 3\hat{k}\right) + \lambda\left(3\hat{i} + 2\hat{j} - 2\hat{k}\right)$$

Thus (b) is correct option.

10. $\int \sqrt{1 + \cos x}\, dx$ is equal to

(a) $2\sin\left(\frac{x}{2}\right) + C$ (b) $\sqrt{2}\sin\left(\frac{x}{2}\right) + C$

(c) $2\sqrt{2}\sin\left(\frac{x}{2}\right) + C$ (d) $\frac{1}{2}\sin\left(\frac{x}{2}\right) + C$

Sol :

$$\int \sqrt{1 + \cos x}\, dx = \int \sqrt{2\cos^2(x/2)}\, dx$$

$$= \sqrt{2}\int \cos(x/2)\, dx$$

$$= \sqrt{2}\frac{\sin(x/2)}{1/2} + C$$

$$= 2\sqrt{2}\sin(x/2) + C$$

Thus (c) is correct option.

11. The family of curves $y = e^{a\sin x}$, where a is an arbitrary constant is represented by the differential equation

(a) $\log y = \tan x\frac{dy}{dx}$ (b) $y\log y = \tan x\frac{dy}{dx}$

(c) $y\log y = \sin\frac{dy}{dx}$ (d) $\log y = \cos x\frac{dy}{dx}$

Sol :

Given curve is $y = e^{a\sin x}$

On taking log both sides, we get

$$\sin x = \frac{\log y}{a}$$

$$\frac{dy}{dx} = e^{a\sin x} \cdot a\cos x$$

$$\frac{dy}{dx} = y\cos x \cdot \frac{\log y}{\sin x}$$

$$y\log y = \tan x\frac{dy}{dx}$$

Thus (b) is correct option.

12. If a line makes angles 90°, 60° and θ with X, Y and Z-axis respectively, where θ is acute angle, then find θ.

(a) 30° (b) 60°

(c) 45° (d) 90°

Sol :

If a line makes angles α, β and γ with the x-axis, y-axis and z-axis, respectively, the we have

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

$$\cos^2 90^\circ + \cos^2 60^\circ + \cos^2\theta = 1$$

$$0 + \left(\frac{1}{2}\right)^2 + \cos^2\theta = 1$$

$$\cos^2\theta = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\cos\theta = \pm\frac{\sqrt{3}}{2}$$

$$= \cos 30^\circ$$

$$\theta = 30^\circ$$

Since $\cos\theta$ cannot be negative as θ is an acute angle.

Thus (a) is correct option.

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As we know that, if E_1 and E_2 are two events, then

$$P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cap E_2)}{P(E_2)},\ 0 \le P(E_2) \le 1$$

Thus (a) is correct option.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Evaluate $\int \frac{e^{2x} - e^{-2x}}{e^{2x} + e^{-2x}} dx$.

OR

Evaluate $\int \frac{\cos\sqrt{x}}{\sqrt{x}} dx$.

Sol :

We have $I = \int \frac{e^{2x} - e^{-2x}}{e^{2x} + e^{-2x}} dx$

Substituting $e^{2x} + e^{-2x} = t$ we have

$$(2e^{2x} - e^{-2x})\,dx = dt \qquad \frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$(e^{2x} - e^{-2x})\,dx = \frac{dt}{2}$$

$$I = \int \frac{e^{2x} - e^{-2x}}{e^{2x} + e^{-2x}} dx = \frac{1}{2}\int \frac{dt}{t} = \frac{1}{2}\log|t| + C$$

$$= \int \frac{e^{2x} - e^{-2x}}{e^{2x} + e^{-2x}} dx = \frac{1}{2}\int \frac{dt}{t} = \frac{1}{2}\log|t| + C$$

$$= \int \frac{e^{2x} - e^{-2x}}{e^{2x} + e^{-2x}} dx = \frac{1}{2}\int \frac{dt}{t} = \frac{1}{2}\log|t| + C$$

Substituting $t = e^{2x} + e^{-2x}$ we have

$$I = \frac{1}{2}\log|e^{2x} + e^{-2x}| + C$$

OR

We have $I = \int \frac{\cos\sqrt{x}}{\sqrt{x}} dx$

Substituting $\sqrt{x} = t$

$$\frac{1}{2\sqrt{x}} dx = dt$$

$$\frac{1}{\sqrt{x}} dx = 2dt$$

$$I = \int \frac{\cos\sqrt{x}}{\sqrt{x}} dx = 2\int \cos t\, dt$$

$$= \int \frac{\cos\sqrt{x}}{\sqrt{x}} dx = 2\int \cos t\, dt$$

$$= 2\sin t + C = 2\sin\sqrt{x} + C$$

Substituting $t = \sqrt{x}$ we have

$$I = 2\sin t + C = 2\sin\sqrt{x} + C$$

22. Write the direction ratios of the vector $3\vec{a} + 2\vec{b}$, where $\vec{a} = \hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = 2\hat{i} - 4\hat{j} + 5\hat{k}$.

OR

Find the unit vector in the direction of the sum of vectors $2\hat{i} + 3\hat{j} - \hat{k}$ and $4\hat{i} - 3\hat{j} + 2\hat{k}$.

Sol :

The given vector $3\vec{a} + 2\vec{b}$ can be given as

$$3\vec{a} + 2\vec{b} = 3(\hat{i} + \hat{j} - 2\hat{k}) + 2(2\hat{i} - 4\hat{j} + 5\hat{k})$$

$$= (3\hat{i} + 3\hat{j} - 6\hat{k}) + (4\hat{i} - 8\hat{j} + 10\hat{k})$$

$$= 7\hat{i} - 5\hat{j} + 4\hat{k}$$

Hence, direction ratios of vectors $3\vec{a} + 2\vec{b}$ are 7, -4 and 4.

OR

We have $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and

$\vec{b} = 4\hat{i} - 3\hat{j} + 2\hat{k}$

Sum of two vectors,

$$\vec{a} + \vec{b} = (2\hat{i} + 2\hat{j} - \hat{k}) + (4\hat{i} - 3\hat{j} + 2\hat{k})$$

$$= 6\hat{i} + \hat{k}$$

Required unit vector

$$\frac{\vec{a} + \vec{b}}{|\vec{a} + \vec{b}|} = \frac{6\hat{i} + \hat{k}}{\sqrt{6^2 + 1^2}} = \frac{6\hat{i} + \hat{k}}{\sqrt{36 + 1}}$$

$$= \frac{6\hat{i} + \hat{k}}{\sqrt{37}} = \frac{6}{\sqrt{37}}\hat{i} + \frac{\hat{k}}{\sqrt{37}}$$

23. If $P(A) = \frac{1}{12}$, $P(B) = \frac{5}{12}$ and $P\left(\frac{B}{A}\right) = \frac{1}{15}$, then find $P(A \cup B)$

Sol :

We have $P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$

$$\frac{1}{15} = \frac{P(A \cap B)}{\frac{1}{12}}$$

$$P(A \cap B) = \frac{1}{180}$$

Also, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$P(A \cup B) = \frac{1}{12} + \frac{5}{12} - \frac{1}{180} = \frac{89}{180}$$

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$$= \pi[\tan^{-1}1 - \tan^{-1}(-1)]$$
$$= \pi\left[\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right]$$
$$= \frac{\pi^2}{2}$$
$$I = \frac{\pi^2}{4}$$

OR

We have $I = \int_0^{\pi} \frac{x \tan x}{\sec x + \tan x}\, dx$ ..(1)

Using the fact $\int_0^a f(x)\, dx = \int_0^a f(a-x)\, dx$ we have

$$I = \int_0^{\pi} \frac{(\pi - x)\tan(\pi - x)}{\sec(\pi - x) + \tan(\pi - x)}$$
$$= \int_0^{\pi} \frac{(\pi - x)\tan x}{\sec x + \tan x}\, dx \quad ...(2)$$

Adding eqs. (1) and (2), we get

$$2I = \int_0^{\pi} \frac{\pi \tan x}{\sec x + \tan x}\, dx$$
$$I = \frac{\pi}{2}\int_0^{\pi} \frac{\tan x(\sec x - \tan x)}{(\sec x + \tan x)(\sec x - \tan x)}\, dx$$
$$= \frac{\pi}{2}\int_0^{\pi} \frac{(\tan x \sec x - \tan^2 x)\, dx}{(\sec^2 x - \tan^2 x)}$$
$$= \frac{\pi}{2}\int_0^{\pi} \frac{\tan x \sec x - \sec^2 x + 1}{1}\, dx \quad \tan^2 x = \sec^2 x - 1$$
$$= \frac{\pi}{2}[\sec x - \tan x + x]_0^{\pi}$$
$$= \frac{\pi}{2}[(\sec \pi - \tan \pi + \pi) - (\sec 0 - \tan 0 + 0)]$$
$$= \frac{\pi}{2}[\sec \pi - \sec 0 - \tan \pi + \tan 0 + \pi - 0]$$
$$= \frac{\pi}{2}[-1 - 1 - 0 + 0 + \pi - 0]$$
$$= \frac{\pi}{2}[\pi - 2]$$

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Solve the differential equation

$$x \log|x| \frac{dy}{dx} + y = \frac{2}{x}\log|x|.$$

OR

Find the particular solution of the differential equation $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$ given that $y = 1$, when $x = 0$.

Sol :

We have $x \log|x| \frac{dy}{dx} + y = \frac{2}{x}\log|x|.$

$$\frac{dy}{dx} + \frac{y}{x \log|x|} = \frac{2\log|x|}{x^2 \log|x|} = \frac{2}{x^2}$$

Which is the a linear differential equation of first order and is of the form $\frac{dy}{dx} = Py = Q$,

Where, $P = \frac{1}{x \log|x|}$ and $Q = \frac{2}{x^2}$

$$\text{IF} = e^{\int P\, dx} = e^{\int \frac{1}{x\log|x|} dx}$$
$$= e^{\log|\log x|} = \log|x|$$

Using $y \cdot (IF) = \int Q \cdot (IF)\, dx + C$ the general solution is

$$y \log|x| = \int \underset{II}{\frac{2}{x^2}} \underset{I}{\log|x|}\, dx$$
$$= 2\left[\log|x| \int \frac{1}{x^2} dx - \int \left(\frac{d}{dx}(\log|x|) \cdot \int \frac{1}{x^2} dx\right) dx\right]$$
$$= 2\left[\log|x| \cdot \left(-\frac{1}{x}\right) - \int \frac{1}{x} \cdot \left(-\frac{1}{x}\right) dx\right]$$
$$= 2\left[-\frac{1}{2}\log|x| + \int \frac{1}{x^2} dx\right]$$

or $y \log|x| = -\frac{2}{x}\log|x| - \frac{2}{x} + C$,

Which is the required solution.

OR

We have $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$

$$= \frac{\frac{y}{x}}{1 + \frac{y^2}{x^2}} \quad ...(1)$$

Which is a homogeneous differential equation as

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

Substituting $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ in eq (1) we have

$$v + x\frac{dv}{dx} = \frac{v}{1 + v^2}$$
$$x\frac{dv}{dx} = \frac{v}{1 + v^2} - v$$
$$x\frac{dv}{dx} = \frac{v - v - v^3}{1 + v^2}$$
$$x\frac{dv}{dx} = -\frac{v^3}{1 + v^2}$$
$$\frac{1 + v^2}{v^3} dv = -\frac{dx}{x}$$

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For strictly increasing or strictly decreasing, put $f'(x) = 0$, we get

$$\begin{aligned} 6x^3 - 12x^2 - 90x &= 0 \\ 6x(x^2 - 2x - 15) &= 0 \\ 6x(x+3)(x-5) &= 0 \\ x &= 0, -3, 5 \end{aligned}$$

Now, we find intervals in which $f(x)$ is strictly increasing or strictly decreasing.

Interval	$f'(x) = 6(x+3)x(x-5)$	**Sing of** $f'(x)$
$x < -3$	$(-)(-)(-)$	$-$ ve
$-3 < x < 0$	$(+)(-)(-)$	$+$ ve
$0 < x < 5$	$(+)(+)(-)$	$-$ ve
$x > 5$	$(+)(+)(+)$	$+$ ve

A function $f(x)$ is said to be strictly increasing, if $f'(x) > 0$ and it is said to be strictly decreasing, if $f'(x) < 0$. So, the given function $f(x)$ is

(i) Strictly increasing in $(-3, 0)$ and $(5, \infty)$.

(ii) Strictly decreasing in $(-\infty, -3)$ and $(0, 5)$.

29. If $e^y(x+1) = 1$, then show that $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$.

Sol :

We have $e^y(x+1) = 1$

Taking log both sides, we get

$$\begin{aligned} \log[e^y(x+1)] &= \log 1 \\ \log e^y + \log(x+1) &= 0 \\ y + \log(x+1) &= 0 \qquad [\because \log e^y = y] \end{aligned}$$

Differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} + \frac{1}{x+1} = 0 \qquad \ldots(1)$$

Again, differentiating both sides w.r.t. x, we get

$$\begin{aligned} \frac{d^2y}{dx^2} - \frac{1}{(x+1)^2} &= 0 \\ \frac{d^2y}{dx^2} - \left(-\frac{dy}{dx}\right)^2 &= 0 \qquad \text{from Eq. (1)} \\ \frac{d^2y}{dx^2} - \left(\frac{dy}{dx}\right)^2 &= 0 \\ \frac{d^2y}{dx^2} &= \left(\frac{dy}{dx}\right)^2 \text{Hence proved.} \end{aligned}$$

30. If $\vec{a} = \hat{i} - \hat{j} + 7\hat{k}$ and $\vec{b} = 5\hat{i} - \hat{j} + \lambda\hat{k}$, then find the value of λ, so that $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are perpendicular vectors.

Sol :

We have $\vec{a} = \hat{i} - \hat{j} + 7\hat{k}$

and $\vec{b} = 5\hat{i} - \hat{j} + \lambda\hat{k}$

Then, $\vec{a} + \vec{b} = (\hat{i} - \hat{j} + 7\hat{k}) + (5\hat{i} - \hat{j} + \lambda\hat{k})$

$= 6\hat{i} - 2\hat{j} + (7 + \lambda)\hat{k}$

and $\vec{a} - \vec{b} = (\hat{i} - \hat{j} + 7\hat{k}) - (5\hat{i} - \hat{j} + \lambda\hat{k})$

$= -4\hat{i} + (7 - \lambda)\hat{k}$

Since, $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$ are perpendicular vectors, we have

$$\begin{aligned} (\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) &= 0 \\ [6\hat{i} - 2\hat{j} + (7+\lambda)\hat{k}] \cdot [-4\hat{i} + (7-\lambda)\hat{k}] &= 0 \\ -24 + (7+\lambda)(7-\lambda) &= 0 \\ 49 - \lambda^2 &= 24 \\ \lambda^2 &= 25 \\ \lambda &= \pm 5 \end{aligned}$$

31. Evaluate $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx$.

OR

Evaluate $\int_0^{\pi} \frac{x \tan x}{\sec x + \tan x} dx$.

Sol :

We have $I = \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx \qquad \ldots(i)$

$$= \int_0^{\pi} \frac{(\pi - x)\sin(\pi - x)}{1 + \cos^2(\pi - x)} dx$$

$$= \int_0^{\pi} \frac{(\pi - x)\sin x}{1 + \cos^2 x} dx \qquad \ldots(ii)$$

Adding Eqs. (i) and (ii), we get

$$2I = \int_0^{\pi} \frac{\pi \sin x}{(1 + \cos^2 x)} dx$$

Substituting $\cos x = t$

$$\begin{aligned} -\sin x\, dx &= dt \\ \sin x\, dx &= -dt \end{aligned}$$

when $x = 0$, then $t = 1$ and $x = \pi$, then $t = -1$

Thus $2I = -\pi \int_1^{-1} \frac{dt}{1 + t^2}$,

$$= \pi \int_{-1}^{1} \frac{dt}{(1 + t^2)}$$

$$= \pi [\tan^{-1} t]_{-1}^{1}$$

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You are required to find out following using matrix method.

(i) Express the total overhead of the service departments in the form of simultaneous equations.

(ii) Express these equations in a matrix form and solve for total overhead of service departments using matrix inverse method.

(iii) Determine the total overhead to be allocated from each of S_1 and S_2 to the production department.

Sol :

Let S_1 be total overhead of service department and S_2 be total overhead of service department.

Then, $S_1 = 20000 + 0.3S_2$

and $S_2 = 40000 + 0.2S_1$

The total overhead of the service departments can be expressed in the form of simultaneous equations as:

$$S_1 - 0.3S_2 = 20000$$

$$-0.2S_1 + S_2 = 40000$$

The above simultaneous equations can be expressed in the matrix form as:

$$\begin{bmatrix} 1 & -0.3 \\ -0.2 & 1 \end{bmatrix}\begin{bmatrix} S_1 \\ S_2 \end{bmatrix} = \begin{bmatrix} 20000 \\ 40000 \end{bmatrix},$$

or $AS = B$

Here $|A| = 1 - 0.06 = 0.94$

Since $|A| \neq 0$ thus A^{-1} exists, so that the unique solution of $AX = B$ is $X = A^{-1}B$.

Here, $A^{-1} = \dfrac{\text{adj.}A}{|A|} = \dfrac{1}{0.94}\begin{bmatrix} 1 & 0.3 \\ 0.2 & 1 \end{bmatrix}$

$$\begin{bmatrix} S_1 \\ S_2 \end{bmatrix} = S = A^{-1}B$$

$$= \frac{1}{0.94}\begin{bmatrix} 1 & 0.3 \\ 0.2 & 1 \end{bmatrix}\begin{bmatrix} 20000 \\ 40000 \end{bmatrix} = \begin{bmatrix} 34043 \\ 46809 \end{bmatrix}$$

$S_1 = ₹\,34043$ and $S_2 = ₹\,46809$

The allocation of overhead from S_1 and S_2 to the production department becomes:

$$[P_1\ P_2\ P_3\ P_4] = [34043][0.3\ \ 0.25\ \ 0.15\ \ 0.1]$$

$$= [10213\ \ 8511\ \ 5106\ \ 3404]$$

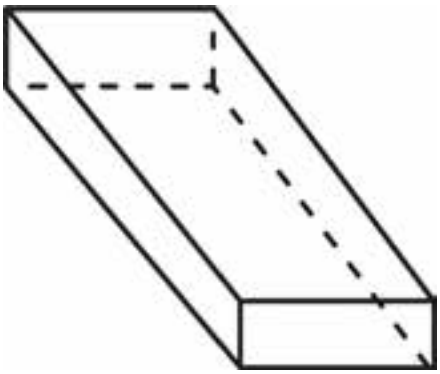
and $$[P_1\ P_2\ P_3\ P_4] = [46809][0.1\ \ 0.35\ \ 0.2\ \ 0.05]$$

$$= [4681\ \ 16383\ \ 9362\ \ 2340]$$

The final allocation becomes :

Department	Total	P_1	P_2	P_3	P_4
S_1	27234	10213	8511	5106	3404
S_2	32766	4681	16383	9362	2340
Total	60000	14894	24894	14468	5744

38. Parallelepiped is the Greek word, which essentially means the object that has a parallel plane. Principally, the Parallelepiped is framed by the six parallelogram sides which bring about the prism or the 3D figure, and it consists of the parallelogram base. It can be categorized as anything but the polyhedron, where 3 sets of the parallel faces are made to combine for framing a three-dimensional (3D) shape that has six faces. The cube, cuboid, and rhomboid are the three exceptional cases. The Rectangular Parallelepiped consists of six faces in a rectangular shape.



The sum of the surface area of a rectangular parallelopied with sides of x, $2x$ and $\frac{x}{3}$ and a shape of radius of y is given to be constant.

On the basis of above information, answer the following questions.

(i) If S is the constant, then find the relation between S, x and y.

(ii) If the combined volume is denoted by V , then find the relation between V, x and y.

(iii) Find the relation between x and y when the volume V is minimum.

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Integrating both sides, we have

$$\int\left(\frac{1}{v^3}+\frac{1}{v}\right)dv = -\int\frac{dx}{x}$$

$$-\frac{1}{2v^2}+\log|v| = -\log|x|+C$$

$$-\frac{x^2}{2y^2}+\log\left|\frac{y}{x}\right| = -\log|x|+C$$

$$-\frac{x^2}{2y^2}+\log|y|-\log|x| = -\log|x|+C$$

$$-\frac{x^2}{2y^2}+\log|y| = C \quad ...(2)$$

Also, it is given that $y=1$, when $x=0$.
From Eq. (2), we have

$$0+\log|1| = C \Rightarrow C=0$$

Substituting $C=0$ in Eq. (2), we get

$$-\frac{x^2}{2y^2}+\log|y| = 0$$

$$\log|y| = \frac{x^2}{2y^2}$$

$$y = e^{\frac{x^2}{2y^2}}$$

Which is the required solution.

33. Maximize $Z=-x+2y$, subject to the constraints:

$x \geq 3,\ x+y\geq 5,\ x+2y\geq 6,\ y\geq 0$

OR

Maximize $Z=x+y$, subject to $x-y\leq -1,\ x+y\leq 0,\ x,\ y\geq 0$.

Sol :

Maximize $Z = -x+2y$

Subject to $x+y \geq 5$ (i)

$x+2y \geq 6$ (ii)

and $x\geq 3,\ y\geq 0$

(i) Region corresponding to $x+y\geq 5$:

Line $x+y=5$			
x	0	5	Point (0,0) is false for $x+y\geq 5$. So, the region is away from the origin.
y	5	0	

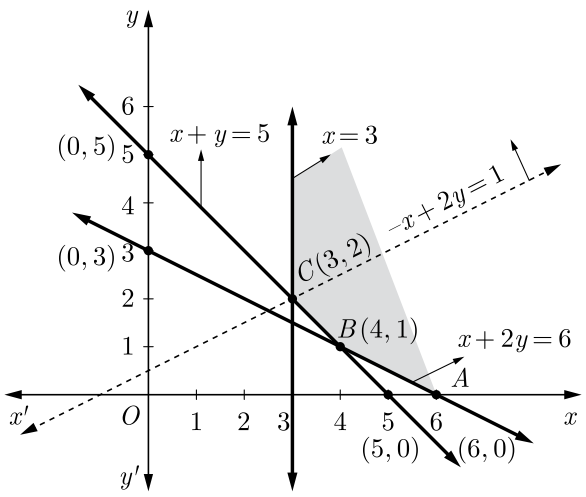
(ii) Region corresponding to $x+2y\geq 6$:

Line $x+2y=6$			
x	0	3	Point (0,0) is false for $x+2y\geq 6$. So, the region is away from the origin.
y	6	0	

(iii) Region corresponding to $x\geq 3$ is away from the origin.

Since $x\geq 3,\ y\geq 0$ the feasible region lies in the first quadrant and away from the origin and other side of the line $x=3$.

Now we draw all line on the graph and find the common area. Clearly, feasible region is CBA. The corner points of the feasible region are $A(6,0)$, $B(4,1)$ and $C(3,2)$.



The values of objective function Z at these points are as follows.

Corner points	$Z=-x+2y$
$A(6,0)$	$Z=-6+2\times 0=-6$
$B(4,1)$	$Z=-4+2\times 1=-2$
$C(3,2)$	$Z=-3+2\times 2=1$

Since the feasible region is unbounded, therefore, $Z=1$ may or may not be the maximum value. For this, we graph the inequality $-x+2y>1$ and check weather the resulting region has points in common with the feasible region or not.

The resulting feasible region has points in common with the feasible region. Hence, $Z=1$ is not the maximum value or Z has no maximum value.

OR

Maximize $Z = x+y$

Subject to $x-y \leq -1$ (i)

$x+y \leq 0$

and $x\geq 0,\ y\geq 0$

(i) Region corresponding to $x-y\leq -1$:

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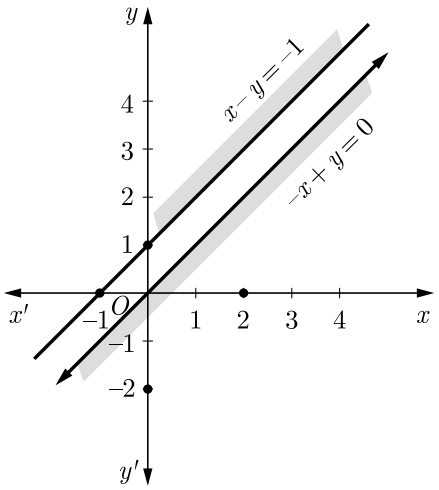
Line $x-y=-1$			
x	0	-1	Point $(0,0)$ is false for $x-y\le -1$. So, the region is away from the origin.
y	1	0	

(ii) Region corresponding to $x+y\le 0$:

Line $x+y=0$			
x	0	2	Point $(0,-2)$ is true for $x+y\le 0$. So, the region includes point $(0,-2)$.
y	0	2	

Since $x\ge 0$, $y\ge 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Here we get no feasible region (constraints are inconsistent). Hence, Z has no maximum value.



34. Evaluate $\int_0^{\pi}\frac{x}{a^2\cos^2 x+b^2\sin^2 x}\,dx$.

Sol :

We have $I=\int_0^{\pi}\frac{x}{a^2\cos^2 x+b^2\sin^2 x}\,dx$ ...(i)

Using the fact $\int_0^a f(x)\,dx=\int_0^a f(a-x)\,dx$ we have

$$I=\int_0^{\pi}\frac{(\pi-x)}{a^2\cos^2(\pi-x)+b^2\sin^2(\pi-x)}\,dx$$

$$=\int_0^{\pi}\frac{(\pi-x)}{a^2\cos^2 x+b^2\sin^2 x}\,dx \qquad ...(ii)$$

Adding Eqs. (i) and (ii), we get

$$2I=\int_0^{\pi}\frac{(x+\pi-x)}{a^2\cos^2 x+b^2\sin^2 x}\,dx$$

$$=\pi\int_0^{\pi}\frac{dx}{a^2\cos^2 x+b^2\sin^2 x}$$

Here, $a^2\cos^2(\pi-x)+b^2\sin^2(\pi-x)=a^2\cos^2 x+b^2\sin^2 x$

Using the fact

$$\int_0^{2a}f(x)\,dx=2\int_0^a f(x)\,dx,$$

if $f(2a-x)=f(x)$

We have $2I=2\pi\int_0^{\pi}\frac{dx}{a^2\cos^2 x+b^2\sin^2 x}$

$$=2\pi\int_0^{\pi}\frac{\sec^2 x}{a^2+b^2\tan^2 x}\,dx$$

Substituting $\tan x=t\Rightarrow \sec^2 x\,dx=dt$

When $x=0$, then $t=\tan 0=0$ and

when $x=\frac{\pi}{2}$, then $t=\tan\frac{\pi}{2}=\infty$.

$$I=\pi\int_0^{\infty}\frac{dt}{a^2+b^2t^2}$$

$$=\pi\int_0^{\infty}\frac{dt}{a^2+(bt)^2}$$

$$=\frac{\pi}{b^2}\int_0^{\infty}\frac{dt}{\left(\frac{a}{b}\right)^2+t^2}$$

$$I=\frac{\pi}{ab}\left[\tan^{-1}\frac{bt}{a}\right]_0^{\infty} \qquad \int\frac{dx}{a^2+x^2}=\frac{1}{a}\tan^{-1}\frac{x}{a}+C$$

$$=\frac{\pi}{ab}[\tan^{-1}\infty-\tan^{-1}0]$$

$$=\frac{\pi}{ab}\left[\frac{\pi}{2}-0\right]=\frac{\pi^2}{2ab}$$

35. Find the values of p, so that the lines

$$l_1:\frac{1-x}{3}=\frac{7y-14}{p}=\frac{z-3}{2}\text{ and}$$

$$l_2:\frac{7-7x}{3p}=\frac{y-5}{1}=\frac{6-z}{5}$$

are perpendicular to each other. Also, find the equation of a line passing through a point $(3,2,-4)$ and parallel to line l_1.

Sol :

Writing the given line in standard form as

$$\frac{x-1}{-3}=\frac{y-2}{\frac{p}{7}}=\frac{z-3}{2}=r_1\text{ (let)} \qquad ...(1)$$

and $\frac{x-1}{\frac{-3p}{7}}=\frac{y-5}{1}=\frac{z-6}{-5}=r_2$ (let) ...(2)

Two lines with DR's a_1, b, c_1 and a_2, b_2, c_2 are perpendicular if

$$a_1a_2+b_1b_2+c_1c_2=0.$$

Thus line (1) and (2) will intersect at right angle, if

$$-3\left(\frac{-3p}{7}\right)+\frac{p}{7}(1)+2(-5)=0$$

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$$\frac{9p}{7}+\frac{p}{7}=10$$

$$\frac{10p}{7}=10 \Rightarrow p=7$$

which is the required value of p.

Also, we know that, the equation of a line which passes through the point (x_1, y_1, z_1) with direction ratios a, b, c is given by

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$$

Since, required line is parallel to line l_1.

So, $a=-3,\ b=\frac{7}{7}=1$ and $c=2$

Now, equation of line passing through the point $(3,2,-4)$ and having direction ratios $(-3,1,2)$ is

$$\frac{x-3}{-3}=\frac{y-2}{1}=\frac{z+4}{2}$$

$$\frac{3-x}{3}=\frac{y-2}{1}=\frac{z+4}{2}$$

SECTION - E

Case study based questions are compulsory.

36. Rajneesh do outsourcing work for companies and runs a form processing agency. He collect form from different office and then extract data and record data on computer. In his office three employees Vikas, Sarita and Ishaan process incoming copies of a form. Vikas process 50% of the forms. Sarita processes 20% and Ishaan the remaining 30% of the forms. Vikas has an error rate of 0.06, Sarita has an error rate of 0.04 and Ishaan has an error rate of 0.03.



Based on the above information answer the following questions.

(i) The total probability of committing an error in processing the form.

(ii) The manager of the company wants to do a quality check. During inspection he selects a form at random from the days output of processed forms. If the form selected at random has an error, the probability that the form is not processed by Vikas.

Sol :

(i) Required probability,

$$= P(A)$$

$$= P(E_1)\,P\left(\frac{A}{E_1}\right)+P(E_2)\,P\left(\frac{A}{E_2}\right)+P(E_3)\,P\left(\frac{A}{E_3}\right)$$

$$= 0.5\times 0.06+0.2\times 0.4+0.3\times 0.3$$

$$= 0.030+0.008+0.009$$

$$= 0.047$$

(ii) Required probability,

$$P\left(\frac{\overline{E_1}}{A}\right)=1-P\left(\frac{\overline{E_1}}{A}\right)$$

$$=1-\left[\frac{P(E_1)\,P\left(\frac{A}{E_1}\right)}{P(E_1)\,P\left(\frac{A}{E_1}\right)+P(E_2)\,P\left(\frac{A}{E_2}\right)+P(E_3)\,P\left(\frac{A}{E_3}\right)}\right]$$

$$=1-\left[\frac{0.5\times 0.6}{0.5\times 0.06+0.2\times 0.04+0.3\times 0.03}\right]$$

$$=1-\frac{0.030}{0.047}$$

$$=1-\frac{30}{47}=\frac{17}{47}$$

37. A manufacturing company has two service departments, S_1, S_2 and four production departments P_1, P_2, P_3 and P_4.

Overhead is allocated to the production departments for inclusion in the stock valuation. The analysis of benefits received by each department during the last quarter and the overhead expense incurred by each department were:

Service Department	**Percentages to be allocated to departments**					
	S_1	S_2	P_1	P_2	P_3	P_4
S_1	0	20	30	25	15	10
S_2	30	0	10	35	20	5
Direct overhead expense ₹ '000	20	40	25	30	20	10

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24. The x-coordinate of point on the line joining the points $P(2,2,1)$ and $Q(5,1,-2)$ is 4. Find its z -coordinate.

Sol :

The equation of the line joining the points $P(2,2,1)$ and $Q(5,1,-2)$ is

$$\frac{x-2}{5-2} = \frac{y-2}{1-2} = \frac{z-1}{-2-1}$$
$$\frac{x-2}{3} = \frac{y-2}{-1} = \frac{z-1}{-3}$$

Since at a point, x-coordinate is 4, we have

$$\frac{4-2}{3} = \frac{z-1}{-3} \Rightarrow z = -1$$

25. Let R is the equivalence relation in the set $A = \{0,1,2,3,4,5\}$ given by $R = \{(a,b):2 \text{ divides } (a-b)\}$. Write the equivalence class $[0]$.

Sol :

We have $R = \{(a,b):2 \text{ divides } (a-b)\}$

and $A = \{0,1,2,3,4,5\}$

Clearly, $[0] = \{b \in A : (0,b) \in R\}$

$= \{b \in A : 2 \text{ divides } (0-b)\}$

$= \{b \in A : 2 \text{ divides } (-b)\}$

$= \{0,2,4\}$

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. If $\begin{vmatrix} 3x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 7 & 3 \end{vmatrix}$, then write the value of x.

Sol :

We have $\begin{vmatrix} 2x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 7 & 3 \end{vmatrix}$

$2x^2 - 40 = 18 - (-14)$

$2x^2 - 40 = 32$

$2x^2 = 72$

$x^2 = 36$

$x = \pm 6$

27. Write the value of $\cos^{-1}\left(\frac{1}{2}\right) - 2\sin^{-1}\left(-\frac{1}{2}\right)$.

Sol :

$$\cos^{-1}\left(\frac{1}{2}\right) - 2\sin^{-1}\left(\frac{-1}{2}\right) = \cos^{-1}\left(\frac{1}{2}\right) + 2\sin^{-1}\left(\frac{1}{2}\right)$$
$$= \frac{\pi}{3} + 2\sin^{-1}\left(\sin\frac{\pi}{6}\right)$$
$$= \frac{\pi}{3} + 2 \times \frac{\pi}{6}$$
$$= \frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$$

28. The volume of a sphere is increasing at the rate of 3 cubic centimetre per second. Find the rate of increase of its surface area, when the radius is 2 cm.

OR

Find the intervals in which the function

$$f(x) = \frac{3}{2}x^4 - 4x^3 - 45x^2 + 51 \text{ is}$$

(i) strictly increasing

(ii) strictly decreasing.

Sol :

Let r be the radius, V be the volume and S be the surface area of sphere.

We have $\frac{dV}{dt} = 3 \text{ cm}^3/\text{s}$

Now $V = \frac{4}{3}\pi r^3$

$$\frac{dV}{dt} = \frac{4}{3}\pi \cdot 3r^2 \cdot \frac{dr}{dt}$$
$$3 = 4\pi \times r^2 \frac{dr}{dt}$$
$$\frac{dr}{dt} = \frac{3}{4\pi r^2} \text{ cm/s} \quad \ldots(1)$$

Now, $S = 4\pi r^2$

$$\frac{dS}{dt} = \frac{d}{dt}(4\pi r^2)$$
$$= 4\pi \times 2r \cdot \frac{dr}{dt}$$
$$= 8\pi r \times \frac{3}{4\pi r^2} \quad \text{Using eq. (1)}$$
$$= \frac{6}{r}$$
$$\left(\frac{dS}{dt}\right)_{r=2} = \frac{6}{2} = 2 \text{ cm}^2/\text{s}$$

OR

We have $f(x) = \frac{3}{2}x^4 - 4x^3 - 45x^2 + 51$

Differentiating both sides w.r.t. x, we get

$$f'(x) = 6x^3 - 12x^2 - 90x$$

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OR

(iv) If at $x = 3y$, volume V is minimum, then find the value of minimum volume and the value of S.

Sol :

(i) We have,

$$S = 2\left(x \times 2x + 2x \times \frac{x}{3} + \frac{x}{3} \times x\right) + 4\pi y^2$$

$$S = 6x^2 + 4\pi y^2 \quad \text{...(i)}$$

(ii) we have,

$$V = \frac{4}{3}\pi y^3 + x \times 2x \times \frac{x}{3}$$

$$V = \frac{4}{3}\pi y^3 + \frac{2}{3}x^3$$

(iii) We have,

$$V = \frac{4}{3}\pi y^3 + \frac{2}{3}x^3 \quad \text{[from part (ii)]}$$

$$= \frac{4}{3}\pi\left(\frac{S - 6x^2}{4\pi}\right) + \frac{2}{3}x^3 \quad [s = 6x^2 + 4\pi y^2]$$

$$\frac{dV}{dx} = \frac{1}{6\sqrt{\pi}} \times \frac{3}{2}(S - 6x^2)^{\frac{1}{2}}(-12x) + \frac{2}{3} \times 3x^2$$

$$= -\frac{3}{\sqrt{\pi}}(S - 6x^2)^{\frac{1}{2}} \times + 2x^2$$

For minimum, $\frac{dV}{dx} = 0$

$$-\frac{3}{\sqrt{\pi}}(S - 6x^2)^{\frac{1}{2}}x + 2x^2 = 0$$

$$2x^2 = \frac{3x}{\sqrt{\pi}}(S - 6x^2)^{\frac{1}{2}}$$

$$2x^2 = \frac{3x}{\sqrt{\pi}}(4\pi y^2)^{\frac{1}{2}} \quad \text{[using Eq.(i)]}$$

$$2\sqrt{\pi}\,x = 3(4\pi y^2)^{\frac{1}{2}}$$

$$4\pi x^2 = 9 \times 4\pi y^2$$

$$x^2 = 9y^2$$

$$x = 3y$$

OR

(iv)

Minimum value of $V = \frac{4}{3}\pi y^3 + \frac{2}{3}x^3$

$$= \frac{4}{3}\pi\left(\frac{x}{3}\right)^3 + \frac{2}{3}x^3 \quad [x = 3y]$$

$$= \frac{2}{3}x^3\left(1 + \frac{2\pi}{27}\right)$$

Now, when V is minimum,

$$S = 6x^2 + 4\pi y^2$$

$$= 6x^2 + 4\pi\left(\frac{x}{3}\right)^2 \quad [x = 3y]$$

$$= 6x^2 + \frac{4}{9}\pi x^2$$

$$= 2x^2\left[3 + \frac{2}{9}\pi\right]$$

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Sample Paper 15

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

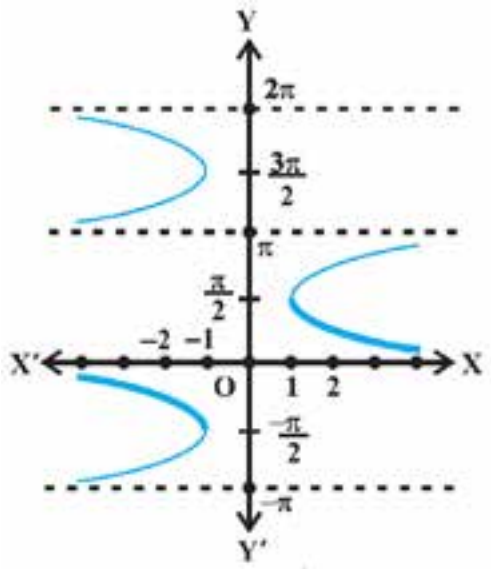
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the domain of the function $y = \text{cosec}^{-1}x$ as shown in the graph?



(a) $(-\infty,\infty)$ (b) $(-1,1)$

(c) $(-\infty,-1] \cup [1,\infty)$ (d) $[0,\pi]$

Sol :

The graph is drawn only for X-values less than or equal to –1 and greater than or equal to 1. Between –1 and 1, the curve does not exist. This shows that the function is defined only in the intervals $(-\infty,-1]$ and $[1,\infty)$.

Thus (c) is correct option.

2. If $\sin(x+y) = \log(x+y)$, then dy/dx is equal to

(a) 2 (b) -2

(c) 1 (d) -1

Sol :

We have, $\sin(x+y) = \log(x+y)$

Differentiating w.r.t. x, we get

$$\cos(x+y)\left(1+\frac{dy}{dx}\right) = \frac{1}{x+1}\left(1+\frac{dy}{dx}\right)$$

$$\left[\cos(x+y) - \frac{1}{x+y}\right]\left(1+\frac{dy}{dx}\right) = 0$$

$$\frac{dy}{dx} = -1$$

$$\left[\because \cos(x+y) - \frac{1}{x+y} \neq 0\right]$$

Thus (d) is correct option.

3. The area bounded by the parabola $y^2 = 8x$ and its latusrectum is

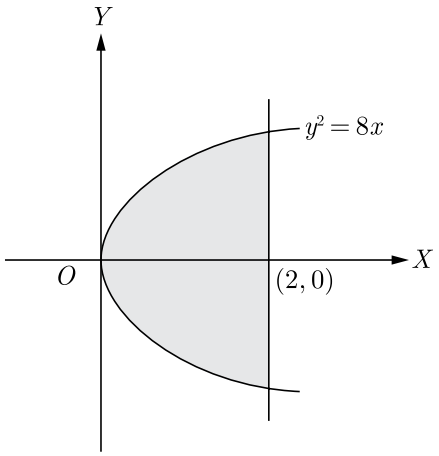
(a) 16/3 sq units (b) 32/3 sq units

(c) 8/3 sq units (d) 64/3 sq units

Sol :

$$\text{Required area} = 2\int_0^2 \sqrt{8x}\,dx = 4\sqrt{2}\left[\frac{x^{3/2}}{3/2}\right]_0^2$$

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$$= 4\sqrt{2}\left[\frac{2\sqrt{2}}{3/2}\right] = \frac{32}{3} \text{ sq units}$$

Thus (b) is correct option.

4. Evaluate $\int \frac{x^3 - x^2 + x - 1}{x - 1} dx$

(a) $\frac{x^3}{3} + x + C$ (b) $\frac{x^3}{4} + x + C$

(c) $\frac{x^3}{5} + x + C$ (d) $\frac{x^3}{6} + x + C$

Sol :

Let $I = \int \frac{x^3 - x^2 + x - 1}{x - 1} dx$

$$= \int \frac{x^2(x-1) + 1(x-1)}{x-1} dx$$

$$= \int \frac{(x-1)(x^2+1)}{x-1} dx$$

$$= \int (x^2 + 1) dx = \int x^2 dx + \int 1 \, dx$$

$$= \frac{x^3}{3} + x + C$$

Hence, $I = \frac{x^3}{3} + x + C$

Thus (a) is correct option.

5. If $\vec{a}$ and $\vec{b}$ are two unit vectors inclined at an angle $\pi/3$, then the value of $|\vec{a} + \vec{b}|$ is

(a) equal to (b) greater than 1

(c) equal to (d) less than 1

Sol :

Given,

$$|\vec{a}| = |\vec{b}| = 1 \text{ and } \theta = \pi/3$$

Now, $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2|\vec{a}||\vec{b}|\cos\theta$

$$= 1^2 + 1^2 + 2 \times 1 \times 1 \times \cos\frac{\pi}{3}$$

$$= 1 + 1 + 2 \times \frac{1}{2} = 1 + 1 + 1 = 3$$

$$|\vec{a} + \vec{b}| = \sqrt{3}$$

$$|\vec{a} + \vec{b}| > 1$$

Thus (b) is correct option.

6. The point of intersection of lines

$$\frac{x-5}{3} = \frac{y-7}{-1} = \frac{z+2}{1}$$

and $\frac{x+3}{-36} = \frac{y-3}{2} = \frac{z-6}{4}$ is

(a) $(5, 7, -2)$ (b) $(-3, 3, 6)$

(c) $(2, 10, 4)$ (d) $\left(21, \frac{5}{3}, \frac{10}{3}\right)$

Sol :

Let

$$\frac{x-5}{3} = \frac{y-7}{-1} = \frac{z+2}{1} = k \quad \text{[say]}$$

Any point on the line is $P(3k+5, -k+7, k-2)$.

To determine the intersection point, equation second satisfying the point

$P(3k+5, -k+7, k-2)$.

$$\frac{3k+5+3}{-36} = \frac{-k+7-3}{2} = \frac{k-2-6}{4}$$

$$\frac{3k+8}{-36} = \frac{-k+4}{2} \Rightarrow k = \frac{16}{3}$$

Point is P is $\left(21, \frac{5}{3}, \frac{10}{3}\right)$.

Thus (d) is correct option.

7. The area bounded by $y = \log x$, X-axis and ordinates $x = 1$, $x = 2$ is

(a) $\frac{1}{2}(\log 2)^2$ (b) $\log(2/e)$

(c) $\log(4/e)$ (d) $\log 4$

Sol :

$$\text{Required area} = \int_1^2 \log x \, dx$$

$$= [x \log x - x]_1^2$$

$$= 2\log 2 - 1$$

$$= \log 4 - \log e$$

$$= \log\left(\frac{4}{e}\right)$$

Thus (c) is correct option.

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$$\frac{a}{-5} = \frac{b}{2} = \frac{c}{1}$$

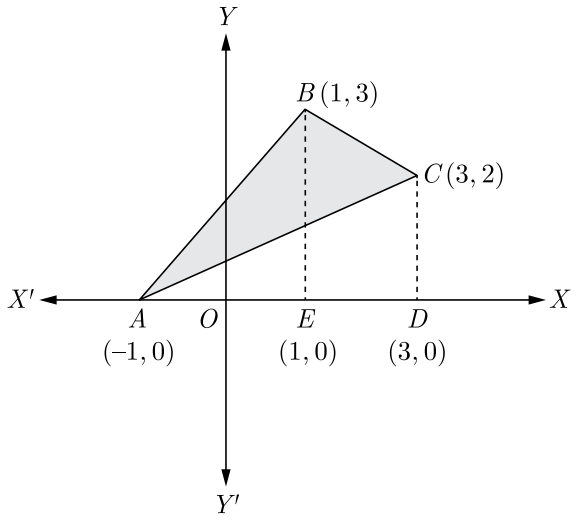
Hence, the required equation of the plane is

$$\begin{aligned} -5(x-1)+2(y+1)+1(z-1) &= 0 \\ -5x+5+2y+2+z-1 &= 0 \\ 5x-5-2y-2-z+1 &= 0 \\ 5x-2y-z &= 6 \end{aligned}$$

34. Using integration, find the area of the region bounded by the triangle whose vertices are $(-1,0)$, $(1,3)$ and $(3,2)$.

Sol :

Let $A(-1,0)$, $B(1,3)$ and $(3,2)$ be the vertices of the ΔABC.



Equation of AB is

$$\begin{aligned} y-3 &= \frac{3-0}{1+1}(x+1) \\ y &= \frac{3}{2}(x+1) \end{aligned}$$

Equation of BC is

$$\begin{aligned} y-3 &= \frac{2-3}{3-1}(x-1) \\ y-3 &= -\frac{1}{2}(x-1) \\ y &= -\frac{1}{2}(x-1)+3 \\ y &= \frac{-x+7}{2} \end{aligned}$$

Equation of AC is

$$y-0 = \frac{2-0}{3+1}(x+1) \Rightarrow y = \frac{1}{2}(x+1)$$

Required area of the shaded region is

Area of ABEA + Area of BCDEB − Area of $ACDA$

$$\begin{aligned} &= \int_{-1}^{1} y_{AB}\,dx + \int_{1}^{3} y_{BC}\,dx - \int_{-1}^{3} y_{AC}\,dx \\ &= \int_{-1}^{1} \frac{3}{2}(x+1)\,dx + \int_{1}^{3} \frac{-x+7}{2}\,dx - \int_{-1}^{3} \frac{1}{2}(x+1)\,dx \\ &= \frac{3}{2}\left[\frac{x^2}{2}+x\right]_{-1}^{1} + \frac{1}{2}\left[-\frac{x^2}{2}+7x\right]_{1}^{3} - \frac{1}{2}\left[\frac{x^2}{2}+x\right]_{-1}^{3} \\ &= \frac{3}{2}\left[\left(\frac{1}{2}+1\right)-\left(\frac{1}{2}-1\right)\right] + \frac{1}{2}\left[\left(-\frac{9}{2}+21\right)-\left(-\frac{1}{2}+7\right)\right] \\ &\qquad -\frac{1}{2}\left[\left(\frac{9}{2}+3\right)-\left(\frac{1}{2}-1\right)\right] \\ &= \frac{3}{2}\left[\frac{3}{2}+\frac{1}{2}\right] + \frac{1}{2}\left[\frac{33}{2}-\frac{13}{2}\right] - \frac{1}{2}\left[\frac{15}{2}+\frac{1}{2}\right] \\ &= \frac{3}{2}\times 2 + \frac{1}{2}\times 10 - \frac{1}{2}\times 8 \\ &= 3+5-4 = 4 \text{ sq. units} \end{aligned}$$

Hence, the required area = 4 sq. units.

OR

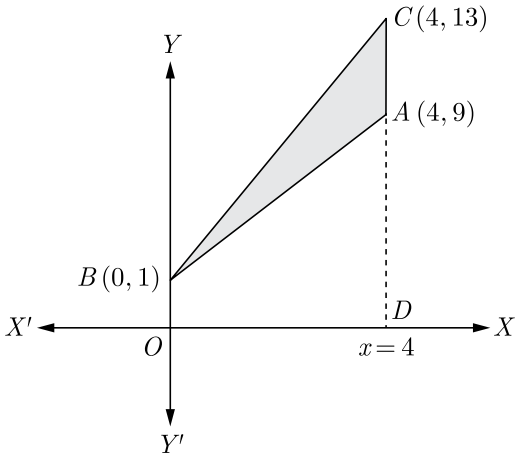
Using integration, find the area of the triangular region whose have the equation $y = 2x+1$, $y = 3x+1$ and $x = 4$.

Sol :

Let the equation of sides of ΔABC be

$$\begin{aligned} AB &: y = 2x+1 \\ BC &: y = 3x+1 \end{aligned}$$

and $$AC : x = 4$$



Solving the equation of AB and AC, we get

$$x = 4 \text{ and } y = 2\times 4+1 = 9$$

Coordinates of A are $(4,9)$.

Solving equation of BC and AC, we get

$$x = 4, \text{ and } y = 3\times 4+1 = 13$$

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$$-1-x-x^2+2x+2x^2+2x^3-1+x-x^2$$

$$=\frac{-2x+2x^2-2x^3}{(1+x+x^2)^2}$$

$$=\frac{2x^2-2}{(1+x+x^2)^2}=\frac{2(x-1)(x+1)}{(1+x+x^2)^2}$$

For maxima and minima, $\frac{dy}{dx}=0$

$$2(x-1)(x+1)=0$$

$$x=1,\ x=-1$$

$$(1+x+x^2)^2(4x)-(2x^2-2)$$

$$\frac{d^2y}{dx^2}=\frac{2(1+x+x^2)(1+2x)}{(1+x+x^2)^4}$$

At $x=1$,

$$\frac{d^2y}{dx^2}=\frac{(1+1+1)^2(4)-2(2-2)(1+1+1)(1+2)}{(1+1+1)^4}$$

$$=\frac{9\times 4}{81}=\frac{4}{9}>0 \text{ (minima)}$$

At $x=-1$,

$$\frac{d^2y}{dx^2}=\frac{(1-1+1)^2(-4)-2(2-2)(1-1+1)(1-2)}{(1-1+1)^4}$$

$$=\frac{-4}{1}=-4<0 \text{ (maxima)}$$

Thus, the minimum value of the function at $x=1$ is

$$y=\frac{1-1+1}{1+1+1}=\frac{1}{3}$$

Thus (d) is correct option.

14. $\int\sqrt{1+\cos x}\,dx$ is equal to

(a) $2\sin\left(\frac{x}{2}\right)+C$

(b) $\sqrt{2}\sin\left(\frac{x}{2}\right)+C$

(c) $2\sqrt{2}\sin\left(\frac{x}{2}\right)+C$

(d) $\frac{1}{2}\sin\left(\frac{x}{2}\right)+C$

Sol :

$$\int\sqrt{1+\cos x}\,dx=\int\sqrt{2\cos^2(x/2)}\,dx$$

$$=\sqrt{2}\int\cos(x/2)\,dx$$

$$=\sqrt{2}\frac{\sin(x/2)}{1/2}+C$$

$$=2\sqrt{2}\sin(x/2)+C$$

Thus (c) is correct option.

15. If $A=\begin{bmatrix}3 & 1\\ -1 & 2\end{bmatrix}$, then

(a) $A^2+7A-5I=O$ (b) $A^2-7A+5I=O$

(c) $A^2+5A-7I=O$ (d) $A^2-5A+7I=O$

Sol :

We have, $A=\begin{bmatrix}3 & 1\\ -1 & 2\end{bmatrix}$

$$A^2=\begin{bmatrix}3 & 1\\ -1 & 2\end{bmatrix}\begin{bmatrix}3 & 1\\ -1 & 2\end{bmatrix}=\begin{bmatrix}8 & 5\\ -5 & 3\end{bmatrix}$$

and $5A=\begin{bmatrix}15 & 5\\ -5 & 10\end{bmatrix}$

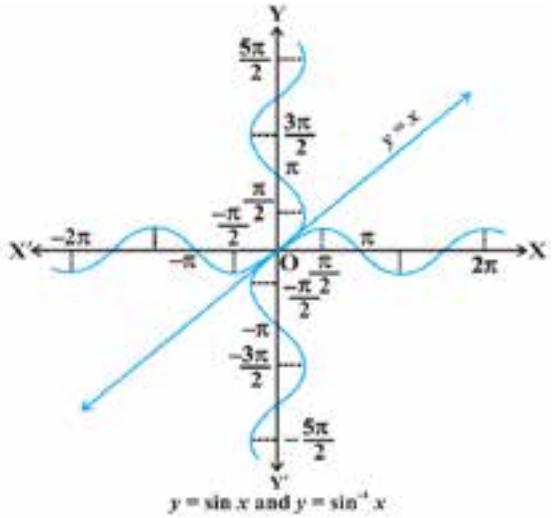
Now, $A^2=5A-7I$

$$=\begin{bmatrix}8 & 5\\ -5 & 3\end{bmatrix}-\begin{bmatrix}15 & 5\\ 5 & 10\end{bmatrix}+\begin{bmatrix}7 & 0\\ 0 & 7\end{bmatrix}$$

$$=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}=O$$

Thus (d) is correct option.

16. Which vertical lines act as asymptotes for the graph of $y=\tan x$?



(a) $x=\pi$ and $x=2\pi$

(b) $x=-\pi/2,\pi/2,3\pi/2$

(c) $x=-\pi,0,\pi$

(d) $x=-\pi/4,\pi/4,3\pi/4$

Sol :

The graph shows vertical dotted lines where the curve shoots upward and downward without touching them. These lines are vertical asymptotes of the tangent function, occurring at $x=(2n+1)\pi/2$.

Thus (b) is correct option.

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8. It is given that the events A and B are such that $P(A)=\frac{1}{4}$, $P(A/B)=\frac{1}{2}$ and $P(B/A)=\frac{2}{3}$. Then, $P(B)$ is equal to

(a) $\frac{1}{2}$ (b) $\frac{1}{6}$

(c) $\frac{1}{3}$ (d) $\frac{2}{3}$

Sol :

We know that,

$$P\left(\frac{A}{B}\right)=\frac{P(A\cap B)}{P(B)}$$

and $$P\left(\frac{B}{A}\right)=\frac{P(B\cap A)}{P(A)}$$

$$P(B)=\frac{P\left(\frac{B}{A}\right)\cdot P(A)}{P\left(\frac{A}{B}\right)}=\frac{\left(\frac{2}{3}\right)\cdot\left(\frac{1}{4}\right)}{\left(\frac{1}{2}\right)}=\frac{1}{3}$$

Thus (c) is correct option.

9. If $\vec{a}\cdot\vec{b}=-|\vec{a}||\vec{b}|$, then the angle between a and b is

(a) 45° (b) 180°

(c) 90° (d) 60°

Sol :

Given,

$$\vec{a}\cdot\vec{b}=-|\vec{a}||\vec{b}|$$

$$|\vec{a}||\vec{b}|\cos\theta=-|\vec{a}||\vec{b}|$$

$$\cos\theta=-1\Rightarrow\theta=180^\circ$$

Thus (b) is correct option.

10. Equation of the line passing through $(2,-1,1)$ and parallel to the line $\frac{x-5}{4}=\frac{y+1}{-3}=\frac{z}{5}$ is

(a) $\frac{x-2}{4}=\frac{y+1}{-3}=\frac{z-1}{5}$

(b) $\frac{x-2}{4}=\frac{y+1}{3}=\frac{z-1}{5}$

(c) $\frac{x-2}{-4}=\frac{y+1}{-3}=\frac{z-1}{5}$

(d) None of the above

Sol :

Equation of line passing through (x_1,y_1,z_1) and parallel to the line $\frac{x-x_2}{a}=\frac{y-y_2}{b}=\frac{z-z_2}{c}$ is

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}$$

Equation of line is

$$\frac{x-2}{4}=\frac{y+1}{-3}=\frac{z-1}{5}.$$

Thus (a) is correct option.

11. If A is 3×4 matrix and B is a matrix such that $A'B$ and BA' are both defined, then B is of the type

(a) 4×4 (b) 3×4

(c) 4×3 (d) 3×3

Sol :

Given, A is 3×4 matrix. So, A' is 4×3 matrix. Since, $A'B$ is defined, therefore B should be 3×4. Also, BA' is defined, so B should be 3×4.
Hence, B is the type 3×4.
Thus (b) is correct option.

12. $\int\frac{\sec^2(\sin^{-1}x)}{\sqrt{1-x^2}}dx$ is equal to

(a) $\sin(\tan^{-1}x)+C$ (b) $\tan(\sec^{-1}x)+C$

(c) $\tan(\sin^{-1}x)+C$ (d) $-\tan(\cos^{-1}x)+C$

Sol :

We have

$$I=\int\frac{\sec^2(\sin^{-1}x)}{\sqrt{1-x^2}}dx$$

On Substituting $\sin^{-1}x=t\Rightarrow\frac{dt}{dx}=\frac{1}{\sqrt{1-x^2}}$

$$dt=\frac{1}{\sqrt{1-x^2}}dx$$

$$I=\int\sec^2t\,dt$$

$$=\tan t+C=\tan(\sin^{-1}x)+C$$

Thus (c) is correct option.

13. For all real values of x, the minimum value of $\frac{1-x+x^2}{1+x+x^2}$ is

(a) 0 (b) 1

(c) 3 (d) $\frac{1}{3}$

Sol :

Let $y=\frac{1-x+x^2}{1+x+x^2}$ $(1+x+x^2)(1-x+x^2)$

$$\frac{dy}{dx}=\frac{-(1-x+x^2)\frac{d}{dx}(1+x+x^2)}{(1+x+x^2)^2}$$

$$=\frac{(1+x+x^2)(-1+2x)-(1-x+x^2)(1+2x)}{(1+x+x^2)^2}$$

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Then, $y = Ax$

Differentiating, we get

$$\frac{dy}{dx} = A$$

So given equation can be expressed as

$$y = x\frac{dy}{dx}$$

$$x\frac{dy}{dx} = y$$

Hence, Both Assertion and reason are true and reason is correct explanation for assertion.

Thus (a) is correct option.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Find the vector equation of the line passing through the point $A(1,2,-1)$ and parallel to the line $5x-25=14-7y=35z$.

Sol :

We have $5x-25 = 14-7y = 35z.$

$$\frac{x-5}{1/5} = \frac{2-y}{1/7} = \frac{z}{1/35}$$

$$\frac{x-5}{1/5} = \frac{y-2}{-1/7} = \frac{z}{1/35}$$

Direction ratio of the given line are $\frac{1}{5}, -\frac{1}{7}, \frac{1}{35}$.

Direction ratio of a line parallel to the given line are $\frac{1}{5}, -\frac{1}{7}, \frac{1}{35}$.

The required equation of a line passing through the point $A(1,2,-1)$ and parallel to the given line is

$$\frac{x-1}{1/5} = \frac{y-2}{-1/7} = \frac{z+1}{1/35}$$

OR

The x-coordinate of point on the line joining the points $P(2,2,1)$ and $Q(5,1,-2)$ is 4. Find its z -coordinate.

Sol :

The equation of the line joining the points $P(2,2,1)$ and $Q(5,1,-2)$ is

$$\frac{x-2}{5-2} = \frac{y-2}{1-2} = \frac{z-1}{-2-1}$$

$$\frac{x-2}{3} = \frac{y-2}{-1} = \frac{z-1}{-3}$$

Since at a point, x-coordinate is 4, we have

$$\frac{4-2}{3} = \frac{z-1}{-3} \Rightarrow z = -1$$

22. Show that the function $f(x) = x^3 - 3x^2 + 3x$, $x \in R$ is increasing on R.

Sol :

We know that, a continuous function $y = f(x)$ is said to be increasing on R, if $\frac{dy}{dx} \geq 0$, $\forall\, x \in R$.

We have $y = x^3 - 3x^2 + 3x$

Differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = 3x^2 - 6x + 3$$

$$= 3(x^2 - 2x + 1)$$

$$= 3(x-1)^2$$

Here $3(x-1)^2 \geq 0$ for all real values of x, i.e. $\forall\, x \in R$. Thus $\frac{dy}{dx} \geq 0$, $\forall\, x \in R$. Hence, the given function is increasing on R.

23. Find the general solution of following equation $e^x \tan y\, dx + (1-e^x)\sec^2 y\, dy = 0$

Sol :

Given differential equation is

$$e^x \tan y\, dx + (1-e^x)\sec^2 y\, dy = 0$$

$$e^x \tan y\, dx = (1-e^x)\sec^2 y\, dy$$

$$(1-e^x)\sec^2 y\, dy = -e^x \tan y\, dx$$

Separating the variables, we get

$$\frac{\sec^2 y}{\tan y}dy = -\frac{e^x}{1-e^x}dx$$

Integrating both sides, we get

$$\int \frac{\sec^2 y}{\tan y}dy = \int \frac{e^x}{e^x - 1}dx + C$$

$$\log|\tan y| = \log|e^x - 1| + \log C$$

$$\log|\tan y| = \log|e^x - 1| + \log C$$

$$\log\left|\frac{\tan y}{e^x - 1}\right| = \log C$$

$$\frac{\tan y}{e^x - 1} = C$$

$$\tan y = C(e^x - 1)$$

$\tan y = C(e^x - 1)$ is the required solution.

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or $x = \frac{4y+3}{6y-4} \in B$

$B = R - \left\{\frac{2}{3}\right\}$

or For every value of y except $y = \left\{\frac{2}{3}\right\}$, there is a pre-image $x = \frac{4y+3}{6y-4} = g(y)$.

or $x \in A$

f is onto.

SECTION - E

Case study based questions are compulsory.

36. ICAR-Indian Agricultural Research Institute is an autonomous body responsible for co-ordinating agricultural education and research in India. It reports to the Department of Agricultural Research and Education, Ministry of Agriculture. The Union Minister of Agriculture serves as its president. It is the largest network of agricultural research and education institutes in the world.



ICAR grows vegetables and grades each one as either good or bad for its taste, good or bad for its size, and good or bad for its appearance. Overall 78% of the vegetables have a good taste. However, only 69% of the vegetables have both a good taste and a good size. Also, 5% of the vegetables have both a good taste and a good appearance, but a bad size. Finally, 84% of the vegetables have either a good size or a good appearance.

(i) If a vegetable has a good taste, what is the probability that it also has a good size?

(ii) If a vegetable has a bad size and a bad appearance, what is the probability that it has a good taste?

Sol :

Let T be the event 'good taste', let S be the event 'good size', and let A be the event 'good appearance'.

(i) $P(T) = 0.78$

$P(T \cap S) = 0.69$

$P(T \cap S' \cap A) = 0.05$

$P(S \cup A) = 0.84$

Therefore,

$$P(S \mid T) = \frac{P(T \cap S)}{P(T)} = \frac{0.69}{0.78} = 0.885$$

(ii) Notice that

$$P(S' \cap A') = 1 - P(S \cup A)$$
$$= 1 - 0.84 = 0.16$$

Also,

$$P(T \cap S') = P(T) - P(T \cap S)$$
$$= -0.78 - 0.69 = 0.09$$

So that

$$P(T \cap S' \cap A') = P(T \cap S') - P(T \cap S' \cap A)$$
$$= 0.09 - 0.05 = 0.04$$

Therefore,

$$P(T \mid S' \cap A') = \frac{P(T' \cap S' \cap A')}{P(S' \cap A')} = \frac{0.04}{0.16}$$
$$= 0.25$$

37. Pastry is a dough of flour, water and shortening that may be savoury or sweetened. Sweetened pastries are often described as bakers' confectionery. The word "pastries" suggests many kinds of baked products made from ingredients such as flour, sugar, milk, butter, shortening, baking powder, and eggs.



The Sunrise Bakery Pvt Ltd produces three basic pastry mixes A, B and C. In the past the mix of ingredients has shown in the following matrix:

		Flour	Fat	Sugar
Type	A	5	1	1
	B	6.5	2.5	0.5
	C	4.5	3	2

(All quantities in kg)

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Hence, $f(x)$ is increasing function when $x > -1$ throughout its domain.

27. Show that the modulus function $f : R \to R$, given by $f(x) = |x|$, is neither one-one nor onto, where $|x|$ is x, if x is positive or 0 and $|x|$ is $-x$, if x is negative.

Sol :

Let $x_1 = 3$ and $x_2 = -3$

Given: $f(x) = |x|$

$f(x_1) = |x_1|$

$f(3) = |3| = 3$

and $f(x_2) = |x_2|$

$f(-3) = |-3| = 3$

Clearly, $f(x_1) = f(x_2)$ but $x_1 \neq x_2$

So, f is not one-one function.

Let $f(x) = -2 \in R$, then $|x| = -2$ which is not possible as there is no value of x corresponding to which $|x| = -2$.

So, f is not onto function.

Hence, f is neither one-one nor onto.

OR

Show that the Signum function of $f : R \to R$, given by

$$f(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -1, & \text{if } x < 0 \end{cases}$$

is neither one-one nor onto.

Sol :

Let $x_1 = 3 > 0$ and $x_2 = 2 > 0 \in R$

Then $f(x_1) = f(3) = 1$

and $f(x_2) = f(4) = 1$

Clearly, $f(x_1) = f(x_2)$ but $x_1 \neq x_2$

So, f is not one-one function.

Let $f(x) = 5 \in R$, there is no value of x for which $f(x) = 5$ as the range of f is $\{-1, 0, 1\}$.

So, f is onto function.

Hence, f is neither one-one nor onto function.

28. Find the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$.

Sol :

We have $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$

and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$

Comparing the given equations of lines with standard form we have

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$$

and $$\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}.$$

we get $x_1 = 1,\ y_1 = 2,\ z_1 = 3;$

$a_1 = 2,\ b_1 = 3,\ c_1 = 4$

and $x_2 = 2,\ y_2 = 4,\ z_2 = 5;$

$a_2 = 3,\ b_2 = 4,\ c_2 = 5$

Now $$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = \begin{vmatrix} 2-1 & 4-2 & 5-3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 2 & 2 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$$

$= 1(15-16) - 2(10-12) + 2(8-9)$

$= -1 + 4 - 2 = 1$

Now, $\sqrt{(b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 + (a_1b_2 - a_2b_1)^2}$

$= \sqrt{(3\times5 - 4\times4)^2 + (4\times3 - 5\times2)^2 + (2\times4 - 3\times3)^2}$

$= \sqrt{(15-16)^2 + (12-10)^2 + (8-9)^2}$

$= \sqrt{(-1)^2 + (2)^2 + (-1)^2}$

$= \sqrt{1+4+1} = \sqrt{6}$

$$\text{SD} = \frac{\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}}{\sqrt{(b_1c_2 - b_2c_1)^2 + (c_1a_2 - c_2a_1)^2 + (a_1b_2 - a_2b_1)^2}}$$

$= \frac{1}{\sqrt{6}}$ units,

Which is the required shortest distance.

29. Maximise and minimise $Z = x + 2y$ subject to the constraints

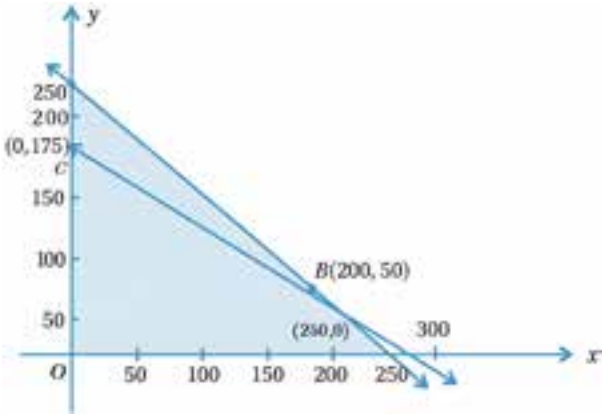
$x + 2y \geq 100$

$2x - y \leq 0$

$2x + y \leq 200$

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17. The corner points of the bounded feasible region of an LPP are O(0,0), A(250,0), B(200,50), and C(0,175). If the maximum value of the objective function $Z = 2ax +$ by occurs at the points A(250,0) and B(200,50), then the relation between a and b is to be determined.



(a) 2a=b (b) 2a=3b

(c) a=b (d) a=2b

Sol :

For the objective function Z=2ax+by, the maximum value occurs at two distinct points, A(250,0) and B(200,50).

This implies that the values of Z at A and B must be equal.

At A(250,0) : Z = 2a(250)+b(0)=500a

At B(200,50): Z = 2a(200)+b(50)=400a+50b

Since both are equal at the maximum,

500a=400a+50b; 100a = 50b; 2a=b

Thus (a) is correct option.

18. The maximum value of xe^{-x} is

(a) e (b) $1/e$

(c) $-e$ (d) $-1/e$

Sol :

Let $f(x) = xe^{-x}$

$$f'(x) = -xe^{-x} + e^{-x} = e^{-x}(1-x)$$

For maxima or minima, put $f'(x) = 0$

$$e^{-x}(1-x) = 0 \Rightarrow x = 1 \qquad [\because\ e^{-x} < 0, \forall x \in R]$$

Further, $f''(x) < 0$ at $x = 1$

Therefore, $f(x)$ attains its maxima at $x = 1$ and the maximum value is $\frac{1}{e}$.

Thus (b) is correct option.

19. **Assertion :** The general solution of $\frac{dy}{dx} + y = 1$ is $ye^x = e^x + c$

Reason : The number of arbitrary constants is in the general solution of the differential equation is equal to the order of differential equation.

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

Sol :

Given that, $\frac{dy}{dx} + y = 1$

$$\frac{dy}{dx} = 1 - y$$

$$\frac{dy}{1-y} = dx$$

$$\int \frac{dy}{1-y} = \int dx$$

$$-\log(1-y) = x$$

$$1 - y = e^{-x}$$

$$ye^x = e^x + c$$

Since, order of differential equation is the number of arbitrary Constants, so given Reason is true.

Hence, Both Assertion and reason are true but reason is not a correct explanation for assertion.

Thus (b) is correct option.

20. **Assertion :** The elimination of four arbitrary constants in $y = (c_1 + c_2 + c_3 e^{c_4})x$ results into a differential equation of the first order $x\frac{dy}{dx} = y$

Reason : Elimination of n arbitrary constants requires in general a differential equation of the nth order.

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

Sol :

Here given Reason is standard property.

So, given Reason is true.

Let $c_1 + c_2 + c_3 e^{c_4} =$ A (constant)

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$$|B| = \begin{vmatrix} 1 & -2 \\ -1 & 3 \end{vmatrix} = 3 - 2 = 1 \neq 0$$

and $$|AB| = \begin{vmatrix} -1 & 5 \\ 5 & -14 \end{vmatrix} = 14 - 25 = -11 \neq 0$$

Thus, A, B and AB are non-singular matrices, so their inverse exists.

Now, $$adj\,(A) = \begin{bmatrix} -4 & -3 \\ -1 & 2 \end{bmatrix}$$

$$adj\,(B) = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

and $$adj(AB) = \begin{bmatrix} -14 & -5 \\ -5 & -1 \end{bmatrix}$$

$$(AB)^{-1} = \frac{1}{|AB|} adj(AB)$$

$$= \frac{1}{-11}\begin{bmatrix} -15 & -5 \\ -5 & -1 \end{bmatrix}$$

$$(AB)^{-1} = \frac{1}{11}\begin{bmatrix} 14 & 5 \\ 5 & 1 \end{bmatrix} \quad \text{...(i)}$$

$$A^{-1} = \frac{1}{|A|} adj(A)$$

$$= \frac{1}{-11}\begin{bmatrix} -4 & -3 \\ -1 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{11}\begin{bmatrix} 4 & 3 \\ 1 & -2 \end{bmatrix}$$

and $$B^{-1} = \frac{1}{|B|} adj(B)$$

$$= \frac{1}{1}\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$$

Now, $$B^{-1}A^{-1} = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}\frac{1}{11}\begin{bmatrix} 4 & 3 \\ 1 & -2 \end{bmatrix}$$

$$= \frac{1}{11}\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 4 & 3 \\ 1 & -2 \end{bmatrix}$$

$$= \frac{1}{11}\begin{bmatrix} 12+2 & 9-4 \\ 4+1 & 3-2 \end{bmatrix}$$

$$= \frac{1}{11}\begin{bmatrix} 14 & 5 \\ 5 & 1 \end{bmatrix}$$

$$= (AB)^{-1} \quad \text{[from Eq. (i)]}$$

Hence, $$(AB)^{-1} = B^{-1}A^{-1}.$$

31. Find $\int \frac{(2x-5)\,e^{2x}}{(2x-3)^3}\,dx$.

Sol :

We have $$I = \int \frac{(2x-5)\,e^{2x}}{(2x-3)^3}\,dx$$

$$= \int \frac{(2x-3-2)\,e^{2x}}{(2x-3)^3}\,dx$$

$$= \int \frac{e^{2x}}{(2x-3)^2}\,dx - 2\int \frac{e^{2x}}{(2x-3)^3}\,dx$$

$$= \int e^{2x}(2x-3)^{-2}\,dx - 2\int e^{2x}(2x-3)^{-3}\,dx$$

Using integration by parts technique we get

$$I = (2x-3)^{-2}\int e^{2x}\,dx - \int\left\{\frac{d}{dx}(2x-3)^{-2}\int e^{2x}\,dx\right\}dx - 2\int e^{2x}(2x-3)^{-3}\,dx$$

$$= (2x-3)^{-2}\frac{e^{2x}}{2} - \int -2(2x+3)^{-3}\times 2\times\frac{e^{2x}}{2}\,dx - 2\int e^{2x}(2x-3)^{-3}\,dx$$

$$= \frac{e^{2x}(2x-3)^{-2}}{2} + 2\int e^{2x}(2x-3)^{-3}\,dx - 2\int e^{2x}(2x-3)^{-3}\,dx$$

$$= \frac{e^{2x}(2x-3)^{-2}}{2} + C$$

OR

Evaluate $\int \frac{x\sin^{-1}x}{\sqrt{1-x^2}}\,dx$.

Sol :

We have $$I = \int \frac{x\sin^{-1}x}{\sqrt{1-x^2}}\,dx$$

Substituting $\sin^{-1}x = t$ we get

$$\frac{1}{\sqrt{1-x^2}}\,dx = dt$$

Thus $$I = \int t\sin t\,dt$$

Using integration by parts, taking t as the first function and $\sin t$ as the second function, we get

$$I = \int \underset{I}{t}\,\underset{II}{\sin t}\,dt$$

$$= t\int \sin t\,dt - \int\left[\frac{d}{dt}(t)\cdot\int \sin t\,dt\right]dt$$

$$= -t\cos t - \int(1\times -\cos t)\,dt$$

$$= -t\cos t + \int \cos t\,dt$$

$$= -t\cos t + \sin t + C$$

$$= -t\sqrt{1-\sin^2 t} + \sin t + C$$

Substituting $t = \sin^{-1}x \Rightarrow x = \sin t$ we have

$$I = -\sin^{-1}x\sqrt{1-x^2} + x + C$$

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$$x,\ y \geq 0$$

Solve the above LPP graphically.

Sol :

We have to minimise and maximise

$$Z = x + 2y$$

Subject to constraints,

$$x + 2y \geq 100 \quad \text{...(i)}$$

$$2x - y \leq 0 \quad \text{...(ii)}$$

$$2x + y \leq 200 \quad \text{...(iii)}$$

and $x \geq 0,\ y \geq 0$

(i) Region corresponding to $x + 2y \geq 100$:

Line $x + 2y = 100$			
x	0	100	Point $(0,0)$ is false for $x + 2y \geq 100$. So, the region is away from the origin.
y	50	0	

(ii) Region corresponding to $2x - y \leq 0$:

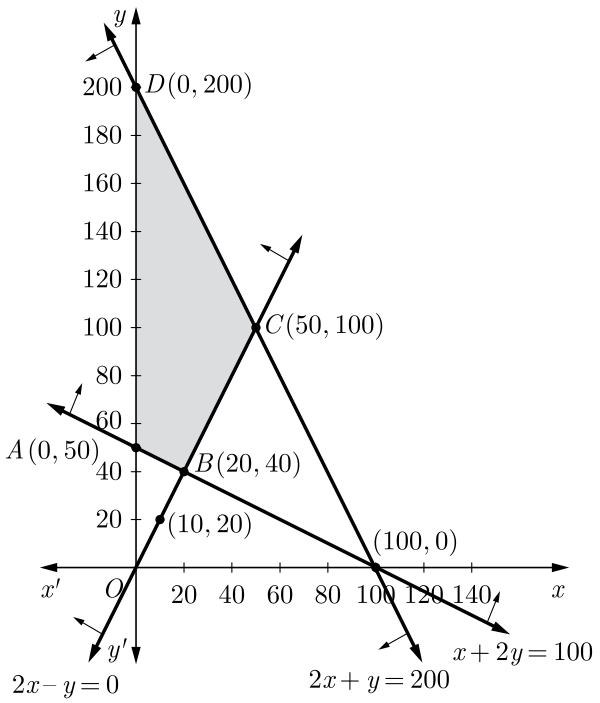
Line $2x - y = 0$			
x	0	10	Point $(0,5)$ is true for $2x - y \leq 0$, thus region includes $(0,5)$ and towards y axis.
y	0	20	

(iii) Region corresponding to $2x + y \leq 200$:

Line $2x + y = 200$			
x	0	100	Point $(0,0)$ is true for $2x + y \leq 200$. So, the region is towards the origin.
y	200	0	

Since $x \geq 0,\ y \geq 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is $ABCDA$. Solving equations $2x - y = 0$ and $x + 2y = 100$, we get $B(20,40)$. Again, solving equations $2x - y = 0$ and $2x + y = 200$, we get $C(50,100)$. The corner points of the feasible region are $A(0,50)$, $B(20,40)$, $C(50,100)$ and $D(0,200)$.



Since, the feasible region is a bounded region, we can check the profit function at all the vertices to find the maxima. The values of Z at corner points are given below

Corner points	$Z = x + 2y$
$A(0,50)$	$Z = 0 + 2 \times 50 = 100$
$B(20,40)$	$Z = 20 + 2 \times 40 = 100$
$C(50,100)$	$Z = 50 + 2 \times 100 = 250$
$D(0,200)$	$Z = 0 + 2 \times 200 = 400$

The maximum value of Z is 400 at $D(0,200)$ and the minimum value of Z is 100 at the points on the line segment joining $A(0,50)$ and $B(20,40)$.

30. If $A = \begin{bmatrix} 2 & 3 \\ 1 & -4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1}A^{-1}$.

Sol :

We have, $A = \begin{bmatrix} 2 & 3 \\ 1 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$

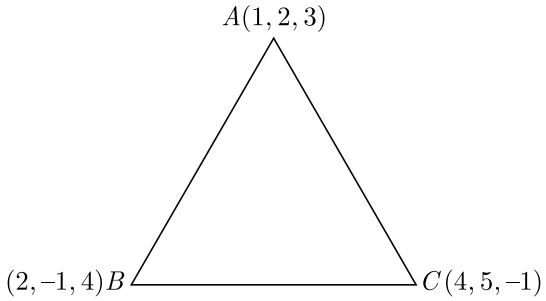
$$AB = \begin{bmatrix} 2 & 3 \\ 1 & -4 \end{bmatrix}\begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2-3 & -4+9 \\ 1+4 & -2-12 \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 5 & -14 \end{bmatrix}$$

Now, $|A| = \begin{vmatrix} 2 & 3 \\ 1 & -4 \end{vmatrix} = -8 - 3 = -11 \neq 0$

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below.



Position vectors of the vectors A, B and C of ΔABC are

$$\overrightarrow{OA} = \hat{i} + 2\hat{j} + 3\hat{k},$$

$$\overrightarrow{OB} = 2\hat{i} - \hat{j} + 4\hat{k}$$

and $$\overrightarrow{OC} = 4\hat{i} + 4\hat{j} - \hat{k}$$

Now, $$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

$$= (2\hat{i} - \hat{j} + 4\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$= \hat{i} - 3\hat{j} + \hat{k}$$

and $$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA}$$

$$= (4\hat{i} + 4\hat{j} - \hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$= (3\hat{i} + 3\hat{j} - 4\hat{k})$$

Now, $$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 1 \\ 3 & 3 & -4 \end{vmatrix}$$

$$= \hat{i}(12 - 3) - \hat{j}(-4 - 3) + \hat{k}(3 + 9)$$

$$= 9\hat{i} + 7\hat{j} + 12\hat{k}$$

and $$\left|\overrightarrow{AB} \times \overrightarrow{AC}\right| = \sqrt{(9)^2 + (7)^2 + (12)^2}$$

$$= \sqrt{81 + 49 + 144} = \sqrt{274}$$

Area of ΔABC is given by

$$\Delta = \frac{1}{2}\left|\overrightarrow{AB} \times \overrightarrow{AC}\right|$$

$$= \frac{1}{2}\sqrt{274} \text{ sq units}$$

OR

If lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$ intersect, then find the value of k and hence, find the equation of the plane containing these lines.

Sol :

Given equation of lines are

$$\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4} \quad ...(1)$$

and $$\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1} \quad ...(2)$$

Since, these lines intersect each other, therefore the shortest distance between them will be zero.

Now, comparing these lines with standard form,

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$$

and $$\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2},$$

we get $$x_1 = 1,\ y_1 = -1,\ z_1 = 1$$

$$x_2 = 3,\ y_2 = k,\ z_2 = 0$$

$$a_1 = 2,\ b_1 = 3,\ c_1 = 4$$

$$a_2 = 1,\ b_2 = 2,\ c_2 = 1$$

Since, two lines are intersect, so shortest distance $= 0$.

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

$$\begin{vmatrix} 3-1 & k+1 & 0-1 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} 2 & k+1 & -1 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

$$2(3-8) - (k+1)(2-4) - 1(4-3) = 0$$

$$2(-5) - (k+1)(-2) - 1(1) = 0$$

$$-10 + 2(k+1) - 1 = 0 \Rightarrow 2(k+1) = 11$$

$$k = \frac{11}{2} - 1 = \frac{9}{2}$$

Let the required equation of plane be

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

$$a(x-1) + b(y+1) + c(z-1) = 0 \quad ...(3)$$

Where, a, b and c are direction ratios of normal and $(1, -1, 1)$ is the point on the line (1).

As, plane contains the intersecting lines, so normal to the plane is perpendicular to both the lines.

$$2a + 3b + 4c = 0$$

and $$a + 2 + c = 0 \qquad a_1a_2 + b_1b_2 + c_1c_2 = 0$$

$$\frac{a}{3-8} = \frac{b}{4-2} = \frac{c}{4-3}$$

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24. If $P(not\ A) = 0.7$, $P(B) = 0.7$ and $P(B/A) = 0.5$, then find $P(A/B)$.

Sol :

We have $P(A') = 0.7,\ P(B) = 0.7$

and $P\left(\frac{B}{A}\right) = 0.5$

Now $P(A) = 1 - P(A')$

$= 1 - 0.7 = 03$

Now, $P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)}$

$0.5 = \frac{P(A \cap B)}{0.3}$

$P(A \cap B) = 0.15$

$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{0.15}{0.7}$

Thus $P\left(\frac{A}{B}\right) = \frac{3}{14}$

25. Maximize $Z = x + y$, subject to $x - y \le -1$, $x + y \le 0$, $x,\ y \ge 0$.

Sol :

Maximize $Z = x + y$

Subject to $x - y \le -1$ ...(i)

$x + y \le 0$

and $x \ge 0,\ y \ge 0$

(i) Region corresponding to $x - y \le -1$:

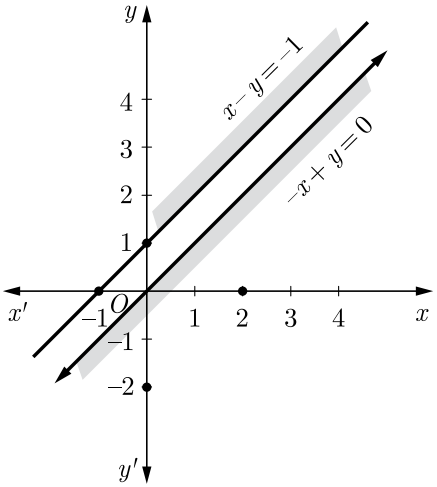
Line $x - y = -1$			
x	0	-1	Point $(0,0)$ is false for $x - y \le -1$. So, the region is away from the origin.
y	1	0	

(ii) Region corresponding to $x + y \le 0$:

Line $x + y = 0$			
x	0	2	Point $(0, -2)$ is true for $x + y \le 0$. So, the region includes point $(0, -2)$.
y	0	2	

Since $x \ge 0,\ y \ge 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Here we get no feasible region (constraints are inconsistent). Hence, Z has no maximum value.



SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. Show that $y = \log(1 + x) - \frac{2x}{2 + x}$, $x > -1$ is an increasing function of x throughout its domain.

Sol :

Here $y = \log(1 + x) - \frac{2x}{2 + x}$, $x > -1$

$\frac{dy}{dx} = \frac{1}{1 + x} - \left[\frac{(2 + x) \cdot 2x \cdot 1}{(2 + x)^2}\right]$

$\frac{dy}{dx} = \frac{1}{1 + x} - \left[\frac{4 + 2x - 2x}{(2 + x)^2}\right]$

$\frac{dy}{dx} = \frac{1}{1 + x} - \frac{4}{(2 + x)^2}$

$= \frac{1}{1 + x} - \frac{4}{(2 + x)^2}$

$= \frac{(2 + x)^2 - 4(1 + x)}{(1 + x)(2 + x)^2}$

$= \frac{4 + x^2 + 4x - 4 - 4x}{(1 + x)(2 + x)^2}$

$= \frac{x^2}{(1 + x)(2 + x)^2}$

For $x > -1,\ x^2 > 0,\ (2 + x)^2 > 0$

and $1 + x > 0$

$f'(x) > 0,\ x > -1$

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Coordinates of C or are $(4,13)$.

Solving equations of AB and BC, we get

$$3x+1 = 2x+1$$

$$x = 0$$

$$y = 1$$

Coordinates of B are $(0,1)$.

$$\text{Required area} = \text{Area of OBCDO} - \text{Area of OBADO}$$

$$= \int_0^4 y_{BC}\,dx - \int_0^4 y_{BA}\,dx$$

$$= \int_0^4 (3x+1)\,dx - \int_0^4 (2x+1)\,dx$$

$$= \left[\frac{3}{2}x^2 + x\right]_0^4 - \left[2\cdot\frac{x^2}{2} + x\right]_0^4$$

$$= \left[\left(\frac{3}{2}\times 16 + 4\right) - (0+0)\right] - [(16+4) - (0+0)]$$

$$= (24+4) - (16+4)$$

$$= 28 - 20 = 8 \text{ sq. units}$$

Hence, the required area $= 8$ sq. units.

35. Let N denote the set of all natural numbers and R be the relation on $N \times N$ defined by $(a,b)R(c,d)$ if $ad(b+c) = bc(a+d)$. Show that R is an equivalence relation.

Sol :

Given $(a,b)R(c,d)$ as $ad(b+c) = bc(a+d)$

$$\forall\, a,b \in N,$$

or $$ab(b+a) = ba(a+b)$$

or $$(a,b)\ R(a,b)$$

R is reflexive. ...(i)

Let $(a,b)R(c,d)$ for $(a,b),(c,d) \in N \times N$

$$ad(b+c) = bc(a+d) \quad \text{...(ii)}$$

Also, $$(c,d)\ R(a,b)$$

$$cb(d+a) = da(c+b)$$ [By commutation of addition and multiplication on N]

R is symmetric. ...(iii)

Let $(a,b)R(c,d)$ and $(c,d)R(e,f)$ for $a,b,c,d,e,f \in N$

$$ad(b+c) = bc(a+d) \quad \text{...(iv)}$$

and $$cf(d+e) = de(c+f) \quad \text{...(v)}$$

Dividing eqn. (iv) by $abcd$ and eqn.(v) by $cdef$

i.e., $$\frac{1}{c}+\frac{1}{b} = \frac{1}{d}+\frac{1}{a}$$

and $$\frac{1}{e}+\frac{1}{d} = \frac{1}{f}+\frac{1}{c}$$

On adding, we get

$$\frac{1}{c}+\frac{1}{b}+\frac{1}{e}+\frac{1}{d} = \frac{1}{d}+\frac{1}{a}+\frac{1}{f}+\frac{1}{c}$$

or, $$af(b+e) = be(a+f)$$

Hence, $$(a,b)\ R(e,f)$$

R is transitive.

From equations (i) , (iii) and (iv), R is an equivalence relation.

OR

Show that the function f in $A = R - \left\{\frac{2}{3}\right\}$ defined as $f(x) = \frac{4x+3}{6x-4}$ is one-one and onto.

Sol :

Given $$f(x) = \frac{4x+3}{6x-4}$$

To show f is one-one :

Let $$f(x_1) = f(x_2),$$

then $$\frac{4x_1+3}{6x_1-4} = \frac{4x_2+3}{6x_2-4}$$

or $$(4x_1+3)(6x_2-4) = (6x_1-4)(4x_2+3)$$

or $$24x_1x_2 - 16x_1 + 18x_2 - 12 = 24x_1x_2 + 18x_1 - 16x_2 - 12$$

or $$-16x_1 + 18x_2 = 18x_1 - 16x_2$$

or $$-16x_1 - 18x_1 = -18x_2 - 16x_2$$

or $$-34x_1 = -34x_2$$

or $$x_1 = x_2$$

of f is one-one.

To show f is onto :

Let, $$y \in B$$

$$y = f(x)$$

or $$y = \frac{4x+3}{6x-4}$$

or $$y(6x-4) = 4x+3$$

or $$6xy - 4y = 4x+3$$

or $$6xy - 4x = 4y+3$$

or $$x(6y-4) = 4y+3$$

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SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. If $y=\tan^{-1}\left(\frac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}\right)$, $x^2\le 1$, then find dy/dx.

Sol :

We have $y=\tan^{-1}\left(\frac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}\right)$ ...(1)

Substituting $x^2=\sin\theta \Rightarrow \theta=\sin^{-1}x^2$ we have

$$\sqrt{1+x^2}=\sqrt{1+\sin\theta}$$
$$=\sqrt{\cos^2\frac{\theta}{2}+\sin^2\frac{\theta}{2}+2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}$$
$$=\sqrt{\left(\cos\frac{\theta}{2}+\sin\frac{\theta}{2}\right)^2}$$

or $\sqrt{1+x^2}=\left(\cos\frac{\theta}{2}+\sin\frac{\theta}{2}\right)$ ...(2)

Similarly

$$\sqrt{1-x^2}=\left(\cos\frac{\theta}{2}-\sin\frac{\theta}{2}\right) \quad ...(3)$$

Adding eq (1) and (2) we have

$$\sqrt{1+x^2}+\sqrt{1-x^2}=2\cos\frac{\theta}{2} \quad ...(4)$$

Subtracting eq (2) from (1) we have

$$\sqrt{1+x^2}-\sqrt{1-x^2}=2\sin\frac{\theta}{2} \quad ...(5)$$

Substituting eq (4) and (5) we have

$$\sqrt{1-x^2}\frac{dy}{dx}=\tan^{-1}\left(\frac{2\cos\frac{\theta}{2}}{2\sin\frac{\theta}{2}}\right)=\tan^{-1}\left(\cot\frac{\theta}{2}\right)$$
$$=\tan^{-1}\left[\tan\left(\frac{\pi}{2}-\frac{\theta}{2}\right)\right]$$
$$=\frac{\pi}{2}-\frac{\theta}{2}$$
$$y=\frac{\pi}{2}-\frac{1}{2}\sin^{-1}x^2$$

Differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx}=-\frac{1}{2}\frac{1}{\sqrt{1-(x^2)^2}}(2x)$$

or $$\frac{dy}{dx}=\frac{-x}{\sqrt{1-x^4}}$$

OR

Find whether the following function is differentiable at $x=1$ and $x=2$ or not.

$$f(x)=\begin{cases}x, & x<1\\ 2-x, & 1\le x\le 2\\ -2+3x-x^2, & x>2\end{cases}$$

Sol :

Given, $f(x)=\begin{cases}x, & x<1\\ 2-x, & 1\le x\le 2\\ -2+3x-x^2, & x>2\end{cases}$

Differentiability at $x=1$

$$\text{LHD}=\lim_{h\to 0}\frac{f(1-h)-f(1)}{-h}$$
$$=\lim_{h\to 0}\frac{(1-h)-[2-(1)]}{-h}=\lim_{h\to 0}\frac{-h}{-h}=1$$
$$\text{RHD}=\lim_{h\to 0}\frac{f(1+h)-f(1)}{h}$$
$$=\lim_{h\to 0}\frac{2-(1+h)-(2-1)}{h}$$
$$=\lim_{h\to 0}\frac{-h}{h}=-1$$

$\text{LHD}\neq\text{RHD}$

So, $f(x)$ is not differentiable at $x=1$.

Differentiability at $x=2$

$$\text{LHD}=\lim_{h\to 0}\frac{f(2-h)-f(2)}{-h}$$
$$=\lim_{h\to 0}\frac{2-(2-h)-(2-2)}{-h}$$
$$=\lim_{h\to 0}\frac{h}{-h}=-1$$
$$\text{RHD}=\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}$$
$$=\lim_{h\to 0}\frac{-2+3(2+h)-(2+h)^2-(2-2)}{h}$$
$$=\lim_{h\to 0}\frac{-2+6+3h-(4+h^2+4h)-0}{h}$$
$$=\lim_{h\to 0}\frac{-h^2-h}{h}$$
$$=\lim_{h\to 0}\frac{-h(h+1)}{h}=-(0+1)=-1$$

$\text{LHD}=\text{RHD}$

So, $f(x)$ is differentiable at $x=2$.

Hence, $f(x)$ is not differentiable at $x=1$, but it differentiable at $x=2$.

33. Using vectors, find the area of the ΔABC, whose vertices are $A(1,2,3)$, $B(2,-1,4)$ and $C(4,5,-1)$.

Sol :

As per question the ΔABC, whose vertices are $A(1,2,3)$, $B(2,-1,4)$ and $C(4,5,-1)$ is shown

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Due to changes in the consumer's tastes it has been decided to change the mixes using the following amendment matrix:

$$\begin{array}{c} \\ \text{Type} \end{array}\begin{array}{c} \begin{matrix} \text{Flour} & \text{Fat} & \text{Sugar} \end{matrix} \\ \begin{bmatrix} 0 & 1 & 0 \\ -0.5 & 0.5 & 0.5 \\ 0.5 & 0 & 0 \end{bmatrix} \end{array}\begin{matrix} A \\ B \\ C \end{matrix}$$

Using matrix algebra you are required to calculate:

(i) The matrix for the new mix:

(ii) The production requirement to meet an order for 50 units of type A, 30 units of type B and 20 units of type C of the new mix;

(iii) The amount of each type that must be made to totally use up 370 kg of flour, 170 kg of fat and 80 kg of sugar that are at present in the stores.

Sol :

(i) Matrix for the new mix : The matrix for the new mix is obtained by adding the amendment matrix to the original mix matrix, i.e.,

$$\begin{bmatrix} 6 & 1 & 1 \\ 6.5 & 2.5 & 0.5 \\ 4.5 & 3 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 \\ -0.5 & 0.5 & 0.5 \\ 0.5 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 2 & 1 \\ 6 & 3 & 1 \\ 5 & 3 & 2 \end{bmatrix}$$

(ii) Production requirement : The production requirement to meet an order for 50 units of type A, 30 units of type B and 20 units of type C of the new mix is given by the matrix product

$$\begin{bmatrix} 50 & 30 & 20 \end{bmatrix}\begin{bmatrix} 5 & 2 & 1 \\ 6 & 3 & 1 \\ 5 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 530 & 250 & 120 \end{bmatrix}$$

Hence, the production requirement to meet a given order of the new mix is 530 kg of flour, 250 kg of fat, 120 kg of sugar.

(iii) Amount of Each Type : Let x_1, x_2 and x_3 be the amount of mixes of type A, B and C respectively. Then, we have the following set of simultaneous equations:

$$5x_1 + 6x_2 + 5x_3 = 370$$

$$2x_1 + 3x_2 + 3x_3 = 170$$

$$x_1 + x_2 + 2x_3 = 80$$

The above system of equations can be written in matrix form as:

$$\begin{bmatrix} 5 & 6 & 5 \\ 2 & 3 & 3 \\ 1 & 1 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 370 \\ 170 \\ 80 \end{bmatrix}$$

or $$AX = B$$

where $$A = \begin{bmatrix} 5 & 6 & 5 \\ 2 & 3 & 3 \\ 1 & 1 & 2 \end{bmatrix},\ X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 370 \\ 170 \\ 80 \end{bmatrix}$$

Since, $$|A| = 5(6-3) - 6(4-3) + 5(2-3)$$
$$= 4 \neq 0,$$

thus A^{-1} exists, so that the unique solution of $AX = B$ is $X = A^{-1}B$.

Here, adj $A = \begin{bmatrix} 3 & -7 & 3 \\ -1 & 5 & -5 \\ -1 & 1 & 3 \end{bmatrix}$

Here, $$A^{-1} = \frac{\text{adj}.A}{|A|}$$
$$= \frac{1}{4}\begin{bmatrix} 3 & -7 & 3 \\ -1 & 5 & -5 \\ -1 & 1 & 3 \end{bmatrix}$$

Now $$X = A^{-1}B$$
$$= \frac{1}{4}\begin{bmatrix} 3 & -7 & 3 \\ -1 & 5 & -5 \\ -1 & 1 & 3 \end{bmatrix}\begin{bmatrix} 370 \\ 170 \\ 80 \end{bmatrix} = \begin{bmatrix} 40 \\ 20 \\ 10 \end{bmatrix}$$

Hence $x = 40, y = 20$ and $z = 10$. Hence, the production of products A, B and C are 40 units, 20 units and 10 units, respectively.

38. Brine is a high-concentration solution of salt in water. In diverse contexts, brine may refer to the salt solutions ranging from about 3.5% up to about 26%. Brine forms naturally due to evaporation of ground saline water but it is also generated in the mining of sodium chloride.



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A tank initially contains 10 gallons of pure water. Brine containing 3 pounds of salt per gallon flows into the tank at a rate of 2 gallons per minute, and the well-stirred mixture flows out of the tank at the same rate.

(i) How much salt is present at the end of 10 minutes?

(ii) How much salt is present in the long run?

Sol :

Suppose $A(t)$ denotes the amount of salt in the tank at any time t. Then the derivative $\frac{dA}{dt}$, the rate of change of the amount of salt at any time t, must satisfy the condition

$$\frac{dA}{dt} = \text{Rate of salt flowing in} - \text{Rate of salt flowing out}$$

But the rate at which salt flows into the tank is given by $(2gal/\min)(3\,lb/\min)$ or 6 pounds per minute. Since the rate at which the solution leaves the tank is the same as the rate at which the brine is poured into it, the tank contains 10 gallons of the mixture at any time t. Since the salt content at any time t is A pounds, the concentration of the salt in the mixture is $(A/10)$ pounds per gallon. Therefore, the rate at which salt flows out of the tank is given by $(2gal/\min)(\frac{A}{10}\,lb/\min)$ or $\frac{A}{5}$ pounds per minute. Therefore, we are led to the differential equation

$$\frac{dA}{dt} = 6 - \frac{A}{5}$$

subject to the condition $A(0) = 0$. The differential equation

$$\frac{dA}{dt} = 6 - \frac{A}{5}$$

$$= \frac{30 - A}{5}$$

is separable. Separating the variables and integrating, we obtain

$$\int \frac{dA}{30 - A} = \int \frac{1}{5}\,dt$$

$$-\ln|30 - A| = \frac{1}{5}t + b$$

$$\ln|30 - A| = -\frac{1}{5}t - b$$

$$30 - A = e^{-b}e^{-t/5}$$

$$A = 30 - Ce^{-t/5}$$

The condition $A(0) = 0$ implies that

$$0 = 30 - C$$

Giving $C = 30$, and so

$$A(t) = 30(1 - e^{-t/5})$$

The amount of salt present after 10 minutes is given by

$$A(10) = 30(1 - e^{-2}) \approx 25.94$$

or 25.94 pounds. The amount of salt present in the long run is given by

$$\lim_{t \to \infty} A(t) = \lim_{t \to \infty} 30(1 - e^{-t/5}) = 30$$

or 30 pounds.

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Sample Paper 16

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

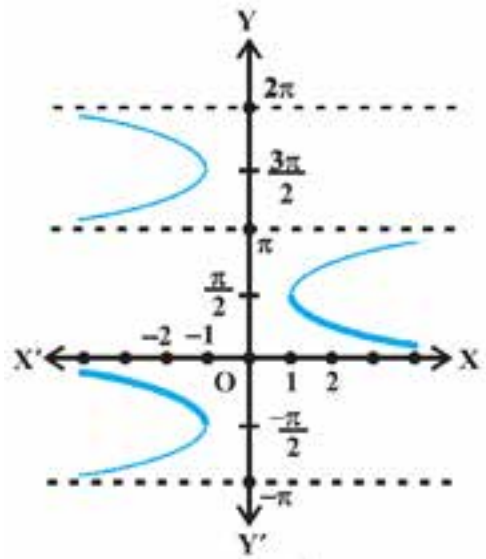
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the range of the function $y = \csc^{-1} x$ as represented in the graph?



(a) $[-\pi/2, \pi/2]$ (b) $[0,\pi]\backslash\{\pi/2\}$

(c) $(-\pi, \pi)$ (d) $(-\infty, \infty)$

Sol :

From the graph, the Y-values extend from 0 up to π, but the line $y = \pi/2$ is missing (the function is not defined there). This confirms that the range is $[0,\pi]$, excluding $\pi/2$.

Thus (b) is correct option.

2. The solution of the differential equation $2x\frac{dy}{dx} - y = 3$ represents

(a) straight line (b) circle

(c) parabola (d) ellipse

Sol :

$$2x\frac{dy}{dx} = y + 3$$

$$2\frac{dy}{y+3} = \frac{dx}{x}$$

$$2\log(y+3) = \log x + \log c$$

$$(y+3)^2 = cx$$

Which is an equation of parabola.

Thus (c) is correct option.

3. The solution of the differential equation $\frac{dy}{dx} + \frac{y}{x} = x^2$ is

(a) $x + y = \frac{x^2}{2} + c$ (b) $x - y = \frac{1}{3}x^3 + c$

(c) $xy = \frac{1}{4}x^4 + c$ (d) $y - x = \frac{1}{4}x^4 + c$

Sol :

Here, $P = \frac{1}{x}$

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and $Q = x^2$

$$\text{IF} = e^{\int \frac{1}{x} dx} = e^{\log x} = x$$

Solution is, $yx = \int x^2 \cdot x \, dx$

$$= \frac{x^4}{4} + c$$

Thus (c) is correct option.

4. Let D, E, F are the mid points of sides BC, CA, AB respectively of ΔABC. Which of the following is true?

(a) $\overrightarrow{AB} = 2\overrightarrow{ED}$ (b) $\overrightarrow{AB} = 2\overrightarrow{DE}$

(c) $\overrightarrow{AB} = \overrightarrow{ED}$ (d) $\overrightarrow{AB} = 2\overrightarrow{DF}$

Sol :

$$\overrightarrow{AB} = 2\overrightarrow{ED} \quad \left(ED \parallel AB \text{ and } ED = \frac{1}{2}AB\right)$$

Thus (a) is correct option.

5. The points (5, 2, 4), $(6, -1, 2)$ and $(8, -7, k)$ are collinear, if k is equal to

(a) -2 (b) 2

(c) 3 (d) -1

Sol :

Since, points are collinear.

Hence, $\begin{vmatrix} 5 & 2 & 4 \\ 6 & -1 & 2 \\ 8 & -7 & k \end{vmatrix} = 0$

$$8(4+4) + 7(10-24) + k(-5-12) = 0$$

$$64 - 98 - 17k = 0$$

$$k = -2$$

Thus (a) is correct option.

6. If A and B are two independent events such that $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{3}$, then P (neither A nor B) is equal to

(a) $\frac{2}{3}$ (b) $\frac{1}{6}$

(c) $\frac{5}{6}$ (d) $\frac{1}{3}$

Sol :

Since, A and B are independent events.

$$P(A \cap B) = P(A) \cdot P(B)$$

$$= \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$$

Now, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$= \frac{1}{2} + \frac{1}{3} - \frac{1}{6} = \frac{2}{3}$$

$$P(\overline{A} \cap \overline{B}) = 1 - P(A \cup B)$$

$$= 1 - \frac{2}{3} = \frac{1}{3}$$

Thus (d) is correct option.

7. Find the values of x, y and z from the following equations $\begin{bmatrix} 4 & x-z \\ 2+y & xz \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ -1 & 10 \end{bmatrix}$.

(a) $x = -5,\ y = 3,\ z = 2$

(b) $x = 5,\ y = -3,\ z = 2$

(c) $x = 5,\ y = 3,\ z = -2$

(d) $x = 5,\ y = -3,\ z = -2$

Sol :

We have $\begin{bmatrix} 4 & x-z \\ 2+y & xz \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ -1 & 10 \end{bmatrix}$

Comparing the corresponding elements, we get

$$x - z = 3 \quad \text{...(i)}$$

$$2 + y = -1$$

$$y = -1 - 2 = -3 \quad \text{..(ii)}$$

and $xz = 10 \quad \text{...(iii)}$

Now, $(x+z)^2 = (x-z)^2 + 4xz$

$$= (3)^2 + 4 \times 10$$

$$= 9 + 40$$

$$(x+z)^2 = 49$$

$$x + z = 7 \quad \text{...(iv)}$$

Adding Eqs. (i) and (iv), we get

$$2x = 10 \Rightarrow x = 5$$

From Eq. (i), we have

$$5 - z = 3 \Rightarrow -z = 3 - 5$$

$$z = 2$$

Hence, $x = 5,\ y = -3$ and $z = 2$.

Thus (b) is correct option.

8. If $y = a\log|x| + bx^2 + x$ has its extreme value at $x = -1$ and $x = 2$, then

(a) $a = 2, b = -1$ (b) $a = 2, b = -\frac{1}{2}$

(c) $a = -2, b = \frac{1}{2}$ (d) $a = -2, b = -\frac{1}{2}$

Sol :

$$y = a\log|x| + bx^2 + x$$

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19. **Assertion:** if $2P(A) = P(B) = \frac{5}{13}$ and $P\left(\frac{A}{B}\right) = \frac{2}{5}$, then $P(A \cup B)$ is $\frac{11}{26}$

Reason: E_1 and E_2 are two events. then

$$P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cup E_2)}{P(E_2)},\ 0 < P(E_2) \le 1$$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

Given that,

$$2P(A) = P(B) = \frac{5}{13}$$

$$P\left(\frac{A}{B}\right) = \frac{P(A \cup B)}{P(B)}$$

$$\frac{2}{5} = \frac{P(A \cup B)}{\frac{5}{13}}$$

$$P(A \cap B) = \frac{2}{5} \times \frac{5}{13} = \frac{2}{13}$$

Again,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{5}{26} + \frac{5}{13} - \frac{2}{13}$$

$$= \frac{5 + 10 - 4}{26} = \frac{11}{26}$$

As we know that, if E_1 and E_2 are two events, then

$$P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cup E_2)}{P(E_2)},\ 0 \le P(E_2) \le 1$$

Hence, Assertion is true, reason is true and reason is a correct explanation for Assertion.
Thus (a) is correct option.

20. Consider, if $u = f(n)$, $v = g(x)$, then the derivative of f with respect to g is $\frac{du}{dv} = \frac{du/dx}{dv/dx}$.

Assertion: Derivative of $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ with respect to $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is 1 for $0 < x < 1$

Reason: $\sin^{-1}\left(\frac{2x}{1+x^2}\right) \ne \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ for $-1 \le 1x \le 1$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

Assertion is true; Reason is false.
For $0 < x < 1$

$$\sin^{-1}\left(\frac{2x}{1+x^2}\right) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

Let $u = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$

and $v = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

Again let $x = \tan\theta$

So, $u = \sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$

$= \sin(\sin 2\theta) = 2\theta$

$\frac{du}{d\theta} = 2$

Now, $v = \cos^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)$

$= \cos^{-1}(\cos 2\theta) = 2\theta$

$\frac{dv}{d\theta} = 2$

Now, $\frac{du}{dx} = \frac{du/d\theta}{dv/d\theta} = \frac{2}{2} = 1$

Hence, assertion is true and reason is false
Thus (c) is correct option.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. If $|\vec{a}| = 8$, $|\vec{b}| = 3$ and $|\vec{a} \times \vec{b}| = 12$, find the angle between $\vec{a}$ and $\vec{b}$.

OR

Write the value of λ, so that the vectors $\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ are perpendicular to each other.

Sol :

Let θ be the angle between $\vec{a}$ and $\vec{b}$.

We have $|\vec{a}| = 8$, $|\vec{b}| = 3$ and $|\vec{a} \times \vec{b}| = 12$

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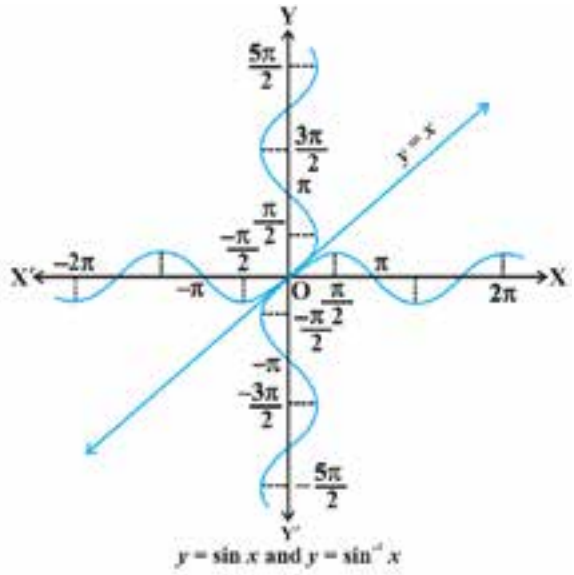
$$= x^2 + 2x + 1 + 1$$

$$= (x+1)^2 + 1 \geq 1$$

So, the range of $f(x)$ is $[1, \infty]$.

Thus (d) is correct option.

16. What is the period of the function $y = \tan x$ as seen in the graph?



$y = \sin x$ and $y = \sin^{-1} x$

(a) π (b) 2π

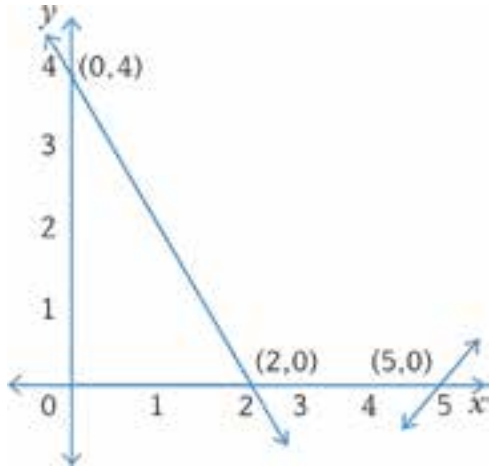
(c) $\pi/2$ (d) Infinite

Sol :

The tangent curve repeats its shape after every interval of π. For example, the part between $-\pi/2$ and $\pi/2$ is exactly repeated between $\pi/2$ and $3\pi/2$. Thus, the period is π.

Thus (a) is correct option.

17. For the LPP shown in the figure, the objective is Minimise $Z = 3x + 2y$. From the graph, the minimum value of Z is obtained at the corner point (2,0). The optimal solution of this LPP:



(a) Does not exist as the feasible region is unbounded.

(b) Does not exist as the inequality 3x+2y<6 does not have any point in common with the feasible region.

(c) Exists as the inequality 3x+2y>6 has infinitely many points in common with the feasible region.

(d) Exists as the inequality 3x+2y<6 does not have any point in common with the feasible region.

Sol :

At the visible vertices, $Z(2,0) = 6, Z(0,4) = 8$, and $Z(5,0) = 15$; thus $(2,0)$ gives the least value. The region is unbounded, but a minimum can still exist. To check, test whether there is any feasible point with a smaller value than 6, i.e., satisfying $3x + 2y < 6$. From the graph, every feasible point lies on or above the line $3x + 2y = 6$; moving to $3x + 2y < 6$ leaves the feasible region immediately. Hence no feasible point satisfies $3x + 2y < 6$, and the minimum $Z\min = 6$ at (2,0) is valid.

Thus (d) is correct option.

18. The solution of the equation $(x^2 - xy)\, dy = (xy + y^2)\, dx$ is

(a) $xy = ce^{-y/x}$ (b) $xy = ce^{-x/y}$

(c) $yx^2 = ce^{1/x}$ (d) none of these

Sol :

$$\frac{dy}{dx} = \frac{xy + y^2}{x^2 - xy}$$

Put $y = vx$

$$v + x\frac{dv}{dx} = \frac{v + v^2}{1 - v}$$

$$x\frac{dv}{dx} = \frac{v + v^2}{1 - v} - v = \frac{2v^2}{1 - v}$$

$$\frac{1 - v}{v^2} dv = 2\frac{dx}{x}$$

$$\left(\frac{1}{v^2} - \frac{1}{v}\right) dv = 2\frac{dx}{x}$$

$$-\frac{1}{v} - \log v = 2\log x + \text{constant}$$

$$-\frac{x}{y} - \log\frac{y}{x} = \log x^2 + \log c_1$$

$$-\frac{x}{y} = \log\left(\frac{y}{x} \cdot x^2 \cdot c_1\right) = \log c_1 xy$$

$$xy = \frac{1}{c_1} e^{-x/y} = ce^{-x/y}$$

$$\left(\text{take}, c_1 = \frac{1}{c}\right)$$

Thus (b) is correct option.

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$$\frac{dy}{dx} = \frac{a}{x} + 2bx + 1$$

$$-a - 2b + 1 = 0$$

and $\quad a + 8b + 2 = 0$

Solving, we get $\quad a = 2,\ b = -\frac{1}{2}$

Thus (d) is correct option.

9. The point of inflexion for the curve $y = x^{5/3}$ is

(a) $(1,1)$ (b) $(0,0)$

(c) $(1,0)$ (d) $(0,1)$

Sol :

Given that, $\quad y = x^{5/3} \Rightarrow \frac{dy}{dx} = \frac{5}{3}x^{2/3}$

Which exists at all points of the domain$= R$

Also, $\quad \frac{dy}{dx} = 0$

$$\frac{5}{3}x^{2/3} = 0 \Rightarrow x = 0$$

When we move from left to right through origin, $\frac{dy}{dx}$ does not change sign.

Hence, y has a point of inflexion at $x = 0, y = 0$.

Thus (b) is correct option.

10. If $\vec{a}$ and $\vec{b}$ are two unit vectors inclined to x-axis at angles 30° and 120°, then $|\vec{a} \times \vec{b}|$ equals

(a) $\sqrt{\frac{2}{3}}$ (b) $\sqrt{2}$

(c) $\sqrt{3}$ (d) 2

Sol :

Since, the angles between $\vec{a}$ and $\vec{b}$ $= 120^\circ - 30^\circ = 90^\circ$

$$\vec{a} \cdot \vec{b} = 0$$

Now, $\quad (\vec{a} + \vec{b})^2 = |\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} \cdot \vec{b}$

$$= 1^2 + 1^2 + 0 = 2$$

$$|\vec{a} + \vec{b}| = \sqrt{2}$$

Thus (b) is correct option.

11. $\int_0^{\pi/2} \left|\cos\left(\frac{x}{2}\right)\right| dx$ is equal to

(a) 1 (b) -2

(c) $\sqrt{2}$ (d) 0

Sol :

$$\int_0^{\pi/2} \left|\cos\left(\frac{x}{2}\right)\right| dx = \int_0^{\pi/2} \cos\left(\frac{x}{2}\right) dx$$

$$= 2\left[\sin\left(\frac{x}{2}\right)\right]_0^{\pi/2}$$

$$= 2\left[\sin\frac{\pi}{4} - \sin 0\right]$$

$$= 2\sin\frac{\pi}{4}$$

$$= 2 \times \frac{1}{\sqrt{2}} = \sqrt{2}$$

Thus (c) is correct option.

12. The point of discontinuous of $\tan x$ are

(a) $n\pi,\ n \in I$ (b) $2n\pi,\ n \in I$

(c) $(2n+1)\frac{\pi}{2},\ n \in I$ (d) None of these

Sol :

The points of discontinuity of $f(x)$ are those points, where $\tan x$ is infinite.

i.e. $\quad \tan x = \tan\infty \Rightarrow x = (2n+1)\frac{\pi}{2},\ n \in I$

Thus (c) is correct option.

13. The area bounded by the curve $y = \frac{1}{2}x^2$, the X-axis and the ordinate $x = 2$ is

(a) $\frac{1}{3}$ sq unit (b) $\frac{2}{3}$ sq unit

(c) 1 sq unit (d) $\frac{4}{3}$ sq unit

Sol :

Required area, $\quad A = \int_0^2 y\,dx$

$$= \int_0^2 \frac{x^2}{2}dx = \left[\frac{x^3}{6}\right]_0^2 = \frac{4}{3} \text{ sq units}$$

Thus (d) is correct option.

14. If A and B are two equivalence relations defined on set C, then

(a) $A \cap B$ is an equivalence relation

(b) $A \cap B$ is not an equivalence relation

(c) $A \cap B$ is an equivalence relation

(d) $A \cap B$ is not an equivalence relation

Sol :

If A and B are equivalence relations, then $A \cap B$ is also an equivalence relation.

Thus (a) is correct option.

15. The range of the function $f(x) = x^2 + 2x + 2$ is

(a) $(1,\infty)$ (b) $(2,\infty)$

(c) $(0,\infty)$ (d) $[1,\infty)$

Sol :

Given, $\quad f(x) = x^2 + 2x + 2$

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If θ is the angle between $\vec{a}$ and $\vec{b}$, the we have

$$|\vec{a}\times\vec{b}| = |\vec{a}||\vec{b}|\sin\theta$$

$$12 = |\vec{a}||\vec{b}|\sin\theta$$

$$\sin\theta = \frac{12}{|\vec{a}||\vec{b}|} = \frac{12}{8\times 3}$$

$$\sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$$

Hence, the required angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$.

OR

We have $\vec{a} = 2\hat{i}+\lambda\hat{j}+\hat{k}$

and $\vec{b} = \hat{i}-2\hat{j}+3\hat{k}$

Since, vectors are perpendicular.

$$\vec{a}\cdot\vec{b} = 0$$

$$(2\hat{i}+\lambda\hat{j}+\hat{k})\cdot(\hat{i}-2\hat{j}+3\hat{k}) = 0$$

$$2-2\lambda+3 = 0$$

$$5-2\lambda = 0 \Rightarrow \lambda = 5/2$$

22. The equation of a line is

$$5x-3 = 15y+7 = 3-10z$$

Write the direction cosines of the line.

Sol :

Given equation of a line is

$$5x-3 = 15y+7 = 3-10z \quad ...(1)$$

Here coefficients of x, y and z are 5, 15 and 10. LCM $(5,15,10) = 30$. Thus dividing by 30 we have eq. (1) becomes

$$\frac{5x-3}{30} = \frac{15y+7}{30} = \frac{3-10z}{30}$$

$$\frac{5(x-\frac{3}{5})}{30} = \frac{15(y+\frac{7}{15})}{30} = \frac{-10(z-\frac{3}{10})}{30}$$

$$\frac{x-\frac{3}{5}}{6} = \frac{y+\frac{7}{15}}{2} = \frac{z-\frac{3}{10}}{-3} \quad ...(2)$$

The standard form of equation is given as

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} \quad ...(3)$$

Comparing the above standard equation with Eq. (2), we get 6, 2, -3 are the direction ratios of the given line.

Now $\sqrt{6^2+2^2+(-3)^2} = \sqrt{49} = 7$

Now, the direction cosines of given line are $\frac{6}{7}$, $\frac{2}{7}$, $\frac{-3}{7}$.

23. Find the vector equation of the line which passes through the point (3,4,5) and is parallel to the vector $2\hat{i}+2\hat{j}-3\hat{k}$.

Sol :

Equation of a line passing through a point with position vector $\vec{a}$ and parallel to a vector $\vec{b}$ is given by

$$\vec{r} = \vec{a}+\lambda\vec{b}$$

Since, line passes through (3,4,5), is

$$\vec{a} = 3\hat{i}+4\hat{j}+5\hat{k}$$

Since, lines is parallel to $2\hat{i}+2\hat{j}-3\hat{k}$, we get

$$\vec{b} = 2\hat{i}+2\hat{j}-3\hat{k}$$

Equation of line is $\vec{r} = \vec{a}+\lambda\vec{b}$, i.e.

$$\vec{r} = (3\hat{i}+4\hat{j}+5\hat{k})+\lambda(2\hat{i}+2\hat{j}-3\hat{k})$$

Which is the required vector equation.

24. Show that the relation R on IR defined as $R = \{(a,b): (a \le b)\}$, is reflexive and transitive but not symmetric.

Sol :

Given a relation $R = \{(a,b): a \le b\}$ on IR (the set of real numbers).

Reflexivity: Since, $a \le a$ is true for all value of $a \in IR$

$(a,a) \in R \forall\, a \in IR$

Hence, the given relation is reflexive.

Transitivity: Let $(a,b) \in R$ be any arbitrary elements.

Then, we have $a \le b$ and $b \le c$

$$a \le b \le c$$

$$a \le c \ (a,c) \in R$$

Hence, the given relation is transitive.

Symmetricity: Note that $(2,3) \in R$ as $2 < 3$

But $(3,2) \notin R$ as $3 \not< 2$

Hence, the given relation is not symmetric.

25. Write the value of $\int \frac{2-3\sin x}{\cos^2 x}\,dx$.

OR

Write the value of $\int \sec x(\sec x+\tan x)\,dx$.

Sol :

We have $I = \int \frac{2-3\sin x}{\cos^2 x}\,dx$

$$= \int\left(\frac{2}{\cos^2 x}-\frac{3\sin x}{\cos^2 x}\right)dx$$

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$$\cos x - \sin x = 0$$

$$\tan x = 1,$$

$$x = \frac{\pi}{4}, \frac{5\pi}{4}, \text{ as } 0 \le x \le 2\pi$$

Now, we find the intervals in which $f(x)$ is strictly increasing or strictly decreasing.

For interval $0 < x < \frac{\pi}{4}$ at $x = \frac{\pi}{6}$,

$$f'(x) = \cos x - \sin x$$

$$= \frac{\sqrt{3}}{2} - \frac{1}{2} = \frac{\sqrt{3}-1}{2} \quad + \text{ve}$$

For interval $\frac{\pi}{4} < x < \frac{5\pi}{6}$ at $x = \frac{\pi}{2}$,

$$f'(x) = \cos x - \sin x$$

$$= 0 - 1 = -1 \quad - \text{ve}$$

For interval $\frac{5\pi}{4} < x < 2\pi$ at $x = \frac{3\pi}{2}$,

$$f'(x) = \cos x - \sin x$$

$$= 0 - (-1) = 1 \quad + \text{ve}$$

Here $f'(x) > 0$ in $\left(0, \frac{\pi}{4}\right)$, $f'(x) < 0$ in $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$ and $f'(x) > 0$ in $\left(\frac{5\pi}{4}, 2\pi\right)$.

Since, $f(x)$ is a trigonometric function, so it is continuous at $x = 0$, $\frac{\pi}{4}$, $\frac{5\pi}{4}$ and 2π.

Hence, the function is

(i) increasing in $\left[0, \frac{\pi}{4}\right]$ and $\left[\frac{5\pi}{4}, 2\pi\right]$.

(ii) decreasing in $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Maximize $Z = 3x + 4y$, subject to the constraints; $x + y \le 4$, $x \ge 0$, $y \ge 0$.

OR

Minimize $Z = -3x + 4y$ subject to the constraints

$$x + 2y \le 8,$$

$$3x + 2y \le 12,$$

$$x \ge 0, \ y \ge 0$$

Sol :

We have to maximize $Z = 3x + 4y$, subject to the constraints :

Maximize $Z = 3x + 4y$

Subject to $x + y \le 4$ ...(i)

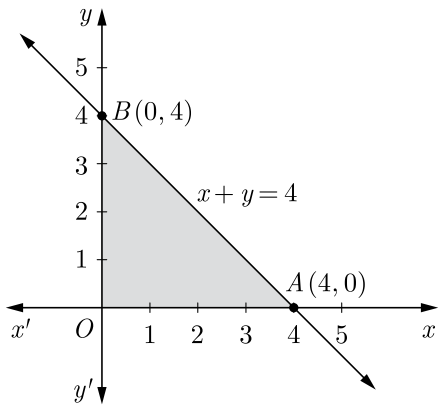
and $x \ge 0, \ y \ge 0$

Region corresponding to $x + y \le 4$:

Line $x + y = 4$			
x	0	4	Point (0,0) is true for $x + y \le 4$. So, the region is towards the origin.
y	4	0	

Since $x \ge 0$, $y \ge 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is $OABO$ which is bounded The corner points of the feasible region are $O(0,0)$, $A(4,0)$ and $B(0,4)$.



Since, the feasible region is a bounded region, we can check the objective function at all the corner to find the maxima. The values of objective function Z at these points are as follows.

Corner points	$Z = 3x + 4y$
$O(0,0)$	$Z = 0 + 0 = 0$
$A(4,0)$	$Z = 3 \times 4 + 4 \times 0 = 12$
$B(0,4)$	$Z = 3 \times 0 + 4 \times 4 = 16$

The maximum value of Z is 16 at $B(0,4)$ and the optimal solution is $x = 0$, $y = 4$.

OR

Minimized $Z = -3x + 4y$

Subjected to $x + 2y \le 8$

$3x + 2y \le 12$,

and $x \ge 0, \ y \ge 0$

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OR

We have $I = \int_{-2}^{2} \frac{x^2}{1+5^x}\,dx$ ...(i)

Using the fact $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$ we have

$$I = \int_{-2}^{2} \frac{(2-2-x)^2}{1+5^{2-2-x}}\,dx$$

$$= \int_{-2}^{2} \frac{x^2}{1+5^{-x}}\,dx$$

or $I = \int_{-2}^{2} \frac{5^{-x}}{5^x+1}x^2\,dx$ ...(ii)

Adding Eqs. (i) and (ii), we get

$$2I = \int_{-2}^{2}\left(\frac{1+5^x}{5^x+1}\right)x^2\,dx$$

$$= \int_{-2}^{2} x^2\,dx$$

or $2I = 2\int_0^2 x^2\,dx$ Even function

$$I = \left[\frac{x^3}{3}\right]_0^3 = \frac{1}{3}(2^3-0) = \frac{8}{3}$$

29. Prove that $\tan\left\{\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\frac{a}{b}\right\}$

$\tan\left\{\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\frac{a}{b}\right\}+\tan\left\{\frac{\pi}{4}-\frac{1}{2}\cos^{-1}\left(\frac{a}{b}\right)\right\} = \frac{2b}{a}$

Sol :

We have

$\tan\left\{\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\frac{a}{b}\right\}+\tan\left\{\frac{\pi}{4}-\frac{1}{2}\cos^{-1}\left(\frac{a}{b}\right)\right\} = \frac{2b}{a}$

Let $\cos^{-1}\frac{a}{b} = \theta$ then $\cos\theta = \frac{a}{b}$. Now substituting in LHS we have

$$\text{LHS} = \tan\left(\frac{\pi}{4}+\frac{\theta}{2}\right)+\tan\left(\frac{\pi}{4}-\frac{\theta}{2}\right)$$

$$= \frac{\tan\frac{\pi}{4}+\tan\frac{\theta}{2}}{1-\tan\frac{\pi}{4}\tan\frac{\theta}{2}} + \frac{\tan\frac{\pi}{4}-\tan\frac{\theta}{2}}{1+\tan\frac{\pi}{4}\tan\frac{\theta}{2}}$$

$$= \frac{1+\tan\frac{\theta}{2}}{1-\tan\frac{\theta}{2}} + \frac{1-\tan\frac{\theta}{2}}{1+\tan\frac{\theta}{2}}$$

$$= \frac{\left(1+\tan\frac{\theta}{2}\right)^2+\left(1-\tan\frac{\theta}{2}\right)^2}{\left(1-\tan\frac{\theta}{2}\right)\left(1+\tan\frac{\theta}{2}\right)}$$

$$= 2\left(\frac{1+\tan^2\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}}\right) \qquad \cos\theta = \frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}$$

$$= \frac{2}{\cos\theta} = \frac{2}{\frac{a}{b}} = \frac{2b}{a}$$

$= \text{RHS}$ Hence proved.

30. If $\begin{vmatrix} 3x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 7 & 3 \end{vmatrix}$, then write the value of x.

Sol :

We have $\begin{vmatrix} 2x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 7 & 3 \end{vmatrix}$

$$2x^2 - 40 = 18 - (-14)$$

$$2x^2 - 40 = 32$$

$$2x^2 = 72$$

$$x^2 = 36$$

$$x = \pm 6$$

31. Show that $y = \log(1+x) - \frac{2x}{2+x}$, $x > -1$ is an increasing function of x, throughout its domain.

OR

Find the intervals in which the function given by $f(x) = \sin x + \cos x$, $0 \le x \le 2\pi$ is

(i) increasing

(ii) decreasing.

Sol :

We have $y = \log(1+x) - \frac{2x}{2+x}$.

Differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = \frac{1}{1+x}(1) - \frac{(2+x)\cdot 2 - 2x\cdot 1}{(2+x)^2}$$

$$= \frac{1}{1+x} - \frac{4+2x-2x}{(2+x)^2}$$

$$= \frac{(2+x)^2 - 4(1+x)}{(1+x)(2+x)^2}$$

$$= \frac{4+x^2+4x-4-4x}{(1+x)(2+x)^2}$$

$$= \frac{x^2}{(1+x)(2+x)^2} \qquad \text{...(i)}$$

Now, x^2, $(2+x)^2$ are always positive, also $1+x > 0$ for $x > -1$. Thus $\frac{dy}{dx} > 0$ for $x > -1$. Hence, function increases for $x > -1$.

OR

We have $f(x) = \sin x + \cos x$.

Differentiating both sides w.r.t. x, we get

$$f'(x) = \cos x - \sin x$$

Now, substituting $f'(x) = 0$ we have

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$$= \int (2\sec^2 x - 3\sec x \tan x)\, dx$$
$$= 2\int \sec^2 x\, dx - 3\int \sec x \tan x\, dx$$
$$= 2\tan x - 3\sec x + C$$

OR

We have $I = \int \sec x(\sec x + \tan x)\, dx$
$$= \int (\sec^2 x + \sec x \tan x)\, dx$$
$$= \int \sec^2 x\, dx + \int \sec x \tan x\, dx$$
$$= \tan x + \sec x + C$$

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. If $y = (\sin x)^x + \sin^{-1}\sqrt{x}$, then find $\frac{dy}{dx}$.

Sol :

We have $y = (\sin x)^x + \sin^{-1}\sqrt{x}$ ...(1)

Let $u = (\sin x)^x$ ...(2)

Then, Eq. (1) becomes, $y = u + \sin^{-1}\sqrt{x}$ ...(3)

Taking log both sides of Eq. (2), we get

$$\frac{1}{u}\frac{du}{dx} = x\frac{d}{dx}(\log \sin x) + \log \sin x \frac{d}{dx}(x)$$
$$\frac{du}{dx} = u\left[x \times \frac{1}{\sin x}\frac{1}{dx}(\sin x) + \log \sin x(1)\right]$$
$$\frac{du}{dx} = (\sin x)^x\left[\frac{x}{\sin x} \times \cos x + \log \sin x\right]$$
[from Eq. (2)]
$$\frac{du}{dx} = (\sin x)^x[x\cot x + \log \sin x] \quad ...(4)$$

Differentiating both sides of Eq. (4) w.r.t. x, we get

$$\frac{dy}{dx} = \frac{du}{dx} + \frac{1}{\sqrt{1-(\sqrt{x})^2}}\frac{d}{dx}(\sqrt{x})$$
$$\frac{dy}{dx} = (\sin x)^x[x\cot x + \log \sin x]$$

27. If θ is the angle between two vectors $\hat{i} - 2\hat{j} + 3\hat{k}$ and $3\hat{i} - 2\hat{j} + \hat{k}$, find $\sin\theta$.

Sol :

We have $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$

and $\vec{b} = 3\hat{i} - 2\hat{j} + \hat{k}$

Here, $|\vec{a}| = \sqrt{1^2 + (-2)^2 + 3^2}$
$$= \sqrt{1+4+9} = \sqrt{14}$$
$$|\vec{b}| = \sqrt{3^2 + (-2)^2 + 1^2}$$
$$= \sqrt{9+4+1} = \sqrt{14}$$

and $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 3 & -2 & 1 \end{vmatrix}$
$$= \hat{i}(-2+6) - \hat{j}(1-9) + \hat{k}(-2+6)$$
$$= 4\hat{i} + 8\hat{j} + 4\hat{k} = 4(\hat{i} + 2\hat{j} + \hat{k})$$
$$|\vec{a} \times \vec{b}| = 4\sqrt{1^2 + 2^2 + 1^2}$$
$$= 4\sqrt{1+4+1} = 4\sqrt{6}$$

Now $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|\sin\theta$

$$\sin\theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|} \quad ...(1)$$
$$= \frac{4\sqrt{6}}{\sqrt{14} \cdot \sqrt{14}}$$
$$= \frac{4\sqrt{6}}{14} = \frac{2\sqrt{6}}{7}$$

28. Evaluate $\int_0^{\pi} |x^3 - x|\, dx$.

OR

Evaluate $\int_{-2}^{2} \frac{x^2}{1+5^x}\, dx$.

Sol :

We have $I = \int_{-1}^{2} |x^3 - x|\, dx$

We observe that,

Since $|x^3 - x| = \begin{cases} (x^3 - x), when -1 < x < 0 \\ -(x^3 - x), when\ 0 \le x < 1 \\ (x^3 - x), when\ 1 \le x < 2 \end{cases}$

Thus

$$I = \int_{-1}^{0}(x^3 - x)\, dx + \int_0^1 -(x^3 - x)\, dx + \int_1^2 (x^3 - x)\, dx$$
$$= \int_{-1}^{0}(x^3 - x)\, dx - \int_0^1 (x^3 - x)\, dx + \int_1^2 (x^3 - x)\, dx$$
$$= \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_{-1}^{0} - \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_0^1 + \left[\frac{x^4}{4} - \frac{x^2}{2}\right]_1^2$$
$$= \left[0 - \left(\frac{1}{4} - \frac{1}{2}\right)\right] - \left[\left(\frac{1}{4} - \frac{1}{2}\right) - 0\right] + \left[\left(\frac{16}{4} - \frac{4}{2}\right) - \left(\frac{1}{4} - \frac{1}{2}\right)\right]$$
$$= -\frac{1}{4} + \frac{1}{2} - \frac{1}{4} + \frac{1}{2} + 4 - 2 - \frac{1}{4} + \frac{1}{2}$$
$$= -\frac{3}{4} + \frac{3}{2} + 2 = \frac{-3+6+8}{4} = \frac{11}{4}$$

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(i) Region corresponding to $x+2y \le 8$:

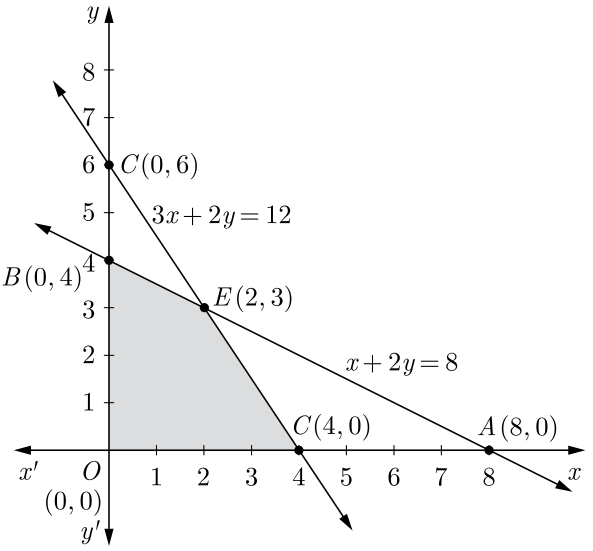
Line $x+2y=8$			
x	0	8	Point $(0,0)$ is true for $x+2y \le 8$. So, the region is towards the origin.
y	4	0	

(ii) Region corresponding to $3x+2y \le 12$:

Line $3x+2y=12$			
x	0	4	Point $(0,0)$ is true for $3x+2y \le 12$. So, the region is towards the origin.
y	6	0	

Since $x \ge 0$, $y \ge 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is $OCEBO$ which is bounded. Solving equations $x+2y=8$ and $3x+2y=12$, we get $E(2,3)$ The corner points of the feasible region are $O(0,0)$, $C(4,0)$, $E(2,3)$ and $B(0,4)$.



Since, the feasible region is a bounded region, we can check the objective function at all the corner to find the maxima. The values of objective function Z at these points are as follows.

Corner points	$Z = -3x+4y$
$O(0,0)$	$Z = 0+0 = 0$
$B(0,4)$	$Z = -3\times 0+4\times 4 = 16$
$E(2,3)$	$Z = -3\times 2+4\times 3 = 6$
$C(4,0)$	$Z = -3\times 4+4\times 0 = -12$

Hence the minimum value of Z is -12 at $C(4,0)$ and the optimal solution is $x=4$; $y=0$.

33. Evaluate $\int_0^{\pi/2} \frac{x \sin x \cos x}{\sin^4 x+\cos^4 x}\,dx$.

Sol :

We have $I = \int_0^{\pi/2} \frac{x \sin x \cos x}{\sin^4 x+\cos^4 x}\,dx$ ...(1)

Using the fact $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ we have

$$I = \int_0^{\pi/2} \frac{(\frac{\pi}{2}-x)\sin(\frac{\pi}{2}-x)\cos(\frac{\pi}{2}-x)}{\sin^4(\frac{\pi}{2}-x)+\cos^4(\frac{\pi}{2}-x)}\,dx$$

$$= \int_0^{\pi/2} \frac{(\frac{\pi}{2}-x)\cos x \sin x}{\cos^4 x+\sin^4 x}\,dx \quad ...(2)$$

Adding Eqs. (1) and (2), we get

$$2I = \frac{\pi}{2}\int_0^{\pi/2} \frac{\cos x \sin x}{\sin^4 x+\cos^4 x}\,dx$$

or $$I = \frac{\pi}{4}\int_0^{\pi/2} \frac{\sin x \cos x}{(\sin^2 x)^2+(1-\sin^2 x)^2}\,dx$$

Substituting $\sin^2 x = t$

$$2\sin x \cos x\,dx = dt$$

$$\sin x \cos x\,dx = \frac{dt}{2}$$

When $x=0$, then $t = \sin 0 = 0$ and

when $x=\frac{\pi}{2}$, then $t = \sin^2\frac{\pi}{2} = 1$

Thus

$$I = \frac{\pi}{4}\int_0^1 \frac{1}{t^2+(1-t)^2}\frac{dt}{2}$$

$$= \frac{\pi}{8}\int_0^1 \frac{1}{t^2+(1+t^2-2t)}\,dt$$

$$= \frac{\pi}{8}\int_0^1 \frac{1}{2t^2-2t+1}\,dt$$

$$= \frac{\pi}{16}\int_0^1 \frac{1}{t^2-t+(\frac{1}{2})^2-(\frac{1}{2})^2+\frac{1}{2}}\,dt$$

$$= \frac{\pi}{16}\int_0^1 \frac{1}{(t-\frac{1}{2})^2+(\frac{1}{2})^2}\,dt$$

$$= \frac{\pi}{16}\frac{1}{\frac{1}{2}}\left[\tan^{-1}\left(\frac{t-\frac{1}{2}}{\frac{1}{2}}\right)\right]_0^1 \qquad \int \frac{1}{x^2+a^2}dx = \frac{1}{a}\tan^{-1}\frac{x}{a}+C$$

$$= \frac{\pi}{8}\left[\tan^{-1}2\left(1-\frac{1}{2}\right)-\tan^{-1}2\left(0-\frac{1}{2}\right)\right]$$

$$= \frac{\pi}{8}[\tan^{-1}(1)-\tan^{-1}(-1)] \qquad \tan^{-1}(-1) = -\tan^{-1}(1) = -\frac{\pi}{4}$$

or $I = \frac{\pi}{8}\left[\frac{\pi}{4}+\frac{\pi}{4}\right] = \frac{\pi^2}{16}$

34. Solve the following differential equation $\operatorname{cosec} x \log|y|\frac{dy}{dx}+x^2y^2=0$.

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$= [6 \ \ 15 \ \ 9 \ \ 5 \ \ 0]$

The total number of posts of each kind in all the offices of branch B are the element of the matrix:

$$B = B_1 + B_2 + B_3$$

$$= [2 \ \ 5 \ \ 3 \ \ 0 \ \ 2] + [2 \ \ 5 \ \ 3 \ \ 0 \ \ 2] + [2 \ \ 5 \ \ 3 \ \ 0 \ \ 2]$$

$$= [6 \ \ 15 \ \ 9 \ \ 0 \ \ 6]$$

(ii) The total number of posts of each kind in all the offices taken together from both branches A and B are the elements of the matrix:

$$A + B = (A_1 + A_2 + A_3) + (B_1 + B_2 + B_3)$$

$$= [12 \ \ 30 \ \ 18 \ \ 5 \ \ 6]$$

37. Mahesh runs a form processing agency. He collect form from different office and then extract data and record data on computer. In his office three employees Vikas, Sarita and Ishaan process incoming copies of a form. Vikas process 50% of the forms. Sarita processes 20% and Ishaan the remaining 30% of the forms. Vikas has an error rate of 0.06, Sarita has an error rate of 0.04 and Ishaan has an error rate of 0.03.



Based on the above information answer the following questions.

(i) The total probability of committing an error in processing the form.

(ii) The manager of the company wants to do a quality check. During inspection he selects a form at random from the days output of processed forms. If the form selected at random has an error, the probability that the form is not processed by Vikas.

Sol :

(i) Required probability,

$$= P(A)$$

$$= P(E_1)\,P\left(\frac{A}{E_1}\right) + P(E_2)\,P\left(\frac{A}{E_2}\right) + P(E_3)\,P\left(\frac{A}{E_3}\right)$$

$$= 0.5 \times 0.06 + 0.2 \times 0.4 + 0.3 \times 0.3$$

$$= 0.030 + 0.008 + 0.009 = 0.047$$

(ii) Required probability,

$$P\left(\frac{\overline{E_1}}{A}\right) = 1 - P\left(\frac{\overline{E_1}}{A}\right)$$

$$= 1 - \left[\frac{P(E_1)\,P\left(\frac{A}{E_1}\right)}{P(E_1)\,P\left(\frac{A}{E_1}\right) + P(E_2)\,P\left(\frac{A}{E_2}\right) + P(E_3)\,P\left(\frac{A}{E_3}\right)}\right]$$

$$= 1 - \left[\frac{0.5 \times 0.6}{0.5 \times 0.06 + 0.2 \times 0.04 + 0.3 \times 0.03}\right]$$

$$= 1 - \frac{0.030}{0.047}$$

$$= 1 - \frac{30}{47} = \frac{17}{47}$$

38. RK Verma is production analysts of a ready-made garment company. He has to maximize the profit of company using data available. He find that $P(x) = -6x^2 + 120x + 25000$ (in Rupee) is the total profit function of a company where x denotes the production of the company.



Based on the above information, answer the following questions.

(i) Find the profit of the company, when the production is 3 units.

(ii) Find $P'(5)$

(iii) Find the interval in which the profit is strictly increasing.

OR

(iv) Find the production, when the profit is maximum.

Sol :

We have $P(x) = -6x^2 + 120x + 25000$

$$P'(x) = -12x + 120$$

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Sol :

Writing the given line in standard form as

$$\frac{x-1}{-3} = \frac{y-2}{\frac{p}{7}} = \frac{z-3}{2} = r_1 \text{ (let)} \qquad ...(1)$$

and $$\frac{x-1}{\frac{-3p}{7}} = \frac{y-5}{1} = \frac{z-6}{-5} = r_2 \text{ (let)} \qquad ...(2)$$

Two lines with DR's a_1, b, c_1 and a_2, b_2, c_2 are perpendicular if

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

Thus line (1) and (2) will intersect at right angle, if

$$-3\left(\frac{-3p}{7}\right) + \frac{p}{7}(1) + 2(-5) = 0$$

$$\frac{9p}{7} + \frac{p}{7} = 10$$

$$\frac{10p}{7} = 10 \Rightarrow p = 7$$

which is the required value of p.

Also, we know that, the equation of a line which passes through the point (x_1, y_1, z_1) with direction ratios a, b, c is given by

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$$

Since, required line is parallel to line l_1.

So, $a = -3$, $b = \frac{7}{7} = 1$ and $c = 2$

Now, equation of line passing through the point $(3, 2, -4)$ and having direction ratios $(-3, 1, 2)$ is

$$\frac{x-3}{-3} = \frac{y-2}{1} = \frac{z+4}{2}$$

$$\frac{3-x}{3} = \frac{y-2}{1} = \frac{z+4}{2}$$

SECTION - E

Case study based questions are compulsory.

36. Publishing is the activity of making information, literature, music, software and other content available to the public for sale or for free. Traditionally, the term refers to the creation and distribution of printed works, such as books, newspapers, and magazines.



NODIA Press is a such publishing house having two branch at Jaipur. In each branch there are three offices. In each office, there are 2 peons, 5 clerks and 3 typists. In one office of a branch, 5 salesmen are also working. In each office of other branch 2 head-clerks are also working. Using matrix notations find:

(i) The total number of posts of each kind in all the offices taken together in each branch.

(ii) The total number of posts of each kind in all the offices taken together from both branches.

Sol :

Consider the three row matrices,

$$A_1 = [2 \;\; 5 \;\; 3 \;\; 5 \;\; 0]$$

$$A_2 = [2 \;\; 5 \;\; 3 \;\; 0 \;\; 0]$$

$$A_3 = [2 \;\; 5 \;\; 3 \;\; 0 \;\; 0]$$

The matrices A_1, A_2, A_3 represent the three offices of one branch (say A), where the elements appearing in the row represents the number of peons, clerks, typists, salesman and head-clerks taken in that order working in the three offices.

Now, consider the three row matrices,

$$B_1 = [2 \;\; 5 \;\; 3 \;\; 0 \;\; 2]$$

$$B_2 = [2 \;\; 5 \;\; 3 \;\; 0 \;\; 2]$$

$$B_3 = [2 \;\; 5 \;\; 3 \;\; 0 \;\; 2]$$

The matrices B_1, B_2, B_3 represents the three offices of other branch (say B), where the elements appearing in the row represents the number of peons, clerks, typists, salesman and head-clerks taken in that order working in the three offices.

The total number of posts of each kind in all the offices of branch A are the elements of the matrix:

$$A = A_1 + A_2 + A_3$$

$$= [2 \;\; 5 \;\; 3 \;\; 5 \;\; 0] + [2 \;\; 5 \;\; 3 \;\; 0 \;\; 0] + [2 \;\; 5 \;\; 3 \;\; 0 \;\; 0]$$

$$= [2+2+2 \; 5+5+5 \; 3+3+3 \; 5+0+0 \; 0+0+0]$$

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OR

Solve the following differential equation.

$$x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) + x;\ x \neq 0$$

Sol :

We have $\operatorname{cosec} x \log|y|\frac{dy}{dx} + x^2y^2 = 0$ ...(1)

$$\operatorname{cosec} x \log|y|\frac{dy}{dx} = -x^2y^2$$

Separating the variables, we get

$$\frac{\log|y|}{y^2}dy = \frac{-x^2}{\operatorname{cosec} x}dx$$

Integrating both sides, we have

$$\int\frac{\log|y|}{y^2}dy = -\int\frac{x^2}{\operatorname{cosec} x}dx$$

$$I_1 = -I_2 \quad ...(2)$$

where, $I_1 = \int\frac{\log|y|}{y^2}dy$

and $I_2 = \int\frac{x^2}{\operatorname{cosec} x}dx$

$$= \int x^2\sin x\ dx$$

Now $I_1 = \int\frac{\log|y|}{y^2}dy$

Substituting $\log y = t \Rightarrow y = e^t$, then $\frac{dy}{y} = dt$ we have

$$I_1 = \int \underset{I}{t}\ \underset{II}{e^{-t}}dt$$

$$= t\int e^{-t}dt - \int\left[\frac{d}{dt}(t)\int e^{-t}dt\right]dt$$

$$= -te^{-t} - \int(-e^{-t})dt$$

$$= -te^{-t} + \int e^{-t}dt$$

$$= -te^{-t} - e^{-t} + C_1$$

$$= -\frac{\log|y|}{y} - \frac{1}{y} + C_1 \quad ...(3)$$

and $I_2 = \int \underset{I}{x^2}\ \underset{II}{\sin x}\,dx$

$$= x^2\int\sin x\,dx - \int\left[\frac{d}{dx}(x^2)\int\sin x\,dx\right]dx$$

$$= x^2(-\cos x) - \int[2x(-\cos x)]\,dx$$

$$= -x^2\cos x + 2\int \underset{I}{x}\ \underset{II}{\cos x}\,dx$$

$$= -x^2\cos x + 2\left[x\int\cos x dx - \int\left\{\frac{d}{dx}(x)\int\cos x dx\right\}dx\right]$$

$$= -x^2\cos x + 2[x\sin x - \int\sin x\,dx]$$

$$= -x^2\cos x + 2x\sin x + 2\cos x + C_2 \quad ...(4)$$

Substituting the values of I_1 and I_2 from Eqs. (3) and (4) in Eq. (2), we get

$$-\frac{\log|y|}{y} - \frac{1}{y} + C_1 = x^2\cos x - 2x\sin x - 2\cos x - C_2$$

$$-\frac{(1+\log|y|)}{y} = x^2\cos x - 2x\sin x - 2\cos x - C_2 - C_1$$

$$-\frac{(1+\log|y|)}{y} = x^2\cos x - 2x\sin x - 2\cos x + C$$

Where, $C = -C_2 - C_1$

Which is the required solution of given differential equation.

OR

We have $x\cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) + x$ ...(1)

Which is a homogeneous differential equation as

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

Substituting $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ in eq (1) we have

$$x\cos v\left[v + x\frac{dv}{dx}\right] = vx\cos v + x$$

$$vx\cos v + x^2\cos v\frac{dv}{dx} = vx\cos v + x$$

$$x^2\cos v\frac{dv}{dx} = x$$

$$\cos v\,dv = \frac{dx}{x}$$

Integrating both sides, we have

$$\int\cos v\,dv = \int\frac{dx}{x}$$

$$\sin v = \log|x| + C$$

$$\sin\left(\frac{y}{x}\right) = \log|x| + C \quad [\text{put } v = \frac{y}{x}$$

Which is the required solution of given differential equation.

35. Find the values of p, so that the lines

$$l_1 : \frac{1-x}{3} = \frac{7y-14}{p} = \frac{z-3}{2} \text{ and}$$

$$l_2 : \frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5}$$

are perpendicular to each other. Also, find the equation of a line passing through a point $(3, 2, -4)$ and parallel to line l_1.

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(i) At $x = 3$ we have,

$$P(3) = -6(3)^2 + 120(3) + 25000$$
$$= -54 + 360 + 25000$$
$$= 25306$$

(ii) $P'(5) = -12 \times 5 + 120$

$$= -60 + 120 = 60$$

(iii) For strictly increasing, $P'(x) > 0$

$$-12x + 120 > 0$$
$$120 > 12x$$
$$x < 10\text{t}$$
$$x \in (0,10)$$

OR

(iv) For maximum profit we put $P'(x) = 0$, i.e

$$0 = -12x + 120$$
$$x = 10$$

Now $P''(x) = -12 < 0$

At $x = 10$, profit function is maximum.

In this book, we have provided solutions along with the papers. To download the same papers in unsolved format for printing, please visit
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Sample Paper 17

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

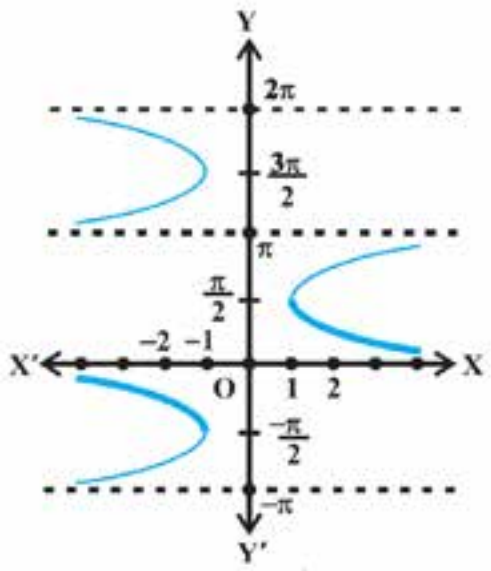
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. At which point does the graph of $y = \csc^{-1} x$ intersect the X-axis?



(a) (0,0)

(b) (1,0)

(c) (-1,0)

(d) The curve does not intersect the X-axis.

Sol :

From the graph, we can see that the curve never passes through the X-axis. This is because for any value of $x, \csc^{-1} x$ is never equal to 0.

Thus (d) is correct option.

2. The figure formed by four points $\hat{i} + \hat{j} + \hat{k}$, $2\hat{i} + 3\hat{j}$, $3\hat{i} - 5\hat{j} - 2\hat{k}$, $\hat{k} - \hat{j}$ is a

(a) Parallelogram

(b) Rectangle

(c) Trapezium

(d) Square

Sol :

Let the vertices be $\overrightarrow{OA} = \hat{i} + \hat{j} + \hat{k}$,

$\overrightarrow{OB} = 2\hat{i} + 3\hat{j}$, $\overrightarrow{OC} = 3\hat{i} + 5\hat{j} - 2\hat{k}$ and

$\overrightarrow{OD} = \hat{k} - \hat{j}$

$\overrightarrow{AB} = \hat{i} + 2\hat{j} - \hat{k}$, $\overrightarrow{BC} = \hat{i} + 2\hat{j} - 2\hat{k}$

$\overrightarrow{CD} = -3\hat{i} - 6\hat{j} + 3\hat{k} = -(\hat{i} + 2\hat{j} - \hat{k})$

$\overrightarrow{DA} = \hat{i} + 2\hat{j} + \hat{k}$

It is clear that, $|\overrightarrow{AB}| \neq |\overrightarrow{BC}| \neq |\overrightarrow{CA}| \neq |\overrightarrow{DA}|$

Also, $\overrightarrow{AB} \,||\, \overrightarrow{CD}$

Hence, figure formed by four points is a trapezium.

Thus (c) is correct option.

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3. If $\vec{a}\cdot\vec{b}=0$ and $\vec{a}+\vec{b}$ makes an angle of 60° with $\vec{a}$, then

(a) $|\vec{a}|=2|\vec{b}|$

(b) $2|\vec{a}|=|\vec{b}|$

(c) $|\vec{a}|=\sqrt{3}|\vec{b}|$

(d) $\sqrt{3}|\vec{a}|=|\vec{b}|$

Sol :

Given,

$$\vec{a}\cdot\vec{b}=0$$

Now, $(\vec{a}+\vec{b})\cdot\vec{a}=|\vec{a}+\vec{b}||\vec{a}|\cos 60^\circ$

$$\vec{a}\cdot\vec{a}+\vec{b}\cdot\vec{a}=|\vec{a}+\vec{b}||\vec{a}|\frac{1}{2}$$

$$|\vec{a}|^2+0=\frac{|\vec{a}+\vec{b}||\vec{a}|}{2}$$

$$2|\vec{a}|=|\vec{a}+\vec{b}|$$

On squaring both sides, we get

$$4|\vec{a}|^2=|\vec{a}|^2+|\vec{b}|^2+2\vec{a}\,\vec{b}$$

$$3|\vec{a}|^2=|\vec{b}|^2+0 \quad [\because |\vec{a}||\vec{b}|\cos\theta=\vec{a}\cdot\vec{b}=0]$$

$$\sqrt{3}|\vec{a}|=|\vec{b}|$$

Thus (d) is correct option.

4. If the direction cosines of a line are $\left(\frac{1}{c},\frac{1}{c},\frac{1}{c}\right)$, then

(a) $0<c<1$ (b) $c>2$

(c) $c=\pm\sqrt{2}$ (d) $c=\pm\sqrt{3}$

Sol :

Since, DC's of a line are $\left(\frac{1}{c},\frac{1}{c},\frac{1}{c}\right)$.

$$\left(\frac{1}{c}\right)^2+\left(\frac{1}{c}\right)^2+\left(\frac{1}{c}\right)^2=1\Rightarrow c^2=3\Rightarrow c=\pm\sqrt{3}$$

Thus (d) is correct option.

5. A coin and six faced die, both unbiased, are thrown simultaneously. The probability of getting a head on the coin and an odd number on the die is

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$

(c) $\frac{1}{4}$ (d) $\frac{2}{3}$

Sol :

Let

$E=$ Event of getting a head on a coin and

$F=$ Event of getting an odd number $\{1,3,5\}$, on a die

$$P(E)=\frac{1}{2}$$

$$P(F)=\frac{3}{6}=\frac{1}{2}$$

Since, E and F are independent events.

$$P(E\cap F)=P(E)\cdot P(F)$$

$$=\frac{1}{2}\times\frac{1}{2}=\frac{1}{4}$$

Thus (c) is correct option.

6. Find the angle between the following pairs of lines:

$$\vec{r}=2\hat{i}-5\hat{j}+\hat{k}+\lambda\left(3\hat{i}+2\hat{j}+6\hat{k}\right)$$

and $\vec{r}=7\hat{i}-6\hat{k}+\mu\left(\hat{i}+2\hat{j}+2\hat{k}\right)$

(a) $\cos^{-1}\left(\frac{19}{21}\right)$ (b) $\cos^{-1}\left(\frac{16}{21}\right)$

(c) $\cos^{-1}\left(\frac{13}{21}\right)$ (d) $\cos^{-1}\left(\frac{11}{21}\right)$

Sol :

The given lines are;

$$\vec{r}=2\hat{i}-5\hat{j}+\hat{k}+\lambda\left(3\hat{i}+2\hat{j}+6\hat{k}\right)$$

and $\vec{r}=7\hat{i}-6\hat{k}+\mu\left(\hat{i}+2\hat{j}+2\hat{k}\right)$

The given lines are parallel to the vectors

$$\vec{b}_1=3\hat{i}+2\hat{j}+6\hat{k}$$

and $\vec{b}_2=\hat{i}+2\hat{j}+2\hat{k}$

$$\vec{b}_1\cdot\vec{b}_2=\left(3\hat{i}+2\hat{j}+6\hat{k}\right)\cdot\left(\hat{i}+2\hat{j}+2\hat{k}\right)$$

$$=3\times1+2\times2+6\times2$$

$$=3+4+12=19$$

$$|\vec{b}_1|=\sqrt{(3)^2+(2)^2+(6)^2}$$

$$=\sqrt{9+4+36}=\sqrt{49}=7$$

$$|\vec{b}_2|=\sqrt{(1)^2+(2)^2+(2)^2}$$

$$=\sqrt{1+4+4}=\sqrt{9}=3$$

Let θ be the angle between the two lines

$$\cos\theta=\frac{\vec{b}_1\cdot\vec{b}_2}{|\vec{b}_1||\vec{b}_2|}$$

$$=\frac{19}{7\cdot3}=\frac{19}{21}$$

Hence, $\theta=\cos^{-1}\left(\frac{19}{21}\right)$.

Thus (a) is correct option.

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(a) Z=8 at (0,4) (b) Z=15 at (5,0)

(c) Z=6 at (2,0) (d) No minimum exists

Sol :

By evaluating Z at all corner points of the feasible region:

At (2,0), Z=6; at (0,4), Z=8; at (5,0), Z=15.

The least value is 6, obtained at (2,0).

Thus (c) is correct option.

18. Solution of $e^{dy/dx} = x$, when $x = 1$ and $y = 0$ is

(a) $y = x(\log x - 1) + 4$

(b) $y = x(\log x - 1) + 3$

(b) $y = x(\log x + 1) + 1$

(d) $y = x(\log x - 1) + 1$

Sol :

Given,

$$e^{dy/dx} = x$$

On taking log both sides, we get

$$\frac{dy}{dx} = \log x \Rightarrow dy = \log x \cdot dx$$

Integrating both sides, we have

$$\int dy = \int \log x \cdot dx$$

$$y = x\log x - \int \frac{x}{x} + C$$

$$= x\log x - x + C \qquad ...(1)$$

Also, it is given that at $x = 1$, $y = 0$

$$0 = (1)\log 1 - 1 + C \Rightarrow C = 1$$

On substituting it in Eq. (1), we get

$$y = x\log x - x + 1$$

$$y = x(\log x - 1) + 1$$

Thus (d) is correct option.

19. If $n > 1$, then

Assertion : $\int_0^{\infty} \frac{dx}{1+x^n} = \int_0^{1} \frac{dx}{(1-x^n)^{1/n}}$

Reason : $\int_a^b f(x)\,dx = \int_a^b f(a+b+x)\,dx$

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

Sol :

As we know that,

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$$

So, given Reason is false.

Now, in L.H.S, put

$$x^n = \tan^2\theta$$

$$nx^{n-1}dx = 2\tan\theta\sec^2\theta d\theta$$

So, $$\int_0^{\infty} \frac{dx}{1+x^n} = \frac{2}{n}\int_0^{\pi/2} \tan^{1-2+2/n}\,d\theta$$

$$= \frac{2}{n}\int_0^{\pi/2} \tan^{(2/n)-1}\theta d\theta$$

In R.H.S, put $x^n = \sin^2\theta$

$$nx^{n-1}dx = 2\sin\theta\cos\theta\,d\theta$$

So, $$\int_0^{1} \frac{dx}{(1-x^n)^{1/n}} = \frac{2}{n}\int_0^{\pi/2} \frac{1}{\cos^{2/n}\theta}\cdot \sin^{2/n-1}\theta\cos\theta\,d\theta$$

$$= \frac{2}{n}\int_0^{\pi/2} \tan^{(2/n\,-1)}\theta\,d\theta$$

Hence, Assertion is true; reason is false.

Thus (c) is correct option.

20. Assertion : $\int_{-\pi/2}^{\pi/2} |\sin x|\,dx = 2$

Reason : $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$, where $c \in a, b)$

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

Sol :

As We know that

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx, \text{ Where } c\,\varepsilon\ (a.b)$$

So, given reason is true.

Now, $\int_{-\pi/2}^{\pi/2} |\sin x|\,dx\ \ \sin x,$

Since, $|\sin x|$ is an even faction

so, $$\int_{-\pi/2}^{\pi/2} |\sin x|\,dx = 2\int_0^{\pi/2} |\sin x|\,dx$$

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$$= \lim_{h \to 0} 4(0+h)+1 = 1$$

LHL $\neq$ RHL

Hence, $f(x)$ is not continuous at $x = 0$ for any value of λ.

Thus (d) is correct option.

13. If gas is being pumped into a spherical balloon at the rate of 30 ft3/min. Then, the rate at which the radius increases, when it reaches the value 15 ft is

(a) $\frac{1}{15\pi}$ ft/min (b) $\frac{1}{30\pi}$ ft/min

(c) $\frac{1}{20}$ ft/min (d) $\frac{1}{25}$ ft/min

Sol :

Let $V = \frac{4}{3}\pi r^3$

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{1}{4\pi r^2}\frac{dV}{dt}$$

$$= \frac{30}{4 \times \pi \times 15 \times 15} \qquad \frac{dV}{dt} = 30, r = 15$$

$$= \frac{1}{30\pi} \text{ ft/min}$$

Thus (b) is correct option.

14. The length of the longest interval, in which $f(x) = 3\sin x - 4\sin^3 x$ is increasing is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$

(c) $\frac{3\pi}{2}$ (d) π

Sol :

We have $f(x) = 3\sin x - 4\sin^3 x = \sin 3x$

Since, $\sin x$ is increasing in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

$$-\frac{\pi}{2} \leq 2x \leq \frac{\pi}{2}$$

$$-\frac{\pi}{6} \leq x \leq \frac{\pi}{6}$$

Thus, the length of interval is $\frac{\pi}{6} - \left(-\frac{\pi}{6}\right) = \frac{\pi}{3}$.

Thus (a) is correct option.

15. If $A(\alpha) = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$, then the matrix $A^2(\alpha)$ is equal to

(a) $A(2\alpha)$ (b) $A(\alpha)$

(c) $A(3\alpha)$ (d) $A(4\alpha)$

Sol :

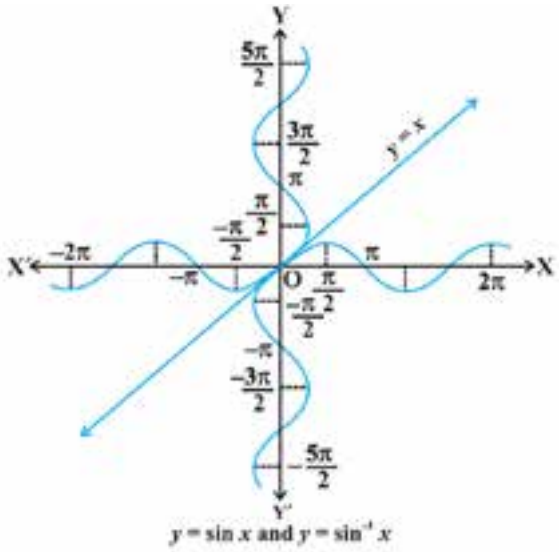
$$A^2(\alpha) = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}\begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2\alpha - \sin^2\alpha & 2\cos\alpha\sin\alpha \\ -2\sin\alpha\cos\alpha & \cos^2\alpha - \sin^2\alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix} = A(2\alpha)$$

Thus (a) is correct option.

16. What happens to the value of $y = \tan x$ as x approaches $\pi/2$ from the left side?



$y = \sin x$ and $y = \sin^{-1} x$

(a) $y \to 0$ (b) $y \to \infty$

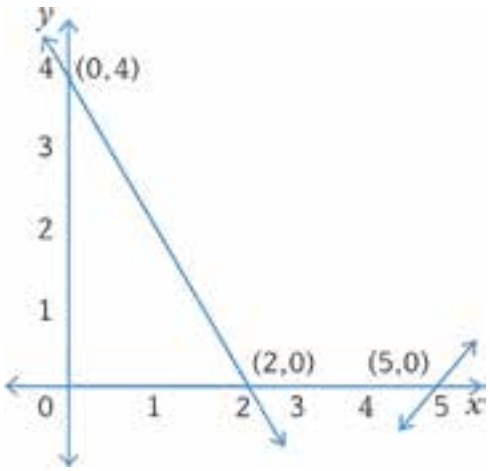
(c) $y + \infty$ (d) Undefined

Sol :

From the graph, as x approaches $\pi/2$ from the left, the tangent curve shoots upwards infinitely. This indicates that the value of $\tan x$ tends to $+\infty$.

Thus (c) is correct option.

17. For the same LPP with objective function $Z = 3x + 2y$, what is the minimum value of Z and at which point does it occur?



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7. If $P(A)=\frac{1}{12}$, $P(B)=\frac{5}{12}$ and $P\left(\frac{B}{A}\right)=\frac{1}{15}$, then $P(A \cup B)$ is equal to

(a) $\frac{89}{180}$ (b) $\frac{90}{180}$

(c) $\frac{91}{180}$ (d) $\frac{92}{180}$

Sol :

$$P\left(\frac{B}{A}\right)=\frac{P(A \cap B)}{P(A)}$$

$$\frac{1}{15}=\frac{P(A \cap B)}{\frac{1}{12}} \Rightarrow P(A \cap B)=\frac{1}{180}$$

Also, $P(A \cup B)=P(A)+P(B)-P(A \cap B)$

$$P(A \cup B)=\frac{1}{12}+\frac{5}{12}-\frac{1}{180}=\frac{89}{180}$$

Thus (a) is correct option.

8. Solution of the differential equation $xdy-ydx=0$ represents a

(a) Parabola (b) Circle

(c) Hyperbola (d) Straight line

Sol :

Given differential equation is

$$x\,dy = y\,dx \Rightarrow \frac{dy}{y}=\frac{dx}{x}$$

$$\int \frac{dy}{y}=\int \frac{dx}{x}$$

$$\log_e y = \log_e x+\log_e C$$

$$y = Cx$$

Which is a straight line.

Thus (d) is correct option.

9. If $\int \frac{x+2}{2x^2+6x+5}dx=P\int \frac{4x+6}{2x^2+6x+5}dx+\frac{1}{2}\int \frac{dx}{2x^2+6x+5}$

Then the value of P is

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$

(c) $\frac{1}{4}$ (d) 2

Sol :

$$\int \frac{x+2}{2x^2+6x+5}dx = \frac{1}{2}\int \frac{4x+6+2}{2x^2+6x+5}dx$$

$$= \frac{1}{4}\int \frac{4x+6}{2x^2+6x+5}dx+\frac{1}{2}\int \frac{dx}{2x^2+6x+5}$$

$$P = \frac{1}{4}$$

Thus (c) is correct option.

10. $\int_0^{\pi/2}\left|\cos\left(\frac{x}{2}\right)\right|dx$ is equal to

(a) 1 (b) -2

(c) $\sqrt{2}$ (d) 0

Sol :

$$\int_0^{\pi/2}\left|\cos\left(\frac{x}{2}\right)\right|dx = \int_0^{\pi/2}\cos\left(\frac{x}{2}\right)dx$$

$$= 2\left[\sin\left(\frac{x}{2}\right)\right]_0^{\pi/2}$$

$$= 2\left[\sin\frac{\pi}{4}-\sin 0\right]$$

$$= 2\sin\frac{\pi}{4}$$

$$= 2\times\frac{1}{\sqrt{2}}=\sqrt{2}$$

Thus (c) is correct option.

11. An integrating factor of the differential equation $x\frac{dy}{dx}+y\log x=xe^x x^{-\frac{1}{2}\log x}(x>0)$ is

(a) $x^{\log x}$ (b) $(\sqrt{x})^{\log x}$

(c) $(\sqrt{e})^{(\log x)^2}$ (d) e^{x^2}

Sol :

Here,

$$\frac{dy}{dx}+y\frac{1}{x}\log x = e^x x^{-(1/2)\log x}$$

$$\text{IF} = e^{\frac{1}{x}\log x\,dx}=e^{\frac{(\log x)^2}{2}}=(\sqrt{e})^{(\log x)^2}$$

Thus (c) is correct option.

12. For the value of λ is the function defined by $f(x)=\begin{cases}\lambda(x^2-2x), & \text{if } x\le 0\\ 4x+1, & \text{if } x<0\end{cases}$ continuous at $x=0$?

(a) 0

(b) 1

(c) 2

(d) Not continuous at $x=0$ for any value of λ

Sol :

Given: $f(x)=\begin{cases}\lambda(x^2-2x), & \text{if } x\le 0\\ 4x+1, & \text{if } x<0\end{cases}$

At $x=0$

$$\text{LHL} = \lim_{x\to 0^-}f(x)=\lim_{x\to 0^-}\lambda(x^2-2x)$$

$$= \lim_{h\to 0}\lambda\left[(0-h)^2-2(0-h)\right]=0$$

$$\text{RHL} = \lim_{x\to 0^+}f(x)=\lim_{x\to 0^+}4x+1$$

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$$= 2\int_0^{\pi/2} |\sin x|\,dx$$

$$= 2\int_0^{\pi/2} \sin x\,dx$$

$$= -(\cos x)0^{\pi/2}$$

$$= 2(0-1) = 2$$

Hence, both Assertion and reason are true bat reason is not correct explanation for assertion.

Thus (b) is correct option.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. If a line makes angles 90°, 135°, 45° with then x, y and z axes respectively, find the direction consines.

Sol :

Let l, m and n be direction cosines of the lines.

We have $\alpha = 90^\circ$, $\beta = 135^\circ$ and $\gamma = 45^\circ$

Now $l = \cos\alpha = \cos 90^\circ = 0$,

$m = \cos\beta = \cos 135^\circ = \dfrac{-1}{\sqrt{2}}$

and $n = \cos\gamma = \cos 45^\circ = \dfrac{1}{\sqrt{2}}$

Hence, the direction cosines of a line are 0, $\dfrac{-1}{\sqrt{2}}$ and $\dfrac{1}{\sqrt{2}}$.

22. If $P(not\ A) = 0.7$, $P(B) = 0.7$ and $P(B/A) = 0.5$, then find $P(A/B)$.

Sol :

We have $P(A') = 0.7$, $P(B) = 0.7$

and $P\left(\dfrac{B}{A}\right) = 0.5$

Now $P(A) = 1 - P(A')$

$= 1 - 0.7 = 03$

Now, $P\left(\dfrac{B}{A}\right) = \dfrac{P(A \cap B)}{P(A)}$

$0.5 = \dfrac{P(A \cap B)}{0.3}$

$P(A \cap B) = 0.15$

$P\left(\dfrac{A}{B}\right) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{0.15}{0.7}$

Thus $P\left(\dfrac{A}{B}\right) = \dfrac{3}{14}$

23. Find $\int \dfrac{\sin^2 x - \cos^2 x}{\sin x \cos x}\,dx$.

OR

Find $\int \dfrac{\sin^2 x - \cos^2 x}{\sin^2 x \cos^2 x}\,dx$.

Sol :

We have $I = \int \dfrac{\sin^2 x - \cos^2 x}{\sin x \cdot \cos x}\,dx$

$= \int \left[\dfrac{\sin^2 x}{\sin x \cdot \cos x} - \dfrac{\cos^2 x}{\sin x \cdot \cos x}\right]dx$

$= \int \left[\dfrac{\sin x}{\cos x} - \dfrac{\cos x}{\sin x}\right]dx$

$= \int (\tan x - \cot x)\,dx$

$= \int \tan x dx - \int \cot x dx$

$= \log|\sec x| - [-\log|\operatorname{cosec} x|] + C$

$= \log|\sec x| + \log|\operatorname{cosec} x| + C$

$= \log|\sec x \cdot \operatorname{cosec} x| + C$

OR

We have $I = \int \dfrac{\sin^2 x - \cos^2 x}{\sin^2 x \cos^2 x}\,dx$

$= \int \dfrac{\sin^2 x}{\sin^2 x \cos^2 x}\,dx - \int \dfrac{\cos^2 x}{\sin^2 x \cos^2 x}\,dx$

$= \int \sec^2 x\,dx - \int \operatorname{cosec}^2 x\,dx$

$= \tan x + \cot x + C$

24. Find the position vector of a point which divides the join of points with position vectors $\vec{a} - 2\vec{b}$ and $2\vec{a} + \vec{b}$ externally in the ratio 2 : 1.

OR

If $\vec{a} = 4\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$, then find a unit vector parallel to the vector $\vec{a} + \vec{b}$.

Sol :

Given position vectors are

$\overrightarrow{OA} = \vec{a} - 2\vec{b}$

and $\overrightarrow{OB} = 2\vec{a} + \vec{b}$

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SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Find the particular solution of the differential equation $\frac{dy}{dx}=\frac{xy}{x^2+y^2}$ given that $y=1$, when $x=0$.

Sol :

We have $$\frac{dy}{dx}=\frac{xy}{x^2+y^2}$$
$$=\frac{\frac{y}{x}}{1+\frac{y^2}{x^2}} \quad ...(1)$$

Which is a homogeneous differential equation as

$$\frac{dy}{dx}=F\left(\frac{y}{x}\right)$$

Substituting $y=vx \Rightarrow \frac{dy}{dx}=v+x\frac{dv}{dx}$ in eq (1) we have

$$v+x\frac{dv}{dx}=\frac{v}{1+v^2}$$
$$x\frac{dv}{dx}=\frac{v}{1+v^2}-v$$
$$x\frac{dv}{dx}=\frac{v-v-v^3}{1+v^2}$$
$$x\frac{dv}{dx}=-\frac{v^3}{1+v^2}$$
$$\frac{1+v^2}{v^3}dv=-\frac{dx}{x}$$

Integrating both sides, we have

$$\int\left(\frac{1}{v^3}+\frac{1}{v}\right)dv=-\int\frac{dx}{x}$$
$$-\frac{1}{2v^2}+\log|v|=-\log|x|+C$$
$$-\frac{x^2}{2y^2}+\log\left|\frac{y}{x}\right|=-\log|x|+C$$
$$-\frac{x^2}{2y^2}+\log|y|-\log|x|=-\log|x|+C$$
$$-\frac{x^2}{2y^2}+\log|y|=C \quad ...(2)$$

Also, it is given that $y=1$, when $x=0$.
From Eq. (2), we have

$$0+\log|1|=C \Rightarrow C=0$$

Substituting $C=0$ in Eq. (2), we get

$$-\frac{x^2}{2y^2}+\log|y|=0$$
$$\log|y|=\frac{x^2}{2y^2}$$
$$y=e^{\frac{x^2}{2y^2}}$$

Which is the required solution.

33. Maximize $Z=3x+4y$, subject to the constraints; $x+y\le 4$, $x\ge 0$, $y\ge 0$.

OR

Minimize $Z=-3x+4y$ subject to the constraints

$$x+2y\le 8,$$
$$3x+2y\le 12,$$
$$x\ge 0,\ y\ge 0$$

Sol :

We have to maximize $Z=3x+4y$, subject to the constraints :

Maximize $Z=3x+4y$

Subject to $x+y\le 4$ ...(i)

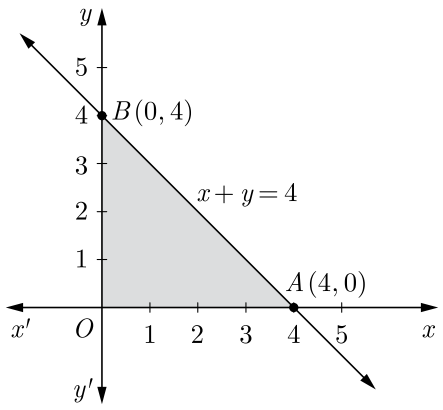
and $x\ge 0,\ y\ge 0$

Region corresponding to $x+y\le 4$:

Line $x+y=4$			
x	0	4	Point $(0,0)$ is true for $x+y\le 4$. So, the region is towards the origin.
y	4	0	

Since $x\ge 0$, $y\ge 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is $OABO$ which is bounded The corner points of the feasible region are $O(0,0)$, $A(4,0)$ and $B(0,4)$.



Since, the feasible region is a bounded region, we can check the objective function at all the corner to find the maxima. The values of objective function Z at these points are as follows.

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$$= 2\sin t + C = 2\sin\sqrt{x} + C$$

Substituting $t = \sqrt{x}$ we have

$$I = 2\sin t + C = 2\sin\sqrt{x} + C$$

28. If $\hat{a}$, $\hat{b}$ and $\hat{c}$ are mutually perpendicular unit vectors, then find the value of $|2\hat{a} + \hat{b} + \hat{c}|$.

Sol :

Given, $\hat{a}$, $\hat{b}$ and $\hat{c}$ are mutually perpendicular unit vector, i.e.

$$\hat{a}\cdot\hat{b} = \hat{b}\cdot\hat{c} = \hat{c}\cdot\hat{a} = 0 \quad ...(1)$$

and $$|\hat{a}| = |\hat{b}| = |\hat{c}| = 1 \quad ...(2)$$

Now, $|2\hat{a} + \hat{b} + \hat{c}|^2 = (2\hat{a} + \hat{b} + \hat{c})\cdot(2\hat{a} + \hat{b} + \hat{c})$

$$= 4(\hat{a}\cdot\hat{a}) + 2(\hat{a}\cdot\hat{b}) + 2(\hat{a}\cdot\hat{c}) + 2(\hat{b}\cdot\hat{a}) + (\hat{b}\cdot\hat{b}) + (\hat{b}\cdot\hat{c}) + 2(\hat{c}\cdot\hat{a}) + (\hat{c}\cdot\hat{b}) + (\hat{c}\cdot\hat{c})$$

$$= 4(|\hat{a}|^2 + 2(0) + 2(0) + 2(0) + |\vec{b}|^2 + (0) + 2(0) + (0) + |\hat{c}|^2$$

$$= 4(1) + 1 + 1 = 6$$

Thus $|2\hat{a} + \hat{b} + \hat{c}| = \sqrt{6}$ as length is always positive.

29. Determine the value of k for which the following function is continuous at $x = 3$:

$$f(x) = \begin{cases} \dfrac{(x+3)^2 - 36}{x-3}, & x \neq 3 \\ k, & x = 3 \end{cases}$$

OR

Determine the value of constant k so that the function $f(x) = \begin{cases} \dfrac{kx}{|x|}, & \text{if } x < 0 \\ 3, & \text{if } x \geq 0 \end{cases}$ is continuous at $x = 0$.

Sol :

We have $$f(x) = \begin{cases} \dfrac{(x+3)^2 - 36}{x-3}, & x \neq 3 \\ k, & x = 3 \end{cases}$$

Let $f(x)$ is continuous at $x = 3$, then we have

$$\lim_{x\to 3} f(x) = f(3)$$

$$\lim_{x\to 3} \frac{(x+3)^2 - 36}{x-3} = k$$

$$\lim_{x\to 3} \frac{(x+3)^2 - 6^2}{x-3} = k$$

$$\lim_{x\to 3} \frac{(x+3-6)(x+3+6)}{x-3} = k$$

$$\lim_{x\to 3} \frac{(x-3)(x+9)}{(x-3)} = k$$

$$\lim_{x\to 3} (x+9) = k$$

$$3 + 9 = k \Rightarrow k = 12$$

OR

Let $$f(x) = \begin{cases} \dfrac{kx}{|x|}, & \text{if } x < 0 \\ 3, & \text{if } x \geq 0 \end{cases}$$

is continuous at $x = 0$

then, $$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^-} f(0-h) = f(0)$$

$$\lim_{h\to 0} f(0+h) = \lim_{h\to 0} \frac{k(-h)}{|-h|} = 3$$

$$\lim_{h\to 0} \left(\frac{-kh}{h}\right) = 3 \Rightarrow \lim_{h\to 0} (-k) = 3$$

$$k = -3$$

30. Find $|AB|$, if $A = \begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix}$.

Sol :

We have $A = \begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix}$

Now, $$AB = \begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0+0 & 0+0 \\ 0+0 & 0+0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$|AB| = 0$$

31. Write the value of $\tan^{-1}\left[2\sin\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right]$.

Sol :

$$\tan^{-1}\left[2\sin\left(2\cos^{-1}\frac{\sqrt{3}}{2}\right)\right]$$

$$= \tan^{-1}\left[2\sin\left\{2\cos^{-1}\left(\cos\frac{\pi}{6}\right)\right\}\right]$$

$$= \tan^{-1}\left[2\sin\left\{2\times\frac{\pi}{6}\right\}\right]$$

$$= \tan^{-1}\left(2\sin\frac{\pi}{3}\right)$$

$$= \tan^{-1}\left(2\cdot\frac{\sqrt{3}}{2}\right)$$

$$= \tan^{-1}(\sqrt{3})$$

$$= \tan^{-1}\left(\tan\frac{\pi}{3}\right) = \frac{\pi}{3}$$

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Let $\overrightarrow{OC}$ be the position vector of a point C which divides the join of points, with position vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$, externally in the ratio $2:1$.

Using externally section formula we have

$$\begin{aligned}\overrightarrow{OC} &= \frac{2\overrightarrow{OB}-1\overrightarrow{OA}}{2-1}\\ &= \frac{2(2\vec{a}+\vec{b})-1(\vec{a}-2\vec{b})}{1}\\ &= 4\vec{a}+2\vec{b}-\vec{a}+2\vec{b}\\ &= 3\vec{a}+4\vec{b}\end{aligned}$$

OR

We have $\vec{a} = 4\hat{i}-\hat{j}+\hat{k}$,

$\vec{b} = 2\hat{i}-2\hat{j}+\hat{k}$

Now, $\vec{a}+\vec{b} = (4\hat{i}-\hat{j}+\hat{k})+(2\hat{i}-2\hat{j}+\hat{k})$

$= 6\hat{i}-2\hat{j}+2\hat{k}$

and $|\vec{a}+\vec{b}| = \sqrt{(6)^2+(-3)^2+(2)^2}$

$= \sqrt{36+9+4}$

$= \sqrt{49} = 7$ units

The unit vector parallel to the vector $\vec{a}+\vec{b}$ is

$$\frac{\vec{a}+\vec{b}}{|\vec{a}+\vec{b}|} = \frac{6\hat{i}-3\hat{j}+2\hat{k}}{7}$$

25. If $R=\{(a,a^3)$: a is a prime number less than $5\}$ be a relation. Find the range of R.

Sol :

Given, $R=\{(a,a^3)$: a is a prime number less than $5\}$

Since 2 and 3 are the prime numbers less than 5. So, a can take value 2 and 3.

Thus, $R = \{(2,2^3),(3,3^3)\}$

$= \{(2,8),(3,27)\}$

Hence, the range of R is $\{8,27\}$.

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. The total cost $C(x)$ associated with the production of x units of an item is given by $C(x)=0.005x^3-0.02x^2+30x+5000$.

Find the marginal cost when 3 units are produced, where by marginal cost we mean the instantaneous rate of change of total cost at any level of output.

OR

The total revenue received from the sale of x units of a product is given by $R(x)=3x^2+36x+5$ in rupees. Find the marginal revenue we mean the rate of change of total revenue with respect to the number of times sold at an instant.

Sol :

We have,

$$C(x) = 0.005x^3-0.02x^2+30x+5000$$

Marginal cost,

$$\begin{aligned}MC(x) &= \frac{d}{dx}C(x)\\ &= \frac{d}{dx}(0.005x^3-0.02x^2+30x+5000)\\ &= 0.005\times 3x^2-0.02\times 2x+30+0\\ &= 0.015x^2-0.04x+30\end{aligned}$$

When 3 units are produced, Marginal cost

$$\begin{aligned}MC(3) &= 0.015(9)-0.04(3)+30\\ &= 0.135-0.12+30 = 30.015\end{aligned}$$

OR

Marginal Revenue,

$$\begin{aligned}MR &= \frac{dR}{dx}\\ &= \frac{d}{dx}(3x^2+36x+5)\\ &= 6x+36\end{aligned}$$

When $x=5$, Marginal Revenue

$$MR = 6\times 5+36 = 66$$

27. Evaluate $\int \frac{\cos\sqrt{x}}{\sqrt{x}}dx$.

Sol :

We have $I = \int \frac{\cos\sqrt{x}}{\sqrt{x}}dx$

Substituting $\sqrt{x} = t \Rightarrow \frac{1}{2\sqrt{x}}dx = dt$

$$\frac{1}{\sqrt{x}}dx = 2dt$$

$$I = \int \frac{\cos\sqrt{x}}{\sqrt{x}}dx = 2\int \cos t\, dt$$

$$= \int \frac{\cos\sqrt{x}}{\sqrt{x}}dx = 2\int \cos t\, dt$$

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Corner points	$Z = 3x + 4y$
$O(0,0)$	$Z = 0 + 0 = 0$
$A(4,0)$	$Z = 3 \times 4 + 4 \times 0 = 12$
$B(0,4)$	$Z = 3 \times 0 + 4 \times 4 = 16$

The maximum value of Z is 16 at $B(0,4)$ and the optimal solution is $x = 0$, $y = 4$.

OR

Minimized $Z = -3x + 4y$

Subjected to $x + 2y \leq 8$

$3x + 2y \leq 12$,

and $x \geq 0$, $y \geq 0$

(i) Region corresponding to $x + 2y \leq 8$:

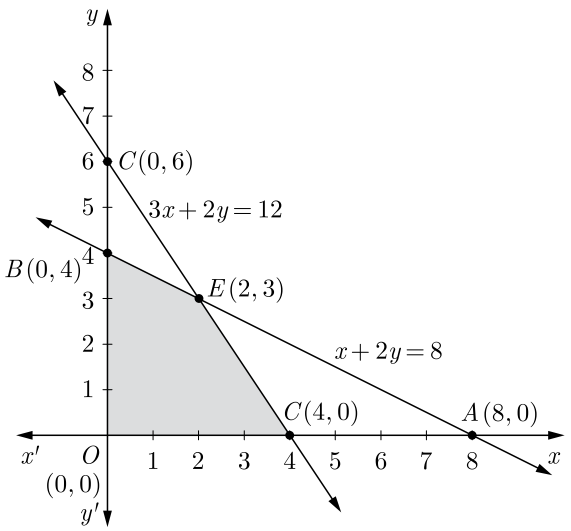
Line $x + 2y = 8$			
x	0	8	Point $(0,0)$ is true for $x + 2y \leq 8$. So, the region is towards the origin.
y	4	0	

(ii) Region corresponding to $3x + 2y \leq 12$:

Line $3x + 2y = 12$			
x	0	4	Point $(0,0)$ is true for $3x + 2y \leq 12$. So, the region is towards the origin.
y	6	0	

Since $x \geq 0$, $y \geq 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is $OCEBO$ which is bounded. Solving equations $x + 2y = 8$ and $3x + 2y = 12$, we get $E(2,3)$ The corner points of the feasible region are $O(0,0)$, $C(4,0)$, $E(2,3)$ and $B(0,4)$.



Since, the feasible region is a bounded region, we can check the objective function at all the corner to find the maxima. The values of objective function Z at these points are as follows.

Corner points	$Z = -3x + 4y$
$O(0,0)$	$Z = 0 + 0 = 0$
$B(0,4)$	$Z = -3 \times 0 + 4 \times 4 = 16$
$E(2,3)$	$Z = -3 \times 2 + 4 \times 3 = 6$
$C(4,0)$	$Z = -3 \times 4 + 4 \times 0 = -12$

Hence the minimum value of Z is -12 at $C(4,0)$ and the optimal solution is $x = 4$; $y = 0$.

34. Find : $\int \frac{3x+5}{x^2+3x-18}\,dx$.

OR

Prove that $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$, hence evaluate $\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x}\,dx$.

Sol :

We have $I = \int \frac{3x+5}{x^2+3x-18}\,dx$ ...(1)

Now $3x + 5 = A\frac{d}{dx}(x^2+3x-18) + B$

or $3x + 5 = A(2x+3) + B$ ...(2)

Comparing the coefficient of x on both side, we have

$2A = 3 \Rightarrow A = \frac{3}{2}$

and comparing the constant on both side, we get

$5 = 3A + B$

$B = 5 - 3A \Rightarrow B = 5 - 3\left(\frac{3}{2}\right) = \frac{1}{2}$

Now $3x + 5 = \frac{3}{2}(2x+3) + \frac{1}{2}$ ...(3)

Now $I = \int \frac{\frac{3}{2}(2x+3)+\frac{1}{2}}{x^2+3x-18}\,dx$

$= \frac{3}{2}\int \frac{2x+3}{x^2+3x-18}\,dx + \frac{1}{2}\int \frac{1}{x^2+3x-18}\,dx$

$= I_1 + I_2$

$I_1 = \frac{3}{2}\log\left|x^2+3x-18\right|$ $\quad \int \frac{f'(x)}{f(x)}dx = \log|f(x)|$

$I_2 = \frac{1}{2}\int \frac{1}{\left(x+\frac{3}{2}\right)^2 - \frac{81}{4}}\,dx$

$= \frac{1}{2}\int \frac{1}{\left(x+\frac{3}{2}\right)^2 - \left(\frac{9}{2}\right)^2}\,dx$

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A transport company uses three types of trucks T_1, T_2 and T_3 to transport three types of vehicles V_1, V_2 and V_3. The capacity of each truck in terms of three types of vehicles is given below:

	V_1	V_2	V_3
T_1	1	3	2
T_2	2	2	3
T_3	3	2	2

Using matrix method find:

(i) The number of trucks of each type required to transport 85, 105 and 110 vehicles of V_1, V_2 and V_3 types respectively.

(ii) Find the number of vehicles of each type which can be transported if company has 10, 20 and 30 trucks of each type respectively.

Sol :

Let x, y and z be the number of trucks of types T_1, T_2 and T_3 respectively, then the given information can be put in the form of simultaneous linear equations as:

$$x + 2y + 3z = 85$$

$$3x + 2y + 2z = 105$$

$$2x + 3y + 2z = 110$$

The above system of linear equations can be written in the matrix form as:

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 2 \\ 2 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 85 \\ 105 \\ 110 \end{bmatrix}$$

or $$AX = B$$

Since, $|A| = 1(4-6) - 2(6-4) + 3(9-4) = 9 \neq 0$, then A^{-1} exist, so that the unique solution of $AX = B$ is $X = A^{-1}B$.

Here, $$A^{-1} = \frac{\text{adj.}A}{|A|}$$

$$= \frac{1}{9}\begin{bmatrix} -2 & 5 & -2 \\ -2 & -4 & 7 \\ 5 & 1 & -4 \end{bmatrix}$$

and $$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = X = A^{-1}B$$

$$= \frac{1}{9}\begin{bmatrix} -2 & 5 & -2 \\ -2 & -4 & 7 \\ 5 & 1 & -4 \end{bmatrix} \begin{bmatrix} 85 \\ 105 \\ 110 \end{bmatrix}$$

$$= \frac{1}{9}\begin{bmatrix} -170 + 525 - 220 \\ -170 - 420 + 770 \\ 425 + 105 - 440 \end{bmatrix}$$

$$= \frac{1}{9}\begin{bmatrix} 135 \\ 180 \\ 90 \end{bmatrix} = \begin{bmatrix} 15 \\ 20 \\ 10 \end{bmatrix}$$

Hence the number of trucks of types T_1, T_2 and T_3 15, 20 and 10 respectively.

Now number of vehicles of types V_1, V_2 and V_3 are

$$V = \begin{matrix} \\ \begin{bmatrix} 10 & 20 & 30 \end{bmatrix} \end{matrix} \begin{matrix} \begin{matrix} V_1 & V_2 & V_3 \end{matrix} \\ \begin{bmatrix} 1 & 3 & 2 \\ 2 & 2 & 3 \\ 3 & 2 & 2 \end{bmatrix} \end{matrix} \begin{matrix} T_1 \\ T_2 \\ T_3 \end{matrix}$$

$$= \begin{matrix} V_1 & V_2 & V_3 \\ \begin{bmatrix} 140 & 130 & 140 \end{bmatrix} \end{matrix}$$

Hence the number of vehicles of types V_1, V_2 and V_3 are 140, 130 and 140, respectively.

38. Hindustan Pencils Pvt. Ltd. is an Indian manufacturer of pencils, writing materials and other stationery items, established in 1958 in Bombay. The company makes writing implements under the brands Nataraj and Apsara, and claims to be the largest pencil manufacturer in India.



Hindustan Pencils manufactures x units of pencil in a given time, if the cost of raw material is square of the pencils produced, cost of transportation is twice the number of pencils produced and the property tax costs ₹ 5000. Then,

(i) Find the cost function $C(x)$.

(ii) Find the cost of producing 21st pencil.

(iii) The marginal cost of producing 50 pencils.

Sol :

(i) Cost function $C(x)$,

The cost function is sum of variable cost and fixed cost.

Thus $$C(x) = x^2 + 2x + 5000$$

(ii) Find the cost of producing 21st pen.

$$C(21) - C(20) = \left[(21)^2 + 2 \times 21 + 5000\right] - \left[(20)^2 + 2 \times 20 + 5000\right]$$

$$= (21)^2 - (20)^2 + 21$$

$$= 41 + 21 = 62$$

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SECTION - E

Case study based questions are compulsory.

36. OYO Rooms, also known as OYO Hotels & Homes, is an Indian multinational hospitality chain of leased and franchised hotels, homes and living spaces. Founded in 2012 by Ritesh Agarwal, OYO initially consisted mainly of budget hotels.



Data analyst at OYO say that during frequent trips to a certain city, a traveling salesperson stays at hotel A 50% of the time, at hotel B 30% of the time, and at hotel C 20% of the time. When checking in, there is some problem with the reservation 3% of the time at hotel A, 6% of the time at hotel B, and 10% of the time at hotel C. Suppose the salesperson travels to this city.

(i) Find the probability that the salesperson stays at hotel A and has a problem with the reservation.

(ii) Find the probability that the salesperson has a problem with the reservation.

(iii) Suppose the salesperson has a problem with the reservation; what is the probability that the salesperson is staying at hotel A?

Sol :

Define the following events:

A = stays at hotel A; B = stays at hotel B;

C = stays at hotel C; and R = problem with the reservation.

Convert all the given percentages into probabilities.

The phrase of the time indicates conditional probability.

$$P(A) = 0.50 \;\; P(B) = 0.30 = P(C) = 0.20$$

$$P(R) \mid A) = 0.03 \;\; P(R \mid B) = 0.06 \;\; P(R \mid C) = 0.10$$

(i) The events A and R are dependent. The likelihood of a problem with a reservation depends on the hotel

$$\begin{aligned} P \mid (A \cap R) &= P(A) \cdot P(R \mid A) \\ &= (0.50)(0.03) \\ &= 0.0150 \end{aligned}$$

The probability of staying at hotel A and having a problem with the reservation is 0.0150.

(ii) $$\begin{aligned} P(R) &= P(A \cap R) + P(B \cap R) + P(C \cap R) \\ &= P(A) \cdot P(R \mid A) + P(R \mid B) \\ &\quad + P(C) \cdot P(R) \mid C) \\ &= (0.50)(0.03) + 90.30(0.06) + (0.20)(0.10) \\ &= 0.0150 + 0.0180 + 0.0200 = 0.0530 \end{aligned}$$

The probability of a problem with the reservation (regardless of the hotel) is 0.0530.

(iii) $$\begin{aligned} P(A \mid R) &= \frac{P(A \cap R)}{P(R)} \\ &= \frac{0.0150}{0.0530} = 0.2830 \end{aligned}$$

The probability that the salesperson stayed at hotel A, given a problem with the reservation, is 0.2830.

37. A car carrier trailer, also known as a car-carrying trailer, car hauler, or auto transport trailer, is a type of trailer or semi-trailer designed to efficiently transport passenger vehicles via truck. Commercial-size car carrying trailers are commonly used to ship new cars from the manufacturer to auto dealerships. Modern car carrier trailers can be open or enclosed. Most commercial trailers have built-in ramps for loading and off-loading cars, as well as power hydraulics to raise and lower ramps for stand-alone accessibility.



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$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right| + C$

$$= \frac{1}{2}\cdot\frac{1}{2\left(\frac{9}{2}\right)}\log\left|\frac{\left(x+\frac{3}{2}\right)-\frac{9}{2}}{\left(x+\frac{3}{2}\right)+\frac{9}{2}}\right| + C$$

$$= \frac{1}{18}\log\left|\frac{x-3}{x+6}\right| + C$$

$$I = I_1 + I_2$$

$$= \frac{3}{2}\log\left|x^2+3x-18\right| + \frac{1}{18}\log\left|\frac{x-3}{x+6}\right| + C$$

OR

We have $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

Consider RHS $\int_0^a f(a-x)\,dx$

Substituting $t = a - x$, then $dt = -dx$

Also, when $x = 0$, then $t = a$ and when $x = a$, then $t = 0$

Thus RHS

$$\int_0^a f(a-x)\,dx = -\int_a^0 f(t)\,dt$$

$$= \int_0^a f(t)\,dt$$

$$= \text{LHS} \qquad \text{Hence proved.}$$

Now, we have

$$I = \int_0^\pi \frac{x\sin x}{1+\cos^2 x}\,dx \qquad ...(i)$$

$$= \int_0^\pi \frac{(\pi-x)\sin(\pi-x)\,dx}{1+\cos^2(\pi-x)}$$

$$= \int_0^\pi \frac{(\pi-x)\sin x\,dx}{1+\cos^2 x}$$

$$= \pi\int_0^\pi \frac{\sin x\,dx}{1+\cos^2 x} - \int_0^\pi \frac{x\sin x\,dx}{1+\cos^2 x}$$

$$= \pi\int_0^\pi \frac{\sin x\,dx}{1+\cos^2 x} - I \qquad \text{[from eq. (i)]}$$

$$2I = \pi\int_0^\pi \frac{\sin x}{1+\cos^2 x}\,dx$$

$$I = \frac{\pi}{2}\int_0^\pi \frac{\sin x}{1+\cos^2 x}\,dx$$

Substituting $\cos x = t \Rightarrow -\sin x\,dx = dt$

Also, when $x = 0$, then $t = 1$ and when $x = \pi$, then $t = -1$

Thus $$I = -\frac{\pi}{2}\int_1^{-1} \frac{dt}{1+t^2}$$

$$= \frac{\pi}{2}\int_{-1}^{1} \frac{dt}{1+t^2}$$

$$= \frac{\pi}{2}[\tan^{-1}t]_{-1}^{1}$$

$$= \frac{\pi}{2}[\tan^{-1}(1) - \tan^{-1}(-1)]$$

$$= \frac{\pi}{2}\left[\frac{\pi}{4} - \left(-\frac{\pi}{4}\right)\right]$$

$$= \frac{\pi}{4}\left[\frac{\pi}{2}\right] = \frac{\pi^2}{4}$$

35. Prove that the line through $A(0,-1,-1)$ and $B(4,5,1)$ intersects the line through $C(3,9,4)$ and $D(-4,4,4)$.

Sol :

The equation of line through $A(0,-1,-1)$ and $B(4,5,1)$ is

$$\frac{x-0}{4-0} = \frac{y+1}{5+1} = \frac{z+1}{1+1},$$

i.e. $$\frac{x}{4} = \frac{y+1}{6} = \frac{z+1}{2} \qquad ...(1)$$

and equation of the through $C(3,9,4)$ and $D(-4,4,4)$ is

$$\frac{x-3}{-4-3} = \frac{y-9}{4-9} = \frac{z-4}{0}$$

i.e., $$\frac{x-3}{-7} = \frac{y-9}{-5} = \frac{z-4}{0} \qquad ...(2)$$

We know that, the lines

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$$

and $$\frac{x-x_2}{a_2} = \frac{y-y_2}{b_2} = \frac{z-z_2}{c_2}$$

Will intersect, if

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ a_2 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

The given lines will intersect, if

$$\begin{vmatrix} 3-0 & 9-(-1) & 4-(-1) \\ 4 & 6 & 2 \\ -7 & -5 & 0 \end{vmatrix} = 0$$

Now, consider,

$$\begin{vmatrix} 3-0 & 9-(-1) & 4-(-1) \\ 4 & 6 & 2 \\ -7 & -5 & 0 \end{vmatrix} = \begin{vmatrix} 3 & 10 & 5 \\ 4 & 6 & 2 \\ -7 & -5 & 0 \end{vmatrix}$$

$$= 3(0+10) - 10(0+14) + 5(-20+42)$$

$$= 30 - 140 + 110 = 0$$

Hence, the given lines intersect.

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The cost of producing 21st pencil is ₹ 62.

(iii) The marginal cost of producing 50 pencils.

$$MC = \frac{dC(x)}{dx}$$

$$= 2x + 2$$

$$(MC)_{x=50} = 2 \times 50 + 2 = 102$$

The marginal cost of producing 50 pencils is ₹ 102.

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Sample Paper 18

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

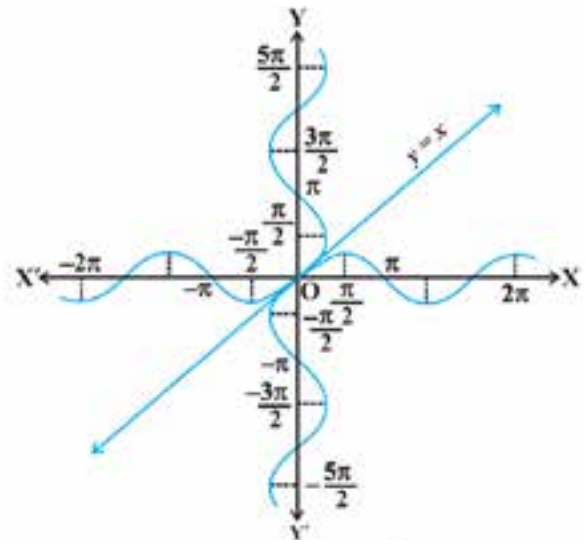
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the value of y when $x = 1$ in the graph of $y = \csc^{-1}x$?



(a) 0

(b) $\pi/2$

(c) π

(d) Undefined

Sol :

From the graph, when $x = 1$, the Y-value corresponds to $y = \pi/2$. This is consistent with the property $\csc^{-1}(1) = \pi/2$.

Thus (b) is correct option.

2. If $|x| \leq 1$, which of the following four is different from the other three?

(a) $\sin(\cos^{-1}x)$ (b) $\cos(\sin^{-1}x)$

(c) $\sqrt{1-x^2}$ (d) $\frac{\sqrt{1-x^2}}{x}$

Sol :

Now, $\sin(\cos^{-1}x) = \cos(\sin^{-1}x)$

$= \sqrt{1-x^2}$

Option (d) is different from the other three.

Thus (d) is correct option.

3. If $f(x) = \log_e(\log_e x)$, then $f'(e)$ is equal to

(a) e^{-1}

(b) e

(c) 1

(d) 0

Sol :

Given,

$$f(x) = \log_e(\log_e x)$$

$$f'(x) = \frac{1}{x\log_e(x)} \Rightarrow f'(e) = \frac{1}{e}$$

Thus (a) is correct option.

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4. If $f(x)=\begin{cases}\frac{3\sin\pi x}{5x}, & x\neq 0\\ 2k, & x=0\end{cases}$ is continuous at $x=0$, then the value of k is

(a) $\frac{\pi}{10}$ (b) $\frac{3\pi}{10}$

(c) $\frac{3\pi}{2}$ (d) $\frac{3\pi}{5}$

Sol :

We have $f(x)=\begin{cases}\frac{3\sin\pi x}{5x}, & x\neq 0\\ 2k, & x=0\end{cases}$

Now, $\lim_{x\to 0} f(x)=\lim_{x\to 0}\left(\frac{3\sin x}{5x}\right)$

$$=\frac{3}{5}\lim_{x\to 0}\left(\sin\frac{\pi x}{\pi x}\right)\times\pi$$

$$=\frac{3}{5}\times 1\times\pi=\frac{3}{5}\pi$$

Also, $f(0)=2k$

Since, $f(x)$ is continuous at $x=0$.

$$f(0)=\lim_{x\to 0} f(x)$$

$$2k=\frac{3}{5}\pi\Rightarrow k=\frac{3\pi}{10}$$

Thus (b) is correct option.

5. The area of the region bounded by the lines $y=mx$, $x=1$, $x=2$ and X-axis is 6 sq units, then m is equal to

(a) 3 (b) 1

(c) 2 (d) 4

Sol :

Given, equation of line is $y=mx$ and bounded by $x=1$, $x=2$ and X-axis.

Required area, $A=\int_1^2 mx\,dx$

$$6=m\left[\frac{x^2}{2}\right]_1^2$$

$$6=m\left(\frac{4}{2}-\frac{1}{2}\right)$$

$$6=mx\frac{3}{2}$$

$$m=4$$

Thus (d) is correct option.

6. The function $f(x)=x^2+bx+c$, where b and c are real constants, describes

(a) One-one mapping

(b) Onto mapping

(c) Not one-one but onto mapping

(d) Neither one-one nor onto mapping

Sol :

Given function is

$$f(x)=x^2+bx+c$$

It is a quadratic equation in x.

So, we will get a parabola either downward or upward.

Hence, it is many-one mapping and not onto mapping.

Hence, it is neither one-one nor onto mapping.

Thus (d) is correct option.

7. The equation of the curve, whose slope at any point different from origin is $y+\frac{y}{x}$, is

(a) $y=cxe^x, c\neq 0$ (b) $y=xe^x$

(c) $xy=e^x$ (d) $y+xe^x=c$

Sol :

By the given condition

$$\frac{dy}{dx}=y+\frac{y}{x}$$

$$\frac{dy}{dx}=y\left(1+\frac{1}{x}\right)$$

$$\frac{dy}{y}=\left(1+\frac{1}{x}\right)dx$$

$$\log y=x+\log x+\log c$$

$$\log\left(\frac{y}{cx}\right)=x$$

$$y=cxe^x$$

Thus (a) is correct option.

8. The differential equation representing the family of curve $y^2=(x+\sqrt{c})$, where c is positive perimeter, is of

(a) Order 1, degree 3

(b) Order 2, degree 2

(c) Degree 3, order 1

(d) Degree 4, order 4

Sol :

$$y^2=2yy_1(x+\sqrt{yy_1})$$

$$(y^2-2xyy_1)^2=4y^2y_1^2\cdot yy_1=4y^3y_1^3$$

Here, degree is 3 and order is 1.

Thus (c) is correct option.

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$$\frac{d^2S}{dx^2} = 2r^2 - 3(2r^2) = -4r^2 < 0$$

Hence, area is maximum when triangle is an isosceles.

Hence proved.

OR

Find the minimum value of $(ax+by)$, where $xy = c^2$.

Sol :

We have $xy = c^2$

Let $f(x) = ax+by$, whose minimum value is required.

Now $f(x) = ax + \frac{bc^2}{x}$ $\quad xy = c^2 \Rightarrow y = \frac{c^2}{x}$

Differentiating both sides w.r.t. x, we get

$$f'(x) = a - \frac{bc^2}{x^2} \quad \text{...(ii)}$$

For maxima or minima, put $f'(x) = 0$

$$a - \frac{bc^2}{x^2} = 0$$

$$a = \frac{bc^2}{x^2}$$

$$x^2 = \frac{bc^2}{a} \Rightarrow x = \pm\sqrt{\frac{b}{a}}\,c$$

Now differentiating both sides of Eq. (ii) w.r.t. x,

$$f''(x) = 0 + \frac{2bc^2}{x^3}$$

At $x = \sqrt{\frac{b}{a}}\,c$, $f''(x) = \frac{2bc^2}{\left(\sqrt{\frac{b}{a}}\,c\right)^3}$ +ve

Hence, $f(x)$ has minimum value at $x = \sqrt{\frac{b}{a}}\,c$.

At $x = -\sqrt{\frac{b}{a}}\,c, f''(x) = \frac{2bc^2}{\left(-\sqrt{\frac{b}{C}}\,c\right)^3}$ −ve

Hence, $f(x)$ has maximum value at $x = -\sqrt{\frac{b}{a}}\,c$.

When $x = \sqrt{\frac{b}{a}}\,c$, then

$$y = \frac{c^2}{x} = \frac{c^2}{\left(\sqrt{\frac{b}{a}}\,c\right)} = \sqrt{\frac{b}{a}}\,c$$

Minimum value of $f(x)$,

$$f(x)_{\min} = a\sqrt{\frac{b}{a}} \cdot c + b\sqrt{\frac{a}{b}} \cdot c$$

$$= \sqrt{ab} \cdot c + \sqrt{ab} \cdot c$$

$$= 2\sqrt{ab} \cdot c$$

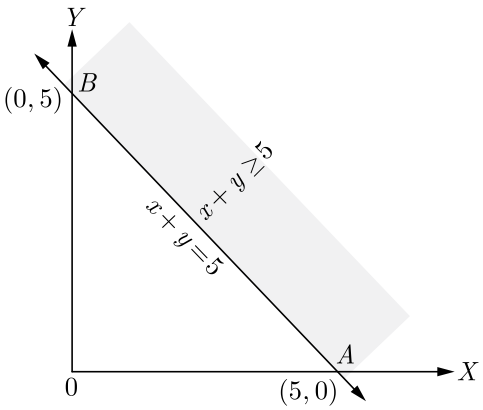
33. Solve graphically.

$x+y \geq 5,\ 2x+3 \geq 3y,\ 0 \leq x \leq 4,\ 0 \leq y \leq 2.$

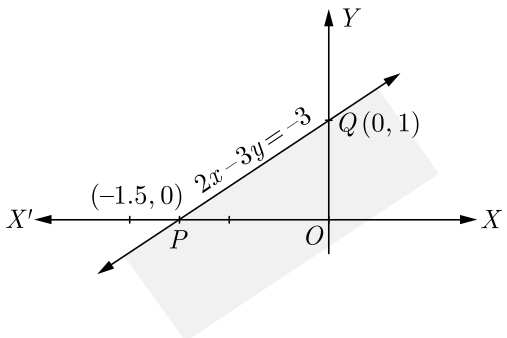
Sol :

First we convert the inequalities $x+y \geq 5$, $2x+3 \geq 3y$, $0 \leq x \leq 4$ and $0 \leq y \leq 2$ into linear equations $x+y = 5$, $2x+3 = 3y$, $x = 4$ and $y = 2$ respectively.

Region represented by $x+y \geq 5$: The line $x+y = 5$ meets the axes at points $A(5,0)$ and $B(0,5)$. Draw a thick line joining A and B. We find that $(0,0)$ does not satisfy the in-equation $x+y \geq 5$. So, the portion not containing the origin represents the solution set of the inequality $x+y \geq 5$.



Region represented by $2x+3 \geq 3y$: The line $2x-3y = -3$ meets the axes at points $P(1.5,0)$ and $Q(0,1)$. Draw a thick line joining P and Q. We find that inequality $2x-3y \geq -3$ is satisfied by $(0,0)$. So, the portion containing the origin represents the solution set of the inequality $2x-3y \geq -3$.



Region represented by $0 \leq x \leq 4$ and $0 \geq y \geq 2$. $0 \leq x \leq 4$ represents the region on the right side of the y-axis and left side of the line $x = 4$ and $0 \leq y \leq 2$ represents the region above the x-axis and below the line $y = 2$.

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Sol :

Let h and r be the height and radius of cylinder.

Given that, $\frac{dr}{dt} = 3\text{m/s}$ and $\frac{dh}{dt} = -4\text{m/s}$

Also, $V = \pi r^2 h$

$$\frac{dV}{dt} = \pi\left[r^2\frac{dh}{dt} + h \cdot 2r\frac{dr}{dt}\right]$$

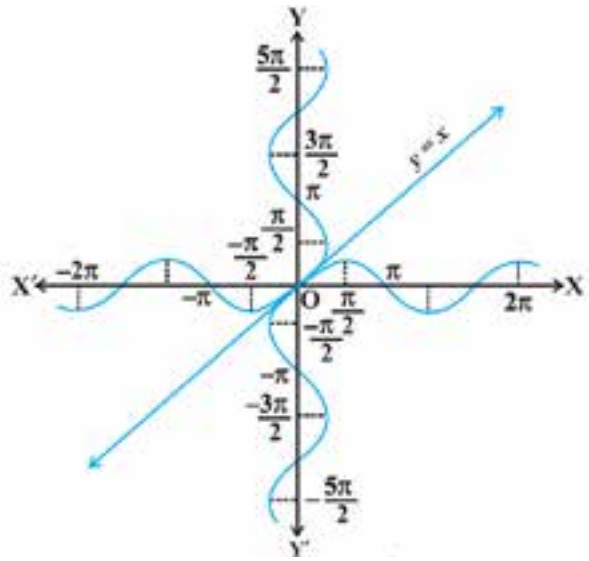
At $r = 4\text{m}$ and $h = 6\text{m}$

$$\frac{dV}{dt} = \pi[-64 + 144]$$

$$= 80\pi\text{m}^3/\text{s}$$

Thus (a) is correct option.

16. Which of the following is the symmetry property of the graph $y = \tan x$ visible in the image?



(a) Symmetry about X-axis

(b) Symmetry about Y-axis

(c) Symmetry about the origin

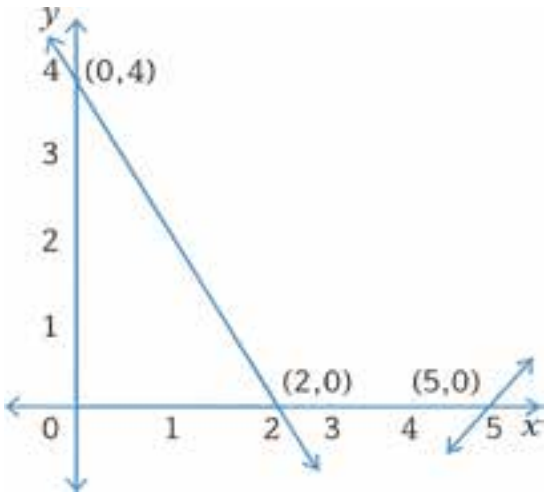
(d) No symmetry

Sol :

The graph of tangent is an odd function, which means $\tan(-x) = -\tan(x)$. The curve in the image confirms this by being symmetric about the origin: the branch in the first quadrant mirrors the branch in the third quadrant.

Thus (c) is correct option.

17. Why does the minimum value of $Z = 3x + 2y$ exist for this problem even though the feasible region is unbounded?



(a) Because the feasible region is bounded by all constraints.

(b) Because the inequality 3x+2y<6 has no point in common with the feasible region.

(c) Because the inequality 3x+2y<6 has infinitely many points in common with the feasible region.

(d) Because every LPP always has both a minimum and maximum solution.

Sol :

The feasible region extends infinitely, but still the minimum exists because the line $3x + 2y = 6$ touches the feasible region at (2,0) and all other feasible points give values ≥ 6. Since no feasible point satisfies $3x + 2y < 6$, the minimum value is valid and finite.

Thus (b) is correct option.

18. If A is a matrix of order 3 such that $A(adj\ A) = 10I$, then $|adj\ A|$ is equal to

(a) 1 (b) 10

(c) 100 (d) $10I$

Sol :

We have $A(adj\ A) = 10I$

$$|A(adj\ A)| = |A|I$$

$$10I = |A|I$$

$$10 = |A|$$

$$|adj\ A| = |A|^{3-1}$$

$$= (10)^2 = 100 \qquad |adj(A)| = |A|^{n-1}$$

where n is the order of square matrix A]

Thus (c) is correct option.

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9. Let $\vec{a}$ and $\vec{b}$ be two non-parallel unit vectors in a plane. If the vectors $(\alpha\vec{a}+\vec{b})$ bisects the internal angle between $\vec{a}$ and $\vec{b}$, then α is equal to

(a) 1 (b) 1/2

(c) 4 (d) 2

Sol :

Take $\alpha = 1$

$\vec{a}+\vec{b}$ bisects the internal angle between $\vec{a}$ and $\vec{b}$ (Since, $\vec{a}$ and $\vec{b}$ are unit vectors)

Thus (a) is correct option.

10. If α, β, γ are the angles which a half ray makes with the positive directions of the axes, then $\sin^2\alpha+\sin^2\beta+\sin^2\gamma$ is equal to

(a) 2 (b) 1

(c) 0 (d) -1

Sol :

Since, $\cos^2\alpha+\cos^2\beta+\cos^2\gamma = 1$

Hence, $1-\sin^2\alpha+1-\sin^2\beta+1-\sin^2\gamma = 1$

$\sin^2\alpha+\sin^2\beta+\sin^2\gamma = 2$

Thus (a) is correct option.

11. Which of the following triplets gives the direction cosines of a line?

(a) $<1,1,1>$ (b) $<1,-1,1>$

(c) $<1,-1,-1>$ (d) $<\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}>$

Sol :

$$l = \frac{1}{\sqrt{3}},\ m = \frac{1}{\sqrt{3}},\ n = \frac{1}{\sqrt{3}}$$

Hence, $l^2+m^2+n^2 = \left(\frac{1}{\sqrt{3}}\right)^2+\left(\frac{1}{\sqrt{3}}\right)^2+\left(\frac{1}{\sqrt{3}}\right)^2$

$= \frac{3}{3} = 1$

Thus (d) is correct option.

12. Solution of $\frac{dy}{dx}+y\sec x = \tan x$ is

(a) $y(\sec x+\tan x) = \sec x+\tan x-x+c$

(b) $y = \sec x+\tan x-x+c$

(c) $y(\sec x+\tan x) = \sec x+\tan x+x+c$

(d) None of the above

Sol :

Here, $P = \sec x$,

$Q = \tan x$

$\text{IF} = e^{\int P\,dx} = e^{\log(\sec x+\tan x)}$

$= \sec x+\tan x$

Solution is,

$$y(\sec x+\tan x) = \int(\sec x\tan x+\tan^2 x)\,dx$$

$$= \int \sec x\tan x\,dx+\int \sec^2 x\,dx-\int dx$$

$$= \sec x+\tan x-x+c$$

Thus (a) is correct option.

13. The function $f(x) = 2-3x$ is

(a) Decreasing

(b) Increasing

(c) Neither decreasing nor increasing

(d) None of the above

Sol :

Let $f(x) = 2-3x$

On differentiating, we get

$f'(x) = -3 < 0$

Hence, $f(x)$ is strictly decreasing function.

Thus (a) is correct option.

14. If $y = \log[\sin(x^2)]$, $0 < x < \frac{\pi}{2}$, then $\frac{dy}{dx}$ at $x = \frac{\sqrt{\pi}}{2}$ is

(a) 0 (b) 1

(c) $\pi/4$ (d) $\sqrt{\pi}$

Sol :

We have $y = \log[\sin(x^2)]$

Differentiating w.r.t. x, we get

$$\frac{dy}{dx} = \frac{1}{\sin x^2}\cdot\cos x^2\cdot 2x$$

$$= 2x\cot x^2$$

$$\left(\frac{dy}{dx}\right)_{x=\frac{\sqrt{\pi}}{2}} = 2\frac{\sqrt{\pi}}{2}\cot\frac{\pi}{4} = \sqrt{\pi}$$

Thus (d) is correct option.

15. The radius of a cylinder is increasing at the rate of 3m/s and its altitude is decreasing at the rate of 4m/s. The rate of change of volume when radius is 4m and altitude is 6m, is

(a) $80\pi\text{m}^3/\text{s}$ (b) $144\pi\text{m}^3/\text{s}$

(c) $80\text{m}^3/\text{s}$ (d) $64\text{m}^3/\text{s}$

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$$
\begin{aligned}
-\log|1-y| &= x+\log C\\
x &= -\log|1-y|-\log C\\
x &= -\big(\log|1-y|+\log C\big)\\
x &= -\log C(1-y)\\
-x &= \log C(1-y)\\
C(1-y) &= e^{-x}\\
1-y &= \frac{1}{C}e^{-x}\\
y &= 1-\frac{1}{C}e^{-x}\\
y &= 1+Ae^{-x}, \text{ where } A=-\frac{1}{C}
\end{aligned}
$$

Hence, $y=1+Ae^{-x}$ is the required solution.

22. If $\vec{a}=4\hat{i}-\hat{j}+\hat{k}$ and $\vec{b}=2\hat{i}-2\hat{j}+\hat{k}$, then find a unit vector parallel to the vector $\vec{a}+\vec{b}$.

Sol :

We have $\vec{a}=4\hat{i}-\hat{j}+\hat{k}$,

$\vec{b}=2\hat{i}-2\hat{j}+\hat{k}$

Now, $\vec{a}+\vec{b}=(4\hat{i}-\hat{j}+\hat{k})+(2\hat{i}-2\hat{j}+\hat{k})$

$=6\hat{i}-2\hat{j}+2\hat{k}$

and $|\vec{a}+\vec{b}|=\sqrt{(6)^2+(-3)^2+(2)^2}$

$=\sqrt{36+9+4}$

$=\sqrt{49}=7$ units

The unit vector parallel to the vector $\vec{a}+\vec{b}$ is

$$\frac{\vec{a}+\vec{b}}{|\vec{a}+\vec{b}|}=\frac{6\hat{i}-3\hat{j}+2\hat{k}}{7}$$

23. Determine the value of constant k so that the function $f(x)=\begin{cases}\dfrac{kx}{|x|}, & \text{if } x<0\\ 3, & \text{if } x\geq 0\end{cases}$ is continuous at $x=0$.

Sol :

Let $f(x)=\begin{cases}\dfrac{kx}{|x|}, & \text{if } x<0\\ 3, & \text{if } x\geq 0\end{cases}$ is continuous at $x=0$

then, $\lim\limits_{x\to 0^-}f(x)=\lim\limits_{x\to 0^-}f(0-h)=f(0)$

$\lim\limits_{h\to 0}f(0+h)=\lim\limits_{h\to 0}\dfrac{k(-h)}{|-h|}=3$

$\lim\limits_{h\to 0}\left(\dfrac{-kh}{h}\right)=3\Rightarrow\lim\limits_{h\to 0}(-k)=3$

$k=-3$

24. Find the maximum value of $\begin{vmatrix}1 & 1 & 1\\ 1 & 1+\sin\theta & 1\\ 1 & 1 & 1+\cos\theta\end{vmatrix}$

Sol :

We have $\Delta=\begin{vmatrix}1 & 1 & 1\\ 1 & 1+\sin\theta & 1\\ 1 & 1 & 1+\cos\theta\end{vmatrix}$

Expanding along first row i.e. R_1, we get

$$
\begin{aligned}
\Delta &= 1\begin{vmatrix}1+\sin\theta & 1\\ 1 & 1+\cos\theta\end{vmatrix}-1\begin{vmatrix}1 & 1\\ 1 & 1+\cos\theta\end{vmatrix}+1\begin{vmatrix}1 & 1+\sin\theta\\ 1 & 1\end{vmatrix}\\
&= [(1+\sin\theta)(1+\cos\theta)-1]-[1+\cos\theta-1]\\
&\qquad +[1-1-\sin\theta]\\
&= 1+\cos\theta+\sin\theta+\sin\theta\cos\theta-1-\cos\theta-\sin\theta\\
&= \sin\theta\cos\theta\\
&= \left|\frac{1}{2}\right|(2\sin\theta\cos\theta)=\frac{1}{2}\sin 2\theta
\end{aligned}
$$

Since maximum value of $\sin 2\theta$ is 1.

$$\Delta_{\max}=\frac{1}{2}\times 1=\frac{1}{2}.$$

OR

For what value of x, $A=\begin{bmatrix}2(x+1) & 2x\\ x & x-2\end{bmatrix}$ is a singular matrix?

Sol :

We have $A=\begin{bmatrix}2(x+1) & 2x\\ x & x-2\end{bmatrix}$

Matrix A is said to be singular, if $|A|=0$.

Thus $\begin{vmatrix}2x+2 & 2x\\ x & x-2\end{vmatrix}=0$

$$
\begin{aligned}
(2x+2)(x-2)-2x^2 &= 0\\
2x^2-2x-4-2x^2 &= 0\\
-2x-4 &= 0\\
x &= -2
\end{aligned}
$$

25. Evaluate $P(A\cup B)$, if $2P(A)=P(B)=\dfrac{5}{13}$ and $P\left(\dfrac{A}{B}\right)=\dfrac{2}{5}$.

Sol :

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19. **Assertion:** Two dice are tossed the following two events A and B are

$$A = \{(x,y): x+y=11\},$$

$$B = \{(x,y): x \neq 5\} \text{ independent events.}$$

Reason: E_1 and E_2 are independent events, then

$$P(E_1 \cap E_2) = P(E_1)\,P(E_2)$$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

Let S be the sample Space having $6 \times 6 = 36$ elements.

Given, $A = \{x,y : x+y=11\}$

$$A = \{(5,6),(6,5)\}$$

So, $P(A) = \dfrac{n(A)}{n(S)}$

$$= \frac{2}{36} = \frac{1}{18}$$

And $B = \{(x,y): x \neq 5\}$

$$= S - \{(1,5),(2,5),(3,5),(4,5),(6,5)\}$$

$$n(B) = 36 - 5 = 31$$

So, $P(B) = \dfrac{n(B)}{n(S)} = \dfrac{31}{36}$

And $A \cap B = \{(x,y): x+y=11, x \neq 5\} = \phi$

$$n(A \cap B) = 0$$

$$P(A \cap B) = \frac{n(A \cap B}{n(S)}$$

$$= \frac{0}{36} = 0$$

Clearly,

$$P(A)\,P(B) = \frac{1}{18} \times \frac{31}{36} \neq P(A \cap B)$$

Therefore A and B are not independent events,
Hence, Assertion is false and reason is true.
Thus (d) is correct option.

20. **Assertion:** If the matrix $\begin{bmatrix} 0 & a & 3 \\ 2 & b & -1 \\ c & 1 & 0 \end{bmatrix}$ is a skew symmetric matrix, then value of $(a-b-c)$ is 1

Reason: A square matrix $A = [a_{ij}]$ is said to be skew-symmetric if $A' = -A$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

Let $A = \begin{bmatrix} 0 & a & 3 \\ 2 & b & -1 \\ c & 1 & 0 \end{bmatrix}$

Since, A is a skew symmetric matrix , So $A' = -A$

$$\begin{bmatrix} 0 & 2 & c \\ a & b & 1 \\ 3 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -a & -3 \\ -2 & -b & 1 \\ -c & -1 & 0 \end{bmatrix}$$

$$2 = -a,\ b = -b,\ 3 = -c$$

$$a = -2,\ 2b = 0,\ c = -3$$

Therefore, $a = -2,\ b = 0$,and $c = -3$

Now,

$$a - b - c = -2 - 0 + 3 = 1$$

Hence, Assertion is true, reason is true and reason is a correct explanation for assertion.
Thus (a) is correct option.

SECTION - B

Section B consists of 5 questions of 2 marks each.

21. Find the general solution of equation $\dfrac{dy}{dx} + y = 1 (y \neq 1)$

Sol :

Given differential equation is

$$\frac{dy}{dx} + y = 1 (y \neq 1)$$

$$\frac{dy}{dx} = 1 - y$$

Separating the variables, we get

$$\frac{dy}{1-y} = dx$$

Integrating both sides, we get

$$\int \frac{dy}{1-y} = \int dx$$

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also, $1+bc=1+(-1)\frac{1}{2}=\frac{1}{2}>0$ Thus $(b,c)\in R$

But, $1+ac=1+(-8)\frac{1}{2}=-3<0$, hence, R is not transitive.

29. If $A=\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$ and I is the identity matrix of order 2, then show that $A^2=4A-3I$. Hence, find A^{-1}.

Sol :

We have, $A=\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$

Now, $A^2=A\cdot A$

$$=\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$$

$$=\begin{bmatrix}4+1 & -2-2\\ -2-2 & 1+4\end{bmatrix}$$

$$A^2=\begin{bmatrix}5 & -4\\ -4 & 5\end{bmatrix} \quad ...(i)$$

Now $4A-3I=4\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}-3\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$

$$=\begin{bmatrix}8 & -4\\ -4 & 8\end{bmatrix}-\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}$$

$$=\begin{bmatrix}5 & -4\\ -4 & 5\end{bmatrix} \quad ...(ii)$$

From Eqs. (i) and (ii), we get

$A^2=4A-3I$ Hence proved...(iii)

Here, $|A|=\begin{vmatrix}2 & -1\\ -1 & 2\end{vmatrix}=4-1=3\neq 0$

Since $|A|\neq 0$, matrix A is non-singular and A^{-1} exists.

Now, pre-multiplying both sides of Eq. (iii) by A^{-1}, we get

$A^{-1}\cdot A^2=A^{-1}\cdot(4A-3I)$

$(A^{-1}\cdot A)\cdot A=4A^{-1}\cdot A-3A^{-1}\cdot I$

$IA=4I-3A^{-1}$

$A=4I-3A^{-1}$

$3A^{-1}=4I-A$

$$A^{-1}=\frac{1}{3}\left(4\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}-\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}\right)$$

$$=\frac{1}{3}\left(\begin{bmatrix}4 & 0\\ 0 & 4\end{bmatrix}-\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}\right)$$

$$A^{-1}=\frac{1}{3}\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}=\begin{bmatrix}\frac{2}{3} & \frac{1}{3}\\ \frac{1}{3} & \frac{2}{3}\end{bmatrix}$$

OR

Show that for the matrix $A=\begin{bmatrix}1 & 1 & 1\\ 1 & 2 & -3\\ 2 & -1 & 3\end{bmatrix}$, $A^3-6A^2+5A+11I=O$. Hence, find A^{-1}.

Sol :

We have, $A=\begin{bmatrix}1 & 1 & 1\\ 1 & 2 & -3\\ 2 & -1 & 3\end{bmatrix}$

Now $A^2=AA$

$$=\begin{bmatrix}1 & 1 & 1\\ 1 & 2 & -3\\ 2 & -1 & 3\end{bmatrix}\begin{bmatrix}1 & 1 & 1\\ 1 & 2 & -3\\ 2 & -1 & 3\end{bmatrix}$$

$$=\begin{bmatrix}4 & 2 & 1\\ -3 & 8 & -14\\ 7 & -3 & 14\end{bmatrix}$$

and $A^3=A^2A$

$$=\begin{bmatrix}4 & 2 & 1\\ -3 & 8 & -14\\ 7 & -3 & 14\end{bmatrix}\begin{bmatrix}1 & 1 & 1\\ 1 & 2 & -3\\ 2 & -1 & 3\end{bmatrix}$$

$$=\begin{bmatrix}8 & 7 & 1\\ -23 & 27 & -69\\ 32 & -13 & 58\end{bmatrix}$$

$A^3-6A^2+5A+11I_3$

$$=\begin{bmatrix}8 & 7 & 1\\ -23 & 27 & -69\\ 32 & -13 & 58\end{bmatrix}-6\begin{bmatrix}4 & 2 & 1\\ -3 & 8 & -14\\ 7 & -3 & 14\end{bmatrix}+5\begin{bmatrix}1 & 1 & 1\\ 1 & 2 & -3\\ 2 & -1 & 3\end{bmatrix}+11\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$=\begin{bmatrix}8-24+5+11 & 7-12+5+0 & 1-6+6+0\\ -23+18+5+0 & 27-48+10+11 & -69+84-15+0\\ 32-42+10+0 & -13+18-5+0 & 58-84+15+11\end{bmatrix}$$

$$=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}=O$$

Thus $A^3-6A^2+5A+11I_3=O$

Now, multiplying both sides by A^{-1}, we get

$A^{-1}(A^3-6A^2+5A+11I_3)=A^{-1}O$

$A^2-6A+5I+11A^{-1}=O$

$11A^{-1}=-A^2+6A-5I$

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A GST form is either filed on time or late, is either from a small or a large business, and is either accurate or inaccurate. There is an 11% probability that a form is from a small business and is accurate and on time. There is a 13% probability that a form is from a small business and is accurate but is late. There is a 15% probability that a form is from a small business and is on time. There is a 21% probability that a form is from a small business and is inaccurate and is late.

(i) If a form is from a small business and is accurate, what is the probability that it was filed on time?

(ii) What is the probability that a form is from a large business?

Sol :

Let T be the event that 'the tax form is filed on time', let S be the event that 'the tax form is from a small business', and let A be the event that 'the tax form is accurate'.

(i) $P(T \cap S \cap A) = 0.11$

$P(T' \cap S \cap A) = 0.13$

$P(T \cap S) = 0.15$

$P(T' \cap S \cap A') = 0.21$

Therefore,

$$P(T \mid S \cap A) = \frac{P(T \cap S \cap A)}{P(S \cap A)}$$

$$= \frac{P(T \cap S \cap A)}{P(T \cap S \cap A) + P(T' \cap S \cap A)}$$

$$= \frac{0.11}{0.11 + 0.13}$$

$$= \frac{11}{24}$$

(ii) $P(S') = 1 - P(S)$

$= 1 - P(T \cap S) - P(T' \cap S)$

$= 1 - P(T \cap S) - P(T' \cap S \cap A) - P(T' \cap S \cap A')$

$= 1 - 0.15 - 0.13 - 0.21 = 0.51$

37. Fertilizer, natural or artificial substance containing the chemical elements that improve growth and productiveness of plants. Fertilizers enhance the natural fertility of the soil or replace chemical elements taken from the soil by previous crops.



The following matrix gives the proportionate mix of constituents used for three fertilisers:

$$\begin{array}{r} \\ \text{I} \\ \text{Fertilisers II} \\ \text{III} \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0.5 & 0 & 0.5 & 0 \\ 0.2 & 0.3 & 0 & 0.5 \\ 0.2 & 0.2 & 0.1 & 0.5 \end{bmatrix} \end{array} \quad \text{Constituents}$$

(i) If sales are 1000 tins (of one kilogram) per week, 20% being fertiliser I, 30% being fertiliser II and 50% being fertiliser III, how much of each constituent is used.

(ii) If the cost of each constituents is Rs 5, Rs 6, Rs 7.5 and Rs 10 per 100 grams, respectively, how much does a one kilogram tin of each fertiliser cost

(iii) What is the total cost per week?

Sol :

The sales of fertilisers I, II and III per week can be expressed as the following matrix:

$$P = 1000 \begin{array}{c} \begin{array}{ccc} \text{I} & \text{II} & \text{III} \end{array} \\ \begin{bmatrix} 0.2 & 0.3 & 0.5 \end{bmatrix} \end{array}$$

$$= \begin{array}{c} \begin{array}{ccc} \text{I} & \text{II} & \text{III} \end{array} \\ \begin{bmatrix} 200 & 300 & 500 \end{bmatrix} \end{array}$$

Let the matrix denote the proportionate mix of constituents used for three fertilisers I, II and III, then

$$Q = \begin{array}{r} \\ \text{I} \\ \text{Fertilisers II} \\ \text{III} \end{array} \begin{array}{c} \begin{array}{cccc} A & B & C & D \end{array} \\ \begin{bmatrix} 0.5 & 0 & 0.5 & 0 \\ 0.2 & 0.3 & 0 & 0.5 \\ 0.2 & 0.2 & 0.1 & 0.5 \end{bmatrix} \end{array} \quad \text{Constituents}$$

The requirement of each constituents given by the matrix product PQ is:

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continuous at $x = 0$, then find the value of a.

Sol :

Given, $f(x) = \begin{cases} \dfrac{1-\cos 4x}{x^2}, & \text{when } x < 0 \\ a, & \text{when } x = 0 \\ \dfrac{\sqrt{x}}{\sqrt{16+\sqrt{x}}-4}, & \text{when } x > 0 \end{cases}$

Since, $f(x)$ is continuous at $x = 0$.

$$(LHL)_{x=0} = (RHL)_{x=0} = f(0)$$

Now,

$$(LHL)_{x=0} = \lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} \frac{1-\cos 4x}{x^2}$$

$$= \lim_{h \to 0} \frac{1-\cos 4(0-h)}{(0-h)^2}$$

[put $x = 0 - h$; when $x \to 0^-$, then $h \to 0$]

$$= \lim_{h \to 0} \frac{1-\cos 4h}{h^2} \quad [\because \cos(-\theta) = \cos\theta]$$

$$= \lim_{h \to 0} \frac{2\sin^2 2h}{h^2} \quad [\because 1 - \cos 2\theta = 2\sin^2\theta]$$

$$= 2\lim_{h \to 0}\left(\frac{\sin 2h}{2h}\right)^2 \times 4$$

$$= 2 \times (1)^2 \times 4 \qquad \left[\because \lim_{x \to 0} \frac{\sin x}{x} = 1\right]$$

$$= 8$$

Now, from Eq. (i), we have

$$(LHL)_{x=0} = f(0)$$

$$8 = a \qquad [\because f(0) = a\text{k (given)}]$$

$$a = 8$$

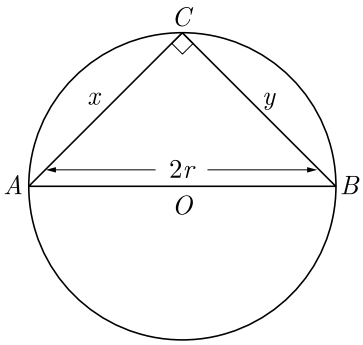
SECTION - D

Section D consists of 4 questions of 5 marks each.

32. AB is the diameter of a circle and C is any point of the circle. Show that the area of ΔABC is maximum, when it is an isosceles triangle.

Sol :

Let $AC = x$, $BC = y$ and r be the radius of circle. Also, $\angle C = 90^\circ$ [$\because$ angle made in semi-circle is 90°]



In ΔABC, we have

$$(AB)^2 = (AC)^2 + (BC)^2$$

$$(2r)^2 = (x)^2 + (y)^2$$

$$4r^2 = x^2 + y^2 \qquad \text{...(i)}$$

Area of ΔABC, $A = \frac{1}{2} x \cdot y$

Squaring both sides, we get

$$A^2 = \frac{1}{4} x^2 y^2$$

Let $A^2 = S$, then, $S = \frac{1}{4} x^2 y^2$

$$= \frac{1}{4} x^2 (4r^2 - x^2) \quad \text{[from Eq. (i)]}$$

$$S = \frac{1}{4}(4x^2 r^2 - x^4)$$

Differentiating both sides w.r.t. x, we get

$$\frac{dS}{dx} = \frac{1}{4}(8r^2 x - 4x^3)$$

For maxima or minima, $\frac{dS}{dx} = 0$

$$\frac{1}{4}(8r^2 x - 4x^3) = 0$$

$$8r^2 x = 4x^3$$

$$8r^2 = 4x^2$$

$$x = \sqrt{2}\, r$$

From Eq. (i), we get

$$y^2 = 4r^2 - 2r^2 = 2r^2$$

$$y = \sqrt{2}\, r$$

Here, $x = y$, so triangle is an isosceles.

Also, $$\frac{d^2 S}{dx^2} = \frac{d}{dx}\left[\frac{1}{4}(8r^2 x - 4x^3)\right]$$

$$= \frac{1}{4}(8r^2 - 12x^2)$$

$$= 2r^2 - 3x^2$$

At $x = \sqrt{2}\, r$,

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$$11A^{-1} = -\begin{bmatrix} 4 & 2 & 1 \\ -3 & 8 & -14 \\ 7 & -3 & 14 \end{bmatrix} + 6\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & -3 \\ 2 & -1 & 3 \end{bmatrix} - 5\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$11A^{-1} = \begin{bmatrix} -4+6-5 & -2+6+0 & -1+6+0 \\ 3+6+0 & -8+12-5 & 14-18+0 \\ -7+12+0 & 3-6+0 & -14+18-5 \end{bmatrix}$$

$$= \begin{bmatrix} -3 & 4 & 5 \\ 9 & -1 & -4 \\ 5 & -3 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{11}\begin{bmatrix} -3 & 4 & 5 \\ 9 & -1 & -4 \\ 5 & -3 & -1 \end{bmatrix}$$

30. Find the position of a point R, which divides the line joining two points P and Q whose position vectors are $2\vec{a}+\vec{b}$ and $\vec{a}-3\vec{b}$ respectively, externally in the ratio $1 : 2$. Also, show that P is the mid-point of line segment RQ.

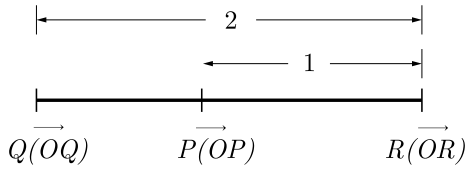
Sol :

Given,

Position vector of P, $\overrightarrow{OP} = 2\vec{a}+\vec{b}$

Position vector of Q, $\overrightarrow{OQ} = \vec{a}-3\vec{b}$

Let $\overrightarrow{OR}$ be the position vector of point R, which divides PQ in the ratio $1 : 2$ externally as shown below



Using external section formula we have

$$\overrightarrow{OR} = \frac{1(\vec{a}-3\vec{b})-2(2\vec{a}+\vec{b})}{1-2}$$

$$= \frac{\vec{a}-3\vec{b}-4\vec{a}-2\vec{b}}{-1}$$

$$= \frac{-3\vec{a}-5\vec{b}}{-1} = 3\vec{a}+5\vec{b}$$

Hence, $\overrightarrow{OR} = 3\vec{a}+5\vec{b}$

Now, we have to show that P is the mid-point of RQ,

i.e. $\overrightarrow{OP} = \frac{\overrightarrow{OR}+\overrightarrow{OQ}}{2}$

We have, $\overrightarrow{OR} = 3\vec{a}+5\vec{b}$, $\overrightarrow{OQ} = \vec{a}-3\vec{b}$

$$\frac{\overrightarrow{OR}+\overrightarrow{OQ}}{2} = \frac{(3\vec{a}+5\vec{b})+(\vec{a}-3\vec{b})}{2}$$

$$= \frac{4\vec{a}+2\vec{b}}{2}$$

$$= \frac{2(2\vec{a}+\vec{b})}{2}$$

$$= 2\vec{a}+\vec{b} = \overrightarrow{OP}$$

Hence, P is the mid-point of line segment RQ.

31. Find the value of k, for which

$$f(x) = \begin{cases} \frac{\sqrt{1+kx}-\sqrt{1-kx}}{x}, & if -1 \le x < 0 \\ \frac{2x+1}{x-1}, & if\ 0 \le x < 1 \end{cases}$$

is continuous at $x = 0$.

Sol :

Let $f(x) = \begin{cases} \frac{\sqrt{1+kx}-\sqrt{1-kx}}{x}, & if -1 \le x < 0 \\ \frac{2x+1}{x-1}, & if\ 0 \le x < 1 \end{cases}$

is continuous at $x = 0$.

Now, $f(0) = \frac{2\cdot 0+1}{0-1} = \frac{1}{-1} = -1$

and $\text{LHL} = \lim_{h\to 0} f(0-h)$

$$= \lim_{h\to 0} \frac{\sqrt{1-kh}-\sqrt{1+kh}}{-h}$$

$$= \lim_{h\to 0} \frac{\sqrt{1-kh}-\sqrt{1+kh}}{-h} \times \frac{(\sqrt{1+kh}+\sqrt{1+kh})}{(\sqrt{1-kh}+\sqrt{1+kh})}$$

$$= \lim_{h\to 0} \frac{(1-kh)-(1+kh)}{-h(\sqrt{1-kh}+\sqrt{1+kh})}$$

$$[\because (a+b)(a-b) = a^2-b^2]$$

$$= \lim_{h\to 0} \frac{-2kh}{-h(\sqrt{1-kh}+\sqrt{1+kh})}$$

$$= \lim_{h\to 0} \frac{2k}{\sqrt{1-kh}+\sqrt{1+kh}} = \frac{2k}{1+1} = \frac{2k}{2} = k$$

Since, $f(x)$ is continuous at $x = 0$.

$$f(0) = LHL \Rightarrow -1 = k$$

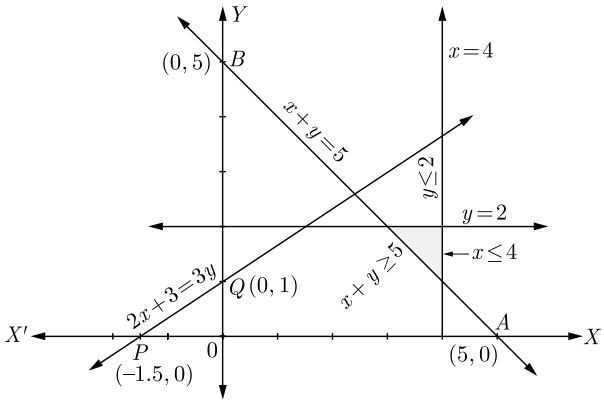
$$k = -1$$

OR

If $f(x) = \begin{cases} \frac{1-\cos 4x}{x^2}, & \text{when } x < 0 \\ a, & \text{when } x = 0 \text{ and } f \text{ is} \\ \frac{\sqrt{x}}{\sqrt{17+\sqrt{x}}-4}, & \text{when } x > 0 \end{cases}$

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Hence, the common region is the shaded, region which is represented by the system of given inequalities. Any point in this shaded region represents the solution of the given inequalities.



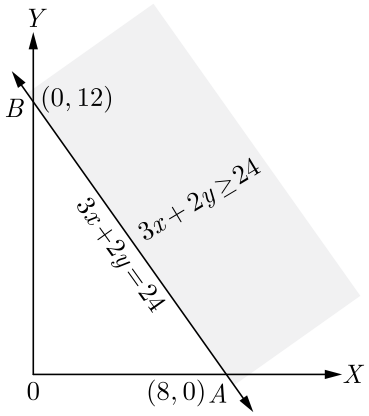
OR

Solve the following system of inequalities graphically:
$3x+2y \geq 24$, $3x+y \leq 15$, $x \geq 4$.

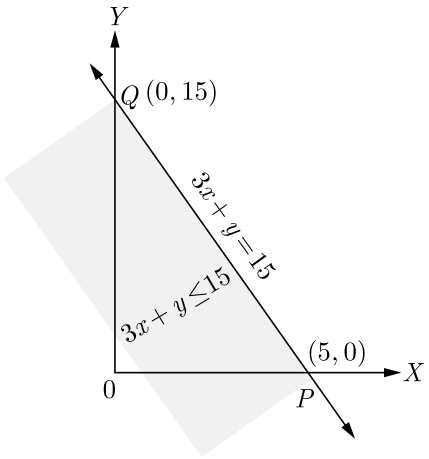
Sol :

First, we convert the inequalities $3x+2y \geq 24$, $3x+y \leq 15$ and $x \geq 4$ into linear equations $3x+2y=24$, $3x+y=15$ and $x=4$ respectively.

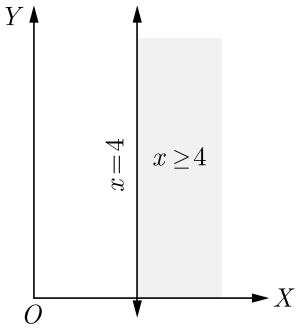
Region represented by $3x+2y \geq 24$: The line $3x+2y=24$ meets the axes at $A(8,0)$ and $B(0,12)$. Draw a thick line joining A and B. We find that (0,0) does not satisfy the in-equation $3x+2y \geq 24$. So, the portion does not containing the origin, represents the solution set of the in-equation $3x+2y \geq 24$.



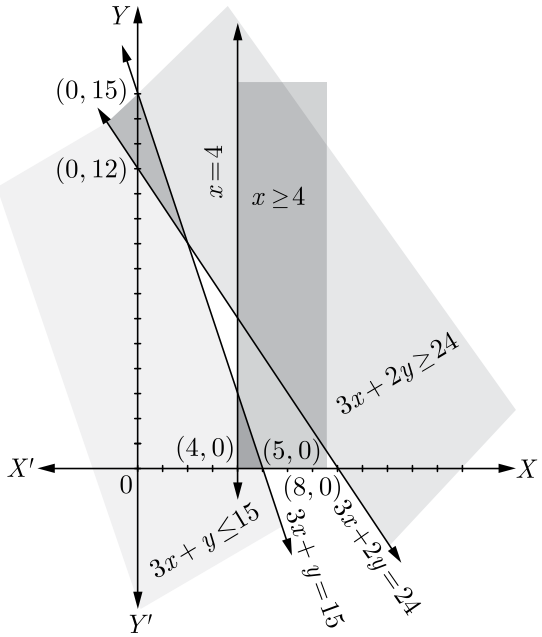
Region represented by $3x+y \leq 15$: The line $3x+y=15$ meets the axes at $P(5,0)$ and $Q(0,15)$. Draw a thick line joining P and Q. We find that (0,0) satisfies the inequality $3x+y \leq 15$. So, the portion containing the origin represents the solution set of the inequality $3x+y \leq 15$.



Region represented by $x \geq 4$: Clearly $x=4$ is a line parallel to y-axis at a distance of 4 units from the origin. Since, (0,0) does not satisfy the inequality $x \geq 4$. So, the portion not containing the origin, is represented by the given inequality.



We find that there is no common portion between the three different shaded regions. Therefore, the system of inequality has no solution.



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We have $2P(A) = P(B) = \frac{5}{13}$

$P(A) = \frac{5}{26},\ P(B) = \frac{5}{13}$

and $P\left(\frac{A}{B}\right) = \frac{2}{5}$

Now $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$

$\frac{2}{5} = \frac{P(A \cap B)}{\frac{5}{13}}$

$P(A \cap B) = \frac{2}{5} \times \frac{5}{13} = \frac{2}{13}$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{5}{26} + \frac{5}{13} - \frac{2}{13}$

$= \frac{5 + 10 - 4}{26} = \frac{11}{26}$

SECTION - C

Section C consists of 6 questions of 3 marks each.

26. Find the general solution of equation $y \log y\, dx - x\, dy = 0$

Sol :

Given differential equation is

$y \log y\, dx - x\, dy = 0$

$x\, dy = y \log y\, dx$

Separating the variables, we get

$\frac{dy}{y \log y} = \frac{dx}{x}$

Integrating both sides, we get

$\int \frac{dy}{y \log y} = \int \frac{dy}{x}$

Put $\log y = t$

$\frac{1}{y} dy = dt$

$\int \frac{dt}{t} = \int \frac{dx}{x}$

$\log|t| = \log|x| + \log|C|$

$\log|\log y| = \log|x| + \log|C|$

$\log|\log y| = \log|xC|$

$\log y = xC$

$y = e^{xC}$

$y = e^{xC}$ is the required solution.

27. Find $\int \frac{x-3}{(x-1)^3} e^x\, dx$.

Sol :

We have

$I = \int \frac{(x-3)}{(x-1)^3} e^x\, dx$

$= \int \frac{e^x (x-1-2)}{(x-1)^3} dx$

$= \int e^x \left\{ \frac{(x-1)}{(x-1)^3} - \frac{2}{(x-1)^3} \right\} dx$

$= \int e^x \left\{ \frac{1}{(x-1)^2} - \frac{2}{(x-1)^3} \right\} dx$

$= \int e^x \cdot \{f(x) + f'(x)\}\, dx,$

where $f(x) = \frac{1}{(x-1)^2}$ and $f'(x) = \frac{-2}{(x-1)^3}$

Thus $I = e^x \cdot f(x) + C$

$= e^x \cdot \frac{1}{(x-1)^2} + C = \frac{e^x}{(x-1)^2} + C$

$= e^x \cdot \frac{1}{(x-1)^2} + C = \frac{e^x}{(x-1)^2} + C$

28. Check whether the relation R in the set R of real numbers, defined by $R = \{(a, b) : 1 + ab > 0\}$, is reflexive, symmetric or transitive.

Sol :

Reflexive :

R is reflexive, as

$1 + a \cdot a = 1 + a^2 > 0 \Rightarrow (a, a) \in R \forall$

$a \in R$

Symmetric :

If $(a, b) \in R$

then, $1 + ab > 0$

$1 + ba > 0$

$(b, a) \in R$

Hence, R is symmetric.

Transitive :

Let $a = -8,\ b = -1$ and $c = \frac{1}{2}$

Since, $1 + ab = 1 + (-8)(-1) = 9 > 0$ thus $(a, b) \in R$

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$$= i(-2-1) - \hat{j}(2-2) + \hat{k}(1+2)$$
$$= -3\hat{i} + 0 + 3\hat{k} = -3\hat{i} + 3\hat{k}$$
$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{(-3)^2 + (3)^2} = \sqrt{9+9}$$
$$= \sqrt{18} = 3\sqrt{2}$$

Shortest distance between the given lines

$$\text{SD} = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$
$$= \left| \frac{(\hat{i} - 3\hat{j} - 2\hat{k}) \cdot (-3\hat{i} + 3\hat{k})}{3\sqrt{2}} \right|$$
$$= \left| \frac{-3-6}{3\sqrt{2}} \right|$$
$$= \left| \frac{-9}{3\sqrt{2}} \right| = \frac{3}{\sqrt{2}}$$

Hence, the required shortest distance $= \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$.

OR

Find the shortest distance between the lines

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$$

and $\quad \frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$.

Sol :

Here, the given equations are

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1} \quad \ldots(1)$$

and $\quad \frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1} \quad \ldots(2)$

From eq. (1), we get

$$\vec{a}_1 = -\hat{i} - \hat{j} - \hat{k} \text{ and } \vec{b}_1 = 7\hat{i} - 6\hat{j} + \hat{k}$$

From eq. (2), we get

$$\vec{a}_2 = 3\hat{i} + 5\hat{j} + 7\hat{k} \text{ and } \vec{b}_2 = \hat{i} - 2\hat{j} + \hat{k}$$

Shortest distance between the lines is given by

$$\vec{a}_2 - \vec{a}_1 = (3\hat{i} + 5\hat{j} + 7\hat{k}) - (-\hat{i} - \hat{j} - \hat{k})$$
$$= 3\hat{i} + 5\hat{j} + 7\hat{k} + \hat{i} + \hat{j} + \hat{k}$$
$$= 4\hat{i} + 6\hat{j} + 8\hat{k}$$
$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 7 & -6 & 1 \\ 1 & -2 & 1 \end{vmatrix}$$
$$= \hat{i}(-6+2) - \hat{j}(7-1) + \hat{k}(-14+6)$$
$$= -4\hat{i} - 6\hat{j} - 8\hat{k}$$
$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{(-4)^2 + (-6)^2 + (-8)^2}$$
$$= \sqrt{16+36+64} = \sqrt{116}$$
$$= 2\sqrt{29}$$
$$\text{SD} = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$
$$= \left| \frac{(4\hat{i} + 6\hat{j} + 8\hat{k}) \cdot (-4\hat{i} - 6\hat{j} - 8\hat{k})}{2\sqrt{29}} \right|$$
$$= \left| \frac{-16-36-64}{2\sqrt{29}} \right|$$
$$= \left| \frac{-116}{2\sqrt{29}} \right|$$
$$= \sqrt{116} = 2\sqrt{29} \text{ units}$$

Hence, the required shortest distance $= 2\sqrt{29}$ units.

SECTION - E

Section E consists of 3 case study based questions of 4 marks each.

36. Goods and Services Tax (GST) is an indirect tax (or consumption tax) used in India on the supply of goods and services. It is a comprehensive, multistage, destination-based tax: comprehensive because it has subsumed almost all the indirect taxes except a few state taxes. Multi-staged as it is, the GST is imposed at every step in the production process, but is meant to be refunded to all parties in the various stages of production other than the final consumer and as a destination-based tax, it is collected from point of consumption and not point of origin like previous taxes.



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34. Write the following functions in the simplest from:

$\tan^{-1}\frac{\sqrt{1+x^2}}{x},\ x \neq 0$

Sol :

We have

$\tan^{-1}\frac{\sqrt{1+x^2}-1}{x},\ x \neq 0$

Put $x = \tan\theta$

$\theta = \tan^{-1}x$

$= \tan^{-1}\left(\frac{\sqrt{1+\tan^2\theta}-1}{\tan\theta}\right)$

$= \tan^{-1}\left(\frac{\sec\theta-1}{\tan\theta}\right)$

$= \tan^{-1}\left(\frac{\frac{1}{\cos\theta}-1}{\frac{\sin\theta}{\cos\theta}}\right)$

$= \tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right)$

$= \tan^{-1}\left(\frac{2\sin^2\theta/2}{2\sin\theta/2\cos\theta/2}\right)$

$[\because \cos 2\theta = 1-2\sin^2\theta \ \ \sin 2\theta = 2\sin\theta\cos\theta]$

$= \tan^{-1}\left(\frac{\sin\theta/2}{\cos\theta/2}\right)$

$= \tan^{-1}\left(\tan\frac{\theta}{2}\right)$

$= \frac{\theta}{2} = \frac{1}{2}\tan^{-1}x$

Hence, $\tan^{-1}\frac{\sqrt{1+x^2}-1}{x} = \frac{1}{2}\tan^{-1}x$

OR

Find the value of $\tan^{-1}\left(\tan\frac{5\pi}{6}\right)+\cos^{-1}\left(\cos\frac{13\pi}{6}\right)$.

Thinking Process

Use the property, $\tan^{-1}\tan x = x,\ x \in \left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ and $\cos^{-1}(\cos x) = x,\ x \in [0,\pi]$ to get the answer.

Sol :

We know that, $\tan^{-1}\tan x = x; \quad x \in \left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ and $\cos^{-1}\cos x = x;\ x \in [0,\pi]$

$\tan^{-1}\left(\tan\frac{5\pi}{6}\right)+\cos^{-1}\left(\cos\frac{13\pi}{6}\right)$

$= \tan^{-1}\left[\tan\left(\pi-\frac{\pi}{6}\right)\right]+\cos^{-1}\left[\cos\left(\pi+\frac{\pi}{6}\right)\right]$

$= \tan^{-1}\left(-\tan\frac{\pi}{6}\right)+\cos^{-1}\left(-\cos\frac{7\pi}{6}\right)$

$[\because \cos(\pi+\theta) = -\cos\theta]$

$= -\tan^{-1}\left(\tan\frac{\pi}{6}\right)+\pi-\left(\cos^{-1}\cos\frac{7\pi}{6}\right)$

$\{\because \tan^{-1}(-x) = -\tan^{-1}x; x \in R$ and $\cos^{-1}(-x) = \pi-\cos^{-1}x; x \in [-1,1]\}$

$= -\tan^{-1}\left(\tan\frac{\pi}{6}\right)+\pi-\cos^{-1}\left[\cos\left(\pi+\frac{\pi}{6}\right)\right]$

$= -\tan^{-1}\left(\tan\frac{\pi}{6}\right)+\pi-\left[\cos^{-1}\left(-\cos\frac{\pi}{6}\right)\right]$

$[\because \cos(\pi+\theta) = -\cos\theta]$

$= -\tan^{-1}\left(\tan\frac{\pi}{6}\right)+\pi-\pi+\cos^{-1}\left(\cos\frac{\pi}{6}\right)$

$[\because \cos^{-1}(-x) = \pi-\cos^{-1}x]$

$= -\frac{\pi}{6}+0+\frac{\pi}{6} = 0$

Not Remember that, $\tan^{-1}\left(\tan\frac{5\pi}{6}\right) \neq \frac{5\pi}{6}$ and $\cos^{-1}\left(\cos\frac{13\pi}{6}\right) \neq \frac{13\pi}{6}$

Since, $\frac{5\pi}{6} \notin \left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ and $\frac{13\pi}{6} \notin [0,\pi]$

35. Find the shortest distance between the lines

$\vec{r} = (\hat{i}+2\hat{j}+\hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k})$

and $\vec{r} = (2\hat{i}-\hat{j}-\hat{k})+\mu(2\hat{i}+\hat{j}+2\hat{k})$

Sol :

Given that:

$\vec{r} = (\hat{i}+2\hat{j}+\hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k}) \quad ...(1)$

Here $\vec{a_1} = \hat{i}+2\hat{j}+\hat{k}$

and $\vec{b_1} = \hat{i}-\hat{j}+\hat{k}$

$\vec{r} = (2\hat{i}-\hat{j}-\hat{k})+\mu(2\hat{i}+\hat{j}+2\hat{k}) \quad ...(2)$

Here $\vec{a_2} = 2\hat{i}-\hat{j}-\hat{k}$

and $\vec{b_2} = 2\hat{i}+\hat{j}+2\hat{k}$

$\vec{a_1}-\vec{a_2} = (2\hat{i}-\hat{j}-\hat{k})-(\hat{i}+2\hat{j}+\hat{k})$

$= 2\hat{i}-\hat{j}-\hat{k}-\hat{i}-2\hat{j}-\hat{k}$

$= \hat{i}-3\hat{j}-2\hat{k}$

$\vec{b_1}\times\vec{b_2} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 2 & 1 & 2 \end{vmatrix}$

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$$PQ = \begin{matrix} & \text{I} & \text{II} & \text{III} \end{matrix}\begin{bmatrix} 200 & 300 & 500 \end{bmatrix}$$

$$= \begin{matrix} & A & B & C & D \\ \text{I} \\ \text{II} \\ \text{III} \end{matrix}\begin{bmatrix} 0.50 & 0 & 0.5 & 0 \\ 0.2 & 0.3 & 0 & 0.5 \\ 0.2 & 0.2 & 0.1 & 0.5 \end{bmatrix}$$

$$= \begin{matrix} A & B & C & D \end{matrix}\begin{bmatrix} 100+600+100 & 0+90+100 & 100+0+50 & 0+150+250 \end{bmatrix}$$

$$= \begin{matrix} A & B & C & D \end{matrix}\begin{bmatrix} 260 & 190 & 150 & 400 \end{bmatrix}$$

Thus, the requirements of constituents are:

$A-260, B-190, C-150, D-400.$

of each constituents is Rs 5, Rs 6, Rs 7.5 e and Rs 10 per 100 grams, i.e., Rs 50, Rs 60, Rs 75 and Rs 100 per 1000 grams (one kilogram) of each constituent, respectively.

Let R denote the cost of constituents A, B, C and D per kilogram, then

$$R = \begin{matrix} A \\ B \\ C \\ D \end{matrix}\begin{bmatrix} 50 \\ 60 \\ 75 \\ 100 \end{bmatrix}$$

The cost of one kilogram tin of each fertiliser I, II and III is given by the matrix product QR is:

$$QR = \begin{matrix} & A & B & C & D \\ \text{I} \\ \text{II} \\ \text{III} \end{matrix}\begin{bmatrix} 0.5 & 0 & 0.5 & 0 \\ 0.2 & 0.3 & 0 & 0.5 \\ 0.2 & 0.2 & 0.1 & 0.5 \end{bmatrix}\begin{matrix} A \\ B \\ C \\ D \end{matrix}\begin{bmatrix} 50 \\ 60 \\ 75 \\ 100 \end{bmatrix}$$

$$= \begin{matrix} \text{I} \\ \text{II} \\ \text{III} \end{matrix}\begin{bmatrix} 25+0+37.5+0 \\ 10+18+0+50 \\ 10+12+7.5+50 \end{bmatrix}$$

$$= \begin{matrix} \text{I} \\ \text{II} \\ \text{III} \end{matrix}\begin{bmatrix} 62.5 \\ 78 \\ 79.5 \end{bmatrix}$$

Thus, the costs per kilogram tin of fertiliser are:

$$\text{I} - ₹\,62.5, \text{II} - ₹\,78, \text{III} - ₹\,79.5.$$

The total cost of fertiliser of 1000 tins (of one kilogram) are needed per week may be calculated by

Either: $$P(QR) = \begin{matrix} & \text{I} & \text{II} & \text{III} \end{matrix}\begin{bmatrix} 200 & 300 & 500 \end{bmatrix}\begin{matrix} \text{I} \\ \text{II} \\ \text{III} \end{matrix}\begin{bmatrix} 62.5 \\ 78 \\ 79.5 \end{bmatrix}$$

$$= \begin{bmatrix} 12500 + 23400 + 39750 \end{bmatrix}$$

$$= \begin{bmatrix} 75650 \end{bmatrix}$$

Hence, the total cost per week is ₹ 75650.

38. Bata India is the largest retailer and leading manufacturer of footwear in India and is a part of the Bata Shoe Organization. Incorporated as Bata Shoe Company Private Limited in 1931, the company was set up initially as a small operation in Konnagar (near Calcutta) in 1932. In January 1934,



The manager of BATA show room at Jaipur determines that the price p (dollars) for each pair of a popular brand of sports sneakers is changing at the rate of

$$p'(x) = \frac{-300x}{(x^2+9)^{3/2}}$$

when x (hundred) pairs are demanded by consumers. When the price is \$75 per pair, 400 pairs $(x=4)$ are demanded by consumers.

(i) Find the demand (price) function $p(x)$.

(ii) At what price will 500 pairs of sneakers be demanded? At what price will no sneakers be demanded?

(iii) How many pairs will be demanded at a price of \$90 per pair?

Sol :

(i) $$p(x) = \int p'(x)\,dx$$

$$= \int \frac{-300x}{(x^2+9)^{3/2}}\,dx$$

Let $u = x^2+9$, then we have $\frac{du}{dx} = 2x$, or $\frac{1}{2}du = x\,dx$.

Thus $$p(x) = -300\int \frac{1}{(x^2+9)^{3/2}}x\,dx$$

$$= -150\int u^{-3/2}\,du$$

$$= \frac{300}{\sqrt{x^2+9}} + C$$

When the price is \$75, 4 hundred pair are demanded, so

$$75 = \frac{300}{\sqrt{(4)^2+9}} + C$$

or $$C = 15$$

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So, $$p(x) = \frac{300}{\sqrt{x^2+9}} + 15$$

(ii) When $x = 5$ hundred,

$$p(5) = \frac{300}{\sqrt{(5)^2+9}} + 15$$

$$= \$66.45 \text{ per pair}$$

When $x = 0$ hundred

, $$p(0) = \frac{300}{\sqrt{0+9}} + 15$$

$$= \$115 \text{ per pair}$$

(iii) At a price of \$90 per pair,

$$90 = \frac{300}{\sqrt{x^2+9}} + 15$$

$$\sqrt{x^2+9} = 4$$

or $$x \approx 2.65 \text{ or } 265 \text{ pairs}$$

In this book, we have provided solutions along with the papers. To download the same papers in unsolved format for printing, please visit
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Sample Paper 19

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

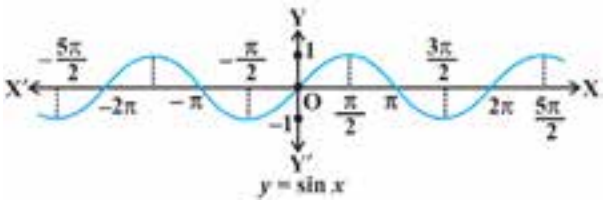
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. At which point does the graph of $y = \sin x$ intersect the origin in the given image?



(a) $x = -\pi$ (b) $x = 0$

(c) $x = \pi$ (d) $x = 2\pi$

Sol :

From the graph, we can see that the sine curve starts from the origin O(0,0). This means that at $x = 0$, the value of $y = \sin 0 = 0$. The curve passes exactly through this central point where the X-axis and Y-axis intersect. None of the other options $(-\pi, \pi, 2\pi)$ coincide with the origin in the graph. Hence, the correct intersection point is at (0,0).

Thus (b) is correct option.

2. Order of the differential equation of the family of all concentric circles centred at (h, k), is

(a) 2 (b) 3

(c) 1 (d) 4

Sol :

$$(x-h)^2 + (y-k)^2 = r^2$$

Here only one arbitrary constant r.

So, order of differential equation is 1.

Thus (c) is correct option.

3. If m and n are the order and degree of the differential equation

$$\left(\frac{d^2y}{dx^2}\right)^5 + 4\frac{\left(\frac{d^2y}{dx^2}\right)^3}{\frac{d^3y}{dx^3}} + \frac{d^3y}{dx^3} = x^2 - 1, \text{ then}$$

(a) $m = 3, n = 2$ (b) $m = 3, n = 3$

(c) $m = 3, n = 5$ (d) $m = 3, n = 1$

Sol :

Given differential equation can be rewritten as

$$\left(\frac{d^2y}{dx^2}\right)^5 \frac{d^3y}{dx^3} + 4\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{d^3y}{dx^3}\right)^2 = (x^2-1)\frac{d^2y}{dx^2}$$

Here, order $m = 3$

Degree $n = 2$

Thus (a) is correct option.

4. The vectors $\vec{a} = 2\hat{i} - 3\hat{j}$ and $\vec{b} = -4\hat{i} + 6\hat{j}$ are

(a) Coincident

(b) Parallel

(c) Perpendicular

(d) Neither parallel nor perpendicular

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Sol :

Given that, $\vec{a} = 2\hat{j} - 3\hat{j}$

and $\vec{b} = -4\hat{i} + 6\hat{j} = -2\vec{a}$

Hence, $\vec{a}$ and $\vec{b}$ are parallel vectors.
Thus (b) is correct option.

5. The area bounded by $y = \log x$, X-axis and ordinates $x = 1$, $x = 2$ is

(a) $\frac{1}{2}(\log 2)^2$ (b) $\log(2/e)$

(c) $\log(4/e)$ (d) $\log 4$

Sol :

Required area, $A = \int_1^2 \log x \, dx$

$= [x \log x - x]_1^2$

$= 2\log 2 - 1$

$= \log 4 - \log e$

$= \log\left(\frac{4}{e}\right)$

Thus (c) is correct option.

6. A straight line which makes an angle of 60° with each of y and z axes, inclined with x-axis at an angle of

(a) 30° (b) 45°

(c) 75° (d) 60°

Sol :

Now, $\cos^2\alpha + \cos^2 60^\circ + \cos^2 60^\circ = 1$

$\cos^2\alpha = 1 - \frac{1}{4} - \frac{1}{4} = \frac{1}{2}$

$\alpha = 45^\circ$

Thus (b) is correct option.

7. It is given that the events A and B are such that $P(A) = \frac{1}{4}$, $P(\frac{A}{B}) = \frac{1}{2}$ and $P(\frac{B}{A}) = \frac{2}{3}$. Then, $P(B)$ is equal to

(a) $\frac{1}{2}$ (b) $\frac{1}{6}$

(c) $\frac{1}{3}$ (d) $\frac{2}{3}$

Sol :

We have $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$

and $P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)}$

$P(B) = \frac{P(\frac{B}{A}) \cdot P(A)}{P(\frac{A}{B})}$

$= \frac{\frac{2}{5} \cdot \frac{1}{4}}{\frac{1}{2}} = \frac{1}{3}$

Thus (c) is correct option.

8. If $\sin^{-1}x = \theta + \beta$ and $\sin^{-1}y = \theta - \beta$, then $1 + xy$ is equal to

(a) $\sin^2\theta + \sin^2\beta$ (b) $\sin^2\theta + \cos^2\beta$

(c) $\cos^2\theta + \cos^2\beta$ (d) none of these

Sol :

We have,

$\sin^{-1}x = \theta + \beta$ and $\sin^{-1}y = \theta - \beta$

$x = \sin(\theta + \beta)$ and $y = \sin(\theta - \beta)$

$1 + xy = 1 + \sin(\theta + \beta)\sin(\theta - \beta)$

$= 1 + \sin^2\theta - \sin^2\beta$

$= \cos^2\beta + \sin^2\theta$

Thus (b) is correct option.

9. Let $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}, \vec{b} = \hat{j}$. The value of $\vec{c}$ for which $\vec{a}, \vec{b}, \vec{c}$ form a right handed system is

(a) $\vec{0}$ (b) $z\hat{i} - x\hat{k}$

(c) $-z\hat{i} + x\hat{k}$ (d) $y\hat{j}$

Sol :

Since, $\vec{a}, \vec{b}$ and $\vec{c}$ vectors form a right handed system, then

$\vec{c} = \vec{a} \times \vec{b}$

$= (x\hat{i} + y\hat{j} + z\hat{k}) \times \hat{j} = x\hat{k} - z\hat{i}$

Thus (c) is correct option.

10. $\int \frac{\sec^2(\sin^{-1}x)}{\sqrt{1 - x^2}} dx$ is equal to

(a) $\sin(\tan^{-1}x) + C$ (b) $\tan(\sec^{-1}x) + C$

(c) $\tan(\sin^{-1}x) + C$ (d) $-\tan(\cos^{-1}x) + C$

Sol :

We have $I = \int \frac{\sec^2(\sin^{-1}x)}{\sqrt{1 - x^2}} dx$

Substituting $\sin^{-1}x = t \Rightarrow \frac{dt}{dx} = \frac{1}{\sqrt{1 - x^2}}$

$dt = \frac{1}{\sqrt{1 - x^2}} dx$

$I = \int \sec^2 t \, dt$

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$$\cos\frac{y}{x} = \log|x| + 1 \qquad \text{[from Eq. (2)]}$$

which is required particular solution of given differential equation.

OR

We have $$\frac{dy}{dx} = 1 + x + y + xy$$

$$\frac{dy}{dx} = 1(1+x) + y(1+x)$$

$$\frac{dy}{dx} = (1+x)(1+y) \qquad ...(1)$$

Separating the variables, we get

$$\frac{1}{(1+y)}dy = (1+x)\,dx \qquad ...(2)$$

Integrating both sides of above equation, we have

$$\int \frac{1}{1+y}dy = \int (1+x)\,dx$$

$$\log|1+y| = x + \frac{x^2}{2} + C \qquad ...(3)$$

Also, given that $y = 0$, when $x = 1$.

Substituting $x = 1$, $y = 0$ in Eq. (3), we get

$$\log[1+0] = 1 + \frac{1}{2} + C \Rightarrow C = -\frac{3}{2}$$

Now, substituting the value of C in Eq. (3), we get

$$\log|1+y| = x + \frac{x^2}{2} - \frac{3}{2}$$

which is the required particular solution of given differential equation.

35. Find the vector and Cartesian equations of the line passing through the point (2,1,3) and perpendicular to the lines

$$\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$$

and $$\frac{x}{-3} = \frac{y}{2} = \frac{z}{5}.$$

Sol :

Any line through the point (2,1,3) can be written as

$$\frac{x-2}{a} = \frac{y-1}{b} = \frac{z-3}{c} \qquad ...(1)$$

where, a, b and c are the direction ratios of line (1).

Now, the line (1) is perpendicular to the lines

$$\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$$

and $$\frac{x-0}{-3} = \frac{y-0}{2} = \frac{z-0}{5}.$$

Direction ratio of these two lines are (1,2,3) and (−3,2,5), respectively.

We know that, if two lines are perpendicular, then

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

So we have $$a + 2b + 3c = 0 \qquad ...(2)$$

and $$-3a + 2b + 5c = 0 \qquad ...(3)$$

In Eqs. (2) and (3), by cross-multiplication, we get

$$\frac{a}{10-6} = \frac{b}{-9-5} = \frac{c}{2+6}$$

$$\frac{a}{4} = \frac{b}{-14} = \frac{c}{8}$$

$$\frac{a}{2} = \frac{b}{-7} = \frac{c}{4} = \lambda \text{ (say)}$$

$$a = 2\lambda,\ b = -7\lambda \text{ and } c = 4\lambda$$

Substituting the values of a, b and c in Eq. (1), we get

$$\frac{x-2}{2\lambda} = \frac{y-1}{-7\lambda} = \frac{z-3}{4\lambda}$$

$$\frac{x-2}{2} = \frac{y-1}{-7} = \frac{z-3}{4}$$

which is the required Cartesian equation of the line.

The vector equation of line which passes through (2,1,3) and parallel to the vector $2\hat{i} - 7\hat{j} + 4\hat{k}$ is

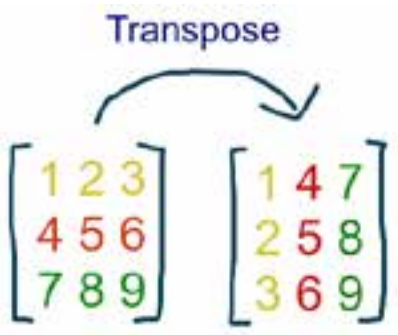
$$\vec{r} = 2\hat{i} + \hat{j} + 3\hat{k} + \lambda(2\hat{i} - 7\hat{j} + 4\hat{k})$$

which is the required vector equation of the line.

SECTION - E

Case study based questions are compulsory.

36. In linear algebra, the transpose of a matrix is an operator which flips a matrix over its diagonal; that is, it switches the row and column indices of the matrix A by producing another matrix, often denoted by A^T. The transpose of a matrix was introduced in 1858 by the British mathematician Arthur Cayley.



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16. In the given figure, the shaded regions are bounded by straight lines and vertical boundaries. Which of the following statements correctly describes them?

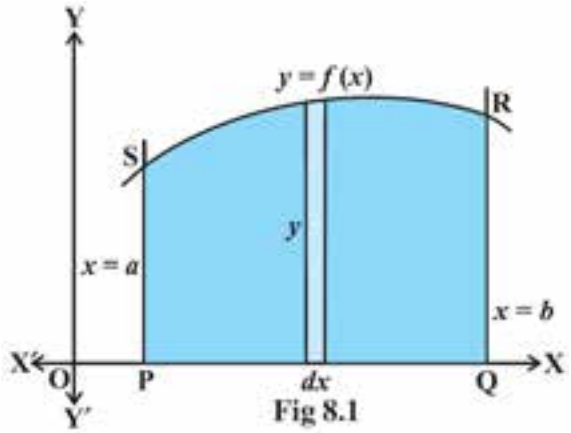


Fig 8.1

(a) Left shaded region lies between x=-1 and x=-2/3, bounded by line segments AB and AC.

(b) Right shaded region lies between x=0 and x=1, bounded by line segments OD and OE.

(c) Both shaded regions together form trapeziums.

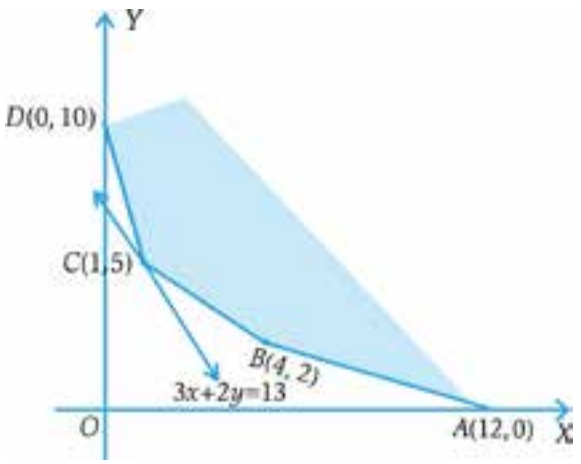
(d) Both (a) and (b) are correct.

Sol :

From the figure, the left shaded region is enclosed by vertical lines x=-1 and x=-2/3, along with segments AB and AC. The right shaded region is enclosed by vertical lines x=0 and x=1, along with segments OD and OE. Both are correctly described in options (a) and (b), making option (d) correct.

Thus (d) is correct option.

17. With the same feasible region, find the minimum value of $z = 4x + y$.



(a) 48 (b) 18

(c) 9 (d) 10

Sol :

Evaluate z=4x+y at the vertices. At A(12,0), z=48; at B(4,2), z=18; at C(1,5), z=9; at D(0,10), z=10. The least of these is 9, achieved at C(1,5). Therefore the minimum value is 9.

Thus (c) is correct option.

18. Inverse of the matrix $\begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$ is

(a) $\begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$ (b) $\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$

(c) $\begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$ (d) $\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$

Sol :

Here, cofactors are

$$C_{11} = \cos 2\theta,\ C_{12} = -\sin 2\theta$$

$$C_{21} = \sin 2\theta \text{ and } C_{22} = \cos 2\theta$$

$$|A| = |\cos^2 2\theta + \sin^2 2\theta| = 1$$

$$A^{-1} = \frac{1}{|A|}\begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}^T$$

$$= \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$$

Thus option (d) is correct.

19. Let A be a 2×2 matrix.

Assertion: adj (adj A) = A.

Reason: $|\text{adj } A| = |A|$.

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

Let $$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$\text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\text{adj (adj } A) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$|\text{adj } A| = ad - bc = |A|$$

Thus assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

Thus (b) is correct option.

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$$= \tan t + C$$

$$= \tan(\sin^{-1} x) + C$$

Thus (c) is correct option.

11. If R is a relation on the set N, defined by $\{(x, y): 2x - y = 10\}$, then R is

(a) Reflexive (b) Symmetric

(c) Transitive (d) None of the above

Sol :

Reflexive: As $20 \in N$ but $(20, 20) \notin R$. So, it is not reflexive.

Symmetric: As $(20, 30) \in R$ but $(30, 20) \notin R$. So, it is not symmetric.

Transitive: As $(20, 30) \in R$, $(30, 50) \in R$ but $(20, 50) \notin R$.

So, it is not transitive.

Thus (d) is correct option.

12. If the function $f(x) = kx^3 - 9x^2 + 9x + 3$ is monotonically increasing in every interval, then

(a) $k < 3$ (b) $k \leq 3$

(c) $k > 3$ (d) $k \geq 3$

Sol :

$$f'(x) = 3kx^2 - 18x + 9$$

$f(x)$ is increasing on every interval, if

$$f'(x) \geq 0 \quad \forall\, x \varepsilon R$$

$$3kx^2 - 18x + 9 \geq 0 \quad \forall\, x \varepsilon R$$

If $k > 0$ and $(-18)^2 - 4(3k)(9) \leq 0$

If $k > 0$ and $324 - 108k \leq 0$

If $k > 0$ and $k \geq 3$

Thus (d) is correct option.

13. If $f(x) = \begin{cases} \dfrac{3\sin \pi x}{5x}, & x \neq 0 \\ 2k, & x = 0 \end{cases}$ is continuous at $x = 0$, then the value of k is

(a) $\dfrac{\pi}{10}$ (b) $\dfrac{3\pi}{10}$

(c) $\dfrac{3\pi}{2}$ (d) $\dfrac{3\pi}{5}$

Sol :

We have $f(x) = \begin{cases} \dfrac{3\sin \pi x}{5x}, & x \neq 0 \\ 2k, & x = 0 \end{cases}$

Now, $\lim\limits_{x \to 0} f(x) = \lim\limits_{x \to 0} \left(\dfrac{3\sin x}{5x}\right)$

$$= \frac{3}{5} \lim_{x \to 0} \left(\sin \frac{\pi x}{\pi x}\right) \times \pi$$

$$= \frac{3}{5} \times 1 \times \pi = \frac{3}{5}\pi$$

Also, $f(0) = 2k$

Since, $f(x)$ is continuous at $x = 0$.

$$f(0) = \lim_{x \to 0} f(x)$$

$$2k = \frac{3}{5}\pi \Rightarrow k = \frac{3\pi}{10}$$

Thus (b) is correct option.

14. The domain of the function $f(x) = \sqrt{\cos x}$ is

(a) $\left[\dfrac{3\pi}{2}, 2\pi\right]$ (b) $\left[0, \dfrac{\pi}{2}\right]$

(c) $[-\pi, \pi]$ (d) $\left[0, \dfrac{\pi}{2}\right] \cup \left[\dfrac{3\pi}{2}, 2\pi\right]$

Sol :

Given, $f(x) = \sqrt{\cos x}$, i.e. $\cos x \geq 0$

But, $-1 \leq \cos x \leq 1$

$$0 \leq \cos x \leq 1$$

i.e. x lies in 1st or IVth quadrant.

$$0 \leq x \leq \frac{\pi}{2}$$

or $$\frac{3\pi}{2} \leq x \leq 2\pi$$

$$x \in \left[0, \frac{\pi}{2}\right] \cup \left[\frac{3\pi}{2}, 2\pi\right]$$

Also, $\cos(-x) = \cos x$

$\left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$ is also the domain of the function.

Thus (d) is correct option.

15. If $f(x) = \begin{cases} \dfrac{\log x}{x - 1}, & \text{if } x \neq 1 \\ k, & \text{if } x = 1 \end{cases}$ is continuous at $x = 1$, then the value of k is

(a) 0 (b) -1

(c) 1 (d) e

Sol :

At $x = 1$, $\lim\limits_{x \to 1^+} f(x) = \lim\limits_{h \to 0} \dfrac{\log(1 + h)}{1 + h - 1}$

$$= \lim_{h \to 0} \frac{\log(1 + h)}{h} = 1$$

As $f(x)$ is continuous at $x = 1$.

$$\lim_{x \to 1^+} f(x) = f(1) \Rightarrow 1 = k$$

Thus (c) is correct option.

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OR

We have $|\vec{a}+\vec{b}| = |\vec{a}|$

Squaring both sides, we get

$$|\vec{a}+\vec{b}|^2 = |\vec{a}|^2$$

$$|\vec{a}|^2 + 2\vec{a}\cdot\vec{b} + |\vec{b}|^2 = |\vec{a}|^2$$

$$2\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{b} = 0$$

$$(2\vec{a}+\vec{b})\cdot\vec{b} = 0$$

$$(2\vec{a}+\vec{b}) \perp \vec{b}$$

24. Two dice are thrown n times in succession. What is the probability of obtaining a doublet six at-least once?

Sol :

p = Probability of getting double six in two dice

$= \frac{1}{6^2} = \frac{1}{36}$ and $q = \frac{35}{36}$

Required probability

= 1 − (Probability of not getting double six)n

$= 1 - \left(\frac{35}{36}\right)^n$

25. Evaluate $\int \frac{(\log x)^2}{x}\,dx$.

OR

Evaluate $\int \frac{e^{\tan^{-1}x}}{1+x^2}\,dx$.

Sol :

We have $I = \int \frac{(\log x)^x}{x}\,dx$

Substituting $\log x = t \Rightarrow \frac{1}{x}\,dx = dt$ we have

$$I = \int \frac{(\log x)^2}{x}\,dx = \int t^2\,dt$$

$$= \int \frac{(\log x)^2}{x}\,dx = \int t^2\,dt$$

$$= \frac{t^3}{3} + C = \frac{(\log x)^3}{3} + C$$

Substituting $t = \log x$ we have

$$I = \frac{t^3}{3} + C = \frac{(\log x)^3}{3} + C$$

OR

We have $I = \int \frac{e^{\tan^{-1}x}}{1+x^2}\,dx$

Substituting $\tan^{-1}x = t \Rightarrow \frac{1}{1+x^2}\,dx = dt$ we have

$$I = \int \frac{e^{\tan^{-1}x}}{1+x^2}\,dx = \int e^t\,dt$$

$$= e^t + C$$

Substituting $t = \tan^{-1}x$ we have

$$I = e^{\tan^{-1}x} + C$$

SECTION - C

Section C consists of 6 questions of 3 marks each.

26. If $x = \cos t(3 - 2\cos^2 t)$ and $y = \sin t(3 - 2\sin^2 t)$, then find the value of $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$.

Sol :

We have $x = \cos t(3 - 2\cos^2 t)$

or $x = 3\cos t - 2\cos^3 t$

Differentiating both sides w.r.t. t, we get

$$\frac{dx}{dt} = 3(-\sin t) - 2(3)\cos^2 t(-\sin t)$$

$$\frac{dx}{dt} = -3\sin t + 6\cos^2 t \sin t \quad \text{...(i)}$$

Also, $y = \sin t(3 - 2\sin^2 t)$

or $y = 3\sin t - 2\sin^3 t$

Differentiating both sides w.r.t. t, we get

$$\frac{dy}{dt} = 3\cos t - 2 \times 3 \times \sin^2 t \cos t$$

$$\frac{dy}{dt} = 3\cos t - 6\sin^2 t \cos t \quad \text{...(ii)}$$

Now, $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$

$$= \frac{3\cos t - 6\cos t \sin^2 t}{-2\sin t + 6\cos^2 t \sin t}$$

$$= \frac{\cos t - 2\cos t \sin^2 t}{-\sin t + 2\cos^2 t \sin t}$$

$$= \frac{\cos t(1 - 2\sin^2 t)}{\sin t(2\cos^2 t - 1)}$$

$$= \frac{\cos t \cdot \cos 2t}{\sin t \cdot \cos 2t}$$

$$= \cot(t)$$

$$\left.\frac{dy}{dx}\right|_{t=\frac{\pi}{4}} = \cot\left(\frac{\pi}{4}\right) = 1$$

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$$= \int_0^{\pi} \frac{-(\pi - x)\tan x}{-\sec x - \tan x}\, dx$$

$$= \int_0^{\pi} \frac{(\pi - x)\tan x}{\sec x + \tan x}\, dx \qquad \ldots(ii)$$

Adding Eqs. (i) and (ii), we get

$$2I = \int_0^{\pi} \frac{\pi \tan x}{\sec x + \tan x}\, dx$$

$$= \pi \int_0^{\pi/2} \frac{\tan x}{\sec x + \tan x}\, dx + \pi \int_{\pi/2}^{\pi} \frac{\tan x}{\sec x + \tan x}\, dx$$

$$= \pi \int_0^{\pi/2} \frac{\sin x}{1 + \sin x} + \pi \int_{\pi/2}^{\pi} \frac{\sin x}{1 + \sin x}\, dx$$

$$= \pi \int_0^{\pi} \frac{\sin x}{1 + \sin x}\, d$$

$$= \pi \int_0^{p} \frac{1 + \sin x - 1}{1 + \sin x}\, dx$$

$$= \pi \int_0^{\pi} \left(\frac{1 + \sin x}{1 + \sin x} - \frac{1}{1 + \sin x}\right) dx$$

$$= \pi \int_0^{\pi} \left(1 - \frac{1}{1 + \sin x}\right) dx$$

$$= \pi \left[\int_0^{\pi} 1\, dx - \int_0^{\pi} \frac{1}{1 + \sin x}\, dx\right]$$

$$= \pi \left[\pi - 2\int_0^{\pi/2} \frac{1}{1 + \sin x}\, dx\right] \qquad \int_0^{2a} f(x) = 2\int_0^{a} f(x)\, dx, \text{ if } f(2a - x) = f(x)$$

$$= \pi \left[\pi - 2\int_0^{\pi/2} \frac{1}{1 + \sin x} \times \frac{1 - \sin x}{1 - \sin x}\, dx\right]$$

$$= \pi \left[\pi - 2\int_0^{\pi/2} \frac{1 - \sin x}{1 - \sin^2 x}\, dx\right]$$

$$= \pi \left[\pi - 2\int_0^{\pi/2} \frac{1 - \sin x}{\cos^2 x}\, dx\right] \qquad 1 - \sin^2 x = \cos^2 x$$

$$= \pi \left[\pi - 2\int_0^{\pi/2} \left(\frac{1}{\cos^2 x} - \frac{\sin x}{\cos^2 x}\right) dx\right]$$

$$= \pi \left[\pi - 2\int_0^{\pi/2} (\sec^2 x - \sec x \tan x)\, dx\right]$$

$$= \pi \{\pi - 2[\tan x - \sec x]_0^{\pi/2}\}$$

$$= \pi \left\{\pi - \lim_{x \to \frac{\pi}{2}} 2(\tan x - \sec x) + 2(\tan 0 - \sec 0)\right\}$$

$$2I = \pi\{\pi - 0 + 2(0 - 1)\} = \pi(\pi - 2)$$

$$I = \frac{\pi}{2}(\pi - 2)$$

33. Maximise $Z = 5x + 3y$ subject to the constraints: $3x + 5y \le 15$; $5x + 2y \le 10$, $x \ge 0$, $y \ge 0$.

OR

Minimize $Z = 3x + 5y$ such that $x + 3y \ge 3$, $x + y \ge 2$, x, $y \ge 0$.

Sol :

Maximize $Z = 5x + 3y$

Subject to $3x + 5y \le 15$ ...(i)

$5x + 2y \le 10$ ...(ii)

and $x \ge 0,\ y \ge 0$

(i) Region corresponding to $3x + 5y \le 15$:

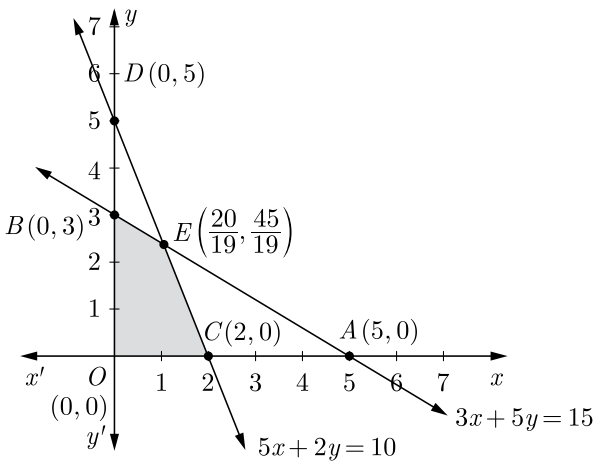
Line $3x + 5y = 15$			
x	0	5	Point $(0,0)$ is true for $3x + 5y \le 15$. So, the region is towards the origin.
y	3	0	

(ii) Region corresponding to $5x + 2y \le 10$:

Line $5x + 2y = 10$			
x	0	2	Point $(0,0)$ is true for $5x + 2y \le 10$. So, the region is towards the origin.
y	5	0	

Since $x \ge 0$, $y \ge 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is $OBECO$ which is bounded. Solving equations $3x + 5y = 15$ and $5x + 2y = 10$, we get $E(\frac{20}{19}, \frac{45}{19})$. The corner points of the feasible region are $O(0,0)$, $B(0,3)$, $E(\frac{20}{19}, \frac{45}{19})$ and $B(2,0)$.



Since, the feasible region is a bounded region, we can check the objective function at all the corner to find the maxima. The values of objective function Z at these points are as follows.

Corner	$Z = 5x + 3y$
$O(0,0)$	$Z = 0 + 0 = 0$
$B(0,3)$	$Z = 5 \times 0 + 3 \times 3 = 9$
$E(\frac{20}{19}, \frac{45}{19})$	$Z = 5 \times \frac{20}{19} + 3 \times \frac{45}{19} = 12.37$
$C(2,0)$	$= 5 \times 2 + 3 \times 0 = 10$

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Let x and y be the lengths of sides of a rectangle. Again, let A denotes its area and P be the perimeter.
Now, area of rectangle,

$$A = xy$$

$$y = \frac{A}{x} \quad ...(i)$$

And $$P = 2(x+y)$$

$$P = 2\left(x+\frac{A}{x}\right) \quad y=\frac{A}{x}$$

Differentiating both sides w.r.t. x, we get

$$\frac{dP}{dx} = 2\left(1-\frac{A}{x^2}\right)$$

For maxima or minima, putting $\frac{dP}{dx}=0$ we have,

$$2\left(1-\frac{A}{x^2}\right) = 0 \Rightarrow 1 = \frac{A}{x^2}$$

$$A = x^2$$

$$xy = x^2 \quad A=xy$$

$$x = y$$

Also, $$\frac{d^2P}{dx^2} = \frac{d}{dx}\left[2\left(1-\frac{A}{x^2}\right)\right]$$

$$= 2\left(\frac{2A}{x^3}\right) = \frac{4A}{x^3} > 0$$

Here, x and A being the side and area of rectangle can never be negative. So, P is minimum.
Hence, perimeter of rectangle is minimum, when rectangle is a square. Hence proved.

31. Evaluate $\int_{\pi/2}^{\pi/2}\frac{\cos x}{1+e^x}dx$.

OR

Find $\int\frac{x^3}{x^4+3x^2+2}dx$.

Sol :

We have $$I = \int_{-\pi/2}^{\pi/2}\frac{\cos x}{1+e^x}dx$$

$$= \int_{-\pi}^{0}\frac{\cos x}{1+e^x}dx + \int_{0}^{\pi/2}\frac{\cos x}{1+e^x}dx$$

In first integral, substituting $x = -t \Rightarrow dx = -dt$
When $x=0$, then $t=0$ and
when $x=-\frac{\pi}{2}$, then $t=\frac{\pi}{2}$.

Thus $$I = \int_{\pi/2}^{0}\frac{\cos t}{1+e^{-t}}(-dt) + \int_{0}^{\pi/2}\frac{\cos x}{1+e^x}dx$$

Using the fact $-\int_a^0 f(x)\,dx = \int_0^a f(x)\,dx$ we have

$$I = \int_0^{\pi/2}\frac{\cos t}{1+\frac{1}{e^t}}dt + \int_0^{\pi/2}\frac{\cos x}{1+e^x}dx$$

$$= \int_0^{\pi/2}\frac{e^t\cos t}{e^t+1}dt\int_0^{\pi/2}\frac{\cos x}{1+e^x}dx$$

Using the fact $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$ we have

$$I = \int_0^{\pi/2}\frac{e^x\cos x}{e^x+1}dx + \int_0^{\pi/2}\frac{\cos x}{1+e^x}dx$$

$$= \int_0^{\pi/2}\frac{(1+e^x)\cos x}{(e^x+1)}dx$$

$$= \int_0^{p/2}\cos x dx$$

$$= [\sin x]_0^{\pi/2}$$

$$= \sin\frac{\pi}{2} - 0 = 1$$

OR

We have $$I = \int\frac{x^3}{x^4+3x^2+2}dx$$

Substituting $x^2 = t \Rightarrow 2x = \frac{dt}{dx} \Rightarrow x\,dx = \frac{dt}{2}$ we have

$$I = \frac{1}{2}\int\frac{t}{t^2+3t+2}dt$$

$$= \frac{1}{2}\left[\int\frac{2}{t+2}dt - \int\frac{1}{t+1}dt\right]$$

$$= \frac{1}{2}[2\log|t+2| - \frac{1}{2}\log|t+1\|] + C$$

$$= \log|t+2| - \frac{1}{2}\log|t+1| + C$$

$$= \log|t+2| - \log\sqrt{t+1} + C$$

$$= \log\left|\frac{t+2}{\sqrt{t+1}}\right| + C$$

$$= \log\left|\frac{x^2+2}{\sqrt{x^2+1}}\right| + C \quad t=x^2$$

SECTION - D

Section D consists of 4 questions of 5 marks each.

32. Evaluate $\int_0^{\pi}\frac{x\tan x}{\sec x+\tan x}dx$.

Sol :

We have $$I = \int_0^{\pi}\frac{x\tan x}{\sec x+\tan x}dx \quad ...(i)$$

Using the fact $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ we have

$$I = \int_0^{\pi}\frac{(\pi-x)\tan(\pi-x)}{\sec(\pi-x)+\tan(\pi-x)}dx$$

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27. If $\vec{a}+\vec{b}+\vec{c}=0$ and $|\vec{a}|=5$, $|\vec{b}|=6$ and $|\vec{c}|=9$, then find the angle between $\vec{a}$ and $\vec{b}$.

Sol :

We have
$$\begin{aligned}\vec{a}+\vec{b}+\vec{c} &= 0\\ \vec{a}+\vec{b} &= -\vec{c}\\ (\vec{a}+\vec{b})^2 &= (-\vec{c})^2\\ (\vec{a}+\vec{b})(\vec{a}+\vec{b}) &= (-\vec{c})\cdot(-\vec{c})\\ \vec{a}\cdot\vec{a}+\vec{a}\cdot\vec{b}+\vec{b}\cdot\vec{a}+\vec{b}\cdot\vec{b} &= \vec{c}\cdot\vec{c}\\ |\vec{a}|^2+2\vec{a}\cdot\vec{b}+|\vec{b}|^2 &= |\vec{c}|^2\\ |\vec{a}|^2+2|\vec{a}||\vec{b}|\cos\theta+|\vec{b}|^2 &= |\vec{c}|^2\end{aligned}$$

Substituting $|\vec{a}|=5$, $|\vec{b}|=6$ and $|\vec{c}|=9$ in above we get
$$\begin{aligned}(5)^2+2\times5\times6\times\cos\theta+(6)^2 &= (9)^2\\ 25+60\cos\theta+36 &= 81\\ 60\cos\theta &= 81-61=20\\ \cos\theta &= \frac{20}{60}=\frac{1}{3}\\ \theta &= \cos^{-1}\left(\frac{1}{3}\right)\end{aligned}$$

28. Solve the following equation for x.
$$\cos(\tan^{-1}x) = \sin\left(\cot^{-1}\frac{3}{4}\right)$$

Sol :

We know that
$$\tan^{-1}x = \cos^{-1}\frac{1}{\sqrt{1+x^2}}$$
and
$$\cot^{-1}y = \sin^{-1}\frac{1}{\sqrt{1+y^2}}$$
Thus
$$\cot^{-1}\frac{3}{4} = \sin^{-1}\frac{4}{5}$$
Now
$$\begin{aligned}\cos(\tan^{-1}x) &= \sin\left(\cot^{-1}\frac{3}{4}\right)\\ \cos\left(\cos^{-1}\frac{1}{\sqrt{1+x^2}}\right) &= \sin\left(\sin^{-1}\frac{4}{5}\right)\\ \frac{1}{\sqrt{1+x^2}} &= \frac{4}{5}\\ 16(x^2+1) &= 25\\ 16x^2 &= 9\\ x &= \pm\frac{3}{4}\end{aligned}$$
But $x=\frac{-3}{4}$ does not satisfy the given equation.

Hence, the required solution is $x=\frac{3}{4}$.

29. Find $|AB|$, if $A=\begin{bmatrix}0 & -1\\ 0 & 2\end{bmatrix}$ and $B=\begin{bmatrix}3 & 5\\ 0 & 0\end{bmatrix}$.

Sol :

We have $A=\begin{bmatrix}0 & -1\\ 0 & 2\end{bmatrix}$ and $B=\begin{bmatrix}3 & 5\\ 0 & 0\end{bmatrix}$

Now,
$$\begin{aligned}AB &= \begin{bmatrix}0 & -1\\ 0 & 2\end{bmatrix}\begin{bmatrix}3 & 5\\ 0 & 0\end{bmatrix}\\ &= \begin{bmatrix}0+0 & 0+0\\ 0+0 & 0+0\end{bmatrix}=\begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}\\ |AB| &= 0\end{aligned}$$

30. Find the intervals in which the function given by $f(x)=x^4-8x^3+22x^2-24x+21$ is
(i) increasing
(ii) decreasing.

OR

Show that of all the rectangles of given area, the square has the smallest perimeter.

Sol :

We have $f(x) = x^4-8x^3+22x^2-24x+21$
Differentiating both sides w.r.t. x, we get
$$\begin{aligned}f'(x) &= 4x^3-24x^2+44x-24\\ &= 4(x^3-6x^2+11x-6)\\ &= 4(x-1)(x^2-5x+6)\\ &= 4(x-1)(x-2)(x-3)\end{aligned}$$
Substituting $f'(x)=0$ we have
$$\begin{aligned}4(x-1)(x-2)(x-3) &= 0\\ x &= 1, 2, 3\end{aligned}$$
So, the possible intervals are $(-\infty,1)$, $(1,2)$, $(2,3)$ and $(3,\infty)$.
For interval $(-\infty,1)$ $f'(x)<0$
For interval $(1,2)$, $f'(x)>0$
For interval $(2,3)<0$
For interval $(3,\infty)$, $f'(x)>0$.
Also, as $f(x)$ is a polynomial function, so it is continuous at $x=1, 2, 3$. Hence,
(i) function increases in $[1,2]$ and $[3,\infty)$.
(ii) function decreases in $(-\infty,1]$ and $[2,3]$.

OR

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Hence the maximum value of Z is 12.37 is at $E(\frac{20}{19}, \frac{45}{19})$ and the optimal solution is $x = \frac{20}{19}$ and $y = \frac{45}{19}$.

OR

Minimize $Z = 3x + 5y$

Subject to $x + 3y \geq 3$ ...(i)

$x + y \geq 2$ ...(ii)

and $x \geq 0,\ y \geq 0$

(i) Region corresponding to $x + 3y \geqslant 3$:

Line $x + 3y = 3$			
x	0	3	Point $(0,0)$ is false for $x + 3y \geqslant 3$. So, the region is away from the origin.
y	1	0	

(ii) Region corresponding to $x + y \geqslant 2$:

Line $x + y = 2$			
x	0	2	Point $(0,0)$ is false for $x + y \geqslant 2$. So, the region is away from the origin.
y	2	0	

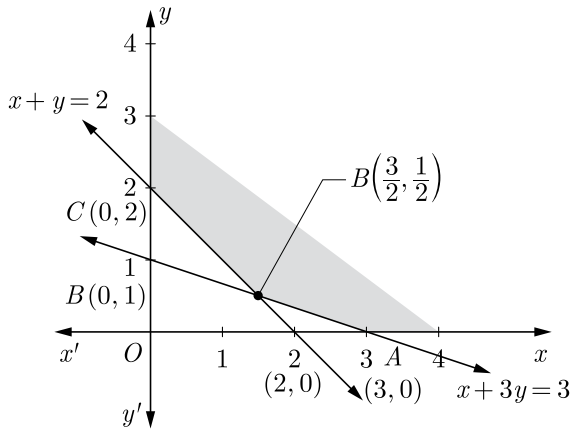
Since $x \geq 0,\ y \geq 0$ the feasible region lies in the first quadrant.

Now we draw all line on the graph and find the common area. Clearly, feasible region is ABC which is open and unbounded region. Solving equations $x + 3y = 3$ and $x + y = 2$, we get $B(\frac{3}{2}, \frac{1}{2})$. The corner points of the feasible region are $A(3,0)$, $B(\frac{3}{2}, \frac{1}{2})$ and $C(0,2)$.

The values of objective function Z at these points are as follows.

Corner points	$Z = 3x + 5y$
$A(3,0)$	$Z = 3 \times 3 + 0 = 9$
$B(\frac{3}{2}, \frac{1}{2})$	$Z = 3 \times \frac{3}{2} + 5 \times \frac{1}{2} = 7$
$C(0,2)$	$Z = 3 \times 0 + 5 \times 2 = 10$

Hence the minimum value of Z is 7 at $B(\frac{3}{2}, \frac{1}{2})$.



34. Find the particular solution of the differential equation

$$x\frac{dy}{dx} - y + x\,\text{cosec}\left(\frac{y}{x}\right) = 0$$

OR

Find the particular solution of the differential equation

$$\frac{dy}{dx} = 1 + x + y + xy,$$

given that $y = 0$ when $x = 1$.

Sol :

Given differential equation is

We have $x\frac{dy}{dx} - y + x\,\text{cosec}\left(\frac{y}{x}\right) = 0$

$$\frac{dy}{dx} - \frac{y}{x} + \text{cosec}\left(\frac{y}{x}\right) = 0$$

$$\frac{dy}{dx} = \frac{y}{x} - \text{cosec}\left(\frac{y}{x}\right) \quad ...(1)$$

Which is a homogeneous differential equation as

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

Substituting $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ in eq. (1) we have

$$v + x\frac{dv}{dx} = \frac{vx}{x} - \text{cosec}\left(\frac{vx}{x}\right)$$

$$v + x\frac{dv}{dx} = v - \text{cosec}\, v$$

$$x\frac{dv}{dx} = -\,\text{cosec}\, v$$

$$\frac{dv}{\text{cosec}\, v} = \frac{-\,dx}{x}$$

Integrating both sides, we have

$$\int \frac{dv}{\text{cosec}\, v} = \int -\frac{dx}{x}$$

$$\int \sin v\, dv = \int -\frac{dx}{x}$$

$$-\cos v = -\log|x| + C$$

$$-\cos\frac{y}{x} = -\log|x| + C$$

$$\cos\frac{y}{x} = (\log|x| - C) \quad ...(2)$$

Also, given that $x = 1$ and $y = 0$.

Substituting above values in Eq. (2), we get

$$\cos 0 = \log|1| - C$$

$$1 = 0 - C \Rightarrow C = -1$$

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20. **Assertion:** The equation of curve passing through (3, 9) which satisfies differential equation $\frac{dy}{dx} = x + \frac{1}{x^2}$ is $6xy = 3x^3 + 29x - 6$

Reason: The solution of differential equation $\left(\frac{dy}{dx}\right)^2 - \left(\frac{dy}{dx}\right)(e^x + e^{-x}) + 1 = 0$ is $y = c_1 e^x + c_2 e^{-x}$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

Sol :

From reason differential equation can be written as

$$\left(\frac{dy}{dx} - e^x\right)\left(\frac{dy}{dx} - e^{-x}\right) = 0$$

$$\frac{dy}{dx} = e^x \text{ or } \frac{dy}{dx} = e^{-x}$$

$$\int dy = \int e^x\,dx \text{ or } \int dy = \int e^{-x}\,dx$$

$$y = e^x + c_1 \text{ or } y = -e^{-x} + c_2$$

$$= c_1 e^x + c_2 e^{-x} + c_2$$

$$= c_1 e^x + c_2 e^{-x}$$

It will satisfy the above equation from assertion.

$$\int dy = \int \left(x + \frac{1}{x^2}\right) dx$$

$$y = \frac{x^2}{2} - \frac{1}{x} + c$$

It passes through (3,9).

$$9 = \frac{9}{2} - \frac{1}{3} + c$$

$$c = 9 - \frac{9}{2} + \frac{1}{3} = \frac{29}{6}$$

$$y = \frac{x^2}{2} - \frac{1}{x} + \frac{29}{6}$$

$$6xy = 3x^3 + 29x - 6$$

Thus assertion is true, Reason is true; Reason is not a correct explanation for Assertion.
Thus (b) is correct option.

SECTION - B

Section B consists of 5 questions of 2 marks each.

21. If a line has direction ratios $(2, -1, -2)$, then what are its direction cosines?

Sol :

If (a, b, c) are DR's of a line then direction cosines of line is given by

$$l = \pm\frac{a}{\sqrt{a^2 + b^2 + c^2}}$$

$$m = \pm\frac{b}{\sqrt{a^2 + b^2 + c^2}}$$

$$n = \pm\frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

Here DR's of the line are $(2, -1, -2)$.

Now $\sqrt{2^2 + (-1)^2 + (-2)^2} = \sqrt{9} = 3$

Direction cosines of the give line are $\left(\frac{2}{3}, \frac{-1}{3}, \frac{-2}{3}\right)$.

22. Let R is the equivalence relation in the set $A = \{0, 1, 2, 3, 4, 5\}$ given by $R = \{(a, b): 2 \text{ divides } (a - b)\}$. Write the equivalence class $[0]$.

Sol :

We have $R = \{(a, b): 2 \text{ divides } (a - b)\}$

and $A = \{0, 1, 2, 3, 4, 5\}$

Clearly, $[0] = \{b \in A : (0, b) \in R\}$

$= \{b \in A : 2 \text{ divides } (0 - b)\}$

$= \{b \in A : 2 \text{ divides } (-b)\}$

$= \{0, 2, 4\}$

23. Write the projection of $(\vec{b} + \vec{c})$ on $\vec{a}$, where $\vec{a} = 2\hat{i} - 2\hat{j} + \hat{k}, \vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$.

OR

If $\vec{a}$ and $\vec{b}$ are to vectors such that $|\vec{a} + \vec{b}| = |\vec{a}|$, the prove that vector $2\vec{a} + \vec{b}$ is perpendicular to vector $\vec{b}$.

Sol :

We have, $\vec{a} = 2\hat{i} - 2\hat{j} + \hat{k}$,

$\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$

and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$

Now $(\vec{b} + \vec{c}) = (\hat{i} + 2\hat{j} - 2\hat{k}) + (2\hat{i} - \hat{j} + 4\hat{k})$

$= 3\hat{i} + \hat{j} + 2\hat{k}$

Now, the projection of $\vec{b} + \vec{c}$ on $\vec{a}$ is given by

$$\frac{(\vec{b} + \vec{c}) \cdot \vec{a}}{|\vec{a}|} = \frac{(3\hat{i} + \hat{j} + 2\hat{k}) \cdot (2\hat{i} - 2\hat{j} + \hat{k})}{\sqrt{2^2 + (-2)^2 + (1)^2}}$$

$$= \frac{6 - 2 + 2}{\sqrt{4 + 4 + 1}} = \frac{6}{3} = 2$$

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If $A=[a_{ij}]$ be a $m \times n$ matrix, then the matrix obtained by interchanging the rows and columns of A is called the transpose of A. A square matrix $A=[a_{ij}]$ is said to be symmetric, $A^T=A$ for all possible values of i and j. A square matrix $A\ [a_{ij}]$ is said to be skew-symmetric, if $A^T=-A$ for all possible values of i and j. Based on the above, information, answer the following questions.

(i) Find the transpose of [1, -2, -5].

(ii) Find the transpose of matrix (ABC).

(iii) Evaluate $(A+B)^T-A$, $A=\begin{bmatrix}0 & 1\\ 2 & -1\end{bmatrix}$ and $B=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$

OR

Evaluate $(AB)^T$, where $A=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ and $B=\begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}$

Sol :

(i) $[1-2-5]^T=\begin{bmatrix}1\\ -2\\ -5\end{bmatrix}$

(ii) $(ABC)^T=C^TB^TA^T$

(iii) $(A+B)=\begin{bmatrix}0 & 1\\ 2 & -1\end{bmatrix}+\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$

$=\begin{bmatrix}1 & 3\\ 5 & 3\end{bmatrix}$

Now, $(A+B)^T-A=\begin{bmatrix}1 & 5\\ 3 & 3\end{bmatrix}-\begin{bmatrix}0 & 1\\ 2 & -1\end{bmatrix}$

$=\begin{bmatrix}1 & 4\\ 1 & 4\end{bmatrix}$

OR

$AB=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}\begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}$

$=\begin{bmatrix}4 & 6\\ 1 & 4\end{bmatrix}$

$(AB)^T=\begin{bmatrix}4 & 1\\ 6 & 4\end{bmatrix}$

37. RK Verma is production analysts of a ready-made garment company. He has to maximize the profit of company using data available. He find that $P(x)=-6x^2+120x+25000$ (in Rupee) is the total profit function of a company where x denotes the production of the company.



Based on the above information, answer the following questions.

(i) Find the profit of the company, when the production is 3 units.

(ii) Find $P'(5)$

(iii) Find the interval in which the profit is strictly increasing.

OR

Find the production, when the profit is maximum.

Sol :

We have $P(x)=-6x^2+120x+25000$

$P'(x)=-12x+120$

(i) At $x=3$ we have,

$P(3)=-6(3)^2+120(3)+25000$

$=-54+360+25000$

$=25306$

(ii) $P'(5)=-12\times 5+120$

$=-60+120$

$=60$

(iii) For strictly increasing, $P'(x)>0$

$-12x+120>0$

$120>12x$

$x<10$

$x\in(0,10)$

OR

For maximum profit we put $P'(x)=0$, i.e.,

$0=-12x+120$

$x=10$

Now $P''(x)=-12<0$

At $x=10$, profit function is maximum.

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38. Measures of joint and survivor life expectancy are potentially useful to those designing or evaluating policies affecting older couples and to couples making retirement, savings, and long-term care decisions. However, couple-based measure of life expectancies are virtually unknown in the social sciences.



The odds against a husband who is 45 yr old, living till he is 70 are. 7 : 5 and the odds against his wife who is now 36, living till she is 61 are 5 : 3.

On the basis of above information, answer the following questions.

(i) Find the probabilities of husband living till 70 and wife living till 61.

(ii) Find the probability P(couple will be alive 25 yr hence).

(iii) Find the probability P(exactly one of them will be alive 25 yr hence).

OR

Find the probability P(none of them will be alive 25 yr hence) and probability P(at-least one of them will be alive 25 yr hence).

Sol :

(i) Let A be the event that the husband will be alive 25 yr hence and B be the event that the wife will be alive 25 yr hence. Then, A and B are independent events such that

$$P(A) = \frac{5}{7+5} = \frac{5}{12}$$

and $$P(B) = \frac{3}{5+3} = \frac{3}{8}$$

Hence, the required probabilities of husband living till 70 and wife living till 61 are $\frac{5}{12}$ and $\frac{3}{8}$, respectively.

(ii) P (couple will be alive 25 yr hence) $= P(A$ and $B)$

$$= P(A \cap B)$$

$$= P(A)\,P(B)$$

$[A$ and B are independent events$]$

$$= \frac{5}{12} \times \frac{3}{8} = \frac{5}{32}$$

(iii) Exactly one of them will be alive 25 yr hence in two mutually exclusive ways:

1. Husband will be alive 25 yr hence and wife will not i.e., $A \cap \overline{B}$
2. Wife will be alive 25 yr hence and husband will not i.e., $\overline{A} \cap B$

P(exactly one will be alive 25 yr hence)

$$P(I \text{ or } II) = P[A \cap \overline{B} \cup (\overline{A} \cup B)]$$

$$= P(A \cap \overline{B}) + P(\overline{A} \cap B)$$

$[A \cap \overline{B}$ and $\overline{A} \cap B$ are mutually exclusive$]$

$$= P(A)\,P(\overline{B}) + P(\overline{A})\,P(B)$$

$[A, B$ are independent events$]$

$$= \frac{5}{12} \times \left(1 - \frac{3}{8}\right) + \left(1 - \frac{5}{12}\right) \times \frac{3}{8}$$

$$= \frac{5}{12} \times \frac{5}{8} + \frac{7}{12} \times \frac{3}{8}$$

$$= \frac{46}{96} = \frac{23}{48}$$

OR

P (none of them will be alive 25 yr hence)

$$= P(\overline{A} \cap \overline{B})$$

$$= P(\overline{A})\,P(\overline{B})$$

$[A, B$ are independent events$]$

$$= \left(1 - \frac{5}{12}\right) \times \left(1 - \frac{3}{8}\right)$$

$$= \frac{7}{12} \times \frac{5}{8} = \frac{35}{96}$$

and P (at least one of them will be alive 25 yr hence)

$$= 1 - P(\overline{A})\,P(\overline{B})$$

$$= 1 - \left(1 - \frac{5}{12}\right)\left(1 - \frac{3}{8}\right)$$

$$= 1 - \frac{35}{96} = \frac{61}{96}$$

****

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Sample Paper 20

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

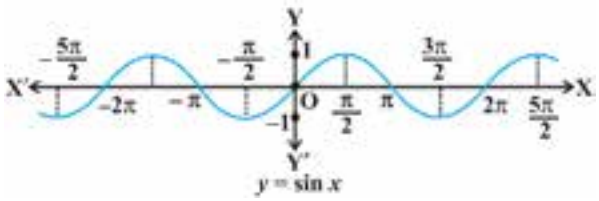
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the maximum value of $y = \sin x$ observed in the graph?



(a) 0

(b) 1

(c) –1

(d) 2

Sol :

In the graph, the sine curve reaches its highest point at several positions, such as $x = \frac{\pi}{2}$ and $x = \frac{5}{2}$. At these points, the curve touches the line parallel to the X-axis marked at $y = 1$. This shows that the maximum vertical displacement of the sine wave is exactly 1. The curve never crosses above this level in the image. Hence, the maximum value represented in the graph is 1.

Thus (b) is correct option.

2. Which of the following is a homogeneous differential equation?

(a) $x^2 y dx - (x^2 + y^2) dy = 0$

(b) $(xy) dx - (x^4 + y^4) dy = 0$

(c) $(2x + y - 3) dy - (x + 2y - 3) dx = 0$

(d) $(x - y) dy = (x^2 + y + 1) dx$

Sol :

A differential equation of the form

$$\frac{dy}{dx} = \frac{a_1 x + b_1 y + c_1}{a_2 x + b_2 y + c_2}$$

is known as homogeneous differential equation.

$$(2x + y - 3) dy - (x + 2y - 3) dx = 0$$

$$\frac{dy}{dx} = \frac{x + 2y - 3}{2x + y - 3}$$

Thus (c) is correct option.

3. A ball thrown vertically upwards according to the formula $s = 13.8t - 4.9t^2$, where s is in metres and t is in seconds. Then its velocity at $t = 1$ sec is

(a) 6m/sec (b) 4 m/sec

(c) 2 m/sec (d) 8 m/sec

Sol :

$$v = \frac{ds}{dt} = 13.8 - 9.8t$$

At $t = 1;\ v = 4\ m/s$

Thus (b) is correct option.

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4. The degree of the equation $\left(\frac{d^2y}{dx^2}\right)^2 - x\left(\frac{dy}{dx}\right)^3 = y^3$ is

(a) 0 (b) 1

(c) 2 (d) 3

Sol :

Degree of differential equation is degree of the highest order occurring in it. Therefore degree of differential equation

$$\left(\frac{d^2y}{dx^2}\right)^2 - x\left(\frac{dy}{dx}\right)^3 = y^3 \text{ is } 2$$

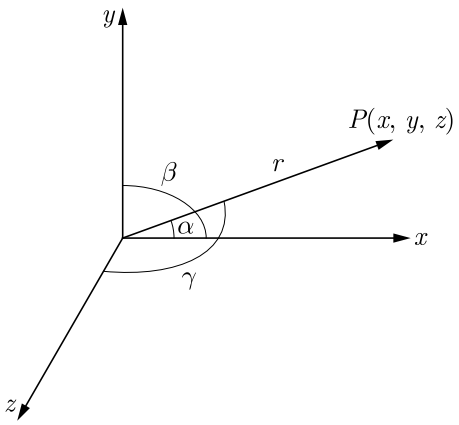
Thus (c) is correct option.

5. If l, m, n are the direction cosines of a straight line then

(a) $l^2 + m^2 - n^2 = 1$ (b) $l^2 - m^2 + n^2 = 1$

(c) $l^2 - m^2 - n^2 = 1$ (d) $l^2 + m^2 + n^2 = 1$

Sol :



From figure, we get

$$l = \cos\alpha = \frac{x}{r}$$

$$m = \cos\beta = \frac{y}{r}$$

$$n = \cos\gamma = \frac{z}{r}$$

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{x^2}{r^2} + \frac{y^2}{r^2} + \frac{z^2}{r^2}$$

$$= \frac{x^2 + y^2 + z^2}{r^2}$$

$$= \frac{r^2}{r^2} = 1$$

Hence, $l^2 + m^2 + n^2 = 1$

Thus (d) is correct option.

6. The direction ratios of the line joining the points (x_1, y_1, z_1) and (x_2, y_2, z_2) are

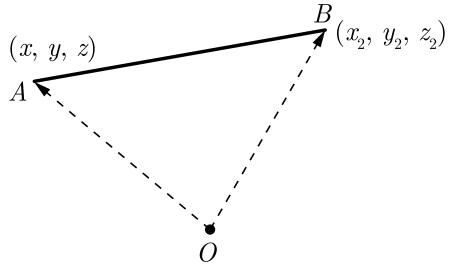
(a) $x_1 + x_2,\ y_1 + y_2,\ z_1 + z_2$

(b) $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

(c) $\frac{x_1 + x_2}{2},\ \frac{y_1 + y_2}{2},\ \frac{z_1 + z_2}{2}$

(d) $(x_2 - x_1),\ (y_2 - y_1),\ (z_2 - z_1)$

Sol :



$$\overrightarrow{OA} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$$

$$\overrightarrow{OB} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

$$= (x_2\hat{i} + y_2\hat{j} + z_2\hat{k}) - (x_1\hat{i} + y_1\hat{j} + z_1\hat{k})$$

$$= (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

Therefore, required Direction ratios are

$$= (x_2 - x_1, y_2 - y_1, z_2 - z_1)$$

Thus (d) is correct option.

7. $\frac{d}{dx}\left[\lim_{x \to a}\frac{x^5 - a^5}{x - a}\right] =$

(a) $5a^4$ (b) $5x^4$

(c) 1 (d) 0

Sol :

$$f(x) = \frac{d}{dx}\left[\lim_{x \to a}\frac{x^5 - a^5}{x - a}\right]$$

$$= \frac{d}{dx}[5 \cdot (a)^{5-1}] \quad \left[\lim_{x \to a}\frac{x^n - a^n}{x - a} = xa^{n-1}\right]$$

$$= \frac{d}{dx}(5a^4)$$

$$= 5\frac{d}{dx}(a^4)$$

$$= 5 \times 0 \quad \left[\frac{d}{dx}(\text{constant}) = 0\right]$$

$$= 0$$

Thus (d) is correct option.

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$$c = 2\pi r \qquad ...(1)$$

Differentiating Both sides of equation (1), we get

$$\frac{dc}{dt} = 2\pi\frac{dr}{dt}$$
$$= 2\pi(0.4)$$
$$= 0.8\pi \text{ cm/s}$$

Thus (b) is correct option.

19. A and B are two events

Assertion : If $P(\overline{A}) = 0.7$, $P(\overline{B}) = 0.5$ and $P(A \cup B) = 0.6$ then $P(A \cap B) = 0.2$

Reason : $P(A \cup B)+P(A \cap B)+P(\overline{A})+P(\overline{B})= 2.5$

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

Sol :

Given,

$$P(\overline{A}) = 0.7$$
$$P(A) = 1 - 0.7 \quad [P(A) + P(\overline{A}) = 1]$$
$$= 0.3$$

Similarly, $P(\overline{B}) = 0.5$

$$P(B) = 1 - 0.5 = 0.5$$
$$P(A \cup B) = 0.6$$

We know that,

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) \qquad ...(1)$$

Putting given values, we get

$$P(A \cap B) = 0.5 + 0.3 - 0.6$$
$$= 0.8 - 0.6 = 0.2$$

Hence statement-I is correct.

$$\text{L.H.S} = P(A \cup B) + P(A \cap B) + P(\overline{A}) + P(\overline{B})$$
$$= P(A) + P(B) - P(A \cap B) + P(A \cap B) + P(\overline{A}) + P(\overline{B})$$

Since $P(E) + P(\overline{E}) = 1$ where E is any event.

$$= 1 + 1$$
$$= 2$$

Thus (c) is correct option.

20. **Assertion :** If $y = x^3\cos x$, then $\frac{dy}{dx} = x^3\sin x + 3x^2\cos x$

Reason : $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$.

(a) Both assertion and reason are true and reason is the correct explanation of assertion.

(b) Both assertion and reason are true, but reason is not the correct explanation of assertion.

(c) Assertion is true, but reason is false.

(d) Assertion is false, but reason is true.

Sol :

$$y = x^3\cos x$$

Differentiating both sides with respect to x using product rule, we get

$$\frac{dy}{dx} = \frac{d}{dx}(x^3\cos x)$$
$$= \cos x\frac{d}{dx}x^3 + x^3\frac{d}{dx}\cos x$$
$$= 3x^2\cos x - \sin x \cdot x^3$$

product rule, $\frac{dy}{dx} = \frac{d}{dx}(uv)$

$$= u\frac{dv}{dx} + v\frac{du}{dx}$$

Statement I is wrong but reason is correct since coefficient of $\sin x$ should be -1.

Thus (d) is correct option.

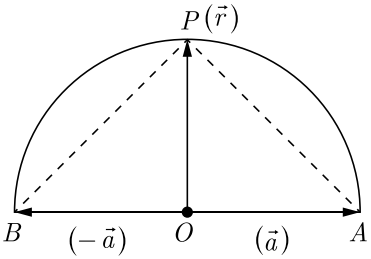
SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Prove, by Vector method, that the angle inscribed in a semi-circle is a right angle.

Sol :

Let O be the centre of semicircle PAB and position vector as follows



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Thus (a) is correct option.

14. $\tan^{-1}x+\cot^{-1}x = ?$

(a) 0 (b) 1

(c) $\frac{\pi}{2}$ (d) $-\frac{\pi}{2}$

Sol :

Let, $\tan^{-1}x = \theta$ ...(1)

$$x = \tan\theta$$

$$x = \cot\left(\frac{\pi}{2}-\theta\right)$$

$$\cot^{-1}x = \frac{\pi}{2}-\theta$$

from eq. (1)

$$\theta+\cot^{-1}x = \frac{\pi}{2}$$

$$\tan^{-1}x+\cot^{-1}x = \frac{\pi}{2}$$

Thus (c) is correct option.

15. Which of the following is the unit matrix of order 3×3?

(a) $\begin{bmatrix}1&0&0\\1&0&0\\1&0&0\end{bmatrix}$ (b) $\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$

(c) $\begin{bmatrix}0&0&1\\0&0&1\\0&0&1\end{bmatrix}$ (d) $\begin{bmatrix}0&1&0\\0&1&0\\0&1&0\end{bmatrix}$

Sol :

A Square matrix having 1 (one) on its Principle diagonal and O (zero) else where is called as unit matrix.

Thus (b) is correct option.

16. In the figure, point A has coordinates:

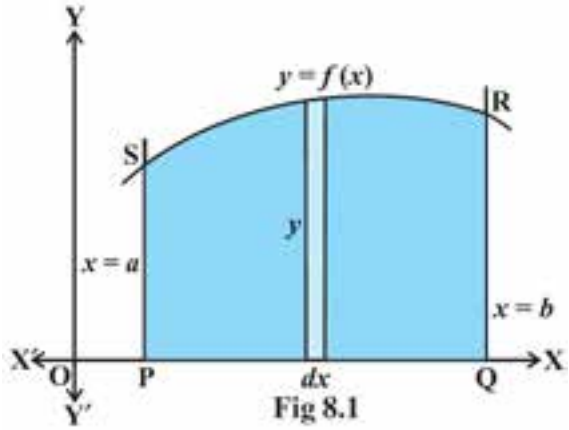


Fig 8.1

(a) (-1,0) (b) (-2/3,0)

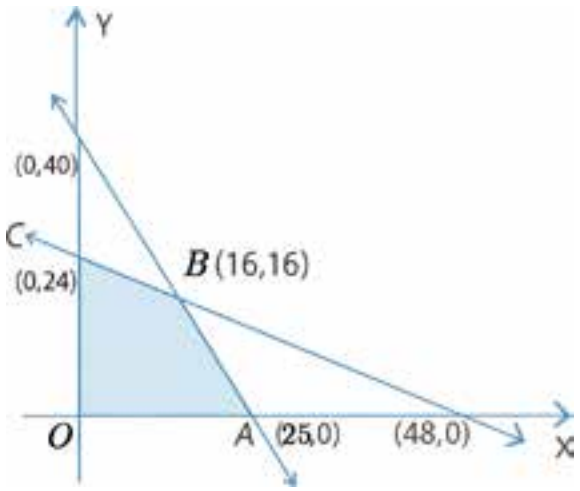
(c) (0,0) (d) (1,0)

Sol :

The figure clearly marks point A on the X-axis with the coordinate (-2/3,0). This is shown directly in the diagram, making option (b) the correct answer.

Thus (b) is correct option.

17. For the given feasible region in the figure, bounded by the lines shown, consider the objective function $z = 2x+y$. The feasible region includes the corner points $O(0,0), A(25,0), B(16,16)$, and $C(0,24)$. At the vertex $B(16,16)$, what is the value of z?



(a) 32 (b) 48

(c) 64 (d) 72

Sol :

To evaluate the value of the objective function at a specific point, we substitute the coordinates of that point into the given equation. At B(16,16), we calculate : $z = 2x+y = 2(16)+16 = 32+16 = 48$

Therefore, the value of the function at this vertex is 48. Since this is exactly equal to option (b), the correct answer is 48.

Thus (b) is correct option.

18. The radius of a circle is increasing at the rate of 0.4 cm/s. The rate of increase of its circumference is

(a) 0.4π cm/s (b) 0.8π cm/s

(c) 0.8 cm/s (d) None of these

Sol :

Given, $\frac{dr}{dt} = 0.4$ cm/s

where, r be the radius of the circle.

Circumference of circle,

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8. The direction cosines of the vector $3\hat{i}-4\hat{j}+12\hat{k}$ is

(a) $\frac{3}{13}, \frac{4}{13}, \frac{12}{13}$ (b) $\frac{3}{13}, \frac{-4}{13}, \frac{12}{13}$

(c) $\frac{3}{\sqrt{13}}, \frac{4}{\sqrt{13}}, \frac{12}{\sqrt{13}}$ (d) $\frac{3}{\sqrt{13}}, \frac{-4}{\sqrt{13}}, \frac{12}{\sqrt{13}}$

Sol :

We have $\vec{a} = 3\hat{i}-4\hat{j}+12\hat{k}$

$|\vec{a}| = \sqrt{3^2+4^2+12^2}$

$= \sqrt{9+16+144}$

$= \sqrt{169} = 13$ units

Thus direction Ratios $\left(\frac{3}{13}, \frac{-4}{13}, \frac{12}{13}\right)$

Thus (b) is correct option.

9. Of all the points of the feasible region, for maximum or minimum of objective functions, the point lies:

(a) inside the feasible region

(b) at the boundary line of the feasible region

(c) vertex point of the boundary of the feasible region

(d) none of the above

Sol :

Thus (c) is correct option.

10. The maximum value of $y = 2x^3 - 21x^2 + 36x - 20$ is

(a) -128 (b) -126

(c) -120 (d) None of these

Sol :

$\frac{dy}{dx} = 6x^2 - 42x + 36$

and $\frac{d^2y}{dx^2} = 12x - 42$

For Max. or min. $\frac{dy}{dx} = 0$

$6x^2 - 42x + 36 = 0 \Rightarrow x = 1, 6$

$\left(\frac{d^2y}{dx^2}\right)_{x=1} = -30\,(-x)$

and $\left(\frac{d^2y}{dx^2}\right)_{x=6} = 392\,(+ve)$

Minimum value at $x = 6$

$= 2 \times 6^3 - 21 \times 6^2 + 36 \times 6 - 20 = -128$

Thus (a) is correct option.

11. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^9 x\, dx = ?$

(a) -1 (b) 0

(c) 1 (d) $\frac{\pi}{2}$

Sol :

Let, $I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^9 x\, dx$

Let, $f(x) = \sin^9 x$

$f(-x) = \sin^9(-x) = [\sin(-x)]^9$

$= (-\sin x)^9 = -\sin^9 x$

$f(-x) = -f(x)$

Hence, $f(x)$ is odd function.

Thus (b) is correct option.

12. What type of a relation is "Less than" in the set of real numbers?

(a) only symmetric

(b) only transitive

(c) only reflexive

(d) equivalence relation

Sol :

Let A be the set of all real numbers, and R be the relation 'less than' i.e., $<$ on A, then R is

1. The relation R is not reflective, since a is not less than a for any natural number a.
2. The relation R is not symmetric, since if $(a, b) \in R$, then a is less than b but b is not less than a, i.e., $(b, a) \notin R$.
3. The relation R is transitive, since $a < b$ and $b < c \Rightarrow a < c$ i.e., $(a, b) \in R$, $(b, c) \in R \Rightarrow (a, c) \in R$.

Thus (b) is correct option.

13. $\int x^5\, dx =$ .....

(a) $\frac{x^6}{6} + k$ (b) $\frac{x^5}{5} + k$

(c) $\frac{x^7}{7} + k$ (d) $\frac{x^8}{8} + k$

Sol :

We know that,

$I = \int x^n\, dx = \frac{x^{n+1}}{n+1} + k$

Here, $n = 5$

Therefore, $I = \frac{x^{5+1}}{5+1} + k = \frac{x^6}{6} + k$

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$\overrightarrow{OA} = \vec{a}$

$\overrightarrow{OB} = -\vec{a}$ (since AB is a diameters of circle

$\overrightarrow{OP} = \vec{r}$

$|\overrightarrow{OA}| = |\overrightarrow{OB}| = |\overrightarrow{OP}| = |\vec{a}| = |\vec{r}|$

= Radius of a given circle ...(1)

$\overrightarrow{AP}$ = Position vector of $\overrightarrow{OP}$-position vector of $\overrightarrow{OA}$

$= \vec{r} - \vec{a}$

$\overrightarrow{BP}$ = Position vector of $\overrightarrow{OP}$-Position vector of $\overrightarrow{OB}$

$= \vec{r} + \vec{a}$

For $\angle BPA$ to be right angled Δ

$\overrightarrow{AP} \cdot \overrightarrow{BP} = 0$

$\overrightarrow{AP} \cdot \overrightarrow{BP} = (\vec{r} - \vec{a}) \cdot (\vec{r} + \vec{a})$

$= \vec{r} \cdot \vec{r} - \vec{a} \cdot \vec{r} + \vec{r} \cdot \vec{a} - \vec{a} \cdot \vec{a}$

$= |\vec{r}|^2 - |\vec{a}|^2$ $\left[\begin{matrix}\vec{a} \cdot \vec{r} = \vec{r} \cdot \vec{a} \\ \vec{r} \cdot \vec{r} = |\vec{r}|^2\end{matrix}\right]$

$\overrightarrow{AP} \cdot \overrightarrow{BP} = 0$ [Using eq. (1)]

Therefore angle inscribed in a semicircle is right angled triangle.

22. If $y = \sqrt{x + \sqrt{x + \sqrt{x + ...\text{to}\,\infty}}}$ then $\frac{dy}{dx}$

OR

If $y = \tan(\sin^{-1}x)$ then find $\frac{dy}{dx}$

Sol :

$$y = \sqrt{x + \underbrace{\sqrt{x + \sqrt{x + ..\text{to}\ \infty}}}_{y}}$$

$$y = \sqrt{x + y}$$

Squaring both sides

$$y^2 = x + y$$

Differentiating both sides

$$2y\frac{dy}{dx} = 1 + \frac{dy}{dx}$$

$$(2y - 1)\frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{2y - 1}$$

OR

$$y = \tan(\sin^{-1}x)$$

Using chain Rule

$$\left[y = f(u) \text{ and } u = g(x) \text{ then} \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}\right]$$

$$\frac{dy}{dx} = \sec^2(\sin^{-1}x) \times \frac{d}{dx}(\sin^{-1}x)$$

$$= \frac{\sec^2 \sin^{-1}x}{\sqrt{1 - x^2}} \quad \left[\begin{matrix}\frac{d}{dx}\tan x = \sec^2 x \\ \frac{d}{dx}\sin^{-1}x = \frac{1}{\sqrt{1 - x^2}}\end{matrix}\right]$$

23. Prove by direction numbers, that the point $(1, -1, 3)$, $(2, -4, 5)$ and $(5, -13, 11)$ are in a straight line.

Sol :

Let three given point be $A(1, -1, 3)$, $B(2, -4, 5)$ $C(5, -13, 11)$

Direction ratio of line joining AB

$= (2 - 1, -4 + 1, 5 - 3)$

$= (1, -3, 2)$

Direction ratio of line joining BC

$= (5 - 2, -13 + 4, 11 - 5) = (3, -9, 6)$

Since direction ratio of AB and BC are proportional

$AB \parallel BC$

But since B is common point between AB and BC

A, B, C are three collinear point.

A, B, C forms a straight line (hence proved)

24. Odds are 8 : 5 against a man, who is 55 years old, living till he is 75 and 4 : 3 against his wife who is now 48, living till she is 68. Find the probability that the couple will be alive 20 years hence.

Sol :

Odd against event $E = \frac{P(E')}{P(E)}$

Let E_1 be event that man will live then

$\frac{P(E_1')}{P(E_1)} = \frac{8}{5}$ (given)

$P(E_1) = \frac{5}{8 + 5} = \frac{5}{13}$ ...(1)

$[P(E') + P(E) = 1]$

Let E_2 be event that his wife will live then

$\frac{P(E_2')}{P(E_2)} = \frac{4}{3}$ (given)

$P(E_2) = \frac{3}{4 + 3} = \frac{3}{7}$ ...(2)

$[P(E_2') + P(E_2) = 1]$

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Since existence of man and his wife is independent of each other.

Therefore E_1 and E_2 are independent event.

Required Probability $= P(E_1 \cap E_2)$

Since Events are independent

$$P(E_1 \cap E_2) = P(E_1) \times P(E_2)$$

$$= \frac{5}{13} \times \frac{3}{7} = \frac{15}{91} \qquad \text{(using 1 and 2)}$$

25. Show that the function $y = ae^{\tan^{-1}x}$ is a solution of the differential equation

$$(1+x^2)\frac{d^2y}{dx^2} + (2x-1)\frac{dy}{dx} = 0$$

OR

Show that the function $y = ax + \frac{b}{a}$ is a solution of the differential equation

$$y = x\frac{dy}{dx} + \frac{b}{\frac{dy}{dx}}$$

Sol :

Given function is $y = ae^{\tan^{-1}x}$

$$\frac{y}{e^{\tan^{-1}x}} = a$$

Differentiating both sides w.r.t. x, we get

$$\frac{(e^{\tan^{-1}x})\left(\frac{dy}{dx}\right) - (y)\left(e^{\tan^{-1}x} \cdot \frac{1}{1+x^2}\right)}{(e^{\tan^{-1}x})^2} = 0$$

$$e^{\tan^{-1}x}\left[\frac{dy}{dx} - \frac{y}{1+x^2}\right] = 0$$

$$\frac{dy}{dx} - \frac{y}{1+x^2} = 0$$

$$\frac{dy}{dx} = \frac{y}{1+x^2}$$

$$(1+x^2)\frac{dy}{dx} = y$$

Differentiating both sides w.r.t. x, we get

$$(1+x^2)\frac{d^2y}{dx^2} + \frac{dy}{dx}(2x) = \frac{dy}{dx}$$

$$(1+x^2)\frac{d^2y}{dx^2} + (2x-1)\frac{dy}{dx} = 0$$

which is the given differential equation.

Hence, given function is a solution of the given differential equation.

OR

Given function is $y = ax + \frac{b}{a}$ ...(1)

Differentiating both sides w.r.t. x, we get

$$\frac{dy}{dx} = a$$

Substituting value of a in (1), we get

$$y = x\frac{dy}{dx} + \frac{b}{\frac{dy}{dx}}$$

which is the given differential equation. Hence, given function is a solution of the given differential equation.

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. If $\vec{a} = \hat{i} - 2\hat{j} - 3\hat{k}$; $\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{c} = \hat{i} + 3\hat{j} - 2\hat{k}$ then find $\vec{a} \times (\vec{b} \times \vec{c})$.

Sol :

Given, $\vec{a} = \hat{i} - 2\hat{j} - 3\hat{k}$

$$\vec{b} = 2\hat{i} + \hat{j} - \hat{k}$$

$$\vec{c} = \hat{i} + 3\hat{j} - 2\hat{k}$$

$$\vec{b} \times \vec{c} = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 3 & -2 \end{bmatrix}$$

$$= (-2+3)\hat{i} - \hat{j}(-4+1) + \hat{k}(6-1)$$

Thus, $\vec{b} \times \vec{c} = 1\hat{i} + 3\hat{j} + 5\hat{k}$

Let, $\vec{b} \times \vec{c} = \vec{p}$

$$\vec{a} \times \vec{p} = \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & -3 \\ 1 & 3 & 5 \end{bmatrix}$$

$$= \hat{i}(-10+9) - \hat{j}(5+3) + \hat{k}(3+2)$$

$$\vec{a} \times \vec{p} = -\hat{i} - 8\hat{j} + 5\hat{k}$$

27. Thus, $\vec{a} \times (\vec{b} \times \vec{c}) = -(\hat{i} + 8\hat{j} - 5\hat{k})$ If $f : R \to R$ is defined by $f(x) = x^2 - 3x + 2$, find $f\{f(x)\}$.

Sol :

Given, $f : R \to R$

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$$f(x) = x^2 - 3x + 2 \quad ...(1)$$

Replacing $x \to f(x)$

$$f\{f(x)\} = \{(f(x))^2 - 3(f(x)\} + 2$$

Putting value of $f(x)$ using equation (1), we get

$$\begin{aligned} f\{f(x)\} &= (x^2 - 3x + 2)^2 - 3(x^2 - 3x + 2) + 2 \\ &= x^4 + 9x^2 + 4 - 6x^3 + 4x^2 - 12x \\ &\quad - 3x^2 + 9x - 6 + 2 \\ &= x^4 - 6x^3 + 10x^2 - 3x \end{aligned}$$

$$\left[(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca\right]$$

$$= x(x^3 - 6x^2 + 10x - 3)$$

28. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then find the value of $A^2 + 3A + 2I$.

OR

Find the values of X and $Y : X + Y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix}$ and $X - Y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$.

Sol :

Given: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Then, $A^2 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$$= \begin{bmatrix} 1+6 & 2+8 \\ 3+12 & 6+16 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} \quad ...(1)$$

$$3A = 3\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix} \quad ...(2)$$

$$2I = 2\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \quad ...(3)$$

Adding (1)+(2)+(3), we get

$$\begin{aligned} A^2 + 3A + 2I &= \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} + \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 7+3+2 & 10+6+0 \\ 15+9+0 & 22+12+2 \end{bmatrix} \\ &= \begin{bmatrix} 12 & 16 \\ 24 & 36 \end{bmatrix} \end{aligned}$$

OR

$$X + Y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} \quad ....(1)$$

$$X - Y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \quad ....(2)$$

After adding equation(1) and (2) we get,

$$\begin{aligned} 2X &= \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 7+3 & 0+0 \\ 2+0 & 5+3 \end{bmatrix} = \begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix} \\ X &= \frac{1}{2}\begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix} \end{aligned}$$

After subtraction equation (2), from equation (1) we get,

$$\begin{aligned} 2Y &= \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 7-3 & 0-0 \\ 2-0 & 5-3 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 2 & 2 \end{bmatrix} \\ Y &= \frac{1}{2}\begin{bmatrix} 4 & 0 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix} \end{aligned}$$

Therefore, $X = \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}$ and $Y = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$

29. If $y = \log \tan\left(\frac{\pi}{4} + \frac{x}{2}\right)$, show that $\frac{dy}{dx} - \sec x = 0$.

Sol :

We have $y = \log \tan\left(\frac{\pi}{4} + \frac{x}{2}\right)$

Differentiating both sides with respect to x we get

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}\left(\log \tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right) \\ &= \frac{1}{\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)} \times \frac{d}{dx}\tan\left(\frac{\pi}{4} + \frac{x}{2}\right) \left[\frac{d}{dx}\log x = \frac{1}{x}\right] \\ &= \frac{1}{\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)} \times \sec^2\left(\frac{\pi}{4} + \frac{x}{2}\right) \times \frac{1}{2} \\ &= \frac{\cos\left(\frac{\pi}{4} + \frac{x}{2}\right)}{\sin\left(\frac{\pi}{4} + \frac{x}{2}\right)} \times \frac{1}{\cos^2\left(\frac{\pi}{4} + \frac{x}{2}\right)} \times \frac{1}{2} \end{aligned}$$

Multiplying and dividing by 2 we get

$$\frac{dy}{dx} = \frac{2}{2\sin\left(\frac{\pi}{4} + \frac{x}{2}\right)\cos\left(\frac{\pi}{4} + \frac{x}{2}\right)} \times \frac{1}{2}$$

We know that

$$2\sin\theta\cos\theta = \sin 2\theta$$

Therefore we get

$$\frac{dy}{dx} = \frac{1}{\sin\left(\frac{\pi}{2} + x\right)}$$

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$$= \frac{1}{9}\sqrt{9x^2+25} + C_1$$

Now, $I_2 = \int \frac{1}{\sqrt{9x^2+25}}\,dx$

$$= \frac{1}{3}\int \frac{1}{\sqrt{x^2+\left(\frac{5}{3}\right)^2}}\,dx$$

$$= \frac{1}{3}\log\left|x+\sqrt{x^2+\left(\frac{5}{3}\right)^2}\right| + C_2$$

$$= \frac{1}{3}\log\left|x+\sqrt{x^2+\frac{25}{9}}\right| + C_2$$

Putting values of I_1 and I_2 in (1), we get

$$I = \frac{5}{9}\sqrt{9x^2+25} + \frac{11}{3}\log\left|x+\sqrt{x^2+\frac{25}{9}}\right| + C$$

OR

We have $I = \int_0^{\frac{\pi}{2}} \log(\tan\theta + \cot\theta)\,d\theta$

$$= \int_0^{\frac{\pi}{2}} \log\left(\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}\right)d\theta$$

$$\left[\tan\theta = \frac{\sin\theta}{\cos\theta} \text{ and } \cot\theta = \frac{\cos\theta}{\sin\theta}\right]$$

$$= \int_0^{\frac{\pi}{2}} \log\left(\frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta}\right)d\theta$$

$$= \int_0^{\frac{\pi}{2}} \log\left(\frac{1}{\sin\theta\cos\theta}\right)d\theta$$

$$= \int_0^{\frac{\pi}{2}} -\log(\sin\theta\cos\theta)\,d\theta$$

$$\left[\log\frac{1}{a} = -\log a\right]$$

$$= -\left[\int_0^{\frac{\pi}{2}} \text{logsin}\,\theta\,d\theta + \int_0^{\frac{\pi}{2}} \text{logcos}\,\theta\,d\theta\right] \quad ...(1)$$

$$[\log xy = \log x + \log y]$$

Let, $\int_0^{\frac{\pi}{2}} \log\sin\theta\,d\theta = I_1$

and $\int_0^{\frac{\pi}{2}} \log\cos\theta\,d\theta = I_2$

$$\int_0^{\frac{\pi}{2}} \log\sin\theta\,d\theta = \int_0^{\frac{\pi}{2}} \text{logsin}\left(\frac{\pi}{2}-\theta\right)d\theta$$

$$= \int_0^{\frac{\pi}{2}} \text{logcos}\,\theta\,d\theta$$

$$\left[\int_a^b \log x\,dx = \int_a^b \log(a+b-x)\,dx\right]$$

$$I_1 = I_2$$

Using Property

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

$$I_1 = \int_0^{\frac{\pi}{2}} \text{logsin}\,\theta\,dx\theta$$

$$= \int_0^{\frac{\pi}{2}} \text{logsin}\left(\frac{\pi}{2}-\theta\right)d\theta$$

$$= \int_0^{\frac{\pi}{2}} \text{logcos}\,\theta\,d\theta$$

$$2I_1 = \int_0^{\frac{\pi}{2}} \text{logsin}\,\theta\,d\theta + \int_0^{\frac{\pi}{2}} \text{logcos}\,\theta\,d\theta$$

$$= \int_0^{\frac{\pi}{2}} \text{logsin}\,\theta\cos\theta\,d\theta$$

$$[\log x + \log y = \log xy]$$

Multiplying and dividing by 2

$$2I_1 = \int_0^{\frac{\pi}{2}} \log\left(\frac{2\sin\theta\cos\theta}{2}\right)d\theta$$

$$= \int_0^{\frac{\pi}{2}} \log\left(\frac{\sin 2\theta}{2}\right)d\theta$$

$$[2\sin\theta\cos\theta = \sin 2\theta]$$

$$= \int_0^{\frac{\pi}{2}} \text{logsin}\,2\theta\,d\theta - \int_0^{\frac{\pi}{2}} \log 2\,d\theta$$

$$= \int_0^{\frac{\pi}{2}} \text{logsin}\,2\theta\,d\theta - \left(\frac{\pi}{2}-0\right)\log 2$$

$$2I_1 = \int_0^{\frac{\pi}{2}} \log\sin 2\theta\,d\theta - \frac{\pi}{2}\log 2 \quad ...(2)$$

Let $I_3 = \int_0^{\frac{\pi}{2}} \log\sin 2\theta\,d\theta$

Substituting $t = 2\theta \Rightarrow dt = 2\,d\theta$ we have

$$I_3 = \frac{1}{2}\int_0^{\pi} \log\sin t\,dt \qquad \left[\theta = 0 \Rightarrow t = 0,\ \theta = \frac{\pi}{2} \Rightarrow t = \pi\right]$$

Using property

$$\int_0^{2a} f(x)\,dx = 2\int_0^a f(a-x) \text{ if } f(2a-x) = f(x)$$

$$I_3 = \frac{1}{2}\int_0^{\pi} \log\sin t\,dt$$

$$= \int_0^{\frac{\pi}{2}} \sin t\,dt = I_1$$

Therefore (1) becomes

$$2I_1 = I_1 - \frac{\pi}{2}\log 2$$

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Therefore, $\tan\alpha = \frac{1}{3} \Rightarrow \alpha = \tan^{-1}\frac{1}{3}$

$\tan\beta = \frac{1}{2} \Rightarrow \beta = \tan^{-1}\frac{1}{2}$

Therefore, $f = 4(\alpha+\beta)$

$= 4\left(\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{2}\right)$

$= 4\left(\tan^{-1}\frac{\frac{1}{3}+\frac{1}{2}}{1-\frac{1}{3}\times\frac{1}{2}}\right)$

$\left[\tan^{-1}A+\tan^{-1}B = \tan^{-1}\left(\frac{A+B}{1-AB}\right)\right]$

$= 4\left(\tan^{-1}\frac{\frac{5}{6}}{\frac{6-1}{5}}\right)$

$= 4\times\tan^{-1}1$

$= 4\times\tan^{-1}\left(\tan\frac{\pi}{4}\right)$

$= 4\times\frac{\pi}{4} = \pi$ Hence proved.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Find the particular solution of the differential equation $\frac{dy}{dx} = \frac{xy}{x^2+y^2}$ given that $y=1$, when $x=0$.

Sol :

We have $\frac{dy}{dx} = \frac{xy}{x^2+y^2}$

$= \frac{\frac{y}{x}}{1+\frac{y^2}{x^2}}$ ...(1)

Which is a homogeneous differential equation as

$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$

Substituting $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ in eq (1) we have

$v + x\frac{dv}{dx} = \frac{v}{1+v^2}$

$x\frac{dv}{dx} = \frac{v}{1+v^2} - v$

$x\frac{dv}{dx} = \frac{v-v-v^3}{1+v^2}$

$x\frac{dv}{dx} = -\frac{v^3}{1+v^2}$

$\frac{1+v^2}{v^3}dv = -\frac{dx}{x}$

Integrating both sides, we have

$\int\left(\frac{1}{v^3}+\frac{1}{v}\right)dv = -\int\frac{dx}{x}$

$-\frac{1}{2v^2}+\log|v| = -\log|x|+C$

$-\frac{x^2}{2y^2}+\log\left|\frac{y}{x}\right| = -\log|x|+C$

$-\frac{x^2}{2y^2}+\log|y|-\log|x| = -\log|x|+C$

$-\frac{x^2}{2y^2}+\log|y| = C$ ...(2)

Also, it is given that $y=1$, when $x=0$.

From Eq. (2), we have

$0+\log|1| = C \Rightarrow C=0$

Substituting $C=0$ in Eq. (2), we get

$-\frac{x^2}{2y^2}+\log|y| = 0$

$\log|y| = \frac{x^2}{2y^2}$

$y = e^{\frac{x^2}{2y^2}}$

Which is the required solution.

33. Evaluate $\int\frac{5x+11}{\sqrt{9x^2+25}}dx$

OR

Prove that : $\int_0^{\pi/2}\log(\tan\theta+\cot\theta)\,d\theta = \pi\log 2$

Sol :

Let $I = \int\frac{5x+11}{\sqrt{9x^2+25}}dx$

$= 5\int\frac{x}{\sqrt{9x^2+25}}dx + 11\int\frac{1}{\sqrt{9x^2+25}}dx$

or $I = 5I_1 + 11I_2$

Now, $I_1 = \int\frac{x}{\sqrt{9x^2+25}}dx$

Putting $\sqrt{9x^2+25} = y$ we have

$\frac{x}{\sqrt{9x^2+25}}dx = \frac{1}{9}dy$

Thus $I_1 = \frac{1}{9}\int 1\,dy$

$= \frac{1}{9}(y)+C_1$

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$$\ln\frac{t-2}{t} = 2\ln x + 2\ln k$$
$$= \ln x^2 + \ln k^2$$
$$= \ln k^2 x^2$$

Thus, $\ln\frac{t-2}{t} = \ln k^2 x^2$

Therefore, $\frac{t-2}{t} = k^2 x^2$

Substituting, $t = y/x$

$$\frac{y/x - 2}{y/x} = k^2 x^2$$
$$\frac{y-2x}{y} = k^2 x^2$$
$$\frac{y-2x}{y} = cx^2 \qquad (k^2 = c)$$

where c is constant of integrating

Formula used:

$$\int \frac{dx}{x} = \ln x + c$$
$$\log x + \log y = \log xy$$
$$a\log x = \log x^a$$

SECTION - E

Case study based questions are compulsory.

36. Pastry is a dough of flour, water and shortening that may be savoury or sweetened. Sweetened pastries are often described as bakers' confectionery. The word "pastries" suggests many kinds of baked products made from ingredients such as flour, sugar, milk, butter, shortening, baking powder, and eggs.



The Sunrise Bakery Pvt Ltd produces three basic pastry mixes A, B and C. In the past the mix of ingredients has shown in the following matrix:

$$\begin{array}{cc} & \begin{matrix}\text{Flour} & \text{Fat} & \text{Sugar}\end{matrix} \\ \text{Type}\begin{matrix}A\\B\\C\end{matrix} & \begin{bmatrix}5 & 1 & 1\\ 6.5 & 2.5 & 0.5\\ 4.5 & 3 & 2\end{bmatrix}\end{array} \quad \text{(All quantities in kg)}$$

Due to changes in the consumer's tastes it has been decided to change the mixes using the following amendment matrix:

$$\begin{array}{cc} & \begin{matrix}\text{Flour} & \text{Fat} & \text{Sugar}\end{matrix} \\ \text{Type}\begin{matrix}A\\B\\C\end{matrix} & \begin{bmatrix}0 & 1 & 0\\ -0.5 & 0.5 & 0.5\\ 0.5 & 0 & 0\end{bmatrix}\end{array}$$

Using matrix algebra you are required to calculate:

(i) The matrix for the new mix:

(ii) The production requirement to meet an order for 50 units of type A, 30 units of type B and 20 units of type C of the new mix;

(iii) The amount of each type that must be made to totally use up 370 kg of flour, 170 kg of fat and 80 kg of sugar that are at present in the stores.

Sol :

(i) Matrix for the new mix : The matrix for the new mix is obtained by adding the amendment matrix to the original mix matrix, i.e.,

$$\begin{bmatrix}6 & 1 & 1\\ 6.5 & 2.5 & 0.5\\ 4.5 & 3 & 2\end{bmatrix} + \begin{bmatrix}0 & 1 & 0\\ -0.5 & 0.5 & 0.5\\ 0.5 & 0 & 0\end{bmatrix} = \begin{bmatrix}5 & 2 & 1\\ 6 & 3 & 1\\ 5 & 3 & 2\end{bmatrix}$$

(ii) Production requirement : The production requirement to meet an order for 50 units of type A, 30 units of type B and 20 units of type C of the new mix is given by the matrix product

$$\begin{bmatrix}50 & 30 & 20\end{bmatrix}\begin{bmatrix}5 & 2 & 1\\ 6 & 3 & 1\\ 5 & 3 & 2\end{bmatrix} = \begin{bmatrix}530 & 250 & 120\end{bmatrix}$$

Hence, the production requirement to meet a given order of the new mix is 530 kg of flour, 250 kg of fat, 120 kg of sugar.

(iii) Amount of Each Type :

Let x_1, x_2 and x_3 be the amount of mixes of type A, B and C respectively. Then, we have the following set of simultaneous equations:

$$5x_1 + 6x_2 + 5x_3 = 370$$
$$2x_1 + 3x_2 + 3x_3 = 170$$
$$x_1 + x_2 + 2x_3 = 80$$

The above system of equations can be written in matrix form as:

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$$I_1 = -\frac{\pi}{2}\log 2$$

Putting values in eq. (1) we get

$$I = -\left[-\frac{\pi}{2}\log 2 - \frac{\pi}{2}\log 2\right]$$

$$= -[-\pi \log 2]$$

Thus, $I = \pi \log 2$

34. Solve the following L.P.P. graphically:

Minimise and Maximise

$$Z = 3x + 5y$$

Subject to the constraints:

$$3x - 4y + 12 \geq 0$$

$$2x - y + 2 \geq 0$$

$$2x + 3y - 12 \geq 0$$

$$x \leq 4$$

$$y \geq 2$$

$$x \geq 0$$

Sol :

The objective function is

$$z = 3x + 5y$$

The constraints are:

$$3x - 4y + 12 \geq 0 \quad ...(1)$$

$$2x - y + 2 \geq 0 \quad ...(2)$$

$$2x + 3y - 12 \geq 0 \quad ...(3)$$

$$x \leq 4 \quad ...(4)$$

$$y \geq 2 \quad ...(5)$$

$$x \geq 0 \quad ...(6)$$

Step I : The line corresponding to constraint Eq. (1) is,

$$3x - 4y + 12 = 0 \quad ...(7)$$

Putting $x = 0$ in Eq. (7), we get

$$y = 3$$

and Putting $y = 0$ in Eq. (7), we get

$$x = -4$$

Hence, This line passes through the points (0, 3) and $(-4, 0)$.

Test of origin:

Putting $x = 0$ and $y = 0$ in Eq. (1), we get

$$0 + 12 \geq 0 \text{ which is true}$$

Hence, The closed half plane containing the origin is the region of constraint Eq. (1). The line corresponding to constraint Eq. (2) is,

$$2x - y + 2 = 0 \quad ...(8)$$

Putting $x = 0$ in Eq. (8), we get

$$y = 2$$

and Putting $y = 0$ in Eq. (8), we get

$$x = -1$$

Hence, This line passes through the point (0, 2) and $(-1, 0)$.

Test of origin:

Putting $x = 0$ and $y = 0$ in Eq. (2), we get

$$0 + 2 \geq 0 \text{ which is true}$$

Hence, The closed half plane containing the origin is the region of constraint Eq. (2). The line corresponding to constraint Eq. (3) is

$$2x + 3y - 12 = 0 \quad ...(9)$$

Putting $x = 0$ in Eq. (9), we get

$$y = 4$$

and Putting $y = 0$ in Eq. (9), we get

$$x = 6$$

Hence, This line passes through the point (0, 4) and (6, 0).

Test of origin:

Putting $x = 0$ and $y = 0$ in Eq. (3), we get

$$0 - 12 \geq 0 \text{ which is not true}$$

Hence, The closed half-plane not containing the origin is the region of constraint Eq. (3).

Also $x \leq 4$ represents the closed half plane on the left of the vertical line $x = 4$ and $y \geq 2$ represents the closed half plane on the right of y-axis i.e., first and 4th quadrants.

The shaded regions is the feasible region of the given L.P.P. and is bounded as shown in figure.

Step II : Corner points of the shaded feasible region are A, B, C, D and E.

To find corner point A, let us solve

$$y = 2$$

and Eq. (9) i.e.,

$$2x + 3y - 12 = 0$$

Putting $y = 2$ in Eq. (9), we get

$$2x + 3(2) - 12 = 0$$

$$2x = 6 \text{ i.e.,}$$

$$x = 3$$

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$$= \frac{1}{\cos x} = \sec x$$

Hence, $\frac{dy}{dx} - \sec x = 0$ Hence proved

30. The volume of a cube is increasing at a rate of 9 cubic centimetres per second. How fast is the surface area increasing when the length of an edge is 10 cm long?

Sol :

Let x be the edge of a cube and V, the volume of the cube at time t, then $V = x^3$.

The rate of change of volume V wrt time

$$\frac{dV}{dt} = \frac{d}{dt}(x^3)$$

$$= 3x^2 \frac{dx}{dt}$$

Also $\frac{dV}{dt} = 9 \text{ cm}^3/\text{sec}$ (Given)

$$3x^2 \frac{dx}{dt} = 9$$

$$\frac{dx}{dt} = \frac{9}{3x^2}$$

$$= \frac{3}{x^2}$$

Let S be the surface area of the cube. Then

$$S = 6x^2$$

$$\frac{dS}{dt} = \frac{d}{dt}(6x^2)$$

$$= \frac{d}{dx}(6x^2) \cdot \frac{dx}{dt}$$

$$= 12x \times \frac{3}{x^2}$$

$$= \frac{36}{x} \qquad \left[\frac{dx}{dt} = \frac{3}{x^2}\right]$$

Thus $\left(\frac{dS}{dt}\right)_{x=10\text{ cm}} = \frac{36}{10} = 3.6$

Hence, the rate of increase in surface area is 3.6 cm^2/sec

31. Prove that:

$\tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1}\frac{a}{b}\right) + \tan\left(\frac{\pi}{4} - \frac{1}{2}\cos^{-1}\frac{a}{b}\right) = \frac{2b}{a}$

OR

Prove that $4(\cot^{-1}3 + \text{cosec}^{-1}\sqrt{5}) = \pi$.

Sol :

Given :

$\tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1}\frac{a}{b}\right) + \tan\left(\frac{\pi}{4} - \frac{1}{2}\cos^{-1}\frac{a}{b}\right) = \frac{2b}{a}$

Let $\frac{1}{2}\cos^{-1}\frac{a}{b} = \theta$

or, $\cos^{-1}\frac{a}{b} = 2\theta$

or, $\frac{a}{b} = \cos 2\theta$

$$\text{L.H.S.} = \tan\left(\frac{\pi}{4} + \theta\right) + \tan\left(\frac{\pi}{4} - \theta\right)$$

$$= \frac{\tan\frac{\pi}{4} + \tan\theta}{\tan\frac{\pi}{4} - \tan\theta} + \frac{\tan\frac{\pi}{4} - \tan\theta}{\tan\frac{\pi}{4} + \tan\theta}$$

$$= \frac{1 + \tan\theta}{1 - \tan\theta} + \frac{1 - \tan\theta}{1 + \tan\theta}$$

$$= \frac{(1 + \tan\theta)^2 + (1 - \tan\theta)^2}{(1 - \tan\theta)(1 + \tan\theta)}$$

$$= \frac{1 + \tan^2\theta + 2\tan\theta + 1 + \tan^2\theta - 2\tan\theta}{1 - \tan^2\theta}$$

$$\left[\begin{matrix}(a+b)^2 = a^2 + b^2 + 2ab;\\ (a+b)(a-b) = a^2 - b^2\end{matrix}\right]$$

$$= \frac{2(1 + \tan^2\theta)}{(1 - \tan^2\theta)}$$

$$= \frac{2}{\cos 2\theta} \qquad \left[\frac{1 - \tan^2\theta}{1 + \tan^2\theta} = \cos 2\theta\right]$$

$$= \frac{2}{\frac{a}{b}}$$

$$= \frac{2b}{a} = \text{R.H.S.}$$

Since, L.H.S. = R.H.S. Hence Proved

OR

Let, $f = 4(\cot^{-1}3 + \text{cosec}^{-1}\sqrt{5})$

and $\cot^{-1}3 = \alpha$

or $\cot\alpha = 3$

$$\tan\alpha = \frac{1}{3}$$

and $\text{cosec}^{-1}\sqrt{5} = \beta$

or $\text{cosec}\,\beta = \sqrt{5}$

$$\tan\beta = \frac{1}{\cot\beta}$$

$$= \frac{1}{\sqrt{\text{cosec}^2\beta - 1}}$$

$$= \frac{1}{\sqrt{5-1}} = \frac{1}{2}$$

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Substituting values of y and $\frac{dy}{dx}$ from (2) and (3) in equation (1) we get

$$V + x\frac{dV}{dx} = \frac{x + Vx}{x - Vx}$$

$$= \frac{x(1+V)}{x(1-V)} = \frac{1+V}{1-V}$$

$$x\frac{dV}{dx} = \frac{1+V}{1-V} - V$$

$$= \frac{1+V-V(1-V)}{1-V}$$

$$= \frac{1+V-V+V^2}{1-V}$$

Thus, $$x\frac{dV}{dx} = \frac{1+V^2}{1-V}$$

Separating variables x and V in L.H.S and R.H.S, we get

$$\left(\frac{1-V}{1+V^2}\right)dV = \frac{dx}{x}$$

$$\frac{dV}{1+V^2} - \frac{VdV}{1+V^2} = \frac{dx}{x}$$

$$\frac{dV}{1+V^2} - \frac{2V\,dV}{2(1+V^2)} = \frac{dx}{x}$$

Integrating both sides

$$\int \frac{dV}{1+V^2} - \frac{1}{2}\int \frac{2VdV}{(1+V^2)} = \int \frac{dx}{x}$$

$$\tan^{-1}V - \frac{1}{2}\ln(1+V^2) = \ln x + \ln k$$

$$\tan^{-1}V = \ln x + \ln k + \frac{1}{2}\ln\,(1+V^2)$$

$$= \ln kx \times (1+V^2)^{1/2}$$

Substituting, $V = \frac{y}{x}$, we get,

$$\tan^{-1}\frac{y}{x} = \ln kx\sqrt{1+\frac{y^2}{x^2}}$$

$$= \ln kx\sqrt{\frac{x^2+y^2}{x}}$$

$$= \ln k\sqrt{x^2+y^2}$$

$$\tan^{-1}\frac{y}{x} - \ln k\sqrt{x^2+y^2} = 0$$

$$\tan^{-1}\frac{y}{x} - \ln k - \ln\sqrt{x^2+y^2} = 0$$

$$\tan^{-1}\frac{y}{x} - \ln\sqrt{x^2+y^2} = \ln k$$

$$\tan^{-1}\frac{y}{x} - \ln\sqrt{x^2+y^2} = C \qquad (\ln k = c)$$

Formula used:

$$\int \frac{1}{1+x^2} = \tan^{-1}x + c$$

$$\ln x + \ln y = \ln xy$$

$$\int \frac{f^1(x)}{f(x)}dn = \ln f(x) + c$$

OR

We have $$\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2} \qquad ....(1)$$

Let, $$\frac{y}{x} = t \qquad ....(2)$$

$$y = xt$$

Differentiating both sides with respect to x we get

$$\frac{dy}{dx} = \frac{d}{dx}(xt)$$

$$= t\frac{d}{dx}x + x\frac{dt}{dx}$$

Thus, $$\frac{dy}{dx} = t + x\frac{dt}{dx} \qquad ....(2)$$

Substituting values from (2) and (3) in equation (1) we get,

$$t + x\frac{dt}{dx} + t = t^2$$

$$x\frac{dt}{dx} = t^2 - 2t$$

Separating t and x on L.H.S. and R.H.S. sides we get

$$\frac{dt}{t^2-2t} = \frac{dx}{x}$$

$$\frac{dt}{t(t-2)} = \frac{dx}{x}$$

Multiplying and dividing by 2 on L.H.S.

$$\frac{2dt}{2t(t-2)} = \frac{dx}{x} \qquad ....(4)$$

$$2 = t - (t-2)$$

Substituting in equation (4) we get

$$\frac{t-(t-2)}{2t(t-2)}dt = \frac{dn}{x}$$

$$\frac{dt}{2(t-2)} - \frac{dt}{2t} = \frac{dx}{x}$$

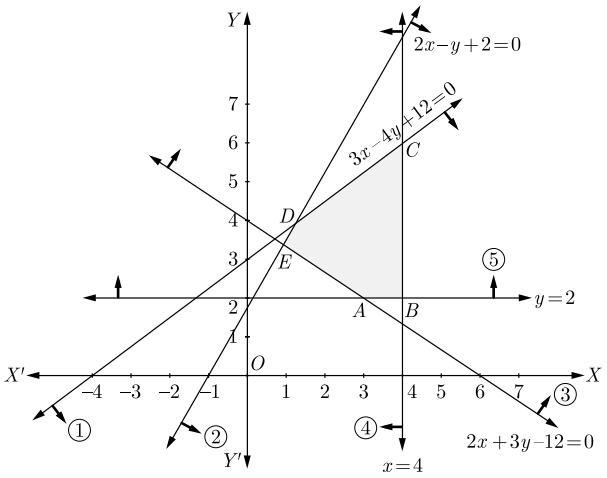
Integrating both sides we get

$$\frac{1}{2}\ln(t-2) - \frac{1}{2}\ln t = \ln x + \ln k$$

$$\frac{1}{2}\ln\frac{(t-2)}{t} = \ln x + \ln k$$

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Hence, A is (3, 2).



To find corner point i.e, vertex B, we are to solve the lines $x = 4$ and $y = 2$

Hence, Point B is (4, 2)

To find corner point C, let us solve $x = 4$ and line Eq. (7) i.e., $3x - 4y + 12 = 0$

Putting $x = 4$ in Eq. (7), we get

$$3(4) - 4y + 12 = 0$$

$$-4y = -24$$

i.e., $$y = 6$$

Hence, point C is (4, 6)

To find corner point D, let us solve lines Eq. (7) and Eq. (8)

$$3x - 4y + 12 = 0$$

and $$2x - y + 2 = 0$$

$2 \times \text{Eq.}(7) - 3 \times \text{Eq.}(8)$ gives

$$-5y + 18 = 0$$

$$y = \frac{18}{5}$$

Putting $y = \frac{18}{5}$ in Eq. (8), we get

$$2x - \frac{18}{5} + 2 = 0$$

$$2x = \frac{18}{5} - 2$$

$$= \frac{18 - 10}{5} = \frac{8}{5}$$

$$x = \frac{4}{5}$$

Hence, D is $\left(\frac{4}{5}, \frac{18}{5}\right)$.

To find corner point (i.e., vertex) E, let us solve lines Eq. (8) and Eq. (9),

$$2x - y + 2 = 0 \quad ...(8)$$

$$2x + 3y - 12 = 0 \quad ...(9)$$

Eq. (8) − Eq. (9) gives

$$-4y + 14 = 0$$

$$y = \frac{7}{2}$$

Putting $y = \frac{7}{2}$ in Eq. (8), we get

$$2x - \frac{7}{2} + 2 = 0$$

$$2x = \frac{7}{2} - 2 = \frac{3}{2}$$

$$x = \frac{3}{4}$$

Hence, Point E is $\left(\frac{3}{4}, \frac{7}{2}\right)$.

Step III :

Let us find the values of objective function $Z = 3x + 5y$ at corner points.

At A (3,2);	$Z = 3(3) + 5(2) = 19$
At B (4,2);	$Z = 3(4) + 5(2) = 22$
At C (4,6);	$Z = 3(4) + 5(6) = 42$
At $D\left(\frac{4}{5}, \frac{18}{5}\right)$;	$Z = 3\left(\frac{4}{5}\right) + 5\left(\frac{18}{5}\right) = \frac{12 + 90}{5} = \frac{102}{5}$
At $E\left(\frac{3}{4}, \frac{7}{2}\right)$;	$Z = 3\left(\frac{3}{4}\right) + 5\left(\frac{7}{2}\right) = \frac{9 + 70}{4} = \frac{79}{4}$

Hence, Maximum value of Z is 42 at C (4, 6) and minimum value of Z is 19 at A (3,2).

35. Solve the differential equation :

$$(x - y)\,dy - (x + y)\,dx = 0$$

OR

Solve the differential equation $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$.

Sol :

$$(x - y)\,dy - (x + y)\,dx = 0$$

$$(x - y)\,dy = (x + y)\,dx$$

$$\frac{dy}{dx} = \frac{x + y}{x - y} \quad ....(1)$$

Let, $$y = Vx \quad ....(2)$$

Differentiating both sides with respect to x we get

$$\frac{dy}{dx} = V\frac{dx}{dx} + x\frac{dV}{dx}$$

$$\frac{dy}{dx} = V + x\frac{dV}{dx} \quad ....(3)$$

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$$\begin{bmatrix}5 & 6 & 5\\ 2 & 3 & 3\\ 1 & 1 & 2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=\begin{bmatrix}370\\ 170\\ 80\end{bmatrix}$$

or $AX = B$

where $A=\begin{bmatrix}5 & 6 & 5\\ 2 & 3 & 3\\ 1 & 1 & 2\end{bmatrix}$, $X=\begin{bmatrix}x\\ y\\ z\end{bmatrix}$ and $B=\begin{bmatrix}370\\ 170\\ 80\end{bmatrix}$

Since, $|A| = 5(6-3)-6(4-3)+5(2-3)$

$= 4 \neq 0$,

thus A^{-1} exists, so that the unique solution of $AX = B$ is $X = A^{-1}B$.

Here, adj $A=\begin{bmatrix}3 & -7 & 3\\ -1 & 5 & -5\\ -1 & 1 & 3\end{bmatrix}$

Here, $A^{-1}=\frac{\text{adj.}A}{|A|}=\frac{1}{4}\begin{bmatrix}3 & -7 & 3\\ -1 & 5 & -5\\ -1 & 1 & 3\end{bmatrix}$

Now $X = A^{-1}B$

$$=\frac{1}{4}\begin{bmatrix}3 & -7 & 3\\ -1 & 5 & -5\\ -1 & 1 & 3\end{bmatrix}\begin{bmatrix}370\\ 170\\ 80\end{bmatrix}=\begin{bmatrix}40\\ 20\\ 10\end{bmatrix}$$

Hence $x = 40, y = 20$ and $z = 10$. Hence, the production of products A, B and C are 40 units, 20 units and 10 units, respectively.

37. A steel can, tin can, tin, steel packaging, or can is a container for the distribution or storage of goods, made of thin metal. Many cans require opening by cutting the "end" open; others have removable covers. They can store a broad variety of contents: food, beverages, oil, chemicals, etc.



A tin can manufacturer a cylindrical tin can for a company making sanitizer and disinfector. The tin can is made to hold 3 litres of sanitizer or disinfector.

Based on the above information, answer the following questions.

(i) If r be the radius and h be the height of the cylindrical tin can, find the surface area expressed as a function of r.

(ii) Find the radius that will minimize the cost of the material to manufacture the tin can.

(iii) Find the height that will minimize the cost of the material to manufacture the tin can.

OR

(iv) If the cost of the material used to manufacture the tin can is ₹100/m² find the minimum cost. $\sqrt[3]{\frac{1500}{\pi}} \approx 7.8$

Sol :

(i) Given, r cm is the radius and h cm is the height of required cylindrical can.

Given that,

$$3\,l = 3000\,\text{cm}^3 \qquad (1l = 1000\,\text{cm}^3$$

$$\pi r^2 h = 3000$$

$$h = \frac{3000}{\pi r^2}$$

Now, the surface area, as a function of r is given by

$$S(r) = 2\pi r^2 + 2\pi rh$$

$$= 2\pi r^2 + 2\pi r\left(\frac{3000}{\pi r^2}\right)$$

$$= 2\pi r^2 + \frac{6000}{r}$$

(ii) $$S(r) = 2\pi r^2 + \frac{6000}{r}$$

$$S'(r) = 4\pi r - \frac{6000}{r^2}$$

For critical points, we put $S'(r) = 0$

$$\frac{4\pi r^3 - 6000}{r^2} = 0$$

$$r^3 = \frac{6000}{4\pi}$$

$$r = \left(\frac{1500}{\pi}\right)^{\frac{1}{3}}$$

Also, $$S''(r) = 4\pi + \frac{12000}{r^3}$$

$$S''(r)\big|_{r=\sqrt[3]{\frac{1500}{\pi}}} = 4\pi + \frac{12000 \times \pi}{1500}$$

$$= 4\pi + 8\pi = 12\pi > 0$$

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(iii) The cost of material for the tin can is minimized when $r = \sqrt[3]{\frac{1500}{\pi}}$ cm and the height is

$$\frac{3000}{\pi\left(\sqrt[3]{\frac{1500}{\pi}}\right)^2} = 2\sqrt[3]{\frac{1500}{\pi}} \text{ cm}$$

OR

(iv) Minimum surface area,

$$= \frac{2\pi r^3 + 6000}{r}$$

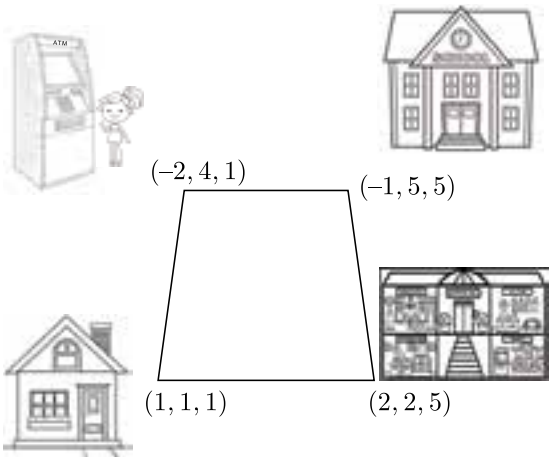
$$= \frac{2\pi \cdot \frac{1500}{\pi} + 6000}{\sqrt[3]{\frac{1500}{\pi}}}$$

Cost of 1 m^2 material $= ₹100$

Cost of 1 cm^2 material $= ₹\frac{1}{100}$

Minimum cost $= ₹\frac{1153.84}{100} = ₹11.538$

38. Lavanya starts walking from his house to shopping mall. Instead of going to the mall directly, she first goes to ATM, from there to her daughter's school and then reaches the mall. In the diagram, using co-ordinate geometry the location of each place is given.



Based on the above information, answer the following questions.

(i) What is the distance between House and ATM?

(ii) What is the distance between ATM and school?

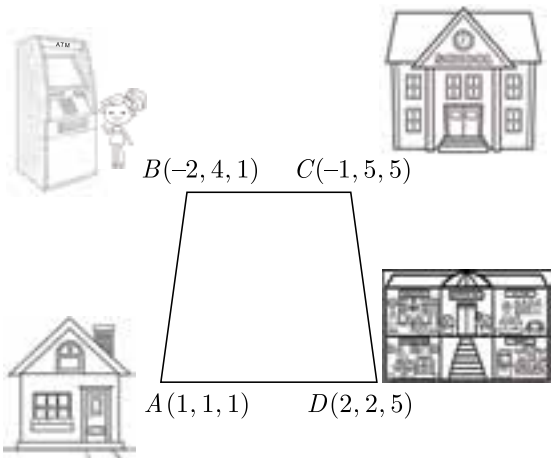
(iii) What is the total distance travelled by Lavanya?

OR

(iv) What is the extra distance travelled by Lavanya in reaching the shopping mall?

Sol :

We can redraw the given diagram as below.



(i) $\overrightarrow{AB} = (-2\hat{i} + 4\hat{j} + \hat{k}) - (\hat{i} + \hat{j} + \hat{k})$

$= -3i + 3j$

$\overrightarrow{AB} = \sqrt{(-3)^2 + 3^2}$

$= \sqrt{9+9} \ \sqrt{18}$

$= 3\sqrt{2}$

Distance between House (A) and ATM (B) is $3\sqrt{2}$ units.

(ii) $\overrightarrow{BC} = (-\hat{i} + 5\hat{j} + 5\hat{k}) - (2\hat{i} + 4\hat{j} + \hat{k})$

$= \hat{i} + \hat{j} + 4\hat{k}$

$|\overrightarrow{BC}| = \sqrt{1^2 + 1^2 + 4^2}$

$= \sqrt{1+1+16}$

$= \sqrt{18} = 3\sqrt{2}$

Distance between ATM (B) and School (C) is $3\sqrt{2}$ units.

(iii) $\overrightarrow{CD} = (2\hat{i} + 2\hat{j} + 5\hat{k}) - (-\hat{i} + 5\hat{j} + 5\hat{k})$

$= (3\hat{i} - 3\hat{j})$

$|\overrightarrow{CD}| = \sqrt{3^2 + (-3)^2}$

$= \sqrt{9+9} = 3\sqrt{2}$

Distance between School (C) and Shopping mall (D) is $3\sqrt{2}$ units.

Total distance travelled by Lavanya

$= |\overrightarrow{AB}| + |\overrightarrow{BC}| + |\overrightarrow{CD}|$

$= (3\sqrt{2} + 3\sqrt{2} + 3\sqrt{2})$ units.

$= 9\sqrt{2}$

OR

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(iv) Distance between house and shopping mall is $|\overrightarrow{AD}|$

Now, $|\overrightarrow{AD}| = \hat{i} + \hat{j} + 4\hat{k}$

$$|\overrightarrow{AD}| = \sqrt{1^2 + 1^2 + 4^2}$$

$$= \sqrt{1 + 1 + 16}$$

$$= \sqrt{18} = 3\sqrt{2}$$

Thus, extra distance travelled by Lavanya is reaching shopping mall $= (9\sqrt{2} - 3\sqrt{2})$ units $= 6\sqrt{2}$ units.

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